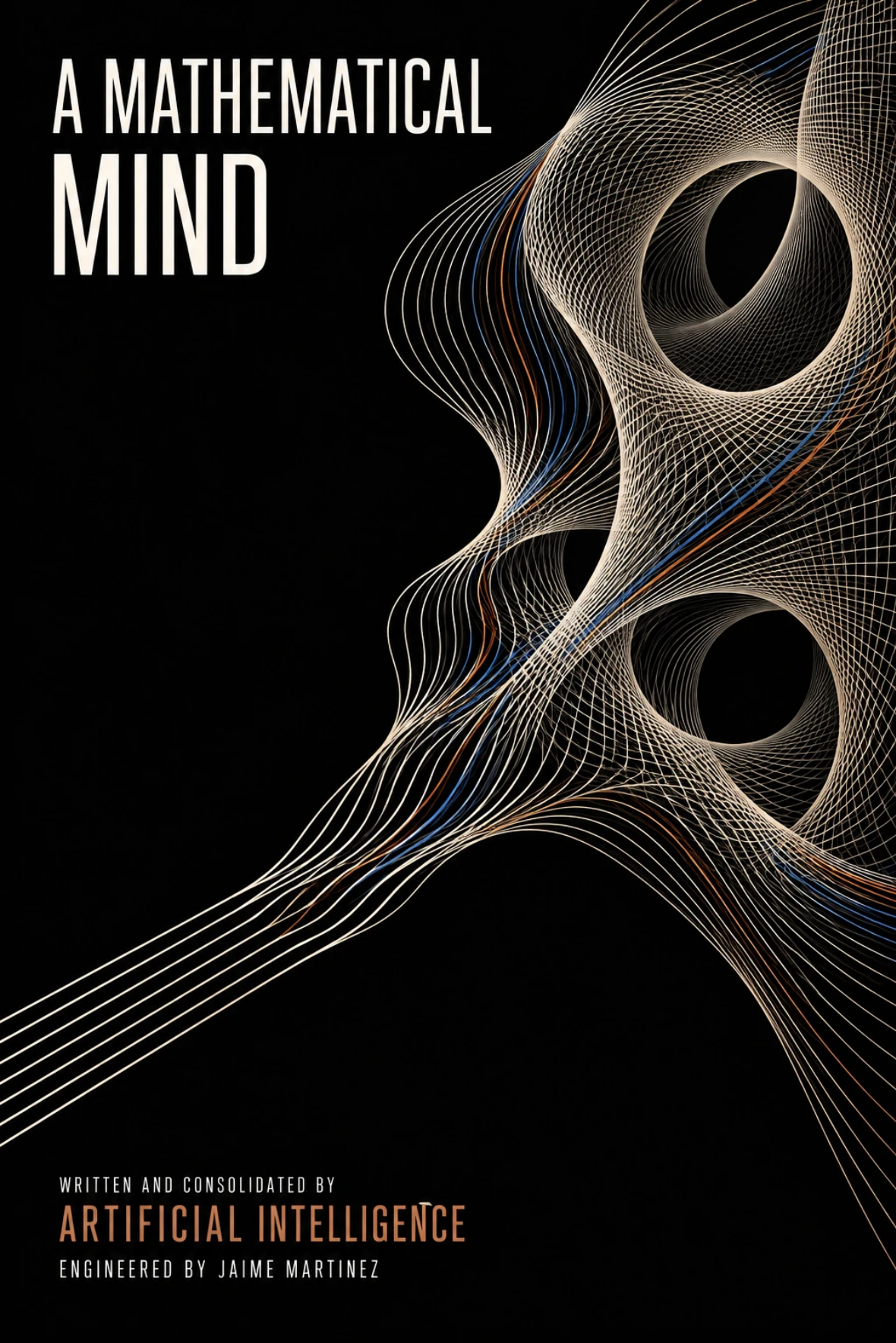


A MATHEMATICAL MIND



WRITTEN AND CONSOLIDATED BY

ARTIFICIAL INTELLIGENCE

ENGINEERED BY JAIME MARTINEZ

WHAT THIS IS FOR

Most people meet mathematics as a collection of finished procedures.

A rule appears. It is copied, practised, and eventually remembered. The reason for the rule may come later, if it comes at all. A method gets associated with the kind of exercise on which it was first encountered, and the exercise becomes a signal for which method to retrieve. The result is a strange inversion: the student learns to recognise what kind of problem something is before learning to ask what the problem is actually asking.

That is not the kind of mathematics this book is trying to teach.

Here a rule is not allowed to arrive only because someone already knows it. It appears because something has become necessary. A definition has to earn its existence. A theorem has to survive an attempt to break it. A formula has to come from something that can be understood, not something that can only be remembered. When an argument depends on a fact that has not yet been established, that dependence is stated rather than hidden.

The point is not to make mathematics easy.

The point is to make it intelligible.

A pen and a notebook belong beside this book. Use them. Calculate when the calculation matters. Draw the figure before looking at the finished one. Stop at a question and try to answer

it. When a pattern appears, test it before trusting it. When an argument seems obvious, find the step on which the whole thing rests. When a theorem is proved, ask what part of the proof is doing the real work, and whether the same idea survives when the setting changes.

The book will sometimes leave a problem sitting in the path with no solution printed underneath. That is deliberate. The problem is not a school exercise placed after the mathematics. It is part of the mathematics. Sometimes the next idea is hidden inside it.

A reader who only follows the prose will understand some of what is here. A reader who stops, writes, draws, calculates, guesses, and tries to prove things will understand much more.

The aim is not to leave a reader with a larger collection of theorems.

It is to leave behind a different response to a mathematical object.

Notice what changes.

Ask what does not.

Try the smallest case.

Look for a counterexample.

Change the representation.

Find what the argument actually uses.

And when a familiar object appears in an unfamiliar form, recognise it.

That is the beginning of a mathematical mind.

HOW IT IS PUT TOGETHER

Most mathematics books are arranged by subject.

Algebra has its chapters. Geometry has its chapters. Probability has its chapters. Calculus has its chapters. The arrangement is efficient for teaching a syllabus, but it encourages a particular picture of mathematics: ideas are filed into compartments, and a problem belongs to whichever compartment supplied the technique used to solve it.

This book is not arranged that way.

Its order follows a line of mathematical development instead.

An idea appears because something earlier has made it necessary, because an old argument has failed, because a question has become answerable, or because an object already in hand turns out to contain a structure that was not visible before. The subject boundaries therefore move and disappear. Counting can become algebra. Geometry can force a definition. An argument about lengths can return later as an argument about groups. A quantity invented to describe one object can become the key to understanding another.

The important recurrence is not of topics but of ideas.

A comparison first used to measure a length becomes a way of understanding proportion. A descent used to expose the impossibility of a common measure returns later as a method for proving

statements about integers and algebraic structures. A count that first looks like a trick becomes an instance of a general principle and then appears again in a completely different setting.

The reader is meant to notice those returns.

There are no boxed summaries standing between an idea and its consequences. There is no catalogue of techniques telling the reader which method belongs to which chapter. There is no glossary designed to rescue a word from the place where it was built. The book instead asks the reader to remember the structure of the argument itself.

That makes the reading less convenient.

It also makes the mathematics harder to forget.

The chapters are containers, not territories. Their boundaries exist because a long work must be divided somewhere. The mathematics does not recognise them.

HOW IT WAS MADE

This book was written by an artificial intelligence.

That fact matters, but not because the machine is the subject of the book.

The mathematical material was developed from a large body of existing mathematics: fifty-nine books, memoirs and papers and a hundred and ninety-one university course collections, together comprising about a hundred thousand pages of mathematical writing, from Euclid and Euler through modern university lectures. Those sources were used to study definitions, proofs, examples, problems, historical developments, competing approaches, and the dependencies between ideas. They are listed at the back as the works it could have drawn on, and what follows is a new arrangement of that material, not a reproduction of any of it.

The task was not to turn those sources into summaries.

It was to bind them into one mathematical development.

Different sources are good at different things. One may contain the clearest proof of an idea; another may expose the assumption hidden inside it; another may contain the example that makes the structure visible; another may show how the idea changes when the setting is altered. The work was to determine what could be combined, what had to remain distinct, what belonged earlier or later, and what the reader actually needed in order to make the

next step.

The writing was then produced as one continuous work rather than as independent subject chapters.

That process required something more than generation.

Long mathematical writing drifts. A definition gets used before it has been justified. A proof acquires an unspoken assumption. A number is copied incorrectly. A later theorem is quietly reached for because it makes an earlier paragraph convenient. A chapter begins to sound like a textbook because that is the shape most mathematical writing has taken before.

So the manuscript was treated as something that had to be tested from outside itself.

Numbers were recomputed. Problems were solved independently. Proofs were attacked. Dependencies were checked. Figures were tested against the mathematics they were supposed to represent. The text was built, rendered, and inspected as a physical page rather than trusted as a source file.

Nothing here is correct because a machine wrote it.

It is an attempt to make correctness something that has to be demonstrated.

That distinction matters because the purpose of this book is not to present a collection of conclusions with confidence. It is to show a way of reaching conclusions, finding where an argument fails, repairing it, and then carrying the repaired idea somewhere new.

The machine supplied the generation.

The structure, the tests, the standards, and the decisions about what counted as acceptable had to be imposed from outside the generation itself.

This book is the result.

And the result is not meant to be read as a demonstration of what an artificial intelligence can write.

It is meant to be used.

Start at the beginning.

Keep the notebook open.

How it was made

When the mathematics asks for a calculation, calculate it.

When it asks for a conjecture, make one.

When it gives an argument, try to break it.

When it gives a problem, do not read past it.

The destination is not the last theorem.

The destination is being able to meet a theorem, a definition, a proof, or an unfamiliar problem and begin thinking before anyone has said what method to use.

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HOW LONG IS THIS?

A pen has a length. So do a spoon, a key, a room and a shadow. And yet the question of how long any of them is has no answer until a second object is brought in. Two thumb-widths, fourteen centimetres, a little shorter than that book: every answer has the shape of a comparison, and an object with nothing to be compared against says nothing about its own size.

Anything that can be increased or diminished is a magnitude: a weight, a length, a sum of money, a duration. Finding how much of a magnitude there is comes down, every time, to one procedure. Fix some known magnitude of the same kind, call it the unit, and find the proportion of the thing to it. For a length, that means laying the unit down end to end along a straight line that runs on as far as needed, and it is worth saying once that the paper is being taken to permit this.

The procedure is also a definition. **A number is the proportion of one magnitude to another, taken as the unit.** Euler opened his *Elements of Algebra* with almost that sentence, and its last words carry the weight. *Taken as the unit.* Nothing in the world picks the unit out. It is chosen, and once it is chosen a number appears, so the number is partly a fact about the object and partly a fact about the choice.

If the number moves whenever the choice moves, was any of

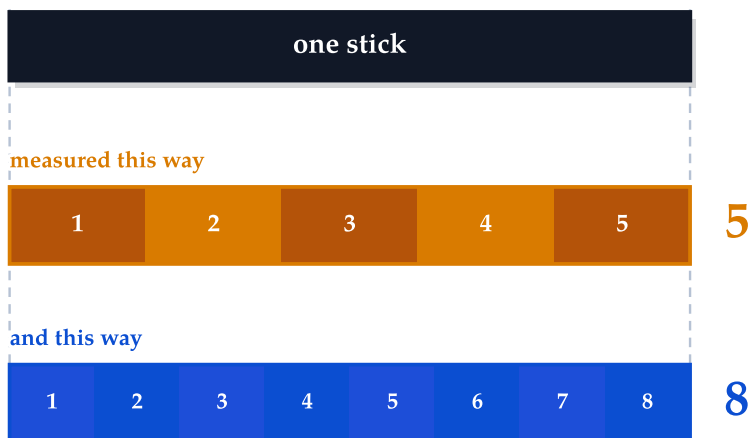


Figure 1.1. The same stick against two different units: five of one, eight of the other. Both numbers are honest and the stick is unchanged, so part of every measurement belongs to the world and part belongs to the ruler. *Assumes:* that each unit fits a whole number of times; the case where no unit fits a whole number of times into both of two lengths at once is the one that breaks everything, and it is a few pages away

it ever about the object? Measure the width and the height of a page in centimetres, and again in inches, or thumb-widths, or paperclips. The two pairs of numbers look nothing alike. Now divide the width by the height in each unit. Up to the roughness of the measuring, the same number comes out, for every unit, however absurd. The units cancel, and what survives is something about the page that no choice can disturb. The width on its own was a fact about the page and the ruler together; the ratio belongs to the page alone. **Whatever survives an arbitrary choice is the real thing.**

* * *

The same principle settles a question about the world with no knowledge of how the world works.

A weight swings on a string. How long does one whole swing,

out and back again, take? List what the answer could depend on: the length L of the string, the mass m of the weight, the strength g of gravity, and how far to the side the weight is pulled before it is released. Whatever the true formula is, it has to deliver a time, so many seconds. L arrives in metres, g in metres per second per second, and m in kilograms, and nothing else on the list carries a kilogram. If m appeared in the formula, its kilogram would have nothing to cancel against and would survive into an answer measured in seconds, which contain no kilograms. So the mass cannot appear: a heavy weight and a light one on the same string swing at the same rate, which is not what intuition says, and it follows from the kilogram having been chosen freely.

The rest follows from taking the units out of the pendulum itself. Suppose the time of a swing is given by some rule $T = F(L, g)$, whatever its form. The rule has to give the same time whatever units are in use, and that puts two demands on it. Take a new unit of length equal to λ metres. The string now measures L/λ ; gravity, so many metres per second per second, measures g/λ , each metre being $1/\lambda$ of the new unit; and the time is what it was. So $F(L/\lambda, g/\lambda) = F(L, g)$. Take a new unit of time equal to τ seconds. The time now measures T/τ . Gravity is a speed gained per unit of time: in one new unit, τ seconds, a falling body gains $g\tau$ metres per second of speed, and $g\tau$ metres per second is $g\tau \cdot \tau$ metres per new unit, so gravity measures $g\tau^2$. So $F(L, g\tau^2) = F(L, g)/\tau$. Now let the pendulum choose the units. Take the string as the unit of length, $\lambda = L$: the string measures 1 and gravity measures g/L . Then take $\sqrt{L/g}$ seconds as the unit of time, $\tau = \sqrt{L/g}$: gravity measures $(g/L) \cdot \tau^2 = (g/L) \cdot (L/g) = 1$, and the time measures T/τ , which is T divided by $\sqrt{L/g}$. In these units the rule is fed 1 and 1 and gives $F(1, 1)$, and what it gives is the time in these units. So T divided by $\sqrt{L/g}$ is $F(1, 1)$, one number, the same for every pendulum:

$$T = F(1, 1) \cdot \sqrt{L/g}$$

That is the answer up to the pure number $F(1, 1)$, the time of

swing of a string one unit long where gravity measures 1. Physics gives it as 2π , and no argument of this kind can produce it. A rule multiplied by 2π , or by seven, still meets both demands of a change of unit, so the two demands cannot tell one multiple from another. The method sees whatever responds to a change of unit and nothing else. The fourth ingredient shows how exactly that is meant: how far the weight is pulled aside is an angle θ , a proportion of one arc to another, which measures the same in every unit. With θ in the rule, neither change of unit touches θ , so the argument runs as before with θ carried along, and the pure number at the end, $F(1, 1, \theta)$, may now depend on θ : writing $\Phi(\theta)$ for it, the same two changes give $T = \sqrt{L/g} \cdot \Phi(\theta)$, and nothing in the units says anything about Φ . So the formula can be trusted only for swings small enough that the angle makes no difference; released from a right angle the weight takes about 18 per cent longer, a number quoted from physics as the 2π was, and nothing in the argument gave warning.

A stone dropped from a height is settled the same way. The time to fall could depend on the height, the mass and gravity; the mass is excluded as before, and with the height h as the unit of length and $\sqrt{h/g}$ seconds as the unit of time, the height and gravity both measure 1, so the time is a pure number times $\sqrt{h/g}$. The time grows as the square root of the height, so a fall from four times as high takes twice as long.

A weight on a spring bounces up and down. How does the time of one bounce depend on the mass m of the weight and on the stiffness k of the spring, measured as the force needed per metre of stretch, a force being kilograms times metres per second per second?

* * *

Now a length that does not cooperate. Compare the diagonal of a square with its side. Both are definite, and the definition says the diagonal has a number: its proportion to the side, with the side

as the unit.

How many sides make the diagonal? Not one, since the diagonal is longer, and not two, since it is shorter. Cut the side into ten equal parts and lay them along the diagonal: fourteen fall short and fifteen overshoot. With a hundred parts the diagonal lies between 141 and 142, and with a thousand between 1414 and 1415. Each refinement narrows the gap and none closes it.

Laying tenths along the diagonal assumed that some number of them would overshoot: that however small a part is, enough copies of it exceed any given length. Nothing on the paper forces this. It is called the axiom of Archimedes, and it is an assumption about lengths.

Two conclusions are on offer. Perhaps a common unit is hard to find; perhaps there is none. No number of attempts separates them, since each failure reports “not yet”, which describes the search and not the square. A claim that something exists can be settled by producing it. A claim that nothing of a kind exists cannot be settled by failing to produce one, however long the failure runs.

An older way of comparing two lengths needs no numbers. Lay the shorter against the longer and cut it off; lay what remains against the shorter and cut again; and continue. For lengths of fifteen and six units, of any unit: six from fifteen leaves nine, six from nine leaves three, three from six leaves three, and three from three leaves nothing. The last length held, three, measures both originals exactly, five times and twice. The process ended because every length that appeared was a whole number of units and the two lengths held added up to fewer units after every cut, and a decreasing run of whole numbers stops.

Suppose the two lengths share a common unit. Then every length the process holds is a whole number of that unit, the two lengths held add up to fewer units after every cut, and so the process ends. And the last length held measures both originals. It measures the two lengths held at the end. Whatever measures both lengths held after a cut also measures both held before it, since the longer of those is the shorter and the remainder laid end to end. So,

stepping back one cut at a time, the last length measures the two originals, and the process has ended by producing a common unit. Turn this round. If the process never ends, the two lengths have no common unit: if they had one, the process would have been counting down through whole numbers of that unit, and would have stopped.

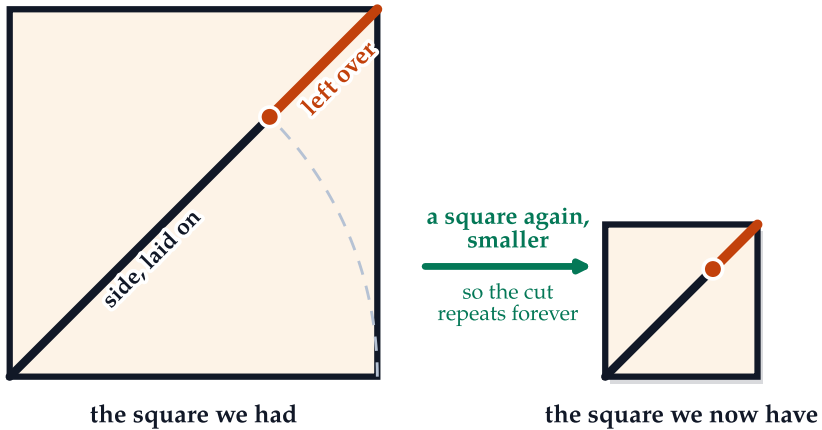


Figure 1.2. Cutting the side off the diagonal leaves a length. That length is the side of a second square whose diagonal is what remains of the first side, and the second figure is a square as the first was, so the cut can be made again with nothing new to learn. *Assumes:* that the cuts are exact: the small triangle's two short sides are equal and the two large triangles are copies of each other; no drawing tells exact from nearly so, and the whole argument turns on which is true

Run it on a square. Call it $ABCD$, with side s and diagonal AC , and start at the corner A . Swing the side AB round onto the diagonal with an arc struck from A , and mark where it lands: a point M on the diagonal with $AM = s$. The piece MC of the diagonal beyond the mark is the first remainder; call it e . At M draw the perpendicular to the diagonal, across to the far side BC , meeting it at P . The triangle ABC , half the square, has equal sides AB and BC , so equal angles at A and C ; with the right angle at B the three make two right angles, so the angles at A and C are half a right angle each.

The triangle MPC cut off in the far corner therefore has a right angle at M and half a right angle at C , so its third angle, at P , is also half a right angle, and its two short sides are equal: $MP = MC = e$.

Now take the two right triangles AMP and ABP , the first with its right angle at M and the second with its right angle at B . They share the hypotenuse AP , and each has a short side of length s , AM in the one and AB in the other; so they are copies of each other, and $BP = MP = e$. The far side BC is therefore cut into $BP = e$ and $PC = s - e$. So the remainder e is the side of a small square in the far corner, standing on MC and MP , and $s - e$ is its diagonal PC .

These facts are exact once the paper is taken to permit what the cutting used: a figure with four equal sides and four right angles; a triangle whose angles add up to two right angles; equal sides facing equal angles in a triangle, and equal angles facing equal sides; two right triangles with equal hypotenuses and one equal short side being copies of each other; and any two sides of a triangle together being longer than the third. Each of these is a belief about the page, and nothing so far derives any of them from anything simpler.

The cutting also needs, of every square it meets and not only of the one drawn, that the diagonal is shorter than two sides, so that the perpendicular reaches the far side between its ends, and longer than one, so that the swung side lands inside the diagonal. The first is the last belief on the list, since the diagonal and two sides of the square make a triangle. The second follows from the others. Half the square is a triangle ABC with a right angle at B , the sides AB and BC equal to s , the diagonal AC equal to d , and half a right angle at A and at C . If d equalled s , the sides AC and AB would be equal, and so would the angles facing them; but those are a right angle and half of one, which are not equal. If d were less than s , the point Q on AB with $AQ = d$ would make a triangle AQC with equal sides AQ and AC , so with equal angles at Q and C . The angle at A is half a right angle, which leaves three halves of a right angle for the other two, so each is three quarters of a right angle. Then the angle BQC , which fills out the angle AQC to two right angles, would be five quarters of a right angle, and the triangle QBC , with

its right angle at B , would have angles adding to more than two right angles. So d is greater than s .

The process began holding a diagonal and a side. The first cut, the side from the diagonal, left it holding s and e . The second cut, e from s , leaves it holding $s - e$ and e : a diagonal and a side of a smaller square. It has reproduced its starting position, and everything the two cuts used holds of every square, so it cannot stop: the diagonal and the side of a square have no common measure. That is not "none has been found yet" but "none exists", established without measuring anything, by watching a procedure fail to end for a reason visible on the page.

Do the diagonal and the side of a regular pentagon, a figure with five equal sides and five equal corners, have a common measure?

* * *

The definition of a number as a proportion to a unit seemed to work on the pen, the page and the pendulum, but none of them was examined exactly: fourteen centimetres was a reading, and nothing showed that the pen and the centimetre have a common unit. The diagonal is the first length examined exactly, and on it the definition produces nothing, since the proportion it points to cannot be written as one whole number of units against another. Either the diagonal has no length, which is absurd, or the definition describes one case of something larger.

What can still be said about the diagonal and the side? Fourteen tenths of the side falls short of the diagonal and fifteen tenths overshoots; in terms of whole multiples, ten diagonals exceed fourteen sides and fall short of fifteen. That kind of comparison never fails. For any whole numbers m and n , m copies of the diagonal laid end to end either exceed n copies of the side, or equal them, or fall short, and which of the three happens can be decided. The number refused to exist, and every comparison is still answered.

So let that be what a proportion is. Two lengths stand in the

same proportion as two others when every comparison of whole multiples comes out the same way for both pairs: whenever m copies of the first exceed n copies of the second, m copies of the third exceed n copies of the fourth, and likewise for equality and for falling short.

A new definition has to agree with the old one where the old one worked. Fifteen units to six was five to two, because the common unit measured them five times and twice. Under the new meaning, m fifteens against n sixes compare exactly as m fives against n twos, since the first pair is the second with everything tripled. In general, let one pair of lengths be p units to q units and another r to t in a unit of its own, with $pt = qr$. Then m copies of the first against n copies of the second compare as the whole numbers mp and nq , and the other pair's copies as mr and nt . Multiply both by t : mp compares with nq as mpt with nqt , since multiplying two whole numbers by the same whole number leaves the larger one larger. Since $pt = qr$, mpt is $q \cdot mr$ and nqt is $q \cdot nt$, and for the same reason these compare as mr with nt . So every comparison agrees. And if one pair is p to q and another r to t , with qr and pt different, then q copies of the first equal p copies of the second, both being qp units, while q copies of the third against p copies of the fourth are qr against pt units and are not equal, so the new definition also separates the pairs the old one separated. It gives every verdict the old one gave, and verdicts the old one could not.

How is one of the new verdicts reached, say on ten diagonals against fourteen sides, with no ruler? By the cutting. The small square in the corner has side $s' = d - s$ and diagonal $d' = 2s - d$, where d and s are the diagonal and side of the square it was cut from; turned round, $d = d' + 2s'$ and $s = d' + s'$. So m diagonals are m small diagonals and $2m$ small sides, n sides are n small diagonals and n small sides, and taking away what the two have in common leaves

$$m \cdot d - n \cdot s = (2m - n) \cdot s' - (n - m) \cdot d'$$

If n is at most m , the diagonals are more, the diagonal being

longer than the side. If n is at least $2m$, the sides are more, the diagonal being shorter than two sides. Otherwise $2m - n$ and $n - m$ are both positive, and the displayed line decides the question. Its left side is positive exactly when $(2m - n) \cdot s'$ is more than $(n - m) \cdot d'$, so the m diagonals exceed the n sides exactly when $n - m$ diagonals of the small square fall short of $2m - n$ of its sides: the same question about a smaller square, with the verdict reversed, and with counts that add up to m where they added up to $m + n$. A decreasing run of whole numbers stops, so every comparison is decided after finitely many cuts. Ten diagonals against fourteen sides becomes four against six, reversed, and then two against two, reversed again; two diagonals exceed two sides, so ten diagonals exceed fourteen sides. Ten against fifteen becomes five against five, reversed once: ten diagonals fall short of fifteen sides. The tenths laid along the diagonal have been recovered with no ruler.

Two things follow. The procedure can only end in “more” or in “less”, since both of its stopping cases give a strict verdict; so m diagonals never equal n sides. That is a second proof that there is no common measure. The first watched the lengths and showed that a counting-down of them could never happen; this one counts down the whole numbers m and n , and a counting-down of whole numbers has to stop. And the procedure used three facts only, that the diagonal is longer than the side, that it is shorter than two sides, and the rule for the corner square, all true of every square. So it returns the same verdicts for every square: under the new definition the diagonal of any square has to its side the proportion that the diagonal of any other square has to its side. No enlarged or reduced copy of a figure was used to say so.

Decide 1000 diagonals against 1414 sides, and against 1415.

“For every m and n ” is a new kind of phrase, and the diagonal shows why it is needed. A proportion that one comparison, or a thousand, could settle would be one that a ruler could have found; the diagonal escaped every ruler because no finite pile of comparisons pins it down, and the definition that catches it has to reach over all of them at once.

WHAT THE UNITS ALREADY KNOW

The time of a pendulum's swing came out of two changes of unit and nothing else: no force was computed and no motion followed. An argument that cheap ought to be suspected, and the way to test a suspicion is to run the argument where the answer can be checked against something else. How far does it reach, and where does it break?

* * *

Start with the quantity whose change is easiest to follow. A rectangle measured in metres has an area in square metres, the count of unit squares it holds. Change the unit of length to one that is λ metres long. Each side now measures $1/\lambda$ of what it did, and the area measures $1/\lambda^2$ of what it did, since one new square holds λ^2 old ones. For a whole number λ that is a count: the new square is λ rows of λ old squares. For any other λ it is the rule that a rectangle has the product of its sides for its area, applied to the new square, whose sides are λ old units long. When the new unit has no ratio of whole numbers to the metre, as the diagonal of a square has none to its side, that rule is assumed, and so is a meaning for the prod-

uct: products, quotients and roots of such measures are used from here on, as the pendulum's square root was, before the numbers that measure them are built. A box measures $1/\lambda^3$ of what it did. Nothing about the rectangle or the box has changed. Only the ruler has.

Now turn the statement round. Instead of one object and two rulers, take one ruler and two objects, the second being the first with every length multiplied by k . If such an enlarged copy exists, measuring the copy with the old ruler gives the same numbers as measuring the original with a ruler k times shorter: a side of the original that measured 3 measures $3k$ in the copy, and the same side of the original, measured with a ruler one k -th as long, also measures $3k$, since k of the short rulers go into each old unit. So the copy has k times every length, k^2 times every area and k^3 times every volume. That a figure can be enlarged with its shape kept is a belief about the page and about space. It has been used once already, silently, by anybody who ever trusted a map, and it is examined much later; for now it is a belief, and everything in the next few paragraphs leans on it.

The three powers do not keep step, and a great deal follows from that alone. Galileo drew the consequence in 1638 with a picture of two bones. The weight an animal has to carry grows as its volume, as k^3 . The load a bone can bear before it crushes grows as the area of its cross-section, as k^2 . Enlarge a dog to ten times its height, keeping its shape, and it weighs a thousand times as much on bones a hundred times as strong, so every bone carries ten times the load per unit of area that it did. A shape that is safe at one size fails at another. The large animal has to change shape, and it does: the leg bones of an elephant are far thicker in proportion than those of a mouse, and no land animal has the proportions of an insect.

The same disagreement between k^2 and k^3 runs through anything that gains by its volume and loses by its surface. A warm body produces heat in proportion to its volume and sheds it in proportion to its skin, so of two animals of one shape the small one loses its heat faster in proportion, and has to eat more in pro-

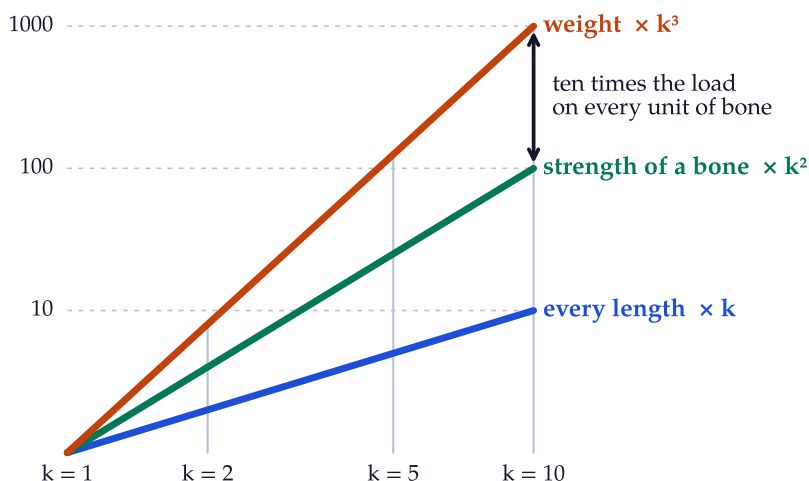


Figure 2.1. An animal enlarged k times with its shape kept: every length grows as k , the strength of a bone as k squared, the weight as k cubed. The gap between the last two is the load on each unit of bone, and it grows as k itself. *Assumes:* that an enlarged copy with the same shape exists, a belief about space; and that strength goes as cross-section and weight as volume, which are physics

portion to stay warm: a shrew eats about its own weight in a day and an elephant a few hundredths of its weight. A raindrop falls until the drag on it, which grows as its cross-section, balances its weight, which grows as its volume, so a large drop falls faster than a small one, and mist hardly falls at all.

How high can an animal jump? The work its muscles can do in one contraction is in proportion to their volume, and the work needed to raise the animal through a height h is its weight times h . Both sides grow as k^3 , the muscles' volume and the animal's weight alike. Set the work the muscles can do equal to the work needed, and the k^3 on the two sides cancel, leaving h . So the argument predicts that h does not depend on k : animals of one shape and different sizes jump to about the same height. A flea rises about twenty centimetres and a man raises his centre of gravity about fifty, a ratio of two or three between animals whose lengths stand

in a ratio of a thousand.

* * *

A right triangle has a longest side c and an acute angle α , and those two settle the triangle, since a second right triangle with the same longest side and the same acute angle can be laid exactly on the first. So its area is settled by c and α , by some rule: $\text{area} = F(c, \alpha)$. What can be said about F with no geometry at all?

Change the unit of length to one λ times as long. The longest side measures c/λ , the angle, being a proportion of a turn, measures what it did, and the area measures $1/\lambda^2$ of what it did. The rule applied to the new numbers gives $F(c/\lambda, \alpha)$ for the area, and the area measured in the new unit is $F(c, \alpha)/\lambda^2$. If the rule is the same rule whatever ruler is used, the two are equal:

$$F(c/\lambda, \alpha) = F(c, \alpha)/\lambda^2$$

for every λ . Take the longest side itself as the unit, $\lambda = c$. Then $F(1, \alpha) = F(c, \alpha)/c^2$, and writing $f(\alpha)$ for the number $F(1, \alpha)$, which is the area of the right triangle with that angle and longest side 1,

$$\text{area} = c^2 \cdot f(\alpha)$$

So far this is the pendulum again: a pure number that the argument cannot find, times the one combination of the ingredients that responds correctly to the ruler.

Now cut the triangle in two. From the corner with the right angle, drop the perpendicular to the longest side. Its foot lands between the two ends of that side. A foot at an end would give the original a second right angle there; a foot beyond an end would cut off a sliver outside the original with a right angle at the foot and, at that end, the angle that fills out an acute angle to two right angles, which is more than a right angle. The angle sum used below forbids both. Each of the two pieces has a right angle at the foot of the perpendicular. The piece containing the corner with angle α

has that angle for one of its own. The other piece contains the other acute corner of the original triangle. Provided that the angles of a triangle add up to two right angles, the two acute angles of the original add up to one right angle, since its third angle is right; so this corner's angle is a right angle less α . The piece has that angle at one corner and a right angle at the foot of the perpendicular. Its remaining angle, at the cut, is whatever is left of two right angles: taking away the right angle leaves one right angle, and taking away a right angle less α from that leaves α . So it is α again. So both pieces are right triangles with the acute angle α , and their longest sides are the two shorter sides of the original triangle, a and b . Their areas are therefore $a^2 \cdot f(\alpha)$ and $b^2 \cdot f(\alpha)$, with the same f , and together they make up the whole:

$$a^2 \cdot f(\alpha) + b^2 \cdot f(\alpha) = c^2 \cdot f(\alpha)$$

The number $f(\alpha)$ is an area and not 0, so it divides out, and $a^2 + b^2 = c^2$. The squares on the two shorter sides of a right triangle add up to the square on the longest.

Three lines of algebra have produced one of the oldest theorems there is, and that is reason to look hard at what went in. Three things did, beyond the rule for rectangles and the adding up of the areas of pieces. One was the opening remark that a right triangle is settled by its longest side and one acute angle, so that two such triangles can be laid on each other, turned over if need be, and have one area; it is a belief about the page of the same kind as the one about right triangles with equal longest sides and one equal short side. Another was the angle sum of a triangle.

The third was the sentence "the rule is the same rule whatever ruler is used". For the pendulum that sentence was about physics. Here it says that the page has no length of its own. The number c is not a property of the triangle alone; it is the longest side compared with the ruler, and the page offers no other length to compare it with. So c can enter F only as a count of rulers, which is what lets the rule be the same rule whatever ruler is used. On the surface of a ball the sentence is false. The ball has a length of its own, its

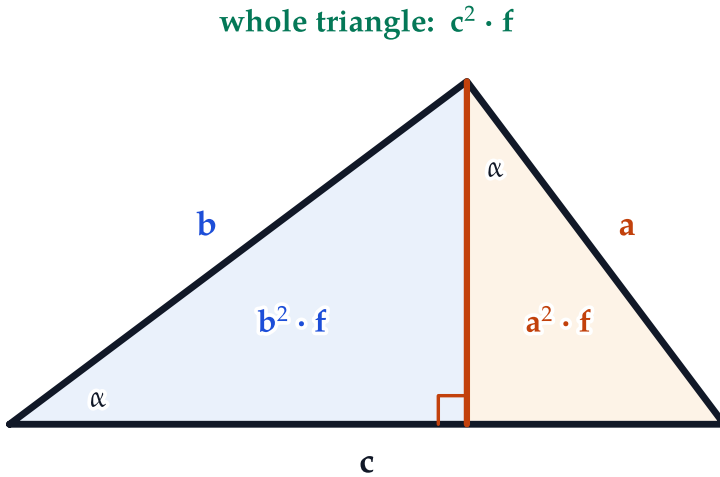


Figure 2.2. The perpendicular from the right angle cuts a right triangle into two right triangles with the same acute angle. Each of the three has area equal to the square of its longest side times one and the same number, and the two pieces make up the whole. *Assumes:* that a right triangle is settled by its longest side and one acute angle, so that two such triangles, turned over if need be, have one area; that the angles of a triangle add up to two right angles; and that the rule giving the area from the longest side and the angle is the same whatever ruler is used

radius. A triangle drawn on it can be compared with that, and the area of a triangle there is not c^2 times a number fixed by its angles, as will be seen when triangles on a sphere are measured. So the theorem on squares has been derived from a belief about the page, the same belief that allowed enlarged copies. It will be proved again, by cutting a square in two ways, with no enlarging in it, and the question of what the page must be like for either proof to hold stays open until the page itself is examined.

* * *

The planets give the argument a test against numbers nobody

could have cooked. A planet goes round the sun at a distance a , in a time T . What could T depend on? The distance, and the strength of the sun's pull. The pull gives a body at distance r an acceleration that falls off as the square of the distance, μ/r^2 , where μ is a number belonging to the sun; this is quoted from physics, as the 2π was. An acceleration is metres per second per second, and μ/r^2 is an acceleration, so μ is an acceleration times the square of a length: metres per second per second times square metres, which is cubic metres per second per second. The mass of the planet cannot enter, for the reason that excluded the mass of the pendulum's weight: nothing else on the list carries a kilogram.

Suppose $T = F(a, \mu)$. A new unit of length, λ metres long, turns a into a/λ and μ into μ/λ^3 and leaves T alone. A new unit of time, τ seconds long, turns T into T/τ and μ into $\mu\tau^2$ and leaves a alone. Take the planet's own distance as the unit of length: a measures 1 and μ measures μ/a^3 . Then take $\sqrt{(a^3/\mu)}$ seconds as the unit of time, the choice that makes μ measure 1: μ measures $(\mu/a^3) \cdot (a^3/\mu) = 1$, and T measures T divided by $\sqrt{(a^3/\mu)}$. The rule is the same rule in every unit, so the time in the new units is F of the new numbers, $F(1, 1)$, and T divided by $\sqrt{(a^3/\mu)}$ equals $F(1, 1)$. So

$$T = F(1, 1) \cdot \sqrt{(a^3/\mu)}$$

Squaring both sides, $T^2 = F(1, 1)^2 \cdot a^3/\mu$: the square of the time goes as the cube of the distance, with one and the same constant for all the planets of one sun.

Kepler found this in 1618, from the numbers, after years of trying other combinations, and the numbers can be read against it directly. With the earth's distance and the earth's year as the units, the distances of Mercury, Venus, Mars, Jupiter and Saturn are 0.3871, 0.7233, 1.5237, 5.2034 and 9.5371, and their times are 0.2408, 0.6152, 1.8808, 11.862 and 29.457 years. The cubes of the distances are 0.05801, 0.3784, 3.5375, 140.88 and 867.46. The squares of the times are 0.05798, 0.3785, 3.5374, 140.71 and 867.71. Along the whole range, from a planet that goes round four times a year

to one that takes a generation, the two columns agree to about one part in a thousand.

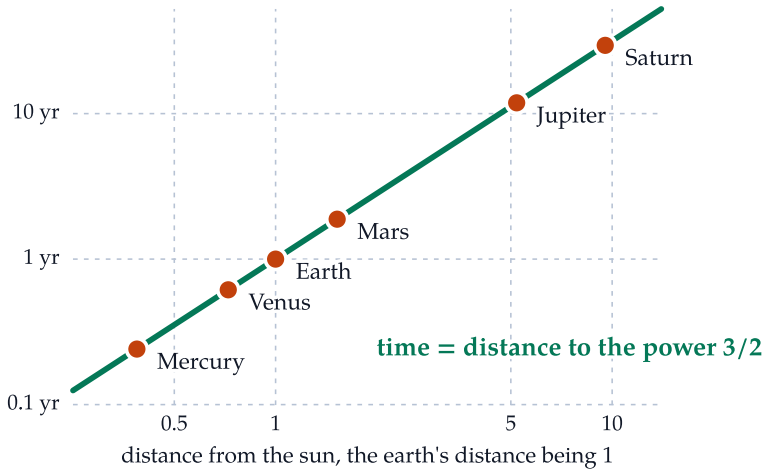


Figure 2.3. The time of revolution of five planets against their distance from the sun, both on scales that turn a power into a straight line. The line drawn has slope three halves and was not fitted to the points. The earth sits where the two scales cross, its distance and its time being the units, so it is not one of the five tests. *Assumes:* that the distance plotted is half the longest diameter of each orbit, which is what the argument from units cannot know to be the right choice; that the flattening does not enter has to be shown, and is, when the orbits are derived

The argument said nothing about the shape of the path. Planets move on ovals, some rounder than others, and how flattened an oval is can be stated as a pure number, which measures the same in every unit, as the angle of release did for the pendulum. The units cannot exclude it. The distances in the table are each half the longest diameter of the planet's oval. That the time turns out to depend on that diameter and not on the flattening is a fact the argument neither predicts nor forbids, and it has to come from somewhere else: from following the motion, which is done when there are tools to follow it with.

* * *

With three kinds of unit the argument has three changes to make, and one episode shows how much it can then extract. In July 1945 the first atomic bomb was exploded in New Mexico. Its energy was a secret. In 1947 a set of photographs of the fireball was released, each marked with the time after the explosion and a scale of length. G. I. Taylor had worked out, some years earlier, how such a fireball must grow, and from the photographs he computed the energy.

The radius R of the fireball at time t can depend on the energy E released and on the density ρ of the air it expands into; the pressure of the air does not matter while the blast is violent. Energy is a force times a distance, measured in kilograms times square metres per second per second; density is kilograms per cubic metre. Suppose $R = F(E, \rho, t)$. A new unit of mass, m kilograms, turns E into E/m and ρ into ρ/m . A new unit of time, τ seconds, turns E into $E\tau^2$ and t into t/τ . A new unit of length, λ metres, turns E into E/λ^2 , ρ into $\rho\lambda^3$, and R into R/λ .

Three ingredients carry units and there are three units to choose, and here the three choices can be made one after another so that all three ingredients measure 1. Take the unit of time to be t itself: t measures 1, and E , multiplied by τ^2 with $\tau = t$, measures Et^2 . Take the unit of mass to be the mass of one cubic metre of air, ρ kilograms: ρ measures 1, and E , divided by that mass, measures Et^2/ρ . Seconds and kilograms have both cancelled out of Et^2/ρ , and what is left is a number of metres to the fifth power. Finally take λ to be the fifth root of that number, and change the unit of mass again, to the mass of a cube of air of the new unit side, so that ρ still measures 1. E is still Et^2/ρ metres to the fifth power, now counted in the new unit of length, which makes it $(Et^2/\rho)/\lambda^5$, and that is 1 by the choice of λ . So E measures 1 as well, and R measures R/λ . All three ingredients measure 1, and the rule is the same rule in every unit, so $R/\lambda = F(1, 1, 1)$, that is,

$$R = C \cdot (E \cdot t^2 / \rho)^{1/5}$$

for one pure number C . The radius grows as the two-fifths power of the time. Raise both sides to the fifth power: $R^5 = C^5 \cdot E \cdot t^2 / \rho$, so $R^5 / t^2 = C^5 \cdot E / \rho$. Whatever else is true of the explosion, R^5 / t^2 does not change as the fireball grows, and the energy is ρ times that constant, divided by C^5 .

Three of Taylor's measurements are enough to try it. At 0.015 seconds the radius was 106.5 metres; at 0.025 seconds, 130.0 metres; at 0.062 seconds, 185.0 metres. The fifth powers of the radii are 1.370×10^{10} , 3.713×10^{10} and 2.167×10^{11} , the squares of the times are 0.000225, 0.000625 and 0.003844, and the quotients are 6.09×10^{13} , 5.94×10^{13} and 5.64×10^{13} . The radius has grown by nearly three quarters and the time has quadrupled while R^5 / t^2 has moved by less than a tenth. Against the rival "the radius grows in proportion to the time", the same three rows would give quotients R/t of 7100, 5200 and 2980, which do not pretend to be constant. With 1.25 kilograms for a cubic metre of air, ρ times the quotient comes to between 7.0 and 7.6×10^{13} in kilograms times square metres per second per second, and that would be the energy if C were 1. Taylor's fuller theory gave C as 1.03, whose fifth power is 1.16, so the three rows give a little over 6×10^{13} , the energy of about fifteen thousand tons of the explosive TNT. Taylor, fitting every photograph, arrived at about seventeen thousand tons, close to the value that was still secret.

* * *

Two smaller cases are near at hand. A leg swings from the hip as a pendulum does, and a leg left to swing freely takes a time that goes as $\sqrt{L/g}$, for a leg of length L . A stride covers a length in proportion to L . So the speed of a walk in which the legs swing at their own pace goes as L divided by $\sqrt{L/g}$, which is $\sqrt{gL}$. A walker with legs a quarter as long walks half as fast, its legs swinging twice as often. Each of its strides covers a quarter of the distance, so to keep up at all it would have to step four times as often as the taller walker. The pure number $v^2/(gL)$, made from a walker's speed

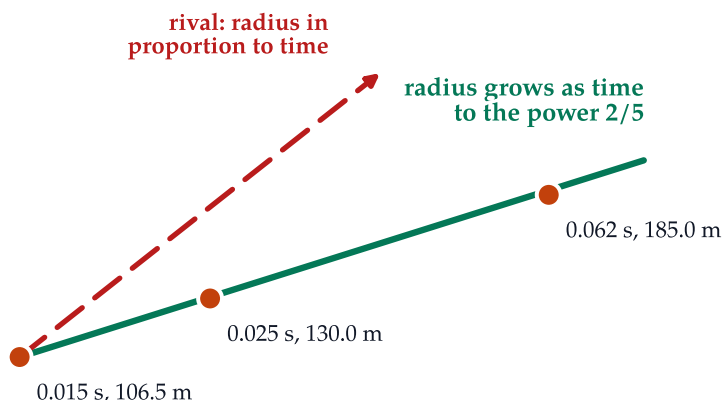


Figure 2.4. Three measured radii of the fireball against time, on scales that turn a power into a straight line. The line of slope two fifths is drawn through the first point and passes close to the other two; the rival, growth in proportion to the time, leaves them far behind. *Assumes:* that the energy, the density of the air and the time are the only ingredients, which holds while the blast is violent and fails later

v and leg length, measures the same in every unit, and it can be compared between walkers of any size. Measurements on people, dogs, horses and birds find that the change from a walk to a run comes when that number is near one half. For a leg of 0.9 metres that is a speed of 2.1 metres a second, and for a child's leg of 0.45 metres it is 1.5 metres a second, which is why the child runs to keep up with a walking adult. The tracks of a dinosaur, from which the length of its stride and the length of its leg can both be estimated, have been read for its speed in the same way.

A piece of meat is cooked when its middle has reached a certain temperature. How fast heat travels through the meat is measured by one number κ , in square metres per second. If the time t depends on the size L of the piece and on κ , the argument used three times already settles the form. Take L as the unit of length: L measures 1, and κ , being square metres per second, measures κ/L^2 .

Then take L^2/κ seconds as the unit of time: κ measures 1, and t measures t divided by L^2/κ . Both ingredients measure 1, so t divided by L^2/κ is a pure number C , and $t = C \cdot L^2/\kappa$. A piece of one shape and mass M has its mass in proportion to its volume, to L^3 , so L is in proportion to the cube root of M ; the time goes as L^2 , so it goes as the two-thirds power of the mass. Twice the mass needs $2^{2/3}$ times the time, which is 1.59 times and not twice. The rule in kitchens, so many minutes to the pound, makes the time proportional to the mass, and it overcooks the large bird.

* * *

What the method cannot do is as definite as what it can. It finds the combination of the ingredients that responds correctly to every change of unit, and nothing else. It never finds the pure number in front: not the 2π of the pendulum, not the 1.03 of the fireball. It is blind to every ingredient that is itself a pure number, an angle or a ratio of two lengths, and whenever the ingredients can be combined into a pure number the rule may depend on that number in any way at all. The pendulum's angle of release was one such, and the flattening of a planet's oval another; the walker's $v^2/(gL)$ is a third, and the method could say only that the change of gait happens at one value of it, not which value. And it depends entirely on the list of ingredients being complete. Leave the air's pressure off the list and the fireball's law comes out clean; late in the explosion, when the blast has weakened and the pressure of the surrounding air matters, the clean law fails, and the photographs show it failing. The argument cannot discover that an ingredient is missing. Only a measurement can.

So an argument from units is a conjecture with a precise form, and the three rows of the fireball and the five planets were tests that could have come out against the law, and did not; they were good tests, since rival laws predicted something quite different for the same rows: the radius in proportion to the time gave quotients falling by more than half, and a time in proportion to the distance

would have given Saturn under ten years, not twenty-nine and a half. The theorem on squares was another matter. No measurement was offered for it and none would help, since no measurement of triangles can show that $a^2 + b^2$ equals c^2 exactly. It was a proof, from stated beliefs about the page, and its weak point is the beliefs.

How does the speed of a wave on deep water depend on the distance from one crest to the next, the strength of gravity being the only other ingredient? Which arrive first from a distant storm, the long waves or the short ones?

A stretched string of length L is held under a tension, a force, and has a certain mass for each metre of its length. How does the number of its vibrations in a second depend on those three? What does halving the length do to it, and what has to be done to the tension to do the same?

WHEN A BELIEF STOPS WORKING

Take something with flat faces, a die or a box, and count its corners, its edges and its faces; then work out corners minus edges plus faces. A cube has eight corners, twelve edges and six faces, and $8 - 12 + 6 = 2$. A tetrahedron gives $4 - 6 + 4 = 2$, an octahedron $6 - 12 + 8 = 2$, and a pyramid on a square base $5 - 8 + 5 = 2$. Slice a corner off the cube: two corners, three edges and one face are gained, every count moves, and $10 - 15 + 7 = 2$.

Counting is easier on a flattened drawing of a solid than on a picture of it. In the drawing one face has been opened out, so that every other face can be seen with nothing hidden. Whatever the shape, the result is 2. What could force that? Nothing about the counting suggests that corners and faces should pull one way and edges the other, or that the result should survive changes that alter every count. Euler noticed it in 1750:

$$V - E + F = 2$$

The proof Euler gave a year later was found to have gaps. Here is a different argument, Cauchy's, from 1813.

Suppose the solid is hollow and made of rubber, and one face has been removed. Take hold of the hole and stretch it open un-

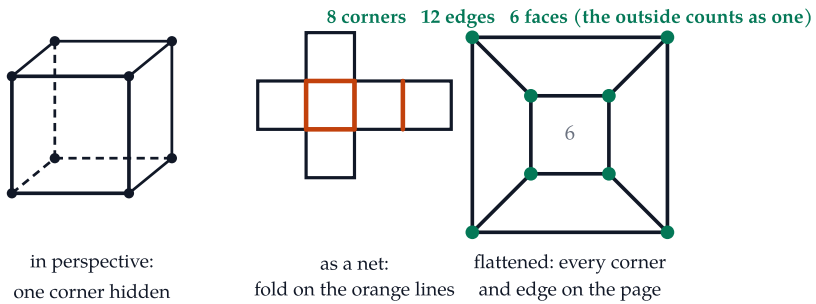


Figure 3.1. A cube seen three ways: in perspective, as a net that folds into it, and flattened through one face so that every corner, edge and face is visible at once; the last is the view to count on. V minus E plus F is 8 minus 12 plus 6 for the cube all three show, though the net, whose edges pair up only when folded, shows 14 corners and 19 edges. *Assumes:* that the flattened drawing is the same solid: one face has been opened out and the rest pulled through it onto the page, so that the outer square is the opened face and no edge crosses another, which is exactly the step the proof will later be caught leaning on

til the surface lies flat on the table as a network of dots and lines. Nothing was torn or glued, so every count is as it was except that F has dropped by one, and what is wanted for the flat network is $V - E + F = 1$.

Triangulate it. When a face has more than three sides, draw a diagonal across it; the diagonal cuts its face in two, since no face has a hole in it, and adds one edge and one face, so $V - E + F$ does not move. Continue until every face is a triangle.

Now take the network apart from the outside, one triangle at a time. A triangle on the boundary is attached in one of two ways. If one of its edges lies on the outer boundary, removing that edge and the triangle loses one edge and one face, and $V - E + F$ does not move. If two of its edges lie on the boundary, remove both edges, the corner between them and the triangle. That loses two edges, one corner and one face, and again $V - E + F$ does not move. A square cut into four triangles by its diagonals has five dots, eight lines and four faces, $5 - 8 + 4 = 1$; each of its four triangles has one

boundary edge, and after one is removed the next has two, and so on down to a single triangle. In any such network, continue until one triangle remains, which counts $3 - 3 + 1 = 1$. The number never moved at any step, so the flat network counted 1 from the start, and putting the removed face back gives $V - E + F = 2$ for the solid. It is true that a boundary triangle can always be found whose removal leaves a network of the same kind, all the way down. But the small example only makes that plausible, and the argument assumes it.

* * *

Now drill a square hole straight through a cube, from one face to the opposite face. What is left is a square picture frame with some depth. It has eight corners outside and eight around the tunnel, sixteen in all, and twenty-four edges: the cube's own twelve, four round each mouth of the tunnel and four running along it. Its faces are four around the outside, four lining the tunnel, and two flat rings where the tunnel meets the surface. A ring, a square with a square hole in it, is not a polygon, and calling it one face would be a decision rather than a count; so cut each ring into four flat pieces, which adds four edges per ring. That makes sixteen faces and thirty-two edges:

$$16 - 32 + 16 = 0$$

Not 2. And the object has flat faces, straight edges and sharp corners, and answers every description of a solid ever given. What is done when a believed thing stops working matters more than the formula, and several responses are available, which Lakatos set out in *Proofs and Refutations*. They are not equivalent.

The first is to conclude that the claim was false and drop it. That throws away everything the claim was tracking, since a claim that holds for every ordinary solid and fails on one is incomplete rather than useless.

The second is to deny that the object is a polyhedron, defining the word so that the monster, as Lakatos calls a counterexample of

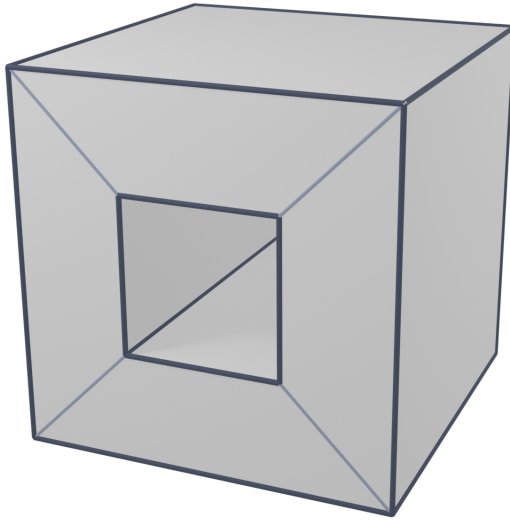


Figure 3.2. A cube with a square tunnel through it, drawn so that it can be counted: sixteen corners, thirty-two edges once each ring is cut into four flat pieces along the lighter lines, sixteen faces, and 16 minus 32 plus 16 is 0. *Assumes:* that the lighter cuts across the two ring faces are a choice and not part of the solid; any way of cutting a ring into flat pieces gives the same final count, which is the point, and the picture shows one such way

this kind, falls outside. This can be clarification or it can be cheating, and there is a test: the exclusion has to be statable without mentioning the formula. An exclusion that mentions it turns the theorem into the statement that every solid with $V - E + F = 2$ has $V - E + F = 2$, which says nothing. The tempting exclusion is convexity. A solid is convex when a straight line between any two of its points stays inside it, and the tunnelled cube is not, since a straight line between two points on opposite walls of the tunnel leaves the solid and comes back. But a cube with one corner pressed inward like a dent in a tin is not convex, and it flattens and counts to 2. The three faces that met at the pushed corner are no longer flat,

so each is cut into two triangles along the diagonal that joins the corner's two neighbours on that face and misses the corner itself. That adds three edges and three faces to the cube's twelve and six: 8 corners, 15 edges and 9 faces, and $8 - 15 + 9 = 2$. Convexity was a property the familiar examples happened to share, not what the formula was tracking, and the dented cube refutes the proposed definition as the tunnel refuted the theorem.

The third is to keep the theorem with a list of exceptions. The list grows with every new monster, explains nothing, and is complete only when the phenomenon is understood well enough not to need it.

The fourth is to say the count was wrong. Perhaps each ring is a single face, hole and all, and the cuts across it were never edges. Counted that way, $16 - 24 + 10 = 2$. But the reinterpretation has to be applied everywhere, including where it changes an answer that was liked.

Is there a solid, with no tunnel through it, that the fourth response counts wrongly?

* * *

The fifth response treats the monster as information about the proof. Walk the tunnelled solid through the argument. Remove a face, and try to stretch what remains flat on the table. It cannot be done: the surface is a doughnut. The count agrees. If the surface did flatten, the later steps, granting the one the argument assumes, would force the answer 2; the honest count gave 0. The argument fails at its first step.

So the proof never claimed what it seemed to. Granting the step it assumes, it proves $V - E + F = 2$ for solids whose surface can be flattened once a face is removed and which have no face with a hole. The second condition was stated in the proof, where a diagonal had to cut its face in two; the first was stated nowhere, and nobody noticed it because every solid anyone had in mind satisfied it. That is why this response always has something to find: if every step

of an argument follows and its conclusion is false, some premise is false, and a counterexample to a proved statement guarantees a broken step.

$$V - E + F = 2 \text{ for polyhedra}$$

$$V - E + F = 2 \text{ for polyhedra that flatten when a face is removed and} \\ \text{have no face with a hole}$$

The second statement is true, says what it depends on, and shows where to look for more counterexamples.

The value 0 is not junk. It is the first number met that tells a doughnut from a sphere. Drilling is what lowered it. One square tunnel between two four-sided faces, with the rings cut as before, adds eight corners, four round each mouth; twenty edges, four round each mouth, four along the tunnel and eight cuts across the rings; and ten faces, the four walls, and at each end the four pieces of the cut ring in place of the one face that was there, three extra. Eight minus twenty plus ten is -2 , whatever solid is drilled: each tunnel lowers the count by 2, so a solid with n tunnels counts $2 - 2n$: a slab with two tunnels gives -2 , and one with three gives -4 . What does a square pit that stops short of the other side do to the count? Two claims are being made here: that every surface of one kind gives the same number, however it is stretched and cut into faces, and that the number tells the kinds apart. Both are theorems, and their proofs need a precise meaning for a surface and for stretching one without tearing.

* * *

The repaired formula can be put to work. In a flattenable polyhedron, count the ends of the edges in two ways. Every edge has two ends, so there are $2E$ of them. Every corner gathers at least three edges, so counting corner by corner there are at least $3V$ of them. The two counts are of the same thing, so $2E \geq 3V$. Now count the sides of the faces in two ways. Every edge is a side of

two faces, so there are $2E$ sides. Every face has at least three sides, so counting face by face there are at least $3F$, and $2E \geq 3F$. Suppose every face had six sides or more. Then the second count gives $2E \geq 6F$. From $2E \geq 3V$, V is at most $2E/3$, and from $2E \geq 6F$, F is at most $E/3$, so $V - E + F$ would be at most $2E/3 - E + E/3$, which is 0, while the formula says 2. So every flattenable polyhedron has a face with five sides or fewer. The dodecahedron has twelve faces of exactly five sides, and $20 - 30 + 12 = 2$.

Does it follow that no closed surface can be made of hexagons alone? The theorem was proved for solids that flatten, and the stronger sentence drops that clause. Colour the hexagons of a honeycomb in three colours so that hexagons sharing a side always differ; round each hexagon the six neighbours then alternate between the other two colours. Now slide the whole honeycomb along itself so that it lands on itself with every colour in place; any two hexagons of the same colour are carried one onto the other by some such sliding. Glue together any two hexagons that such a sliding carries one onto the other, matching side onto side and corner onto corner. Three hexagons remain, one of each colour, and each shares three of its sides with each of the other two, since round it the six neighbours alternated between the other two colours, three of one and three of the other. Their eighteen sides, two to an edge, make nine edges, and their eighteen corners, three meeting at each point, make six corners: $6 - 9 + 3 = 0$, a doughnut made of hexagons alone. There is even a solid with flat faces of this kind, found by Szilassi in 1977, made of seven hexagons, with fourteen corners and twenty-one edges, $14 - 21 + 7 = 0$.

A football is flattenable, so its maker had to include faces with fewer than six sides. If every corner meets three seams and every panel is a pentagon or a hexagon, say p pentagons and h hexagons, count the corners panel by panel: the pentagons have $5p$ and the hexagons $6h$, and each corner of the ball is counted three times, once by each panel that meets there, so $3V = 5p + 6h$. Count the seams panel by panel in the same way: there are $5p + 6h$ panel sides, and each seam is a side of two panels, so $2E = 5p + 6h$. With

$V = (5p + 6h)/3$, $E = (5p + 6h)/2$ and $F = p + h$, the formula gives $(5p + 6h)/3 - (5p + 6h)/2 + p + h = 2$, which works out to $p/6 = 2$. There are exactly twelve pentagons on every such ball; the usual one has 60 corners, 90 seams and 32 panels, and $60 - 90 + 32 = 2$.

If three seams meet at every corner and the panels may have any numbers of sides, what does the formula force on the numbers of panels of each kind?

The same two counts settle how many flattenable solids can have every face alike and every corner alike. Let every face have p sides and every corner gather q edges. Then $pF = 2E = qV$, by the same two counts, and the formula, with $2E/q$ for V and $2E/p$ for F , reads $2E/q - E + 2E/p = 2$; dividing every term by $2E$, it becomes

$$1/p + 1/q = 1/2 + 1/E$$

so $1/p + 1/q$ exceeds $1/2$. Both p and q are at least 3, and if both were 4 or more the sum would be at most $1/2$; so one of them is 3, and then 1 over the other exceeds $1/2 - 1/3 = 1/6$, so the other is below 6. Five pairs remain, and each fixes E , since $1/E = 1/p + 1/q - 1/2$, and with it every count, $V = 2E/q$ and $F = 2E/p$. Three sides and three edges give $E = 6$ and the tetrahedron's 4, 6 and 4; three and four give 6, 12 and 8, the octahedron; four and three give 8, 12 and 6, the cube; five and three give 20, 30 and 12, the dodecahedron; and three and five give 12 corners, 30 edges and 20 triangular faces, a solid not yet met here, the icosahedron.

That each of the five can be built is a separate matter, settled by building them, and Euclid's *Elements* ends by showing in another way, from the angles that fit round a corner, that there are no more than five. When $1/p + 1/q$ is exactly $1/2$ the count is 0, since the same working with a count c in place of 2 gives $1/p + 1/q = 1/2 + c/2E$, and a sum of exactly $1/2$ forces $c = 0$: hexagons three to a corner, which made the doughnut above and Szilassi's solid, and also squares four to a corner and triangles six to a corner.

* * *

Before the argument, “polyhedron” had an informal meaning and a stock of familiar examples. The definition, closed and flattenable with faces that have no holes, was assembled afterwards from what the proof turned out to need at the places where it broke. Definitions are manufactured under pressure, to save proofs, as the definition of proportion was made from what could still be said after the old one failed.

One more response is rarer and worth keeping. Sometimes the right move is to admit that the question was the wrong size. One formula for all polyhedra does not exist. What exists is a number for each kind of surface, telling the kinds apart: not the theorem that was wanted, but a better question.

A RUN THAT CANNOT GO ON FOREVER

The subtraction at the diagonal would have ended, had the diagonal and the side shared a unit, because a decreasing run of whole numbers stops. The triangles came off the flattened network one at a time, down to a single one, for the same reason. Two proofs about things as unlike as a length and a solid stand on one sentence. It is easier to see what that sentence says by first running the same idea the other way, uphill, from 1 to 2 to 3 and onwards, on claims that can be checked at every step.

* * *

Add the odd numbers from the start: 1 ; $1 + 3 = 4$; $1 + 3 + 5 = 9$; then 16 , 25 , 36 , 49 . The sums are the squares. Is that a theorem, or a pattern that has held as far as anyone looked?

A picture gives the reason. A square of dots three by three has nine dots. The square four by four is made by adding a row along the bottom and a column down the side, meeting in a corner dot, and that hook has three, three and one dots, seven in all, the next odd number. The same hook turns any square into the next: n along the bottom, n down the side and one in the corner make

$2n + 1$, the next odd number in the list; from five by five to six by six the hook has eleven dots. So if some square is the sum of the first so many odd numbers, the next square is that sum plus the next odd number.

That sentence alone proves nothing. It says that whenever a square is a sum of odd numbers, the next one is too, and nothing about whether any square is. The first is: the square 1 is the sum of the first one odd number, 1 itself. Then the hook makes 4 such a sum, then 9, and no square is out of reach, because each square is the one before plus a hook. The claim starts, and it passes from each case to the next, so it holds for every case.

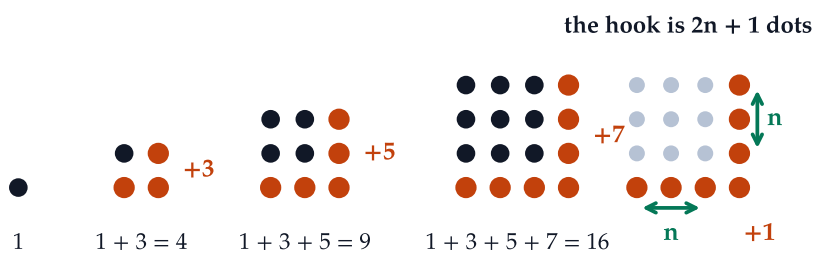


Figure 4.1. A three-by-three square of dots becomes four-by-four by adding a hook of seven along the bottom and the side; the hook that wraps any n -by- n square has n plus n plus one dots, which is the next odd number, and that, with the single dot the climb starts from, proves that the odd numbers sum to the squares. *Assumes:* that the picture of one hook stands for every hook, which is the step the argument then makes explicit: the count $2n + 1$ does not depend on the picture, only on the two sides and the corner

The same climb works on a shape. A chessboard of side 2 with one square removed leaves three squares in an L , which one L -shaped tile of three squares covers. Take a board of side 4 with one square removed. Cut it into four quarters of side 2; one quarter holds the missing square. Place one tile at the centre so that it covers one square of each of the other three quarters, and every quarter is now a board of side 2 with one square missing, which is already done. A board of side 8 with one square removed cuts into four boards of side 4, one tile at the centre does the same, and

each quarter is done. So every board whose side is a power of 2 and which has one square removed, wherever it lies, can be tiled.

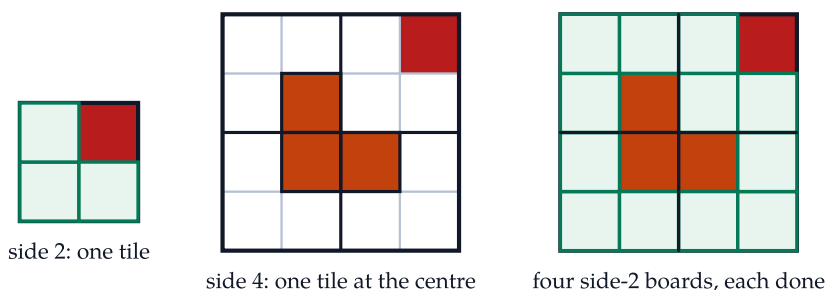


Figure 4.2. A four-by-four board with one square missing: one L-shaped tile placed at the centre takes a square from each of the three quarters that do not hold the gap, and every quarter is then a two-by-two board with one square missing, which one tile covers. *Assumes:* that the missing square can be anywhere; the picture puts it in one place, and the argument does not care where, because the centre tile always faces away from whichever quarter holds the gap

And on a count of moves. Three pegs hold discs of different sizes, and a stack of discs, largest at the bottom, is to be moved from one peg to another, one disc at a time, never putting a larger disc on a smaller. The largest disc has to move at some point, from one peg to another. At that moment none of the smaller discs can be on the peg it leaves, since it was at the bottom there, and none can be on the peg it lands on, since it may not be put on a smaller disc. So the other $n - 1$ discs must all be on the third peg. So n discs can be moved in this many moves: the moves for $n - 1$ discs, to clear the way; one move for the largest; and the moves for $n - 1$ discs again, to bring the rest back on top. No route is shorter. Before the largest disc first moves, the other $n - 1$ must have travelled as a stack from one peg to another, and after its last moves they must travel as a stack again.

One disc takes one move; two take $1 + 1 + 1 = 3$; three take $3 + 1 + 3 = 7$; four take 15; five take 31. If $n - 1$ discs take one less than 2^{n-1} moves, then n discs take twice that plus one; twice 2^{n-1}

is 2^n , so twice one less than 2^{n-1} is two less than 2^n , and plus one is one less than 2^n . The count starts and passes on, so n discs take $2^n - 1$ moves.

If a disc may move only between neighbouring pegs, from the left peg to the middle one or from the middle one to the right one and back, how many moves does it take to carry a stack of n discs from the left peg to the right one?

* * *

The three arguments share one shape. Each shows that the first case holds, and that whenever a case holds, so does the next, and concludes that every case holds. That is *induction*: a claim about every whole number is proved by proving that it starts and that it passes on.

The passing on is where arguments go wrong, and a failure shows how. Here is an argument that all horses in any collection have one colour. One horse has one colour, so the claim starts. Suppose every collection of n horses has one colour, and take $n + 1$ horses. Remove one, and the remaining n have one colour. Put it back and remove a different one, and the remaining n again have one colour. The two collections overlap, and a horse in the overlap has the colour of both, so all $n + 1$ have one colour. Every collection of horses has one colour.

The conclusion is false, so some step fails. The step is “the two collections overlap”. With three horses, removing one leaves two, removing a different one leaves a different two, and the two pairs share a horse. With two horses, removing one leaves one and removing the other leaves the other, and the two collections share nothing. The passing on works for every n except the first: one horse does not pass to two, so two was never reached, and nothing above it either. The case most likely to break the passing on is the smallest one, where the general argument leans on something not yet there.

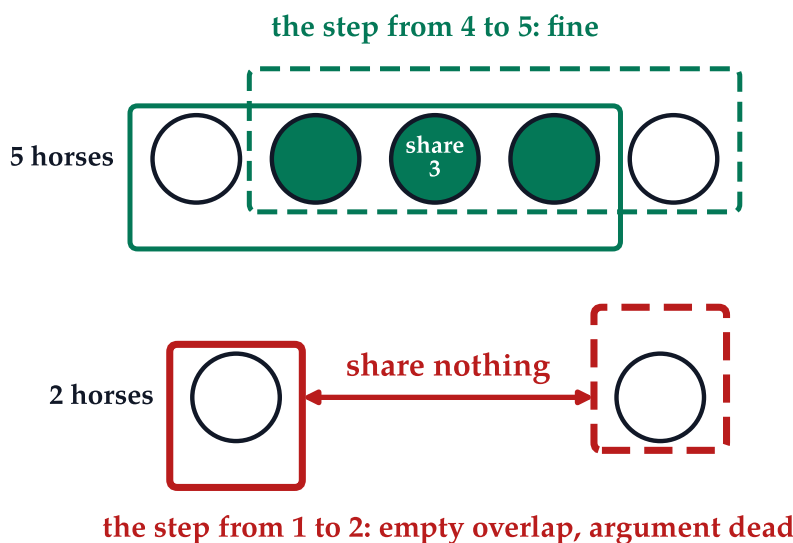


Figure 4.3. The step that passes the claim on compares two groups that leave out different members: with five horses they share three, and the shared ones force one colour, but with two horses the two groups share nothing and the step proves nothing at all. *Assumes:* that sharing a member really does force the two groups to agree, which is true and is not the defect; the defect is that the sharing itself was never checked in the first case, one horse passing to two

* * *

Now the sentence that ran downhill: a decreasing run of whole numbers stops. Suppose some run of whole numbers decreased forever. Call a whole number a bad start when a decreasing run can begin there and never stop; the first number of the supposed run is a bad start. The bad starts form a collection of whole numbers with something in it, so, if every such collection has a smallest member, there is a smallest bad start, call it m . The run that begins at m and never stops takes its first step down to some number below m , and from there the rest of the run continues forever. So that smaller number is a bad start too, and m was not the smallest after

all. The supposition breaks, so no run decreases forever, provided that every collection of whole numbers with something in it has a smallest member.

The two sentences “no decreasing run of whole numbers goes on forever” and “every collection of whole numbers with something in it has a smallest member” say the same thing turned two ways. The argument above gives the first from the second. The second follows from the first: in a collection with something in it but no smallest member, any member has a smaller member below it, and that one has a smaller still, and so on, which is a decreasing run that never stops.

Induction is the same fact a third time. Suppose a claim starts and passes on, but fails for some numbers. The numbers where it fails form a collection with something in it, so it has a smallest member. That member is not the first whole number, since the claim starts. So the number just before it is one where the claim holds; and the claim passes on, so it holds at the next number too, which is the very number where it was supposed to fail.

Conversely, suppose induction, and take a collection with something in it but no smallest member. Consider the claim “no number up to n belongs to the collection”. It starts, since if the first whole number belonged it would be the smallest member. It passes on: if nothing up to n belongs, then $n + 1$ does not belong either, since if it did it would be the smallest member. So by induction the claim holds for every n , and nothing belongs to the collection at all, against the supposition that something does. None of the three is proved from anything more basic. They are part of what the whole numbers are.

* * *

Descent settles a claim about every whole number at once. Call a whole number above 1 *prime* when the only whole numbers dividing it exactly are 1 and itself: 2, 3, 5, 7, 11, Every whole number above 1 has a prime among its divisors. Suppose not, and let n be

the smallest whole number above 1 with no prime divisor. It is not prime, since a prime counts as a prime divisor of itself, so it has a divisor d other than 1 and n , and d lies strictly between them. Since d is smaller than n , some prime p divides d , and then p divides n , since p divides d and d divides n . So n has a prime divisor after all, and no whole number above 1 is the smallest one without a prime divisor. A collection of whole numbers with something in it has a smallest member, so the collection of numbers without a prime divisor has nothing in it. Run as a descent, the same argument reads the other way: a whole number above 1 with no prime divisor has a divisor d strictly between 1 and itself, and d has no prime divisor either, since a prime dividing d would divide the number too. So each such number has a smaller such number below it, and any one of them would start a decreasing run of whole numbers that never stops, which cannot happen.

One consequence is that the primes never run out. Take any finite list of primes, multiply them all together and add 1. The result is above 1, so it has a prime divisor, and that prime is not on the list: a listed prime divides the product, and if it also divided the product plus 1 it would divide their difference, which is 1, and no prime divides 1. So no finite list holds every prime, which is how Euclid proved it. The argument does not say that the product plus 1 is itself prime, although it starts that way: $2 + 1 = 3$, $2 \cdot 3 + 1 = 7$, $2 \cdot 3 \cdot 5 + 1 = 31$, and then 211 and 2311 are all prime, but $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 + 1$ is 30031, which is $59 \cdot 509$. Is there, for every whole number n , a prime above n and no larger than $1 \cdot 2 \cdot 3 \cdots n + 1$?

* * *

The proof about solids can be run as a descent too, and run so that the assumption the tunnel broke is written down where it can be seen. Take a closed surface with a network of corners and edges drawn on it, the faces being the regions into which the edges cut the surface. Suppose two things: that the network is in one piece,

so that any corner can be reached from any other along edges; and that every closed loop of edges cuts the surface into two parts. Then $V - E + F = 2$.

Suppose some edge lies on a closed loop. By the second supposition the loop cuts the surface into two parts. Picture a thin strip of surface running along the loop on one side of it, and another strip along the other side. Neither strip crosses the loop, so each strip lies whole inside one of the two parts. Each of the two parts touches the loop, so each part contains one of the strips; there are two parts and two strips, so the two strips lie in different parts. The region on one side of the edge contains the piece of strip beside it, and the region on the other side contains the piece of the other strip. The two regions lie in different parts of the surface, so they are different regions. Remove the edge. The two regions merge, E and F each drop by 1, and $V - E + F$ does not move. The network is still in one piece, since the rest of the loop still joins the ends of the removed edge, and every loop of the new network was a loop of the old, so both suppositions still hold.

Suppose instead that no edge lies on a loop, and that some edge is left. Walk along edges, never turning straight back. No corner is visited twice, since the first return would close a loop, and there are finitely many corners, so the walk cannot go on forever. It can only be stopped at a corner that offers no edge to leave by except the one it arrived on, that is, a corner with only one edge. Remove that corner and its edge. The edge had the same region on both sides, since walking round its free end passes from one side to the other without crossing any edge, so V and E each drop by 1, F is unchanged, and $V - E + F$ does not move; the network is still in one piece, and no loop has been made.

Each step lowers E , and a decreasing run of whole numbers stops. While any edge is left, one of the two steps applies, so the run can only stop when no edge is left. A network in one piece with no edges is a single corner, and with no edges to cut it the surface is a single region, so the run stops at $V - E + F = 1 - 0 + 1 = 2$. So it was 2 from the start. No triangle had to be found that could be

taken off a boundary, the step the earlier argument assumed.

A solid that flattens when a face is removed and has no face with a hole satisfies both suppositions. A closed loop of its edges becomes a closed curve on the page, and a closed curve on the page has an inside and an outside; that is one more belief about the page. The removed face lies on the outside, so the loop cuts the surface in two. And its network is in one piece, since a face bordering two separate pieces of it would be a face with a hole. On the tunnelled cube the second supposition fails, and visibly. The loop of four edges round a mouth of the tunnel does not cut the surface in two: from one side of it a path runs down the tunnel, out of the other mouth, round the outside and back to the other side. Run the removals on it anyway. The count starts at 0 and ends at 2. Removing a corner with its single edge leaves the count alone, as before, and so does removing an edge on a loop whose two sides are different regions. The one step the failed supposition lets in is the removal of an edge that lies on a loop and yet has one region on both sides: nothing merges, so E drops by 1 and F does not, and the count rises by 1. That is the only step that can move the count, so in whatever order the edges come off, it happens exactly twice.

What does the count give when the edges fall into k separate pieces but every closed loop still cuts the surface in two?

DRAWN WITHOUT CROSSING

Three houses stand in a row, and across the road stand the gasworks, the waterworks and the power station. Each house is to be joined to each of the three by a pipe or a cable of its own, nine connections in all, laid on the ground, and no two of them are to cross. It is an old puzzle, and anybody who tries it gets eight of the nine. The last house cannot reach the last supplier: a loop of earlier connections has shut one of them in and the other out.

Eight out of nine, on every attempt, is evidence that it cannot be done, and evidence of the weak kind: every failure reports on one way of laying the first eight. The connections can be bent at pleasure, so there is no finite list of layouts to work through. A proof has to speak of all of them at once, and the count of corners, edges and faces can.

* * *

The formula was proved for a network in one piece, drawn on a surface in which every closed loop of edges cuts the surface in two. The page is such a surface, provided the outer region, the part of the page that runs off to the margin, is counted as a region like any other. What makes every loop cut the page in two is the statement that a closed curve on the page has an inside and an outside; for

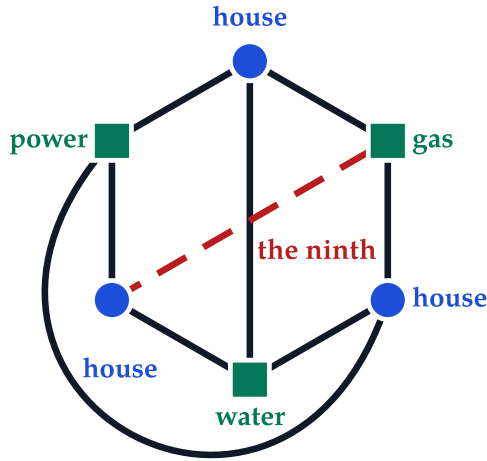


Figure 5.1. Three houses and three suppliers placed alternately round a ring. The six connections between neighbours and one connection across the middle and one round the outside make eight; the ninth would have to cross one of them whichever way it went. *Assumes:* that this is only one way of laying the first eight; the picture shows one attempt failing, and the argument in the text is what rules out all of them

the polygons a network draws, that statement is proved next. So for every network in one piece drawn on the page without crossings,

$$V - E + F = 2$$

where F counts the regions into which the edges cut the page, the outer one included. A square with one diagonal has 4 corners, 5 edges and 3 regions, two triangles and the outside. A single edge has 2 corners, 1 edge and 1 region.

Suppose the nine connections could be laid without crossing. The network has 6 corners and 9 edges, and it is in one piece, so it would cut the page into $F = 2 - 6 + 9 = 5$ regions. Now count the sides of the regions. Walk round the edge of any region, keeping it on the left, until the walk closes up. The corners met are alter-

nately a house and a supplier, since every connection joins a house to a supplier, so the walk has an even number of steps. It does not have two, since two steps would be out along one connection and straight back along the same one, and the walk turns straight back only at a corner that has no other edge to continue along; here every corner has three. A two-step walk bounds a region only when the network is a single edge with the rest of the page round it. So every region has at least four sides, and five regions have at least twenty between them. But every connection has two sides and no more, so nine connections supply eighteen. Twenty sides cannot be found among eighteen, and the ninth connection can never be laid.

Five points, each joined to each of the other four, make ten connections, and the same argument disposes of them. A layout without crossings would have 5 corners and 10 edges, so $2 - 5 + 10 = 7$ regions. Two different points are joined by one connection only, so a walk round a region cannot close in two steps by going out along one connection and back along another; and it cannot turn straight back, as before. So a walk round a region has at least three steps, and seven regions need at least twenty-one sides, while ten connections supply twenty. The margin is one side. With four points nothing stands in the way, and six connections can be laid without crossing, as a triangle with a point inside it joined to the three corners.

Both arguments are one argument, and it gives a general bound. Take a network in one piece on the page, with at least three corners, in which no two corners are joined twice and no corner is joined to itself. Every region has at least three sides, every edge supplies two, so $3F \leq 2E$. Put $F = 2 - V + E$ into that:

$$E \leq 3V - 6$$

And if every closed walk in the network has an even number of steps, as when the corners are of two kinds and every edge joins one kind to the other, then $4F \leq 2E$, and $E \leq 2V - 4$. Five points with ten connections fail the first bound, which allows nine. Six

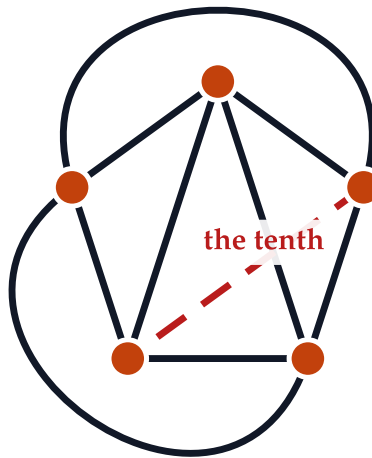


Figure 5.2. Five points with nine of their ten connections laid without crossing: the five sides, two across the inside and two round the outside. The tenth has an end inside a closed loop and an end outside it. *Assumes:* that this is one attempt among endlessly many; the count of corners, edges and regions is what shows that every attempt fails

points of two kinds with nine connections fail the second, which allows eight. A network of a thousand corners can be shown in one line to be impossible to draw, if it has three thousand edges.

The bound is a test that a network can fail, and passing it proves nothing. Take the five points with their ten connections and put a new corner in the middle of each connection: that makes 15 corners and 20 edges, well inside $3V - 6 = 39$, and the network can no more be drawn than before: a drawing of it without crossings, with the ten new corners rubbed out, would be a drawing of the old one without crossings. Kuratowski showed in 1930 that these two are the only things that ever stand in the way: a network that cannot be drawn on the page without crossings always contains the five points, or the three houses and three suppliers, with extra

corners perhaps placed along the connections. His proof is an induction with many cases, of a different kind from anything here, and nothing later leans on it.

Can the three houses be joined to the three suppliers without crossings on the surface of a doughnut?

* * *

The bound $E \leq 3V - 6$ has a consequence that looks too weak to matter. Count the ends of the edges: every edge has two, so the numbers of edges at the corners add up to $2E$, which is at most $6V - 12$. If every corner had six edges or more, the ends would number at least $6V$, which is more than $6V - 12$. So some corner has five edges or fewer. That settles it for a network in one piece with at least three corners, and the remaining cases are quick. A network in one piece with one or two corners has no corner with more than one edge. A network in several pieces has a piece which, drawn alone, is still drawn without crossings; a corner of that piece has the same edges in the whole network as in the piece, so a corner of the piece with five edges or fewer is such a corner of the whole. So

every network drawn on the page without crossings, with no double connections and no corner joined to itself, has a corner with five edges or fewer.

That sentence has been met before in other clothes. In a solid that flattens, some face has five sides or fewer, and the argument for it counted edge-sides where this one counts edge-ends. The two statements are exchanged by a construction that exchanges corners with faces. Given a network on a surface, mark a point inside each region, and for each edge draw a new edge across it, joining the marked points of the two regions on either side. The new network has a corner for each old face, an edge for each old edge, and a face for each old corner, the old corner sitting inside it. Done on a cube, it gives 6 corners, 12 edges and 8 triangular faces: the octahedron.

Done on the octahedron it gives back a cube. The tetrahedron gives a tetrahedron. The solid with twelve pentagons becomes the solid with twenty triangles, five at each corner. $V - E + F$ reads the same forwards and backwards, which is why each of its consequences comes in a pair.

* * *

In October 1852 Francis Guthrie, colouring a map of the counties of England, noticed that four colours were enough to give different colours to every two counties that share a stretch of border, and asked whether four are enough for every map. His brother passed the question to De Morgan, who could not answer it. In 1879 Kempe published a proof that four colours always suffice. It was accepted, and it stood for eleven years.

Turn the map into a network first, by the construction that exchanges corners and faces: a corner for each country, and an edge between two corners when the countries share a stretch of border. Countries that meet only at a point, as four states of America do, are not joined. Every country is taken to be in one piece, and without that the network need not be drawable: number two rows of five squares 1, 2, 3, 4, 5 along the top and 3, 4, 5, 1, 2 along the bottom, and let each number be one country made of two squares. Every two of the five countries then share a border, and their network is the five points with ten connections, which cannot be drawn without crossings. The network is drawn on the page without crossings, and colouring the map has become colouring the corners so that the two ends of every edge differ.

Six colours are always enough, and the proof is a climb on the number of corners. A network with six corners or fewer can be coloured by giving every corner its own colour. Suppose every network with n corners drawn without crossings, with no double connections and no corner joined to itself, can be coloured with six colours, and take one with $n + 1$. It has a corner v with five edges or fewer. Remove v and its edges. What remains has n corners, and

it can be coloured, piece by piece if it has fallen into pieces. Put v back. Its neighbours, five at most, use five colours at most, and a sixth is free for v .

Five colours are always enough, and here the argument has real work to do. Run the same climb with five colours. It goes through unless v has exactly five neighbours and they carry all five colours. Number the neighbours in order round v as n_1, n_2, n_3, n_4 and n_5 , and number the colours so that n_1 has colour 1, n_2 colour 2, and so on. Look at the corners coloured 1 or 3, and the edges between them. They form a network of their own, perhaps in several pieces. In any one piece the two colours can be exchanged, every 1 becoming a 3 and every 3 a 1, and the whole network stays coloured as required: inside the piece every edge still has ends of different colours, and a corner of the piece is joined to nothing coloured 1 or 3 outside the piece, since such a neighbour would belong to the piece.

If n_1 and n_3 lie in different pieces, exchange the colours in the piece containing n_1 . Now n_1 has colour 3, n_3 still has colour 3, and no neighbour of v has colour 1. Give v colour 1.

If n_1 and n_3 lie in one piece, there is a path from n_1 to n_3 along corners coloured 1 and 3 alternately. That path, with the two edges from its ends to v , is a closed loop on the page. The loop passes through v along the edges to n_1 and n_3 , and those two edges divide the circle of neighbours round v in two: going round v in order, n_2 comes between n_1 and n_3 on one side and n_4 on the other. The loop has an inside and an outside, so n_2 is on one side of it and n_4 on the other. Now look at the corners coloured 2 or 4. A path of such corners from n_2 to n_4 would have to cross the loop, and it cannot: it cannot cross at a corner, since the corners of the loop are coloured 1 and 3 or are v itself, and edges do not cross edges. So n_2 and n_4 lie in different pieces of the network of colours 2 and 4. Exchange those colours in the piece containing n_2 , and colour 2 is free for v .

So every map can be coloured with five colours. The paths of two alternating colours are Kempe's, and the proof is a repair of his. For four colours his argument ran the same way and reached a

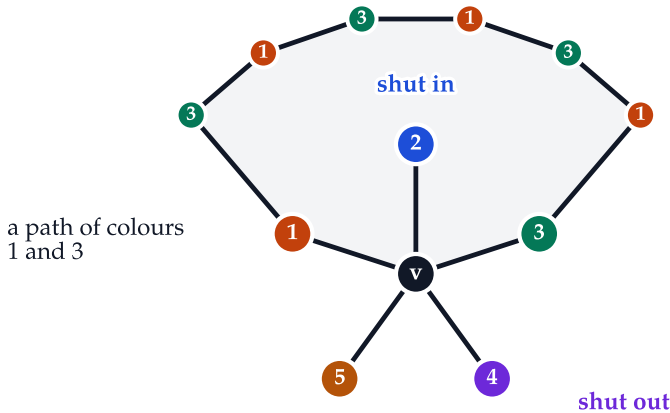


Figure 5.3. The corner v and its five neighbours, carrying five colours. A path of corners coloured alternately 1 and 3 joins the first neighbour to the third and closes a loop through v . The second neighbour is inside the loop and the fourth outside, so no path of colours 2 and 4 joins them. *Assumes:* that a closed loop on the page has an inside and an outside; and the rest of the network, which is not drawn, may be anything at all

last case in which v has five neighbours carrying four colours, one colour appearing twice, on n_2 and n_5 say. If n_1 and n_3 , or n_1 and n_4 , lie in different pieces of the relevant two colours, one exchange frees a colour. If not, there is a loop through v , n_1 and n_3 and another through v , n_1 and n_4 . Kempe then exchanged colours in two pieces at once, one containing n_2 and one containing n_5 , each shut in by one of the loops, to remove the repeated colour from both. In 1890 Heawood drew a map of twenty-five countries on which this fails. The two loops can cross each other at a corner they share, since both pass through corners of n_1 's colour. After the first exchange, a corner that has changed colour can then join what the second loop was supposed to shut in to what it was supposed to shut out. Each exchange is safe alone. The two together are not, and the proof had asserted the safety of the pair on the strength of the safety of each.

Heawood did not know how to mend it for four colours. He showed that the argument, used once and not twice, gives five, which is the theorem above, and the four-colour statement went back to being a conjecture. It was settled in 1976 by Appel and Haken, with the same two ideas: a list of configurations, one of which every map must contain, and a way of reducing each of them. The difference is in the size. The list has not one entry, the corner with five edges or fewer, but nearly two thousand, and the reductions were checked by a computer running for some twelve hundred hours. For eleven years the statement had the standing of a theorem and the evidence of a conjecture, and nothing in how it looked on the page distinguished the two.

* * *

On a doughnut the question was answered completely in 1890, eighty-six years before the page. A network on a doughnut whose regions are all plain patches, none with a hole in it and none going right round the doughnut, counts to $V - E + F = 0$. When some region is not a plain patch the count is larger, never smaller: a triangle drawn on a small patch of the doughnut counts to $3 - 3 + 2 = 2$. That every network on a doughnut gives these counts is one of the facts about surfaces whose proof needs more than is available, and what follows rests on it.

Take a network on the doughnut with no double connections and no corner joined to itself. A piece of it with one or two corners has no corner with more than one edge. For a piece with three corners or more, drawn alone, $3F \leq 2E$ as before. The count is 0 or more, and $F \leq 2E/3$ makes it at most $V - E + 2E/3 = V - E/3$. So $0 \leq V - E/3$, and $E \leq 3V$. The edge-ends number $2E$, at most $6V$, and if every corner had seven edges or more they would number at least $7V$: so some corner has six edges or fewer. The climb that gave six colours on the page gives seven here.

And seven are needed. The solid that Szilassi found has seven faces, each a hexagon. Its fourteen corners and twenty-one edges

count to $14 - 21 + 7 = 0$, so its surface is a doughnut. Each of its faces shares an edge with each of the other six: seven faces make twenty-one pairs, there are twenty-one edges, and no two edges join the same two faces. Seven countries, every two of them neighbours, need seven colours. So on a doughnut seven colours are always enough and sometimes necessary, and the proof is short, while on the page, which looks the simpler surface, the corresponding proof has never been reduced to a size that one person can check.

* * *

A map is drawn by ruling straight lines right across the page, any number of them. How many colours does it need? And if the lines are replaced by circles?

Is there a network, drawn on the page without crossings, with no double connections and no corner joined to itself, in which every corner has exactly five edges? What does the count of corners, edges and faces say about how many corners it has at least?

In a network drawn without crossings, with no double connections, no corner joined to itself and at least four corners, are there always at least four corners with five edges or fewer?

A CURVE HAS AN INSIDE

The formula for networks on the page was proved for surfaces in which every closed loop of edges cuts the surface in two, and the page was declared to be such a surface because a closed curve drawn on it has an inside and an outside. Nothing was offered for that. It is the kind of statement whose truth is so plain on a drawing that the demand for a proof looks like pedantry. Anyone can see that it is true. But what anyone can see is the pencil mark, not the curve the statement is about.

The demand is not pedantry, and two drawings show why. A closed curve wound into a spiral of many turns, with a point marked deep inside the coils, does not announce which side of the curve the point is on; a reader asked to answer by eye will hesitate and then start counting something. And on a surface that is not the page the statement is false: on a doughnut a loop running the short way round cuts nothing in two, and the faces of a network there need not obey the formula. So the statement is about the page in particular, it is what the formula rests on, and it has to be argued.

What has to be argued, and no more, is the case the formula uses. The edges of a network are drawn as straight segments, or as curves that can be replaced by chains of segments meeting where the curves met and nowhere else. So the curve is a closed polygon.

That a drawing with curved edges can be redrawn with straight ones in this way is a statement about curves in the page, and it is taken here from other hands. A polygon here is a finite chain of straight segments, each joined end to end to the next and the last joined to the first, with no two segments meeting except at the ends they share.

A closed polygon with no crossings cuts the rest of the page into exactly two pieces, and it is the boundary of each.

The proof is a count, and nothing will be asked of the count except whether it is odd or even, its *parity*.

* * *

Fix a direction that is parallel to no side of the polygon, which is possible since there are finitely many sides and infinitely many directions. For a point x not on the polygon, send a ray from x in that direction and count how many times it meets the polygon. A side is a straight segment, and a straight ray not parallel to it meets it at most once, so the ray meets each side at most once and the count is finite. Call x *odd* or *even* according to that count.

The count needs one piece of care. The ray may pass through a corner of the polygon, where two sides meet. Count such a corner as one crossing when the two sides leave it on opposite sides of the ray, and as no crossing when they leave on the same side. With that rule the parity of the count is what a ray in a slightly turned direction would find. Turning the direction a little moves the ray off the corner, to one side of it. If the two sides leave the corner on opposite sides of the original ray, the turned ray cuts exactly one of them. If they leave on the same side, the turned ray cuts both of them or neither, according to which side of the corner it passes. That is one crossing where the rule says one, and none or two where the rule says none. The count itself can move by two, and the parity cannot move at all. Parity is what the argument below uses, and nothing else is claimed for the rule.

Now three facts.

Two points close to the middle of a side, one on each side of it, have opposite parities. Take a point y in the middle of a side, off the corners, and two points a little way from y on opposite sides of that side, both closer to it than to any other part of the polygon. The fixed direction is not along the side, so it points across it, from one of its two sides to the other. The ray from the point on the near side crosses the line of the side, and does so close to y , which is in the middle of the side, so it crosses the side itself. The ray from the point on the far side moves away from the side and never meets it. Elsewhere the two rays are parallel and close together, so they meet the same sides, apart from corners, where the rule holds the parity fixed. The side at y makes the counts differ by one, and each corner can change that difference only by an even number, so their parities differ. So both parities occur, and neither the odd points nor the even points are empty.

The parity does not change while a point moves without touching the polygon. Move x a little, and the ray from it shifts a little. A ray that crosses a side away from its ends goes on crossing it after a small enough move, so a crossing can be gained or lost only where the ray slides over a corner. More exactly: for x off the polygon there is a disc about x missing the polygon, and for every point of that disc the ray meets the same sides, except near a corner, where one ray may pass the corner on one side, another on the other side, and a third through it. The rule for corners was made so that the parity is the same in all three cases, and that includes a ray sliding past the outermost corner of the polygon, after which there is nothing left to cross. So parity is the same throughout the disc. Take, then, any path from one point to another that misses the polygon and is cut into finitely many stretches, each stretch lying inside a single such disc. The parity is the same at the two ends of each stretch, so it is the same at the two ends of the path. Every path built below is built in exactly that way, out of finitely many stretches of known size. That a path of any shape missing the polygon can be cut in the same way needs a precise idea of what a path is, which comes

much later; the claim below that no path at all runs from an odd point to an even one rests on that.

Any two points of the same parity can be joined by a path missing the polygon. Two bands are needed first. Fix a direction of travel round the polygon, so that each side has a left and a right for someone walking along it that way. Call a point within a small distance of the polygon, and to the left of the side nearest it, a *left* point, and one to the right a *right* point. The left points together form the left band and the right points the right band. Each band is in one piece, meaning that any two of its points are joined by a path that stays inside the band. Along a single side that is plain, since the left points of that side fill a thin strip beside it. At a corner, the left points of the two sides meeting there lie in the same one of the two angles those sides make at the corner, whether the corner turns left or right, and inside a small disc about the corner that angle is a single wedge; so the strip along one side is joined to the strip along the next. The band therefore runs round the whole chain and closes up. Each band misses the polygon and is in one piece, so by the second fact the parity is the same at every point of it; and by the first fact the two bands have opposite parities.

Now take any point off the polygon and walk from it in the fixed direction. If its ray meets the polygon, the walk reaches a band just before the first crossing, so the starting point is joined to a band by the walk itself. If its ray meets nothing, the walk carries it out past every point of the polygon, into the region outside a circle large enough to hold the polygon. That region is in one piece: to join two of its points, walk from the first in along its radius to just outside the circle, round the circle, and out along the radius of the second. The outer region is itself joined to a band: walk in from a far point along a line that meets the polygon and stop short of it. So every point off the polygon is joined by a path missing the polygon to the left band or to the right band, and since parity never changes along such a path, the band it reaches is the one of its own parity. Two points of the same parity reach the same band, and so each other.

Putting the three together: the points off the polygon fall into

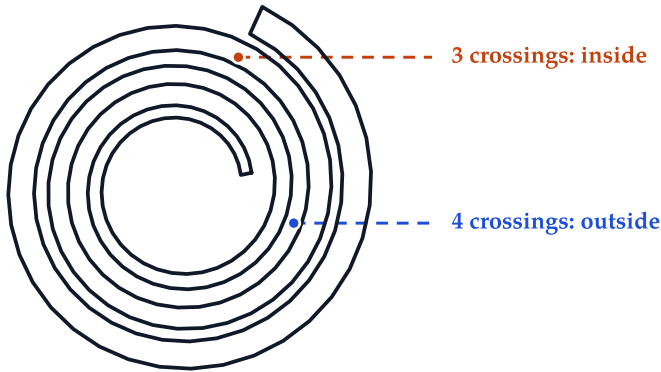
the odd ones and the even ones, both non-empty, neither meeting the other, and each in one piece. No path missing the polygon runs from an odd point to an even one, since parity does not change along such a path, so the two do not merge into a single region; and each is in one piece, so neither splits into two. That is two regions and no more. And every point of the polygon, corners included, has points of both kinds as close as desired: any small disc about it contains left points and right points, and the two bands have opposite parities. So the polygon is the boundary of each region. One of the two is bounded and the other is not. Draw a circle large enough to hold the polygon inside it. A point far out along the fixed direction, past the whole polygon, sends a ray that crosses nothing, so it is even. The points outside that circle form one piece, as was seen above, and parity does not change along a path that misses the polygon, so every point outside the circle is even. The odd points therefore all lie inside the circle, and their region is bounded; the even region reaches past every circle and is not. The odd points are the inside.

* * *

The count answers, for any point and any polygon, which side the point is on, by a procedure a clerk could follow: send a ray, count the crossings, read the parity. That is how a machine decides whether a place lies inside a border, and the corner rule is the part that the first attempt at such a program always gets wrong.

The count also explains the spiral. Winding the curve tightly does not make the question hard in itself; it makes the *eye's* method fail, which was never the method. The ray through the coils crosses the curve many times, and the parity of that many is all that matters.

What the formula needs is both halves of the statement at once, the two regions and the boundary. A closed loop of edges is a closed polygon; its inside and its outside are two regions; each of them runs right up to the loop; and no path from one to the other



the eye cannot say; the count can

Figure 6.1. A closed polygon wound into a spiral, and two points. The eye cannot say which is inside. A ray from each, in a direction along no side, settles it: one ray crosses an odd number of times, and its point is inside; the other crosses an even number, and its point is outside. Parity is the whole answer, and it does not change while a point moves without touching the polygon. *Assumes:* a ray through a corner is counted once when the two sides leave it on opposite sides of the ray and not at all when they leave on the same side; the two rays drawn here were chosen to pass well clear of every corner, so that rule is not in play

misses it. With that, a network on the page satisfies the hypothesis under which $V - E + F = 2$ was proved, and what was assumed there has been shown.

For a general closed curve, not made of segments, the statement is still true and the proof is harder: the ray can meet the curve in infinitely many points, and the parity argument has nothing to count. Jordan stated it in 1887 and gave an argument that later readers found wanting; Veblen gave a proof in 1905 that stood. Nothing later needs the general case, and the polygon is the case the networks present.

A polygon has 12 sides and a ray from a point crosses it at 5 places, none of them a corner. Which side is the point on?

Draw a five-pointed star as a closed polygon of ten sides. How many regions does it cut the page into, and does the theorem say otherwise?

On a doughnut take the loop that runs the short way round. Which step of the proof is the first that cannot be taken?

Give a polygon and a ray that passes through two corners, and check that the corner rule gives the same parity as a ray turned slightly, and find how far the two counts can differ.

From a point inside a convex polygon every ray crosses exactly once. Find a polygon and a point inside it from which no ray crosses exactly once.

WHAT CANNOT CHANGE

Cut two opposite corners off a chessboard. Sixty-two squares are left, and a domino covers two squares, so thirty-one dominoes have the right total. Can they be laid to cover the board exactly? Trying it goes as the three houses went: nearly, every time, with two squares left over that share no side.

The board's colouring answers at once. A domino, wherever it lies, covers one dark square and one light one. So any region that dominoes cover exactly has as many dark squares as light. Opposite corners of a chessboard have the same colour, so the cut board has thirty squares of one colour and thirty-two of the other, and no arrangement of dominoes covers it. The number of layouts that were never tried is enormous, and none of them needs trying.

The argument has a shape that recurs wherever something is to be shown impossible. A process has steps: here, laying a domino. Some quantity attached to the state of the process, here the number of uncovered dark squares less the number of uncovered light ones, is unchanged by every step. If the quantity has one value at the start and would need a different value at the end, the end cannot be reached, by any sequence of steps whatever. Such a quantity is an *invariant*. The search for a way of doing something looks at sequences of steps; the search for an invariant looks at a single step, which is why it can finish.

* * *

Does the colouring tell the whole story? It rules out a board with two squares of one colour removed. If one dark and one light square are removed, wherever they are, can the rest always be covered? An invariant can never answer a question of that kind, since it only forbids. A construction is needed, and Gomory found one. Draw on the board a closed path that passes through all sixty-four squares, each once, moving always from a square to a neighbour across a side, and returning to its start; a path that runs up and down the board like a fence does it. Along the path the colours alternate.

Remove a dark square and a light one. The path falls into two stretches, or into one if the removed squares were neighbours along it. Each stretch begins just after one removed square and ends just before the other. Colours alternate along the path, so the first square of a stretch has the opposite colour to the removed square it follows, and the last square has the opposite colour to the removed square it precedes; the two removed squares differ in colour, so the two ends of the stretch differ in colour too. Along the stretch the colours alternate as well, so the squares in odd places have the colour of the first square and those in even places the other colour; the last square has the other colour, so it stands in an even place, and the stretch has an even number of squares. Dominoes laid along it, each covering two neighbours on the path, cover it exactly.

So for dominoes on the chessboard the invariant was the only thing in the way. That is not to be expected in general, and the two halves of such a statement always need arguments of the two different kinds: an invariant for “never”, a construction for “always”.

Can a board of ten squares by ten be covered exactly by twenty-five tiles, each a straight strip of four squares?

* * *

In the 1880s a puzzle swept America and Europe: fifteen num-

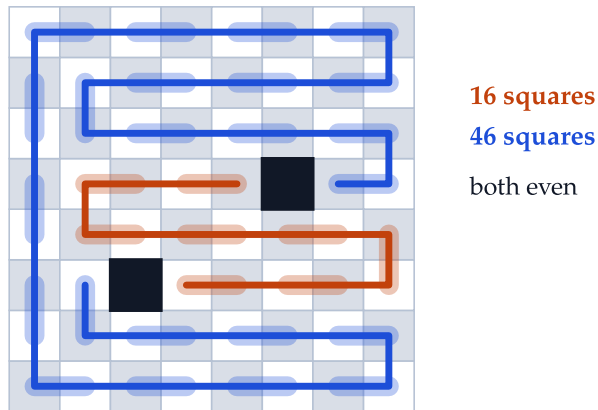


Figure 7.1. A closed path through every square of the board, colours alternating along it. With one dark and one light square removed it falls into two stretches, each with an even number of squares, and dominoes laid along each stretch cover it. *Assumes:* nothing about where the two removed squares are, except that their colours differ; the two shown are one choice among many

bered tiles in a tray of four rows by four, with one place empty, so that a tile next to the empty place can be slid into it. The blocks were tipped into the tray anyhow and were to be slid into order, and about half the time everything came right except that the 14 and the 15 stood exchanged. Rewards were offered for sliding that last position into order, and Sam Loyd, who later claimed the puzzle as his own, spoke of a thousand dollars. Nobody ever collected.

Read the tiles of any position as a list, along the top row from left to right, then the second row, and so on, skipping the empty place. Call a pair of tiles *out of order* when the larger comes before the smaller in the list. The stubborn position has one pair out of order, the 15 and the 14, and the target has none.

Sliding a tile sideways does not change the list at all, since the tile and the empty place exchange places and no tile passes another. Sliding a tile up or down moves it past exactly three tiles in the list, the three that lie between its old place and its new one in reading order. The moved tile makes a pair with each of those three, and

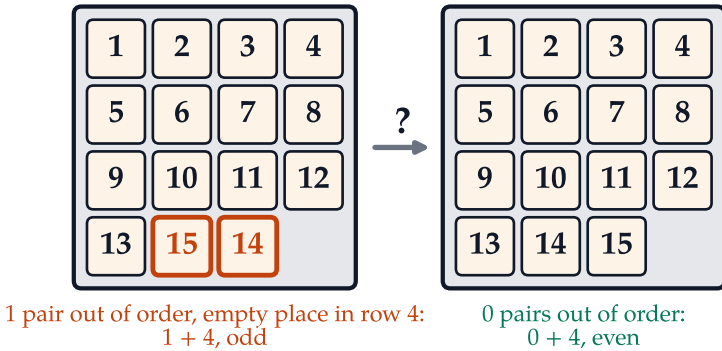


Figure 7.2. The position that would not come right, with the 14 and the 15 exchanged, and the position that was to be reached. The first has one pair of tiles out of order and the second none, and the empty place is in the fourth row in both. *Assumes:* that tiles only slide; lifting two tiles out of the tray and putting each in the other's place changes the evenness of the count

each of those three pairs changes its state, from in order to out of order or back; no other pair changes, since no other two tiles change their order in the list. If some of the three pairs go out of order and the rest come into order, the count of pairs out of order rises by the first number and falls by the second, and the two numbers add up to 3; so the count changes by 3, by 1, by -1 or by -3 , an odd number in every case. The same slide moves the empty place up or down by one row, since the tile and the empty place exchange places. So the sum

the number of pairs out of order + the number of the row that holds the empty place

changes by an even number at a vertical slide, an odd change plus an odd change, and not at all at a sideways one. Whether that sum is even or odd is an invariant. For the target, with the empty place in the fourth row, the sum is $0 + 4$, which is even. For the stubborn position it is $1 + 4$, which is odd. The rewards were safe.

Here too the invariant leaves a question. It splits the positions into two kinds, the even and the odd, and shows that no slide leads from one kind to the other. Are all the even positions reachable from the target, or does some further invariant split them again? That is settled when rearrangements are studied for their own sake.

* * *

A second kind of quantity is as useful as an invariant: one that every step pushes the same way. If a whole number decreases at every step and cannot go below 0, the steps come to an end, since a decreasing run of whole numbers stops. The process is known to stop, though nothing is said about how it goes.

Ten red points and ten blue points are marked on the page, no three of the twenty in a line. Can they be joined in pairs, each red point to a blue one by a straight segment, so that no two segments cross? Join them anyhow. If two segments cross, from red point R to blue point B and from R' to B' , exchange partners: join R to B' and R' to B . The crossing point X splits the old segments, and RB' is shorter than RX and XB' together, $R'B$ shorter than $R'X$ and XB together, since two sides of a triangle are together longer than the third. The two old segments measure $RX + XB$ and $R'X + XB'$, which together come to $(RX + XB') + (R'X + XB)$, and each bracket is longer than the new segment that replaces it. So the exchange makes the total length of the ten segments strictly smaller, the other eight segments being untouched. It may create new crossings. But there are only finitely many ways of pairing the points, so the total length has only finitely many possible values, and a quantity that strictly decreases through a finite set of values cannot decrease forever. The exchanges stop, and they stop only when nothing crosses.

The argument can be compressed: among the finitely many pairings take one of smallest total length; if two of its segments crossed, the exchange would produce a shorter one; so none cross. That is the argument about the smallest whole number with no

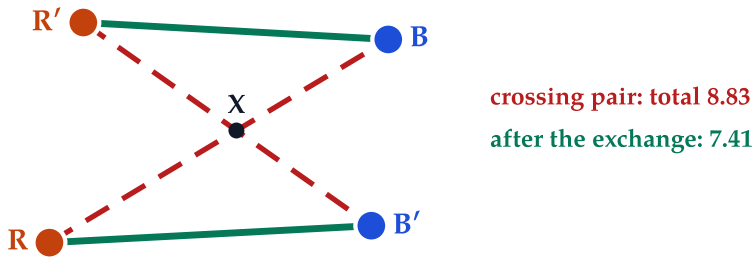


Figure 7.3. Two crossing segments and the two that replace them when the partners are exchanged. Each new segment is one side of a triangle whose other two sides are pieces of the old segments, so the new pair is shorter than the old. *Assumes:* that two sides of a triangle are together longer than the third, and that no three of the points lie in a line, so that the triangles are not flat

prime divisor over again, with lengths in place of whole numbers. One thing was needed that whole numbers give for nothing: the set of possible values had to be finite. A set of whole numbers with something in it always has a smallest member, but a set of lengths can fail to have one, and a finite set of lengths cannot.

In a parliament every member has at most three enemies among the other members, enmity being mutual. Can the members be divided into two houses so that each member has at most one enemy in his own house?

* * *

Games bring the two kinds of quantity together. In the game of Nim there are several piles of counters. Two players move alternately, and a move takes any number of counters, at least one, from a single pile. Whoever takes the last counter wins. With piles of 3, 4 and 5, who wins, and how?

The total number of counters decreases at every move, so every play of the game ends. From that alone it follows that one of

the players has a way of winning whatever the other does, and the proof is a descent. Call a position *won* when the player about to move has a way of winning whatever the other does, and *lost* when the other player has such a way. No position is both, since two such ways played against each other would make both players win. A position from which some move leads to a lost position is won, by making that move; a position from which every move leads to a won position is lost, since whatever is played the other player then has a way of winning. The empty position is lost, since the other has just taken the last counter. If every position with fewer than n counters is won or lost, then so is every position with n counters: each position one move away has fewer counters, so either some one of them is lost or all of them are won. So every position is won or lost, and nothing about Nim was used except that it ends and that a player who cannot move has lost.

That settles nothing in practice, since the number of positions to look through grows very fast. In 1901 Bouton found the rule, and it comes from writing the piles in a different way. Every whole number is a sum of different powers of 2, in exactly one way: 5 is $4 + 1$, 12 is $8 + 4$, 13 is $8 + 4 + 1$. For existence, take away from n the largest power of 2 not exceeding it; what is left is smaller than that power, since otherwise the next power would have fitted, and what is left is dealt with in the same way, a descent. For uniqueness, let 2^k be the largest power of 2 not exceeding n . A sum of different powers of 2 that equals n contains no power above 2^k , since any one such power is already more than n . It must contain 2^k itself: the powers 1, 2, 4, ..., 2^{k-1} add up to $2^k - 1$, one short of 2^k , since 1 is one short of 2 and a total one short of a power, with that power added, is one short of the next; so a sum made only of powers below 2^k falls short of n . So the largest power in any such sum for n has to be the largest power not exceeding n , the same in every sum; taking it away leaves $n - 2^k$, and the rest of the sum is settled in the same way.

Write each pile as a sum of different powers of 2. Call a position *balanced* when every power of 2 occurs in an even number of the

piles, and *unbalanced* otherwise. The piles 3, 4 and 5 are $2 + 1$, 4, and $4 + 1$. The power 4 occurs in two piles, the power 1 in two, and the power 2 in one, so the position is unbalanced. Two facts make balance the whole secret.

Every move from a balanced position unbalances it. A move changes one pile, and two different numbers are never sums of the same powers, so the move changes which powers occur in that pile: some power that was in the pile is no longer in it, or the reverse. That power occurred in an even number of piles before the move, and it now occurs in one pile more or one pile fewer, an odd number.

From every unbalanced position some move balances it. Take the largest power of 2 that occurs in an odd number of piles, 2^k say, and a pile that contains it. In that pile, for each power that occurs in an odd number of piles, switch its presence: remove it from the pile if it is there, and put it in if it is not. Every power above 2^k occurs in an even number of piles, so none of them is switched, and whatever is put in consists of different powers below 2^k , which add up to less than 2^k . The pile loses 2^k , perhaps loses some smaller powers as well, and gains less than 2^k ; so the pile has become smaller, and the change is a legal move. Each switched power now occurs in one pile more or one pile fewer than before, an even number in place of an odd one, and each power that was not switched occurs in as many piles as before, an even number. Afterwards every power occurs in an even number of piles.

The empty position is balanced. A player who always moves to a balanced position is always handed back an unbalanced one, by the first fact. It is never the empty position, since the empty position is balanced; and by the second fact it allows a move that restores the balance. The game ends, so somebody is at last faced with the empty position, and it is not that player. So that player has taken the last counter. The unbalanced positions are the won ones and the balanced positions the lost ones. With 3, 4 and 5, the only offending power is 2, and it lies in the pile of 3; removing it leaves piles of 1, 4 and 5, which is 1, 4, and $4 + 1$, balanced. The first

player wins by taking two counters from the pile of three, and by restoring the balance after every reply.

The total shows that the game ends; balance, which a player can always restore and an opponent must always destroy, shows who wins, and it is an invariant of a new kind, kept not by every step but by every pair of steps.

* * *

A game can be settled without anyone knowing how to play it. A bar of chocolate is a rectangle of squares, and the square in one corner is poisoned. Put the poisoned square at the bottom left. Two players move alternately; a move picks a remaining square and eats it together with every remaining square that lies at least as far up and at least as far to the right as it does, which is what remains of the rectangle stretching away from the poison beyond the chosen square. Whoever eats the poisoned square loses. On a bar of two rows the first player wins by eating the far corner square alone. That leaves the row without the poison one square shorter than the other, and the first player can restore that state after every reply. If the reply eats from the poisoned row, it takes the squares of the other row above its chosen square as well, and leaves the two rows the same length; eating the last square of the other row then makes that row the shorter again. If the reply eats from the other row, it shortens that row alone; eating the square of the poisoned row in the column just beyond the reply's square then does the same. Every restoration shortens the bar, so the opponent is at last handed the poisoned square alone. On most bars of three rows or more nobody knows a rule; a square bar is one of the few exceptions.

And yet the first player has a winning way on every bar of more than one square. The game ends, since the bar gets smaller, so every position is won or lost as before. Suppose the first player's opening position were lost. Then every first move would lead to a position won for the second player, and in particular the move that eats only the far corner square would. The second player would then

have a winning reply to it, a move eating some rectangle stretching away from some square. But every rectangle that stretches away from the poison reaches the far corner, so the squares that reply eats, together with the far corner square already gone, make up a rectangle that stood whole at the very start. The same move was therefore available to the first player then, and it leads to the same position. So the first player could have moved to a position from which the player to move loses, and the opening position was won after all. The argument proves that a winning first move exists and gives no hint of what it is; it steals a strategy it never sees, and proves an existence by making non-existence contradictory, with nothing exhibited.

Draw a few crosses on the page. Each cross has four free ends. A move joins two free ends, of one cross or of two, by a curve that crosses nothing, and then draws a short stroke across the new curve, which adds two new free ends, one on each side. The player who cannot move loses. It looks like a game of skill, in which the page gets crowded and regions are sealed off with their free ends inside.

Every play of it from n crosses lasts exactly $5n - 2$ moves. A move uses two free ends and makes two, so there are always $4n$ free ends. At the start the one region of the page contains the free ends of every cross. A move that joins two separate pieces of the network closes no loop, so it splits no region. The two free ends it used up lay in the region where the move was made, and the two it makes lie there too, so the piece it makes has a free end in every region that either of the old pieces bordered. A move that joins a piece to itself closes a loop and splits a region in two, leaving a new free end of that piece on each side of the new curve; any other piece bordering that region is not met by the curve, so it lies, with its free ends there, wholly on one side, and still has a free end in the half it lies in. Regions the curve does not enter are unchanged. The property held at the start and no move destroys it, so every region always contains a free end of every separate piece that borders it, and in particular at least one free end.

The game ends. A move that joins two separate pieces lowers the number of pieces by one, which can happen $n - 1$ times at most; any other move raises the number of regions by one, and since every region holds one of the $4n$ free ends there are never more than $4n$ regions, so that can happen $4n - 1$ times at most. When no move is possible, no region contains two free ends, since two free ends in one region could be joined; so every region contains exactly one, and there are $4n$ regions. The network is then in one piece, since a region bordered by two separate pieces would hold a free end of each. After m moves there are $n + m$ corners, the n crosses and the m strokes, and $2m$ edges, each move's curve being cut in two by its stroke; the free arms of the crosses and strokes end at no corner and are not edges. The network is in one piece with $4n$ regions, so the formula for corners, edges and regions gives

$$(n + m) - 2m + 4n = 2$$

which gives $m = 5n - 2$. With two crosses the game lasts eight moves, however it is played, and the second player makes the last one. The players' choices change the picture and nothing else.

* * *

Eleven coins lie in a row, all heads up. A move turns over any two neighbouring coins. Can they all be brought tails up? And if a move turns over any three neighbouring coins?

The numbers 1, 2, 3, ..., 20 are written on a board. A move rubs out any two of them, a and b , and writes $a + b + ab$ in their place. After nineteen moves one number is left. What is it?

In Nim with piles of 7, 11, 13 and 15, which moves win?

TWO WAYS OF COUNTING THE SAME THING

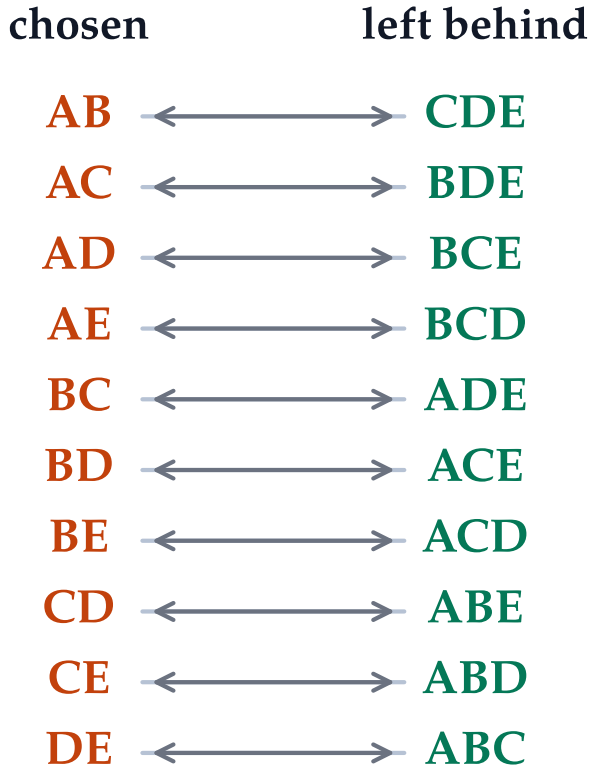
Two objects chosen from five, labelled $A B C D E$, can be listed in full: $AB AC AD AE, BC BD BE, CD CE, DE$. There are ten. Choosing three from the same five gives $ABC ABD ABE ACD ACE ADE, BCD BCE BDE, CDE$: ten again. Why should choosing two out of five have anything to do with choosing three out of five?

Every time two objects are chosen, three are left behind. Choosing AB is leaving CDE . The correspondence is exact in both directions: each choice of two determines one choice of three, each choice of three determines one choice of two, and nothing is left over on either side. So the two collections have the same size, and the argument used no arithmetic.

The argument proves more than the case it was built for. Choosing k from n matches choosing $n - k$ from n , for every n and k , because leaving behind works the same way every time. It holds at the edges too: choosing 0 from 5 has one outcome, choosing nothing, and choosing 5 from 5 has one, choosing everything.

* * *

That correspondence is one case of a technique: **count one col-**



ten each way, and no pair or triple unmatched

Figure 8.1. The ten ways of choosing two of A, B, C, D, E , each joined to the three it leaves behind: every pair meets exactly one triple and every triple exactly one pair, so the two lists have the same length without either being counted. *Assumes:* nothing beyond the listing; the picture is the proof, and it is the reason the identity holds for every n and every k and not only for five and two

lection in two ways, and set the answers equal. It was used on the flattenable solids, where edges were counted by their ends and again by their sides.

It also finds the number of ways to choose. Write n choose k for the number of ways of choosing k things from n . Count the ordered lists of k different things taken from n . The first place can be filled in n ways; whichever thing fills it, the second place can be filled from the $n - 1$ that remain; and so on down to the k -th place, which has $n - k + 1$ things left to choose from. Every way of filling one place goes with every way of filling the next, so the counts multiply, and there are $n(n - 1) \cdots (n - k + 1)$ lists. Now count the same lists another way: choose the k things first, in n choose k ways, and then put them in order. Ordering k things is the same multiplication again, k choices for the first place, $k - 1$ for the second, down to 1 for the last, which gives $k(k - 1) \cdots 1$ orderings, a product written $k!$. So the second way counts $(n \text{ choose } k) \cdot k!$ lists. Both count the same lists, so $(n \text{ choose } k) \cdot k! = n(n - 1) \cdots (n - k + 1)$, and dividing by $k!$,

$$n \text{ choose } k = n(n - 1) \cdots (n - k + 1) / k!$$

For two from five that is $20/2$ and for three from five it is $60/6$, the two tens listed above; for four from seventeen it is $17 \cdot 16 \cdot 15 \cdot 14/24 = 2380$. Two more facts about these numbers come from the same kind of counting. Pick out one of $n + 1$ things and name it. Every choice of k from the $n + 1$ either includes the named thing or leaves it out. The choices that include it are made by choosing the other $k - 1$ from the remaining n things, in $(n \text{ choose } k - 1)$ ways; the choices that leave it out are made by choosing all k from those n , in $(n \text{ choose } k)$ ways. Every choice is in one pile or the other, so $(n + 1 \text{ choose } k) = (n \text{ choose } k) + (n \text{ choose } k - 1)$. Next, count the choices of every size at once. A choice of some things from n is made by deciding, for each of the n things in turn, whether it is in or out: two decisions each, so 2^n choices in all. Sorted by size, those same choices number n choose 0, n choose 1, and so on up to

n choose n . So the numbers n choose k , added over every k , come to 2^n .

Among n people, count the committees with a designated chair, committees of any size from one up. First way: choose the committee, then the chair. A committee of size k can be chosen in " n choose k " ways, and its chair in k ways, which gives $k \cdot (n \text{ choose } k)$, summed over all sizes. Second way: choose the chair first, in n ways; then each of the other $n - 1$ people is on the committee or not, in 2^{n-1} ways. Both counts ran over the same collection, so

$$\text{the sum over } k \text{ of } k \cdot (n \text{ choose } k) = n \cdot 2^{n-1}$$

At $n = 3$ the left side is $1 \cdot 3 + 2 \cdot 3 + 3 \cdot 1 = 12$ and the right side is $3 \cdot 4 = 12$, and listing directly, with people P , Q and R , there are three committees of one, three committees of two with two possible chairs each, and one committee of three with three possible chairs: $3 + 6 + 3 = 12$.

What the identity means is that committees with chairs can be assembled from either end.

Count in two ways the committees, of any size, that have a chair and a secretary who is a different member. What identity results?

A double count can go wrong in one way: the two counts must count the same collection. It is easy to write two plausible counts, set them equal, and never notice that one counted ordered pairs and the other unordered ones, or that one allowed repeats and the other did not. So before counting, the collection should be named in words, with no formula in the sentence.

Choose a committee of size k from n , and then a subcommittee of size j inside it, in $(n \text{ choose } k) \cdot (k \text{ choose } j)$ ways. Or choose the subcommittee of j from all n first, and then the remaining $k - j$ members of the committee from the $n - j$ people left, in $(n \text{ choose } j) \cdot (n - j \text{ choose } k - j)$ ways. Both count pairs of sets, one inside the other, so

$$(n \text{ choose } k) \cdot (k \text{ choose } j) = (n \text{ choose } j) \cdot (n - j \text{ choose } k - j)$$

and at $n = 4, k = 3, j = 2$ both sides are 12. Now change one word, letting the subcommittee include people who are not on the committee. The first count becomes $(n \text{ choose } k) \cdot (n \text{ choose } j)$, which at the same numbers is $4 \cdot 6 = 24$, while the second, unchanged, still gives 12. The identity fails only because the two counts no longer count the same collection.

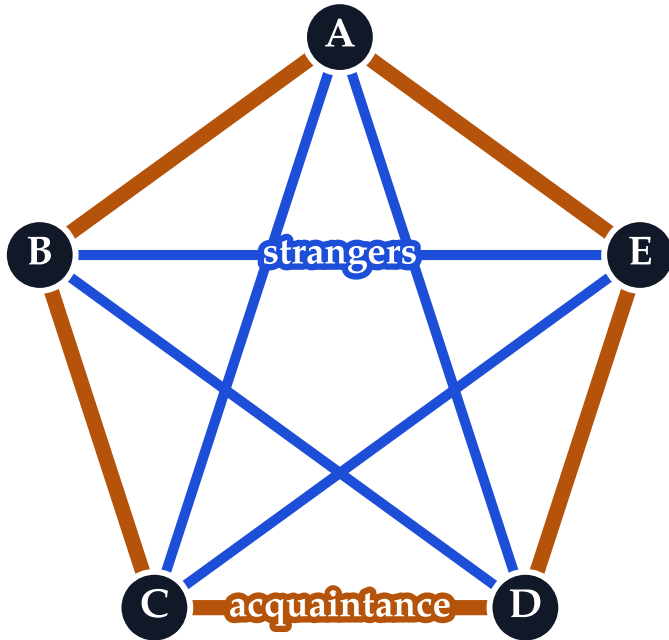
* * *

Among any six people, some three are mutual acquaintances or some three are mutual strangers. That is a statement about every pattern of acquaintance among six people, and there are $2^{15} = 32768$ patterns, since six people make $6 \cdot 5/2 = 15$ pairs and each pair is one thing or the other.

Call one of the six A . She stands in five relationships, each acquaintance or stranger, and five things sorted into two kinds put at least three in one kind, since two kinds holding at most two each account for at most four. Suppose A knows at least three others, B, C and D . If any two of those three know each other, say B and C , then A, B and C are mutual acquaintances. If no two of them do, then B, C and D are mutual strangers. If instead A is a stranger to at least three, the same argument runs with the words exchanged. Thirty-two thousand patterns are settled, and the choice of A was free: any of the six would have served.

Five people are not enough. Seat five in a circle, each acquainted with the two neighbours and nobody else. The acquaintances form a pentagon, which contains no triangle, and the strangers form the five-pointed star joining each person to the two opposite, which contains none either. So the answer is exactly six.

How many people guarantee four mutual acquaintances or four mutual strangers? Start with a lopsided question: among any nine people, some three are mutual acquaintances or some four are mutual strangers. Take one person, A , with eight relationships. If A knows four or more people, then either two of those four know each other, and with A they make three acquaintances, or no two



no one-colour triangle: five is not enough

Figure 8.2. Five people. Amber edges are acquaintance, blue are strangers. Neither pattern contains a triangle, so five is not enough. *Assumes:* that the ten edges are coloured as drawn; the absence of a triangle is argued in the text, since a picture cannot show absence

do, and they are four strangers. If A is a stranger to six or more, then among those six there are three mutual acquaintances, or three mutual strangers, who with A make four. Otherwise A knows at most three of the others and is a stranger to at most five, and three and five only account for all eight relationships if A knows exactly three; and A was any of the nine, so every one of the nine knows exactly three of the others. But then counting the ends of acquaintances gives $9 \cdot 3 = 27$, and every acquaintance has two ends, so the count of ends is twice the number of acquaintances and must be even. It is 27, which is odd, so the last case cannot happen. That count of ends is a double count.

Can eight people be arranged so that no three are mutual acquaintances and no four are mutual strangers?

Now among eighteen people take A , with seventeen relationships, at least nine of one kind. If A knows nine, then among those nine are three mutual acquaintances, who with A make four, or four mutual strangers. If A is a stranger to nine, the same argument with the words exchanged finishes. So eighteen people suffice.

Seventeen do not. Seat seventeen people at positions 0 to 16 round a circle, and let two be acquainted exactly when their positions differ by 1, 2, 4, 8, 9, 13, 15 or 16, the remainders that square numbers leave on division by 17: 1, 4, 9 and 16 are squares, and 25, 36, 49 and 64 leave 8, 2, 15 and 13. Counted the other way round the circle, a difference d becomes $17 - d$, and the eight pair off in that way, 1 with 16, 2 with 15, 4 with 13 and 8 with 9, so whether two people are acquainted does not depend on which way round their difference is counted. There are 2380 groups of four, and three symmetries of the pattern make checking them unnecessary.

Moving everybody the same number of places round the circle changes no difference, so it keeps acquaintances acquainted. Doubling every position does too, once a doubled position past 16 is read by its remainder on division by 17, so that 9 goes to seat 1 and 13 to seat 9: a difference d becomes $2d$, and doubling the eight differences gives 2, 4, 8, 16, 18, 26, 30 and 32, which on division by 17 leave 2, 4, 8, 16, 1, 9, 13 and 15, the same eight. Tripling every posi-

tion exchanges the two kinds: the eight differences become 3, 6, 12, 24, 27, 39, 45 and 48, which leave 3, 6, 12, 7, 10, 5, 11 and 14, exactly the eight differences of strangers; and those eight, 3, 5, 6, 7, 10, 11, 12 and 14, become 9, 15, 18, 21, 30, 33, 36 and 42, which leave 9, 15, 1, 4, 13, 16, 2 and 8, the eight differences of acquaintances. So four mutual strangers, tripled, would be four mutual acquaintances, and it is enough to show that there are none of those.

Doubling carries the difference 1 to 2, 4, 8, 16, 15, 13 and 9 in turn and then back to 1, a cycle through all eight, so enough doublings carry any of the eight differences to 1. Take any acquainted pair. Enough doublings bring its difference to 1, so the pair then sits at some positions p and $p + 1$, and moving everybody p places back puts it at 0 and 1. Doublings and shifts keep acquaintances acquainted, so four mutual acquaintances containing the pair are carried to four mutual acquaintances that include 0 and 1.

A common acquaintance of 0 and 1 is a position x whose difference from 0, which is x , and whose difference from 1, which is $x - 1$, are both among the eight; the positions that pass are 2, 9 and 16. Their differences from each other, 7, 14 and 7, belong to strangers, so no two of them are acquainted. Four mutual acquaintances that include 0 and 1 would need two more people, each acquainted with 0, with 1 and with each other; both would have to be among 2, 9 and 16, and no two of those are acquainted. So there are no four mutual acquaintances that include 0 and 1, hence none at all, and three short computations have replaced 2380 checks. They use that a whole number has one and only one remainder on division by 17, which is proved where division with remainder is taken up. So the answer is exactly eighteen.

The two steps that gave nine and eighteen are one step, and it climbs. Say that m people force (a, b) when every pattern of acquaintance among m people contains a group of mutual acquaintances of size a , or a group of mutual strangers of size b . Any b people force $(2, b)$, since either some two are acquainted or all b are strangers, and in the same way $(a, 2)$ is forced by any group of size a . And if m_1 people force $(a - 1, b)$ and m_2 people force $(a,$

$b - 1$), then $m_1 + m_2$ people force (a, b) . One of them, A , has $m_1 + m_2 - 1$ relationships, so A knows at least m_1 of the others or is a stranger to at least m_2 , since $m_1 - 1$ of one kind and $m_2 - 1$ of the other account for only $m_1 + m_2 - 2$. Among m_1 acquaintances of A there are $a - 1$ mutual acquaintances, who with A make a , or b mutual strangers; among m_2 strangers to A , the same with the words exchanged.

The rule $(n + 1 \text{ choose } k) = (n \text{ choose } k) + (n \text{ choose } k - 1)$ climbs in the same way, and it supplies a number of people that always works: $(a + b - 2 \text{ choose } a - 1)$ people force (a, b) . Check the edges first. When a is 2 the number is $(b \text{ choose } 1) = b$, and when b is 2 it is the number of ways of choosing $a - 1$ from a , which is a , since each such choice leaves out one; those are the two numbers of people found above. Now suppose the claim holds one step down, so that $(a + b - 3 \text{ choose } a - 2)$ people force $(a - 1, b)$ and $(a + b - 3 \text{ choose } a - 1)$ people force $(a, b - 1)$. Then their sum forces (a, b) , and by the rule their sum is $(a + b - 2 \text{ choose } a - 1)$, which is what the claim says. It gives 6 for $(3, 3)$, 10 for $(3, 4)$, 20 for $(4, 4)$ and 70 for $(5, 5)$.

For $(3, 4)$ the formula gives 10, but the count of ends showed that 9 people already force it, and then $9 + 9 = 18$ people force $(4, 4)$, the eighteen found above. Carrying those on, $5 + 9 = 14$ people force $(3, 5)$, $14 + 18 = 32$ force $(4, 5)$, and $32 + 32 = 64$ force $(5, 5)$. In general $(2k - 2 \text{ choose } k - 1)$ people force (k, k) . That number is one of the numbers $2k - 2 \text{ choose } j$, which added over every j come to 2^{2k-2} , so it is at most $2^{2k-2} = 4^{k-1}$. And a number that forces the conclusion still forces it when people are added, since a pattern among more people holds a pattern among any fewer of them inside it. Hence 4^{k-1} people always contain k mutual acquaintances or k mutual strangers.

For five mutual acquaintances or five mutual strangers, nobody knows the answer. It lies between 43 and 46; the lower bound has stood since 1989, and the upper bound was brought down to 46 in 2024. The arguments that settled six and eighteen do not reach it, and the ways of dividing the pairs among 43 people are far too

many to search one by one.

THE TRIANGLE OF CHOICES

The numbers n choose k obey a rule with no multiplication in it: sorting the choices of k things from $n + 1$ by whether one named thing is among them gave

$$(n + 1 \text{ choose } k) = (n \text{ choose } k) + (n \text{ choose } k - 1)$$

So the whole family can be written down by addition alone. Put a 1 at the top, for the one way of choosing nothing from nothing, and let each row begin and end with 1 and have every other entry equal to the sum of the two entries above it:

$$\begin{array}{c}
 1 \\
 1 \ 1 \\
 1 \ 2 \ 1 \\
 1 \ 3 \ 3 \ 1 \\
 1 \ 4 \ 6 \ 4 \ 1 \\
 1 \ 5 \ 10 \ 10 \ 5 \ 1 \\
 1 \ 6 \ 15 \ 20 \ 15 \ 6 \ 1
 \end{array}$$

The entry in row n at place k , both counted from 0, is n choose k . The triangle was drawn in China and in Persia centuries before Pascal wrote his treatise on it in 1654, and nearly everything in it can be seen in two ways: as a fact about sums of neighbours, provable

by climbing from row to row, and as a fact about choices, provable by counting one collection twice. The second way is usually shorter and always says more.

Each row reads the same in both directions, because choosing k things is leaving $n - k$. Each row adds up to twice the row before. Every inner entry of the new row is the sum of a pair of neighbours in the old row. An inner entry of the old row serves in two such pairs, once as the left-hand member and once as the right-hand member. The 1 at an end serves in one pair only, but it is also copied to the end of the new row, so it too is used twice; the lone 1 of the top row is in no pair and is copied to both ends. Adding up the new row therefore adds up every entry of the old row twice; so row n adds up to 2^n , the number of ways of deciding, for each of n things, whether it is in or out.

And if the entries of any row below the top are given alternate signs, they add up to 0: for row 4, $1 - 4 + 6 - 4 + 1 = 0$. For a row with an even number of entries the symmetry explains it: the entry at place k and its equal at place $n - k$ carry opposite signs, so the row cancels in pairs. In a row with an odd number of entries, such as row 4, the mirror entries carry the same sign and nothing cancels that way, so a reason is needed, and the reason that follows covers every row.

Fix one of the n things, call it a , and pair every choice that contains a with the same choice with a taken out. The two choices of a pair differ in size by one, so one has an even number of things and the other an odd number, and every choice belongs to exactly one pair. So the choices of even size are as many as the choices of odd size. That is what the alternating sum says: the entries with a plus sign, at places 0, 2, 4 and so on, count the choices of even size, the entries with a minus sign count the choices of odd size, and the two totals are equal, so the sum is 0.

* * *

Multiply out $(a + b)^4$, which is $(a + b)(a + b)(a + b)(a + b)$. Ev-

ery term of the result comes from taking the a or the b from each of the four brackets and multiplying the four letters taken. The term a^2b^2 arises once for each way of deciding which two brackets supply the b , so it arises 4 choose 2 times, and

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

The coefficients are row 4 of the triangle, and nothing about the number four was used. For every n , the coefficient of $a^{n-k}b^k$ in $(a + b)^n$ is the number of ways of choosing which k of the n brackets supply the b , which is n choose k . That is the *binomial theorem*, and the numbers n choose k are called binomial coefficients because of it.

The two facts about rows are the theorem at particular values. With $a = b = 1$ every product $a^{n-k}b^k$ is 1, so $(a + b)^n$ multiplied out is the sum of row n , while $(a + b)^n$ itself is 2^n . With $a = 1$ and $b = -1$ the products are 1 and -1 by turns, so the multiplied-out form is row n with alternate signs, while $(a + b)^n$ itself is 0^n , which is 0 for every n above 0. With $a = 10$ and $b = 1$ it explains a curiosity: $11^2 = 121$, $11^3 = 1331$ and $11^4 = 14641$ are rows of the triangle read as numbers, because $(10 + 1)^4$, by the theorem, is $1 \cdot 10^4 + 4 \cdot 10^3 + 6 \cdot 10^2 + 4 \cdot 10 + 1$, and a number written like that has 1, 4, 6, 4, 1 as its digits. The same happens for every n as long as no coefficient exceeds 9. At 11^5 the row 1, 5, 10, 10, 5, 1 has to carry, and 11^5 is 161051.

The theorem also gives quick and honest estimates. For a positive x , every term of $(1 + x)^n$ is positive, so stopping early gives a bound from below. With $a = 1$ and $b = x$ the theorem reads $(1 + x)^n = 1 + nx + (n \text{ choose } 2)x^2 + \dots + x^n$, and n choose 2 is $n(n - 1)/2$. So $(1 + x)^n$ is at least $1 + nx$, and at least $1 + nx + n(n - 1)x^2/2$. A sum of money growing by one part in a hundred each year for ten years grows by the factor 1.01^{10} , which is $1 + 0.1 + 0.0045 + 0.00012$ and a little more, about 1.10462; the first two terms are the interest without compounding, and the rest is what compounding adds. The bound $1 + nx$ shows that the powers of any number above 1 pass every bound. Such a number

is $1 + x$ for some positive x ; its n th power $(1 + x)^n$ is at least $1 + nx$, and nx passes every bound by the axiom of Archimedes.

* * *

A town has its streets laid out as a grid. How many shortest routes lead from one corner of the town to the opposite corner, five blocks to the east and three to the north? A shortest route is a sequence of eight blocks, five of them eastward and three northward, in some order, and it is settled by saying which three of the eight are the northward ones. So there are 8 choose 3 routes, which is 56.

The grid turns the triangle into geography. Write at each street corner the number of shortest routes that reach it from the starting corner. A route arrives from the west or from the south, so each number is the sum of its western and southern neighbours, which is the rule of the triangle, tilted so that its rows run along the diagonals of the town.

A route to the corner n blocks east and n blocks north is $2n$ blocks long, and there are $(2n \text{ choose } n)$ of them. After its first n blocks such a route stands at a corner k east and $n - k$ north, for some k . These halfway corners lie along a diagonal line, and the route meets that line at this one moment and no other. There are $(n \text{ choose } k)$ ways to arrive at the corner k east and $n - k$ north. From there n blocks remain, of which $n - k$ must be east, so there are $(n \text{ choose } n - k)$ ways to go on, and $n \text{ choose } n - k$ is $n \text{ choose } k$ again. Any way of arriving can be followed by any way of going on, so $(n \text{ choose } k)^2$ routes pass through that corner. So, sorting the routes by where they cross,

$$(n \text{ choose } 0)^2 + (n \text{ choose } 1)^2 + \dots + (n \text{ choose } n)^2 = (2n \text{ choose } n)$$

For $n = 3$ that reads $1 + 9 + 9 + 1 = 20$. The same sorting works for any corner. Let the corner lie $a + b$ blocks from the start, n of them east, so that $(a + b \text{ choose } n)$ routes lead to it. Sort the routes by where they stand after their first a blocks. A route that has gone

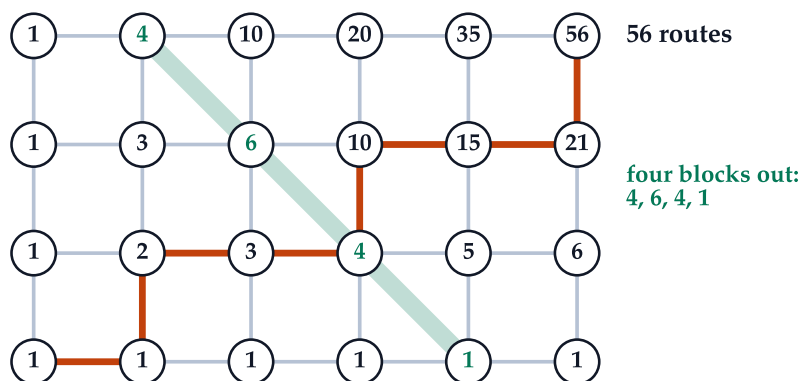


Figure 9.1. At each corner of the town, the number of shortest routes that reach it from the corner at the bottom left. Each number is the sum of its neighbours to the west and the south, and the corners at one distance from the start carry one row of the triangle, cut short here where the row runs off the edge of the town. *Assumes:* that a shortest route only ever goes east or north, so that the last block of a route arrives from the west or from the south and from nowhere else

k blocks east by then has arrived in one of $(a \text{ choose } k)$ ways. It has b blocks left, of which $n - k$ must be east, so it can go on in $(b \text{ choose } n - k)$ ways. So the sum over k of $(a \text{ choose } k)(b \text{ choose } n - k)$ is $(a + b \text{ choose } n)$. Said with people instead of blocks, it is the choosing of n from a men and b women, sorted by the number of men chosen. The identity of squares is the case $a = b = n$.

Sorting routes by their last northward block gives another family. A route to the corner $k + 1$ north and $n - k$ east is $n + 1$ blocks long. It goes north for the last time along some block, after which it only goes east. Say j blocks come before that last northward block. The other k northward blocks are all among those j , so the j blocks end at the corner $j - k$ east and k north, and there are $j \text{ choose } k$ ways of walking them. After that nothing is left to decide: one block north, then east to the end. The number j can be anything from k to n , and each route belongs to one value of j only. So

$$(k \text{ choose } k) + (k + 1 \text{ choose } k) + \dots + (n \text{ choose } k) = (n + 1 \text{ choose } k + 1)$$

In the triangle this is a line of entries running down a diagonal, with their sum sitting one step off the end of the line: $1+3+6+10 = 20$. With $k = 1$ it says $1 + 2 + \dots + n = (n + 1 \text{ choose } 2)$, which is $n(n + 1)/2$. With $k = 2$ it adds up the numbers $(j \text{ choose } 2)$. Now $j \text{ choose } 2$ is $j(j - 1)/2$, so twice it is $j^2 - j$, and j^2 is twice $(j \text{ choose } 2)$ plus j . Adding that up for j from 1 to n gives the sum of the first n squares as $2(n + 1 \text{ choose } 3) + (n + 1 \text{ choose } 2)$, which is $n(n + 1)(2n + 1)/6$.

Sums of cubes and higher powers follow the same way, each power of j being first written as a combination of the numbers $j \text{ choose } 1$, $j \text{ choose } 2$, $j \text{ choose } 3$, and so on. Such a combination always exists, because multiplying $j \text{ choose } i$ by j gives $(i + 1) \cdot (j \text{ choose } i + 1) + i \cdot (j \text{ choose } i)$. To see it, count the ways of choosing i things from j and then pointing at one of the j things, which is $j \cdot (j \text{ choose } i)$. If the thing pointed at lies outside the choice, the two together make a choice of $i + 1$ things with one of them marked, and there are $(i + 1) \cdot (j \text{ choose } i + 1)$ of those. If it lies inside, there is a choice of i things with one marked, and there are $i \cdot (j \text{ choose } i)$ of those. So once a power of j is written as a combination of the numbers $j \text{ choose } i$, multiplying every piece by j writes the next power the same way.

One more kind of choice reduces to the same count. A shop sells n kinds of sweet, and k sweets are to be bought, with repeats allowed and the order not mattering. Lay the purchase out in a row, kind by kind, with a divider between each kind and the next. With three kinds, two sweets of the first kind, none of the second and one of the third make the row sweet, sweet, divider, divider, sweet. In general the row has k sweets and $n - 1$ dividers, in some order. Every order of those $n + k - 1$ objects is a purchase, and different orders are different purchases. An order is settled by saying which k of the $n + k - 1$ places hold sweets. So there are $(n + k - 1 \text{ choose } k)$ purchases. A set of dominoes has one tile for each way of choosing two numbers from 0 to 6 with a repeat allowed, which

is a purchase of $k = 2$ sweets from $n = 7$ kinds. The formula makes that $(8 \text{ choose } 2)$, which is 28, and a set of dominoes has 28 tiles.

* * *

Twelve people vote one at a time, six for each of two candidates, A and B . In how many of the possible orders of voting is A never behind during the count? As a route on the grid, with a vote for A as a block east and a vote for B as a block north, the question asks for the routes from corner to corner of a square town of side 6 that never go above the diagonal.

Count the bad routes instead, the ones that at some moment have B ahead. A bad route touches the line one step above the diagonal, the line of corners with one more block north than east. Take the first corner where it touches that line, and reflect everything after that corner in the line, so that each later block east becomes a block north and the reverse. Before the touching corner the route is untouched. Write n for the side of the town, here 6. At the touching corner the route has gone some number j of blocks east and $j + 1$ north, so it still has $n - j$ blocks east and $n - j - 1$ blocks north to go. The reflection swaps those two numbers. So the reflected route ends not at the corner n east and n north but at its mirror image, $n - 1$ east and $n + 1$ north.

Every route from the start to that mirror corner has to cross the line, since it starts below the line and ends above it. Reflecting such a route after its first touching gives a bad route. The two reflections undo each other, so each bad route is paired with one route to the mirror corner and each route to the mirror corner with one bad route. So the bad routes are exactly as many as all the routes to the mirror corner, $(2n \text{ choose } n + 1)$, and the good routes number

$$(2n \text{ choose } n) - (2n \text{ choose } n + 1)$$

This simplifies. Choosing k things from m can be done in $m(m - 1) \cdots (m - k + 1)/k!$ ways, so the number $(2n \text{ choose } n)$ is $2n(2n - 1) \cdots (n + 1)/n!$. The same formula for $(2n \text{ choose } n + 1)$

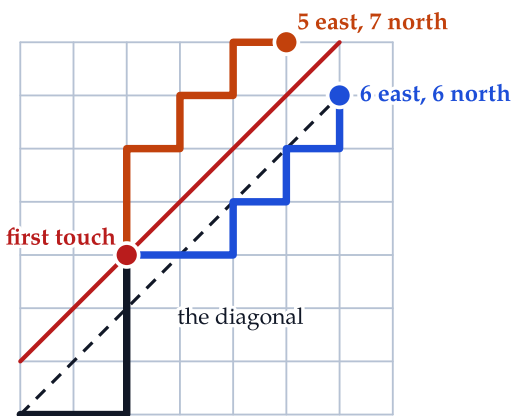


Figure 9.2. A count in which B gets ahead, drawn as a route that touches the line one step above the diagonal. Reflecting everything after the first touch in that line carries the end of the route to the mirror image of the far corner, and every route to that mirror corner arises this way from one bad route. *Assumes:* that the reflection starts at the first touch, which the reflected route shares with the original, so that reflecting again after that same corner gives the original back

has one more factor on top, which is n , and one more factor below, which is $n + 1$. So $(2n \text{ choose } n + 1)$ is $n/(n + 1)$ times $(2n \text{ choose } n)$. Taking it away leaves one part in $n + 1$: the good routes number $(2n \text{ choose } n)/(n + 1)$. For $n = 1, 2, 3, 4, 5$ and 6 that gives $1, 2, 5, 14, 42$ and 132 . Of the 924 possible counts of the twelve votes, 132 never have A behind, exactly one in seven.

These are the *Catalan numbers*, and they count many collections that do not look like elections. A product of $n + 1$ letters can be bracketed so that every multiplication has two factors: for four letters, $((ab)c)d$, $(a(bc))d$, $(ab)(cd)$, $a((bc)d)$ and $a(b(cd))$, five ways. First write in the outermost pair of brackets, so that every multiplication has a pair of its own: $((ab)c)d$ becomes $((ab)c)d$, and $(a(bc))d$ becomes $(a(bc))d$. Now read from left to right, writing a block east for each opening bracket and a block north for each letter except the last. The first bracketing gives east, east, east, north, north, north. The second gives east, east, north, east, north, north.

The route never goes above the diagonal.

Each letter except the last completes the first factor of exactly one multiplication: in $((a(bc))d)$ the letter a completes the first factor of $a(bc)$, the letter b completes the first factor of bc , and the letter c completes $a(bc)$, which is the first factor of the whole product. The bracket of that multiplication was opened before the letter arrived, so every block north answers an earlier block east of its own. Nothing is lost by dropping the closing brackets and the last letter, since a bracket closes at the moment its second factor is complete and at no other; so the route gives back the bracketing. Since each multiplication joins two pieces into one, $n + 1$ letters take n multiplications, and the route has n blocks each way. Every route of n blocks each way that never goes above the diagonal also comes from a bracketing. Write a bracketing as (XY) , with X and Y its two factors. Its route is a block east for the outer bracket, then the route of X , then a block north for the last letter of X , then the route of Y ; during the route of X it stays at least one step east of the diagonal, so that block north is where it first comes back to the diagonal. Now take any route that never goes above the diagonal and has some blocks. Its first block is east. Cut it there and at the block north that first brings it back to the diagonal: the stretch in between and the stretch after are shorter routes of the same kind. Cutting those in turn, and so on, ends at routes with no blocks, which come from single letters; putting the pieces back together as (XY) at every stage then builds a bracketing whose route is the one taken. Euler met the same numbers in 1751 counting the ways of cutting a polygon with $n + 2$ corners into triangles by diagonals that do not cross: a square in 2 ways, a pentagon in 5, a hexagon in 14.

Why is the number of ways of cutting a polygon of $n + 2$ corners into triangles equal to the number of bracketings of $n + 1$ letters?

* * *

How many of the numbers from 1 to 100 are divisible by 2, by 3 or by 5? There are 50 multiples of 2, 33 multiples of 3 and 20

multiples of 5, but $50 + 33 + 20$ counts a number such as 6 twice and 30 three times. Take away the numbers counted twice: 16 multiples of 6, 10 multiples of 10 and 6 multiples of 15. Now 30 has been counted three times and removed three times, so it has to be put back, with the other multiples of 30, of which there are 3. The answer is

$$50 + 33 + 20 - 16 - 10 - 6 + 3 = 74$$

and so the other 26 numbers from 1 to 100 are divisible by none of 2, 3 and 5. Since $30 = 2 \cdot 3 \cdot 5$, those are the numbers that share no factor with 30.

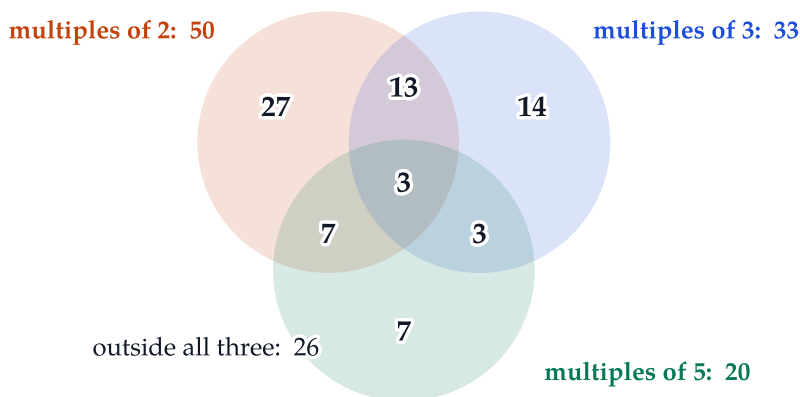


Figure 9.3. The numbers from 1 to 100 sorted by which of 2, 3 and 5 divide them. The seven regions hold 27, 14, 7, 13, 7, 3 and 3 numbers, 74 in all, and 26 lie outside; adding the three circles counts the overlaps twice and the middle three times. *Assumes:* nothing beyond the count itself; the sizes of the circles are not to scale

The pattern is general: add the sizes of the collections, subtract the sizes of the overlaps of two, add the overlaps of three, and so on with alternating signs. To see that it counts everything once, follow a single object that belongs to exactly m of the collections. It is counted m times in the first round, once for each collection it belongs to. In the second round it is taken away once for each pair

of its m collections, which is $(m \text{ choose } 2)$ times. In the third it is put back once for each three of them, $(m \text{ choose } 3)$ times. So in all it is counted

$$(m \text{ choose } 1) - (m \text{ choose } 2) + (m \text{ choose } 3) - \dots$$

times, and that is 1. The alternating sum of the whole of row m is 0, which says $(m \text{ choose } 0) - (m \text{ choose } 1) + (m \text{ choose } 2) - \dots = 0$. Moving everything except the first entry to the other side, $(m \text{ choose } 1) - (m \text{ choose } 2) + (m \text{ choose } 3) - \dots$ is equal to $m \text{ choose } 0$, which is 1. The identity about alternating rows was the whole proof. This way of counting is called *inclusion and exclusion*.

Ten people leave their hats at the door, and the hats are handed back at random, every one of the $10!$ ways of handing them back counting for as much as any other. In what fraction of those ways does nobody get his own hat? Count the ways of handing back the hats in which somebody does. For each person take the collection of arrangements in which that person gets his own hat: it has $9!$ members, since the other nine hats can be handed out in any of their $9!$ ways. For two people, $8!$ arrangements give both their own hats, and there are $(10 \text{ choose } 2)$ pairs of people; and so on, down to all ten people, who all get their own hats in one arrangement only. That fits the pattern $9!, 8!, \dots, 1!$ if $0!$, the number of ways of handing out no hats at all, is taken to be 1, as it is from now on. By inclusion and exclusion the number of arrangements in which at least one person gets his own hat is

$$(10 \text{ choose } 1) \cdot 9! - (10 \text{ choose } 2) \cdot 8! + (10 \text{ choose } 3) \cdot 7! - \dots$$

Now $(10 \text{ choose } k)$ is $10 \cdot 9 \cdots (10 - k + 1)/k!$, and multiplying its top by $(10 - k)!$ supplies the remaining factors of $10!$, so $(10 \text{ choose } k) \cdot (10 - k)!$ is $10!/k!$, and the sum is $10!/1! - 10!/2! + 10!/3! - \dots$. Taking it away from all $10!$ arrangements, the number in which nobody gets his own hat is

$$10! \cdot (1 - 1/1! + 1/2! - 1/3! + \dots + 1/10!)$$

which is 1334961 out of 3628800, a chance of 0.367879. For four people the count is 9 out of 24, for five it is 44 out of 120, and for six 265 out of 720, chances of 0.375, 0.3667 and 0.3681. The chance hardly depends on the number of people: it closes in, from alternate sides and very fast, on a number near 0.3679, which will be met again in a quite different place.

Five different presents are to be given to three children so that every child gets at least one. In how many ways can it be done?

How many of the numbers from 1 to 360 share no factor with 360?

In how many ways can twenty coins, all alike, be shared among four people so that each gets at least one?

WHAT A CONFIRMATION IS WORTH

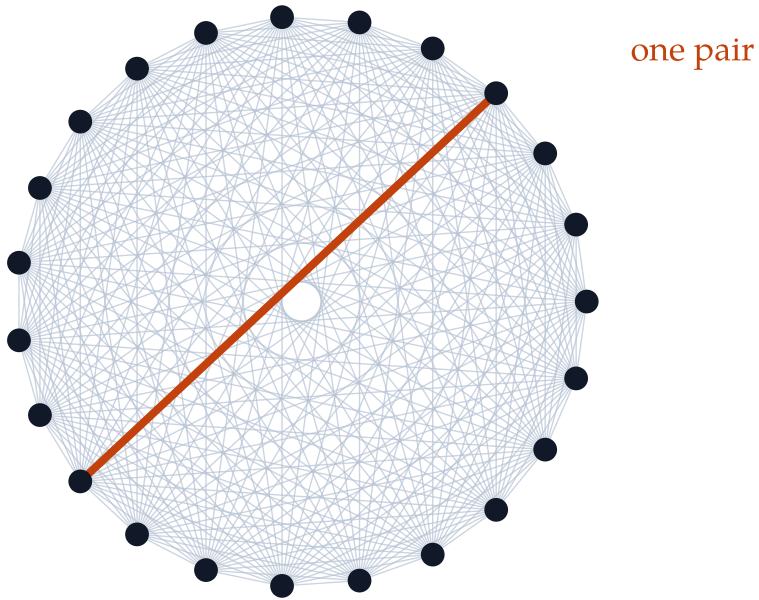
Twenty-three people are in a room. Is it more likely than not that two of them share a birthday? The usual guess is no, and the guess has a source: 23 is a small fraction of 365, and a shared birthday among so few feels like a coincidence. That is a conjecture, and counting tests it.

A chance, throughout, is a share of cases: list the cases that could arise, all of them equally likely, count the ones the description allows, and divide by the whole number of cases. Count the ways the birthdays can fall with no two alike. The first person has any of 365 days, the second any of the other 364, the third any of 363, and so on to the twenty-third, who has 343 left. Multiply those numbers and divide by 365^{23} , the number of ways the birthdays can fall with no restriction, treating every arrangement as equally likely. The ratio is 0.4927, and that is the chance that no two share a birthday. Either no two share or some two do, so the chance that some two share a birthday is $1 - 0.4927 = 0.5073$, a little over one half. With twenty-two people it is 0.4757, so twenty-three is exactly where the chance crosses one half.

The guess compared 23 with 365. But a shared birthday is a fact about a pair of people, and twenty-three people form 253 pairs,

which is not a small fraction of 365.

How many people, besides one particular person, must be in a room before it is more likely than not that one of them shares that particular person's birthday?



23 people, 253 pairs

Figure 10.1. Twenty-three people around a circle and every pair among them joined: 253 pairs, each a separate chance of a shared birthday, which is the count the instinct that compared 23 against 365 never made. *Assumes:* that the 253 chances depend on one another, since three pairs forming a triangle constrain one another: if two of them share a birthday, so does the third; the count of pairs is a re-description that dissolves the surprise, and the exact chance in the text is computed without it

* * *

Chances also give averages, and an average needs a definition. A die is thrown until a six appears; on average, how many throws does that take? The average is the sum of each possible number of throws times its chance: one throw with chance $1/6$, two with chance $(5/6)(1/6)$, and so on without end. A shorter route notices that after a throw that is not a six the situation is exactly the starting situation, so if m is the average, then m is one throw plus, with chance $5/6$, another m : $m = 1 + (5/6)m$, and $m = 6$.

The short route assumes that m is a finite number. The equation alone cannot guarantee that: an endless wait also satisfies it, endless being one throw plus five sixths of endless. The long route shows that the average is finite, and that it is 6. Take the term for a wait of k throws, which is k times the chance of that wait, and read it as that chance counted k times: once for each j from 0 to $k - 1$, that is, once for each j that the wait is longer than. Now add up the first n terms by j instead of by k . For a given j , the chances collected are those of waits of $j + 1$, $j + 2$, and so on up to n throws, which together make the chance that the wait is longer than j throws but not longer than n . The wait is longer than j throws exactly when the first j throws all miss, which has chance $(5/6)^j$, the misses multiplying as the birthdays did; so what is collected for j is $(5/6)^j - (5/6)^n$. Regrouped by j , then, the first n terms of the long route add up to the sum, over j from 0 to $n - 1$, of $(5/6)^j - (5/6)^n$.

The sum $G = 1 + 5/6 + \dots + (5/6)^{n-1}$ is found by subtracting: $G - (5/6)G = 1 - (5/6)^n$, so $G = 6 - 6(5/6)^n$. The sum over j is G less n copies of $(5/6)^n$, so the first n terms add up to $6 - 6(5/6)^n - n(5/6)^n$, which is $6 - (n + 6)(5/6)^n$. That grows with n , since every further term adds something, and it stays below 6. The shortfall $(n + 6)(5/6)^n$ is multiplied at each further step by $(n + 7)/(n + 6) \cdot 5/6$, which is at most $11/12$ once n is 4 or more. A quantity multiplied again and again by at most $11/12$ becomes as small as anyone likes, since the powers of $12/11$ pass every bound; so the first n terms come as close to 6 as anyone likes. What it means for a sum with no last term to have a value is settled once limits are.

The average of a sum is the sum of the averages, whether or not

the pieces depend on each other. On a finite list of outcomes this is one total counted two ways. For each outcome and each piece, take the value of that piece in that outcome times the chance of the outcome, and add all of these products. Added outcome by outcome, the pieces within an outcome combine first into the value of the sum there, and the total is the average of the sum. Added piece by piece, each piece collects its own values times their chances over all the outcomes, which is its own average, and the total is the sum of the averages. Nothing in either count asks how the pieces depend on each other.

Twenty people, ten men and ten women, sit in a row of twenty seats in an order drawn by lot. How many neighbouring pairs, on average, are a man next to a woman? The nineteen neighbouring pairs depend on each other, since neighbouring pairs share a person, and it does not matter. The number of mixed pairs is a sum of nineteen quantities, each 1 or 0, and each has average equal to the chance that its pair is mixed, because a quantity that is 1 with some chance and 0 otherwise has average 1 times that chance. For one pair of seats, the left seat holds a man with chance $10/20$, and then the right seat holds one of the 10 women among the 19 people remaining with chance $10/19$; a woman on the left and a man on the right has the same chance. So the chance that the pair is mixed is $2 \cdot (10/20) \cdot (10/19) = 10/19$. So the average number of mixed pairs is $19 \cdot 10/19 = 10$.

Throw the die until every face has appeared. The first new face takes one throw. The second, with five of six faces new, takes $6/5$ throws on average, by the argument for the six: a throw fails to show a new face with chance $1/6$, so the average m satisfies $m = 1 + (1/6)m$, and $m = 6/5$; the third takes $6/4$, then $6/3$, $6/2$ and $6/1$. Adding the six averages uses the rule for the average of a sum. The rule was argued on a finite list of outcomes, and these waits have no longest case, so their list of outcomes has no end. Here the rule rests on sums with no last term, as the average of a single wait did. The total is $6 \cdot (1 + 1/2 + 1/3 + 1/4 + 1/5 + 1/6)$, which is 14.7 throws. With N faces it is N times $1 + 1/2 + \dots + 1/N$, and that sum is about

2.93 for ten terms, 5.19 for a hundred and 7.49 for a thousand. It grows slowly, and a table cannot show whether it grows without bound or approaches a ceiling.

* * *

An average can show that a pattern exists without building it. Colour every pair among n people as acquaintance or stranger by tossing a coin for each pair. A particular group of five has ten pairs, and all ten come out the same colour in two of the 2^{10} equally likely outcomes, a chance of 1 in 512. The number of groups of five that come out one colour is, like the number of mixed pairs in the row of seats, a sum of quantities each 1 or 0, one for each group, and its average is the sum of their chances.

Here the one total counted two ways is a number of tickets: write one ticket for every combination of a way the coins can fall and a group of five that comes out one colour in that way. Count the tickets group by group: each of the n choose 5 groups comes out one colour in 1 of every 512 ways the coins can fall, so the number of tickets is $(n \text{ choose } 5)/512$ times the number of ways the coins can fall. Count them way by way: the tickets belonging to a way of falling are exactly its groups of five with one colour. All the ways are equally likely, so on average a way of falling has $(n \text{ choose } 5)/512$ groups of five with one colour. If that average is below 1, some way the coins fall produces no such group, since an average below 1 cannot be made of numbers all at least 1; and a colouring with no one-coloured group of five has no five mutual acquaintances and no five mutual strangers. For $n = 11$ the average is $462/512$, below 1; for $n = 12$ it is $792/512$.

An average above 1 forces nothing: it does not make every way of falling contain such a group. Birthdays show it. Among 28 people there are 378 pairs, each sharing a birthday with chance $1/365$, and averages add whatever the dependence between the pairs, so the average number of pairs sharing a birthday is $378/365$, above 1. Yet the chance that no two share is 0.3455, so more than a third

of the time there is no sharing pair at all; among 23 the average is 253/365, about 0.69, against a chance of 0.507 that some pair shares.

Eleven people is a weak result for groups of five, where the smallest number of people that always contains five of one kind is at least 43. But as the size of the group increases, the number of people this averaging argument can colour grows faster than the number any known construction can colour.

For groups of k , one colour needs $k(k-1)/2$ tosses to agree, a chance of 2 in $2^{k(k-1)/2}$, and n choose k is at most $n^k/k!$. So the average number of one-coloured groups of k among n people is at most $(n^k/k!) \cdot 2^{1-k(k-1)/2}$, and when n is at most $2^{k/2}$, so that n^k is at most $2^{k^2/2}$, that is at most $2^{1+k/2}/k!$, the exponents $k^2/2$ and $1-k(k-1)/2$ adding to $1+k/2$. This is below 1 for every k from 3 on: squared, the claim is $4 \cdot 2^k < (k!)^2$, which at $k=3$ says $32 < 36$, and passing from k to $k+1$ doubles the left side and multiplies the right by $(k+1)^2$, at least 16. So for every k from 3 on, the pairs among any number of people up to $2^{k/2}$ can be coloured with no k of one kind.

For groups of twenty, one colour needs 190 tosses to agree, and among 1024 people the average is about 7 in 10^{16} : the pairs among 1024 people can be coloured with no twenty mutual acquaintances and no twenty mutual strangers, while the climb from nine and eighteen says that 4^{19} people, 274,877,906,944 of them, always contain twenty of one kind. Erdős gave this argument in 1947. It produces no colouring.

The argument also counts some colourings twice: adding the chances group by group counts a colouring with two one-coloured groups twice, so the sum is larger than the chance that some group has one colour. A sum that is still below 1 is enough, since the chance it bounds is smaller still. A double count goes wrong when its two counts are of different collections. Here the collections do differ, one-coloured groups on one side and colourings that contain a one-coloured group on the other, but the difference is known and runs in the safe direction.

* * *

Two children, and the elder is a boy. The chance that both are boys is one half, since the younger is a coin. Two children, and at least one is a boy. The chance that both are boys is now one third: of the three equally likely families with a boy, boy-boy, boy-girl and girl-boy, one has two boys. In each case a fact is given and a chance asked for in its light, and the operation is the same: strike out every possibility the fact excludes and count what is left. "The elder is a boy" strikes out two of the four families; "at least one is a boy" strikes out one.

The chance of E given F is the count of what survives F and also satisfies E , over the count of what survives F . Divide both counts by the whole number of cases, which leaves their ratio unchanged, and this becomes the chance of both divided by the chance of F . Turned round, the chance of both is the chance of F times the chance of E given F . That is how the birthday count was built. Share the 365^{23} underneath among the twenty-three numbers multiplied on top, one 365 to each, and the ratio becomes a product of factors $365/365$, $364/365$, $363/365$ and so on. The factor $364/365$ is the second person's chance of avoiding the first birthday; $363/365$ is the third person's chance of avoiding both earlier birthdays, given that those two differ; and so on, each factor a chance given that all the earlier birthdays avoided each other.

E and F are *independent* when the chance of E given F is the chance of E : knowing F changes nothing. The chance of both is the chance of F times the chance of E given F , so for independent E and F it is the chance of F times the chance of E . The coin tosses and the birthdays were treated as independent, which is what allowed multiplying chances. That is a definition, and whenever it is applied it is also a claim about the world, which the arithmetic cannot check.

Toss two coins. Let E be that the first shows heads, F that the second does, and G that the two agree. Each has chance one half, and any two are independent: given the first is heads, the second is

heads with chance one half, and agrees with the first with chance one half, and the same for F and G . Three events are independent when two things hold: every two of them are independent, and knowing that any two of them happened changes nothing about the chance of the third. These three fail the second condition. E and F together make G certain, where its chance alone is one half, and the chance of all three is $1/4$, not the $1/8$ that multiplying three halves would give. Independence of every pair is not independence of the three.

	HH	HT	TH	TT
E: first H	yes	yes	no	no
F: second H	yes	no	yes	no
G: agree	yes	no	no	yes

E and F both: only HH. E and G both: only HH. F and G both: only HH.

All three: only HH, one outcome in four, not one in eight.

Figure 10.2. Two coins, four equally likely outcomes, and three events: E the first shows heads, F the second does, G the two agree. Each event holds in two of the four outcomes, each pair of events holds together in exactly one, so every pair is independent; but E and F together force G , so the three are not. Assumes: that the four outcomes are equally likely, which is the modelling choice the text names; the arithmetic of the table is exact once that is granted

* * *

The chance of both E and F can be computed either way round, so the chance of E given F times the chance of F equals the chance of F given E times the chance of E . Call the ratio of the chance of something to the chance of its failing the *odds* on it, so that a chance

of a third is odds of 1 to 2 and a chance of 0.95 is odds of 19 to 1.

Now write that identity twice. With H in the place of F it says that the chance of H given E , times the chance of E , equals the chance of E given H times the chance of H . With the alternative to H in the place of F it says the same with ' H is false' for H throughout. Divide the first by the second. On the left the chance of E cancels, leaving the chance of H given E over the chance that H is false given E : the odds on H in the light of E . On the right the chance of H over the chance that H is false is the odds on H before the evidence, and it is multiplied by the chance of E given H over the chance of E given that H is false.

So the odds of a hypothesis H against its alternative, in the light of evidence E , are the odds before the evidence multiplied by one ratio, the chance of E if H is true over the chance of E if H is false. Evidence acts through that ratio alone. Evidence equally likely either way changes nothing, however consistent with H it is.

Suppose a court hears that a piece of evidence is consistent with a defendant's guilt. If the evidence would have been just as likely from an innocent person, the ratio is 1 and it moves nothing. **Evidence supports a hypothesis only if it is more likely under that hypothesis than under its rival**, and the ratio says by how much.

A disease affects one person in two hundred. A test detects it 95 times out of 100 when it is present and gives a false alarm once in 100 when it is not. Someone tests positive. What is the chance that this person has the disease? The figure that comes to mind is ninety-five percent. Among two hundred people, one has the disease and is detected with chance 0.95; of the other 199, one in a hundred gives a false positive, 1.99 people. Of those who test positive, 0.95 out of $0.95 + 1.99$ have the disease, which is 32 percent. The ratio gives the same: the odds before the test were 1 to 199, the ratio for a positive result is 0.95 to 0.01, and the product is odds of 95 to 199, about one to two. Odds of 95 to 199 are a chance of 95 in $95 + 199$, which is $95/294$, the same 32 percent.

Test the same person again, and suppose the second result is positive too. If the two results are independent among the ill and,

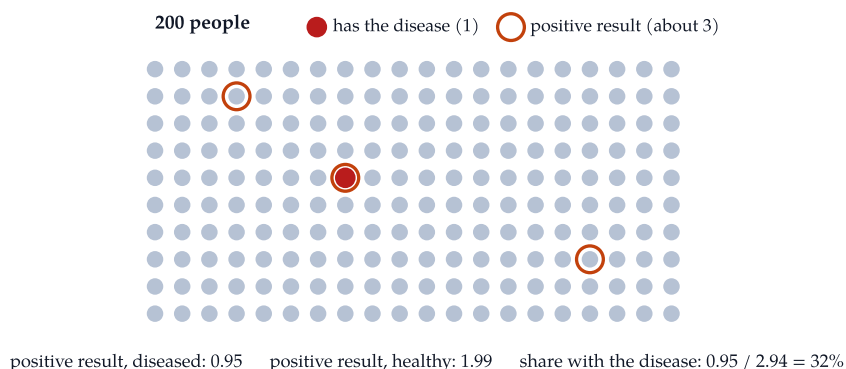


Figure 10.3. Two hundred people, one with the disease: the test gives that one a positive result with chance 0.95 and gives a false positive result to one healthy person in a hundred, so about two of the 199 healthy also get a positive result; of everyone with a positive result, roughly one in three has the disease. *Assumes:* that the one-in-two-hundred rate and the test's two error rates are what the text says they are; the picture draws whole people, so it rounds the diseased person's chance 0.95 of a positive result up to one and the 1.99 healthy people with a positive result to two, while the arithmetic in the text rounds neither

separately, among the healthy, the ratios multiply: 0.95^2 to 0.01^2 , odds of 9025 to 199, a chance of $9025/9224$, about 0.978. But if the result is fixed by the person rather than by the occasion, so that a second test on the same person repeats the first whether that person is ill or healthy, then the second positive was certain under both hypotheses, its ratio is 1, and the chance stays at $95/294$. The arithmetic is the same in both cases, and which case holds is a claim about the world.

* * *

Five solids gave $V - E + F = 2$, and the formula was conjectured on that evidence. The five were a cube, a tetrahedron, an octahedron, a pyramid and a cube with one corner sliced off: all convex, all small, all the kind of solid drawn first. Under the rival hypothe-

sis that the formula holds for tidy shapes and fails elsewhere, five tidy shapes giving 2 was certain, so each confirmation had a ratio of 1: both hypotheses predicted it. The tunnelled cube had a ratio the other way that no pile of confirmations could balance: under “always 2”, a count of 0 has chance 0.

A claim that something exists is settled by exhibiting it. A claim that nothing of a kind exists is not settled by failing to exhibit one, and the ratio says how much each failure counts: very little for confirmations that resemble each other, and a great deal for one that the rival hypothesis would have made surprising. So the case to test next is not the one most like the cases already checked, but the one the rival makes least likely.

THE LONG RUN

A coin is tossed a hundred times. The average number of heads is fifty, since each toss contributes a half and averages add. But fifty is not what happens. The chance of exactly fifty heads is $(100 \text{ choose } 50)/2^{100}$, which is about 0.08, so more than nine times out of ten the count is something else. How far from fifty does it usually fall, and what happens to that distance when the hundred tosses become ten thousand, or a million? The average cannot say. A quantity that is always 50, and one that is 0 or 100 with equal chances, have the same average.

* * *

What is wanted is a measure of spread, and the workable one is an average again. For a quantity X with average a , the *variance* of X is the average of $(X - a)^2$, the squared distance from the average, and the *deviation* of X is the square root of its variance, which is measured in the same units as X . The squaring makes distances on either side count alike, and it has a second merit that decides the matter: variances of independent quantities add, as averages do, while the average of the distance itself, unsquared, obeys no such rule: counting 1 for a head and 0 for a tail, it is $1/2$ for one toss of a fair coin, and $1/2$ again, not 1, for the number of heads in two

tosses.

Two quantities are independent when every event described in terms of the one is independent of every event described in terms of the other, as with the results of two different tosses. For independent X and Y , the average of the product XY is the product of the averages. The average of XY is the sum, over every pair of values x and y , of xy times the chance that X is x and Y is y ; by independence that chance is the product of the two separate chances, so the sum factors into the sum of x times its chance, multiplied by the sum of y times its chance.

Now let X have average a and Y average b . The variance of $X + Y$ is the average of $((X - a) + (Y - b))^2$, which multiplies out into the average of $(X - a)^2$, plus the average of $(Y - b)^2$, plus twice the average of $(X - a)(Y - b)$. The quantities $X - a$ and $Y - b$ are independent, as X and Y are, so by what was shown a moment ago the last average is the product of the averages of $X - a$ and of $Y - b$. The average of $X - a$ is $a - a$, which is 0, and likewise for $Y - b$, so the cross term vanishes. For more than two quantities the square carries one such cross term for each pair, and each vanishes in the same way, so the rule asks nothing beyond the independence of each pair on its own. So

the variance of a sum of independent quantities is the sum of their variances.

For one toss, count 1 for a head and 0 for a tail. The average is $1/2$, and the distance from it is $1/2$ whichever way the coin falls, so the variance is $1/4$. For n tosses the variance of the number of heads is therefore $n/4$, and the deviation is $\sqrt{n}/2$. A hundred tosses have a deviation of 5 heads; ten thousand tosses, 50; a million, 500. For a coin that shows heads with chance p , one toss has average p . With chance p the toss counts 1, at distance $1 - p$ from the average, and with chance $1 - p$ it counts 0, at distance p . So the variance is $p(1 - p)^2 + (1 - p)p^2$. Both terms contain $p(1 - p)$, and what is left over is $(1 - p) + p$, which is 1, so the variance is $p(1 - p)$, and n tosses have variance $np(1 - p)$.

That square root says two things. The deviation grows: in a million tosses the count of heads is typically some hundreds away from half a million, not some units. And it grows more slowly than the number of tosses, so as a fraction of n it shrinks, as $1/(2\sqrt{n})$: a twentieth for a hundred tosses, a two-hundredth for ten thousand, a two-thousandth for a million. To make the proportion of heads ten times steadier takes a hundred times as many tosses.

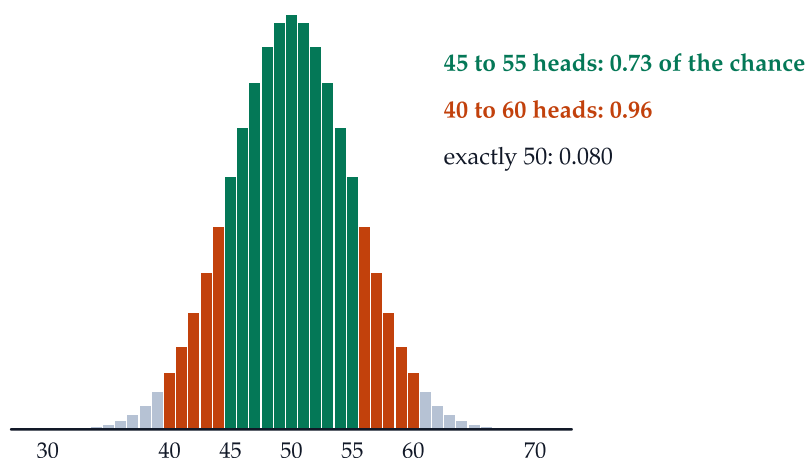


Figure 11.1. The chance of each number of heads in a hundred tosses of a fair coin. The average is 50 and the deviation 5: nearly three quarters of the chance lies between 45 and 55 heads, and all but about a thirtieth between 40 and 60. *Assumes:* that the tosses are independent and the coin fair; the shares within one and two deviations are computed by adding the bars, not taken from any general law

* * *

The variance is an average, and an average by itself guarantees nothing about any one outcome. A small inequality turns it into a guarantee. Let X have average a and variance v , and let t be any positive number. Then

the chance that X differs from a by t or more is at most v/t^2 .

The variance is the sum, over all outcomes, of the chance of the outcome times $(X - a)^2$. Throw away every outcome where X is within t of a : the sum gets no larger. In every outcome that remains, $(X - a)^2$ is at least t^2 . So v is at least t^2 times the total chance of those outcomes. Those outcomes are exactly the ones where X differs from a by t or more, so dividing by t^2 gives the claim. This is Chebyshev's inequality. It holds for every quantity that has a variance, and the price of that generality is that the bound it gives is usually far larger than the true chance.

For ten thousand tosses, v is 2500. The chance that the number of heads differs from 5000 by 200 or more is at most $2500/200^2$, which is $1/16$. The true chance, which can be computed by adding up binomial coefficients, is about 0.00007. The inequality gives away a factor of nearly a thousand.

Let the coin show heads with chance p , and let it be tossed n times. Name any tolerance ε , however small. The proportion of heads differs from p by ε or more exactly when the count differs from np by $n\varepsilon$ or more. The count has average np and variance $np(1 - p)$, so the inequality, with $n\varepsilon$ for t , puts the chance of that at no more than $np(1 - p)/(n\varepsilon)^2$. One n cancels, leaving $p(1 - p)/(n\varepsilon^2)$, which is at most $1/(4n\varepsilon^2)$, since $p(1 - p)$ is never more than $1/4$. So

for every tolerance ε and every positive δ , there is a number of tosses beyond which the chance that the proportion of heads is within ε of p is more than $1 - \delta$.

Any n above $1/(4\varepsilon^2\delta)$ will do: for such n the bound $1/(4n\varepsilon^2)$ on the chance of missing p by ε or more is below δ , so the chance of being within ε of p is more than $1 - \delta$. This is the *law of large numbers*. Jakob Bernoulli proved it, with much more labour and a stronger estimate, in a book published in 1713, eight years after his death, and he wrote that he had thought about it for twenty years. It is the reason a chance, which was defined as a share of equally

likely cases, can be measured by an experiment: toss the coin often enough and the proportion of heads is, with a chance as near to certainty as anyone asks, as near to p as anyone asks. With $\varepsilon = 0.01$ and $\delta = 0.05$ the bound asks for fifty thousand tosses; the truth is that about ten thousand are enough, the difference being what the inequality gives away.

The law does not say that after a run of heads a tail becomes more likely. The coin has no memory: after ten heads in a row the chance of a head is one half, since the tosses are independent, and independence means exactly that knowing the earlier tosses changes nothing. Nor does the law say that an excess of heads is later paid back by an excess of tails. It says that the excess is swamped. The excess of heads over half the tosses is typically about $\sqrt{n}/2$, and it grows; the proportion settles down only because n grows faster. A gambler who expects the wheel to owe him something has the law backwards.

* * *

A gambler with 10 coins plays against an opponent with 90. At each round a fair coin is tossed and the loser pays the winner one coin, and they play until one of them has all 100. What is the gambler's chance of ending with everything?

Let $w(k)$ be the chance of ending with all N coins when starting with k . Then $w(0) = 0$ and $w(N) = 1$, and in between, the first round leads with equal chances to $k + 1$ or to $k - 1$ coins, after which the game starts afresh, so

$$w(k) = (w(k + 1) + w(k - 1))/2$$

Double both sides and move one $w(k)$ across, and the relation reads $w(k + 1) - w(k) = w(k) - w(k - 1)$: each step up in k raises w by the same amount. N equal steps lead from 0 to 1, so each is $1/N$, and $w(k) = k/N$. The gambler with 10 coins out of 100 wins everything with chance $1/10$. His average gain is then $(1/10) \cdot 90 - (9/10) \cdot 10$, which is 0, as it should be in a fair game, and the shape

of the play is a small chance of a large gain against a large chance of a small loss.

The argument assumed that the game ends. Could it go on forever? In any stretch of N consecutive rounds there is a chance of $1/2^N$ that the gambler wins every round, which would end the game whatever he held at the start of the stretch. So the chance that the game survives m such stretches is at most $(1 - 1/2^N)^m$, and the powers of a number below 1 fall below every positive number, since the powers of its reciprocal pass every bound. The chance of an endless game is below every positive number, so it is 0.

How long does the game last? Let $d(k)$ be the average number of rounds. It is a finite number, and that has to be checked before anything is done with it. Write c for $1 - 1/2^N$, so that the chance that the game is still going after m stretches of N rounds is at most c^m . Regroup the average as was done for the wait for a six: it is the sum, over j , of the chance that the game lasts more than j rounds. The first N of those chances are at most 1 each, the next N at most c each, the N after that at most c^2 each, and so on. So its running totals, which only grow, stay below $N(1 + c + c^2 + \dots)$, hence below $N/(1 - c)$, which is $N \cdot 2^N$. That totals which only grow and never pass a fixed bound close in on a definite value, as the totals for the wait for a six closed in on 6, is settled once limits are.

Then $d(0) = d(N) = 0$, and the first round costs one round and leads to $k + 1$ or $k - 1$, so $d(k) = 1 + (d(k + 1) + d(k - 1))/2$. Doubling both sides and moving one $d(k)$ across gives $d(k + 1) - d(k) = (d(k) - d(k - 1)) - 2$: the steps now go down by 2 each time k goes up by 1. The function $k(N - k)$ does exactly that, since its step from k to $k + 1$ is $N - 2k - 1$, and it vanishes at both ends. No other function does both.

Suppose two functions satisfy the relation and agree at the ends, and look at their difference. The 1s cancel in it, so the difference satisfies the first relation, the one with equal steps, and it is 0 at both ends. Equal steps that lead from 0 back to 0 are all 0, so the difference is 0 throughout. So $d(k) = k(N - k)$. With 10 coins against 90 the game lasts 900 rounds on average, and between two players

with 50 each it lasts 2500.

Against a casino the opponent's fortune is in effect without limit. With k fixed and N growing, the chance k/N of ever winning everything falls below every positive number, so against an infinitely rich opponent, in a perfectly fair game, the gambler is ruined with chance 1. No unfairness is needed to explain why casinos do not go bankrupt, and they add some anyway. At a roulette table with 37 slots, a bet on red wins with chance $p = 18/37$ and loses with chance $q = 19/37$. The relation becomes $w(k) = p \cdot w(k+1) + q \cdot w(k-1)$. Since $p + q = 1$, the left side is $p \cdot w(k) + q \cdot w(k)$, and moving terms across gives $p(w(k+1) - w(k)) = q(w(k) - w(k-1))$. So each step $w(k+1) - w(k)$ is q/p times the one before. Write r for q/p , which is $19/18$, and s for the first step, $w(1) - w(0)$. The steps are then s, sr, sr^2 , and so on, and $w(k)$ is the sum of the first k of them, $s(1 + r + \dots + r^{k-1})$, which the subtraction trick for powers turns into $s(r^k - 1)/(r - 1)$. All N steps together must add up to $w(N)$, which is 1, so $s(r^N - 1)/(r - 1) = 1$. Dividing the first sum by the second removes both s and $r - 1$, and gives

$$w(k) = (r^k - 1)/(r^N - 1)$$

A gambler who starts with 50 coins and means to stop at 100 has $r^{50} = 14.93$, and his chance of getting there one coin at a time is $(r^{50} - 1)/(r^{100} - 1)$. Since $r^{100} - 1$ is $(r^{50} - 1)(r^{50} + 1)$, that is $1/(r^{50} + 1)$, about 0.063. If he stakes all 50 on a single spin his chance is $18/37$, about 0.486. A disadvantage of one part in thirty-seven per coin staked, which looks small, is compounded by every round of timid play until it is nearly a certainty of ruin; the bold player exposes his money to it once.

* * *

Two players toss a fair coin twenty times, one scoring the heads and the other the tails. How often does the lead change hands? Intuition expects the lead to pass back and forth, and expects a player

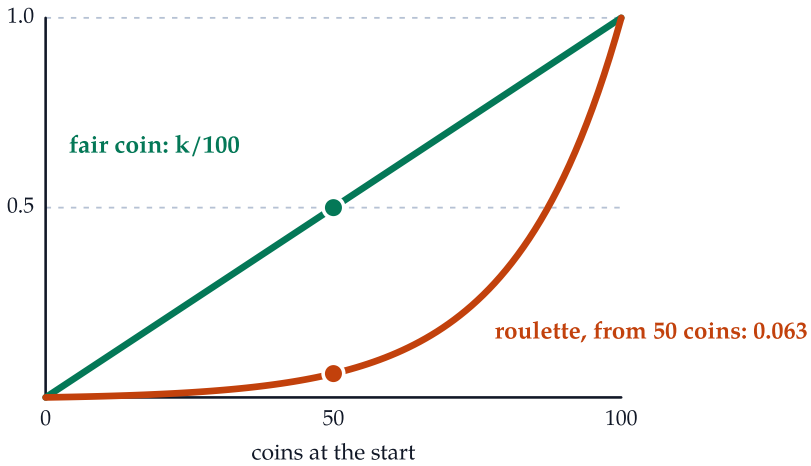


Figure 11.2. The chance of reaching 100 coins before losing everything, against the number of coins held at the start, playing one coin at a time. In the fair game it is a straight line. With the roulette table's 18 chances in 37 it stays near nothing until the start is already close to the goal. *Assumes:* one coin staked at every round and no stopping early; a player who stakes more at once faces a different curve

to be ahead about half the time. Both expectations are wrong, and the count that shows it is the reflection of routes.

Draw the game as a route that goes up one step for a head and down one step for a tail. The chance that the scores are level after $2n$ tosses is $u = (2n \text{ choose } n)/2^{2n}$: for $2n = 2, 4, 6$ and 20 that is $1/2, 3/8, 5/16$ and 0.176 .

Now count the routes of $2n$ steps that stay strictly above the level line after the start. Such a route goes up at the first step, and then, from height 1, makes $2n - 1$ steps without touching height 0. Take first the routes of $2n - 1$ steps from height 1 to a height h , with no restriction. They number $(2n - 1 \text{ choose } j)$, where j is the number of upward steps needed. Now take those among them that touch 0, and reflect the part of each before its first touch. That matches them with all the routes from height -1 to h . Such a route needs one more upward step, so these number $(2n - 1 \text{ choose } j + 1)$. So

the routes that end at h without touching 0 number $(2n - 1 \text{ choose } j) - (2n - 1 \text{ choose } j + 1)$.

Add over all the possible final heights $h = 2, 4, 6, \dots$, that is, over $j = n, n + 1, \dots, 2n - 1$: the sum collapses, each term cancelling part of the next, and leaves $(2n - 1 \text{ choose } n)$, which is half of $(2n \text{ choose } n)$. Routes that stay strictly below the line are as many. The routes that are never level after the start therefore number $(2n \text{ choose } n)$ in all, which is exactly the number of routes that are level at the end, and both are counted out of the same 2^{2n} routes. So

the chance that the scores are never level during the first $2n$ tosses is exactly the chance that they are level at toss $2n$.

In twenty tosses of a fair coin, the chance that one player takes the lead at the first toss and keeps it throughout is 0.176, more than one game in six. A long lead is no evidence of skill or of a bent coin; it is what a fair coin does. Two further facts come out of the same reflection of routes. The first is about changes of lead, and the second is about how the time is shared; the second was published by Chung and Feller in 1949. Both are stated here. The counting that comes next proves the second, and proves the first as far as the chance of no change of lead at all; the rest of the first rests on a count of Feller's, taken on his word. Of all the possible numbers of changes of lead in a long game the likeliest is none at all, the next likeliest one, and so on: in twenty tosses the chances of no change, one, two and three are 0.35, 0.29, 0.19 and 0.10. The second fact: of all the ways the twenty tosses can be shared between the time one player is ahead and the time the other is, the even share, ten and ten, is the least likely, with chance 0.06, while the chance that a named player is never once behind is 0.176. The proportion of heads still settles toward a half, while the difference between the scores, which is twice the excess of heads over half the tosses, wanders off by about the square root of the number of tosses either way and takes a long time to come back.

* * *



Figure 11.3. One game of two thousand tosses of a fair coin, from a fixed list of random numbers. Above, heads minus tails, which wanders and seldom returns to level; below, the proportion of heads, which settles toward one half. Both are the same tosses. *Assumes:* that one game can stand for the behaviour of games in general, which it cannot; the counts in the text are what establish how common long leads are

One more game shows the average itself failing. A coin is tossed until it shows heads. If the first head comes at toss k , the player is paid 2^k coins: 2 if at once, 4 if at the second toss, and so on. What is a fair price for a ticket? The average payment is the sum of 2^k times the chance $1/2^k$, which is $1 + 1 + 1 + \dots$, a sum that passes every bound. A price is fair when, over many games, neither side comes out ahead, and for a game whose payment per play settles near its average, as the proportion of heads settles near p , that price is the average payment. By that reckoning no price is too high. And yet half the time the game pays 2 coins, three times in four it pays 4 or less, and nobody would give 100 coins for a ticket.

Nicolaus Bernoulli posed this in 1713, and it is known by the name of the city where his cousin Daniel published a discussion of it, St Petersburg. There is no mistake in it. The average was defined as a sum, and a sum with no last term need not have a value;

when it has none, the quantity has no average, and every theorem with “the average” in it, the law of large numbers included, is silent about it. Played over and over, the game shows what the missing average means. Over 2^m games, about half pay 2, a quarter pay 4, and so on down to about one game that pays 2^m , and each of those m groups brings in about 2^m coins in all, the first being $2^m/2$ games at 2 coins each, the second $2^m/4$ games at 4 coins each, and so on. The total is about $m \cdot 2^m$ coins from 2^m games; so the winnings per game come to about m , which grows with the number of games played and settles nowhere. The fair price for a ticket depends on how many tickets are to be bought.

In a hundred tosses of a fair coin, the inequality says that the chance of 60 or more heads, or 40 or fewer, is at most how much? The true value is 0.057.

A die is thrown 600 times. What are the average and the deviation of the number of sixes? Between what limits does the inequality put that number with a chance of at least 0.9?

A gambler at the roulette table starts with 20 coins, stakes one coin on red at every spin, and stops when he has 40 or nothing. What is his chance of reaching 40, and how does it compare with staking all 20 on one spin?

HOW LONG IS ONE PLAYER AHEAD

Two facts about the coin-tossing game were quoted and not proved: that the likeliest number of changes of lead in a long game is none at all, and that of all the ways the time can be shared between the two players the even share is the least likely. Both sound wrong. Both come from counting routes and reflecting them, and the game has already supplied those tools. The one exception is a count of routes by their crossings, which is taken from Feller.

Write the game as a route. Each toss is a step of the route, up for a head and down for a tail, and after n tosses the route stands at S_n , the excess of heads over tails. A route of n steps is one of the 2^n strings of ups and downs, all equally likely, and S_n is 0 exactly when the players are level. Levels can happen only after an even number of tosses, and

the chance of being level after $2n$ tosses is $u_{2n} = (2n \text{ choose } n) / 4^n$

the middle entry of the row divided by the row's total. That number is close to $1/\sqrt{(\pi n)}$, which is where the circle enters a question about coin tosses; the estimate comes out of Wallis's product for π and is proved when that product is reached. Being level after a million tosses has chance about $1/1253$.

The reflection of routes says this. Take routes of m steps from a height a above 0 to a height b above 0. Those that touch 0 somewhere are matched, one for one, with all routes from $-a$ to b . Given a route from a that touches 0, turn the part before its first touch of 0 upside down, so that every up step in that part becomes a down step and every down step an up step. The turned part now starts at $-a$ and still arrives at 0 at the same moment, and the rest of the route goes on to b unchanged. In the other direction, every route from $-a$ to b has to touch 0 on its way from below the line to above it, and turning the part before its first touch upside down gives back a route from a . So the number that touch 0 is the number of routes from $-a$ to b , and the number that avoid 0 is the difference of two counts that the triangle of choices supplies.

* * *

The first thing needed was met when the routes that never touch the level line were counted; what follows puts it into the notation the harder count will use.

*The chance that the players are never level in the first $2n$ tosses is u_{2n} ,
the same as the chance that they are level at the end.*

Write $p(m, x)$ for the chance that a route of m steps ends at height x . By symmetry, the chance of never being level is twice the chance of staying above 0 throughout. The routes of $2n$ steps that stay above 0 were counted by the reflection when this fact was first met: there are $(2n - 1 \text{ choose } n)$ of them. A route of $2n - 1$ steps ends at height 1 when n of its steps are up and $n - 1$ down, so $p(2n - 1, 1) = (2n - 1 \text{ choose } n)/2^{2n-1}$, and the chance of staying above 0 throughout is $(2n - 1 \text{ choose } n)/2^{2n}$, which is $p(2n - 1, 1)/2$. The chance of never being level is therefore $p(2n - 1, 1)$, which is the largest entry of row $2n - 1$ divided by 2^{2n-1} . Row $2n - 1$ has two equal largest entries side by side, and the middle entry of row $2n$ is their sum, so it is twice the largest entry of row $2n - 1$. Dividing the middle entry of row $2n$ by 2^{2n} therefore gives

the same number as dividing the largest entry of row $2n - 1$ by 2^{2n-1} , and the chance of never being level is exactly u_{2n} .

A third chance follows at once, the chance of never going *below* 0, which is a different event again: a route may touch the line without crossing it. Put an extra up step in front of a route of $2n$ steps that never goes below 0. Every height is raised by 1, so the result is a route of $2n + 1$ steps that stays above 0. Every route of $2n + 1$ steps that stays above 0 arises in this way from exactly one route: its first step has to be up, and dropping that step leaves a route of $2n$ steps that never goes below 0. So the two kinds are equal in number. Next, a route of $2n$ steps that stays above 0 ends at an even height, so at a height of 2 at least, and whichever way its next step goes it is still above 0. Each such route has two continuations that stay above, so the routes of $2n + 1$ steps that stay above are twice as many as the routes of $2n$ steps that do. Putting the two counts together, the routes of $2n$ steps that never go below are twice as many as those that stay above. Both are counted among the same 2^{2n} routes, so the chance of never going below is twice the chance of staying above, and the chance of staying above was found to be half of u_{2n} .

The chance of never going below 0 in $2n$ tosses is u_{2n} as well.

* * *

Now the sharing of the time. Say that the route is *above* during a step when either end of that step is above 0, so that every step is either above or below, and let K be the number of the $2n$ steps that are above. Between one level moment and the next the route keeps to one side, so the steps of that stretch are all above or all below. Level moments come only after an even number of tosses, so each such stretch has even length, and so does the stretch after the last level moment, because the game ends after $2n$ tosses. K is the sum of the lengths of the stretches that are above, so K is always even.

An exactly even share, $K = n$, is therefore possible only when n is itself even, which for twenty tosses it is.

The chance that $K = 2k$ is $u_{2k} \cdot u_{2n-2k}$, for every k from 0 to n .

At $k = n$ the statement is that the chance of being above the whole time is u_{2n} . Every one of the $2n$ steps is above exactly when the route never reaches -1 , since a step into -1 has neither end above 0, so that chance is the one just found for never going below; and $k = 0$ is its mirror. For the rest, k lies strictly between 0 and n , so the route has steps above and steps below, and to pass from one side to the other it has to be level at some moment. Split on the first moment the players are level, say after $2r$ tosses. Before that moment the route stays on one side, each side with chance $1/2$, and the chance that the first level is at $2r$ is written f_{2r} . If the route is above for those first $2r$ steps, the remaining $2n - 2r$ steps start level again and must contribute $2k - 2r$ above; if below, they must contribute $2k$. So

the chance that $K = 2k$ is the sum over r of $f_{2r} \cdot$ (the chance of $2k - 2r$ above in $2n - 2r$) $/ 2$ plus the sum over r of $f_{2r} \cdot$ (the chance of $2k$ above in $2n - 2r$) $/ 2$

Suppose the formula holds for every shorter game. In a game of $2n - 2r$ tosses the chance of $2k - 2r$ above is then $u_{2k-2r} \cdot u_{2n-2k}$, since $2n - 2k$ of its steps are below. The factor u_{2n-2k} is the same for every r , so it comes out in front, and r runs only from 1 to k , because $2k - 2r$ cannot be negative. So the first sum becomes u_{2n-2k} times half the sum of $f_{2r} \cdot u_{2k-2r}$ over r from 1 to k . In the same way the chance of $2k$ above in a game of $2n - 2r$ tosses is $u_{2k} \cdot u_{2n-2k-2r}$, the factor u_{2k} comes out in front, and the second sum becomes u_{2k} times half the sum of $f_{2r} \cdot u_{2n-2k-2r}$ over r from 1 to $n - k$. Both inner sums are instances of one identity. A route that is level at time $2m$ has a first level at some time $2r$, no later than $2m$. The chance that the first level comes at $2r$ is f_{2r} . From there the route starts level again and has to be level once more $2m - 2r$ steps later, which has

chance u_{2m-2r} , and that chance is $u_0 = 1$ when $r = m$. Adding over the possible values of r ,

$$u_{2m} = \text{the sum of } f_{2r} \cdot u_{2m-2r} \text{ over } r \text{ from } 1 \text{ to } m$$

which says only that the first level happens somewhere. So the first sum is $u_{2n-2k} \cdot u_{2k}/2$ and the second is $u_{2k} \cdot u_{2n-2k}/2$, and together they give $u_{2k} \cdot u_{2n-2k}$. The descent is complete.

What the formula says is the surprise. The numbers $u_{2k}u_{2n-2k}$ are largest at the two ends, where one factor is $u_0 = 1$, and smallest in the middle, and the reason is in how each level chance follows from the one before. By the formula for choosing, the middle entry of row $2j + 2$ is $(2j + 2)(2j + 1)/(j + 1)^2$ times the middle entry of row $2j$, and the row total is 4 times as large, so u_{2j+2} is u_{2j} times $(2j + 1)/(2j + 2)$, which is $1 - 1/(2j + 2)$. Moving the share from $2k$ to $2k + 2$ therefore multiplies u_{2k} by $1 - 1/(2k + 2)$ and divides u_{2n-2k} by $1 - 1/(2n - 2k)$. The product falls exactly when the first of these is the smaller, that is when $2k + 2$ is less than $2n - 2k$, which is while $2k + 1$ is less than n : the chances fall all the way from either end to the middle. For twenty tosses the chance of the even share, ten steps each way, is 0.0606, while the chance that one named player is ahead for the whole game is 0.1762, nearly three times as large. In a game of ten thousand tosses the chance that one named player is ahead for nine thousand nine hundred of them or more is 0.064, about ten times the chance that the two shares come within fifty tosses of even, which is 0.0065.

Since u_{2k} is close to $1/\sqrt{(\pi k)}$, the chance that the share is $2k$ is close to $1/(\pi\sqrt{(k(n - k))})$. Adding the chances from $k = 0$ to $k = xn$ gives the chance that the share is no more than the fraction x of the whole time. The approximation $1/(\pi\sqrt{(k(n - k))})$ is $1/n$ times the height of the curve $1/(\pi\sqrt{(t(1 - t))})$ at $t = k/n$, so as n grows the sum approaches the area under that curve from 0 to x , and that area is $2/\pi$ times the angle whose sine is $\sqrt{x}$. The sine of an angle is defined later, when the turning ray is taken up, and the area under a curve later still, and the limit waits on both. The angle whose sine is a given number is called the arcsine of that

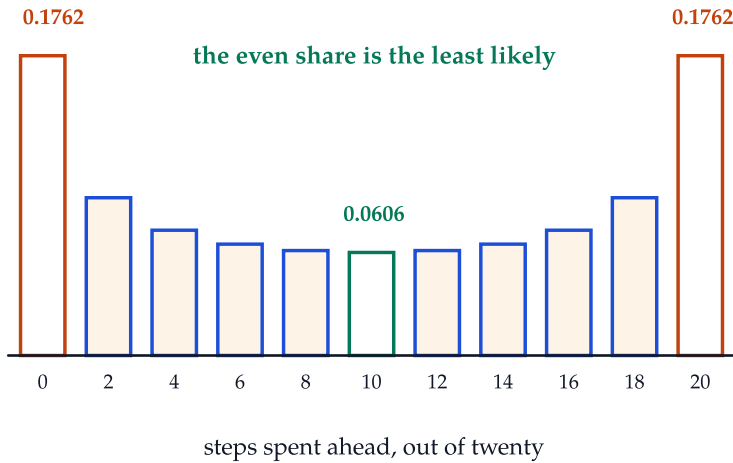


Figure 12.1. The chance, in twenty tosses, that a named player is ahead for exactly $2k$ of the twenty steps. The bars rise toward the two ends and sink in the middle: the even share, ten steps each, has chance 0.0606, while that player leading the whole way has chance 0.1762, almost three times as much. Expecting the middle to be likeliest is the mistake the law of large numbers invites. *Assumes:* the chance of a share $2k$ is $u(k) \cdot u(n - k)$ with $u(j)$ the chance of being level after $2j$ tosses, proved in the text; the bars were also counted directly over all 1048576 routes

number, and the arcsine law is the name this result goes by. The chance that one named player leads for less than a fifth of the time is about 0.295, and the chance of leading for either less than a fifth or more than four fifths is about 0.59: the two extremes together are likelier than the whole middle.

The changes of lead go the same way, and they can be counted. A change of lead is a level moment with the route above just before it and below just after, or the other way about. A game has none exactly when the route keeps to one side of the line: never below, or never above. For suppose the route reaches both a positive height and a negative one. Take the first moment at which it is negative. If that is not the first step, the route was level one step before, because a route is negative for the first time only by stepping down from 0.

One step before that it stood at 1 or at -1 , and it cannot have stood at -1 , because that would have been an earlier negative moment. So it stood at 1, and that level moment has the route above before it and below after: a change of lead. If instead the first step is down, do the same with the first moment at which the route is positive. The two cases, never below and never above, cannot both hold, because the first step already takes the route off the line to one side or the other. By symmetry the two cases have the same chance. So the chance of no change of lead is twice the chance of never going below.

The chance of never going below 0 is u_{2n} , which equals $p(2n - 1, 1)$, so the chance of no change of lead in $2n$ tosses is $2 \cdot p(2n - 1, 1)$, twice the chance that a route of $2n - 1$ steps ends at height 1. The same reflection, pushed further, counts the games with more changes of lead, and the count comes to this:

$$\text{the chance of exactly } c \text{ changes of lead in } 2n \text{ tosses is } 2 \cdot p(2n - 1, 2c + 1)$$

The case $c = 0$ is what was settled above; the count for larger c is Feller's, published in 1957, and is taken here on his word. Everything else follows from the shape of a row of the triangle. The entries of a row rise to the middle and fall away on either side of it: by the formula for choosing, $(m \text{ choose } j + 1)$ is $(m \text{ choose } j)$ times $(m - j)/(j + 1)$, a factor above 1 while $j + 1$ is less than $m - j$ and below 1 once $j + 1$ is more. In row $2n - 1$ the height 1 belongs to one of the two largest entries, next to the middle, and the heights 3, 5 and onward belong to entries further and further from it. So the entries fall, and the chances fall as c grows: the likeliest number of changes of lead in any game is none. For twenty tosses the chances of 0, 1, 2 and 3 changes are 0.3524, 0.2883, 0.1922 and 0.1035, and the first of them says that no change of lead at all is twice as likely as ending level and twice as likely as one named player leading the whole way. Read the other way, the count says that exactly c changes of lead have the same chance as a route of $2n - 1$ steps ending at $2c + 1$ or at $-(2c + 1)$. The height at which such a route

ends has average 0 and deviation $\sqrt{(2n - 1)}$, so its distance from 0 is typically of the order of $\sqrt{n}$, and the number of changes has the same chances as that distance less 1, halved: in a game of $2n$ tosses the average number of changes grows like $\sqrt{n}$, not like n .

None of this contradicts the law of large numbers: the proportion of heads settles toward a half because the excess is of the order of $\sqrt{n}$ while the number of tosses is n , and the lead belongs to the excess.

In a game of six tosses, list the shares of time and check the formula against a count of all 64 routes.

The chance of never being level in $2n$ tosses equals the chance of being level at the end. Is there a matching of the two collections of routes that shows it directly?

What is the average number of returns to level in $2n$ tosses, and how does it grow?

A game is played until the players are level again. What is the chance that this takes more than $2n$ tosses, and what is the average number of tosses it takes?

For twenty tosses, which is likelier: that the lead changes hands exactly once, or that it changes four times or more? By how much?

EIGHT WAYS TO PUT DOWN A SQUARE

Take something square, a coaster or a piece of card, and mark its corners 1, 2, 3, 4 on both sides so that any move can be read off afterwards. It can be put back into its outline in exactly eight ways, the eight *symmetries* of the square: leave it alone; turn it a quarter, a half or three quarters; or flip it about the vertical line through its middle, the horizontal line, or either diagonal. There are no others. A move is settled by where it sends the corners; corner 1 can go to any of four places, then corner 2, its neighbour, must go to one of the two places next to that, and then the card is fixed: corner 4 must take the other place next to corner 1, corner 3 takes the place that is left, and which side faces up is settled with them. So there are at most four times two, eight, moves. The eight listed moves put the corners in eight different arrangements, so they are eight different moves, and they are all there are.

Turn a quarter, then flip about the vertical line, and read the corners: the result is the flip about a diagonal, which neither ingredient looked like. Number the places of the outline 1 at the top left, 2 at the top right, 3 at the bottom right and 4 at the bottom left, and let the quarter turn carry each corner on to the next place. The flip about the vertical line exchanges places 1 and 2, and exchanges

places 3 and 4. Now follow each corner through the turn and then the flip. The corner that began at place 1 is turned to 2 and flipped back to 1; the corner from 3 is turned to 4 and flipped back to 3; the corner from 2 is turned to 3 and flipped to 4; the corner from 4 is turned to 1 and flipped to 2. So the corners that began at places 1 and 3 are back there and the other two have changed places: the flip about the diagonal through 1 and 3.

In the other order, the flip and then the turn, the corner from 1 is flipped to 2 and turned to 3, the corner from 3 is flipped to 4 and turned to 1, while the corner from 2 is flipped to 1 and turned back to 2 and the corner from 4 is flipped to 3 and turned back to 4. This time it is the corners at 2 and 4 that come back: the flip about the other diagonal.

Two moves in succession always give one of the eight, since doing one and then the other still leaves the square sitting in its outline, and the count above found exactly eight ways for it to sit there. Every move can be undone by a move on the list: a quarter turn by three quarters and three quarters by a quarter, and a half turn or any flip by itself. Doing nothing is on the list.

A fourth property is easy to miss. Three moves in succession can be grouped two ways: the first two together and then the third, or the first and then the last two together. The results agree, and the reason has nothing to do with squares. Each move is a rule sending every point of the card somewhere, and in either grouping each point goes where the first rule sends it, then where the second sends that, then where the third sends that. This is *associativity*, and it holds for anything made of rules applied in succession. It does not hold for every operation: $8 - (5 - 2)$ is 5 while $(8 - 5) - 2$ is 1.

* * *

The whole numbers with their negatives, under addition, have the same four properties. The sum of two is one of them; grouping does not matter; each is undone by adding its negative; and adding 0 changes nothing. That system is infinite, and in it the order of two

operations never matters, while on the square it does: a quarter turn and then a flip is not the flip and then the quarter turn. So the two systems agree on four properties and disagree on a fifth, and any description that fits both must leave the fifth out.

A third system is small. Take a clock with four hours, 0, 1, 2 and 3, and four moves: forward 0, forward 1, forward 2 and forward 3, each carrying the hand on by that many hours. Forward 1 then forward 3 is forward 4, which returns the hand to where it was, so it is forward 0. Any two of the four moves combine to one of the four, forward 0 does nothing, every move is undone by the move that completes the circle (forward 1 by forward 3, forward 3 by forward 1, forward 0 and forward 2 each by itself), and grouping does not matter.

Set the table of the clock beside the table of the four turns of the square, where q , h and t are the quarter, half and three-quarter turns and 0 is no turn. Each cell holds the move of its row followed by the move of its column:

+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

◦	0	q	h	t
0	0	q	h	t
q	q	h	t	0
h	h	t	0	q
t	t	0	q	h

They are the same table once forward 0 is matched with no turn, forward 1 with a quarter turn, forward 2 with a half turn and forward 3 with three quarters. Check the cell that looks least likely to agree, row 3 and column 3 of the first table against row t and column t of the second. Forward 3 then forward 3 is forward 6, which is forward 2. Three quarters of a turn twice is one and a half turns,

which lands at a half turn. And forward 2 is matched with the half turn, so the two cells agree.

-----matched-----

clock: forward, then forward

	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

square: turn, then turn

	0	q	h	t
0	0	q	h	t
q	q	h	t	0
h	h	t	0	q
t	t	0	q	h

3 + 3 = 6 = 2 on the clock; t then t is a half turn

Figure 13.1. The four moves of a four-hour clock hand and the four turns of a square, each combined with itself in every way: matching forward 0, 1, 2, 3 with no turn, quarter, half, three quarters makes the tables identical, and the marked cell is the one to check, where three quarters twice must land on a half. Assumes: that the matching was chosen and could have been chosen differently; forward 1 with three quarters and forward 3 with a quarter also works, which the text returns to

Every true statement about how the turns combine becomes a true statement about how the clock moves combine when each turn is replaced by its partner, and back again. No question about combining alone tells the two systems apart. What they share is a shape. The shape remains when both materials, turning a card and counting round a circle, are thrown away. And whatever is true of the shape is true wherever the shape appears.

A *group* is that shape: a collection of moves that combine to moves of the collection, in which grouping does not matter, every move can be undone, and one move, the *identity*, does nothing. The property that the order of two moves never matters is called *commutativity*. It is not in the definition, because the square does not

have it.

Neither size nor commutativity settles the shape. A rectangle that is not a square can be put back into its outline in four ways: leave it, turn it half way round, flip it about its vertical line, or flip it about its horizontal line. Each of the four, done twice, leaves the rectangle as it was, and any two of the three that move it combine to the third. It is a group of four in which order never matters, like the clock.

But no matching of the rectangle's moves with the clock respects combination in the way the matching of the clock with the turns does, sending the combination of any two moves to the combination of their partners, which is what the two tables agreeing cell by cell says. The reason takes three steps. First, in any group the move that does nothing is the only move which, done twice, gives itself. For suppose some move done twice gives that same move, and follow both sides by the move that undoes it: on one side the second doing is cancelled and the move alone is left, on the other side the move is cancelled and nothing is left, so the move was doing nothing all along.

Second, a matching that respects combination carries a move done twice to its partner done twice, so the partner of the move that does nothing, done twice, gives itself, and by the first step it is the move that does nothing. Third, a move undoes itself exactly when, done twice, it gives the move that does nothing; its partner done twice then gives the partner of doing nothing, which is doing nothing, so the partner undoes itself as well. The matching sends moves that undo themselves to moves that undo themselves. All four moves of the rectangle undo themselves, and on the clock only forward 0 and forward 2 do, so four different moves would need four different partners where only two are to be had.

* * *

A picture carries an argument the table does not. Draw a dot for each of the eight moves of the square, and choose two moves, the

quarter turn and the vertical flip. From every dot draw an arrow to the dot reached by turning a quarter next, and a link to the dot reached by flipping next.

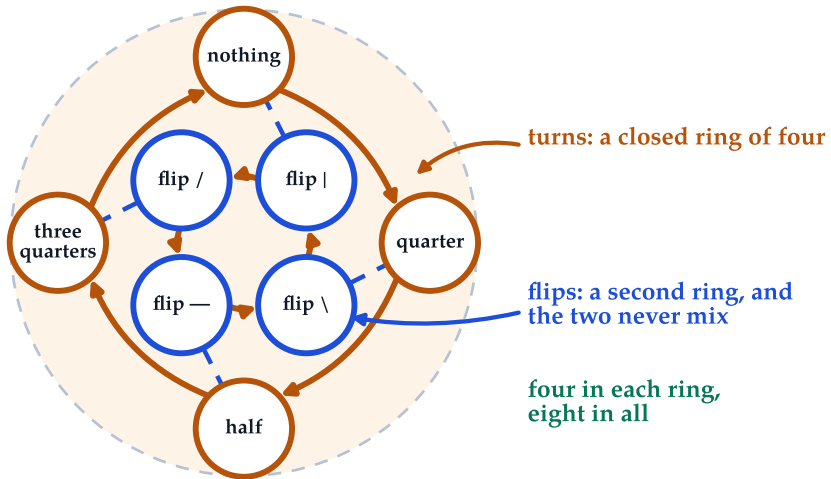


Figure 13.2. The eight ways of putting a square back into its outline. Amber arrows are a quarter turn, blue links are a flip. The turns close into two separate rings of four, and the rings do not mix, the tiling from which the size of a subgroup is later shown to divide the size of the group. *Assumes:* that following an arrow means doing that move next rather than first; reverse the convention and the turn arrows among the flips run the other way round, and the quarter and three-quarter turns trade their diagonal flips

Follow only the turn arrows from any dot: they run through four dots and return to the start. Begin again from a dot outside that ring, and the arrows run through the other four. The eight dots fall into two rings of four that do not mix. The four turns form a group of their own, a *subgroup*, and the picture shows it cutting the whole group into blocks of equal size. If that always happens, the group is filled, with no overlap, by some whole number of blocks, each the size of the subgroup, so the size of a subgroup divides the size of the group. Three does not divide eight, so no subgroup of the square's moves has three moves in it. The picture shows

that cutting for one subgroup of one group, which is not a proof. The identity and the vertical flip form a subgroup of two, and the picture shows its blocks as well: each link joins a dot to the dot reached by flipping next, and the links cut the eight into four pairs.

In how many ways can a regular pentagon be put back into its outline, and do its turns cut that group into blocks of equal size?

Two systems were called the same because an exact matching respecting combination exists between them. The matching was chosen: forward 1 with a quarter turn. Match forward 1 with three quarters instead, and forward 3 with a quarter, and the two tables still agree cell by cell. So the two systems are the same in more than one way. Go from the clock to the turns by the first matching and come back by the second. The result is a matching of the clock with itself that respects combination, and it sends forward 1 to forward 3 and forward 3 to forward 1. The same can be done with any matching of the clock with the turns that respects combination: going out by it and coming back by the first matching gives a matching of the clock with itself that respects combination, and that one followed by the first matching is the matching begun with. So every matching of the clock with the turns is the first one preceded by such a matching of the clock with itself.

How they are the same has structure of its own: the matchings of a system with itself that respect combination form a group. Two such matchings in succession respect combination: the first sends a combined with b to the combination of where it sends a and b , and the second sends that combination to the combination of where it sends those two. A matching run backwards does too, since if a goes to x and b to y , then a combined with b goes to x combined with y , so the backwards matching sends x combined with y to a combined with b , and a and b are where it sends x and y : it sends a combination to the combination of where it sends the parts. Matching each element with itself is one of them, and grouping does not matter, as for any rules applied in succession.

How many matchings of the four-hour clock with itself respect combination, and how many matchings of the rectangle's four

moves with themselves?

REARRANGEMENTS

The eight ways of putting down a square were read off its corners: a move was settled by where it sent the corners 1, 2, 3 and 4, and the eight moves were eight of the ways of rearranging four labels. There are twenty-four ways in all: four places for the first label, three for the second, two for the third and one for the last, and $4 \cdot 3 \cdot 2 \cdot 1$ is 24. A symmetry of the square has to keep neighbouring corners next to each other, and sixteen of the twenty-four rearrangements do not. So the group of the square sits inside a larger group, the group of all rearrangements of four things. Groups of rearrangements have no geometry in them at all, and they govern which tray positions of the fifteen puzzle can be reached, how often a pack of cards must be shuffled to come back to its order, and, much later, which equations can be solved.

A *rearrangement* of n things is a rule that sends each of them to one of them, no two to the same one. Performing one rearrangement and then another is a rearrangement; doing nothing is one; and each can be undone. The n things are sent to n different things, and n different things out of n must be all of them. So every thing has exactly one thing sent to it, and the rule that sends each thing back to the one it came from is again a rearrangement, the one that undoes the first. Grouping does not matter, as for any rules applied in succession. So the rearrangements of n things form a group.

They are counted as the four labels were: there are n choices for where the first thing goes, $n - 1$ left for the second, and so on down to 1 for the last, so the group has $n \cdot (n - 1) \cdots 2 \cdot 1 = n!$ members. For $n = 3$ it has six members, and they are the six symmetries of a triangle with equal sides, every rearrangement of three corners being a symmetry there. For $n = 4$ it has 24, and for a pack of 52 cards it has $52!$, a number of 68 digits.

* * *

A rearrangement of nine things might be given by a table,

1	2	3	4	5	6	7	8	9
4	6	3	9	8	2	7	5	1

each thing in the top row being sent to the thing below it. The table hides what the rearrangement does. Follow one thing instead: 1 goes to 4, 4 goes to 9, and 9 goes back to 1. Start again from a thing not yet met: 2 goes to 6, and 6 back to 2. Then 5 goes to 8 and 8 to 5, while 3 and 7 stay where they are. The rearrangement is written

$$(1\ 4\ 9)(2\ 6)(5\ 8)$$

each bracket a *cycle*, read round and round, with the things that stay put left out. Following a thing always closes up into a cycle in this way. There are only finitely many things, so the path from a thing a must at some step arrive at something already met. The first thing met twice has to be a itself. Suppose it were some other thing b on the path. The first time b was reached, it was reached from the thing before it on the path; the second time it is reached from a different thing, one met later. That makes two things sent to b , and a rearrangement sends no two things to the same one. The starting thing a is the only thing on the path that was not reached from anything, so it is the only one the path can come back to. Different cycles share nothing: a thing settles the next thing on its path, and so its whole cycle, so two cycles with one thing in common are the same cycle. So every rearrangement is a *product* of cycles that move

different things, that is, those cycles performed one after another, and apart from the order in which the cycles are written, in one way only.

The cycles tell at once how many times a rearrangement has to be repeated before everything is back in place. A cycle of length 3 comes back after 3 repetitions, and again after 6, 9, 12, ...; a cycle of length 2 after 2, 4, 6, The rearrangement above comes back when all its cycles do, for the first time after 6 repetitions, the least common multiple of 3, 2 and 2. The *order* of a rearrangement is the least common multiple of the lengths of its cycles.

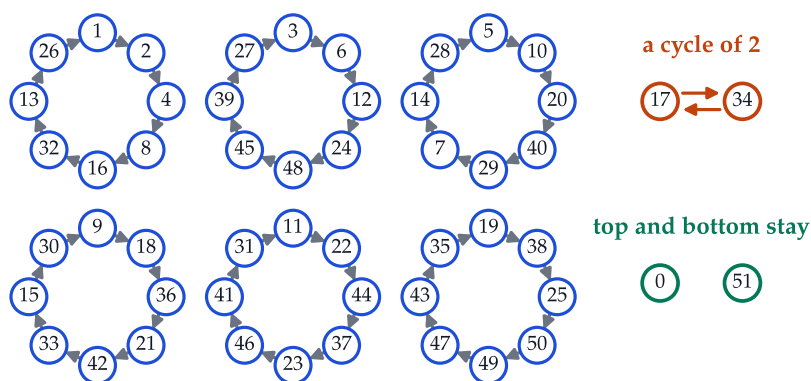


Figure 14.1. The perfect shuffle of 52 cards drawn as its cycles, each place joined to the place its card goes to. Six cycles of eight places, one of two, and the top and bottom cards, which stay: the least common multiple of the lengths is 8, so eight shuffles restore the pack. *Assumes:* that the cut is exact and the riffle perfect, card by card, which is a feat of the hands and not of the mathematics

A pack of 52 cards is cut exactly in half and the halves are riffled together perfectly, the cards falling alternately from each half, with the old top card staying on top. How many such shuffles restore the pack? Number the places from 0 at the top to 51 at the bottom. The card at place k of the top half goes to place $2k$; the card at place $26 + k$ of the bottom half goes to place $2k + 1$. Both halves obey one rule. For the top half, $2k$ is below 51 already. For the bottom half, twice the place $26 + k$ is $52 + 2k$, which is 51 more than $2k + 1$. So in

both halves a card goes to the place that twice its own place leaves as its remainder on division by 51, except that the bottom card, for which that remainder would be 0, stays at 51. The top and bottom cards are cycles of length 1. The card at place 17 goes to 34 and the card at 34 goes to 68, which leaves 17: a cycle of length 2. The card at place 1 goes to 2, 4, 8, 16, 32, then 64, which leaves 13, then 26, then 52, which leaves 1: a cycle of length 8.

Eight shuffles double a place eight times over, which multiplies it by 256. So they carry the card at place k to the place that $256k$ leaves on division by 51. Now 256 is $5 \cdot 51 + 1$, so $256k$ is a multiple of 51 plus k , and $256k$ and k leave the same remainder: every card is home after eight shuffles. A card is home only after a multiple of the length of its cycle, so every length divides 8, and is 1, 2, 4 or 8. A card in a cycle of length 1, 2 or 4 would be home after four shuffles, which multiply its place by 16. Then $16k$ and k would leave the same remainder, and 51 would divide their difference, $15k$. Since 51 is $3 \cdot 17$ and $15k$ is $3 \cdot 5k$, 17 would divide $5k$, and so it would divide 7 times as much, $35k$. But $35k$ is $34k + k$, and 17 divides $34k$, so 17 would divide k . Among the places 0 to 50 that leaves 0, 17 and 34, and place 51 stays by the exception. Those four are the places already traced, so the other forty-eight lie in cycles of length 8, six such cycles in all. So eight perfect shuffles restore the pack and no fewer do, a fact that card players who can perform the shuffle have used for a long time.

If the halves are riffled the other way, so that the old top card ends second, the rule is that place k goes to the place that $2k + 1$ leaves on division by 53. Numbering the places from 1 to 52 instead of from 0 to 51 makes that plainer. The old place k is the new place $k + 1$. The card there goes to the old place that $2k + 1$ leaves on division by 53, and its new number is one more; since that old place is at most 51, one more is what $2k + 2$ leaves on division by 53, and $2k + 2$ is twice $k + 1$. So in the new numbering a place is doubled, and the result is replaced by what it leaves on division by 53. It never leaves 0, which would be no place at all: twice a place is an even number between 2 and 104, and the one multiple of 53 in that

range, 53 itself, is odd. So t shuffles send the place y to what 2^t times y leaves, and the card at place 1 runs through the remainders of 2, 4, 8 and onward, its cycle having the length of the smallest t for which 2^t leaves 1.

Now 2^{13} is 8192, which is $154 \cdot 53 + 30$, and $30 \cdot 30$ is 900, which is $17 \cdot 53 - 1$; so 2^{26} , which is 2^{13} times itself, leaves what $30 \cdot 30$ leaves, namely 52, one short of a multiple of 53. And 2^{52} is 2^{26} times itself, so it leaves what $52 \cdot 52$ leaves; a number one short of a multiple of 53, times itself, is one more than a multiple of 53, so 2^{52} leaves 1.

Let d be the smallest exponent above 0 for which 2^d leaves 1, so that d doublings bring every place home. Divide 52 by d : 52 is a number of d 's plus a remainder r below d . Doubling 52 times is then doubling d times, over and over, and after that r more times. Each run of d doublings brings every place home, and so do all 52 doublings, so the last r doublings must leave every place where it is: 2^r leaves 1 as well. But r is below d , and d was the smallest exponent above 0 that leaves 1. So r is 0, and d divides 52. The divisors of 52 below 52 are 1, 2, 4, 13 and 26, and the corresponding powers leave 2, 4, 16, 30 and 52, none of them 1. So d is 52, one cycle takes in all 52 places, and the pack returns after 52 shuffles and not before.

Which rearrangement of twelve things has the largest order, and what is that order?

* * *

Cycles answer a question about chance that seems at first to have no good answer. A hundred drawers hold the numbers 1 to 100, one in each, in an order chosen by lot. A hundred players, numbered 1 to 100, enter the room one at a time; each may open fifty drawers, must leave everything as it was, and cannot speak to those who follow. The players win only if every one of them finds his own number. A player who opens fifty drawers at random succeeds with chance $1/2$, and a hundred players doing so all succeed with chance $1/2^{100}$. Can they do better?

Let player k begin with drawer k , and then open the drawer

whose number he has just found, and so on. The drawers are a rearrangement, drawer k holding the number that k is sent to, and the player is following the cycle through k ; he meets his own number at the moment the cycle closes. So he succeeds exactly when his cycle has length at most 50, and all the players succeed exactly when the rearrangement has no cycle longer than 50. How likely is that? A rearrangement of 100 things can have at most one cycle of a length m above 50, since two of them would need more than 100 things. The rearrangements that have one are counted in three steps. Choose the m things of the long cycle, in $(100 \text{ choose } m)$ ways, which is $100!$ divided by $m!$ and by $(100 - m)!$. Arrange them in a cycle: there are $m!$ ways of writing the m things in a row inside a bracket, and each cycle is written m times over, once for each thing it can start from, so there are $m!/m = (m - 1)!$ cycles. Rearrange the remaining $100 - m$ things in any way, in $(100 - m)!$ ways. In the product the $(100 - m)!$ cancels, and $(m - 1)!$ over $m!$ is $1/m$, so the product is $100!/m$. All $100!$ rearrangements are equally likely, so the chance of a cycle of length exactly m is $1/m$, and the chance that the players win is

$$1 - (1/51 + 1/52 + \dots + 1/100)$$

which is 0.3118. Nothing a player learns helps any other player, and yet the failures have been tied together: either nobody fails, or everyone on the one long cycle fails, more than fifty players at once.

* * *

The simplest rearrangements exchange two things and leave the rest alone. Call such a rearrangement a *swap*. Every rearrangement can be performed by swaps: put the right thing in the first place by one swap, then the right thing in the second place by another, and so on. A cycle of length m takes $m - 1$ swaps. Swap a with b , then a with c , and follow each thing: a goes to b and is not touched again; b goes to a , and the second swap carries it on to c ; c waits through the first swap and then goes to a . So the two swaps carry

a to b , b to c and c to a , a cycle of length 3, and in the same way each further swap with a lengthens the cycle by one. A rearrangement can be performed by swaps in many ways, using different numbers of them. What cannot vary is whether that number is even or odd.

Write the things in a list, and count the pairs of places whose entries are out of order, the larger before the smaller. This is the count that decided the fifteen puzzle, where a slide either left the list alone or carried one tile past three others; here any two entries may be exchanged.

Exchange the entries in places i and j . A pair of places that includes neither i nor j keeps both its entries, and so keeps its state. The pair of places i and j itself reverses its state. For any other place k , look at the two pairs that k forms with i and with j . If k lies outside the stretch from i to j , it is on the same side of both places, and the two entries it is compared with have traded places. So the pair of k with i is now in the state that the pair of k with j was in, and the other way about: the two pairs exchange their states, and between them the number out of order is unchanged.

If k lies between i and j , its entry is compared with the same two entries as before, but each of them is now on the other side of it. Suppose the entry at k is larger than both of them, or smaller than both. Then the pair of places i and k is in the same state whichever of the two entries sits at i , and likewise the pair of places k and j , so neither pair changes its state. Suppose instead that the entry at k lies between the other two in size. Then each of the two pairs is in order with one of those entries and out of order with the other, so both pairs reverse their states. Between them the number out of order changes by 2, by -2 or by 0. The pair of places i and j changes the count by 1 or by -1 , and every other place k changes it by an even number. So one swap changes the count of pairs out of order by an odd number.

The list in its proper order has count 0, and each swap changes the count by an odd number, turning an even count into an odd one and an odd count into an even one. So a rearrangement performed by an even number of swaps produces a list with an even count,

and one performed by an odd number of swaps produces a list with an odd count. But the list a rearrangement produces does not depend on which swaps were used to perform it, and its count cannot be both even and odd. So no rearrangement can be performed both by an even number of swaps and by an odd number.

Call a rearrangement *even* or *odd* accordingly. Performing two rearrangements in succession adds the numbers of swaps, so even then even is even, odd then odd is even, and even then odd, in either order, is odd, which is how the numbers 1 and -1 multiply. A cycle of length m , being $m - 1$ swaps, is even when m is odd; the rearrangement $(1\ 4\ 9)(2\ 6)(5\ 8)$ is even: the cycle of length 3 is two swaps and each cycle of length 2 is one, four swaps in all.

Fix one swap, and pair each rearrangement with the one obtained by performing that swap after it. Performing the swap a second time gives the first rearrangement back, so the $n!$ rearrangements fall into pairs, and each pair has one even member and one odd member. So exactly half of the $n!$ rearrangements are even. The even ones form a subgroup, since even then even is even and the undoing of an even rearrangement is performed by the same swaps in the reverse order.

Every even rearrangement can be performed by cycles of length 3. It is a product of an even number of swaps, and the swaps can be taken in pairs. A pair that repeats one swap does nothing. A pair sharing one thing is a cycle of length 3: swapping a with b and then a with c is the cycle carrying a to b , b to c and c to a . A pair sharing nothing is two such cycles: swapping a with b and then c with d has the same effect as the cycle $(a\ c\ d)$, carrying a to c , c to d and d to a , followed by the cycle $(a\ c\ b)$.

* * *

The fifteen puzzle can now be finished. The invariant showed that a position with the empty place in the bottom right corner can be reached from the target only if its list of tiles has an even count of pairs out of order, which says that the rearrangement of the fifteen

tiles is even. Are all of those reached?

Slide the empty place once round a block of two places by two, through its four places and back to where it started. Three tiles share the block with it, and each has been pushed one place round: the block's three tiles have undergone a cycle of length 3, and nothing else has moved. Now take any three tiles anywhere in the tray. Some sequence of slides, call it M , brings them into one block of two by two together with the empty place. A tile is moved one place in any direction by bringing the empty place round to the place the tile is to move into, by a route that does not pass through the tile, and then sliding the tile across. So the three tiles can be brought one at a time to a corner of the tray. A tile already parked in the corner, or two tiles parked in the corner and next to it, do not block the routes needed for the next, since the rest of the tray stays in one piece with room to go round. Perform M , then the trip round the block, then M backwards, slide by slide. Every tile other than the three is back where it was, and the three have been cycled among their original places, one way round; going round the block in the other direction cycles them the other way round. So every cycle of length 3 on the tiles can be produced by slides that return the empty place to its corner, hence every product of such cycles, hence every even rearrangement.

So the positions with the empty place in its corner that can be reached are exactly the even ones, half of the $15!$ rearrangements of the tiles, 653,837,184,000 positions. As with dominoes on a chess-board with one dark and one light square removed, the invariant was the whole truth, and it took a construction to show it.

* * *

Which of the twenty-four rearrangements of the corners of a square are even, and which of the eight symmetries of the square are among them?

A perfect shuffle of the first kind is performed on a pack of 32 cards. After how many shuffles does the pack return to its order?

In how many rearrangements of six things does every cycle have length 2, no thing staying in place?

If ten drawers and ten players follow the rule of the hundred, each opening at most five drawers, what is the chance that all succeed?

A NUMBER THAT CANNOT EXIST

Cardano had published a formula for cubic equations like $x^3 = 15x + 4$ in the *Ars Magna* of 1545, hard won and with a disreputable history. It says that a root of $x^3 = px + q$ is

writing s for $q^2/4 - p^3/27$, the sum of the cube roots of $q/2 + \sqrt{s}$ and of $q/2 - \sqrt{s}$

With $p = 15$ and $q = 4$, $q^2/4 = 4$ and $p^3/27 = 125$, so the number under the square root is $4 - 125 = -121$. There is no square root of -121 , since a number times itself is never negative. And yet $x = 4$ is a root: $4^3 = 64$ and $15 \cdot 4 + 4 = 64$. The equation has an ordinary whole-number solution, and the formula that should produce it has walked into a wall.

* * *

Where does the formula come from? Look for a root of $x^3 = px + q$ as a sum of two numbers, $x = u + v$. Multiplying out,

$$(u + v)^3 = u^3 + v^3 + 3uv(u + v)$$

Since $u + v$ is x , the right side is $3uv$ times x , plus $u^3 + v^3$. Set that beside $px + q$: the two agree, and $x = u + v$ satisfies $x^3 = px + q$, as soon as $u^3 + v^3 = q$ and $3uv = p$.

The second demand says $uv = p/3$, and cubing both sides makes the product of u^3 and v^3 equal to $p^3/27$. So u^3 and v^3 are two numbers with sum q and product $p^3/27$, and two numbers with a known sum and product can be found. They lie at equal distances on either side of half their sum, which here is $q/2$. If that distance is t , the two numbers are $q/2 + t$ and $q/2 - t$, and their product is $(q/2 + t)(q/2 - t) = q^2/4 - t^2$. The product has to be $p^3/27$, so $t^2 = q^2/4 - p^3/27$. Call that number s , so that t is $\sqrt{s}$. Then u^3 and v^3 are $q/2 + \sqrt{s}$ and $q/2 - \sqrt{s}$, and u and v are their cube roots, taken so that $uv = p/3$. That is the formula, and the pairing of the two cube roots is part of it.

For $x^3 = 15x + 4$ the demand is for two numbers with sum 4 and product 125, and $s = 4 - 125 = -121$. No two numbers have that sum and that product: the distance t of each from half their sum would need $t^2 = -121$, and no number has a negative square. The wall stands exactly there.

Either the formula fails when the number under the root is negative, or it is right and the wall is not where it seems. Bombelli tested the second. He wrote $\sqrt{-121}$ as $11\sqrt{-1}$, treated $\sqrt{-1}$ as an object obeying the ordinary rules of algebra without asking what it was, and pushed it through the computation. The formula asks for the cube root of $2 + 11\sqrt{-1}$. Guess that it has the form $a + b\sqrt{-1}$ for whole numbers a and b or their negatives, and cube the guess:

$$(a + b\sqrt{-1})^3 = a^3 + 3a^2b\sqrt{-1} + 3ab^2(\sqrt{-1})^2 + b^3(\sqrt{-1})^3$$

The only property $\sqrt{-1}$ has is that its square is -1 , so $(\sqrt{-1})^2 = -1$ and $(\sqrt{-1})^3 = -\sqrt{-1}$, and

$$(a + b\sqrt{-1})^3 = (a^3 - 3ab^2) + (3a^2b - b^3)\sqrt{-1}$$

Matching this with $2 + 11\sqrt{-1}$ asks for $a^3 - 3ab^2 = 2$ and $3a^2b - b^3 = 11$, and $a = 2$, $b = 1$ give $8 - 6 = 2$ and $12 - 1 = 11$. So a cube

root of $2 + 11\sqrt{-1}$ is $2 + \sqrt{-1}$. The derivation of the formula demands cube roots u and v of $2 + 11\sqrt{-1}$ and $2 - 11\sqrt{-1}$ with $uv = p/3$, which is 5. The same computation with $b = -1$ shows that $2 - \sqrt{-1}$ cubes to $2 - 11\sqrt{-1}$. The product is right as well: $(2 + \sqrt{-1})(2 - \sqrt{-1}) = 4 - (\sqrt{-1})^2 = 5$. So $2 + \sqrt{-1}$ and $2 - \sqrt{-1}$ are such a pair. Their sum is 4, because the two copies of $\sqrt{-1}$ cancel, and 4 is the root that was there all along. It had to be a root.

The step of the derivation that makes $u + v$ a root, $(u + v)^3 = u^3 + v^3 + 3uv(u + v)$, used nothing but multiplying out brackets, and those are among the rules $\sqrt{-1}$ was allowed to obey. So that step holds for these u and v as it holds for numbers: here $u^3 + v^3 = (2 + 11\sqrt{-1}) + (2 - 11\sqrt{-1}) = 4$ and $uv = 5$, and any u and v meeting those two demands, whatever they are made of, have a sum $u + v$ that satisfies $x^3 = 15x + 4$. Whether these numbers have other cube roots, and where they lie, is settled once multiplication is shown to add angles. The route from an ordinary equation to an ordinary answer passed through a place where nothing exists, by rules proved only for numbers, and arrived correctly.

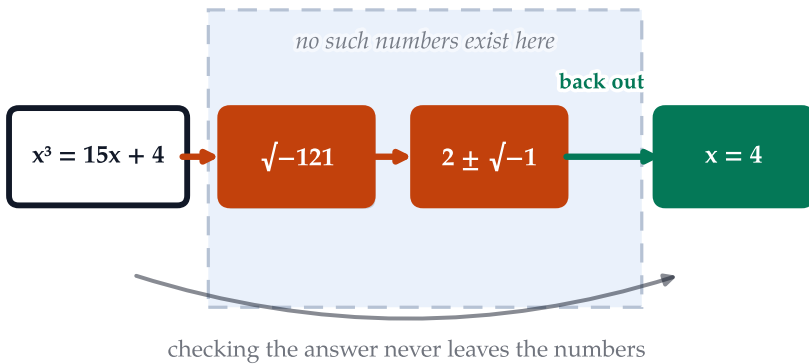


Figure 15.1. The equation and its root are both perfectly ordinary. The formula's route between them leaves the numbers entirely, spends three steps somewhere nothing exists, and comes back with the right answer. *Assumes:* that the rules of algebra keep working on an object nobody has constructed; that assumption is unjustified at this point and the arithmetic depends on it completely

The other roots are as ordinary as the first. Dividing out $x - 4$ gives $x^3 - 15x - 4 = (x - 4)(x^2 + 4x + 1)$, and $x^2 + 4x + 1 = (x + 2)^2 - 3$ vanishes at $-2 + \sqrt{3}$ and at $-2 - \sqrt{3}$; cubing confirms it, since $(-2 + \sqrt{3})^3 = -26 + 15\sqrt{3} = 15(-2 + \sqrt{3}) + 4$. All three roots lie on the number line, and the formula reaches for them through $\sqrt{-121}$.

A second cubic tests whether that was a coincidence. For $x^3 = 6x + 4$, $q^2/4 = 4$ and $p^3/27 = 8$, so the number under the root is -4 , and the formula asks for the cube root of $2 + 2\sqrt{-1}$. The guess $1 + \sqrt{-1}$ looks right, its partner is $1 - \sqrt{-1}$, and their sum is 2; but $2^3 = 8$ while $6 \cdot 2 + 4 = 16$. The guess was wrong, and cubing it shows how: $(1 + \sqrt{-1})^2 = 1 + 2\sqrt{-1} - 1 = 2\sqrt{-1}$, and multiplying by $1 + \sqrt{-1}$ again gives $2\sqrt{-1} - 2$, which is $-2 + 2\sqrt{-1}$. The part without $\sqrt{-1}$ is -2 where 2 was wanted, so $1 + \sqrt{-1}$ is not a cube root of $2 + 2\sqrt{-1}$. The guess $-1 + \sqrt{-1}$ has square $1 - 2\sqrt{-1} - 1 = -2\sqrt{-1}$, and cube $2\sqrt{-1} + 2 = 2 + 2\sqrt{-1}$. Its partner $-1 - \sqrt{-1}$ cubes to $2 - 2\sqrt{-1}$ in the same way, and $(-1 + \sqrt{-1})(-1 - \sqrt{-1}) = 1 - (\sqrt{-1})^2 = 2$, which is $p/3$. So the cube roots, correctly paired, are $-1 + \sqrt{-1}$ and $-1 - \sqrt{-1}$. Their sum is -2 , and $(-2)^3 = -8 = 6 \cdot (-2) + 4$. The route through the nonexistent delivered a true root a second time.

What does the formula give for $x^3 = 15x - 22$, and is it a root?

* * *

What was missing was not an argument that $\sqrt{-1}$ may be used, but a thing for $\sqrt{-1}$ to be: something built from what already exists, whose square is -1 . No number has square -1 , and searching for one is hopeless. Work instead with pairs of numbers, fractions for now, and define how two pairs multiply:

$$(a, b) \times (c, d) = (ac - bd, ad + bc)$$

Multiply $(0, 1)$ by itself: $0 \cdot 0 - 1 \cdot 1 = -1$ and $0 \cdot 1 + 1 \cdot 0 = 0$, so the product is $(-1, 0)$. Pairs add place by place, $(a, b) + (c, d) = (a + c, b + d)$. Pairs of the form $(x, 0)$ then behave as the numbers do. The

rule gives $(x, 0) \times (y, 0) = (xy - 0, 0 + 0) = (xy, 0)$, and adding gives $(x, 0) + (y, 0) = (x + y, 0)$. So the pairs contain a faithful copy of the numbers, and an object, $(0, 1)$, whose square is the pair playing the part of -1 . Nothing impossible happened: still no number has square -1 , but an object built from two numbers has a square that behaves as -1 does. The same thing has happened before. The diagonal had no ratio to the side among counts of a unit, and a proportion was built from comparisons of multiples to supply one. Now $\sqrt{-1}$ is not among the numbers, and it has been built from pairs.

Why that multiplication? Suppose pairs are to multiply so that the laws of algebra survive: multiplication distributes over addition, neither order nor grouping matters, the numbers sitting at $(x, 0)$ behave as before, and $(b, 0)$ times $(0, 1)$ is $(0, b)$. Write i for $(0, 1)$, and write the number a for the pair $(a, 0)$. Adding in each place, $(a, b) = (a, 0) + (0, b)$, and the last demand says that $(0, b)$ is $(b, 0)$ times $(0, 1)$, which is bi . So every pair (a, b) is $a + bi$. Then, multiplying out the brackets as the laws allow,

$$(a + bi)(c + di) = ac + adi + bci + bd \cdot i^2$$

and everything is settled except the last term, which depends on i^2 . Demanding $i^2 = -1$ gives $(ac - bd) + (ad + bc)i$, the rule. So the rule is forced by keeping the algebra and demanding $i^2 = -1$. The rule in turn keeps the laws. Swapping two pairs does not change $ac - bd$ or $ad + bc$. Multiplying (a, b) by the sum of (c, d) and (e, f) , which is $(c + e, d + f)$, gives $a(c + e) - b(d + f)$ in the first place. That is $(ac - bd) + (ae - bf)$, the first places of the two separate products added together. The second place splits in the same way, because every term of the rule takes one entry from each factor. So (a, b) times a sum is the sum of the two products. And (a, b) , (c, d) and (e, f) multiplied in either grouping give the same pair,

$$(ace - adf - bcf - bde, acf + ade + bce - bdf)$$

as multiplying out both groupings shows.

Demand $i^2 = +1$ instead. The rule becomes $(a + bi)(c + di) = (ac + bd) + (ad + bc)i$, which is consistent arithmetic. But $(1 + i)(1 - i) = 1 - i^2 = 0$: two things, neither of them 0, multiply to 0. Among numbers that never happens, and cancelling depends on it. Cancelling is the step from $ab = ac$, with a not 0, to $b = c$. It works because $ab = ac$ says that $a(b - c) = 0$, and if a is not 0 the only way out is $b - c = 0$. Where two things other than 0 can multiply to 0, that way out is no longer the only one. Demanding $i^2 = 0$ gives a third arithmetic, and the same defect shows at once: i is not 0, and i times i is 0. **The choice of i^2 settles everything, and the choices give different worlds.**

* * *

A pair of numbers is a point of the plane, reached from a fixed point, the origin, by going the first number along the horizontal, to the right, or to the left if it is negative, and then the second number up the page, or down if it is negative. Multiplication can be watched on points. The number i lies at distance 1 from the origin, a quarter turn from the horizontal. Its square, -1 , lies at distance 1, a half turn from the horizontal. In this one case, multiplying two points multiplied their distances, 1 times 1 giving 1, and added their angles, a quarter turn and a quarter turn giving a half. The number $1 + i$ lies at the far corner of the unit square, at an eighth of a turn, at distance equal to the diagonal. Its square is $1 + 2i + i^2 = 2i$, at a quarter turn and distance 2. The angle doubled, from an eighth of a turn to a quarter. The distance was multiplied by itself, because the diagonal times itself is 2, the distance of $2i$. To see that, cut two unit squares along their diagonals: the four triangles reassemble into the square on the diagonal, so that square has area 2, taking area in its everyday sense. (What area means, for curved regions above all, is laid down when areas under curves are measured.) Distances multiplied and angles added a second time.

Two instances, and of the kind anyone tries first, make an observation, not a theorem. Before the observation can become a theo-

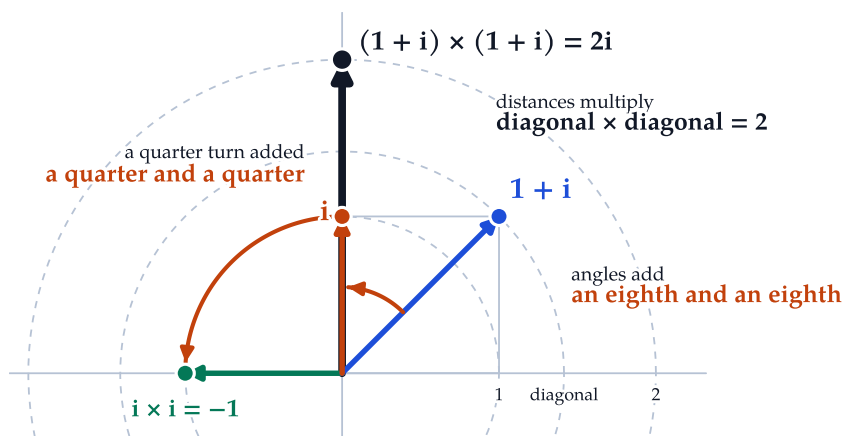


Figure 15.2. Multiplying i by itself and $1 + i$ by itself multiplies the distances from the origin and adds the angles; that this holds for every pair of points is proved once angles are measured by the sine and cosine. *Assumes:* that the angle is measured from the same starting line for both, and that the origin is fixed; move the origin and none of this survives

rem, “angles add” needs a meaning for angles that are not quarters and eighths of a turn. The number $3 + 4i$ has an untidy angle. Its distance from the origin is the longest side of the right triangle with legs 3 and 4, and a cutting like the one that measured the diagonal of $1 + i$ measures the longest side of every right triangle. That is the theorem on squares, derived once already from the belief that allowed enlarged copies. The proof promised then, by cutting a square in two ways, follows, and it enlarges nothing.

Call the two shorter sides of a right triangle a and b , and the longest c . A square of side $a + b$ can be divided in two ways: into four copies of the triangle around a tilted square of side c , and into the same four triangles around two squares of sides a and b . Take the four triangles away from each division. In the first, what is left over is the tilted square, of area c^2 . In the second, what is left over is the two squares, of areas a^2 and b^2 . The same square less the same four triangles leaves the same area, areas again in their everyday sense, so $c^2 = a^2 + b^2$: the squares on the two shorter sides add up to the square on the longest. The tilted figure has four sides of

length c , and it is a square because its corners are right angles. At each corner, two triangles meet along a side of the big square, one bringing one of its acute angles and the other bringing the other. The two acute angles of a right triangle add up to one right angle, since the angles of a triangle add up to two right angles. Along the straight side there are two right angles in all, so one right angle is left for the corner of the tilted figure. That the angles of a triangle add up to two right angles is one of the beliefs about the page listed where the diagonal of the square was cut.

So the distance of $3 + 4i$ from the origin is 5, since $9 + 16 = 25$. Its square is $9 - 16 + 24i = -7 + 24i$, at distance 25, since $49 + 576 = 625$. The distances multiplied. Whether the angle doubled cannot yet be asked properly.

That distances multiply is a theorem already. For any two pairs,

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$$

since on the right the cross terms $-2abcd$ and $+2abcd$ cancel and what remains is $a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2$, which is what the left side gives when its brackets are multiplied out. The left side is the product of the squared distances of (a, b) and (c, d) , and the right side is the squared distance of their product. That disposes of the defect that spoiled the two rival choices. A pair other than $(0, 0)$ has squared distance above 0. Multiply two such pairs: by the identity, the squared distance of the result is one number above 0 times another, and so is above 0. But $(0, 0)$ has squared distance 0, so the result is never $(0, 0)$. Cancelling survives here.

The identity also makes facts about whole numbers. The squared distance of $2 + i$ is $4 + 1 = 5$, and that of $3 + 2i$ and of $3 - 2i$ is $9 + 4 = 13$. So each of the products $(2 + i)(3 + 2i) = 4 + 7i$ and $(2 + i)(3 - 2i) = 8 - i$ has squared distance $5 \cdot 13 = 65$, and 65 is a sum of two squares in two ways, $16 + 49$ and $64 + 1$.

And the square of $a + bi$ is $(a^2 - b^2) + 2abi$. Its distance is the distance of $a + bi$ multiplied by itself, which is $a^2 + b^2$. A point's distance from the origin is the longest side of the right triangle whose shorter sides are its two entries, so, when a is larger than b and b

is above 0, $a^2 - b^2$, $2ab$ and $a^2 + b^2$ are the sides of a right triangle: $2 + i$ gives 3, 4, 5; $3 + 2i$ gives 5, 12, 13; and $4 + 3i$ gives 7, 24, 25, the sides already met in the square of $3 + 4i$, which is $-7 + 24i$. In how many ways is $1105 = 5 \cdot 13 \cdot 17$ a sum of two squares?

What multiplying by i does to the whole plane can be said now. By the rule, $(0, 1) \times (a, b) = (0 \cdot a - 1 \cdot b, 0 \cdot b + 1 \cdot a) = (-b, a)$. The point (a, b) sits at the far corner of the rectangle whose horizontal side is a and whose vertical side is b , and $(-b, a)$ sits at the far corner of that same rectangle stood on end: the side a now runs up the page from the origin, and the side b runs from the origin to the left. The rectangle has been turned about the origin through a right angle. The same holds when a or b is negative and the rectangle lies in another corner of the plane: a side that ran to the right now runs up, one that ran up now runs to the left, one that ran to the left now runs down, and one that ran down now runs to the right, each turned a quarter of the way round in the same direction. So multiplying by i turns every point a quarter of the way round. Multiplying by i twice is a half turn, which is multiplication by -1 ; three times is three quarters; four times brings every point back. The numbers $1, i, -1$ and $-i$, multiplied together in every way, give the table of the four-hour clock and of the four turns of the square: i times i is -1 as a quarter turn twice is a half turn, and $-i$ times $-i$ is i^2 , which is -1 , as three quarters of a turn made twice is a turn and a half and lands on a half turn. The turns are not something these numbers represent; they are what the numbers do.

THE ROTATING RAY

What is the angle of $3 + 4i$, and does squaring $3 + 4i$ double it? The words need a meaning first. An angle so far has been a thing seen, not a thing computed with. What can be computed with is a ratio, and an angle is a ratio in disguise.

* * *

Take a right triangle and one of its acute angles, α . Three lengths are in play: the side opposite α , the side next to it, and the hypotenuse. A second right triangle, of any size, with the same angle α , has the same angles and so the same shape at a different size, and the ratio of any two of its sides equals the ratio of the corresponding sides of the first. So the ratio of the opposite side to the hypotenuse depends on α alone, and it is the *sine* of α ; the ratio of the adjacent side to the hypotenuse is the *cosine*.

That rests on the sentence “the same angles, so the same shape at different sizes”. It has not been proved, and nothing so far proves it. It is a belief about the page, and whether it is a theorem or an assumption is a question examined later. Everything below leans on it. The theorem on squares leans in the same way on the angle sum of a triangle, which is the same belief in another form.

The right triangle with legs 3 and 4 has hypotenuse 5, since

$9 + 16 = 25$. The angle opposite the side 3 has sine $3/5$ and cosine $4/5$, and the angle opposite the side 4 has sine $4/5$ and cosine $3/5$: the two acute angles trade sine for cosine. The two acute angles of a right triangle together make one right angle, so each is what the other lacks of a right angle, its complement, and “cosine” is short for the sine of the complement. In every right triangle, the square of the sine plus the square of the cosine is a single ratio: the square of the opposite side plus the square of the adjacent side, over the square of the hypotenuse. The square on the hypotenuse is the sum of the squares on the other two sides, so the top of that ratio equals the bottom, and the ratio is 1. For the 5-12-13 triangle, $25/169$ added to $144/169$ is 1.

* * *

The sine was defined for acute angles only. A right triangle with two right angles does not exist, but in right triangles with hypotenuse 1 whose angle α grows toward a right angle, the opposite side grows toward the hypotenuse and the adjacent side shrinks toward nothing. So the sine of a right angle is *defined* to be 1 and its cosine 0, the values that keep the pattern. The other end is settled the same way. As the angle shrinks towards nothing the opposite side shrinks towards nothing and the adjacent side grows towards the hypotenuse, so the sine of no angle at all is defined to be 0 and its cosine 1. With those two the whole quarter turn from nothing to a right angle has sines and cosines.

The law of cosines says that in a triangle with sides a , b and c and angle C between a and b ,

$$c^2 = a^2 + b^2 - 2ab \cdot \cos C$$

For an acute angle C , put C at the origin and the side a along the horizontal, and drop the perpendicular from the far end of b . Its foot lies at distance $b \cos C$ from C , and its height is $b \sin C$, so $c^2 = (a - b \cos C)^2 + (b \sin C)^2 = a^2 - 2ab \cos C + b^2(\cos^2 C + \sin^2 C)$,

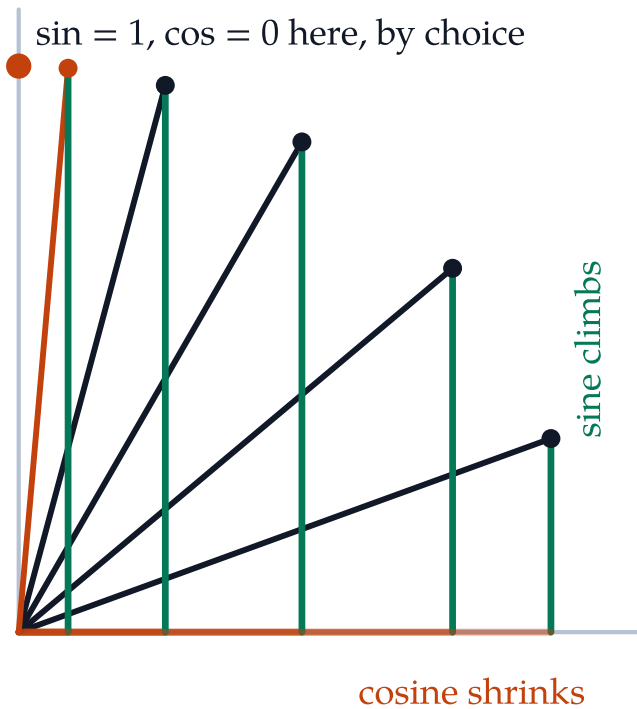


Figure 16.1. Right triangles with hypotenuse 1 and angles growing towards a right angle: the opposite side, the sine, climbs towards 1 and the adjacent side, the cosine, shrinks towards 0, and the values at the right angle itself are chosen so the pattern survives. *Assumes:* that the ratios are the same in every triangle with the same angle, which is the similarity belief the text has just written down as unproved

which is $a^2 + b^2 - 2ab \cos C$. For a right angle the cosine is 0 and this is the theorem on squares.

For an obtuse angle C the foot lies outside the triangle, on the line of the side a continued beyond C . Let C' be the acute angle that, added to C , makes two right angles; it is the angle between b and that continuation. The foot lies at distance $b \cos C'$ beyond C , and the height is $b \sin C'$. The horizontal leg of the right triangle with hypotenuse c is now $a + b \cos C'$ where before it was $a - b \cos C$, and the same computation gives $c^2 = (a + b \cos C')^2 + (b \sin C')^2 = a^2 + b^2 + 2ab \cos C'$. Define the cosine of an obtuse angle to be minus the cosine of the angle completing it to two right angles, and one formula covers every triangle. Define the sine of an obtuse angle to be the sine of that completing angle. The area of a triangle is half the base times the height, a rule belonging to the same everyday sense of area. With a as the base, the height found above is $b \sin C$ for an acute angle C and $b \sin C'$ for an obtuse one, so with this definition the area is $\frac{1}{2}ab \sin C$ for every angle C .

Take the angle that is two thirds of two right angles, which is 120° in the old counting that cuts a right angle into ninety parts. A right triangle has no obtuse angle, so the cosine of 120° , which is $-1/2$, comes from the definition alone.

The $1/2$ itself comes from an equilateral triangle of side 2, whose three equal angles are each two thirds of a right angle, 60° . The perpendicular from a corner to the opposite side cuts it into two right triangles with equal hypotenuses and a shared side, copies of each other, so the base is halved, and so is the angle at the top corner. Each half is a right triangle with hypotenuse 2, a side of 1 along the base, an angle of 60° at the base and an angle of 30° at the top. Its height h satisfies $1 + h^2 = 4$, so $h = \sqrt{3}$. Reading the ratios from the 60° corner and then from the 30° corner gives $\cos 60^\circ = \sin 30^\circ = 1/2$ and $\sin 60^\circ = \cos 30^\circ = \sqrt{3}/2$, and by the definitions for obtuse angles $\cos 120^\circ = -1/2$ and $\sin 120^\circ = \sqrt{3}/2$. The extended formula can be tested on whole numbers: sides 3 and 5 with 120° between them give $c^2 = 9 + 25 + 15 = 49$, a third side of exactly 7. The formula also turns the theorem on squares round. If $a^2 + b^2 =$

c^2 in some triangle, then $2ab \cos C = 0$, so $\cos C = 0$; the cosine of an acute angle is positive and that of an obtuse angle negative, so C is a right angle. In the same way $a^2 + b^2$ exceeds c^2 exactly when C is acute.

* * *

What is $\sin(\alpha + \beta)$, in terms of the sines and cosines of α and β ? Take a point P at distance 1 from the origin O on the ray at angle $\alpha + \beta$, and drop the perpendicular from P to the ray at angle α , meeting it at Q . The triangle OQP has a right angle at Q , hypotenuse 1 and angle β at O , so $OQ = \cos \beta$ and $QP = \sin \beta$. The height of P above the horizontal is $\sin(\alpha + \beta)$. For an acute $\alpha + \beta$ that is the definition of the sine, since $OP = 1$, and for a right or an obtuse $\alpha + \beta$ it is the value the sine was defined to have. That height can also be reached in two steps, from O to Q and then from Q to P . The step from O to Q runs a length OQ along the ray at angle α , so it rises by OQ times $\sin \alpha$, and that is the height of Q . The step from Q to P is perpendicular to that ray. A ray tilted by α away from the horizontal has its perpendicular tilted by α away from the vertical, so a step of length QP along the perpendicular rises by QP times $\cos \alpha$. The height of P is the sum of the two rises. So

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

The same figure can be read along the horizontal. The step from O to Q carries a distance OQ times $\cos \alpha$ to the right. The step from Q to P , tilted by α away from the vertical, leans back over the ray and carries a distance QP times $\sin \alpha$ to the left, and that is where the minus sign comes from. The horizontal position of P is $\cos(\alpha + \beta)$, with distances to the left of O counted as negative, which is what the negative cosine of an obtuse angle was defined to match. So

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

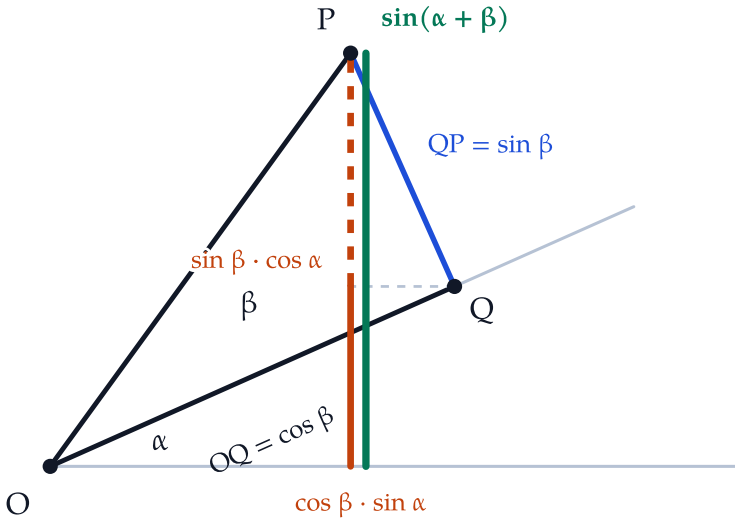


Figure 16.2. One height counted twice: P sits at distance 1 on the ray at angle $\alpha + \beta$, so its height above the base is $\sin(\alpha + \beta)$; the same vertical is also the height of Q , which is $\cos \beta$ times $\sin \alpha$, plus the extra that the segment QP contributes, which is $\sin \beta$ times $\cos \alpha$, because QP is perpendicular to the ray at angle α . Assumes: that QP , being perpendicular to a line at angle α , leans at angle α from the vertical, which is the one geometric fact the two-piece count uses beyond the definitions

That is a double count: one length counted whole and in two pieces.

The number $3 + 4i$ lies at distance 5, and its angle α has sine $4/5$ and cosine $3/5$. If squaring adds the angle to itself, the square lies at angle 2α , with sine $2 \cdot (4/5) \cdot (3/5) = 24/25$ and cosine $9/25 - 16/25 = -7/25$. Distances multiply, so the square lies at distance $5 \cdot 5 = 25$, and its coordinates are $25 \cdot (-7/25) = -7$ and $25 \cdot (24/25) = 24$. The prediction is the point $-7 + 24i$. Multiplying out $(3 + 4i)(3 + 4i)$ as pairs gives $9 - 16 + 24i$, which is $-7 + 24i$ as well. For this number, squaring did double the angle.

The agreement is general. Write one number, at distance r and angle α , as $r(\cos \alpha + i \sin \alpha)$, and another as $s(\cos \beta + i \sin \beta)$, and

multiply them as pairs multiply:

$$rs \cdot [(\cos \alpha \cos \beta - \sin \alpha \sin \beta) + i(\sin \alpha \cos \beta + \cos \alpha \sin \beta)]$$

By the two identities the bracket is $\cos(\alpha + \beta) + i \sin(\alpha + \beta)$, so the product lies at distance rs and angle $\alpha + \beta$, for angles α and β less than a right angle. The lengths and ratios r , s , $\sin \alpha$ and $\cos \alpha$ need not be fractions, while the pairs so far were built from fractions. The numbers that measure such lengths are called the real numbers, and pairs of real numbers come when the real numbers are built. The multiplication rule for pairs, forced by keeping the algebra and demanding $i^2 = -1$, is the addition formula for sine and cosine in another notation.

* * *

Name the corners of any triangle A , B and C , with side a opposite A , side b opposite B and side c opposite C . Drop the perpendicular from C to the line of the side c ; when the angle at A or at B is obtuse it meets that line beyond a corner rather than between the two. The perpendicular is a side of two right triangles, one with hypotenuse b and the angle at A , the other with hypotenuse a and the angle at B . Read in the first, its length is $b \sin A$; read in the second, it is $a \sin B$. When the angle at A or at B is obtuse, the right triangle formed uses the angle completing it to two right angles, whose sine was defined to be the same, so both readings hold in every case. So $b \sin A = a \sin B$, and dividing both sides by $\sin A \cdot \sin B$ gives $a/\sin A = b/\sin B$. The perpendicular from another corner brings in the third side in the same way. That is the law of sines: $a/\sin A = b/\sin B = c/\sin C$, or, with each ratio turned over, $\sin A/a = \sin B/b = \sin C/c$.

* * *

Angles larger than two right angles have not been given sines, and yet the plane has them: $-i$ lies at three quarters of a turn,

and multiplying by $-i$ repeatedly goes round more than once. If distances multiply and angles add for every number of the plane, the sine and cosine must be defined for every angle, however far round.

The definition uses a ray. Turn a ray from the origin, starting along the horizontal, through an angle θ , as far as needed and in either direction. Where it crosses the circle of radius 1, the horizontal coordinate is $\cos \theta$ and the vertical one is $\sin \theta$. For acute, right and obtuse angles this agrees with the earlier definitions, which entitles it to replace them.

The addition formulas hold for all angles too. Every angle is some number of quarter turns, forward or back, together with an angle less than a right angle, and a quarter turn forward is multiplication by i , a quarter turn back multiplication by $-i$. So every point of the unit circle is a point at an angle less than a right angle, multiplied by i or by $-i$ once for each of its quarter turns.

Now multiply two points of the unit circle. The factors of a product can be taken in any order, so the product is the product of the two points at angles less than a right angle, multiplied by all the factors i and $-i$ that the two original points carried. The product of the two points at angles less than a right angle lies at the sum of their angles, by the figure and the multiplication above. The factors i and $-i$ then turn it through the quarter turns of both original points. So the product of the two original points is the point at the sum of their angles. Multiplying the two points as pairs and reading off the coordinates of that point gives the two formulas, for all angles.

What do the two formulas give for $\cos 3\theta$ in terms of $\cos \theta$?

With angles adding, the question left open at the cubic has an answer: what are the cube roots of a number of the plane? Let ω be the point of the unit circle at a third of a turn, $\omega = -1/2 + (\sqrt{3}/2)i$. Its cube lies at a whole turn, so $\omega^3 = 1$, and multiplying as pairs agrees: $\omega^2 = -1/2 - (\sqrt{3}/2)i$, and $\omega^2 \cdot \omega = 1/4 + 3/4 = 1$. Take a number z at distance r and angle θ . Cubing a number cubes its distance and triples its angle, so the number at distance the cube

root of r and angle $\theta/3$ has cube z . So do the two numbers at the same distance a third of a turn and two thirds of a turn further round: tripling their angles gives θ plus one whole turn and θ plus two whole turns, and a whole turn brings a ray back to the same direction. There are no others. A cube root of z must lie at that distance, and three times its angle must be θ plus a whole number of turns. So its angle is $\theta/3$ plus a whole number of thirds of a turn, and since three thirds of a turn bring a ray back to where it was, only three directions arise: $\theta/3$, a third of a turn beyond it, and two thirds of a turn beyond it. So a nonzero number has exactly three cube roots, and if z_0 is one of them the others are $z_0\omega$ and $z_0\omega^2$.

The cube roots of $2 + 11i$ are therefore $2 + i$, $(2 + i)\omega$ and $(2 + i)\omega^2$, and those of $2 - 11i$ are $2 - i$, $(2 - i)\omega$ and $(2 - i)\omega^2$. The formula for the cubic adds a cube root of $2 + 11i$ to a cube root of $2 - 11i$, and its derivation allows only pairs of cube roots whose product is 5. Since $(2 + i)(2 - i) = 5$ and $\omega^3 = 1$, the pairs are $2 + i$ with $2 - i$, $(2 + i)\omega$ with $(2 - i)\omega^2$, and $(2 + i)\omega^2$ with $(2 - i)\omega$; the other six products are 5ω or $5\omega^2$. The first pair has sum 4. For the second, $\omega + \omega^2 = -1$ and $\omega - \omega^2 = \sqrt{3}i$, so the sum is $2(\omega + \omega^2) + i(\omega - \omega^2) = -2 - \sqrt{3}i$, and the third pair gives $-2 + \sqrt{3}i$ in the same way. The formula, with its pairing, delivers all three roots that factoring found. A cube root of a general distance needs the real numbers, which are still to be built; the only cube root of a distance that this example needed was the cube root of $5\sqrt{5}$, the distance of $2 + 11i$, and it is $\sqrt[3]{5}$, the distance of $2 + i$.

The object trigonometry studies is a ray turning about a point, and the triangle was the part of it that could be drawn first; the last widening of the definition lets every nonzero number of the plane be written as a distance and a turn.

GEOMETRY BY MULTIPLICATION

A turn is now a multiplication: multiplying by a point of the unit circle turns the plane, and by a point at distance r from the origin turns it and stretches it by r , for every angle. How much of geometry can be done by arithmetic?

* * *

Write z for the point of the unit circle at angle θ , which is $\cos \theta + i \sin \theta$. Multiplying z by itself adds the angle to itself, so z^2 lies at angle 2θ , z^3 at 3θ , and, climbing from each power to the next, z^n lies at angle $n\theta$:

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

This is de Moivre's formula, and it turns every question about multiples of an angle into the multiplying out of a bracket, which the binomial theorem does. Write c and s for the cosine and sine. Then $(c + is)^3 = c^3 + 3c^2(is) + 3c(is)^2 + (is)^3$, and since $i^2 = -1$ and $i^3 = -i$, the real part is $c^3 - 3cs^2$ and the imaginary part $3c^2s - s^3$. So $\cos 3\theta = c^3 - 3cs^2$, which with $s^2 = 1 - c^2$ is $4c^3 - 3c$, and $\sin 3\theta = 3s - 4s^3$. The same computation gave Bombelli his cube

root: $a + b\sqrt{-1}$ cubed had real part $a^3 - 3ab^2$ and imaginary part $3a^2b - b^3$, and matching it to a given number was, without anyone knowing it, the problem of cutting an angle in three.

Multiplied out, $(c + is)^n$ is a sum of whole-number multiples of a power of c times a power of (is) . An even power of (is) is real, since $i^2 = -1$: it is an even power of s with a sign in front. An odd power of (is) is a real number times i . So in the real part of $(c + is)^n$ only even powers of s occur, and each s^2 can be replaced by $1 - c^2$. So $\cos n\theta$ is a polynomial in $\cos \theta$, for every n :

$$\begin{aligned}\cos 2\theta &= 2c^2 - 1 \\ \cos 3\theta &= 4c^3 - 3c \\ \cos 4\theta &= 8c^4 - 8c^2 + 1 \\ \cos 5\theta &= 16c^5 - 20c^3 + 5c\end{aligned}$$

Each can be had from the two before without any multiplying out. The addition formulas give $\cos(n+1)\theta = \cos n\theta \cos \theta - \sin n\theta \sin \theta$ and $\cos(n-1)\theta = \cos n\theta \cos \theta + \sin n\theta \sin \theta$, and adding the two removes the sines: $\cos(n+1)\theta + \cos(n-1)\theta = 2 \cos \theta \cos n\theta$. So the next polynomial is $2c$ times the last, less the one before. Multiplying by $2c$ raises the degree by one and doubles the leading coefficient, and the polynomial taken away has lower degree, so it cannot touch the top term. So the leading coefficient doubles each time, and the polynomial for $\cos n\theta$ has degree n . Take the cosine of 20° . Three times 20° is 60° , whose cosine is $1/2$, so $c = \cos 20^\circ$ satisfies $4c^3 - 3c = 1/2$, and doubling makes it the cubic $8c^3 - 6c = 1$ with whole-number coefficients. Whether a length satisfying such a cubic can be constructed with ruler and compass is a question that can be asked exactly, and is answered much later.

* * *

Which numbers z of the plane satisfy $z^n = 1$? The distance of z^n from the origin is the n -th power of the distance of z , and the only length whose n -th power is 1 is 1, so z is at distance 1. The angle of z^n is n times the angle of z , and the number 1 lies at angle 0, which

is the same direction as any whole number of turns and no other angle. So n times the angle of z is a whole number of turns, say k of them, and the angle of z is k/n of a turn. The values $k = 0, 1, \dots, n - 1$ give n different points, and any other k repeats one of them. So there are exactly n solutions, the n -th roots of unity, and they are the corners of the regular polygon of n sides with one corner at 1.

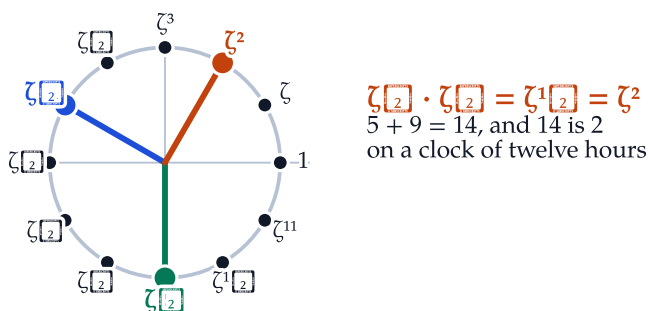


Figure 17.1. The twelve solutions of z to the twelfth power equals 1, at the hours of a clock face. Multiplying the point at hour 5 by the point at hour 9 adds the angles and lands at hour 14, which is hour 2: the roots of unity multiply as the hours add. *Assumes:* that multiplying two points of the unit circle adds their angles, which was proved for all angles; the hours are counted anticlockwise from 1, unlike a clock's

Write ζ for the first corner after 1, the point at $1/n$ of a turn. The corners are $1, \zeta, \zeta^2, \dots, \zeta^{n-1}$, and $\zeta^n = 1$. Multiplying two of them adds the exponents, and an exponent of n or more is reduced by n , since ζ^n is 1. So the n -th roots of unity, under multiplication, are the clock with n hours under addition, with ζ^k matched to the hour k : the fourth roots of unity $1, i, -1$ and $-i$ were the first case, the four turns of the square. Every regular polygon carries its own clock.

The corners of a regular polygon centred at the origin balance: their sum is 0. There is a proof with no computation in it. Let S be the sum $1 + \zeta + \zeta^2 + \dots + \zeta^{n-1}$. Multiply S by ζ : each term becomes the next, and the last becomes ζ^n , which is 1, the first. So $\zeta S = S$, that is, $(\zeta - 1)S = 0$. The product of two numbers of the plane is 0 only when one of them is, because distances multiply:

the distance of a product from the origin is the product of the two distances, and two lengths that are not 0 do not multiply to 0. Since ζ is not 1, the factor $\zeta - 1$ is not 0. So $S = 0$. The picture says the same thing. Multiplying by ζ turns the polygon by one corner and moves it onto itself, so its balance point is unmoved by that turn, and the only point unmoved by a turn is the centre.

From that one line come exact values that no drawing could give. Take $n = 5$. The five corners are $1, \zeta, \zeta^2, \zeta^3$ and ζ^4 , at angles of $0^\circ, 72^\circ, 144^\circ, 216^\circ$ and 288° , and the last two are the mirror images of ζ^2 and ζ in the horizontal axis. A point and its mirror image have the same real part, so ζ and ζ^4 each give $\cos 72^\circ$ to the real part of the sum, and ζ^2 and ζ^3 each give $\cos 144^\circ$. The sum is 0, so its real part is 0:

$$1 + 2 \cos 72^\circ + 2 \cos 144^\circ = 0$$

and $\cos 144^\circ = 2 \cos^2 72^\circ - 1$, by the polynomial for $\cos 2\theta$. Write c for $\cos 72^\circ$. The equation becomes $1 + 2c + 2(2c^2 - 1) = 0$, which is $4c^2 + 2c - 1 = 0$. Its two solutions are $(-1 + \sqrt{5})/4$ and $(-1 - \sqrt{5})/4$, and c is positive, so c is $(\sqrt{5} - 1)/4$, which is 0.30902.

The side of the regular pentagon inscribed in the unit circle is the distance from 1 to ζ , whose square is $(1 - c)^2 + s^2 = 2 - 2c$, and its diagonal is the distance from 1 to ζ^2 , whose square is $2 - 2 \cos 144^\circ$. With the value of c , the square of the side is $2 - 2c = (5 - \sqrt{5})/2$. Since $\cos 144^\circ = 2c^2 - 1 = -(1 + \sqrt{5})/4$, the square of the diagonal is $(5 + \sqrt{5})/2$. The square of the ratio of the diagonal to the side is therefore $(5 + \sqrt{5})/(5 - \sqrt{5})$, which is $(3 + \sqrt{5})/2$, and that is the square of $(1 + \sqrt{5})/2$. So the ratio of the diagonal to the side is $(1 + \sqrt{5})/2$, the number 1.61803 whose square is itself plus 1. Call it φ .

Dividing $\varphi^2 = \varphi + 1$ by φ gives $\varphi = 1 + 1/\varphi$. Take the side of the pentagon as 1, so that the diagonal is φ . The equation says that the diagonal is one side and a remainder $1/\varphi$. The side stands to that remainder as 1 to $1/\varphi$, which is φ again, the proportion in which the diagonal stood to the side. So cutting the side from the diagonal leaves two lengths in the same proportion as the two before. That

is the subtraction that reproduces itself after a single cut, in the pentagon, and ϕ , like the proportion of the diagonal of the square to its side, is the proportion of no two whole numbers.

* * *

An n -th root can now be taken of any nonzero number of the plane, as cube roots were. Take a number w at distance r and angle θ . The number at distance the n -th root of r and angle θ/n has w as its n -th power, since taking the n -th power raises the distance to its n -th power and multiplies the angle by n . Multiply that number by any n -th root of unity and the n -th power is still w , since the n -th power of the root of unity is 1. There are no others, since an n -th root of w must lie at that distance, and n times its angle must be θ plus a whole number of turns, which leaves those n angles. And 0 has only itself for a root, because distances multiply. So a nonzero w has exactly n n -th roots, the corners of a regular polygon centred at the origin. That lengths have n -th roots at all is a fact about the real numbers, taken for granted here as square roots have been.

A polynomial with a root gives up a factor. For any numbers z and r of the plane, and each k ,

$$z^k - r^k = (z - r)(z^{k-1} + z^{k-2}r + \dots + zr^{k-2} + r^{k-1})$$

as multiplying out shows, every product on the right cancelling against a neighbour except the first and the last. If p is a polynomial and $p(r) = 0$, then $p(z)$ is the same as $p(z) - p(r)$. Subtract term by term: the term of $p(z)$ that is a multiple of z^k , less the matching term of $p(r)$, is the same multiple of $z^k - r^k$, and the constant terms cancel. So $p(z)$ is a sum of multiples of the differences $z^k - r^k$, each of which has the factor $z - r$. So $p(z) = (z - r) \cdot q(z)$ for a polynomial q of degree one less. Suppose p has a second root r' different from r . Putting $z = r'$ in $p(z) = (z - r) \cdot q(z)$ gives $(r' - r) \cdot q(r') = 0$. A product in the plane is 0 only when a factor is, and $r' - r$ is not 0, so $q(r') = 0$, and q in turn gives up the factor $z - r'$. Each further root removes a factor in the same way and lowers the degree by

one. After n roots what is left has degree 0: it is a constant that is not 0, and it has no roots. So a polynomial of degree n has at most n roots, and if it has n of them, $r_1, \dots, r_n$, it is its leading coefficient times $(z - r_1)(z - r_2) \dots (z - r_n)$.

The polynomial $z^n - 1$ has its n roots in hand, so

$$z^n - 1 = (z - 1)(z - \zeta)(z - \zeta^2) \dots (z - \zeta^{n-1})$$

Divide both sides by $z - 1$. On the left, by the identity above with $r = 1$, the quotient is $1 + z + z^2 + \dots + z^{n-1}$. So

$$1 + z + z^2 + \dots + z^{n-1} = (z - \zeta)(z - \zeta^2) \dots (z - \zeta^{n-1})$$

for every z other than 1, where the division makes sense. That is more than $n - 1$ values of z , and both sides are polynomials of degree $n - 1$. If the two sides differed anywhere, their difference would be a polynomial of degree at most $n - 1$, not the zero polynomial, with more than $n - 1$ roots, and that cannot happen. So the two sides agree for every z , the value 1 included. Now put $z = 1$. The left side is n . The right side is the product of the numbers $1 - \zeta^k$, whose sizes are the distances from the corner at 1 to each of the other corners. Sizes multiply, so:

in a regular polygon of n sides inscribed in a circle of radius 1, the distances from one corner to all the others multiply to n .

In a square they are $\sqrt{2}$, 2 and $\sqrt{2}$, and the product is 4. In a regular hexagon they are 1, $\sqrt{3}$, 2, $\sqrt{3}$ and 1, and the product is 6. In a pentagon they are two sides and two diagonals, and the product of a side and a diagonal is $\sqrt{5}$.

* * *

Points of the plane can be added as well as multiplied. Adding b to every point raises each coordinate by the matching coordinate of b , which slides the whole plane along the arrow from 0 to b . So $a + b$ is where a lands under that slide: the path runs from 0 out

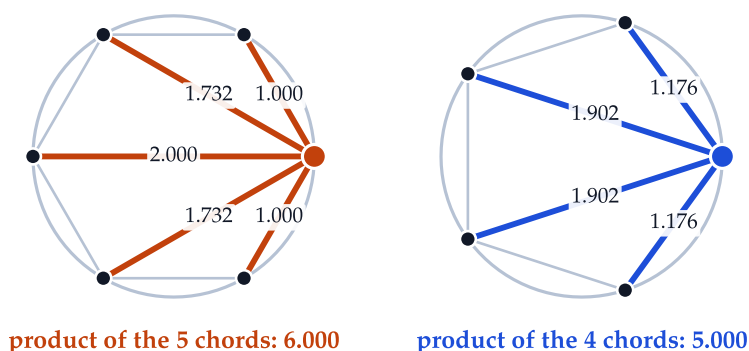


Figure 17.2. The chords from one corner of a regular hexagon and of a regular pentagon inscribed in the circle of radius 1, with their lengths. The five chords of the hexagon multiply to 6 and the four of the pentagon to 5. Assumes: two cases of a statement about every n ; the lengths printed are rounded, and the products are exact only for the exact lengths

to a and then along a copy of the arrow to b . The points $0, a, a + b$ and b are the four corners of a figure whose two sides from 0 are the arrows to a and to b , and whose other two sides are copies of those arrows. That turns whole figures into algebra. The slide that carries a to b is the slide by $b - a$. The midpoint of a and b is reached from a by half of it, so it is $a + (b - a)/2$, which is $(a + b)/2$.

The corners a, b and c of a triangle balance about the point p at which the arrows from p to them add to 0, as the corners of a regular polygon balance about its centre: $(a - p) + (b - p) + (c - p) = 0$, which makes $p = (a + b + c)/3$. To see where it lies, start at the corner a and go two thirds of the way to the midpoint $(b + c)/2$ of the opposite side. The point reached is $a + (2/3)((b + c)/2 - a)$, which multiplies out to $(a + b + c)/3$. That expression treats the three corners alike, so starting from b or from c instead reaches the same point. So the three lines from corners to opposite midpoints pass through one point, two thirds of the way along each. The turn about a point p through an angle whose point on the unit circle is u sends z to $p + u(z - p)$: shift p to the origin, turn, shift back.

When is a triangle with corners x, y, z , named anticlockwise,

equilateral? When the side from x to z is the side from x to y turned through a sixth of a turn: $z - x = \rho(y - x)$, where ρ is the point of the unit circle at 60° , which is $1/2 + (\sqrt{3}/2)i$. Collect the terms on one side: $(\rho - 1)x - \rho y + z = 0$. Two facts about ρ tidy this up. First, $\rho^2 = \rho - 1$, as multiplying out confirms, and ρ^2 is the point ω at a third of a turn, so the multiplier of x is ω . Second, ρ at 60° lies directly opposite ω^2 at 240° , so $-\rho$ is ω^2 . The condition now reads $\omega x + \omega^2 y + z = 0$. Multiply it through by ω^2 , and use $\omega^3 = 1$ and $\omega^4 = \omega$: the condition becomes

$$x + \omega y + \omega^2 z = 0$$

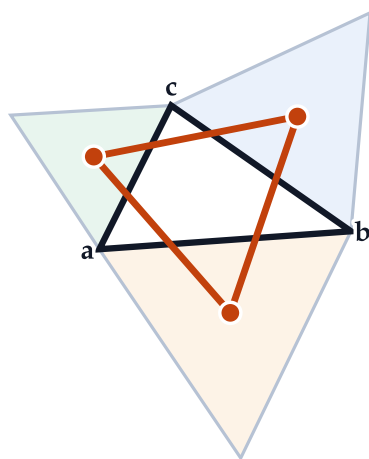
a single equation, in which the three corners appear with the three cube roots of unity as coefficients.

On each side of any triangle, build an equilateral triangle outwards. The centres of the three new triangles are the corners of an equilateral triangle, whatever the shape of the first. The statement carries Napoleon's name, on no good evidence that he found it. Let the triangle have corners a , b and c , anticlockwise. The centre of the equilateral triangle built outwards on the side from a to b lies on the perpendicular through the midpoint of that side, at a distance of the side's length divided by $2\sqrt{3}$, since the height of an equilateral triangle is $\sqrt{3}/2$ of its side and the centre lies a third of the way up. The corners run anticlockwise, so on the way from a to b the triangle lies to the left, and away from the triangle is to the right. Turning $b - a$ through a right angle to the right is a quarter turn clockwise, which is multiplying it by $-i$. The result $-i(b - a)$ is as long as the side, so dividing it by $2\sqrt{3}$ gives the step from the midpoint to the centre, and the other two sides go the same way. So the three centres are

$$(a + b)/2 - i(b - a)/(2\sqrt{3}), (b + c)/2 - i(c - b)/(2\sqrt{3}), \\ (c + a)/2 - i(a - c)/(2\sqrt{3})$$

Put them into $x + \omega y + \omega^2 z$. The result is a multiple of a , plus a multiple of b , plus a multiple of c , and the three multipliers can be computed separately. Write λ for $1/2 + i/(2\sqrt{3})$ and λ' for $1/2$

$-i/(2\sqrt{3})$. Collecting the a and the b in the first centre turns it into $\lambda a + \lambda' b$, and in the same way the other two are $\lambda b + \lambda' c$ and $\lambda c + \lambda' a$. So $x + \omega y + \omega^2 z$ is $(\lambda a + \lambda' b) + \omega(\lambda b + \lambda' c) + \omega^2(\lambda c + \lambda' a)$. The multiplier of a is $\lambda + \omega^2 \lambda'$, that of b is $\lambda' + \omega \lambda$, and that of c is $\omega \lambda' + \omega^2 \lambda$. Now λ is the point at distance $1/\sqrt{3}$ and angle 30° , and λ' the point at the same distance and angle -30° . Multiplying by ω or ω^2 turns a point without changing its distance. So $\omega^2 \lambda'$ lies at angle $240^\circ - 30^\circ = 210^\circ$, directly opposite λ and at the same distance from the origin. Two such points add to 0, and the first multiplier is 0. In the second, $\omega \lambda$ lies at 150° , opposite λ' at -30° . In the third, $\omega \lambda'$ lies at 90° and $\omega^2 \lambda$ at 270° . All three multipliers vanish, so the equation holds for every a, b and c , and the centres form an equilateral triangle.



sides of the triangle of centres:
3.4644, 3.4644, 3.4644

Figure 17.3. An irregular triangle with an equilateral triangle built outwards on each side. The centres of the three are joined, and the triangle they make has three equal sides, whatever triangle is started from. *Assumes:* one triangle drawn and measured; the proof is the computation in the text, which never refers to a particular triangle

Every map $z \mapsto uz + w$ with u on the unit circle is a slide or a turn, and every slide and every turn is such a map. A slide has $u = 1$. A turn about p has $w = p - up$. Conversely, suppose u is not

1. A point p is left in place by $z \mapsto uz + w$ when $p = up + w$, and exactly one point satisfies that, namely $w/(1 - u)$. For that p the equation says $w = p - up$, which is the w of the turn about p . So the map is the turn about that point.

Performing $z \mapsto uz + w$ and then $z \mapsto u'z + w'$ gives $z \mapsto u'u z + (u'w + w')$, a map of the same kind. The map with $u = 1$ and $w = 0$ does nothing, and $z \mapsto uz + w$ is undone by $z \mapsto (z - w)/u$, of the same kind again, since $1/u$ lies on the unit circle. So these maps form a group.

When a turn through α about one point is followed by a turn through β about another, the combined map has the multiplier $u'u$, which is the point of the unit circle at angle $\alpha + \beta$. If that point is not 1, the combined map is a turn through $\alpha + \beta$ about some third point. If $\alpha + \beta$ is a whole number of turns, the multiplier is 1 and the combined map is a slide. The order matters, since $u'w + w'$ and $uw' + w$ need not agree: a quarter turn about the origin followed by the slide by 1 is $z \mapsto iz + 1$, while the slide followed by the quarter turn is $z \mapsto iz + i$. This group, like the group of the square, does not commute.

* * *

For any four points a, b, c and d , multiplying out shows that

$$(a - b)(c - d) + (a - d)(b - c) = (a - c)(b - d)$$

The size of a sum of two numbers of the plane is at most the sum of their sizes. The sum is reached from 0 by going out to the first number and then along the slide by the second, so the two sizes and the size of the sum are the three sides of a triangle, and any two sides of a triangle are together longer than the third, which was among the beliefs about the page the opening pages listed. Now take sizes on both sides of the identity. Sizes multiply, and the size of $a - b$ is the distance from a to b . So the right side has size the distance from a to c times the distance from b to d , and that is at most the sum of the sizes of the two products on the left. So for

any four points, the product of the two diagonals of the four-sided figure with corners a, b, c, d is at most the sum of the products of the two pairs of opposite sides. Ptolemy, in the second century, proved that for four points taken in order on a circle the two are equal, and built his table of chords on it, which was the table of sines of antiquity. For a rectangle, with sides p and q and both diagonals equal to r , his equality reads $r^2 = p^2 + q^2$.

What is $\cos 6\theta$ as a polynomial in $\cos \theta$? And in the polynomial for $\cos n\theta$, what are the leading coefficient and the constant term, for every n ?

The regular polygon of seven sides has corners $1, \zeta, \dots, \zeta^6$. What equation with whole-number coefficients does twice the cosine of a seventh of a turn satisfy?

Squares are built outwards on the four sides of any parallelogram. What figure do their centres form?

What is the sum of the squares of the distances from one corner of a regular polygon of n sides, inscribed in a circle of radius 1, to all the other corners?

A MACHINE RUN BACKWARDS

Two lengths were compared by taking the smaller from the larger as many times as it would go, then taking the remainder from the smaller, and so on, and for the side and the diagonal of a square the process never stopped. On two whole numbers it must stop, since the remainders are whole numbers and they decrease. Run on whole numbers, the process is called Euclid's algorithm. What does it compute?

Dividing 252 by 198 leaves 54. Dividing 198 by 54 gives three of them and leaves 36. Dividing 54 by 36 leaves 18, and dividing 36 by 18 leaves nothing. The last remainder that is not 0 is 18. On 667 and 551 the remainders are 116, 87 and 29, and 87 is exactly three 29s, so the process stops at 29.

Each line uses division with remainder, which deserves a proof. For whole numbers a and b with b positive, a can be written in exactly one way as some number of b s plus a remainder smaller than b ; fifty is seven sevens and one over. The pair exists by descent. Subtract b from a again and again, for as long as what is left is at least b . Each subtraction lowers what is left, and a decreasing run of whole numbers must stop, so at some point what is left is below b . That last value is the remainder, and the number of subtractions is the quotient. The pair is unique. Suppose a is some number of b s plus r , and also some number of b s plus s , with r and s both below

b and s at least r . Taking the first expression from the second shows that $s - r$ is a multiple of b . But $s - r$ is smaller than b , and the only multiple of b smaller than b is 0. So $s = r$, and then the two counts of bs agree as well.

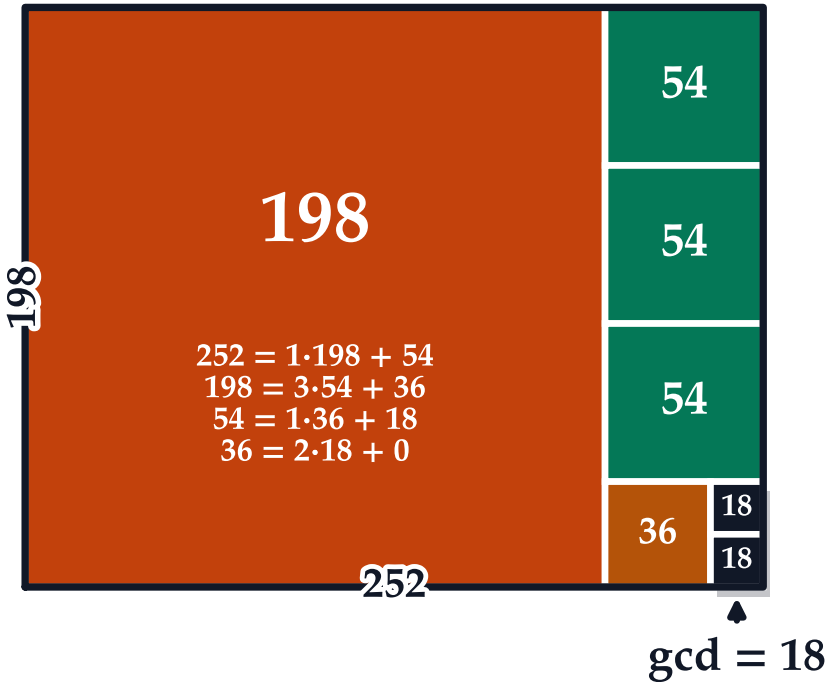


Figure 18.1. A 252 by 198 rectangle tiled by the largest square that fits at each stage: one of side 198, three of side 54, one of side 36 and two of side 18, so Euclid's algorithm stops at 18 and the side of the last square is the greatest common divisor. *Assumes:* that the leftover rectangle at each stage has exactly the same common divisors as the one before, which is the invariant proved in the text and is the only reason the last square answers the original question

At each line the pair (a, b) is replaced by (b, r) , where r is the remainder, and the two pairs have the same common divisors: whatever divides a and b divides r , which is a minus a multiple of b , and whatever divides b and r divides a , which is a multiple of b plus r . The common divisors do not change while the numbers

shrink. The process stops at a pair $(g, 0)$, and since every number divides 0, the common divisors of g and 0 are exactly the divisors of g . So the common divisors of the starting numbers are exactly the divisors of g . One of those is g itself, so g divides both starting numbers, and everything that divides both starting numbers divides g . That is more than the name *greatest common divisor* says: g is not only the largest of the common divisors but the one they all divide. For 667 and 551 the answer can be checked from the other side, since 667 is twenty-three 29s and 551 is nineteen 29s, and 23 and 19 have no common divisor but 1.

Which pair of whole numbers below 100, the larger divided by the smaller first, makes the process run through the most divisions?

* * *

Run the process backwards. Each remainder was defined by one line of arithmetic, so it can be written in terms of the two numbers above it, and substituting upward writes the last remainder in terms of the starting numbers. For 252 and 198: 18 is $54 - 36$; since $36 = 198 - 3 \cdot 54$, this is $4 \cdot 54 - 198$; and since $54 = 252 - 198$, it is $4 \cdot 252 - 5 \cdot 198$. Indeed $4 \cdot 252 = 1008$ and $5 \cdot 198 = 990$. For 667 and 551 the same substitution gives $29 = 5 \cdot 667 - 6 \cdot 551$, and $3335 - 3306 = 29$.

That always works. Call the two starting numbers the first two remainders. Each later remainder is the one two lines above it minus a multiple of the one a line above. Call a number a combination of the starting numbers when it is an integer multiple of one plus an integer multiple of the other, negative multiples allowed, as $18 = 4 \cdot 252 - 5 \cdot 198$ is. If the two remainders above some line are such combinations, so is the remainder on that line, because taking a multiple of one combination from another leaves a combination. And the first two are such combinations, being the starting numbers themselves. The claim passes from each pair of neighbouring remainders to the next, so it holds on every line, and in particular

the greatest common divisor of two numbers is a combination of them with integer coefficients.

That turns the greatest common divisor into an instrument. Suppose a prime p divides a product ab but does not divide a . The only divisors of p are 1 and p . The greatest common divisor of p and a divides p , so it is 1 or p , and it is not p , since p does not divide a . So it is 1, and by the result above $1 = px + ay$ for some integers x and y , one of which may be negative. Multiply by b : $b = pbx + aby$. The first term is a multiple of p , and so is the second, since p divides ab . So p divides b :

a prime that divides a product of two numbers divides one of them.

Look at what the proof used. Nothing about 1 being small came into it. What it used is that 1 had been built as a combination of p and a , which Euclid's algorithm supplied.

* * *

Every whole number above 1 is a product of primes, in only one way apart from order. The two halves of that statement are of different kinds.

That a factorisation exists is a descent. A number above 1 is either prime, and is its own factorisation, or a product of two smaller numbers above 1. If some number had no factorisation, there would be a smallest such number; it is not prime, so it is a product of two smaller numbers, each of which has a factorisation, and putting those together factorises it. So there is no such number. Here the argument reaches a number from all the numbers below it, not only from the one immediately below. Arguing from a smallest failure allows that, because every number below the smallest failure has a factorisation.

Uniqueness needs the lemma about primes. Suppose a number has two factorisations into primes. The first prime of the first list divides the number, and the number is the product of the second

list. The lemma speaks of a product of two numbers, so apply it again and again: the prime divides either the first prime of the second list or the product of the rest, and in the second case either the next prime or the product of what remains, and so on down the list. So it divides one of the primes of the second list, and a prime that divides a prime equals it, since the only divisors of a prime are 1 and itself. Remove the matched pair from both lists and repeat. After each removal the two remaining products are still equal. So neither list runs out before the other: if one did, its product would be 1, and the primes left over on the other side would multiply to 1, which primes cannot do. So the two lists are the same primes in some order. For 60, the factorisation through $6 \cdot 10$ and the one through $4 \cdot 15$ both come to two 2s, a 3 and a 5.

Unique factorisation reproves that no fraction has square 2. In $a^2 = 2b^2$ with positive whole numbers a and b , count the 2s in the factorisation of each side: the left side has an even number of them, twice the number in a , and the right side an odd number, one more than twice the number in b . A number cannot have two factorisations with different numbers of 2s.

A descent proves it a third way. A search for positive whole numbers x and y with $x^2 = 2y^2$ finds none with both up to 20. Suppose solutions exist, and take one with the smallest x . Then x^2 is even, so x is even, say $x = 2u$, and $4u^2 = 2y^2$ gives $y^2 = 2u^2$: so y and u are a solution, and y is smaller than x , since y^2 is half of x^2 . A solution smaller than the smallest cannot exist, so there is none.

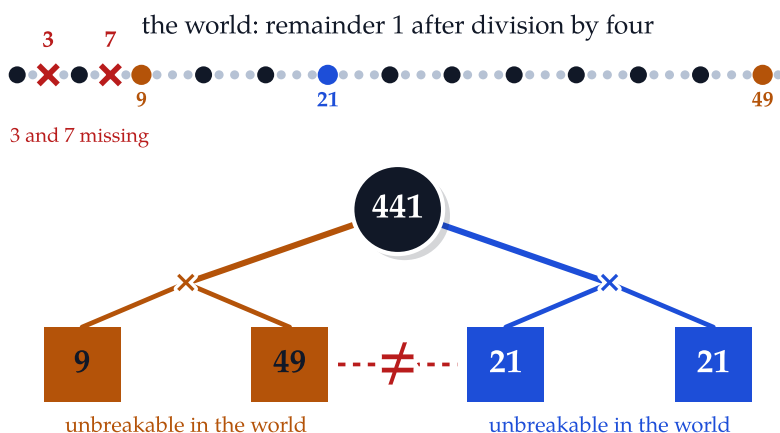
Near misses keep coming, though: 3 and 2 give 9 against 8, and 7 and 5 give 49 against 50. One step makes each of them from the one before. From a pair (x, y) pass to $(x + 2y, x + y)$. Since $(x + 2y)^2 - 2(x + y)^2 = -(x^2 - 2y^2)$, the step keeps the size of $x^2 - 2y^2$ and changes its sign. From $(1, 1)$, where $x^2 - 2y^2$ is -1 , the step gives $(3, 2)$, $(7, 5)$, $(17, 12)$, $(41, 29)$, $(99, 70)$, $(239, 169)$ and $(577, 408)$, with $x^2 - 2y^2$ alternately 1 and -1 . Undone, the step sends (x, y) to $(2y - x, x - y)$, and that is the cutting of the square written in whole numbers: a square of side y and diagonal x would have a corner square of side $x - y$ and diagonal $2y - x$. No square

has whole numbers for its side and diagonal, and these pairs come as near as whole numbers can. Are these all the pairs of positive whole numbers with $x^2 - 2y^2$ equal to 1 or -1 ?

* * *

Unique factorisation can fail, and a small world shows it. Take only the numbers that leave remainder 1 on division by 4: 1, 5, 9, 13, 17, 21, The product of two of them again leaves remainder 1, since $(4m + 1)(4n + 1) = 4(4mn + m + n) + 1$, so the world is closed under multiplication: a product of members is a member. Call a member above 1 *irreducible* in the world when it is not a product of two smaller members above 1. The number 9 is irreducible in the world, since its only splitting is $3 \cdot 3$, and 3 is not a member; so are 21, which splits only as $3 \cdot 7$, and 49, which splits only as $7 \cdot 7$. And $441 = 9 \cdot 49 = 21 \cdot 21$: one member with two factorisations into irreducibles that cannot be matched, since 9, 21 and 49 are different. A member can even have three: $4389 = 21 \cdot 209 = 33 \cdot 133 = 57 \cdot 77$. All six factors are irreducible in the world: each is a product of two primes that leave remainder 3, as $21 = 3 \cdot 7$ and $209 = 11 \cdot 19$ are, so its only splitting is into those two primes, and neither of them is a member.

Walking through the proof of uniqueness in this world shows where it dies. The existence argument survives, and so does the pairing off. What fails is the lemma: 9 is irreducible and divides $21 \cdot 21$ without dividing 21. The lemma's proof built 1 as a combination with integer coefficients by Euclid's algorithm, which subtracts, and the world is not closed under subtraction, since $5 - 1 = 4$ is not a member, so the proof cannot begin there. It dies even earlier, at "the only divisors of p are 1 and p ". Among the members that sentence is true of 9, and 9 and 21 have no common divisor above 1. But Euclid's algorithm works among all the integers, and there 9 has the divisor 3 as well. Run on 21 and 9, the algorithm leaves $21 - 2 \cdot 9 = 3$, so the greatest common divisor it computes is 3, which is not a member, and the combination it



no piece on the left equals a piece on the right, so nothing pairs off

Figure 18.2. In the world of numbers leaving remainder one on division by four, 441 splits two genuinely different ways into pieces that cannot be split further, so uniqueness of factorisation is a theorem and not a law of nature. Assumes: that irreducibility is judged inside the world, not among all whole numbers: 9, 21 and 49 all factor in the integers, and it is the missing members 3 and 7 that make them unbreakable here

builds is 3, not 1. The number 9 is irreducible among the members and is not prime among the integers. Among the whole numbers the proof used two things at that step, so constantly available that they were never stated: the common divisors Euclid's algorithm finds lie in the same world as the numbers, 1 among them, and the world allows subtraction.

Which members of that world below 1000 have more than one factorisation into irreducibles?

Closing the world under subtraction is not enough. The positive even numbers are closed under multiplying and under taking a smaller from a larger. An even number is irreducible among them when it is not a product of two even numbers. A product of two even numbers is a multiple of 4, and a multiple of 4 is 2 times an even number, so the irreducible ones are exactly those that leave remainder 2 on division by 4. And $36 = 2 \cdot 18 = 6 \cdot 6$, where 2,

18 and 6 are all irreducible. The lemma about primes fails here as well. Among the even numbers 6 divides $2 \cdot 18$, since $2 \cdot 18 = 6 \cdot 6$ and the other factor, 6, is even. But 6 divides neither 2 nor 18 there: 18 is 6 times 3, and 3 is not even. Subtraction is there, and 1 is not.

A world with 1 and with subtraction can still fail. Take the numbers $a + bw$ with integers a and b , where w is a new object with $w^2 = -5$, added and multiplied as the pairs with $i^2 = -1$ were. Call $a^2 + 5b^2$ the size of $a + bw$. Sizes multiply,

$$(a^2 + 5b^2)(c^2 + 5d^2) = (ac - 5bd)^2 + 5(ad + bc)^2$$

as the distances did for the pairs with $i^2 = -1$, so the size of a divisor divides the size of the number it divides. No size is 2 or 3, since $a^2 + 5b^2$ is a square when b is 0 and at least 5 otherwise. Now $6 = 2 \cdot 3$, and also $6 = (1 + w)(1 - w)$, since $1 - w^2 = 6$. The sizes of 2, 3, $1 + w$ and $1 - w$ are 4, 9, 6 and 6. Split any one of the four into two factors. The sizes of the two factors are whole numbers that multiply to 4, 9 or 6. The ways to do that are $1 \cdot 4$ and $2 \cdot 2$, then $1 \cdot 9$ and $3 \cdot 3$, then $1 \cdot 6$ and $2 \cdot 3$. Since 2 and 3 are not sizes, only 1 with 4, 1 with 9 and 1 with 6 remain. So one factor has size 1, which forces $b = 0$ and $a^2 = 1$, making that factor 1 or -1 . None of the four splits into two factors that are more than a sign, so all four are irreducible. And the two factorisations differ by more than signs, since 2 is not $1 + w$ or $1 - w$ or the negative of either. These are two factorisations that cannot be matched, in a world that has 1, has negatives, and is closed under adding, subtracting and multiplying.

Walk through the proof again. The number 2 divides the product $(1 + w)(1 - w)$ and divides neither factor, since $1 + w = 2(c + dw)$ would need $2c = 1$. The common divisors of 2 and $1 + w$ are 1 and -1 only: the size of a common divisor has to divide both 4 and 6, so it is 1 or 2, and 2 is not a size, and the only numbers of size 1 are 1 and -1 . But 1 is not a combination of 2 and $1 + w$: multiplying out $2(x + yw) + (1 + w)(z + tw)$ gives $(2x + z - 5t) + (2y + z + t)w$, whose two coordinates differ by the even number $2x - 2y - 6t$, while those of 1 differ by 1. Euclid's algorithm does not even start, because its first line fails. Dividing $1 + w$ by 2 would

mean choosing a quotient $c + dw$ so that the remainder $(1 + w) - 2(c + dw)$ has a smaller size than the divisor 2, whose size is 4. But that remainder has size $(1 - 2c)^2 + 5(1 - 2d)^2$, and since $1 - 2c$ and $1 - 2d$ are odd, each square is at least 1 and the size is at least 6, never below 4. Uniqueness rests on division with a remainder smaller than the divisor, and none of these three worlds has it.

The failure made history. On 1 March 1847 Lamé announced to the Paris Academy a proof of Fermat's claim that $x^n + y^n = z^n$ has no solution in positive whole numbers when n is above 2. His proof factorised in a world of numbers built from an n -th root of unity, and assumed that factorisation there is unique. Liouville questioned the assumption at the same session, and in May a letter from Kummer was read: he had shown three years earlier that it fails.

* * *

The expression $n^2 + n + 41$ is prime for $n = 0, 1, 2$, and for every n up to 39, forty values in a row. At $n = 40$ it is 1681, which is $41 \cdot 41$. The forty confirmations were worth little. The weight of a confirmation is the ratio of its chances under the two rival hypotheses. Under the rival "prime for a while and then not", each prime value was nearly as likely as under "prime always". So the ratio was close to 1, and each confirmation shifted the odds by almost nothing.

The failure could have been predicted. If a polynomial f with integer coefficients takes a value divisible by p at an input a , then its value at $a + p$ is divisible by p too. The reason is that $f(a + p)$ is $f(a)$ plus the difference $f(a + p) - f(a)$. That difference is a sum of terms, each an integer coefficient times some $(a + p)^k - a^k$, and each $(a + p)^k - a^k$ is divisible by $(a + p) - a$, which is p . That rests on $x^k - y^k$ being divisible by $x - y$, which holds for $k = 1$ and passes from k to $k + 1$, since $x^{k+1} - y^{k+1} = x(x^k - y^k) + y^k(x - y)$. For $n^2 + n + 41$ at $a = 0$ the value is 41, so the value at 41 is a multiple of 41, and indeed $41^2 + 41 + 41 = 1763 = 41 \cdot 43$; a composite

value was certain to appear by $n = 41$, and the first one comes one step earlier, at $n = 40$.

No polynomial with integer coefficients, other than a constant, is prime at every whole-number input. One more fact is needed first, and it is already in hand: a polynomial of degree d at least 1 has at most d roots, since each root gives up a factor and lowers the degree by one, so it is 0 at only finitely many integers. Now suppose f is not constant and is prime at every whole number, and let $p = f(0)$. Since $f(0)$ is divisible by p , so is $f(p)$, by the fact about a and $a + p$, and then $f(2p)$, and so on: for every whole number k , $f(kp)$ is divisible by p , and it is prime, so it equals p . Then $f - p$, which is not constant, is 0 at the infinitely many inputs $0, p, 2p, \dots$, which is impossible.

THE SUBTRACTION WRITTEN DOWN

Euclid's algorithm was run for the sake of its last line, the greatest common divisor, and the numbers it passed on the way were thrown away. Keep them. Dividing 252 by 198 goes once and leaves 54; dividing 198 by 54 goes 3 times and leaves 36; 54 by 36 goes once and leaves 18; and 36 by 18 goes twice and leaves nothing. The quotients are 1, 3, 1 and 2. What do they know?

Each line can be read as a statement about a fraction. The first says that $252/198$ is 1 and $54/198$ more, and $54/198$ is 1 divided by $198/54$. The second says that $198/54$ is 3 and $36/54$ more, and so on down. Put together,

$$252/198 = 1 + 1/(3 + 1/(1 + 1/2))$$

an expression in which the four quotients appear and nothing else does. It is called a *continued fraction*, and it is written $[1; 3, 1, 2]$ for short. The common factor 18 has disappeared, since the quotients are the same for 252 and 198 as for 14 and 11. Every fraction has such an expansion, because the algorithm always ends, and the expansion is the record of how the two numbers measure each other: the larger holds the smaller once, the remainder goes into the smaller 3 times, the new remainder goes once into the old one,

and the last goes twice into the one before and finishes. The subtraction of lengths that would not end on the diagonal of the square was keeping such a record too, and nobody wrote it down.

* * *

Cut the expansion off early and work out what is left. $[1]$ is 1; $[1; 3]$ is $1 + 1/3 = 4/3$; $[1; 3, 1]$ is $1 + 1/(3 + 1) = 5/4$; and the whole of it is $14/11$. These are the *convergents* of $14/11$:

$$1, 4/3, 5/4, 14/11$$

As decimals they are 1, 1.3333, 1.25 and 1.2727: the first three fall alternately below and above the target, closing in on it, and the last is the target itself. Working each one out from the bottom of its expression upwards is laborious, and there is a rule that produces them from the top. Write the quotients as $a_0, a_1, a_2, \dots$ and the convergents as $p_0/q_0, p_1/q_1, \dots$. Then

$$p_k = a_k \cdot p_{k-1} + p_{k-2}, \quad q_k = a_k \cdot q_{k-1} + q_{k-2}$$

each new numerator being the new quotient times the last numerator, plus the one before, and the same for denominators. Starting from $p_0/q_0 = 1/1$ and $p_1/q_1 = 4/3$, the quotient 1 gives $(1 \cdot 4 + 1)/(1 \cdot 3 + 1) = 5/4$, and the quotient 2 gives $(2 \cdot 5 + 4)/(2 \cdot 4 + 3) = 14/11$.

The rule holds for every expansion, and the proof is a climb. For the length of the proof the quotients are allowed to be any positive numbers and not only whole ones, because the climb will put a quotient that is not whole in the last place. Nothing is lost by this: the rule is a statement about the expression and not about whole numbers. It can be checked by hand for the first three convergents. Suppose it holds for all expansions with k quotients after the first. An expansion with one more quotient, ending $\dots, a_k, a_{k+1}$, has the same value as the shorter expansion ending $\dots, a_k + 1/a_{k+1}$, since that is what the last two layers of the expression say. Apply the rule to the shorter one, whose earlier convergents are unchanged:

$$((a_k + 1/a_{k+1}) \cdot p_{k-1} + p_{k-2}) / ((a_k + 1/a_{k+1}) \cdot q_{k-1} + q_{k-2})$$

Multiply top and bottom by a_{k+1} . The top becomes $a_{k+1}(a_k p_{k-1} + p_{k-2}) + p_{k-1}$. The bracket is p_k , by the rule as supposed, so the top is $a_{k+1} p_k + p_{k-1}$. The bottom goes the same way and is $a_{k+1} q_k + q_{k-1}$. That is the rule at the next step.

From the rule comes an identity that carries everything else. Work out $p_k q_{k-1} - p_{k-1} q_k$ using the rule for p_k and q_k : the terms with a_k cancel, and what is left is $-(p_{k-1} q_{k-2} - p_{k-2} q_{k-1})$, the same expression one step earlier with its sign changed. The first two convergents are $a_0/1$ and $a_0 + 1/a_1$, which is $(a_0 a_1 + 1)/a_1$, so for them the expression is $(a_0 a_1 + 1) \cdot 1 - a_0 \cdot a_1$, which is 1. So

$$p_k \cdot q_{k-1} - p_{k-1} \cdot q_k = 1 \text{ or } -1, \text{ alternately}$$

For 14/11 the three instances are $4 \cdot 1 - 1 \cdot 3 = 1$, then $5 \cdot 3 - 4 \cdot 4 = -1$, then $14 \cdot 4 - 5 \cdot 11 = 1$.

The last instance, $4 \cdot 14 - 5 \cdot 11 = 1$, multiplied through by 18 says $4 \cdot 252 - 5 \cdot 198 = 18$: the greatest common divisor written as a combination of the two numbers, which was obtained before by running the algorithm backwards, substituting each line into the one above. The backward run was the rule for convergents in disguise, and the coefficients it found, 4 and 5, are the denominator and the numerator of the last convergent but one, $5/4$. To find a multiple of 24 that is 1 more than a multiple of 67, expand $67/24$ as $[2; 1, 3, 1, 4]$, with convergents 2, 3, $11/4$, $14/5$ and $67/24$. The identity gives $67 \cdot 5 - 14 \cdot 24 = -1$, and that is the answer: $14 \cdot 24$ is 336, and $5 \cdot 67$ is 335.

Divide the identity by $q_k q_{k-1}$:

$$p_k/q_k - p_{k-1}/q_{k-1} = \pm 1/(q_k \cdot q_{k-1})$$

so successive convergents differ by exactly one over the product of their denominators. That is the smallest amount by which two fractions with those denominators can differ at all, since the difference of any two such fractions is a whole number divided by that product.

How far is a convergent from the number x being expanded? Stop the expansion after a_k and let t stand for the value of everything that remains, so that x is $[a_0; a_1, \dots, a_k, t]$. This t is more than 1. It begins with the quotient a_{k+1} , which is at least 1. If more quotients follow, they add something to that, and t is more than 1. If a_{k+1} is the last quotient of the algorithm, then t is a_{k+1} itself, and a last quotient is at least 2: the last division leaves nothing, and it divides a remainder by the next one, which is smaller, so the smaller goes in at least twice. The rule, which allowed a last entry that is not whole, gives $x = (t p_k + p_{k-1}) / (t q_k + q_{k-1})$, and subtracting p_k/q_k and using the identity,

$$x - p_k/q_k = \pm 1 / (q_k \cdot (t q_k + q_{k-1}))$$

with the sign alternating from each k to the next. Three facts can be read off. The convergents lie alternately below and above x . Since t is at least a_{k+1} , the bracket is at least $a_{k+1} q_k + q_{k-1}$, which by the rule is q_{k+1} . So a convergent is within $1/(q_k q_{k+1})$ of x , and a large quotient coming next means a very close convergent now. And each convergent is closer than the one before: the same subtraction for p_{k-1}/q_{k-1} gives $t / (q_{k-1}(t q_k + q_{k-1}))$, which is larger than the distance of p_k/q_k by the factor $t q_k/q_{k-1}$. That factor is more than 1, since t is more than 1 and q_k is at least q_{k-1} : the denominators begin 1 and a_1 , and from there the rule only adds to them.

Now let a/b be any fraction, and suppose it lies strictly between two successive convergents p_{k-1}/q_{k-1} and p_k/q_k . Its distance from the first is a whole number, not 0, divided by $b q_{k-1}$, so it is at least $1/(b q_{k-1})$; and it is less than the whole gap, $1/(q_k q_{k-1})$. So $1/(b q_{k-1})$ is less than $1/(q_k q_{k-1})$, which makes $b q_{k-1}$ more than $q_k q_{k-1}$, and b exceeds q_k : a fraction that gets between two successive convergents has a larger denominator than either. From that, for every convergent after the first,

a fraction closer to x than the convergent p_k/q_k has a denominator larger than q_k .

Suppose first that it lies on the same side of x as p_k/q_k , or is x itself. The next convergent is x or lies on the other side of x , so x lies between this convergent and the next, and the closer fraction lies there with it. So it is either the next convergent, or strictly between the next convergent and this one. Either way its denominator is at least q_{k+1} , which exceeds q_k . If it lies on the other side, the side of p_{k-1}/q_{k-1} , then it is closer to x than p_k/q_k is, and p_k/q_k is closer to x than p_{k-1}/q_{k-1} is, by the third of the facts above. So it is closer to x than p_{k-1}/q_{k-1} is. It therefore lies between p_{k-1}/q_{k-1} and x , and so strictly between p_{k-1}/q_{k-1} and p_k/q_k , and its denominator exceeds q_k . The first convergent has no predecessor for the second case to use, and the claim can fail for it: for $[1; 1, 1, \dots]$ the whole number 2 is closer than the first convergent, 1. The convergents after the first are, in an exact sense, the best approximations that small denominators can buy.

* * *

Nothing in the procedure requires the two magnitudes to be whole numbers. Lay the shorter of two lengths along the longer as often as it will go, then the remainder along the shorter, and so on, and record how many times each goes. If the lengths have a common measure, the process ends and the record is the expansion of a fraction. If they have none, the process never ends, and the record is an endless list of whole numbers.

For the diagonal and the side of a square the list can be read off the cutting. The side, s , goes once into the diagonal and leaves a remainder e . The small square in the corner has e for its side and $s - e$ for its diagonal; so e goes into s once with $s - e$ left, and into $s - e$, its own diagonal, once more with a remainder e' , the first remainder of the small square. So e goes twice into s . After that the small square repeats what the large one did: e' goes twice into e , and so on forever. The record for the diagonal and the side is

$[1; 2, 2, 2, 2, \dots]$

and its convergents, by the rule, are $1, 3/2, 7/5, 17/12, 41/29, 99/70, 239/169, 577/408, \dots$. These are the near misses of the descent, the pairs with $x^2 - 2y^2$ equal to 1 or -1 , which came from the step from (x, y) to $(x + 2y, x + y)$. The step and the rule agree. Apply the step to the pair $(x + 2y, x + y)$: the new first member is $(x + 2y) + 2(x + y)$, which is $2(x + 2y) + x$. That is twice the first member that came before it, plus the first member before that, and that is the rule with the quotient 2. The new second member, $(x + 2y) + (x + y)$, is $2(x + y) + y$ in the same way, and both lists begin with 1/1 and 3/2. So the fact about convergents applies to these pairs. The fraction 99/70, which is 1.414286 against the diagonal's 1.414214, is closer to the diagonal than any fraction with a denominator of 70 or less. And any fraction lying between it and the diagonal lies strictly between it and the next convergent, 239/169, so it has a denominator above 169.

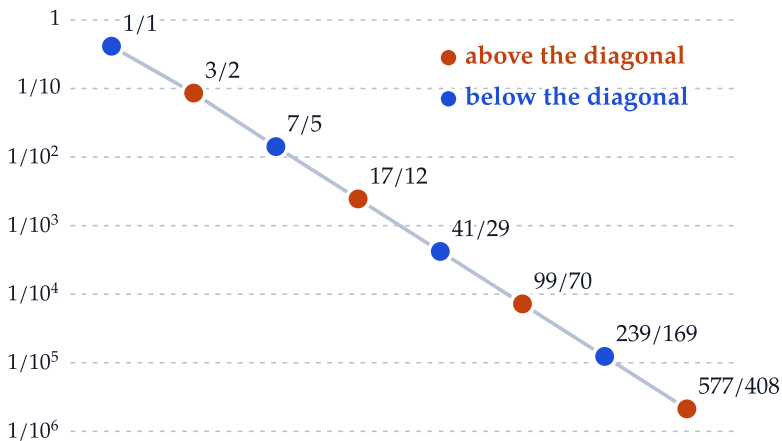


Figure 19.1. The distance of each convergent of $[1; 2, 2, 2, \dots]$ from the diagonal of the unit square, on a scale where each line down is ten times closer. They fall alternately above and below, and each is closer than the last by a factor that settles near 5.83. Assumes: the diagonal taken as 1.41421356237, which is accurate far beyond the last point drawn

For the diagonal and the side of a regular pentagon the cutting

reproduced itself after every single cut, so the record is

$$[1; 1, 1, 1, 1, \dots]$$

the simplest endless expansion there can be. Its convergents, by the rule with every quotient 1, are $1, 2, 3/2, 5/3, 8/5, 13/8, 21/13, \dots$, each numerator being the sum of the two before it: $1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, \dots$. These are the numbers that Leonardo of Pisa, called Fibonacci, met in 1202 in a problem about breeding rabbits.

An endless record is a complete description of the proportion of the two lengths, in the sense that was given to proportion when the diagonal first refused a number. The convergents lie alternately below and above, and any two successive ones differ by $1/(q_k q_{k-1})$, which falls below every fraction however small, the denominators increasing without end. If two pairs of lengths have the same record, they have the same convergents, so both proportions lie in every one of those closing gaps. Suppose some whole-number comparison came out differently for the two pairs. It would put a fraction strictly between the two proportions, and with it a definite distance between them. But the gaps fall below every distance, so some gap is smaller than that one, and a gap smaller than the distance between two proportions cannot contain both. So no comparison comes out differently. So the list of quotients names the proportion. It does what the decimal $1.41421\dots$ does, without the arbitrary choice of ten, and it shows at a glance what a decimal hides: a record that ends is a fraction, a record that never ends is not, and the diagonal of the square, whose decimal digits show no pattern at all, has the most regular record imaginable.

* * *

The pentagon's record explains an old observation about the algorithm. Which pairs of numbers make it run longest? Below 100 the pair is 89 and 55, with nine divisions, and they are neighbours in the list $1, 1, 2, 3, 5, 8, \dots$. On two neighbours of that list every quotient is 1 until the last. Each number in the list is the sum of the

two before it, so the number before it goes into it once and leaves the number before that: 89 is 55 and 34 more, 55 is 34 and 21 more. The algorithm walks back down the list one step at a time. No quotient can be smaller than 1, so no pair can make the remainders shrink more slowly.

Precisely: if the algorithm takes n divisions on a pair with smaller member b , then b is at least the number in place $n + 1$ of the list; nine divisions need a smaller number of at least 55, the tenth. Run the algorithm backwards from its end: the last remainder that is not 0 is at least 1, the one before is at least 2, being a multiple of it and larger, and each earlier one is at least the sum of the two that follow, since it holds the next at least once with the one after left over. Those are the list's own rules, starting from its second and third places: the last remainder that is not 0 is at least the number in place 2, the one before it at least the number in place 3, and so on back. With n divisions the smaller member b stands $n - 1$ steps before that last remainder, so b is at least the number in place $n + 1$. Lamé proved this in 1844. Reynaud in 1811 and Finck in 1841 had already put cruder limits on the number of divisions; these are among the earliest measurements of how long a procedure takes.

The list grows fast. Write F_n for the number in place n . From the second place on, five steps along the list multiply a number by more than ten, since $F_{n+5} = 8F_n + 5F_{n-1}$, which is above $10F_n$ because $2F_{n-1}$ is at least F_n , so that $4F_{n-1}$ is at least $2F_n$ and $5F_{n-1}$ is more than $2F_n$. So F_7 , which is 13, is above 10, F_{12} is above 100, and F_{5d+2} is above every number of d digits. If the smaller number b has d digits and the algorithm takes n divisions, then F_{n+1} is at most b , which is below F_{5d+2} . The list never decreases, so the place $n + 1$ comes before the place $5d + 2$, and n is at most $5d$: the number of divisions is never more than five times the number of digits of the smaller number.

* * *

The length of the year, from one spring to the next, is 365.2422 days, and a calendar has to approximate the awkward part by a fraction with a small denominator: so many leap days in so many years. The expansion of 365.2422 is $[365; 4, 7, 1, 3, \dots]$, and its convergents after 365 itself are

$$365 \frac{1}{4}, 365 \frac{7}{29}, 365 \frac{8}{33}, 365 \frac{31}{128}$$

The first is the calendar of Julius Caesar, one leap year in every four, which runs slow by a day in about 128 years; between the year 325, when the date of Easter was tied to the start of spring, and 1582, when the calendar was reformed, the spring had drifted ten days. The third, eight leap years in thirty-three, was proposed in Persia in the eleventh century, and it is off by a day in about 4500 years. The calendar in use now has 97 leap years in 400, and $97/400$ is not a convergent: it was chosen because centuries are convenient, and at 0.2425 it is a little worse than $8/33$, which is 0.24242.

A clockmaker has the same problem. In 1680 Huygens set out to build a model of the planets driven by toothed wheels, and needed two wheels whose numbers of teeth were in the proportion of Saturn's time round the sun to the earth's, which by his figures was 77708431 to 2640858. The expansion is $[29; 2, 2, 1, 5, 1, 4, \dots]$, with convergents 29, $59/2$, $147/5$, $206/7$ and $1177/40$. He cut wheels of 206 teeth and 7 teeth. The error of $206/7$ is about one part in ten thousand, and the fact about convergents says that every pair of wheels that does better has more than 7 teeth on the smaller wheel, as the next convergent does, with 40.

For the proportion of the circumference of a circle to its diameter, 3.14159265..., the expansion begins $[3; 7, 15, 1, 292, \dots]$ and the convergents are 3, $22/7$, $333/106$, $355/113$ and $103993/33102$. The second was known to Archimedes. The fourth, found in China by Zu Chongzhi in the fifth century, is correct to six decimal places, and the reason is the large quotient that follows it: a convergent is within $1/(q_k q_{k+1})$ of its target, here within $1/(113 \cdot 33102)$, which is less than 3 parts in ten million.

* * *

An expansion that repeats is the root of a quadratic equation. If x is $[1; 2, 2, 2, \dots]$, then x is 1 plus one divided by $[2; 2, 2, \dots]$. That expansion is x with its leading 1 raised to 2, so it is $1 + x$. So $x = 1 + 1/(1 + x)$, which multiplies out to $x^2 = 2$. If x is $[1; 1, 1, \dots]$, then $x = 1 + 1/x$, so $x^2 = x + 1$, the equation of the pentagon's proportion. In the same way $[2; 4, 4, 4, \dots]$ satisfies $x^2 = 5$, and $[1; 1, 2, 1, 2, \dots]$ satisfies $x^2 = 3$. The general case turns the rule for convergents on the repeating tail. Let the quotients repeat from the place after a_k onward, with a period of m quotients, and let t be the value of everything from that place on. Take k at least 1 and m at least 2, since the rule was proved for expansions with at least two quotients after the first: a repetition can always be said to start a place later, and a period of one quotient, such as the 5, 5, 5, ... below, can be counted two at a time. Then t is the number whose expansion is that period followed by t again, so the rule gives $t = (t \cdot P + P') / (t \cdot Q + Q')$, where P/Q is the last convergent of the period taken on its own and P'/Q' is the one before it, all four being whole numbers. Multiplying up,

$$Q \cdot t^2 + (Q' - P) \cdot t - P' = 0$$

and Q is at least 1, so t is the root of a quadratic equation with whole-number coefficients. The same rule gives $x = (t \cdot p_k + p_{k-1}) / (t \cdot q_k + q_{k-1})$, which solves for t as $(p_{k-1} - x \cdot q_{k-1}) / (x \cdot q_k - p_k)$; putting that into the equation for t and clearing the denominators leaves a relation in x with whole-number coefficients, of degree 2 at most. It is not $0 = 0$, with every coefficient 0: dividing that by $(x \cdot q_k - p_k)^2$ would make the equation for t hold at the value of t that comes from every x other than p_k/q_k , and different values of x give different values of t , since x is recovered from t by the rule; but an equation of degree 2 has at most two roots. It is not a constant other than 0, which nothing satisfies, and it is not of degree 1, since an expansion that repeats does not stop and so x is no fraction. So every repeating expansion is the root of a quadratic equation with whole-

number coefficients. The expansion $[1; 2, 3, 4, 5, 5, 5, \dots]$, whose tail stands in no simple relation to x , comes out by this route as a root of $199x^2 - 565x + 401$. Lagrange proved the converse in 1770: the expansion of every root of a quadratic equation with whole-number coefficients, not itself a fraction, sooner or later repeats. For cube roots nothing of the kind is known. The expansion of the cube root of 2 begins $[1; 3, 1, 5, 1, 1, 4, 1, 1, 8, \dots]$, and nobody has found a pattern in it or proved that there is none, or even settled whether its quotients stay below some bound.

What are the expansions of $13/8$ and of $21/13$, and what does the identity for convergents say about any three neighbours of the list $1, 1, 2, 3, 5, 8, 13, \dots$?

The fraction $99/70$ is closer to the diagonal of the unit square than any fraction with a denominator of 70 or less. Is it the closest of all those with a denominator below 100, and on which side of the diagonal would a fraction that beats it have to lie?

What is the number whose expansion is $[3; 3, 6, 3, 6, 3, 6, \dots]$?

A lunar month is 29.5306 days. Which numbers of months come closest to a whole number of years of 365.2422 days, and what has that to do with a calendar in which a year of thirteen months is put in seven times in nineteen years?

THE SMALLEST SOLUTION

A Babylonian clay tablet written about 1800 BC, now in New York and catalogued as Plimpton 322, lists fifteen rows of whole numbers. In the first row are 119 and 169, in the third 4601 and 6649, in the fourth 12709 and 18541. In every row, once four slips of the scribe are put right, the difference of the two squares is itself a square: $169^2 - 119^2 = 120^2$, and $6649^2 - 4601^2 = 4800^2$. The scribe had right-angled triangles with whole-number sides, 119, 120, 169 and 4601, 4800, 6649, a thousand years before Pythagoras, and numbers of that size are not found by trying. What did the scribe know?

Call three positive whole numbers with $x^2 + y^2 = z^2$ a *triple*. Multiplying a triple through by any number gives another, so it is enough to find those whose three numbers have no common divisor; and then no two of them have one either. For a prime dividing two of the numbers divides their squares, and so divides the square of the third, which is the sum or the difference of those two squares; a prime that divides a square divides the number itself, so it would divide all three.

The first step looks at remainders on division by 4. An even number $2k$ has square $4k^2$, which leaves 0; an odd number $2k + 1$ has square $4k^2 + 4k + 1$, which leaves 1. So a sum of two odd squares leaves 2 and is not a square. In a triple without common

divisor x and y are not both even, since 2 would divide both; and they are not both odd, since $x^2 + y^2$ is a square. So one is even and one is odd; say y is the even one. Then x is odd, and $z^2 = x^2 + y^2$ is an odd number plus an even one, so z is odd too.

The second step factorises. Write $y^2 = z^2 - x^2 = (z - x)(z + x)$. Both factors are even, so put $z - x = 2a$ and $z + x = 2b$, and $y = 2c$; then $c^2 = ab$. Any common divisor of a and b divides their sum, which is z , and their difference $b - a$, which is x ; and x and z have no common divisor, so a and b have none. And here unique factorisation does the work. In the factorisation of c^2 every prime occurs an even number of times. Each of those primes divides a or b but not both, so all its occurrences lie in one of them, and therefore in a every prime occurs an even number of times, and likewise in b :

when two positive whole numbers with no common divisor have a square for their product, each of them is a square.

So $b = m^2$ and $a = n^2$, and $c^2 = ab = m^2n^2$ gives $c = mn$. Now z is the sum $a + b$, x is the difference $b - a$, and y is $2c$:

$$x = m^2 - n^2, y = 2mn, z = m^2 + n^2$$

Every triple without common divisor has this form. Here m is larger than n , and the two have no common divisor, because a and b have none. One of them is even and the other odd: they are not both even, having no common divisor, and two odd numbers would make $m^2 - n^2$ even, while x is odd. And every such pair gives a triple, because $(m^2 - n^2)^2 + (2mn)^2 = (m^2 + n^2)^2$ is an identity, true of any two numbers.

m	n	$m^2 - n^2$	$2mn$	$m^2 + n^2$
2	1	3	4	5
3	2	5	12	13
4	1	15	8	17
4	3	7	24	25
5	2	21	20	29
5	4	9	40	41
6	1	35	12	37
7	2	45	28	53
75	32	4601	4800	6649

The scribe's rows come from the pairs 12 and 5, 75 and 32, 125 and 54, numbers whose only prime divisors are 2, 3 and 5, which are the numbers that divide cleanly in arithmetic done in sixties. How the table was made is argued over, but whoever made it could pass from a pair to a triple.

* * *

There is a second way to the same formula, and it shows what the formula is. Divide a triple through by z^2 : the point $(x/z, y/z)$ lies on the circle $x^2 + y^2 = 1$, and its coordinates are fractions. So every triple gives a point of the circle with fraction coordinates, in the quarter where both coordinates are positive. Every such point gives a triple back: write its two fractions over a common denominator z , and the numerators are x and y . The points with fraction coordinates in the other quarters come with these, by changing the signs and exchanging the two coordinates. One such point is in plain view, the point $(-1, 0)$. Draw the line through it with slope t , the line that rises t for every 1 it runs to the right, which is the line $y = t(x + 1)$, and ask where else it meets the circle. Substituting, $x^2 + t^2(x + 1)^2 = 1$. Bring the 1 across and write $x^2 - 1$ as $(x + 1)(x - 1)$; then both terms contain the factor $x + 1$, and the equation is $(x + 1)((x - 1) + t^2(x + 1)) = 0$. One solution is the known point, $x = -1$. The other makes the second bracket 0, which says $x(1 + t^2) = 1 - t^2$, and its y comes from $y = t(x + 1)$:

$$x = (1 - t^2)/(1 + t^2), y = 2t/(1 + t^2)$$

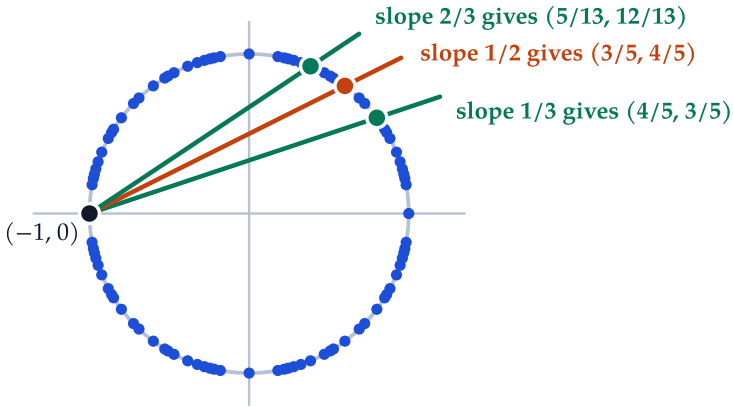


Figure 20.1. The circle of radius 1 and three lines through the point $(-1, 0)$, with slopes $1/3$, $1/2$ and $2/3$. Each meets the circle again at a point whose coordinates are fractions, and the small dots are all such points whose denominators are below 70. *Assumes:* that the curve has degree two and that one point with fraction coordinates is known; the circle of radius $\sqrt{3}$ has none, and the construction never starts

When t is a fraction, so are x and y ; and a point of the circle with fraction coordinates, joined to $(-1, 0)$, gives a line whose slope $y/(x + 1)$ is a fraction. So the points with fraction coordinates correspond exactly to the fraction slopes. Put $t = n/m$ and clear denominators, and the point is $((m^2 - n^2)/(m^2 + n^2), 2mn/(m^2 + n^2))$: the Babylonian formula, found without factorising anything. The slope $1/2$ gives $(3/5, 4/5)$, and the slope $32/75$ gives the scribe's $4601/6649$ and $4800/6649$.

The method used nothing about the circle except that it is given by an equation of the second degree with whole-number coefficients, and that one point on it with fraction coordinates was known. Substituting the line into such an equation gives a quadratic for x , say $ax^2 + bx + c = 0$, whose coefficients are fractions when t is. The two solutions of a quadratic add up to $-b/a$, a fraction made from its coefficients. To see it, let r be one solution,

and subtract $ar^2 + br + c = 0$ from $ax^2 + bx + c = 0$. That leaves $a(x^2 - r^2) + b(x - r) = 0$, which is $(x - r)(a(x + r) + b) = 0$, so the other solution is $-r - b/a$, and the two add to $-b/a$. One solution being a fraction, the other is too. So on any curve of the second degree, one point with fraction coordinates gives them all. In one direction, a line through it with a fraction slope, if it meets the curve a second time, meets it in a point with fraction coordinates. In the other direction, take any other point of the curve with fraction coordinates and join it to the first. The joining line has a fraction slope, so the point is one of those already found; the only exception is a point directly above or below the first, where the joining line is vertical and has no slope.

But the first point has to exist, and it may not. The circle $x^2 + y^2 = 3$ has no point with fraction coordinates at all. Clearing denominators, such a point would give whole numbers with $a^2 + b^2 = 3c^2$ and c positive. Take such numbers with c as small as possible. On division by 3 a square leaves 0 or 1, since $(3k + 1)^2$ and $(3k - 1)^2$ are both multiples of 3 plus 1. A sum of two squares therefore leaves $0 + 0$, $0 + 1$ or $1 + 1$, that is 0, 1 or 2, and it is a multiple of 3 only when both squares leave 0, which happens only when 3 divides both a and b . Then 9 divides a^2 and b^2 , so 9 divides their sum $3c^2$; so 3 divides c^2 , and 3 divides c . Dividing all three numbers by 3 gives a solution with a smaller c , and there is no smaller one than the smallest. So there are two circles with the same centre, of radius 1 and of radius $\sqrt{3}$. On the first, the points with fraction coordinates crowd every arc. For the squared distance between the points where the chords of slopes t and s land multiplies out to $4(t - s)^2 / ((1 + t^2)(1 + s^2))$, which is at most $4(t - s)^2$, so the two points are no farther apart than twice the difference of the slopes, and the landing point moves only a little when t moves a little. So every arc holds all the points that come from some stretch of slopes, and no stretch of slopes is without a fraction. On the second circle there is not one such point.

* * *

The argument about $x^2 + y^2 = 3$ is a descent, of the same shape as the proof that no fraction has square 2. Fermat, who named the method *infinite descent* and was proud of it, used it on much harder equations, and one of his descents survives because for once he wrote down how it goes. The question he wrote it for comes after its nearest relative, which yields to the same method.

No two fourth powers of positive whole numbers add up to a square: $x^4 + y^4 = z^2$ has no solution.

Suppose there are solutions, and take one with z as small as possible. Then x and y have no common divisor: a prime p dividing both would give p^4 dividing z^2 , so p^2 dividing z , and $(x/p, y/p, z/p^2)$ would be a smaller solution. So x^2, y^2 and z form a triple without common divisor; say y is even. By the formula there are m and n , without common divisor, with

$$x^2 = m^2 - n^2, y^2 = 2mn, z = m^2 + n^2$$

The first of these says $x^2 + n^2 = m^2$, another triple without common divisor, in which x is odd; so n is the even one, and the formula applies a second time: there are r and s , without common divisor, with

$$x = r^2 - s^2, n = 2rs, m = r^2 + s^2$$

Now use the middle equation of the first three: $y^2 = 2mn = 4rs(r^2 + s^2)$, so $rs(r^2 + s^2)$ is the square of $y/2$. No two of the numbers r, s and $r^2 + s^2$ have a common divisor. For r and s have none; and a prime dividing r and $r^2 + s^2$ would divide s^2 , and so s , and in the same way a prime dividing s and $r^2 + s^2$ would divide r . A product of positive whole numbers without common divisors, two by two, is a square only when each is, by the same count of primes as before. So $r = a^2, s = b^2$ and $r^2 + s^2 = c^2$, and that says

$$a^4 + b^4 = c^2$$

a new solution of the same equation. And c is at most c^2 , which is m , which is less than $m^2 + n^2$, which is z . A smaller solution than the smallest has been produced, so there are none.

Since z^4 is the square of z^2 , a solution of $x^4 + y^4 = z^4$ would be a solution of the equation above; so $x^4 + y^4 = z^4$ has no solution in positive whole numbers. This is the case $n = 4$ of the claim that $x^n + y^n = z^n$ has no solution when n is above 2. Fermat wrote the claim in the margin of his copy of Diophantus, adding that the margin was too small for the proof. Lamé believed in 1847 that he had proved the claim, and the case of fourth powers is the only one for which an argument of Fermat himself is known. The claim was proved for every n by Andrew Wiles in 1994, by methods that have little in common with this one except that they too start from a supposed solution: they build a curve from it and show that no such curve can exist.

The descent that Fermat wrote down was for a statement about triangles: no right-angled triangle with whole-number sides has a square number for its area. Enlarging a triangle multiplies its area by a square, so it is enough to treat the triangles whose sides have no common divisor. Such a triangle has sides $m^2 - n^2$, $2mn$ and $m^2 + n^2$, and its area, half the product of the two shorter sides, is $mn(m - n)(m + n)$. No two of those four numbers have a common divisor, because m and n have none and one of them is even. So if the area were a square, each of the four would be a square, and with $m = a^2$ and $n = b^2$ the product $(m - n)(m + n)$, which is $a^4 - b^4$, would be a square too. It comes down to a second equation, $x^4 - y^4 = z^2$. The descent that Fermat wrote shows that it has no solution in positive whole numbers: like the one above, it passes through the formula for triples from a supposed solution to a smaller one. It is not carried out here, and the statement rests on his argument; the last questions below return to it.

* * *

Descent can also find everything, when there is something to

find. The near misses to a fraction with square 2 were the pairs with $x^2 - 2y^2$ equal to 1 or -1 , and one step, from (x, y) to $(x + 2y, x + y)$, made an endless chain of them from $(1, 1)$: $(3, 2)$, $(7, 5)$, $(17, 12)$, $(41, 29)$, Are there any others?

Let (x, y) be any pair of positive whole numbers with $x^2 - 2y^2$ equal to 1 or -1 , and suppose y is more than 1. Then $x^2 = 2y^2 \pm 1$ lies above y^2 and below $4y^2$, so x lies between y and $2y$, and the pair $(2y - x, x - y)$, the step undone, consists of positive whole numbers. Its second member $x - y$ is less than y . And it is again a solution, because $(2y - x)^2 - 2(x - y)^2$ multiplies out to $2y^2 - x^2$: the undoing, like the step, reverses the sign of $x^2 - 2y^2$. So from any solution with y above 1 the undoing leads to a solution with a smaller y , and it can be repeated until y is 1, where x^2 is 2 ± 1 , and x is 1. Every solution, traced back, arrives at $(1, 1)$; so every solution lies on the chain that the step makes from $(1, 1)$.

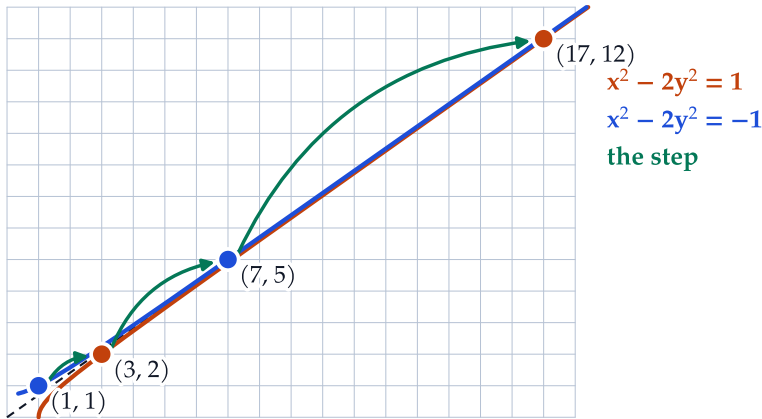


Figure 20.2. The curves $x^2 - 2y^2 = 1$ and $x^2 - 2y^2 = -1$, which hug the line through the origin with slope $1/\sqrt{2}$, and the whole-number points on them. The step from (x, y) to $(x + 2y, x + y)$ hops from one curve to the other, and undone it leads every point back to $(1, 1)$. Assumes: the first four points only; the next, $(41, 29)$, is off the page, and the curves are by then closer to the line than the thickness of the ink

The step is a multiplication. Write a pair (x, y) as the number x

$+ y\sqrt{2}$. Then

$$(x + y\sqrt{2})(1 + \sqrt{2}) = (x + 2y) + (x + y)\sqrt{2}$$

so the chain is the list of powers of $1 + \sqrt{2}$. The quantity $x^2 - 2y^2$ is $(x + y\sqrt{2})(x - y\sqrt{2})$, and it behaves under multiplication as the sizes $a^2 + 5b^2$ did: for any whole number D ,

$$(x^2 - Dy^2)(u^2 - Dv^2) = (xu + Dyv)^2 - D(xv + yu)^2$$

which can be checked by multiplying out, the cross terms $2Dxuyv$ cancelling. Brahmagupta stated it in India in 628. It says that solutions of $x^2 - Dy^2 = 1$ can be multiplied: from two of them, (x, y) and (u, v) , comes a third, $(xu + Dyv, xv + yu)$. From $(3, 2)$, where $9 - 8 = 1$, multiplied by itself, comes $(17, 12)$; and $(17, 12)$ multiplied by $(3, 2)$ gives $(99, 70)$. In the same way one solution of $x^2 - Dy^2 = 1$ in positive whole numbers gives infinitely many, because both members grow at every multiplication.

* * *

Is there always one? The equation $x^2 - Dy^2 = 1$, for a positive whole number D that is not a square, carries the name of Pell, an Englishman who had nothing to do with it. For $D = 2$ the smallest solution is $(3, 2)$, for $D = 3$ it is $(2, 1)$, for $D = 7$ it is $(8, 3)$. In 1657 Fermat challenged the mathematicians of Europe, those of England above all, to solve it for every D , and to Frénicle in Paris he proposed $D = 61$. The smallest solution is

$$x = 1766319049, y = 226153980$$

and no search by trial finds it. Bhaskara had found it in India five centuries earlier, by a method of repeated composition with Brahmagupta's identity.

Where should one look? A solution makes x/y very close to $\sqrt{D}$, because

$$x/y - \sqrt{D} = (x^2 - Dy^2)/(y(x + y\sqrt{D})) = 1/(y(x + y\sqrt{D}))$$

which is less than $1/(2y^2)$, because x and $y\sqrt{D}$ are each more than y , so that $x + y\sqrt{D}$ is more than $2y$. Fractions that close to a number are rare, and the expansion into a continued fraction produces the fractions that are closest for their denominators. More is true, and it is Legendre's: he proved in 1798 that a fraction within $1/(2y^2)$ of a number, y being its denominator, is always one of that number's convergents, and the search that follows rests on his proof. So every solution is a convergent of $\sqrt{D}$, and since the denominators of the convergents increase, the first convergent that gives 1 is the smallest solution. For $\sqrt{7}$ the expansion is $[2; 1, 1, 1, 4, 1, 1, 1, 4, \dots]$, with convergents $2, 3, 5/2, 8/3$, and $8^2 - 7 \cdot 3^2 = 1$. For $\sqrt{61}$ it is $[7; 1, 4, 3, 1, 2, 2, 1, 3, 4, 1, 14, \dots]$, repeating after the 14, and the convergents give $p^2 - 61q^2$ the values $-12, 3, -4, 9, -5, 5, -9, 4, -3, 12, -1$ and then the same again with the signs reversed; the value 1 is reached at the twenty-second convergent, which is the pair above. The pair at the eleventh, $(29718, 3805)$, has $29718^2 - 61 \cdot 3805^2 = -1$, and multiplied by itself with the identity it gives the solution.

That a solution always exists was proved by Lagrange in 1768. The proof that follows is Dirichlet's, from 1842, and its main tool is the observation that if more objects are put into boxes than there are boxes, some box holds two.

First, $\sqrt{D}$ is not a fraction, by the count of primes: D , not being a square, has some prime an odd number of times, and in $a^2 = Db^2$ that prime would occur an even number of times on the left and an odd number on the right. Now let N be any whole number above 1. Take the $N + 1$ numbers $0, \sqrt{D}, 2\sqrt{D}, \dots, N\sqrt{D}$, and from each remove the whole number below it, leaving a part that is at least 0 and less than 1. Cut the stretch from 0 to 1 into N equal pieces. There are $N + 1$ parts and N pieces, so two parts, from $i\sqrt{D}$ and $j\sqrt{D}$ say, with i less than j , fall in one piece, and differ by less than $1/N$. Put $q = j - i$. Then $q\sqrt{D}$ differs from some whole number p by less than $1/N$, where q is at least 1 and at most N , and p is positive because $q\sqrt{D}$ is more than 1:

$$|p - q\sqrt{D}| < 1/N, \text{ and so also } < 1/q$$

For such a pair p is less than $q\sqrt{D} + 1$, so $p + q\sqrt{D}$ is less than $2q\sqrt{D} + 1$. Multiplying $|p - q\sqrt{D}|$ by $p + q\sqrt{D}$ gives $|p^2 - Dq^2|$, and so

$$|p^2 - Dq^2| < (1/q)(2q\sqrt{D} + 1), \text{ which is at most } 2\sqrt{D} + 1$$

There are infinitely many such pairs. For given any one of them, $|p - q\sqrt{D}|$ is not 0, because $\sqrt{D}$ is not a fraction; and an N with $1/N$ below it produces a pair that is closer, hence different.

So infinitely many pairs of positive whole numbers have $p^2 - Dq^2$ among the finitely many whole numbers between $-2\sqrt{D} - 1$ and $2\sqrt{D} + 1$. Boxes again: some one value k is taken by infinitely many pairs, and k is not 0. Among those pairs, sort by the remainders that p and q leave on division by the size of k . There are only k^2 possible pairs of remainders, so two different pairs (p, q) and (r, s) have

$$p^2 - Dq^2 = r^2 - Ds^2 = k, \text{ with } p - r \text{ and } q - s \text{ both multiples of } k$$

Multiply the first by the second with its sign changed, that is $(p + q\sqrt{D})$ by $(r - s\sqrt{D})$. By the identity, the numbers $X = pr - Dqs$ and $Y = qr - ps$ have $X^2 - DY^2$ equal to k times k . But X differs from $p^2 - Dq^2$, which is k , by $p(r - p) - Dq(s - q)$, a multiple of k ; so X is a multiple of k . And $Y = q(r - p) - p(s - q)$ is a multiple of k . Dividing both by k ,

$$(X/k)^2 - D(Y/k)^2 = 1$$

in whole numbers. And Y is not 0. If it were, then $p/q = r/s$, so $p = \lambda r$ and $q = \lambda s$ for some positive fraction λ ; then $p^2 - Dq^2$ is λ^2 times $r^2 - Ds^2$, and since both are k , λ is 1 and the two pairs are the same pair. Taking sizes, $(|X/k|, |Y/k|)$ is a solution in positive whole numbers.

* * *

One solution gives infinitely many, by multiplication, and as with $D = 2$ the smallest gives them all. Among the solutions in

positive whole numbers, let (x_1, y_1) be the one for which $x_1 + y_1\sqrt{D}$ is smallest, and call that number ϵ . There is a smallest, because only finitely many pairs of positive whole numbers have $x + y\sqrt{D}$ below any given bound. For a solution in positive whole numbers, $x + y\sqrt{D}$ is more than 1. Its partner $x - y\sqrt{D}$ is 1 divided by it, because the product of the two is $x^2 - Dy^2$, which is 1; so the partner lies between 0 and 1. Conversely, suppose whole numbers u and v have $u^2 - Dv^2 = 1$ and $u + v\sqrt{D}$ more than 1. Then $u - v\sqrt{D}$, which is 1 divided by that, lies between 0 and 1. Now u is half the sum of the two numbers $u + v\sqrt{D}$ and $u - v\sqrt{D}$, and $v\sqrt{D}$ is half their difference. The sum is positive, and the difference is positive because the first number is above 1 and the second below 1. So u and v are both positive.

Let $\sigma = x + y\sqrt{D}$ be any solution in positive whole numbers. The powers $\epsilon, \epsilon^2, \epsilon^3, \dots$ increase without bound, so σ lies from some power up to but not reaching the next: $\epsilon^n \leq \sigma < \epsilon^{n+1}$. Multiply σ by the partner of ϵ^n , which is 1 divided by ϵ^n . The result is again a solution, by the identity, and dividing $\epsilon^n \leq \sigma < \epsilon^{n+1}$ through by ϵ^n shows that it is at least 1 and less than ϵ . If it were more than 1, its two whole numbers would be positive, by the converse above, and it would be a solution in positive whole numbers smaller than the smallest. So it is 1, and σ is ϵ^n :

every solution of $x^2 - Dy^2 = 1$ in positive whole numbers is a power of the smallest one.

For $D = 2$ the smallest is $3 + 2\sqrt{2}$, the square of $1 + \sqrt{2}$, and the solutions are every second member of the chain. Archimedes is said to have sent the mathematicians of Alexandria a problem about the cattle of the sun, whose conditions come down to $x^2 - 4729494y^2 = 1$ with y a multiple of 9314. The smallest herd that meets them was worked out in 1880: the number of cattle has 206545 digits.

How many right-angled triangles with whole-number sides have a side of 60 next to the right angle?

Which whole numbers are at once a square and a triangular number $1 + 2 + \dots + n$, such as 36, and is there a last one?

What is the step that produces every solution of $x^2 - 3y^2 = 1$ from the smallest, and why does $x^2 - 3y^2 = -1$ have no solution at all?

The difference of two fourth powers, $x^4 - y^4$, is never a square when x and y are different positive whole numbers. Why not?

What is the smallest solution of $x^2 - 13y^2 = 1$, and at which convergent of $\sqrt{13}$ does it appear?

HOW MANY PRIMES

Every whole number above 1 is a product of primes in one way only apart from order, so the primes are the atoms of multiplication. How many atoms are there, and where are they?

There is no last one. Take any finite list of primes whatever, multiply them all together and add 1. The result leaves remainder 1 on division by each prime of the list, so none of them divides it. But it is a number above 1, and so it has a prime divisor; that prime is not on the list. No finite list holds all the primes. The proof is in Euclid, and it does not say that the product plus 1 is itself prime. From the list 2, 3, 5, 7, 11, 13 it makes 30031, which is $59 \cdot 509$; what it guarantees is that 59 and 509 are new.

The argument can be aimed. On division by 4 an odd prime leaves 1 or 3. Are there infinitely many of each kind? Take any list of primes that leave 3, multiply them all, multiply by 4, and this time subtract 1. The result N leaves 3 on division by 4. It is odd, so its prime divisors are odd, and they cannot all leave 1, since numbers that leave 1 have products that leave 1: that was the closed world in which factorisation failed. So some prime divisor of N leaves 3, and it is not on the list, since a prime of the list divides four times the product, and if it divided N as well it would divide the difference between them, which is 1. From the list 3, 7, 11 the number is $923 = 13 \cdot 71$, and 71 is the new prime that leaves 3.

For the primes that leave 1 the same plan fails. From the list consisting of 5 alone, four times the product, plus 1, is 21, which is $3 \cdot 7$: a number that leaves 1 can be made entirely of primes that leave 3, two at a time, and nothing forces a new prime of the wanted kind. There are infinitely many of those too, but the proof has to know which primes can divide a number of the form $n^2 + 1$, and that belongs to the arithmetic of remainders. Dirichlet proved in 1837 that every progression $a, a + d, a + 2d, \dots$ in which a and d have no common divisor contains infinitely many primes. His proof used sums over all the whole numbers and the calculus, and with it began the use of the continuous to study the discrete.

* * *

To find the primes below 100, write down the numbers from 2 to 99 and strike out the multiples of 2 beyond 2, then of 3 beyond 3, then of 5 and of 7. Everything left is prime. Nothing beyond 7 needs striking. A number below 100 that is not prime is a product of two numbers above 1, and the smaller of the two is below 10, since two factors of 10 or more would give a product of 100 or more. That smaller factor has a prime divisor, which is also below 10 and so is one of 2, 3, 5 and 7, and it divides the number. This is the sieve of Eratosthenes, who also measured the earth. It leaves 25 primes below 100.

The survivors look scattered. There are primes two apart, 11 and 13, 59 and 61, 101 and 103, and 1000000000061 and 1000000000063; nobody knows whether such pairs go on forever, though Zhang proved in 2013 that infinitely many pairs of primes differ by less than 70 million, a bound since brought down to 246. There are also deserts. From 114 to 126 there are thirteen numbers in a row and no prime among them, and a desert of any length can be made to order: the numbers $n! + 2, n! + 3, \dots, n! + n$ are divisible by 2, 3, $\dots, n$ in turn, because $n!$ is a multiple of each of 2, 3, $\dots, n$, and a multiple of a number added to that number is again a multiple of it. Each of them is larger than its divisor, so none is prime, which

How many primes

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

struck by 2

struck by 3

struck by 5

struck by 7

bold: the 25 primes

Figure 21.1. The numbers up to 100 after striking out the multiples of 2, 3, 5 and 7. Each struck number is shaded by the prime that strikes it first, and the 25 numbers left standing are the primes. Nothing beyond 7 is needed, since $11 \cdot 11$ is already past 100. *Assumes:* that 1 is set aside by agreement, being neither prime nor a product of primes

makes $n - 1$ numbers in a row without a prime.

Counted in bulk, the scatter gives way to something smooth.

below	primes	one number in
10	4	2.50
100	25	4.00
1000	168	5.95
10000	1229	8.14
100000	9592	10.43
1000000	78498	12.74
10000000	664579	15.05
100000000	5761455	17.36
1000000000	50847534	19.67

The last column is the first divided by the second, and from the fourth row down it rises by 2.29, 2.31, 2.31, 2.31 and 2.31. Each

further digit makes the primes rarer by the same amount: among numbers of d digits, about one in $2.3d$ is prime. Gauss noticed this at fifteen, from tables of primes. The number is 2.302585..., and it arises far from here, as the area of the region under the curve $y = 1/t$ between $t = 1$ and $t = 10$. That such a region has an area at all has still to be shown, and the area is built much later. That the primes should thin out according to that area was proved in 1896, by Hadamard and by de la Vallée Poussin independently, with the calculus of the numbers of the plane and with ideas nothing here prepares. What can be proved with bare hands is the rough truth of it, and the proof goes through a single entry of the triangle of choices.

* * *

How many times does a prime p divide $n!$? Among the numbers $1, 2, \dots, n$, count the multiples of p , then the multiples of p^2 , then of p^3 , and so on until the counts are 0, and add the counts. A number divisible by p exactly e times is a multiple of p , of p^2 , ..., of p^e , so it is counted e times, once for each factor p that it brings to the product. The sum of the counts is therefore the number of times p divides $n!$. Up to 100 there are 20 multiples of 5 and 4 multiples of 25, so 5 divides 100! twenty-four times, and 2 divides it more often than that. Each zero at the end of a number is a factor 10, which is one 2 together with one 5, and the 5s run out first; so 100! ends in exactly 24 zeros.

Now take the middle entry of row $2n$ of the triangle, the number of ways of choosing n things from $2n$. By the formula for n choose k it is $2n(2n-1) \cdots (n+1)/n!$, and multiplying top and bottom by $n!$ supplies the missing factors of $(2n)!$ in the top:

$$C = (2n)! / (n! \cdot n!)$$

The number of times p divides C is the number of times it divides $(2n)!$, less twice the number of times it divides $n!$, since $n!$ stands twice in the bottom. Each of those numbers is a sum of

counts, one count for each power p^j . So the number of times p divides C is a sum with one term for each power p^j : the count of multiples of p^j up to $2n$, less twice the count up to n . Look at one term. Divide n by p^j : say n is q multiples of p^j and r over, with r less than p^j , so that the count up to n is q . Then $2n$ is $2q$ multiples and $2r$ over, and $2r$ is less than $2p^j$, so the count up to $2n$ is $2q$ if $2r$ falls short of p^j , and $2q + 1$ if it does not. Taking away twice q leaves 0 or 1, so each term of the sum is 0 or 1. And a term is 0 once p^j exceeds $2n$, there being then no multiples to count. So the number of times p divides C is at most the number of powers of p up to $2n$. If p divides C exactly e times, then at least e of the powers $p, p^2, p^3, \dots$ are up to $2n$, and p^e is among them, which says:

the power of any prime that divides C is at most $2n$.

Three kinds of prime can now be told apart. A prime between n and $2n$ has one multiple up to $2n$, itself, since twice it is beyond $2n$; so it divides $(2n)!$ once and $n!$ not at all, and it divides C exactly once. A prime p above $2n/3$ and at most n does not divide C , when n is at least 5. Since $3p$ is above $2n$, its multiples up to $2n$ are p and $2p$, two of them, and up to n there is p alone; two, less twice one, is 0, so the term for p itself is 0. Its square is above $4n^2/9$, which is at least $2n$ once n is at least 5, so every higher power of p is beyond $2n$ and those terms are 0 as well. And a prime above the square root of $2n$ divides C at most once: its square is beyond $2n$, so only the term for the prime itself can be 1. For $n = 30$,

$$C = 118264581564861424$$

$$= 2^4 \cdot 7 \cdot 11 \cdot 17 \cdot 19 \cdot 31 \cdot 37 \cdot 41 \cdot 43 \cdot 47 \cdot 53 \cdot 59$$

with every prime from 31 to 59 present once, 23 and 29 absent, and nothing present to a power above 60.

How large is C ? Row $2n$ adds up to 2^{2n} , which is 4^n , so C is less than 4^n . And C is the largest entry of the row. By the formula for n choose k , passing from place k to place $k + 1$ of row $2n$ multiplies the entry by $(2n - k)/(k + 1)$, and that factor is above 1 as long as

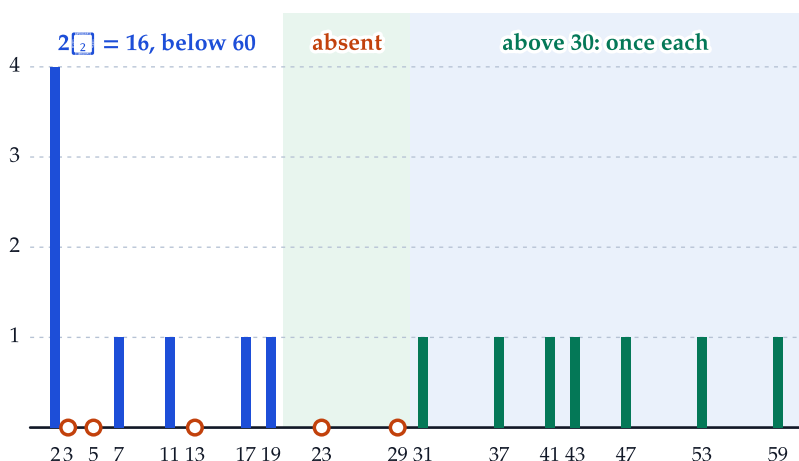


Figure 21.2. The number of times each prime up to 60 divides the number of ways of choosing 30 things from 60. Every prime above 30 divides it once; 23 and 29, which lie above two thirds of 30, do not divide it; and no prime power that divides it exceeds 60. *Assumes: one entry of one row; the three zones are proved in the text for every row, from $2n = 10$ on*

k is below n ; so the entries climb up to the middle, and by the symmetry of the row they fall after it. Put the two 1s at the ends of the row together as a single 2, and the row consists of $2n$ numbers none larger than C . Those $2n$ numbers still add up to 4^n , so $2n$ times C is at least 4^n . So

$$4^n / (2n) \leq C < 4^n$$

The primes between n and $2n$ all divide C , and no two of them are the same prime, so their product divides C and is less than 4^n . If there are k of them, their product is at least n^k , each of them being above n , and so n^k is less than 4^n . With n a million: 4^n has 602060 digits and n^k has $6k + 1$, so k is below 100344. The true number of primes between one million and two million is 70435. In the other direction, C is a product of powers of the primes up to $2n$, each power at most $2n$. If there are t of those primes, C is at most

$(2n)^t$, and C is at least $4^n/(2n)$, so $(2n)^{t+1}$ is at least 4^n . With n a million, $2n$ is below 10^7 , so 10^{7t+7} exceeds 4^n , a number of 602060 digits. A number of that many digits is at least 10^{602059} , so $7t+7$ is more than 602059, and t is at least 86008. There are in fact 148933 primes below two million. Chebyshev found bounds of this kind in 1850, tighter ones by the same means, and they were the first proof that the proportion of primes among the numbers of d digits lies between two fixed multiples of $1/d$.

* * *

One more bound of the same kind is needed.

The product of all the primes up to n is less than 4^n .

For $n = 1$ and $n = 2$ the products are 1 and 2. Suppose the statement holds below n . If n is even and above 2, it is not prime, and the product is the same as for $n - 1$, which is less than 4^{n-1} . If n is odd, say $n = 2m + 1$, look at the entry $(2m + 1)!/(m! \cdot (m + 1)!)$ of row $2m + 1$. It occurs twice in that row, side by side in the middle, and the row adds up to $2 \cdot 4^m$, so the entry is at most 4^m . Every prime from $m + 2$ to $2m + 1$ is one of the numbers multiplied together in the top, and it is too large to divide any of the numbers multiplied together in $m!$ or in $(m + 1)!$; it divides the top and not the bottom, so the product of those primes divides the entry and is at most 4^m . The primes up to $m + 1$ have product less than 4^{m+1} , by the supposition. Multiplying, the primes up to $2m + 1$ have product less than 4^{2m+1} .

In 1845 Bertrand, having checked it up to three million, guessed that between any number and its double there is always a prime. Chebyshev proved it in 1850. The proof that follows was found in 1931 by Erdős, who was eighteen.

For every whole number n at least 1 there is a prime above n and at most $2n$.

Suppose that for some n , at least 5, there is none. A prime that divides C divides $(2n)!$, so it is at most $2n$; and none lies above n and at most $2n$, so every prime that divides C is at most n . The primes above $2n/3$ and at most n were shown not to divide C , so every prime that divides C is at most $2n/3$. Split them at the square root of $2n$. Those up to the square root are s in number. There cannot be more primes up to the square root than there are whole numbers up to it, so s is at most the square root of $2n$, and s^2 is at most $2n$. Each of these primes contributes a power of at most $2n$, so together they contribute at most $(2n)^s$. Those above the square root divide C at most once each, so together they contribute at most the product of all the primes up to $2n/3$, which is less than 4^m , where m is $2n/3$ rounded down. With the lower bound for C ,

$$4^n / (2n) \leq C < (2n)^s \cdot 4^m$$

Cube both sides: $4^{3n} / (2n)^3$ is less than $(2n)^{3s}$ times the cube of 4^m . Since $3m$ is at most $2n$, the cube of 4^m is at most 4^{2n} , so $4^{3n} / (2n)^3$ is less than $(2n)^{3s} \cdot 4^{2n}$. Divide both sides by 4^{2n} , which leaves $4^n / (2n)^3$ on the left and $(2n)^{3s}$ on the right, and then multiply both sides by $(2n)^3$. The result is that 4^n is less than $(2n)^{3s+3}$. Writing x for $2n$, so that 4^n is 2 to the power x ,

$$2^x < x^{3s+3}$$

On the left stands a power of 2 whose exponent is x , and on the right a power of x whose exponent is at most about three times the square root of x , so the left side doubles x times over while the right side does not keep up, and for large x this cannot be. To make that exact, let 4^k be the power of 4 with $4^k \leq x < 4^{k+1}$. Then s^2 is at most x , which is less than 4^{k+1} , the square of 2^{k+1} ; so s is less than 2^{k+1} . On the right side the base x is less than 4^{k+1} and the exponent $3s+3$ is less than $3 \cdot 2^{k+1} + 3$, so the right side is less than 4^{k+1} raised to the power $3 \cdot 2^{k+1} + 3$. Now 4^{k+1} is 2 raised to the power $2(k+1)$, so that bound is 2 raised to the power $2(k+1)(3 \cdot 2^{k+1} + 3)$, an exponent that multiplies out to $6(k+1)(2^{k+1} + 1)$. The left side is at least 2 raised to the power 4^k . Both sides are now powers of 2,

and the smaller power has the smaller exponent, so the inequality requires

$$4^k < 6(k+1)(1+2^{k+1})$$

For $k = 7$ that reads $16384 < 12336$, which is false, and it stays false for every larger k , because each step from k to $k+1$ multiplies the left side by 4, while on the right $k+1$ becomes $k+2$, a gain of at most one part in eight when k is 7 or more, and $1+2^{k+1}$ becomes $1+2^{k+2}$, which is less than double; so the right side is multiplied by less than 3. So k is at most 6, which makes x less than $4^7 = 16384$, and n , which is half of x , less than 8192.

So any n that has no prime above it and at most its double is below 8192. And below 8192 the primes

2, 3, 5, 7, 13, 23, 43, 83, 163, 317, 631, 1259, 2503, 5003, 9973

settle the matter by themselves. Each is less than twice the one before. For $n = 1$ the prime 2 serves. Given any other n below 8192, take the first prime of the list that exceeds n ; the one before it is at most n , and this one is less than twice the one before, so this one is above n and less than $2n$. There is no exception anywhere.

* * *

Is $n!$ ever a square? Let p be the largest prime up to n . There is a prime above p and at most $2p$, and it is larger than n , p being the largest up to n ; so $2p$ is larger than n . Then p divides exactly one of the numbers $1, 2, \dots, n$, namely itself, and divides it once. So p divides $n!$ exactly once. A square contains every prime an even number of times, and once is not even, so $n!$ is not a square for any n above 1.

It is not known whether there is always a prime between two successive squares, though the numbers from n^2 to $(n+1)^2$ are about $2n$ in number and, when n^2 has d digits, about one in $2.3d$ of them should be prime. It is not known whether every even number above 2 is a sum of two primes, as Goldbach guessed in 1742,

though it has been checked beyond $4 \cdot 10^{18}$. The primes are defined by multiplication, and questions that ask about them under addition are, nearly all of them, open.

The numbers $2 + 1$, $2 \cdot 3 + 1$, $2 \cdot 3 \cdot 5 + 1$, and so on, are prime at first. Which of the first eight are?

Are there infinitely many primes that leave remainder 5 on division by 6, and can the argument be run for remainder 2 on division by 3?

With how many zeros does $1000!$ end, and how many times does 7 divide it?

Can $1 + 1/2 + 1/3 + \dots + 1/n$ be a whole number for some n above 1?

Can the numbers from 1 to $2n$ always be paired off so that each pair adds up to a prime?

ARITHMETIC THAT FORGETS

Several arguments have turned on a leftover and nothing else. The count of 2s in $a^2 = 2b^2$ mattered only for being even or odd. The failure of $n^2 + n + 41$ turned on divisibility by 41, whatever the sizes of the values. Both would be shorter with an arithmetic that treats two numbers as the same whenever they leave the same remainder. The construction needs a check that it is allowed, and the check is clearest on the fractions.

* * *

A fraction is a pair of whole numbers, the second one not 0, with a rule for when two pairs count as the same: a/b and a'/b' are the same when $b' = a' b$. So $1/2$, $2/4$ and $3/6$ are one fraction written three ways. Barring 0 below the line is what keeps the rule honest: admit $0/0$ and the rule counts it as the same as $1/2$, since $0 \cdot 2 = 1 \cdot 0$, and as the same as $1/3$, since $0 \cdot 3 = 1 \cdot 0$, while $1/2$ and $1/3$ are different. With no bottom 0, two pairs that are the same as a third are the same as each other: if $ab' = a'b$ and $a'b'' = a''b'$, then $ab'' \cdot b' = a'b \cdot b'' = a''b \cdot b'$, and the middle bottom b' is not 0, so it can be cancelled, leaving $ab'' = a''b$. With $0/0$ in the middle, that cancelling is the step that fails. Everything done with fractions is done by choosing one way of writing each, computing with those,

and reading the result as a fraction.

That can go wrong. Combine two fractions by adding the tops and adding the bottoms, so that $1/2$ with $1/3$ gives $2/5$. The rule is definite, and the result always lies between the two fractions it came from. Now combine $2/4$ with $1/3$ by the same rule: the answer is $3/7$. But $2/4$ is the fraction $1/2$, so the same two fractions were combined twice, giving $2/5$ once and $3/7$ the other time, and those are different fractions. The rule depends on how the fractions are written, so it is not an operation on fractions at all.

Ordinary addition does not have this defect, and that is a theorem. Adding $1/2$ and $1/3$ gives $5/6$, and adding $2/4$ and $1/3$ gives $10/12$, which is $5/6$. In general, if a/b and a'/b' are the same fraction, so that $ab' = a'b$, and c/d and c'/d' are the same, so that $cd' = c'd$, then

$$(ad + bc) \cdot b'd' = ab' \cdot dd' + cd' \cdot bb' = a'b \cdot dd' + c'd \cdot bb' = bd \cdot (a'd' + b'c')$$

which says that $(ad + bc)/bd$ and $(a'd' + b'c')/b'd'$ are the same fraction.

* * *

Every integer leaves exactly one remainder on division by a fixed positive n . For a number that is not negative, this is division with remainder as before. A negative a is handled by lifting it first: choose k with $a + kn$ not negative and divide that number instead. If $a + kn = qn + r$, then $a = (q - k)n + r$, so a leaves the remainder r . That no integer leaves two different remainders is proved as before. Two integers are *congruent* when they leave the same remainder, which happens exactly when their difference is a multiple of n . In one direction, two numbers with the same remainder are each that remainder plus a multiple of n , and subtracting removes the remainder. In the other, if the difference of two numbers is a multiple of n , so is the difference of their remainders; two numbers from 0 to $n - 1$ differ by less than n , so that difference is 0. The second

form is the useful one, since it treats the two numbers alike and says nothing about how large a remainder may be. On division by 12 the numbers 5, 17, 29 and 89 are congruent, their differences being multiples of 12, and so is -7 , which is $5 - 12$. Congruence sorts the integers into n families, the family of a number being everything congruent to it, and the families on division by 12 are the hours of a clock.

Congruence behaves like equality in three ways. Every number is congruent to itself, since 0 is a multiple of n . If one number is congruent to another, the second is congruent to the first, since the negative of a multiple is a multiple. And congruence passes along a chain, since the sum of two multiples is a multiple. Those three properties are what allow speaking of *the* family of a number. A number lies in its own family because it is congruent to itself. If two families share a single member, then, turning congruences round where needed and passing along the chain, every member of one is congruent to every member of the other, so the two families are one family. A number therefore lies in exactly one family.

Adding and multiplying families is done by choosing a member of each, computing, and taking the family of the answer. The rule of adding tops and adding bottoms showed that a definition of this kind needs checking: a different choice of members must lead to the same family. If a is congruent to a' and b to b' , then $a + b$ is congruent to $a' + b'$, since the difference of the sums is the sum of the differences; and ab is congruent to $a'b'$, since $ab - a'b' = b(a - a') + a'(b - b')$, a sum of multiples of n . So the operations do not depend on the members chosen.

Gauss introduced the notation in 1801: $a \equiv b \pmod{n}$, read " a is congruent to b modulo n ", says that n divides $a - b$. The sign echoes equality because congruence may be treated as equality, and the parenthesis names the modulus. On division by 12, $17 \equiv 5 \pmod{12}$ and $5 \cdot 5 \equiv 1 \pmod{12}$. Computation becomes light, since numbers can be reduced at every step: 7^4 modulo 12 is found by squaring 7 to get 49, which is congruent to 1 because $49 = 4 \cdot 12 + 1$, and squaring again to get 1, rather than forming 2401 and dividing.

A polynomial f with integer coefficients is built from its input and fixed integers by additions and multiplications alone, and each of those turns congruent numbers into congruent numbers. Since $a + kn \equiv a \pmod{n}$, it follows that $f(a + kn) \equiv f(a) \pmod{n}$ for every k . A power is a repeated multiplication, so powers need no argument of their own. For $f(x) = x^2 + x + 41$ modulo 41, $f(41k) \equiv f(0) = 41 \equiv 0$, and since $40 \equiv -1$, $f(40) \equiv f(-1) = 1 - 1 + 41 = 41 \equiv 0$. So 41 divides $f(40)$, which is the failure at 40, found without computing 1681.

And no fraction has square 2: write such a fraction in lowest terms, as a/b with the smallest positive bottom it allows (a negative bottom turns positive when both signs are changed), which exists because every collection of whole numbers with something in it has a smallest member; a and b then have no common factor d above 1, since a/d over b/d would be the same fraction with a smaller bottom. Its square being 2 says $a^2 = 2b^2$, so $a^2 \equiv 0 \pmod{2}$, so a is even, since an odd a has $a \equiv 1$ and then $a^2 \equiv 1$; say $a = 2u$; then $4u^2 = 2b^2$ and $b^2 = 2u^2$, so b is even as well, which lowest terms forbids.

Another remainder does it too. Adding b^2 to both sides of $a^2 = 2b^2$ gives $a^2 + b^2 = 3b^2 \equiv 0 \pmod{3}$. A square is 0 or 1 modulo 3, since $0^2 = 0$, $1^2 = 1$ and $2^2 = 4 \equiv 1$, and it is 0 only for a multiple of 3. Two squares add to 0 modulo 3 only when both are 0, the other possible sums being 1 and 2. So 3 divides both a and b , which lowest terms forbids again.

Which of 3, 5 and 7 can be written as $x^2 - 2y^2$ with integers x and y ?

* * *

Checking a product by its digits is this arithmetic. On division by 9, ten is congruent to one, so every power of ten is congruent to one. A number is its digits multiplied by powers of ten and added, 4321 being $4 \cdot 1000 + 3 \cdot 100 + 2 \cdot 10 + 1$, so replacing each power of ten by one shows that a number is congruent to the sum of its

digits. The product of 4321 and 1234 is claimed to be 5332114. The digits of 4321 add to 10 and then to 1, and so do those of 1234, so the product must be congruent to $1 \cdot 1$. The digits of 5332114 add to 19, then to 10, then to 1, and the check passes.

If the answer had been written 5332141, with the last two digits exchanged, the digit sum would be unchanged and the check would pass. If a digit had been mistyped, giving 5332714, the digits would add to 25 and then to 7, and the check would fail. Changing a digit d to d' in any place changes the number by $d' - d$ times a power of ten, which is congruent to $d' - d$. The check misses the change only when $d' - d$ is a multiple of 9, and for two different digits that happens only when one is 0 and the other is 9. So the check catches every change of a single digit except a 0 changed to a 9 or a 9 to a 0. It catches no change of the order of the digits, since reordering leaves the digit sum as it was: the check sees only the family of the number. A failed check proves an error. Set “right” beside the rival “wrong, with the same digit sum”: a pass is certain under either, so the ratio of chances is 1 and the pass is no evidence at all.

A second check sees what this one cannot. On division by 11, ten is congruent to -1 , so the powers of ten are alternately 1 and -1 , and a number is congruent to the alternating sum of its digits taken from the right. For 4321 that is $1 - 2 + 3 - 4 = -2$, and for 1234 it is $4 - 3 + 2 - 1 = 2$, so the product must be congruent to -4 , which is congruent to 7, since $-4 + 11 = 7$. For 5332114 the alternating sum is $4 - 1 + 1 - 2 + 3 - 3 + 5 = 7$, and the check passes. For 5332141 it is 1, and for 5332714 it is 13, which is 2: both errors are caught. Exchanging two neighbouring digits a and b that differ replaces $10a + b$ by $10b + a$ at some place, and so changes the number by $9(b - a)$ times a power of ten. The prime 11 divides none of the three factors: not 9; not $b - a$, which is not 0 and lies between -9 and 9; and not the power of ten, whose only prime factors are 2 and 5. The lemma about primes says that a prime dividing a product divides one of its factors, so 11 does not divide the change: the eleven check catches every such exchange. Against

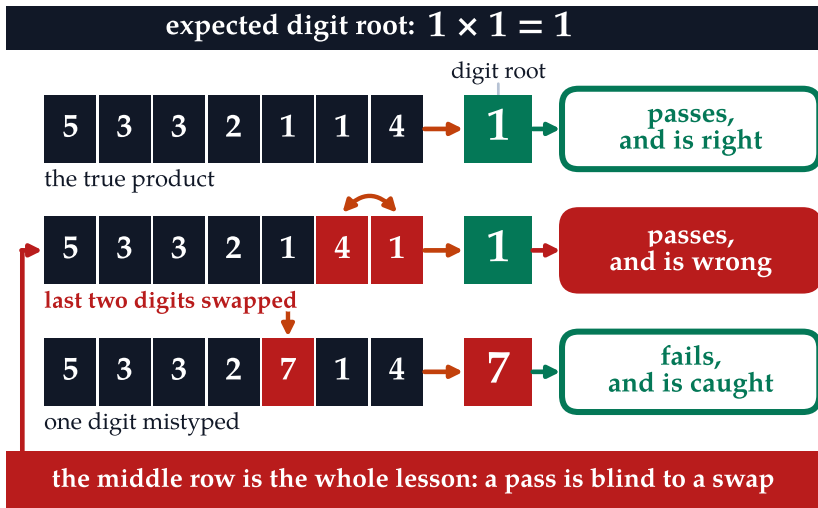


Figure 22.1. The digit check applied to three candidate answers: it convicts the mistyped digit and acquits the transposed pair, because rearranging digits cannot change their sum, so a pass rules out almost every single mistyped digit and is no evidence at all against a transposition. *Assumes:* that the true product is 5332114, established independently; the check never certifies an answer, it only ever removes one

the rival “two neighbouring digits exchanged” a pass has chance 0, where under the nine check it had chance 1.

Which exchanges of two different digits does the eleven check miss?

* * *

Addition, subtraction and multiplication carry over to families. Division is another matter. Dividing by a means undoing multiplication by a , so the question is whether some b has $ab \equiv 1$. On division by 12, 5 has such a partner, since $5 \cdot 5 = 25 \equiv 1$. The number 6 has none. A partner b would have $6b \equiv 1$, which says that $6b - 1$ is a multiple of 12. But every multiple of 6 is even and so is every multiple of 12, so $6b - 1$ is odd and never a multiple of 12.

The difference is that 5 shares no factor with 12 and 6 does.

A number a has a partner on division by n exactly when a and n have no common factor above 1. If their greatest common divisor is 1, the process run backwards gives integers x and y with $ax + ny = 1$, and read as a statement about families, where ny is 0, this says $ax \equiv 1$. If a and n share a factor d above 1, there is no partner. A partner b would make $ab - 1$ a multiple of n , say $ab - 1 = mn$, and then $1 = ab - mn$ would be a combination of a and n . Every combination of a and n is a multiple of d , and 1 is not. For 17 on division by 43: 43 is two 17s and 9, 17 is one 9 and 8, and 9 is one 8 and 1. Backwards, $1 = 9 - 8 = 9 - (17 - 9) = 2 \cdot 9 - 17 = 2(43 - 2 \cdot 17) - 17 = 2 \cdot 43 - 5 \cdot 17$. So $-5 \cdot 17 \equiv 1$, and -5 is in the family of 38; indeed $17 \cdot 38 = 646 = 15 \cdot 43 + 1$.

A problem recorded in China some sixteen centuries ago, in the Sunzi Suanjing, asks for a number of things that leaves 2 when counted by threes, 3 when counted by fives and 2 when counted by sevens. How many things are there?

On division by 12 the numbers with partners are 1, 5, 7 and 11, and each is its own partner, since 25, 49 and 121 all leave remainder 1. On division by 7 the numbers with partners are 1 to 6, and most are not their own partners: 2 pairs with 4 and 3 with 5. In the multiplication table on division by 7, the row of each nonzero family contains every nonzero family exactly once. In the table on division by 12, zeros appear where neither factor is zero, for instance at $3 \cdot 4$ and at $2 \cdot 6$: two families, neither of them 0, multiply to 0, as $(1 + i)(1 - i)$ did in the arithmetic with $i^2 = -1$.

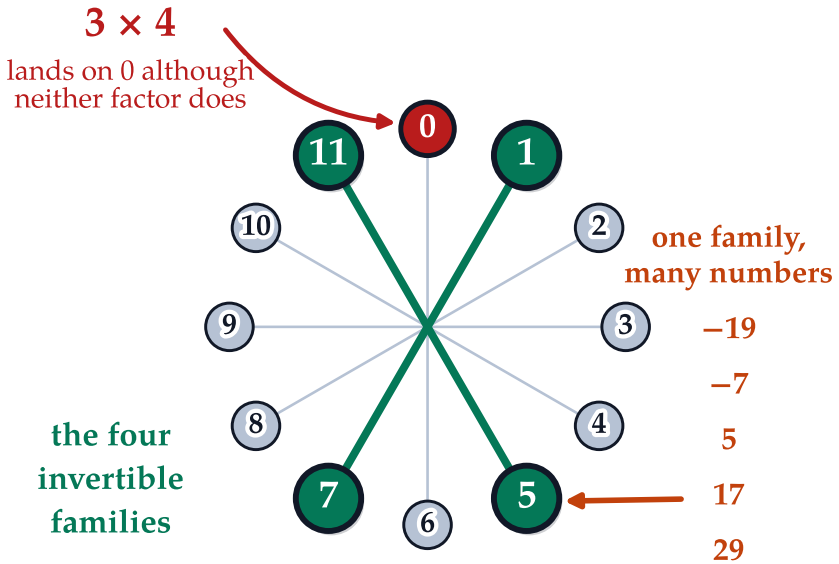


Figure 22.2. Division by twelve sorts every integer into twelve families; the four filled spokes are the families with multiplicative partners, and they are exactly the ones sharing no factor with twelve. *Assumes:* that the families are the objects being added and multiplied, which is legitimate only because the operations were shown not to depend on which member is picked off a spoke

TWO CLOCKS AT ONCE

The problem from the Sunzi Suanjing asks for a number that leaves 2 when counted by threes, 3 when counted by fives and 2 when counted by sevens. The old text gives the answer, 23, and also a recipe: for each unit left over by threes put down 70, for each left over by fives put down 21, for each left over by sevens put down 15, add, and take away 105 as often as possible. Here that is $2 \cdot 70 + 3 \cdot 21 + 2 \cdot 15 = 233$, and $233 - 210 = 23$. Why does the recipe work, and would it work for any three remainders?

Look at what the three numbers are. The number 70 is a multiple of 5 and of 7 that leaves 1 on division by 3. The number 21 is a multiple of 3 and of 7 that leaves 1 on division by 5, and 15 is a multiple of 3 and of 5 that leaves 1 on division by 7. So in $a \cdot 70 + b \cdot 21 + c \cdot 15$, seen modulo 3, the last two terms vanish and the first is a ; modulo 5 only b survives, and modulo 7 only c . Each of the three numbers speaks to one modulus and is silent to the others. And $105 = 3 \cdot 5 \cdot 7$ is silent to all three, so taking it away changes nothing. The recipe works for any remainders a , b and c .

Where do 70, 21 and 15 come from? The number for the modulus 3 has to be a multiple of 5 and of 7, so it is a multiple of 35, and it has to leave 1 on division by 3. The multiples of 35 are 35, 70, 105, ..., which modulo 3 are 2, 1, 0, ..., so the one wanted is 70. Put generally, the one wanted is 35 times the partner of 35 modulo 3:

the partner is 2, and $35 \cdot 2 = 70$. In the same way 21 is 21 times the partner of 21 modulo 5, and 15 is 15 times the partner of 15 modulo 7; both of those partners happen to be 1. A partner exists because 35 and 3 have no common divisor. That is the only thing the recipe needs.

* * *

Two facts about numbers without a common divisor are used throughout, and both come from writing 1 as a combination. Let m and n have no common divisor, so that $1 = mx + ny$ for some integers x and y . First, if m divides a product kn , then m divides k : multiply the combination by k to get $k = mkx + kny$, in which both terms are multiples of m , the second because m divides kn . This is the lemma about primes again, with the prime replaced by any number that shares nothing with n . Second, if m and n both divide N , then mn divides N . Write $N = ns$. Then m divides ns , so by the first fact m divides s , say $s = mt$, and $N = mnt$ is a multiple of mn .

If m and n have no common divisor, then for every pair of remainders a and b there is exactly one number from 0 to $mn - 1$ that leaves a on division by m and b on division by n .

There are two proofs, and they are of different kinds. The first counts. If two numbers from 0 to $mn - 1$ leave the same pair of remainders, their difference is divisible by m and by n , so by mn ; and a difference of two numbers in that range is smaller than mn , so it is 0. So the mn numbers leave mn different pairs. But there are only mn pairs in all, m choices for a and n for b . So every pair occurs, once. This proof shows that the number exists and gives no hint of how to find it, short of trying all mn candidates.

The second proof builds the number. Let u be n times the partner of n modulo m , so that $u \equiv 1 \pmod{m}$ and $u \equiv 0 \pmod{n}$, and let v be m times the partner of m modulo n . Then $au + bv$ leaves a on division by m , because modulo m the term au is a and the term bv is 0; in the same way it leaves b on division by n . Its remainder on

division by mn is the number wanted. It is the only one: the counting proof began by showing that two numbers from 0 to $mn - 1$ with the same pair of remainders are equal, and that step counted nothing, so it can be used here. The partners come from Euclid's algorithm, so the number can be found for moduli of hundreds of digits, where no search could ever finish.

	0	1	2	3	4	5	6	7	8
0	0	28	20	12	4	32	24	16	8
1	9	1	29	21	13	5	33	25	17
2	18	10	2	30	22	14	6	34	26
3	27	19	11	3	31	23	15	7	35

4 and 9: every pair of remainders, once

	0	1	2	3	4	5
0	0, 12		8, 20		4, 16	
1		1, 13		9, 21		5, 17
2	6, 18		2, 14		10, 22	
3		7, 19		3, 15		11, 23

4 and 6: half the pairs, twice

Figure 23.1. Each number is written in the row of its remainder on division by 4 and the column of its remainder on division by 9, on the left, or by 6, on the right. With 4 and 9 the numbers from 0 to 35 fill every cell once. With 4 and 6, which share the divisor 2, the numbers from 0 to 23 land two to a cell, and half the cells stay empty. *Assumes: two small examples; the text proves that the left-hand pattern occurs whenever the two moduli have no common divisor, and only then*

The statement is known as the Chinese remainder theorem. With three or more moduli, no two with a common divisor, it follows by taking them two at a time. Combine the conditions modulo m and modulo n into a single condition modulo mn , and then combine that with the condition for a third modulus r . The second step needs mn and r to have no common divisor, and they have none: a prime that divided both r and mn would divide m or n , by the lemma about primes, and r has no common divisor with either. The theorem has long been a working tool of calendar makers, because a calendar is a set of clocks of different sizes turning together. In 1583 Scaliger combined three cycles then used for dating, one of 28 years after which weekdays return to the same dates, one of 19 years after which new moons do, and a Roman tax cycle of 15 years. No two of 28, 19 and 15 have a common divisor,

so the three positions together determine a year within a span of $28 \cdot 19 \cdot 15 = 7980$ years. Scaliger asked when all three cycles were last at their first year together. Count the years from such a year, calling it year 1; then year N stands in each cycle at the place given by its remainder on division by the cycle's length, a remainder of 0 meaning the last place. The year AD 1 stood tenth in the cycle of 28, second in the cycle of 19 and fourth in the cycle of 15, and the one number from 1 to 7980 that leaves 10, 2 and 4 on division by 28, 19 and 15 is 4714. So the three cycles were last together at their first year 4713 years before AD 1, which is 4713 BC, there being no year 0. Astronomers still number the days from the first of January of that year.

What if the moduli do have a common divisor? A number that leaves 1 on division by 4 is odd, and a number that leaves 2 on division by 6 is even, so no number does both. Each of the two clocks tells whether the number is odd or even, and on that point they have to tell the same thing. In general two clocks overlap in what they say, and what they say in common must agree.

Let g be the greatest common divisor of m and n . If x leaves a on division by m and b on division by n , then $a - b$ is $(x - b) - (x - a)$, a multiple of n less a multiple of m . Since g divides both m and n , g divides $a - b$. And that condition is enough. Write g as a combination, $g = mx' + ny'$, and let $a - b = gt$. Then $a - b = gt = mx't + ny't$, and moving $mx't$ to the left gives $a - mx't = b + ny't$. That number differs from a by a multiple of m and from b by a multiple of n , so it is a solution.

Two solutions differ by a number that m divides and n divides, that is, by a common multiple of m and n . Let L be the least common multiple. Every common multiple is a multiple of L : its remainder on division by L is again a common multiple and is smaller than L , so it is 0. So any two solutions differ by a multiple of L . Adding or taking away L turns a solution into a solution, so some solution lies among the numbers from 0 to $L - 1$; and two solutions in that range differ by less than L , so they are equal. There is exactly one solution from 0 to $L - 1$. For the remainders 1 on division by 4 and

3 on division by 6 the answer is 9, and then 21, 33, ..., at steps of 12.

Periodical cicadas spend 13 or 17 years underground, according to their kind, and come out together in millions. One explanation offered for the prime numbers is this arithmetic. Take a predator or parasite whose own numbers rise and fall with a period of 2, 3, 4 or 6 years. Each of those periods divides 12, so once its peak had fallen on an emergence of a 12-year cicada it would fall on every later one. With 13 years it is otherwise. A cycle of c years, where c is not itself a multiple of 13, has no common divisor with 13, so its peak coincides with the emergence only once in $13c$ years. And a 13-year brood and a 17-year brood in the same wood emerge together once in 221 years.

* * *

The theorem says more than that a number can be recovered from its remainders. Adding and multiplying respect congruence for each modulus separately, so a sum or a product can be read off clock by clock: what a sum shows on each clock is the sum of the two readings on that clock, and what a product shows is their product. A number modulo 60 is a reading of three clocks, with 3, 4 and 5 hours, and arithmetic modulo 60 is arithmetic on the three clocks independently:

*47 is (2, 3, 2), 23 is (2, 3, 3), and $47 \cdot 23$ is (4, 9, 6), which is (1, 1, 1)
and indeed $47 \cdot 23 = 1081$, which is $18 \cdot 60 + 1$*

A question about a large modulus falls apart into questions about small ones. Which numbers have a square that ends in the number itself? The digit 5 does, since 25 ends in 5, and 6 does; with two digits, $25^2 = 625$ and $76^2 = 5776$. The condition for numbers of k digits is $x^2 \equiv x$ modulo 10^k , that is, 10^k divides $x(x - 1)$. Now 10^k is 2^k times 5^k , two numbers without common divisor, so the condition is the same as two conditions at once: 2^k divides $x(x - 1)$, and 5^k does. Since x and $x - 1$ have no common divisor, 5 cannot divide both, so all the factors 5 of the product lie

in one of them. Then 5^k divides x or divides $x - 1$, which says that x is 0 or 1 modulo 5^k ; likewise x is 0 or 1 modulo 2^k . That gives four pairs of remainders, and by the theorem four solutions, for every k . The pairs (0, 0) and (1, 1) are the numbers 0 and 1. The other two are

*5 and 6, 25 and 76, 625 and 376, 0625 and 9376, 90625 and 09376,
890625 and 109376*

and indeed $376^2 = 141376$ and $9376^2 = 87909376$. The two numbers at each stage add up to $10^k + 1$. Their pairs of remainders, (0, 1) and (1, 0), add to (1, 1), so the sum leaves 1 on division by 10^k ; and since each of the two numbers is above 1 and below 10^k , the sum is above 1 and below $2 \cdot 10^k$, where the only number leaving 1 is $10^k + 1$. Their product ends in k zeros, because the pairs multiply to (0, 0). And each stage keeps the digits of the one before and adds a digit on the left, because a solution modulo 10^{k+1} is also a solution modulo 10^k , with the same pair of remainders, and there is only one of those, so its last k digits are the solution of the stage before. So there are two endless strings of digits, ...890625 and ...109376, running off to the left, each of which is its own square to as many places as are kept, and whose product is 0 to as many places as are kept, though neither is 0. Arithmetic on strings that run to the left is a sound arithmetic. The last k digits of a sum or a product depend only on the last k digits of the terms, so every digit of the result is settled after finitely many steps. And each rule of arithmetic holds for the strings because it holds modulo every 10^k . Those two strings, and their product 0, exist because 10 is 2 times 5.

* * *

The theorem is a statement about chance as well. Pick a number at random from 1 to 15. The chance that 3 divides it is $5/15$, the chance that 5 divides it is $3/15$, and the chance that both do is $1/15$, which is the product. Divisibility by 3 and divisibility by 5 are independent in the exact sense the word was given for tosses

of a coin, and so are any two statements about the remainder on division by 3 and the remainder on division by 5: every pair of remainders occurs once in 15, so knowing one remainder leaves the other as uncertain as it was.

How likely are two numbers, chosen at random, to have no common divisor? They fail if some prime divides both. For a prime p the chance that it divides the first number is $1/p$, and the same for the second, so the chance that it divides both is $1/p^2$, and the chance that it does not divide both is $1 - 1/p^2$. For different primes these events are independent, by the theorem. So the chance that no prime divides both should be the product

$$(1 - 1/4)(1 - 1/9)(1 - 1/25)(1 - 1/49)(1 - 1/121) \dots$$

The primes up to 100 bring it to 0.6090, and further primes barely move it; it settles at 0.6079. Among the million pairs of numbers from 1 to 1000, exactly 608383 have no common divisor. The argument is a sketch, since choosing at random from all the whole numbers has not been given a meaning, but the number is right: Dirichlet proved in 1849 that among the pairs of numbers from 1 to N the proportion with no common divisor approaches it as N grows. Euler showed, by methods of the calculus, that this product is 6 divided by the square of 3.14159..., the proportion of the circumference of a circle to its diameter.

The same independence counts the families with partners. Write $\phi(n)$ for the number of families with partners modulo n , that is, the number of whole numbers from 1 to n that have no common divisor with n ; Euler introduced the count in 1763. A number has no common divisor with mn exactly when it has none with m and none with n , since a prime dividing mn divides m or n , and a divisor of m or of n divides mn . Now let m and n have no common divisor. The theorem matches each number from 0 to $mn - 1$ with its pair of remainders. As shown above, the number has no common divisor with mn exactly when it has none with m and none with n . Whether it has one with m depends only on its

remainder modulo m , since the number and m have the same common divisors as that remainder and m , as in Euclid's algorithm; in the same way, whether it has one with n depends only on its remainder modulo n . So the families with partners modulo mn correspond to the pairs made of one such family modulo m and one modulo n , and

$$\varphi(mn) = \varphi(m) \cdot \varphi(n) \text{ when } m \text{ and } n \text{ have no common divisor.}$$

For a power of a prime the count is immediate: of the numbers from 1 to p^k , those with a common divisor with p^k are the multiples of p , of which there are p^{k-1} , so $\varphi(p^k) = p^k - p^{k-1}$. Now $360 = 8 \cdot 9 \cdot 5$, and no two of these prime powers have a common divisor, so $\varphi(360) = \varphi(8) \cdot \varphi(9) \cdot \varphi(5) = 4 \cdot 6 \cdot 4 = 96$. That answers the earlier question of how many numbers from 1 to 360 share no factor with 360, and adding and subtracting the overlapping counts of multiples of 2, 3 and 5 gives the same number: $360 - 180 - 120 - 72 + 60 + 36 + 24 - 12 = 96$. The two methods agree in general. Each prime power contributes $p^k(1 - 1/p)$, so

$$\varphi(n) = n(1 - 1/p)(1 - 1/q) \dots$$

over the different primes $p, q, \dots$ that divide n

and multiplying out the brackets gives $n - n/p - n/q + n/pq - \dots$, which is the count by overlaps term for term.

List the fractions $1/12, 2/12, \dots, 12/12$ and put each in lowest terms:

$$1/12, 1/6, 1/4, 1/3, 5/12, 1/2, 7/12, 2/3, 3/4, 5/6, 11/12, 1/1$$

The denominators that appear are the divisors of 12. For a given divisor d , every fraction c/d in lowest terms with c from 1 to d is in the list, since d divides 12 and so c/d is some number of twelfths; and the list holds no other fraction with denominator d . There are $\varphi(d)$ of them. Counting the twelve fractions by their denominators gives $\varphi(12) + \varphi(6) + \varphi(4) + \varphi(3) + \varphi(2) + \varphi(1) = 4 + 2 + 2 + 2 + 1 + 1 = 12$, and nothing was special about 12:

the values of φ at the divisors of n add up to n .

A basket of eggs, in a problem that Brahmagupta set, leaves one egg over when the eggs are taken out 2 at a time, and also 3, 4, 5 or 6 at a time, and none over when they are taken out 7 at a time. What is the smallest number of eggs it can hold?

Which numbers leave 5 on division by 6, 5 on division by 10 and 2 on division by 15, and what changes if the last remainder is 5?

In base twelve, how many numbers of k digits have squares ending in themselves, and in which bases is the answer only 0 and 1?

Which numbers n have $\varphi(n) = 12$, and why is $\varphi(n)$ even for every n above 2?

For which moduli n is $x^3 \equiv x \pmod{n}$ for every x ?

WHEN DIVISION SURVIVES

On division by 12, three times four is zero, although neither factor is. Cancellation, the rule that $ab = ac$ with a not 0 forces $b = c$, fails too: $3 \cdot 4$ and $3 \cdot 8$ are both 0, and 4 and 8 are different families. The two failures are one failure. The equation $ab = ac$ says the same as $a(b - c) = 0$, and b differs from c exactly when $b - c$ is not 0. So cancelling a fails exactly when a multiplies some nonzero family to 0.

A nonzero family that multiplies some nonzero family to 0 is a *zero divisor*. On division by 12 the zero divisors are 2, 3, 4, 6, 8, 9 and 10, and the others are 1, 5, 7 and 11, the families sharing no factor with 12, which are exactly the families with partners. For a nonzero family a on division by n , one of two things happens. If a shares no factor with n , it has a partner, and it is not a zero divisor, since multiplying $ab = 0$ by the partner gives $b = 0$. If a shares a factor d above 1 with n , write $a = dk$; then a times n/d is k times n , a multiple of n , and so is 0, while n/d is not 0, being a whole number from 1 to $n - 1$; so a is a zero divisor. Every nonzero family has a partner or is a zero divisor, never both, and the greatest common divisor decides which.

For the modulus itself: if $n = rs$ with r and s above 1, then r and s are nonzero families whose product is 0. If n is prime, every nonzero family shares no factor with n , so all have partners and

none is a zero divisor. A prime modulus, cancellation, the absence of zero divisors, and partners for every nonzero family are four descriptions of one situation. Cancellation also explains why the row of each nonzero family a in the table on division by 7 contains each nonzero family once. Two equal entries in that row would say $ab = ac$, and cancelling a would force $b = c$, so the six entries are all different. None of them is 0, there being no zero divisors. Six different nonzero entries are the six nonzero families, each once.

under division by 7

	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

every row is a shuffle
no zeros, cancellation holds

under division by 12

	1	2	3	4	5	6	7	8	9	10	11
1	1	2	3	4	5	6	7	8	9	10	11
2	2	4	6	8	10	0	2	4	6	8	10
3	3	6	9	0	3	6	9	0	3	6	9
4	4	8	0	4	8	0	4	8	0	4	8
5	5	10	3	8	1	6	11	4	9	2	7
6	6	0	6	0	6	0	6	0	6	0	6
7	7	2	9	4	11	6	1	8	3	10	5
8	8	4	0	8	4	0	8	4	0	8	4
9	9	6	3	0	9	6	3	0	9	6	3
10	10	8	6	4	2	0	10	8	6	4	2
11	11	10	9	8	7	6	5	4	3	2	1

seventeen leaks
a product of two nonzero families is zero

Figure 24.1. Multiplication of the nonzero families under division by seven and by twelve: on the left every row is a rearrangement of the same six entries, on the right the marked cells hold a product of two nonzero families that comes to zero. *Assumes:* that reading a row as a rearrangement is the same claim as cancellation holding, which is proved in the text and not visible in the grid itself

* * *

Familiar rules of algebra are consequences of cancellation, and they fail without it. A quadratic equation has at most two solutions, and the usual reason is a chain: if $x^2 = 1$ then $(x - 1)(x + 1) = 0$, so one factor is 0, so x is 1 or -1 . The step from a zero product to

a zero factor is the absence of zero divisors. With that step, every quadratic $ax^2 + bx + c$ with a not 0 has at most two roots. Let r be a root. Subtracting $ar^2 + br + c = 0$ from $ax^2 + bx + c = 0$ leaves $a(x^2 - r^2) + b(x - r) = 0$, and since $x^2 - r^2 = (x - r)(x + r)$, this is $(x - r)(a(x + r) + b) = 0$. A root other than r makes the first factor nonzero, so it must make the second factor 0. If two roots s and t both did that, subtracting $a(t + r) + b = 0$ from $a(s + r) + b = 0$ would give $a(s - t) = 0$, and since a is not 0, $s = t$. So besides r there is at most one more root.

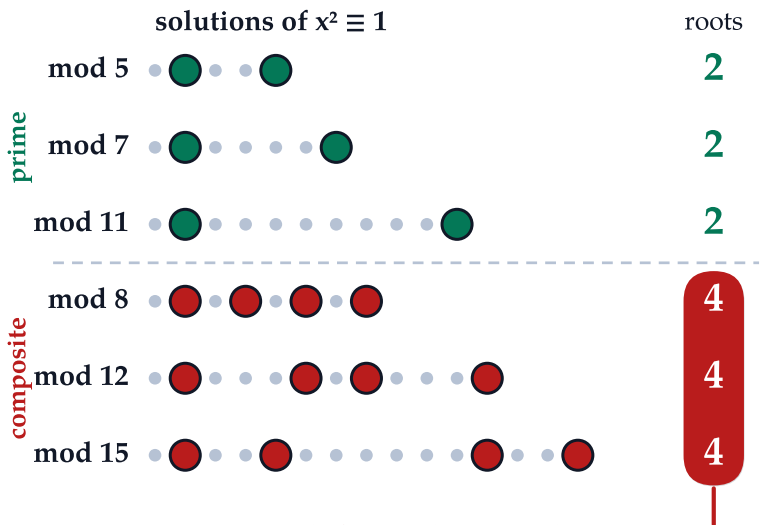
On division by 8 the equation $x^2 = 1$ has four solutions, 1, 3, 5 and 7, since 9, 25 and 49 leave remainder 1. The argument breaks exactly at the predicted step: for $x = 3$ the two factors are 2 and 4, whose product 8 is 0, while neither is 0. When a familiar result fails in a new setting, the thing to do is walk its proof through in that setting one step at a time. Some step is the first to fail, and that step names the hypothesis the result needed.

The self-partnered families on division by 12 are explained the same way. A family with a partner on division by 12 is an odd number x not divisible by 3. Then $x - 1$ and $x + 1$ are consecutive even numbers, so one of them is divisible by 4 and their product by 8; and of the three consecutive numbers $x - 1$, x and $x + 1$ one is divisible by 3, and it is not x . So $(x - 1)(x + 1)$ is divisible by 12, which says $x^2 \equiv 1$. On division by 24 the same argument works, since the product is divisible by 8 and by 3; on division by 5 it fails, since $2 \cdot 2 = 4$.

For which moduli above 1 is every family with a partner its own partner?

On division by a prime the self-partnered families are as few as they can be: $x^2 \equiv 1$ forces $x \equiv 1$ or $x \equiv -1$, there being no zero divisors. So in the product $1 \cdot 2 \cdot 3 \cdots (p - 1)$ every family from 2 to $p - 2$ can be paired off with its partner, a different family in the same range, and each pair multiplies to 1. What remains is $1 \cdot (p - 1)$, and $p - 1$ is the family of -1 , so

$$(p - 1)! \equiv -1 \pmod{p} \text{ for every prime } p$$



a degree-two equation with four solutions, three times over

Figure 24.2. Solutions of x squared equal to 1 under six different moduli: two for each prime drawn and four for each composite drawn, so the rule that a quadratic has at most two roots is a fact about the system rather than about the equation. *Assumes:* that the extra roots come from zero divisors and not from some accident of these six moduli, which is what the accompanying proof establishes and the picture cannot

the primes 2 and 3 being checked directly, since $1! \equiv -1$ modulo 2 and $2! \equiv -1$ modulo 3. On division by 13 the pairs are 2 with 7, 3 with 9, 4 with 10, 5 with 8 and 6 with 11, and $12! + 1 = 479001601 = 13 \cdot 36846277$. The converse holds as well. If n is composite, a divisor d of n strictly between 1 and n is one of the factors of $(n-1)!$, so d divides $(n-1)!$ and cannot divide the next number, $(n-1)! + 1$. If n divided $(n-1)! + 1$, so would its divisor d ; so n does not divide it either. So $(n-1)! \equiv -1$ is an exact test for primes, and a useless one, the factorial taking far longer to compute than a search for divisors. On division by 12, where 1, 5, 7 and 11 are all their own partners, the product $1 \cdot 5 \cdot 7 \cdot 11 = 385$ is congruent to 1, not -1 . Waring published the theorem in 1770, crediting it to his student John Wilson, with no proof; Lagrange proved it the next year.

* * *

The proofs just given used little about remainders: two operations obeying the usual rules, the absence of zero divisors, and, for the factorial, a partner for every nonzero family and only finitely many families. They run unchanged in any system with addition and multiplication obeying the usual rules, provided every element other than 0 has a partner and the elements are finite in number. The partners are needed to pair off the factors of the factorial, and the finiteness is needed for the product of all the nonzero elements to be a finite product. A system with two such operations, in which 1 is not 0 and every nonzero element has a partner under multiplication, is called a *field*. Three are in hand: the remainders on division by a prime p , written $\mathbb{Z}/p$; the fractions, written $\mathbb{Q}$; and the pairs $a + bi$ of fractions with $i^2 = -1$, which form a field because a nonzero pair times $a - bi$ gives $a^2 + b^2$, a nonzero fraction, so that $(a - bi)/(a^2 + b^2)$ is its partner. With $i^2 = +1$ the same step gives $a^2 - b^2$, which is 0 for the pair $1 + i$: then $(1 + i)(1 - i) = 0$ with neither factor 0, so $1 + i$ is a zero divisor, has no partner, and there is no field. $\mathbb{Z}/n$ is a field exactly when n is prime.

* * *

On division by 7 the six families with partners form a group under multiplication: any two multiply to another of them, grouping does not matter since it does not for the integers, each has its partner among them, and 1 changes nothing. Inside it, 1, 2 and 4 form a group of their own, since $2 \cdot 2 = 4$, $2 \cdot 4 = 8 \equiv 1$ and $4 \cdot 4 = 16 \equiv 2$. In a finite group the powers of any element form a subgroup. The powers $a, a^2, a^3, \dots$ cannot all be different, so two must agree, say $a^i = a^j$ with i below j , and cancelling a^i gives $a^{j-i} = 1$. So some power of a is 1; let a^m be the first. The powers $a, a^2, \dots, a^m$ are all different, since two of them agreeing would, by the same cancelling, put a power earlier than a^m at 1. After a^m the powers repeat, and the partner of a^k is a^{m-k} , since the two multiply to $a^m = 1$. The number

m of distinct powers is the *order* of a . The families 1 and 6 form a subgroup too, since $6 \cdot 6 = 36 \equiv 1$. So subgroups of sizes 3 and 2 sit inside a group of six, as the four rotations of a square sat inside its eight symmetries.

Multiply each of 1, 2, 4 by 3: the results 3, 6, 12 reduce to 3, 6, 5, exactly the three families the subgroup does not contain. The subgroup and this translate account for all six elements, each once. Multiply 1 and 6 by 2 to get 2 and 5, and by 3 to get 3 and 4: three blocks of two tile the same six elements differently.

the six families with partners on division by 7, tiled two ways

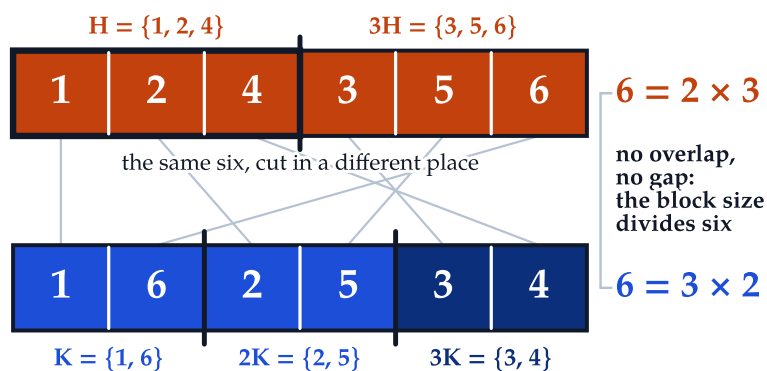


Figure 24.3. The six families with partners on division by 7, tiled twice, first by the two cosets of the subgroup $\{1, 2, 4\}$ and then by the three cosets of the subgroup $\{1, 6\}$, so the same six fall into 2 blocks of 3 and into 3 blocks of 2. Assumes: that every coset really has the same size as the subgroup, which the picture shows for one case and the argument must establish in general

A translate of a subgroup H by an element g , the collection of g combined with each member of H , is a *coset*, and the cosets always tile the group. Each coset has as many elements as H , since $gh = gv$ forces $h = v$ after combining both with the partner of g . And two cosets are either identical or share nothing. Suppose x is in the coset of g and in the coset of k , so that $x = gh = kh'$ for members h and h' of H . Combining both sides of $gh = kh'$ with the partner of h

gives g as k combined with h' and then with the partner of h . Now h' combined with the partner of h is a member of H , call it u , so $g = ku$. Then g combined with any member w of H is k combined with uw , and uw is again a member of H . So the coset of g lies inside the coset of k , and the same argument with g and k exchanged gives the reverse. Every element lies in its own coset, the identity being a member of H . So for a finite group, the size of a subgroup divides the size of the group, the group being a whole number of blocks of equal size. This is Lagrange's theorem.

* * *

The order of an element is the size of the subgroup of its powers, so it divides the size of the group. If an element a has order m and the group has $N = mq$ elements, then a combined with itself N times is a^m combined with itself q times, and a^m is the identity. So combining any element with itself as many times as the group has elements gives the identity. On division by 7 the orders of the six families are 1, 2, 3, 3, 6 and 6, all divisors of 6. The powers of 3 are 3, 2, 6, 4, 5 and 1, so 3 has order 6 and every family with a partner is a power of 3; the same is true of 5, while 2 and 4 run through three families each and 6 through two. In the group of the $p - 1$ families with partners on division by a prime p , every element raised to the power $p - 1$ is therefore 1:

$$a^{p-1} \equiv 1 \pmod{p} \text{ for every prime } p \text{ and every } a \text{ not divisible by } p.$$

A second proof counts strings. For a whole number a at least 1, the strings of length p in an alphabet of a letters number a^p , and a of them use a single letter. Rotating a string by p places returns it. If rotating it by some k from 1 to $p - 1$ also returned it, so would rotating it by any combination of k and p with integer coefficients. Now k , being below the prime p , shares no factor with it, and the greatest common divisor of two numbers is a combination of them, so 1 is such a combination; so rotating by one place would return it, and the string would be a single letter repeated. So a string using

more than one letter has p different rotations: if rotating by i places and by j places gave the same string, with i below j and both from 0 to $p - 1$, then rotating by $j - i$ places, a number from 1 to $p - 1$, would return the string.

The $a^p - a$ strings using more than one letter therefore fall into blocks of exactly p , each string grouped with its rotations. So p divides $a^p - a$, which is $a \cdot (a^{p-1} - 1)$. When p does not divide a , the lemma about primes gives that p divides $a^{p-1} - 1$. The count was made for a at least 1, and that is enough: an integer a not divisible by p is congruent to its remainder, a whole number from 1 to $p - 1$, and congruent numbers have congruent powers.

The theorem can prove that a number is composite without finding a factor. Since $2^{12} = 4096 = 45 \cdot 91 + 1$, the power 2^{12} is 1 modulo 91. Since $90 = 7 \cdot 12 + 6$, the power 2^{90} is the seventh power of 2^{12} times 2^6 , which is $2^6 = 64$ modulo 91, not 1. Were 91 prime, the theorem would make 2^{90} equal to 1 modulo 91; so 91 is not prime, and the argument never found the factors 7 and 13. The converse of the theorem fails, though. Since $2^{10} = 1024 = 3 \cdot 341 + 1$, the power 2^{340} is 1 modulo 341, and yet $341 = 11 \cdot 31$, as Sarrus noticed in 1819. A group shows why. The families with partners modulo any n form a group under multiplication, the partner of ab being the partner of b times the partner of a ; in that group modulo 341 the family of 2 has order 10, and 10 happens to divide 340. Another base exposes 341: modulo 31, $3^{30} \equiv 1$ and $340 = 11 \cdot 30 + 10$, so $3^{340} \equiv 3^{10}$, which is 25 modulo 31. A number that is 1 modulo 341 is also 1 modulo 31, since 31 divides 341; so 3^{340} is not 1 modulo 341. Some composite numbers pass with every base that has a partner. Why is $a^{560} \equiv 1$ on division by $561 = 3 \cdot 11 \cdot 17$ for every a sharing no factor with 561?

A group with a prime number of elements consists of the powers of any element other than the identity. The order of that element divides the prime and is not 1, so it is the prime itself, and the element has as many different powers as the group has elements.

Which subgroups does the group of the twelve families with partners on division by 13 have, and what are their sizes?

* * *

Does every divisor of the size of a group occur as the size of a subgroup? No. The rotations of a regular tetrahedron form a group of twelve with no subgroup of six.

The rotations first. A rotation carrying the tetrahedron onto itself sends corners to corners. A corner is the one kind of point of the solid that lies between no two other points of it, and a motion carrying the solid onto itself keeps that description, since a point lies between two others exactly when its distances to them add up to their distance apart, and the motion keeps distances. There is the identity; for each of the four corners there are two rotations, by a third and by two thirds of a turn, about the axis through that corner and the centre of the opposite face, making eight; and for each of the three pairs of opposite edges there is a half turn about the axis through their midpoints, making three. That is twelve, and no two of them send the corners to the same places. There are no others, and performing one after another gives one of them.

The reason for both is one fact. Number the corners 1 to 4, and say that a motion turns the solid over when the images of corners 1, 2 and 3, seen from the image of corner 4, run the other way round from 1, 2 and 3 seen from 4, as they do after a reflection in a plane; a rotation never does this, even a half turn that brings the bottom to the top. The fact is that a motion that carries the solid onto itself, keeps distances and does not turn the solid over is settled by where it sends corners 1 and 2, and they can go to only twelve ordered pairs of corners. So there are at most twelve such motions. The twelve rotations are such motions and are all different, so they are all of them; and one rotation followed by another is again such a motion, so it is one of the twelve.

It remains to establish the fact. For if two such motions send corners 1 and 2 to the same pair, performing the first and then undoing the second keeps distances, does not turn the solid over, and carries the solid onto itself. It therefore carries corners to corners, so it leaves in place the one point at equal distance from all four, the

centre; and it leaves corners 1 and 2 in place. That is three points not on a line. Such a motion leaves every point of their plane in place, since a point of that plane is settled by its distances to the three. A point off the plane has the same distances to the three as its mirror image in the plane and as no other point, so it stays or goes to its mirror image. And the points off the plane all do the same thing: if some went to their mirror images while others stayed, the distance between a point that went and a point that stayed would change. So the motion is the identity or the reflection in the plane, and the reflection turns the solid over. So it is the identity, and the first motion is the second.

Suppose a subgroup H of six rotations existed. It would tile the group into two blocks, H and its complement. Take any rotation g . If g is outside H , the coset of g has six elements, none of them in H , since gh in H would put g in H ; so the coset of g is the complement. If g is in H , so is g^2 . If g is outside H and g^2 were outside H too, then g^2 would be in the complement, which is the coset of g , so $g^2 = gh$ for some h in H , and cancelling g gives $g = h$, which is in H , against the choice of g . So g^2 lies in H whether g is in H or not. Now each rotation by a third of a turn is the square of the rotation by two thirds about the same axis, and each rotation by two thirds is the square of the one by a third, so all eight of these rotations are squares and lie in H . Eight do not fit in six, and no subgroup of six exists. So the converse of Lagrange's theorem, that every divisor is the size of some subgroup, is false.

Two more facts follow from tiling. The rotations of a square are four of its eight symmetries, and their other coset is the four flips. And every group of four elements is one of two. The order of each element divides four. If some element has order 4, the group is its powers. Otherwise every element other than the identity has order 2, so every element is its own partner. Take two different elements a and b other than the identity. Their product is not a , since cancelling a would make b the identity, and likewise it is not b ; and it is not the identity, since that would make b the partner of a , which is a itself. So the product is the remaining element, and

When division survives

the whole table is forced.

POWERS THAT COME BACK

Divide 1 by 7 by long division. Ten sevenths is 1 and 3 over; thirty is 4 sevens and 2 over; twenty is 2 and 6 over; sixty is 8 and 4 over; forty is 5 and 5 over; fifty is 7 and 1 over. The remainder is back at 1, where it started, so everything repeats:

$$1/7 = 0.142857142857142857...$$

Each step multiplied the remainder by 10 and reduced it modulo 7. So the remainders 1, 3, 2, 6, 4, 5 are the powers of 10 modulo 7, and the length of the repeating block is the number of steps before a power of 10 comes back to 1. That number has a name already: it is the order of the family of 10 in the group of families with partners modulo 7.

The decimal for $1/n$, when n has no common divisor with 10, repeats from the start, and the length of its block is the order of 10 modulo n .

So the length divides the size of that group, by Lagrange's theorem, and for a prime p it divides $p - 1$. The blocks for 3, 11, 13, 17, 19, 37, 41 and 101 have lengths 1, 2, 6, 16, 18, 3, 5 and 4, and each divides the prime less 1. When the block has length k , 10^k is 1 modulo the prime, which says that the prime divides $10^k - 1$, the number written with k nines. In the other direction, a prime that

divides $10^k - 1$ has $10^k \equiv 1$, so its remainder is back at 1 after k steps. The remainders go round one cycle, as long as the block, and come back to 1 only after a whole number of turns, so the length of the block divides k . So the primes with a block of length 6 or a divisor of 6 are the prime divisors of $999999 = 3^3 \cdot 7 \cdot 11 \cdot 13 \cdot 37$. The block itself is the quotient: the six steps of the long division of 1 by 7 divide 10^6 by 7, with the block 142857 as quotient and 1 over, so $10^6 = 7 \cdot 142857 + 1$, which is $999999 = 7 \cdot 142857$.

The other sevenths use the same wheel. The decimal for $3/7$ begins the division at remainder 3, so it is the same cycle of digits entered one place later, $0.428571\dots$, and in the same way $2/7$, $6/7$, $4/7$ and $5/7$ enter at the other places. Each block is also a multiple of 142857: the six steps from remainder 3 divide $3 \cdot 10^6$ by 7 and leave 3 over, so the block of $3/7$ is $(3 \cdot 10^6 - 3)/7$, which is $3 \cdot 999999/7 = 3 \cdot 142857$, and the same holds for every numerator. That is why the multiples

142857, 285714, 428571, 571428, 714285, 857142

of 142857 by 1, 2, 3, 4, 5 and 6 are one string of digits turned round.

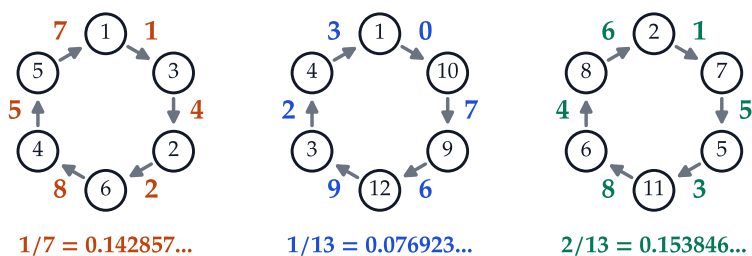


Figure 25.1. Long division by 7 and by 13 drawn as wheels. The circles hold the remainders, and each arrow carries the digit written when that remainder is multiplied by 10 and divided. Every seventh starts somewhere on the one wheel. The thirteenths need two wheels: the powers of 10 and that subgroup multiplied through by 2. Assumes: division in base ten; in another base the wheels have other sizes, since the order of the base is what sets them

With 13 the block has length 6, not 12: the powers of 10 modulo 13 are 1, 10, 9, 12, 3 and 4, and $1/13 = 0.076923\dots$. The fractions $10/13$, $9/13$, $12/13$, $3/13$ and $4/13$ turn that block round. But 2 is not among those remainders, and $2/13 = 0.153846\dots$ runs on a second wheel, through the remainders 2, 7, 5, 11, 6 and 8, shared by the other five thirteenthths. The first wheel is the subgroup of the powers of 10, and the second is its coset, the subgroup multiplied through by 2.

Add the two halves of the block for $1/7$: $142 + 857 = 999$. For $1/13$, $076 + 923 = 999$. For $1/17$, whose block is 0588235294117647, the halves 05882352 and 94117647 add up to 99999999. Let p be a prime whose block has an even length $2h$. The power 10^h is a square root of 1 modulo p , since its square is 10 to the power $2h$, which is 1. It is not 1, because the order of 10 is $2h$ and no smaller power of 10 is 1. And modulo a prime the only square roots of 1 are 1 and -1 : if p divides $x^2 - 1$, which is $(x - 1)(x + 1)$, then p divides one of the two factors, because there are no zero divisors modulo a prime. So $10^h \equiv -1$.

Each step multiplies the remainder by 10, so h steps multiply a remainder r by 10^h , which turns it into $-r$, and the remainder between 1 and $p - 1$ in that family is $p - r$. Now one step of the division, from remainder r , writes a digit d and leaves a next remainder r' , and the three are tied by $10r = pd + r'$. The step h places later starts from remainder $p - r$, writes some digit e , and leaves $p - r'$, so its line is $10(p - r) = pe + (p - r')$. Add the two lines: the r and r' cancel and leave $10p = p(d + e) + p$, so $d + e = 9$. Every digit and the digit half a block later add up to 9.

When the block for a prime p has the full length $p - 1$, the remainders run through every family from 1 to $p - 1$, and 10 is said to be a *primitive root* modulo p . Below 100 this happens for 7, 17, 19, 23, 29, 47, 59, 61 and 97. Whether it happens for infinitely many primes is not known. Artin conjectured in 1927 that it does, for a definite proportion of the primes, about 37 in every 100.

* * *

For a modulus that is not prime, the group of families with partners has $\varphi(n)$ members, and the same appeal to Lagrange's theorem gives what Euler proved in 1763:

$$a^{\varphi(n)} \equiv 1 \pmod{n} \text{ whenever } a \text{ and } n \text{ have no common divisor.}$$

For $n = 100$ there are $\varphi(100) = 40$ families with partners, so 3^{40} ends in 01 and the last two digits of the powers of 3 repeat after at most 40 steps. Since 1000 is $25 \cdot 40$, the power 3^{1000} is 3^{40} raised to the power 25, and it ends in 01 as well. The last three digits of 7^{9999} can be read off without computing anything large: $\varphi(1000) = 400$ divides 10000, so 7^{10000} ends in 001. But 7^{10000} is 7 times 7^{9999} , so 7^{9999} is the number that multiplies 7 to give 1 modulo 1000, the partner of 7. That partner is 143, since $7 \cdot 143 = 1001$, and so 7^{9999} ends in 143.

Such computations need a way to find a power modulo n without ever writing the power out in full. To raise 65 to the power 17 modulo 3233, square four times, reducing each time:

$$65^2 \equiv 992, 65^4 \equiv 1232, 65^8 \equiv 1547, 65^{16} \equiv 789, 65^{17} \equiv 789 \cdot 65 \equiv 2790$$

Five multiplications did the work of sixteen. In general, write the exponent in base two. Start from the number being raised, which accounts for the leading digit 1. Then take the remaining digits in order: at each digit square the running result, and where the digit is 1 multiply by the original number as well, reducing modulo n every time. The running result is always the number raised to the part of the exponent read so far: squaring doubles that part, which writes a 0 after its base-two digits, and the extra multiplication adds 1, which turns that 0 into 1. The exponent 17 is 10001 in base two, and the rule gives the four squarings and the one multiplication carried out above. For the exponent 2753, which is 101011000001 in base two, that is eleven squarings and four more multiplications in place of 2752 multiplications; for an exponent of three hundred decimal digits it is at most two thousand multiplications, where multiplying one at a time could not be finished in the life of the

universe. Raising to a power modulo n is easy. Undoing it is another matter, and on that difference rests a good part of modern commerce.

* * *

Until 1976 every cipher known in public required sender and receiver to share a secret key in advance. In 1977 Rivest, Shamir and Adleman published a cipher in which the key for enciphering is public, printed for anyone to use, and only the key for deciphering is secret. Cocks had found the same scheme in 1973 at the British signals agency, which kept it secret until 1997.

Choose two large primes p and q , different from each other, and let $n = pq$. Then $\varphi(n) = (p - 1)(q - 1)$, since p and q have no common divisor. Choose an exponent e with no common divisor with $\varphi(n)$, and let d be its partner modulo $\varphi(n)$, found by Euclid's algorithm run backwards. Publish n and e ; keep d , p and q secret. A message is a number m below n . To encipher it, anyone computes $c = m^e$ modulo n . To decipher, the owner of d computes c^d modulo n , and recovers m .

Here is why. Since $ed \equiv 1$ modulo $\varphi(n)$, write $ed = 1 + k\varphi(n)$, where $\varphi(n) = (p - 1)(q - 1)$. Work modulo p first. If p divides m , then m to the power ed and m are both 0 modulo p . If not, m to the power $p - 1$ is 1 modulo p . Now $k\varphi(n)$ is $(p - 1)$ times $k(q - 1)$, so m to the power $k\varphi(n)$ is m to the power $p - 1$ raised again to the power $k(q - 1)$, which is a power of 1 and so is 1 too. And m to the power ed is m times m to the power $k\varphi(n)$, which is m times 1. Either way m to the power ed and m agree modulo p , and by the same argument they agree modulo q . So p and q both divide the difference $m^{ed} - m$, and since p and q are different primes, their product pq divides it too:

$$m^{ed} \equiv m \pmod{n} \text{ for every } m.$$

With $p = 61$ and $q = 53$: $n = 3233$ and $\varphi(n) = 60 \cdot 52 = 3120$. Take $e = 17$; its partner modulo 3120 is $d = 2753$, since $17 \cdot 2753$ is

46801, which is $15 \cdot 3120 + 1$. The message 65 is enciphered as $65^{17} \equiv 2790$, the computation above, and the receiver finds 2790 to the power 2753 to be 65 modulo 3233.

What would an eavesdropper need? The direct way is to find d , and d is easy to find from $\varphi(n)$. But knowing $\varphi(n)$ is the same as knowing the factors. From $n = pq$ and $\varphi(n) = (p-1)(q-1)$ comes $p + q = n + 1 - \varphi(n)$, and two numbers with known sum and product are the roots of a quadratic: for $n = 3233$ and $\varphi(n) = 3120$ the sum is 114, and $x^2 - 114x + 3233 = 0$ gives $x = 57 \pm 4$, that is, 61 and 53. So finding $\varphi(n)$ is as hard as factoring n . Whether c can be turned back into m by some road that touches neither d nor the factors is not known either way, and the security of the method rests on the belief that it cannot. And factoring is hard, as far as anyone knows. A computer multiplies two primes of three hundred digits faster than a person could notice. For the reverse, the best methods known would need longer than the age of the universe; the largest number of this kind factored in public so far, in 2020, had 250 digits and took about 2700 years of processor time. No proof exists that a fast method cannot be found, and in 1994 Shor showed that a computer built on quantum principles, if one large enough can be made, would factor quickly.

The same keys sign documents. The owner raises a message to the secret power d ; anyone can raise the result to the public power e and see the message reappear, which only the owner of d could have arranged.

* * *

The scheme needs primes of hundreds of digits, and they are found by picking numbers at random and testing them. By the count of primes, about one number in $2.3d$ among those of d digits is prime, so a few hundred tries are enough. But how is a number of three hundred digits tested, when division by every prime up to its square root is out of the question?

The congruence $a^{p-1} \equiv 1$ modulo a prime p , proved twice al-

ready, was stated by Fermat in a letter of 1640, with no proof attached, and carries his name. It is a test that needs no factors: if a is a number from 1 to $n - 1$ and a to the power $n - 1$ is not 1 modulo n , then n is composite. When n is composite, how many bases expose it? Look only at the bases with partners modulo n . Those that fail to expose n , the ones whose power $n - 1$ is 1 modulo n , form a subgroup: if a and b both give 1 in the test, so do ab and the partner of a . By Lagrange's theorem the size of a subgroup divides the size of the group, and a divisor of a number, if it is not the whole number, is at most half of it. So a subgroup is the whole group or at most half of it. So if even one base with a partner exposes n , at least half of them do, and ten random bases all failing to expose it would be a coincidence of one chance in a thousand. The trouble is the composites that no such base exposes, of which 561 is the smallest. Carmichael found them in 1910, and in 1994 Alford, Granville and Pomerance proved that there are infinitely many.

A sharper test catches those too, and the extra ingredient is the fact that made the halves of a decimal block add up to nines: modulo a prime, 1 has no square roots but 1 and -1 . Write $n - 1$ as 2^s times an odd number t , and compute a to the power t , then square repeatedly, s times, arriving at a to the power $n - 1$. If n is prime, the chain ends in 1, and the entry before the first 1, if there is one, is a square root of 1 and so must be -1 . A chain that reaches 1 from anything else, or never reaches 1, proves n composite. Take $n = 341$ with $a = 2$. The plain test is fooled there, since $2^{340} \equiv 1$ modulo 341 although $341 = 11 \cdot 31$. Here $340 = 4 \cdot 85$, and the chain is

$$2^{85} \equiv 32, 32^2 \equiv 1$$

so 32 is a square root of 1 that is neither 1 nor -1 , and 341 is composite. The chain does more than show that 341 is composite: it finds a factor. Since 341 divides $32^2 - 1 = 31 \cdot 33$ and divides neither 31 nor 33, some of the prime factors of 341 go into 31 and the rest into 33. So 341 and 31 have a common divisor that is more than 1 and less than 341, and Euclid's algorithm on 341 and 31 produces it, the

factor 31. For $n = 561$ with $a = 2$: $560 = 16 \cdot 35$, and the chain 2^{35} , squared and squared again, is

$$263, 166, 67, 1$$

so 67 is a square root of 1 that is neither 1 nor -1 , and the number 561 is composite. As before, 561 divides $67^2 - 1 = 66 \cdot 68$ and divides neither factor, so it shares a divisor greater than 1 with 66, and the greatest common divisor of 66 and 561 is the factor 33.

Rabin proved in 1980 that for every odd composite n at least three quarters of the bases from 1 to $n - 1$ expose it in this sharper test; the proof is a count in the group of families with partners, along the lines of the half proved above, and is not given here. There are no exceptions like 561. In the language of evidence, each random base that fails to expose n multiplies the odds in favour of “prime” by at least 4, and forty such bases multiply them by 4^{40} , which is more than 10^{24} . A random number of three hundred digits starts at odds of about 700 to 1 against, since about one in 700 of them is prime. Multiplying those odds by more than 10^{24} leaves odds of more than 10^{21} to 1 in favour, so the remaining chance of error is below one in 10^{21} , smaller than the chance that the computer has made a mistake in its arithmetic. This is how the primes in everyday use are certified. A test that is never wrong and always fast was found in 2002 by Agrawal, Kayal and Saxena; in practice the random test is still the one used.

For which primes does the decimal of $1/p$ have a block of length exactly 8?

Why does the decimal for $1/91$ have a block of length 6, and what is the length of the block for $1/(17 \cdot 19)$?

With the public key $n = 2773$ and $e = 17$, the message received is 948. What was sent?

If a number x is known with $x^2 \equiv 1$ modulo n , and x is neither 1 nor -1 modulo n , how does n come apart, and how many such x are there when n is a product of three different odd primes?

What are the last two digits of 7 raised to the power 7^7 ?

WHICH NUMBERS ARE SQUARES

Modulo 7 the squares of 1, 2, 3, 4, 5 and 6 are 1, 4, 2, 2, 4 and 1, so the families that are squares are 1, 2 and 4, and the families 3, 5 and 6 are not squares of anything. Modulo 11 the squares are 1, 3, 4, 5 and 9; modulo 13 they are 1, 3, 4, 9, 10 and 12. Among the whole numbers the squares thin out. Modulo an odd prime they are exactly half of the nonzero families, and the question of which half turns out to have one of the most surprising answers in arithmetic.

Why half? The families x and $-x$ have the same square, and they are different families because p is odd. No third family shares that square: $x^2 = c$ is a quadratic equation, and modulo a prime a quadratic has at most two roots. So the $p - 1$ nonzero families fall into pairs with a common square, and there are $(p - 1)/2$ squares.

The squares form a subgroup of the group of nonzero families, a product of squares being a square and the partner of a square being the square of the partner. That subgroup is half of the group, so it has one other coset, which is the rest: the non-squares are the squares multiplied through by any one non-square. Now multiply the whole group through by a non-square c . The squares go to the non-squares. Multiplication by c sends different families to differ-

ent families, and the images of the squares already fill every one of the non-squares. So a non-square times c cannot be a non-square as well, and the non-squares have nowhere to go but the squares:

$$\begin{aligned} \text{square} \cdot \text{square} &= \text{square}, \text{square} \cdot \text{non-square} = \text{non-square}, \text{non-square} \\ &\cdot \text{non-square} = \text{square} \end{aligned}$$

Modulo 7, the non-squares 3 and 5 multiply to 15, which is 1, and 3 and 6 multiply to 18, which is 4. The third rule is false among the whole numbers, where $2 \cdot 3 = 6$ is not a square, and it is the one that gives the subject its shape. Legendre's symbol records the rules: (a/p) , which is not a fraction, stands for 1 when a is a square modulo the odd prime p and for -1 when it is not, a not being divisible by p . The three rules say that $(ab/p) = (a/p)(b/p)$.

* * *

A tool is needed first. The bound of two roots for a quadratic was proved by factorising, and the same proof works in every degree.

Over a field, a polynomial of degree d has at most d roots.

If r is a root of f , then $f(x) = f(x) - f(r)$, and that is a sum of terms $c(x^k - r^k)$, each of which has $x - r$ as a factor, as was shown when polynomials were evaluated at a and $a + p$. So $f(x) = (x - r)g(x)$ with g of degree $d - 1$. Any other root s gives $(s - r)g(s) = 0$. A field has no zero divisors, and $s - r$ is not 0, so $g(s) = 0$. So the other roots of f are roots of g . If g , of degree $d - 1$, has at most $d - 1$ roots, then f has at most those and r , which makes at most d : the bound passes from degree $d - 1$ to degree d . Modulo 8, where there are zero divisors, the polynomial $x^2 - 1$ of degree two has the four roots 1, 3, 5 and 7, as was seen.

With that bound the group of nonzero families modulo a prime can be taken apart completely. Take $p = 13$ and sort the twelve families by their orders:

order 1: 1

order 2: 12

order 3: 3, 9

order 4: 5, 8

order 6: 4, 10

order 12: 2, 6, 7, 11

The numbers of families of each order are 1, 1, 2, 2, 2 and 4, which are $\varphi(1)$, $\varphi(2)$, $\varphi(3)$, $\varphi(4)$, $\varphi(6)$ and $\varphi(12)$. That is a theorem.

Modulo a prime p , for every divisor d of $p - 1$ there are exactly $\varphi(d)$ families of order d . In particular there are families of order $p - 1$.

Suppose some family a has order d . Its d powers $1, a, a^2, \dots$, all different, are roots of $x^d = 1$, and that equation has at most d roots, so they are all of its roots. Any family of order d is a root, so it is one of those powers, a^k . Which powers have order d ? Let g be the greatest common divisor of k and d . The power a^k raised to the power j is a to the power kj , and since a has order d , that is 1 exactly when d divides kj . Dividing both numbers by g , that is when d/g divides $(k/g)j$. The numbers d/g and k/g have no common divisor, so that is when d/g divides j . So a^k has order d/g , and it has order d exactly when g is 1. The powers are a to the exponents $0, 1, \dots, d - 1$, and the exponents among these with no common divisor with d are counted by $\varphi(d)$. That makes $\varphi(d)$ of them. So the number of families of order d is either 0 or $\varphi(d)$. But every one of the $p - 1$ families has some order d dividing $p - 1$, so the counts add up to $p - 1$; and the values $\varphi(d)$ at the divisors of $p - 1$ also add up to $p - 1$, as the fractions in lowest terms showed. If any count were 0 in place of $\varphi(d)$, the total would fall short. None is.

A family g of order $p - 1$ is a *primitive root*: its powers are all the nonzero families. Modulo 13 the primitive roots are 2, 6, 7 and 11. The proof does not say which families they are, and no formula for the smallest one is known; Gauss, who gave this proof in 1801, could offer nothing better for finding one than organised

trial. With a primitive root in hand, every nonzero family is g^k for an exponent k that is determined modulo $p - 1$, and multiplying families is adding exponents: the group of nonzero families modulo p is the group of the clock with $p - 1$ hours, in disguise. And passing from g^k back to k is the problem of undoing a power, for which no fast method is known; the scheme by which Diffie and Hellman, in 1976, let two strangers agree on a secret key over an open line rests on that difficulty.

* * *

Which families are squares can now be said in a formula. Write P for $(p - 1)/2$, where p is an odd prime.

a^P is 1 modulo p when a is a square, and -1 when it is not.

First, a to the power P can only be 1 or -1 . Its square is a to the power $2P$, which is a to the power $p - 1$, which is 1; and the only square roots of 1 modulo a prime are 1 and -1 . If $a = b^2$, then a to the power P is b to the power $p - 1$, which is 1. So all P squares are roots of $x^P = 1$. That equation has at most P roots, so the squares are all of its roots. A non-square is then not a root: raised to the power P it is not 1, and the only other value it can take is -1 . Euler found this criterion. In Legendre's notation it says $(a/p) \equiv a^P$, and with quick powering it decides whether a number of a hundred digits is a square modulo a prime of a hundred digits, without finding the square root.

For $a = -1$ the criterion can be read off at sight: (-1) to the power P is 1 when P is even and -1 when P is odd. And $P = (p - 1)/2$ is even exactly when $p - 1$ is a multiple of 4.

-1 is a square modulo an odd prime p exactly when p leaves 1 on division by 4.

Modulo 5, $2^2 = 4$ is -1 ; modulo 13, $5^2 = 25$ is -1 ; modulo 17, $4^2 = 16$ is -1 ; and modulo 3, 7, 11 and 19 no square is -1 . When p leaves

1, a square root of -1 can even be written down. In the congruence $(p-1)! \equiv -1$, write each factor above P as the negative of one of the factors from 1 to P : $p-1$ is -1 , $p-2$ is -2 , and so on down to $P+1$, which is $-P$. That brings in P minus signs. Since p leaves 1 on division by 4, P is even, so the signs cancel, and the congruence becomes $(P!)^2 \equiv -1$. Modulo 13, $6! = 720$ is 5, and 5^2 is -1 .

That settles a question left open about primes. There are infinitely many primes that leave 1 on division by 4. Take any list of them, multiply them all together and double the result; call that number n , and let $N = n^2 + 1$. Since n is even, N is odd, so any prime q that divides N is odd. It has $n^2 \equiv -1$ modulo q , so -1 is a square modulo q , and by the result above q leaves 1 on division by 4. And q is not on the list, since the primes of the list divide n^2 and not N .

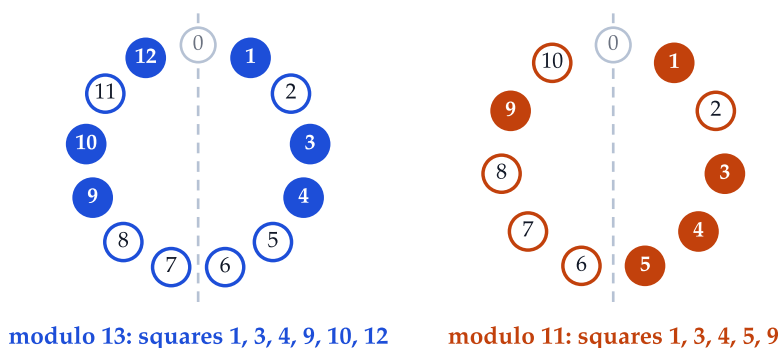


Figure 26.1. The nonzero families modulo 13 and modulo 11 on their clocks, with the squares filled in. Taking the negative is the reflection in the vertical line. Modulo 13, where -1 is a square, the squares are their own mirror image; modulo 11, where it is not, the mirror image of the squares is the non-squares. Assumes: two primes, one leaving 1 and one leaving 3 on division by 4; the text proves that the remainder on division by 4 decides between the two pictures

* * *

When is 2 a square? It is one modulo 7, 17, 23, 31, 41, 47, 71, 73, 79, 89 and 97, and not modulo 3, 5, 11, 13, 19, 29, 37, 43, 53, 59, 61, 67 and 83. On division by 8 the primes of the first list all leave 1 or 7, and those of the second all leave 3 or 5.

The proof computes 2 to the power P by multiplying the even numbers. Take $p = 11$, where $P = 5$. The even numbers 2, 4, 6, 8, 10 are $2 \cdot 1, 2 \cdot 2, 2 \cdot 3, 2 \cdot 4, 2 \cdot 5$, so their product is $2^5 \cdot 5!$. On the other hand, modulo 11 the ones above 5 are negatives of small odd numbers: 6 is -5 , 8 is -3 , 10 is -1 . So the five even numbers are 2, 4, -5 , -3 , -1 , which apart from signs are 1, 2, 3, 4, 5 again, and their product is $(-1)^3 \cdot 5!$. Comparing, $2^5 \cdot 5! \equiv -5!$, and cancelling $5!$, $2^5 \equiv -1$: so 2 is not a square modulo 11, by the criterion. In general, take the even numbers up to $p - 1$ and replace each one above P by its negative. An even number $2j$ above P is the negative of $p - 2j$, and $p - 2j$ is odd and no more than P . So the ones left alone are the even numbers up to P , and the ones replaced become, apart from sign, the odd numbers up to P , each once: together they are 1, 2, ..., P in some order with some signs. As with 11, the product of the even numbers is $2^P \cdot P!$ counted one way, and counted the other it is $P!$ with one factor -1 for each number replaced. Cancelling $P!$ gives

$$2^P \equiv (-1)^m,$$

where m is the number of even numbers above P and below p .

Counting them in the four cases: when p is $8k + 1$, P is $4k$ and m is $2k$; when p is $8k + 3$, P is $4k + 1$ and m is $2k + 1$; when p is $8k + 5$, P is $4k + 2$ and m is $2k + 1$; when p is $8k + 7$, P is $4k + 3$ and m is $2k + 2$. So m is even in the first and last cases:

2 is a square modulo an odd prime p exactly when p leaves 1 or 7 on division by 8.

Here is a use. The numbers $2^p - 1$ with p prime are the first place where large primes are looked for, and $2^{11} - 1 = 2047$ is not one: it is $23 \cdot 89$. The factor 23 could have been predicted. It is a prime of the form $2p + 1$ with $p = 11$, and it leaves 7 on division by 8, so 2

is a square modulo 23. By the criterion, 2 to the power $(23 - 1)/2$, which is 2^{11} , is then 1 modulo 23; and that says 23 divides $2^{11} - 1$. In the same way 47 divides $2^{23} - 1$, and 167 divides $2^{83} - 1$, a number of 25 digits.

* * *

For -1 and for 2, whether the number is a square modulo p depends only on the remainder of p on division by 4 or by 8, although each prime has its own multiplication table and nothing in the definition suggests a pattern. What happens for 3, for 5, for 7?

Euler collected the evidence, and it shows the same thing. The number 3 is a square modulo 11, 13, 23, 37, 47, 59, 61, 71, 73, 83 and 97, which are the primes below 100 leaving 1 or 11 on division by 12. The number 5 is a square modulo 11, 19, 29, 31, 41, 59, 61, 71, 79 and 89, which are the primes below 100 ending in 1 or 9. Looking for the rule behind such lists, Euler and then Legendre found that it concerns two different odd primes at once, and that it is a symmetry. Ask whether p is a square modulo q , and whether q is a square modulo p : two questions about two different multiplication tables, with no visible connection.

If p or q leaves 1 on division by 4, the two questions have the same answer. If both leave 3, they have opposite answers.

In symbols, with $P = (p - 1)/2$ and $Q = (q - 1)/2$, the law says $(p/q)(q/p) = (-1)^{PQ}$. It is the law of quadratic reciprocity. Legendre's attempted proof assumed the theorem about primes in progressions, which was not proved until long afterwards. Gauss found the first proof in 1796, at eighteen, after a year of effort, called it the golden theorem, and gave seven more proofs in the course of his life. More than two hundred are now known. The one that follows was found by Rousseau in 1991, and it needs only what is already in hand: the recovery of a number from its two remainders, the product $(p - 1)! \equiv -1$, the criterion, and the sorting of a group into cosets.

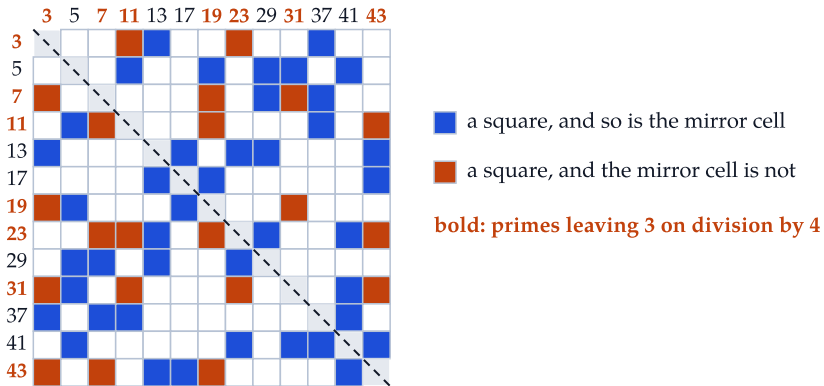


Figure 26.2. A filled cell says that the prime at the left is a square modulo the prime at the top. The primes in plain type leave 1 on division by 4, and their rows repeat their columns. Between two primes in bold, which leave 3, the cell and its mirror image across the diagonal always disagree. *Assumes: the odd primes up to 43; the law proved in the text says the pattern never breaks*

Let p and q be different odd primes, and let G be the group of pairs (a, b) , where a is a nonzero family modulo p and b a nonzero family modulo q , multiplied place by place. It has the two-element subgroup consisting of $(1, 1)$ and $(-1, -1)$, and its cosets are the pairs of pairs $\{(a, b), (-a, -b)\}$. Choose one pair from each coset, in any way, and multiply all the chosen pairs together. A different choice replaces some of the factors (a, b) by $(-a, -b)$, and each replacement multiplies the product by $(-1, -1)$. Two such factors cancel, so a different choice changes the product at most by the factor $(-1, -1)$: the product is determined up to a sign, the same sign in both places. The proof makes the choice in two ways and compares.

First choice: the pairs (a, b) with a anything from 1 to $p - 1$ and b from 1 to Q . The second entry of $(-a, -b)$ is $q - b$, and of b and $q - b$ exactly one lies from 1 to Q . So exactly one of (a, b) and $(-a, -b)$ has its second entry in that range, and this does take one pair from each coset. In the first place of the pair, each a occurs Q times,

so the product is $(p-1)!$ to the power Q , which is $(-1)^Q$. In the second place of the pair, each b from 1 to Q occurs $p-1$ times, so the product is $Q!$ to the power $p-1$, which is $(Q!)^2$ to the power P . And $(Q!)^2$ is known: writing the upper half of the factors of $(q-1)!$ as negatives of the lower half, as before, gives $(q-1)! \equiv (-1)^Q (Q!)^2$, and since $(q-1)! \equiv -1$, $(Q!)^2 \equiv (-1)^{Q+1}$. Raised to the power P , that is $(-1)^{PQ+P}$. The first choice therefore gives the product

$$((-1)^Q, (-1)^P \cdot (-1)^{PQ})$$

Second choice: p and q have no common divisor, so by the theorem on two remainders each pair (a, b) is the pair of remainders of exactly one number k from 1 to $pq-1$, and the numbers k that arise are those divisible by neither p nor q . The negative of the pair of k is the pair of $pq-k$. Of k and $pq-k$ exactly one is below $pq/2$, so the numbers k below $pq/2$, divisible by neither prime, take one pair from each coset too. The first place of their product is the product of these numbers modulo p .

The numbers up to $(pq-1)/2 = Qp + P$ that are not divisible by p form Q full runs 1, 2, ..., $p-1$ modulo p , and a last half run 1, 2, ..., P . Their product modulo p is $(p-1)!$ to the power Q , times $P!$, which is $(-1)^Q \cdot P!$. From these the multiples of q have to be removed. Written the other way round, $(pq-1)/2$ is $Pq + Q$, so the multiples of q that do not exceed it are $q, 2q, \dots, Pq$, with product $q^P \cdot P!$. Dividing, the $P!$ cancels, and what is left is $(-1)^Q$ divided by q^P . By the criterion q^P is (q/p) , which is 1 or -1 and so is its own partner: dividing by it is the same as multiplying by it. So the product is $(-1)^Q (q/p)$ in the first place, and by the same computation with the primes exchanged, $(-1)^P (p/q)$ in the second:

$$((-1)^Q \cdot (q/p), (-1)^P \cdot (p/q))$$

The two products agree up to one sign, the same in both places. Comparing first places, the sign is (q/p) . Comparing second places with that sign, $(-1)^P \cdot (p/q) = (q/p) \cdot (-1)^P \cdot (-1)^{PQ}$, and cancelling $(-1)^P$ leaves $(p/q) = (q/p) \cdot (-1)^{PQ}$. And $(-1)^{PQ}$ is -1

exactly when P and Q are both odd, which is when p and q both leave 3 on division by 4.

The law turns the question of whether a number is a square modulo a prime into a run of divisions like those of Euclid's algorithm: split the number into primes, settle any factor of 2 by the remainder of the prime on division by 8, turn each odd prime's symbol over by the law, replace the top number by its remainder on division by the bottom one, and repeat with the smaller numbers that appear. Is 219 a square modulo the prime 383? Since $219 = 3 \cdot 73$, the symbol is $(3/383)(73/383)$. Both 3 and 383 leave 3 on division by 4, so $(3/383) = -(383/3)$. Since 383 leaves 2 on division by 3, that is $-(2/3)$; and $(2/3)$ is -1 , because 3 leaves 3 on division by 8. So $(3/383) = 1$. And 73 leaves 1, so $(73/383) = (383/73)$, which is $(18/73)$ since 383 leaves 18 on division by 73. That is $(2/73)(9/73) = (2/73)$, 9 being a square, and $(2/73)$ is 1 because 73 leaves 1 on division by 8. So 219 is a square modulo 383, and nothing was squared to find that out. In fact $169^2 = 28561 = 74 \cdot 383 + 219$.

It also explains the lists. Since 5 leaves 1 on division by 4, 5 is a square modulo p exactly when p is a square modulo 5, which is when p leaves 1 or 4 on division by 5; the odd primes that do so end in 1 or 9.

For which primes p is 3 a square, and for which is -3 ?

Which primes can divide a number of the form $n^2 + n + 1$?

What is the product of all the squares among the nonzero families modulo p ?

Can a product of two different primes p and q pass the test $a^{pq} - 1 \equiv 1$ with every base a that shares no factor with it?

Is 1001 a square modulo the prime 9907?

A WORLD WHERE DIVISION WORKS

Which whole numbers are a sum of two squares? Up to 50 they are

1, 2, 4, 5, 8, 9, 10, 13, 16, 17, 18, 20, 25, 26, 29, 32, 34, 36, 37, 40, 41, 45,
49, 50

counting 0 as a square, so that $9 = 9 + 0$. Among the primes, 2, 5, 13, 17, 29, 37 and 41 are on the list, and 3, 7, 11, 19, 23, 31, 43 and 47 are not. Half of that is explained at once. A square leaves 0 or 1 on division by 4, so a sum of two squares leaves 0, 1 or 2, and never 3: no number that leaves 3 is a sum of two squares, prime or not. The other half is a theorem. On Christmas Day 1640 Fermat wrote to Mersenne that every prime that leaves 1 on division by 4 is a sum of two squares, in exactly one way. He gave no proof. Euler found one in 1749, after working at it on and off for seven years.

The identity that makes sizes multiply shows where to look. Since $a^2 + b^2$ is $(a + bi)(a - bi)$, a number is a sum of two squares exactly when it is the size of some $a + bi$ with whole-number a and b , where from now on the size of $a + bi$ means $a^2 + b^2$, the square of its distance from 0, and asking which numbers are sums of two squares is asking how whole numbers factorise when numbers of the form $a + bi$ are allowed as factors: $5 = (2 + i)(2 - i)$, 13

$= (3 + 2i)(3 - 2i)$. The numbers $a + bi$ with a and b integers are called the *Gaussian integers*. They are closed under adding, subtracting and multiplying, like the numbers $a + bw$ with $w^2 = -5$, in which factorisation into irreducibles turned out not to be unique. There the trouble was traced to its root: division with a remainder smaller than the divisor was not available. Here it is.

* * *

The size of $a + bi$ is $a^2 + b^2$, the square of its distance from 0 in the plane, and sizes multiply. Suppose a Gaussian integer has a partner under multiplication, a second Gaussian integer with which its product is 1. Then the two sizes, both whole numbers, multiply to the size of 1, which is 1, so each of them is 1. Size 1 means $a^2 + b^2 = 1$, which happens only when one of a and b is 0 and the other is 1 or -1 . So there are four such numbers, 1, -1 , i and $-i$, the *units*. Multiplying by a unit turns the plane by a whole number of quarter turns, and factorisations that differ only by units are counted as the same, as $6 = 2 \cdot 3 = (-2)(-3)$ are for ordinary integers.

For Gaussian integers α and β , with β not 0, there are Gaussian integers γ and ρ with $\alpha = \beta\gamma + \rho$ and the size of ρ at most half the size of β .

Divide in the field of numbers $a + bi$ with fraction coordinates: $\alpha/\beta = x + yi$, with x and y fractions. Round x and y to nearest integers m and n , so that each is off by at most $1/2$, and put $\gamma = m + ni$. Then $\alpha/\beta - \gamma$ is $(x - m) + (y - n)i$, with both coordinates at most $1/2$ either way, so its size is at most $1/4 + 1/4 = 1/2$. The remainder $\rho = \alpha - \beta\gamma$ is β times that number, and sizes multiply, so the size of ρ is at most half the size of β .

The picture says the same. The multiples of β are the corners of a grid of squares, the grid of all Gaussian integers turned and enlarged by the multiplication by β . Any point α of the plane lies in one of the squares, and no point of a square is farther from the nearest corner than the centre is, which is half a diagonal away.

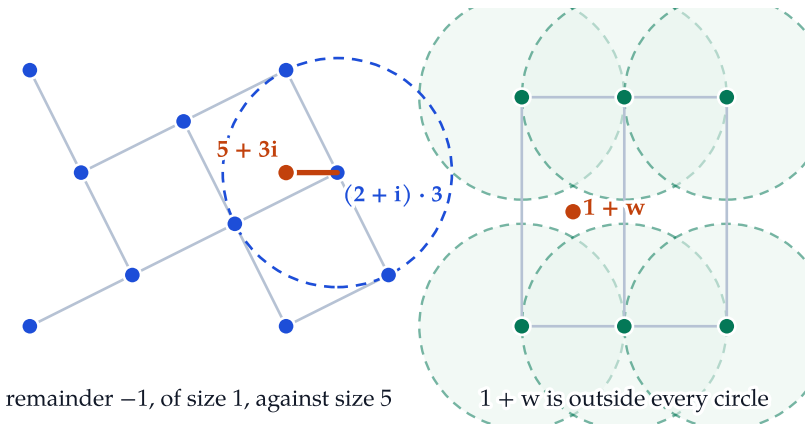


Figure 27.1. On the left, the multiples of $2+i$ form a grid of squares, and any point lies within half a diagonal of a corner, which is less than a side: the remainder is smaller than the divisor. On the right, the multiples of 2 in the numbers $a+bw$ with $w^2 = -5$ form a grid of rectangles, and the centre of a rectangle is farther from every corner than the length of the divisor, shown by the circles. Assumes: the plane picture of $a+bw$, with w drawn as the point at height $\sqrt{5}$ on the vertical axis

Half a diagonal, squared, is half the square of the side, and the square of the side is the size of β , so the remainder has size at most half the size of β , as before.

The same picture shows what goes wrong in the world of the numbers $a+bw$ with $w^2 = -5$. Draw $a+bw$ as the point a across and $b\sqrt{5}$ up, so that too the size, $a^2 + 5b^2$, is the square of the distance from 0. The multiples of β are then the corners of a grid of rectangles, of sides 1 and $\sqrt{5}$ times the length of β . The square of the diagonal of such a rectangle is $1+5=6$ times the size of β , so the square of half the diagonal is $6/4$ of the size of β . The centre of the rectangle is therefore farther from every corner than the length of β . Take $\beta = 2$, where that centre is $1+w$, a number of the world itself. Its distance from each of 0, 2, $2w$ and $2+2w$ has square 6, above the size 4 of 2, so dividing $1+w$ by 2 leaves no remainder smaller than the divisor, whichever multiple of 2 is subtracted. That is why division failed there.

With division, everything that was proved for whole numbers from division returns, by the same proofs. Euclid's algorithm runs, the sizes of the remainders decreasing until a remainder is 0. For $11 + 3i$ and $1 + 8i$:

$$11 + 3i = (1 + 8i)(1 - i) + (2 - 4i)$$

$$1 + 8i = (2 - 4i)(-2 + i) + (1 - 2i)$$

$$2 - 4i = (1 - 2i) \cdot 2$$

In the first line the exact quotient $(11 + 3i)/(1 + 8i)$ is found by multiplying top and bottom by $1 - 8i$, which turns the bottom into the size 65. It is $(35 - 85i)/65$, about $0.54 - 1.31i$, which rounds to $1 - i$. The last remainder that is not 0, which is $1 - 2i$, divides both numbers, and every common divisor divides it. It is $-i$ times $2 + i$, and indeed $11 + 3i = (2 + i)(5 - i)$ and $1 + 8i = (2 + i)(2 + 3i)$. Run backwards, the algorithm writes the greatest common divisor as a combination of the two numbers with Gaussian integer coefficients. Call a Gaussian integer π *prime* when it is not a unit and its only divisors are units and unit multiples of itself. So if a prime π divides a product, then π divides one of the factors. The proof is the one given for whole numbers, word for word. And uniqueness of factorisation follows as it did there:

every Gaussian integer other than 0 and the units is a product of Gaussian primes, in only one way apart from the order of the factors and multiplication by units.

That a factorisation exists at all is a descent on the size. A Gaussian integer other than 0 that is neither a unit nor prime is a product of two factors, neither of them a unit. Neither factor has size 1, and the two sizes multiply to the size of the product, so each factor has smaller size than the product. A factor that is not prime splits again in the same way, and since sizes are positive whole numbers the splitting cannot go on forever: it stops with every factor prime.

* * *

What are the Gaussian primes? A Gaussian integer whose size is an ordinary prime is a Gaussian prime. If it were a product of two factors, neither a unit, its size would be the product of their sizes, two whole numbers larger than 1, and an ordinary prime is no such product. So $1 + i$, $2 + i$, $3 + 2i$ and $4 + i$ are prime, with sizes 2, 5, 13 and 17. What becomes of the ordinary primes?

A prime q that leaves 3 on division by 4 stays prime. If $q = \alpha\beta$ with neither factor a unit, then the sizes give $q^2 = (\text{size of } \alpha)(\text{size of } \beta)$, with neither size 1, so both sizes are q , and q would be a sum of two squares, which it is not.

A prime p that leaves 1 on division by 4 does not stay prime, and this is where the earlier work is used. For such a prime -1 is a square modulo p : there is a whole number n with p dividing $n^2 + 1$. In the Gaussian integers $n^2 + 1 = (n + i)(n - i)$. If p were a Gaussian prime it would divide one of the factors. But p divides neither, since $(n + i)/p$ and $(n - i)/p$ have the coordinate $1/p$ or $-1/p$, which is not an integer. So p is not a Gaussian prime: $p = \alpha\beta$ with neither a unit. The sizes multiply to p^2 , so each is p . Writing $\alpha = a + bi$,

$$p = a^2 + b^2$$

and the first half of the theorem of Christmas Day is proved, the half that says such a prime is a sum of two squares at all. The proof is also a method. Let δ be the greatest common divisor of p and $n + i$. Since δ divides p , its size divides p^2 , the size of p , so that size is 1, p or p^2 . It is not p^2 : the other factor of p would then have size 1, δ would be a unit times p , and p would divide $n + i$. It is not 1: δ would then be a unit, and some combination $p\kappa + (n + i)\lambda$ would equal 1. Multiplied by $n - i$, that reads $p(n - i)\kappa + (n^2 + 1)\lambda = n - i$, and p divides both terms on the left, so p would divide $n - i$. So the size of δ is p . That divisor is the $a + bi$ wanted, and Euclid's algorithm finds it. For $p = 13$, a square root of -1 is 5, and Euclid's algorithm on 13 and $5 + i$ takes two lines,

$$13 = (5 + i) \cdot 2 + (3 - 2i), 5 + i = (3 - 2i)(1 + i)$$

so the greatest common divisor is $3 - 2i$, and $13 = 9 + 4$. For $p = 1009$ a square root of -1 is 469 , and three lines of the same kind lead from 1009 and $469 + i$ to $-28 + 15i$, and $1009 = 28^2 + 15^2$. For a prime of a hundred digits the square root of -1 comes from a quick power. Take any a that is not a square modulo p . By the criterion, a to the power $(p - 1)/2$ is -1 modulo p , and that is the square of a to the power $(p - 1)/4$, which is therefore a square root of -1 . Half of all a are not squares, and one quick power tests each, so a few tries produce one. Euclid's algorithm then finishes the work in a few hundred lines. No search among squares could.

The way is unique. If $p = a^2 + b^2 = c^2 + d^2$, then $(a + bi)(a - bi)$ and $(c + di)(c - di)$ are two factorisations of p into Gaussian primes, the factors having prime size. By uniqueness $c + di$ is a unit times $a + bi$ or $a - bi$, and that says that c and d are a and b apart from order and signs.

The prime 2 is $i(1 - i)^2$, the square of a Gaussian prime apart from a unit. So the ordinary primes behave in three ways: 2 becomes a square, the primes that leave 1 split into two Gaussian primes, mirror images of each other in the horizontal axis, and the primes that leave 3 stay prime. The two mirror images are different primes and not unit multiples of each other: $a - bi$ is a unit times $a + bi$ only when a or b is 0 or a is b or $-b$, and then $a^2 + b^2$ would be a square or twice a square, not an odd prime. Every Gaussian prime is a unit multiple of one of these. The size of a Gaussian prime π is π times its mirror image, so π divides its own size, a whole number larger than 1 and so a product of ordinary primes. Being prime, π divides one of those ordinary primes. That ordinary prime has been factorised into Gaussian primes above, and by uniqueness π is a unit multiple of one of the factors.

* * *

Now the first question can be answered for every number.

A positive whole number is a sum of two squares exactly when every prime that leaves 3 on division by 4 occurs in it an even number of times.

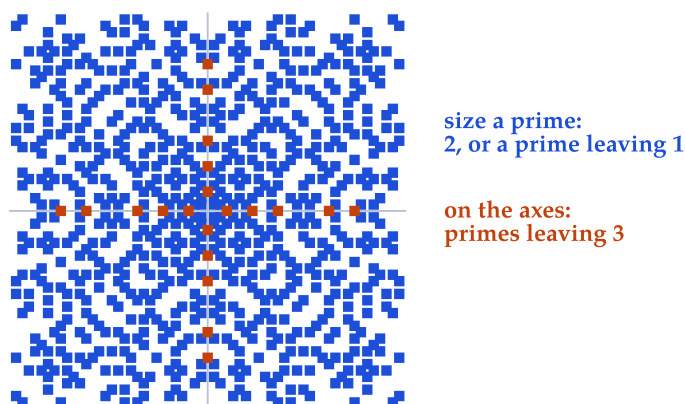


Figure 27.2. The Gaussian primes near the origin. On the two axes lie the ordinary primes that leave 3 on division by 4, which stay prime. Every other point shown has a prime number for its size, and the ordinary primes that leave 1 appear as eight points each, a point with its mirror images and quarter turns. *Assumes:* coordinates up to 30; no pattern in the picture beyond its eightfold symmetry has been proved

Such a number is a product of 2s, of primes that leave 1, and of squares q^2 , and each of those is a sum of two squares, q^2 being $q^2 + 0$. A product of sums of two squares is a sum of two squares, since sizes multiply: the product of the sizes of two Gaussian integers is the size of their product. Conversely let $n = a^2 + b^2$ and let q be a prime that leaves 3 and divides n . Then q , a Gaussian prime, divides $(a + bi)(a - bi)$, so it divides one factor. The quotient of $a + bi$ or $a - bi$ by the whole number q has coordinates a/q and b/q or $-b/q$, and it is a Gaussian integer only when both are integers. So q divides both a and b . So q^2 divides n , and n/q^2 is the sum of the squares of a/q and b/q . Repeating the argument on n/q^2 removes the factors q two at a time until none are left, so there were an even number. For example $1000 = 2^3 \cdot 5^3$ is a sum of two squares, $30^2 + 10^2$ and also $26^2 + 18^2$. And $2024 = 8 \cdot 11 \cdot 23$ is not, while $2025 = 3^4 \cdot 5^2$ is: $27^2 + 36^2$.

The number of ways can be read from the factorisation as well. The number $65 = 5 \cdot 13$ is $1 + 64$ and also $16 + 49$, and

the two ways come from a choice. In the Gaussian integers 65 is $(2 + i)(2 - i)(3 + 2i)(3 - 2i)$. If $a + bi$ has size 65, then $(a + bi)(a - bi)$ is 65, so by uniqueness $a + bi$ is, apart from a unit, a product of some of these four primes, and $a - bi$ is the product of their mirror images. Between them the two products must use each of the four primes exactly once. So $a + bi$ takes one factor from each mirror pair, and its own mirror image takes the others:

$$(2 + i)(3 + 2i) = 4 + 7i, (2 + i)(3 - 2i) = 8 - i$$

and the remaining two choices give the mirror images of these. With three primes that split there are four choices apart from mirror images, and $1105 = 5 \cdot 13 \cdot 17$ is a sum of two squares in four ways: $4^2 + 33^2$, $9^2 + 32^2$, $12^2 + 31^2$ and $23^2 + 24^2$. Count every $a + bi$ of size n separately, so that signs and order count. Then for a number n that is a sum of two squares at all, the number of ways is $4(e_1 + 1)(e_2 + 1) \cdots$, where $e_1, e_2, \dots$ are the numbers of times that the primes leaving 1 occur in n . The reason is the same choice. A prime that occurs e times puts e copies of each of its two mirror factors into n . The number $a + bi$ may take any number k of copies of the first, from 0 to e , and must then take $e - k$ copies of the second, its mirror image taking the rest. That is $e + 1$ choices for each such prime. The other primes of n leave no choice: a prime that leaves 3 is its own mirror image, and $1 - i$ is a unit times its mirror image $1 + i$, so $a + bi$ and $a - bi$ hold equally many copies of each, and $a + bi$ must take exactly half of those in n . At the end any of the four units may be multiplied in.

The formula for the right-angled triangles with whole-number sides is one line of this arithmetic. Square a Gaussian integer: $(m + ni)^2 = (m^2 - n^2) + 2mn \cdot i$. The size of the square is the square of the size, so $(m^2 - n^2)^2 + (2mn)^2 = (m^2 + n^2)^2$. The triangles come from squaring. The longest side of a triangle whose sides have no common divisor is a number $m^2 + n^2$, with m and n without common divisor and one of them even. Such a number is a product of primes that leave 1 on division by 4. It is odd, so 2 does not divide it. And a prime that leaves 3 and divides $m^2 +$

n^2 divides both m and n , as was seen above, which cannot happen when m and n have no common divisor. The number $65 = 5 \cdot 13$ is the smallest product of two different primes that leave 1, and it is the first hypotenuse to serve two such triangles: 16, 63, 65 and 33, 56, 65.

* * *

Other forms can be treated the same way, when their world has division. Take the numbers $a + b\sqrt{-2}$, where $\sqrt{-2}$ stands for a new object whose square is -2 , with size $a^2 + 2b^2$. Rounding the two coordinates of a quotient leaves an error of size at most $1/4 + 2/4$, which is less than 1, so there is division with a smaller remainder, and with it unique factorisation. The argument for two squares, run again, shows that an odd prime modulo which -2 is a square is of the form $a^2 + 2b^2$, and by the rules for -1 and for 2 those are the primes that leave 1 or 3 on division by 8. No other odd prime is of the form: when $a^2 + 2b^2$ is odd, a is odd and a^2 leaves 1 on division by 8, while $2b^2$ leaves 0 or 2, so the sum leaves 1 or 3. So the odd primes of the form $a^2 + 2b^2$ are those that leave 1 or 3 on division by 8: $3 = 1 + 2$, $11 = 9 + 2$, $17 = 9 + 8$, $19 = 1 + 18$, $41 = 9 + 32$, $43 = 25 + 18$.: $3 = 1 + 2$, $11 = 9 + 2$, $17 = 9 + 8$, $19 = 1 + 18$, $41 = 9 + 32$, $43 = 25 + 18$.

The same world settles a question of another kind. The number 26 lies between a square and a cube, 25 and 27. Fermat asserted that no other number does: that $y^2 + 2 = x^3$ has the single solution $y = 5$, $x = 3$ in positive whole numbers. Here is the proof. First y is odd, since if y were even, x would be even too, and $y^2 + 2 = x^3$ would leave 2 on division by 4 on the left and 0 on the right. Factorise the left side:

$$(y + \sqrt{-2})(y - \sqrt{-2}) = x^3$$

The two factors have no common divisor other than 1 and -1 , the only units here. A common divisor would divide their difference, $2\sqrt{-2}$, which is $-(\sqrt{-2})^3$. Now $\sqrt{-2}$ is prime, its size being 2,

so by unique factorisation the only divisors of $-(\sqrt{-2})^3$ are units times powers of $\sqrt{-2}$. A common divisor that is not a unit would therefore be a multiple of $\sqrt{-2}$, and $\sqrt{-2}$ would divide $y + \sqrt{-2}$, hence y . Taking sizes, 2 would divide y^2 , which is odd. Two numbers with no common divisor whose product is a cube: by unique factorisation each is a cube, apart from a unit. For in the factorisation of x^3 every prime occurs a multiple of 3 times, and all the copies of a prime go to the same one of the two factors, since the factors share no prime. So in each factor every prime occurs a multiple of 3 times, which makes that factor a unit times a cube. And the units 1 and -1 are cubes themselves. So

$$y + \sqrt{-2} = (a + b\sqrt{-2})^3 = (a^3 - 6ab^2) + (3a^2b - 2b^3)\sqrt{-2}$$

Comparing the coefficients of $\sqrt{-2}$ gives $1 = b(3a^2 - 2b^2)$. So b is 1 or -1 . If $b = 1$, then $3a^2 - 2 = 1$ and a is 1 or -1 ; if $b = -1$, then $3a^2 - 2 = -1$, which no integer satisfies. Then $y = a^3 - 6ab^2 = a^3 - 6a$ is -5 or 5 , and $x^3 = 27$. There is no other solution. The step on which it turned, from a product that is a cube to factors that are cubes, is the step that uniqueness of factorisation allows, and without uniqueness it fails.

With $w^2 = -5$, where uniqueness fails, the pattern of primes changes in a way that can be seen in the data. Apart from 2 and 5, the primes modulo which -5 is a square are, by the rule for -1 and the law of reciprocity applied to 5, those that leave 1, 3, 7 or 9 on division by 20. The argument that worked for two squares, if it could be run here, would make every one of these primes of the form $a^2 + 5b^2$. It cannot be run, because its conclusion is false. The primes that leave 3 or 7 on division by 20, such as 3, 7, 23, 43 and 47, all leave 3 on division by 4, while $a^2 + 5b^2$ leaves the same as $a^2 + b^2$ on division by 4, which is never 3. So those primes are not of the form $a^2 + 5b^2$.

In the data the primes that leave 1 or 9 are of the form, $29 = 9 + 20$, $41 = 36 + 5$, $61 = 16 + 45$, $89 = 9 + 80$, and the doubles of the others are: $6 = 1 + 5$, $14 = 9 + 5$, $46 = 1 + 45$. Both patterns hold for every such prime. Euler observed them and

Lagrange proved them, by a theory of forms that is not developed here. The failure of unique factorisation in that world, first met as a curiosity about the number 6, is the reason a second family of primes appears. Measuring that failure, for every world of this kind, is where the theory of numbers went next, with Gauss, Kummer and Dedekind.

Which of the numbers 2021, 2022, ..., 2030 are sums of two squares?

Carry out Euclid's algorithm for $7 + 9i$ and $3 - 5i$. What is their greatest common divisor, and what are its size and the sizes of the two numbers?

In how many ways is 5^k a sum of two squares, counting signs and order, and which of those ways have a and b without a common divisor?

The numbers $a + b\omega$, where ω is the cube root of 1 with $\omega^2 + \omega + 1 = 0$, have size $a^2 - ab + b^2$. Is there division with a smaller remainder, and which primes have the form $a^2 - ab + b^2$?

The squares 4 and 121 are each 4 less than a cube. Are there any others?

BUILDING A WORLD BY COLLAPSING ONE

The families on division by 12 replaced infinitely many integers by twelve objects, and arithmetic on them was allowed because it did not depend on the members chosen. The same move can be tried inside any group, with the cosets of a subgroup as the families.

A subgroup tiles its group into cosets, so treat each coset as a single object, and combine two cosets by choosing an element of each, combining those, and taking the coset of the answer. Collapse the families on division by 12 by the subgroup made of 0, 4 and 8. There are four blocks: 0, 4, 8; then 1, 5, 9; then 2, 6, 10; then 3, 7, 11. Adding the block of 1 to the block of 2 through the members 1 and 2 gives 3; through the members 5 and 10 it gives 15, which is 3 on division by 12; either way the answer lies in the block of 3.

That always works here. Replace a chosen member a by $a + t$, where t is 0, 4 or 8, and add the second chosen member b : since addition does not depend on order, $(a + t) + b$ is $(a + b) + t$, and adding t to a number does not take it out of its block. Changing b works the same way, and changing both members adds two elements of the subgroup, whose sum is again 0, 4 or 8. The four blocks with this addition form the clock with four hours.

When order matters the argument does not go through. Write

the twelve of $Z/12$, cut into four blocks by the subgroup $\{0, 4, 8\}$

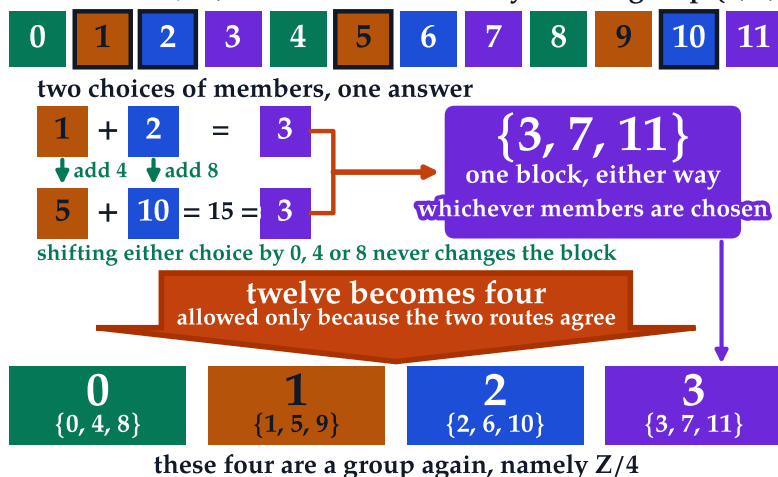


Figure 28.1. The twelve of $Z/12$ coloured into four blocks by the subgroup $N = \{0, 4, 8\}$ and collapsed into the four-element group $Z/4$, with one addition carried out twice from different members, so that the block of the answer does not depend on the choice. *Assumes:* that this works because N is closed under the operation and is normal, as every subgroup is in a group where order does not matter; for a general subgroup of a general group the collapse can fail and the picture would look identical

“ g then h ” for performing g and then h . Fix a subgroup H , and let the block of g be the collection of all “ g then h ” with h in H . Replace a chosen member g by “ g then h ”, a member of the same block, and combine with a second chosen member k : the result is “ g then h then k ”, and it should lie in the block of “ g then k ”. Now “ h then k ” is the same as “ k then s ”, where s is “ k backwards, then h , then k ”: in “ k , then k backwards, then h , then k ” the first two steps cancel and leave “ h then k ”. So the result is “ g then k then s ”, which lies in the block of “ g then k ” exactly when s is in H . Changing the second chosen member instead, from k to “ k then h ”, changes the result to “ g then k then h ”, which lies in the block of “ g then k ” whatever H is.

The element s , written $k^{-1}hk$, is h seen after relabelling by k : it

undoes k , performs h , and performs k again. So where h carries one position to another, s carries the place k sends the first to the place k sends the second. On a square, the flip in the vertical axis seen after relabelling by a quarter turn is the flip in the horizontal axis, and a quarter turn seen after relabelling by any flip is a three-quarter turn. A subgroup is *normal* when every relabelling of each of its members is again a member: $k^{-1}hk$ lies in H for every h in H and every k in the group. The collapse works exactly when the subgroup is normal, since that is word for word what the check above needed.

An answer that does not depend on the members chosen is not yet enough for a group, and three more checks are needed, each a line. Grouping does not matter for blocks, because it does not matter for their members. The block of the identity changes nothing. The block of the reverse of g undoes the block of g . So the blocks of a normal subgroup form a group, the *collapse* of the group by the subgroup.

* * *

Inside the eight symmetries of a square, look for the symmetries that commute with every symmetry. Such elements are unchanged by every relabelling: if h commutes with k , then “ k backwards, then h , then k ” is “ k backwards, then k , then h ”, which is h . That rules out six of the eight, since relabelling by a flip exchanges the quarter turn and the three-quarter turn, and relabelling by a quarter turn exchanges the flips in the two axes and, in the same way, the flips in the two diagonals. The half turn carries each corner to the opposite corner, and every symmetry carries opposite corners to opposite corners, so the half turn followed by any symmetry sends each corner where that symmetry followed by the half turn does, and a symmetry is settled by where it sends the corners. The identity and the half turn are therefore the symmetries that commute with every symmetry. So they form a normal subgroup, and the collapse has four blocks, since eight divided by two is four.

It is one of the two groups of four. Every symmetry performed twice is the identity or the half turn, and both of those lie in the block of the identity, so every block combined with itself gives the block of the identity. The collapse is therefore not the clock with four hours but the group in which every element is its own partner.

The identity, the half turn, and the flips in the two axes through the midpoints of opposite sides form a subgroup of the eight symmetries of a square. Is it normal, and if so, what is the collapse?

A triangle with three equal sides has six symmetries, counted as for the square: corner 1 can go to any of three places and corner 2 to either of the others, which gives the identity, the turns by a third and by two thirds of a turn, and three flips, each keeping one corner in place. Inside the six, the identity and a single flip form a subgroup of two, with three cosets, and it is not normal: the flip that exchanges corners 1 and 2, relabelled by the turn that carries 1 to 2, 2 to 3 and 3 to 1, is the flip that exchanges 2 and 3. The collapse fails concretely. The block containing the rotation by a third of a turn also contains “rotation then flip”. Combining that block with itself through “rotation” and “rotation” gives “rotation then rotation”; through “rotation then flip” and “rotation” it gives “rotation then flip then rotation”. With the turn and the flip named above, that combination carries 1 to 2, back to 1 and on to 2; it carries 2 to 3, where the flip leaves it, and on to 1; and it carries 3 to 1, then to 2, and back to 3. It is the flip itself. The flip lies in the block of the identity, and “rotation then rotation”, a turn by two thirds, does not. One combination of blocks has two answers.

Is every group with six elements either the powers of one of its elements or a copy of the six symmetries of a triangle?

* * *

A rearrangement is even or odd according to whether it is performed by an even or an odd number of swaps, and half of the twenty-four rearrangements of four objects, twelve, are even.

The rotations of a tetrahedron are exactly the even rearrange-

ments of its four corners. Each rotation rearranges the corners: a turn by a third or two thirds of a turn moves three corners in a cycle, which is two swaps; a half turn about an edge axis swaps two pairs, which is two swaps. So all twelve rotations are even, they rearrange the corners in twelve different ways, and there are only twelve even rearrangements: the rotations use up every one of them.

Inside those twelve, the identity and the three half turns, the rearrangements that swap two pairs, form a subgroup: each undoes itself, and performing two different ones gives the third. It is normal, and the reason is a general fact about relabelling. If k carries the object a to $k(a)$, then " k backwards, then h , then k " carries $k(a)$ back to a , on to $h(a)$, and on to $k(h(a))$: it does to the renamed objects exactly what h does to the originals. So a rearrangement that swaps two pairs, relabelled, swaps two pairs. For instance, let h swap 1 with 2 and 3 with 4, and let k carry 1 to 2, 2 to 3 and 3 to 1, leaving 4 alone; the relabelled rearrangement swaps 2 with 3 and 1 with 4. The same fact is why the triangle's single flip was not normal: relabelled, it is a flip of different corners.

The collapse has three blocks, since twelve divided by four is three. One is the subgroup itself, the identity with the three half turns. For the others, take a third of a turn about one corner axis: combining it with each of the four members of the subgroup gives four rotations, one about each corner axis, and they fill a block. The block that undoes it holds the reverses of those four, the four turns by two thirds, and the two blocks are different, since a third of a turn and two thirds about the same axis differ by a third of a turn, which is not a half turn and not the identity. And a group of three elements consists of the powers of any element other than the identity. A group of twelve with no subgroup of six collapses onto a group of three.

* * *

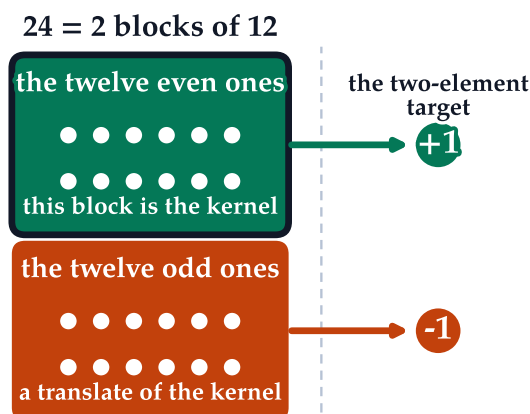
Collapsing created a rule that sends each element to its block, and the rule respects the operation, since the block of a combina-

tion is the combination of the blocks. A map from one group to another that respects the operation, with no requirement that it be reversible or reach everything, is a *homomorphism*, and homomorphisms are common. Reduction to remainders is one. Sending each symmetry of a square to whether or not it turns the square over is another, into a group of two. Sending each rearrangement of four objects to $+1$ if it is even and to -1 if it is odd is a third, the *sign*, and it respects the operation because performing one rearrangement after another adds the numbers of swaps, and even and odd then combine as $+1$ and -1 multiply.

Two questions organise a homomorphism: what is sent to the identity, and how much of the target is reached. What is sent to the identity is the *kernel*. It is a subgroup, and two facts about every homomorphism show it. The identity goes to the identity: the identity combined with itself is the identity, so its image combined with itself is that same image, and cancelling one copy leaves the identity of the target. And the reverse of each element goes to the reverse of its image, since the image of g combined with the image of the reverse of g is the image of the identity, which is the identity. So the identity is in the kernel; two of its members combined go to the identity combined with itself, which is the identity; and the reverse of a member goes to the reverse of the identity, which is the identity again. For the sign, the kernel is the twelve even rearrangements, the target has two elements, and $24 = 2 \cdot 12$.

A subgroup with only two cosets is normal. Let H be such a subgroup; relabelling a member of H by a member of H stays in H , so take k outside H . The collection of " k then h ", for h in H , is the coset of k , which is not H , so it is everything outside H . The collection of " h then k ", for h in H , is everything outside H as well, for three reasons. None of its members lies in H : if " h then k " were a member m of H , then k would be " h backwards, then m ", a member of H . Different h give different members, by cancelling k . So the collection has as many members as H , and that is the number of elements outside H , the two cosets having the same size; a collection of that many elements outside H is all of them. So for each h in H ,

" h then k " is " k then h' " for some h' in H , and " k backwards, then h , then k " is " k backwards, then k , then h' ", which is h' , a member of H . The twelve even rearrangements have two cosets among the twenty-four, and the four turns of a square have two cosets among its eight symmetries, so both are normal.



knowing what collapses to nothing fixes how much collapses everywhere

Figure 28.2. The sign map sends each of the twenty-four rearrangements of four objects to plus one or minus one, and the two blocks landing on the two destinations each hold exactly twelve, because every block is a translate of the one landing on the identity. *Assumes:* that the two blocks really are the same size, which is the coset argument and not something the picture establishes; a map can send blocks of unequal size only by failing to respect the operation

The count $24 = 2 \cdot 12$ is Lagrange's tiling again. If g and h have the same image, then " h backwards, then g " is sent to the identity, so it lies in the kernel, and g is h combined with a member of the kernel; conversely, combining h with a member of the kernel does not move its image. So the elements sent to any one target form a coset of the kernel, and all have the size of the kernel: the size of the group is the size of the kernel times the number of targets reached.

The two constructions describe one thing. Collapsing by a nor-

mal subgroup comes with the homomorphism sending each element to its block, whose kernel is the subgroup. And the kernel of every homomorphism is normal. If h is sent to the identity, then “ k backwards, then h , then k ” is sent to the reverse of the image of k , then the identity, then the image of k , and that combination is the identity; so every relabelling of a member of the kernel is again in the kernel. Kernels and normal subgroups are the same subgroups. The image of a homomorphism is a copy of the collapse by its kernel: sending the block of g to the image of g matches blocks with reached targets one for one, as the cosets above showed, and it respects the operation. For the sign, collapsing twenty-four rearrangements by the twelve even ones gives two blocks, a copy of the target. For reduction modulo 12, the kernel is the multiples of 12, and the collapse is the twelve families.

* * *

Not every natural map respects the operation. Sending each symmetry of a square to that symmetry performed twice is a rule with one answer for each symmetry, and it is not a homomorphism. A quarter turn followed by the flip in the vertical axis is a flip in a diagonal, which performed twice is the identity; but the quarter turn performed twice is the half turn, the vertical flip performed twice is the identity, and the half turn followed by the identity is the half turn. The combination is sent to the identity, while the combination of the two images is the half turn, so the rule does not respect the operation.

And the image of a homomorphism is a subgroup of the target, since it contains the identity and the combinations and reverses of images. Its size divides the size of the group, being a copy of the collapse by the kernel, whose number of blocks is the size of the group divided by the size of the kernel; and it divides the size of the target, being a subgroup. A homomorphism from a group of twelve into a group of twenty reaches a number of elements dividing both 12 and 20, which is 1, 2 or 4; sending each hour of the

twelve-hour clock to five times that hour on the twenty-hour clock reaches four. The map is allowed because changing an hour by 12 changes five times it by 60, a multiple of 20, and it respects addition because $5(a + b) = 5a + 5b$.

How many homomorphisms are there from the twelve-hour clock to the eighteen-hour clock, and how many elements does each reach?

A GROUP AT WORK

Colour each corner of a square black or white. There are sixteen ways to do it, and a fair count gives fewer, because a colouring with one black corner is the same colouring whichever corner it is. Counted that way there are six: no black corners; one; two that are neighbours; two that are opposite; three; four. The sixteen fall into families of sizes 1, 4, 4, 2, 4 and 1.

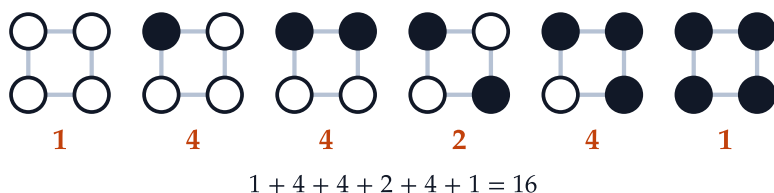


Figure 29.1. One colouring from each of the six families into which the symmetries of the square sort the sixteen colourings of its corners, with the number of colourings in each family. The sizes 1, 4, 4, 2, 4 and 1 all divide 8, the number of symmetries. *Assumes:* that turning the square over is allowed; with rotations only, the count is still six

What decided that two colourings are the same was the group of the eight symmetries of the square. Each symmetry carries every colouring to a colouring, and two colourings were called the same when some symmetry carries one to the other. Groups were met

as collections of symmetries, and here a group is doing what symmetries do, moving the members of some other collection about. A group *acts* on a collection when each element of the group moves the things of the collection, the identity moving nothing, and “ g then h ” moving each thing as g does and then as h does.

The things that a given thing x can be carried to make up the *orbit* of x . If g carries x to y , then g backwards carries y to x , so each of x and y is in the orbit of the other. Whatever y can be carried to, x can be carried to as well, by going to y first; and whatever x can be carried to, y can be carried to, by going back to x first. So the orbit of x and the orbit of y are the same. It follows that a thing lying in two orbits has each of them as its own orbit, and the two are one. So the orbits tile the collection as cosets tile a group: two of them are identical or share nothing. The elements of the group that leave x where it is make up the *stabiliser* of x , which is a subgroup, since performing two of them, or undoing one, leaves x where it is.

The size of the orbit of x , times the size of the stabiliser of x , is the size of the group.

Let y be in the orbit, and let g be one element that carries x to y . If s is in the stabiliser, then “ s then g ” also carries x to y . And every element g' that carries x to y is of that kind: “ g' then g backwards” carries x to y and back, so it is some s in the stabiliser, and g' is “ s then g ”. Different s give different elements “ s then g ”, since g can be cancelled. So the elements that carry x to y are as many as the stabiliser has members, whatever y is. Now every element of the group carries x to one member of the orbit and to no other, so the group is sorted into blocks, one block for each member of the orbit, and every block has the size of the stabiliser. That is the statement: the size of the group is the size of the orbit times the size of the stabiliser. The blocks are the cosets of the stabiliser, with the fixed element g written last instead of first, and the cancellation is the one that gave all cosets one size. For the colouring with two black corners that are neighbours, the stabiliser is the identity and one flip, and the orbit has $8/2 = 4$ members. For two opposite black

corners the stabiliser has four members, the identity, the half turn and the flips in the two diagonals, and the orbit has two.

The rotations of the tetrahedron were counted this way before the theorem had a name: a corner can go to any of 4 places, and 3 rotations keep it in place, which makes 12. For a cube, a face can be carried to any of the 6 faces, and the rotations that keep a face in its place are the 4 turns about the axis through its centre. There are no others: such a rotation sends a chosen corner of the face to one of the four corners of the face, the four turns send it to four different ones, and two such rotations that send it to the same corner are the same rotation, since performing the first and undoing the second leaves in place that corner, the centre of the face and the centre of the cube, three points not on a line, and the argument made for the tetrahedron shows that such a motion is the identity. That makes 24 rotations. Counting by corners gives $8 \cdot 3$ and by edges $12 \cdot 2$, and they agree, as they must. A regular dodecahedron, with its 12 faces of five sides, has $12 \cdot 5 = 60$.

What group is the group of the 24 rotations of the cube? The cube has four long diagonals, joining opposite corners, and each rotation rearranges the four. There are 24 rearrangements of four things, which suggests that the rotations are all of them. Apart from the identity, a rotation of the cube turns it about one of three kinds of axis: through the centres of two opposite faces, along a long diagonal, or through the midpoints of two opposite edges. Here are the rotations, by kind, with what each does to the diagonals.

rotation	number	the four diagonals
identity	1	none moved
quarter turn, face axis	6	all four in a cycle
half turn, face axis	3	swapped in two pairs
third of a turn, diagonal	8	one kept, three cycled
half turn, edge axis	6	two swapped, two kept

The five kinds account for $1 + 6 + 3 + 8 + 6 = 24$ rotations, and only the identity keeps every diagonal in place. Send each rotation to its rearrangement of the diagonals. One rotation and then

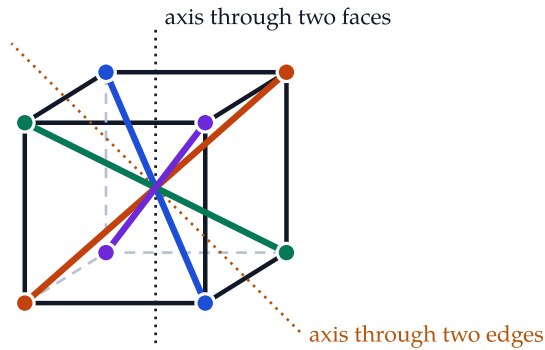


Figure 29.2. A cube and its four long diagonals, drawn in four colours. A quarter turn about the vertical axis through two faces carries each diagonal to the next; a third of a turn about a diagonal keeps that one and cycles the other three; a half turn about the axis through the midpoints of two opposite edges swaps two and keeps two. *Assumes: a drawing in slanted projection, in which lengths along the depth are shortened*

another rearranges the diagonals as the first does and then as the second does, so this is a homomorphism. Its kernel, the rotations that move no diagonal, is the identity alone, and the rotations sent to any one rearrangement are as many as the kernel has members, which is one. So the 24 rotations go to 24 different rearrangements, and that is all of them. The rotations of a cube are the rearrangements of four things, realised in space, and the table is the list of the 24 rearrangements by the lengths of their cycles.

* * *

How many orbits are there? For the corners of the square the six were found by looking. For the faces of a cube painted in three colours, there are $3^6 = 729$ paintings and 24 rotations, and looking will not do.

The number of orbits is the average, over the elements of the group, of the

number of things each element leaves in place.

Count in two ways the pairs (g, x) in which g leaves x in place. Element by element, the total is the sum of the numbers of things left in place, which is what the formula averages. Thing by thing, x contributes the size of its stabiliser, which is the size of the group divided by the size of the orbit of x . In an orbit with k members, then, each member contributes a k -th of the size of the group, and the k of them together contribute the size of the group, whatever k is. So the total is the size of the group times the number of orbits, and dividing by the size of the group gives the formula. Cauchy knew it in 1845 and Frobenius proved it in 1887; it is usually named after Burnside, who put it in a book.

A symmetry leaves a colouring in place exactly when corners that it carries to one another have the same colour. So each cycle of corners gets one colour, chosen freely, a corner that stays put counting as a cycle of one, and the number of colourings left in place is a product of 2 s, one for each cycle. For the corners of the square: the identity leaves all 16 colourings in place. A quarter turn leaves a colouring in place only when all four corners have one colour: 2. The half turn needs opposite corners to match: 4. A flip in an axis through the midpoints of two sides swaps the corners in two pairs: 4. A flip in a diagonal keeps two corners and swaps the other two: 8. There are two quarter turns, one half turn and two flips of each sort, which with the identity are the eight symmetries. Taking the identity and the turns first and then the flips, the average is

$$(16 + 2 + 4 + 2 + 4 + 4 + 8 + 8)/8 = 48/8 = 6$$

With a colours in place of two, each count becomes a power of a , the exponent being the number of cycles into which the symmetry sorts the corners, and the number of colourings is $(a^4 + 2a^3 + 3a^2 + 2a)/8$, which is 21 for three colours.

For the faces of the cube, read the cycles off the five kinds of rotation. The identity leaves 6 cycles of faces. A quarter turn keeps the two faces on its axis and moves the other four in a cycle: 3 cycles. A half turn about the same axis: 4 cycles. A third of a turn about a

long diagonal moves the three faces at each end in a cycle: 2 cycles. A half turn about an axis through edges swaps the faces in three pairs: 3 cycles. So with a colours the number of different cubes is

$$(a^6 + 6a^3 + 3a^4 + 8a^2 + 6a^3)/24$$

which is 10 for two colours and 57 for three.

A formula of this kind has to give a whole number, and sometimes that is news. Take strings of p beads on a circle, p a prime, in a colours, with the p rotations as the group. The identity leaves all a^p strings in place, and each of the other $p - 1$ rotations leaves in place only the a strings of a single colour. A string left in place by a rotation by k places has one colour at a place, at the place k further on, at the place k beyond that, and so on; and when p is prime those steps of k reach every place on the circle, so the string has one colour throughout. So the number of different strings is $(a^p + (p - 1)a)/p$. The top is $a^p - a$ with pa added, and p divides pa , so for the count to be a whole number p must divide $a^p - a$. The proof of that congruence by rotating strings was this count of orbits.

Chemistry supplied a use before the mathematics was in order. In 1865 Kekulé proposed that the six carbon atoms of benzene form a ring, a regular hexagon with one hydrogen atom at each corner. Replace some of the hydrogens by chlorine: how many different compounds should there be? The hexagon has 12 symmetries, and the ways of choosing k corners out of six fall, under them, into

$$1, 1, 3, 3, 3, 1, 1$$

orbits for k from 0 to 6. For $k = 2$ the three are the pairs of corners that are neighbours, one apart, and opposite. Chemists had found exactly three dichlorobenzenes, and in time exactly the twelve chlorinated benzenes that the ring predicts, the orbits for k from 1 to 6, and the agreement of such counts was part of the evidence that the ring is real. In 1937 Pólya turned the method into a systematic tool for counting chemical compounds.

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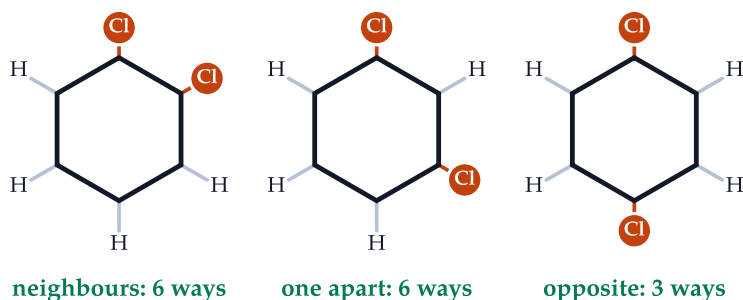


Figure 29.3. The fifteen ways of choosing two corners of a regular hexagon fall into three families under its twelve symmetries: neighbours, one apart, and opposite. They are the three dichlorobenzenes, and chemists have never found a fourth. *Assumes:* that the benzene ring has the full symmetry of the regular hexagon, which was the point at issue

The size of a subgroup divides the size of the group, and the converse is false: the twelve rotations of the tetrahedron have no subgroup of six. For a prime divisor the converse holds, and the proof is the rotation of strings again, with the group itself supplying the beads.

If a prime p divides the size of a finite group, the group has an element of order p .

Let the group have n elements. Consider the strings $(g_1, g_2, \dots, g_p)$ of p elements of the group whose combination, in that order, is the identity. The first $p - 1$ entries can be chosen freely and the last is then forced, being the combination of the others, taken backwards, so there are n^{p-1} such strings. Rotating such a string, moving g_1 from the front to the back, gives another one: if “ g_1 then the rest” is the identity, then “the rest” is g_1 backwards, and “the rest then g_1 ” is the identity too. So the clock with p hours acts on these strings by rotation.

The size of an orbit times the size of a stabiliser is the size of the clock, which is p , so each orbit has 1 or p members, those being the divisors of p . An orbit with one member is a string that rotation

does not change, so its entries are all equal: it is a string $(g, g, \dots, g)$, which belongs to the collection exactly when g combined with itself p times is the identity. The orbits with p members account for a multiple of p strings, and the whole collection has n^{p-1} , a multiple of p because p divides n . Taking the first from the second, the number of orbits with one member is a multiple of p , and that number is the number of elements g with g^p equal to the identity. It is not 0, since the identity is one of them. So there are at least $p - 1$ others, and each of those has order p .

Let m be the order of such a g , the first number of times g combined with itself gives the identity. Then m is at most p , and m is not 1, because g is not the identity. Write $p = qm + r$ with r below m . Combining g with itself qm times gives the identity, and so does p times, so r times gives the identity as well; since m was the first such number and r is below m , that forces $r = 0$. So m divides the prime p , and m is p . Cauchy proved the theorem in 1845; this proof was found by McKay in 1959.

A group also acts on itself, by relabelling: k moves h to “ k backwards, then h , then k ”. The orbits of this action are called *classes*. The stabiliser of h consists of the elements k for which “ k backwards, then h , then k ” is h , which are the elements that commute with h . A normal subgroup is a subgroup that relabelling does not move out of itself, which says that it is made up of whole classes. And the size of each class divides the size of the group, because a class is an orbit, and the size of an orbit times the size of a stabiliser is the size of the group.

Suppose the size of a group is a power of a prime p . The classes have sizes that divide that power, so each has 1 member or a multiple of p . The sizes add up to the size of the group, a multiple of p , so the number of classes with one member is a multiple of p ; and there is at least one, the identity. An element h alone in its class is one that no relabelling moves: “ k backwards, then h , then k ” is h for every k , which says that “ h then k ” is “ k then h ”. So an element alone in its class commutes with everything. So a group whose size is a power of a prime has at least p elements that commute

with everything. If the size is p^2 , more follows. The elements that commute with everything are the elements lying in the stabiliser of every element under relabelling, and performing two of them, or undoing one, stays inside every stabiliser; so they form a subgroup Z . Its size divides p^2 and is at least p , so it is p or p^2 . Suppose it were p , and take g outside Z . The elements that commute with g form a subgroup that contains Z and also g , so it has more than p members, and its size divides p^2 , so it is the whole group; but then g commutes with everything and belongs to Z . So Z is the whole group:

every group with p^2 elements is commutative.

The groups with four elements were found to be commutative by making their tables. The same holds for 9, 25 and 49 elements, and no table was made.

* * *

Suppose h carries u to v , and k carries u to u' and v to v' . Then “ k backwards, then h , then k ” carries u' back to u , on to v , and on to v' : where h carried u to v , the relabelled rearrangement carries u' to v' . So relabelling a rearrangement renames the objects in its cycles and leaves the lengths of the cycles as they were. And two rearrangements with the same cycle lengths are relabellings of each other: write the cycles of one above the cycles of the other, equal lengths together, and let k carry each object to the one below it. So in the group of all rearrangements of a collection, the classes are given by the cycle lengths. For four things the classes have 1, 6, 3, 8 and 6 members, as in the table of the cube. A normal subgroup has to be a union of classes that includes the identity, with a size dividing 24; whether such a union is a subgroup still has to be checked. From 1, 6, 3, 8, 6 the only such totals are 1, $1 + 3 = 4$, $1 + 3 + 8 = 12$ and 24. The subgroup of four is the identity with the three double swaps, and the subgroup of twelve is the even rearrangements; both were met, and now it is known that there are no others.

Now take the sixty even rearrangements of five things. By cycle lengths they are the identity, the 20 cycles of three, the 15 double swaps, and the 24 cycles of five. Within the group of all 120 rearrangements those are classes. In the group of sixty they need not be: the relabelling that carries one cycle of five, say, to another may be an odd rearrangement, and the group of sixty has only the even ones. So the classes of the group of sixty have to be computed afresh, from stabilisers.

A cycle of three, such as the one carrying 1 to 2, 2 to 3 and 3 to 1, has a class of 20 among the 120, so $120/20 = 6$ rearrangements commute with it: its own three powers, each with or without the swap of 4 and 5. Three of the six are even. So in the group of sixty its stabiliser has three members and its class has $60/3 = 20$: the cycles of three remain one class. A double swap has a class of 15 among the 120, so $120/15 = 8$ rearrangements commute with it. One of the eight is a single swap, either of the two swaps the double swap is made of, and it is odd. Pair each of the eight with the one obtained by performing that swap after it. The result is again one of the eight, since the eight make up a stabiliser, and performing the swap a second time gives the first back, so the eight fall into pairs, and, as with all rearrangements, each pair has one even member and one odd member. So 4 of the eight are even, and the class in the group of sixty has $60/4 = 15$.

A cycle of five has a class of 24 among the 120, so 5 rearrangements commute with it, and those are its own powers, all even. In the group of sixty its stabiliser still has five members, and its class has only $60/5 = 12$. The same holds for every cycle of five, and there are 24 of them, so the cycles of five break into two classes of twelve. The classes of the sixty even rearrangements have

1, 15, 20, 12, 12

members. A normal subgroup would be a union of some of these, including the 1, with a total that divides 60. The possible totals are 1, 13, 16, 21, 25, 28, 33, 36, 40, 45, 48 and 60, and of these only 1 and 60 divide 60.

The sixty even rearrangements of five things have no normal subgroup except the identity alone and the whole group.

Such a group is called *simple*: collapsing it gives back either the group itself or the group of one element, and every homomorphism out of it is one-to-one or sends everything to the identity. No smaller group is simple, apart from the clocks with a prime number of hours: a check of the sizes below 60, one size at a time, finds in every other group a normal subgroup that is neither the identity alone nor the whole group, and that check, which is in the literature, is relied on here and not repeated. The sixty rotations of the dodecahedron are a copy of this group: five cubes can be inscribed in the solid, each rotation rearranges the five, and the rearrangements that arise are the sixty even ones; this too is a classical fact, relied on here as others have proved it. The even rearrangements of four things collapsed onto a group of three, and those of three things are a group of three already. At five the collapsing stops, and a good deal will turn on that.

In how many ways can the faces of a tetrahedron be painted with a colours, two paintings being the same when a rotation carries one to the other?

How many different dice can be made, with the numbers 1 to 6 on the faces of a cube in any arrangement?

How many necklaces of six beads in two colours are there when a necklace may be turned round, and how many when it may also be turned over?

The rotations of a cube also rearrange its three axes through the centres of opposite faces. Which rotations keep all three axes in place, and what does that say about the group of 24?

What are the sizes of the classes among the 120 rearrangements of five things, and which unions of them are normal subgroups?

AREA BEFORE LIMITS

A rectangle a units by b units, with a and b whole numbers, contains ab unit squares, which can be counted. A region with a curved edge contains no whole number of anything, and no count gives its area. What can be said about area before any curve is measured comes to three rules. The area of a region is never negative. If a region is cut into two pieces that do not overlap, the area of the whole is the sum of the areas of the pieces. A rectangle with sides a and b has area ab . Together the first two say that a region lying inside another has at most the area of the other: the larger region is the smaller one together with what is left over, and what is left over has an area that is not negative.

The rules say what is meant by area, and only the third puts a number in. For sides that are fractions the third rule is still a count: a rectangle with sides $3/2$ and $5/3$ contains, in half-units by third-units, fifteen small rectangles, each a sixth of a unit square, so its area is fifteen sixths, which is $3/2$ times $5/3$. For a side that has no ratio of whole numbers to the unit, such as the diagonal of the unit square, no equal part of the unit fits into the side a whole number of times, so there are no small rectangles to count and the rule is assumed. Even the product of two such lengths has no meaning as a number until the real numbers are built. The rectangles are those of the flat plane, with four right angles, and that they tile the plane

without gaps is taken as given.

* * *

A staircase can be measured. Cut the interval from 0 to 1 into four equal pieces and raise over each piece a rectangle of some height. The region under the flat-topped steps is four rectangles side by side, and its area is a sum of four products of width and height, with no limit and no curve involved. A function that is constant on each piece of a cut-up interval is a *step function*, and the area under it is a finite sum.

That sum seems to depend on how the interval was cut. If a step of width $1/4$ is described instead as two steps of width $1/8$ at the same height, the sum has more terms, and its value is the same, since a rectangle of width $1/4$ has the area of two rectangles of width $1/8$ and the same height. Given any two cuttings of an interval, make a third that cuts at every point where either of them cuts, a common refinement of the two. Compared with the first description, the third only splits steps into narrower steps of the same height, so it gives the same sum; compared with the second, the same is true. So any two descriptions of one staircase give one number.

Now a curve. Take the parabola, the graph of x^2 from 0 to 1. Cut the interval into n equal pieces. Over each piece the parabola rises from left to right, so the rectangle whose height is its value at the left end lies under the curve, and the rectangle whose height is its value at the right end lies over it. The lower staircase lies inside the region and the upper staircase contains it, so whatever the area of the region is, if it has one, it lies between the two staircase areas.

Both staircase areas can be computed exactly. The pieces have width $1/n$ and the right-hand heights are $(1/n)^2, (2/n)^2, \dots, (n/n)^2$, so the upper staircase has area $(1^2 + 2^2 + \dots + n^2)/n^3$. The left-hand heights are $0, (1/n)^2, \dots, ((n-1)/n)^2$, the same list with a 0 put in front and $(n/n)^2$ dropped, so the lower staircase has area $(1^2 + 2^2 + \dots + (n-1)^2)/n^3$, the same sum without its last term. With

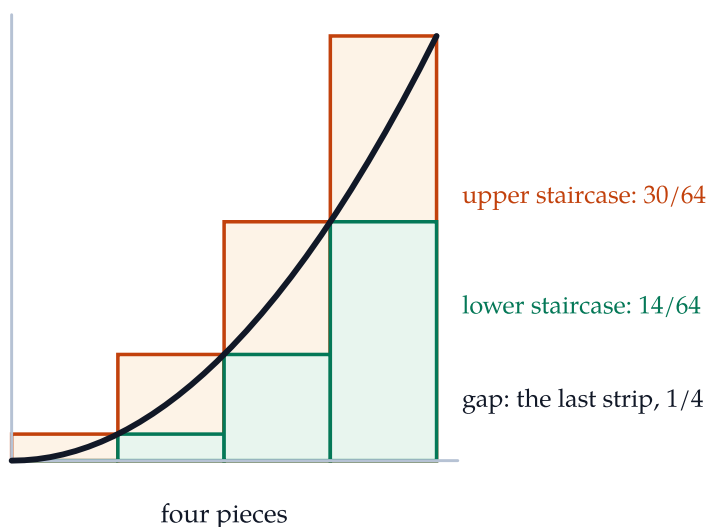


Figure 30.1. The region under the parabola from 0 to 1 trapped on four pieces: the lower staircase, with each step at the curve's left-hand height, lies inside the region and has area $14/64$; the upper staircase, at the right-hand heights, contains it and has area $30/64$; the difference is the last strip alone, and it shrinks as the pieces multiply. *Assumes:* that a region containing another has at least its area, and that the staircases' areas are sums of rectangles, which are the three rules for area and nothing more; that the two areas close on a single number is exactly what is not assumed

$n = 4$ the lower area is $14/64$ and the upper is $30/64$. The difference is always the last strip, $(n/n)^2/n$, which is $1/n$.

The sum of the first n squares was found once already, from the diagonals of the triangle of choices, as $2(n+1 \text{ choose } 3) + (n+1 \text{ choose } 2)$. A double count of another kind lands on the same two parts and shows what each of them counts. Count the triples (a, b, c) of whole numbers from 0 to n in which c is larger than both a and b . Sorted by c , there are c choices for a and c for b , so c^2 triples for each c , and $1^2 + 2^2 + \dots + n^2$ in all.

Sort them instead by whether a and b agree. If they do, the triple amounts to two different numbers out of the $n+1$: the larger is c , and the smaller is the common value of a and b . Two numbers

can be chosen out of $n + 1$ in $(n + 1)n/2$ ways, so there are that many triples of this kind. If they do not, the triple amounts to three different numbers out of the $n + 1$: the largest is c , and the other two can be handed to a and b in either order. Three numbers can be chosen out of $n + 1$ in $(n + 1)n(n - 1)/6$ ways, and each choice gives two triples, so there are $2 \cdot (n + 1)n(n - 1)/6$ of this kind. So the sum of the first n squares is $n(n + 1)/2 + (n + 1)n(n - 1)/3$, which is $n(n + 1)(2n + 1)/6$; at $n = 4$ the two parts are 10 and 20, and $1 + 4 + 9 + 16 = 30$.

The upper area is the sum of the first n squares divided by n^3 , which is $(n + 1)(2n + 1)/(6n^2)$; multiplied out it is $(2n^2 + 3n + 1)/(6n^2)$, which is $1/3 + 1/(2n) + 1/(6n^2)$. The lower area uses the sum of the first $n - 1$ squares, which by the same formula is $(n - 1)n(2n - 1)/6$. Divided by n^3 it is $(n - 1)(2n - 1)/(6n^2)$, which is $1/3 - 1/(2n) + 1/(6n^2)$. Every upper staircase has area above $1/3$. Every lower staircase has area below $1/3$, because $1/(6n^2)$ is smaller than $1/(2n)$. And the difference between the two, $1/n$, falls below every tolerance.

So no number other than $1/3$ lies between the areas of all the lower staircases and all the upper ones: any other number differs from $1/3$ by some definite amount. Take n so large that $1/n$ is smaller than that amount. The lower and upper areas for that n have $1/3$ between them and are less than that amount apart, so the other number cannot lie between them as well. If the region under the parabola has an area, the area is $1/3$. The three rules do not say that it has one, since the only number they put in is for rectangles, and the region under a parabola cannot be cut into finitely many rectangles: its top edge is curved. The natural meaning to give is this: a region has an area when exactly one number lies between the areas of all the staircases inside it and all the staircases containing it, and that number is its area.

That meaning speaks of all staircases, not only those on equal pieces, so one check remains. A staircase inside the region lies inside each upper staircase on equal pieces, because that upper staircase contains the whole region. So its area is at most $1/3 + 1/(2n)$

$+ 1/(6n^2)$ for every n , and a number that is at most $1/3$ plus an amount falling below every tolerance is at most $1/3$. In the same way every staircase containing the region has area at least $1/3$. So $1/3$ lies between the areas of all the staircases inside the region and the areas of all the staircases containing it, and no other number does, since none lies even between the staircases on equal pieces. In that sense the area under the parabola from 0 to 1 is $1/3$.

What is the sum of the first n cubes, and what area do the staircases give under the graph of x^3 from 0 to 1?

* * *

Archimedes reached the same number by another route: the region between a parabola and a chord across it has area four thirds of the largest triangle inside it. His proof fills the region with triangles, finitely many at every stage. Triangles, trapezoids and parallelograms are taken here with their everyday areas, in the same way that half the base times the height was used for the rotating ray.

Take the chord of the parabola from (p, p^2) to (q, q^2) . It lies on the line $y = (p + q)x - pq$, and it stands above the curve by $(p + q)x - pq - x^2 = (x - p)(q - x)$. Write $m = (p + q)/2$ for the midpoint and $w = q - p$ for the width of the chord. Then $x - p$ is $w/2 + (x - m)$ and $q - x$ is $w/2 - (x - m)$, so their product is $w^2/4 - (x - m)^2$. That is largest at $x = m$, midway, where it is $w^2/4$. Put a triangle on the chord with its third corner on the curve at (m, m^2) . Its area is the area under the chord less the areas under the two shorter chords, from (p, p^2) to (m, m^2) and from (m, m^2) to (q, q^2) . Each of the three regions is a trapezoid, and the area of a trapezoid is its width times the average of its two upright sides. That gives $(w/2)(p^2 + q^2) - (w/4)(p^2 + m^2) - (w/4)(m^2 + q^2)$, which is $(w/4)(p^2 + q^2 - 2m^2)$. Since $2m^2$ is $(p + q)^2/2$, the bracket $p^2 + q^2 - 2m^2$ is $(q - p)^2/2$, which is $w^2/2$. So the triangle has area $w^3/8$, wherever along the curve the chord lies.

On the chord from $(0, 0)$ to $(1, 1)$ that is $1/8$, which can be

checked directly: the triangle under the chord is $1/2$, the triangle under the shorter chord from $(0, 0)$ to $(1/2, 1/4)$ is $1/16$, the region under the one from $(1/2, 1/4)$ to $(1, 1)$ is a rectangle of $1/8$ and a triangle of $3/16$, and $1/2 - 1/16 - 5/16 = 1/8$. The two shorter chords have width $1/2$ and carry triangles of $1/64$ each, together a quarter of $1/8$; the four chords after them carry a quarter of that again. Stage j puts a triangle on each of 2^j chords of width $1/2^j$. Each of those triangles has area $(1/8)(1/2^j)^3$, which is $(1/8)(1/8)^j$, and 2^j of them come to $(1/8)(1/4)^j$ in all. By the subtraction that summed the powers of $5/6$, the stages from 0 to k add up to

$$(1/8)(1 + 1/4 + \dots + 1/4^k) = (1/6)(1 - 1/4^{k+1})$$

How much of the segment do the triangles leave uncovered? After stage k there are 2^{k+1} slivers, each between the curve and a chord of width $w = 1/2^{k+1}$. Take one sliver, with its chord from (p, p^2) to (q, q^2) and m midway between p and q . The curve stays above the line $y = 2mx - m^2$, since $x^2 - (2mx - m^2) = (x - m)^2$, a square. That line has slope $2m$, which is $p + q$, the slope of the chord, so the two lines are parallel. At $x = m$ the line meets the curve, and the chord is $w^2/4$ above the curve there, so the line lies $w^2/4$ below the chord all the way along. So each sliver lies in a parallelogram of width w and height $w^2/4$, of area $w^3/4$, and all the slivers together have area at most $2^{k+1} \cdot w^3/4$, which is $(1/4)(1/4^{k+1})$.

If the segment has an area, that area is at least $(1/6)(1 - 1/4^{k+1})$ and at most that plus $(1/4)(1/4^{k+1})$, which is $1/6 + (1/12)(1/4^{k+1})$, for every k . Only $1/6$ lies in all these brackets, since $1/4^{k+1}$ falls below every tolerance. The test for having an area is the one already set, with triangles and parallelograms in the place of the staircases. The triangles up to any stage make a figure inside the segment, and the same triangles together with the parallelograms around the slivers make a figure that contains it. Every figure of the first kind has area below every figure of the second kind, and the two kinds come as close together as desired, so exactly one number lies between them. So the segment has area $1/6$, four thirds of its largest triangle, and the region under the curve is $1/2 - 1/6$, which

is $1/3$. Archimedes argued in the same way. He had the finite identity $1 + 1/4 + \dots + 1/4^k + (1/3)(1/4^k) = 4/3$, and he excluded every value but four thirds of the triangle by showing that a larger or a smaller one would contradict some stage; he never added up infinitely many terms.

Area also gives chance a meaning where there is nothing to count. Choose a point at random in the unit square: how likely is it to lie under the parabola? The chance of an event was a ratio of counts, and here both counts are infinite, so that meaning gives no answer. If “at random” means that the chance of landing in a region is its share of the area of the square, the answer is $1/3$. Nothing forces that meaning, and a question about chords shows what happens when the meaning is left unsaid.

Draw a circle and, inside it, the equilateral triangle with its corners on the circle. Choose a chord of the circle at random. How likely is it to be longer than a side of the triangle? Three reasonable meanings of “at random” give three answers.

Fix one end of the chord at a corner of the triangle and let the other end fall anywhere on the circle. The chord is longer than a side exactly when the other end lies on the arc between the two other corners, which is a third of the circle. The answer is one third.

A chord is settled by its midpoint. Turn the circle, of radius r , so that the chord lies level at height d above the centre: it meets $x^2 + y^2 = r^2$ where x^2 is $r^2 - d^2$, so its midpoint is straight above the centre at distance d , and its length is $2\sqrt{(r^2 - d^2)}$. So the closer the midpoint is to the centre, the longer the chord. A side of the triangle cuts off a third of the circle, so the radii to its ends open at 120° ; turned level, the side has its ends at 30° and 150° round from the horizontal, at height $r \sin 30^\circ$, which is $r/2$. So the sides of the triangle are chords whose midpoints lie at half the radius from the centre. So a chord is longer than a side exactly when its midpoint is closer to the centre than half the radius. Choose a direction for the chord, and let its midpoint fall anywhere along a radius perpendicular to that direction. Half of that radius lies within half the radius of the centre. The answer is one half.

Let the midpoint fall anywhere in the disc, sharing out by area. The midpoints within half the radius of the centre fill a disc of half the radius, which has a quarter of the area, since halving every length quarters every rectangle and so every staircase; that a disc has an area at all is settled later, with the areas under decreasing curves. The answer is one quarter.

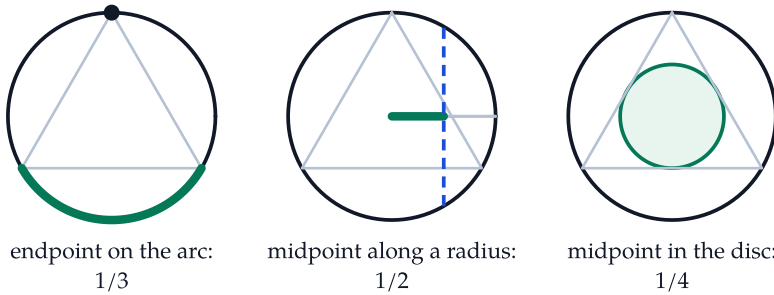


Figure 30.2. Three ways to choose a chord of a circle at random, and whether it beats the side of the inscribed triangle: by its far endpoint on the circle, one third of the arc wins; by the distance of its midpoint from the centre along a radius, half the length wins; by its midpoint anywhere in the disc, a quarter of the area wins. *Assumes:* that a chord is longer than the side exactly when its midpoint lies within half the radius of the centre, which follows from the side sitting at exactly half the radius; and that sharing out an arc, a length or an area evenly are three different acts

None of the three is wrong. “A chord at random” says nothing until it says which quantity is shared out evenly, and the three choices shared out an arc, a length and an area.

AREA BY COUNTING POINTS

Regions under curves were given areas by staircases: a region has an area when exactly one number lies between the totals of the staircases lying inside it and the totals of the staircases containing it. Then, for the parabola, triangles and trapezoids and parallelograms were used with the areas everybody gives them, half the base times the height and the rest. Those areas were never checked against the definition. The formulas are right, and one staircase count shows it.

* * *

Take the line through the origin of slope m above 0, and the region under it from 0 to b . Cut the stretch into n equal pieces, each of width b/n . Over the piece from kb/n to $(k+1)b/n$ the line climbs from height mbk/n to height $m(k+1)b/n$, so the tallest rectangle inside the region over that piece has the lower height and the shortest rectangle covering it has the upper. Adding the inside rectangles,

$$(b/n)(mb/n)(0 + 1 + 2 + \dots + (n-1)) = mb^2(n-1)/(2n)$$

and adding the covering rectangles gives $mb^2(n+1)/(2n)$, the same sum run from 1 to n instead. Both differ from $mb^2/2$ by $mb^2/(2n)$, which falls below every tolerance once n is large

enough, so exactly one number lies between the two families, and it is $mb^2/2$. The check made for the parabola carries over: a staircase inside the region lies inside every covering staircase on equal pieces, and a staircase containing the region contains every inside one, so $mb^2/2$ lies between all the staircases inside and all those containing, and no other number does. That region is the right triangle with legs b along the horizontal and mb upright, so such a triangle has area half the product of its legs. The count carried two restrictions: the slope was above 0, and the stretch began at 0. Both can be lifted at once. Sliding the page sideways carries every rectangle to a rectangle with the same two sides, so it carries the staircases of a region to the staircases of the slid region without changing one area; and reflecting the page left to right does the same. A line of slope below 0 is the mirror image of one above, and a stretch anywhere is a slide of a stretch beginning at 0, so the rule holds for every straight line over every stretch.

Areas of regions set side by side add, which came with the definition. The region under the same line from 0 to b is the region from 0 to a set beside the region from a to b . So the region from a to b has area $mb^2/2 - ma^2/2$, which is $m(b^2 - a^2)/2$, and lifting the line by c adds a rectangle of area $c(b - a)$. Since $b^2 - a^2$ is $(b - a)(b + a)$, the total $m(b^2 - a^2)/2 + c(b - a)$ is $(b - a)$ times the quantity $m(a + b)/2 + c$, and that quantity is the average of the two heights $ma + c$ and $mb + c$. So

the region between a straight line and the horizontal, from a to b , has area $(b - a)$ times the average of the heights at the two ends

which is the trapezoid, and with one of the heights 0 it is a triangle.

A triangle anywhere on the page yields to the same rule. Let its corners be $P = (x_1, y_1)$, $Q = (x_2, y_2)$ and $R = (x_3, y_3)$, labelled so that $x_1 \leq x_2 \leq x_3$, and slide the whole figure upward until it lies above the horizontal, which changes no area. If $x_1 = x_3$ the three corners share a vertical line and there is no triangle, so take x_1 below x_3 .

The vertical line at x_2 cuts the triangle into a left piece and a right piece, either of which may be empty. Each piece lies between two

straight lines, an upper and a lower. The region between them over a stretch is the region under the upper line with the region under the lower line taken away, and both of those are of the kind above. Each has area the width times the average of its two end heights, and the average of the upper heights less the average of the lower heights is the average of the gaps. So the area of the piece is the width times the average of the vertical gaps at the two ends. For the left piece the gap at x_1 is 0, the two lines meeting at P , so its area is $(x_2 - x_1)g/2$, where g is the vertical gap at x_2 between the side from P to R and the corner Q . For the right piece the gap at x_3 is 0, so its area is $(x_3 - x_2)g/2$. Together the triangle has area $(x_3 - x_1)g/2$.

The side from P to R stands at height $y_1 + (y_3 - y_1)(x_2 - x_1)/(x_3 - x_1)$ above x_2 , and g is y_2 minus that height. It is positive when Q stands above that side. When Q stands below, g is negative, and every area written above changes sign together, the two pieces and the whole alike. That is why the size is taken in what follows. Twice the area is the size of $(x_3 - x_1)g$, and multiplying g through by $x_3 - x_1$ gives $(x_3 - x_1)(y_2 - y_1) - (x_2 - x_1)(y_3 - y_1)$. Changing every sign leaves the size alone, so

twice the area of the triangle on P , Q and R is the size of $(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)$

For a triangle standing on a horizontal side, P and R at the same height, the first product vanishes and what is left is the base times the height, so half the base times the height survives the test, as the parabola work assumed.

* * *

For two arrows $u = (u_1, u_2)$ and $v = (v_1, v_2)$ from the origin, write

$$T(u, v) = u_1v_2 - u_2v_1$$

and call it their *twist*. Twice the area of the triangle on P , Q and R is the size of $T(Q - P, R - P)$. Three facts about the twist carry everything below, and each comes from multiplying out: it changes sign when the two arrows trade places, so $T(u, u)$ is 0; it is 0 exactly when the two arrows lie along one line through the origin, since $u_1v_2 = u_2v_1$ says the coordinates are in proportion; and it adds in each slot, $T(u + u', v)$ being $T(u, v) + T(u', v)$, with $T(ku, v)$ equal to $kT(u, v)$.

Travel round the corners of a triangle and add the twists of successive corners, taken as arrows from the origin: $T(P, Q) + T(Q, R) + T(R, P)$. Multiplying out $T(Q - P, R - P)$ gives $T(Q, R) - T(Q, P) - T(P, R) + T(P, P)$. The last term is 0. Trading places turns $-T(Q, P)$ into $T(P, Q)$ and $-T(P, R)$ into $T(R, P)$. What is left is $T(P, Q) + T(Q, R) + T(R, P)$, the sum round the corners, so the two expressions agree. A triangle's area is therefore half the size of a sum taken once round its boundary, and the sign of the sum says which way round it was taken. Label the corners from left to right as before, so that the order P, Q, R gives $T(Q - P, R - P)$, which multiplies out to $-(x_3 - x_1)g$. When Q stands above the side from P to R , g is above 0, and the order that keeps the inside of the triangle on the left is P, R, Q , whose sum is $(x_3 - x_1)g$. When Q stands below, g is below 0, and that order is P, Q, R , whose sum is $-(x_3 - x_1)g$. Either way a triangle travelled with its inside on the left has a sum above 0, and the sum is twice its area. When two polygons are glued along a shared edge, both boundaries travelled with the inside on the left, that edge is travelled one way in the boundary of the first and the other way in the boundary of the second. If its ends are the corners M and N , the first sum holds the term $T(M, N)$ and the second holds $T(N, M)$, and these cancel. What remains of the two sums together is the sum round the boundary of the glued figure, travelled with its inside on the left, and since the areas of the two pieces add, that sum is again twice the area. Any polygon that can be cut into triangles therefore has area half the size of the sum

$$x_1y_2 - x_2y_1 + x_2y_3 - x_3y_2 + \dots + x_ny_1 - x_1y_n$$

the corners taken in the order they come round the boundary. A clerk with the corners of a field and no instrument but arithmetic can read off its area. The corners $(0, 0)$, $(7, 2)$, $(9, 7)$, $(3, 9)$ and $(-1, 4)$ give 112 for the sum, so that field has area 56.

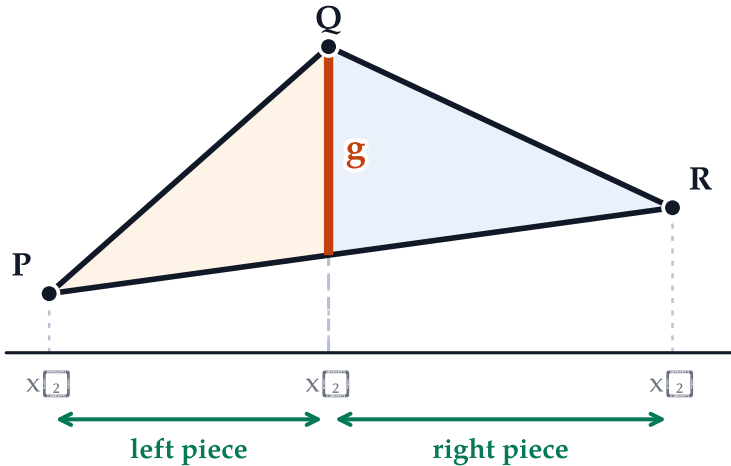
That every polygon can be cut into triangles has still to be shown. A *diagonal* is a segment between two corners that are not neighbours, lying inside the polygon and meeting the boundary only at its two ends. Every polygon with more than three corners has one. Take a corner A that is lowest, and leftmost among the lowest, and let B and C be its neighbours. If the segment from B to C lies inside the polygon it is a diagonal. If it does not, part of the boundary reaches into the triangle ABC . A side of the polygon that reaches in cannot have crossed AB or AC , which are sides of the polygon themselves, so it came in across the segment BC . A straight segment cannot cross BC twice, so that side does not go out again, and it ends at a corner within. So some corner lies inside the triangle ABC .

Of the corners inside take D , the one at the greatest distance from the line BC , and draw the line through D parallel to BC . The part of the triangle ABC between that parallel and A holds no corner but A , by the choice of D , and its boundary is made of pieces of AB and AC together with one piece of the parallel. A side of the polygon entering that part would have to leave it again, and it cannot cross AB or AC , so it would cross the piece of the parallel twice, which a straight segment cannot do. So no side enters, and the segment from A to D , which lies in that part, is a diagonal.

A diagonal breaks the boundary into two chains, and each chain closed by the diagonal is a polygon with no crossings; each has an inside and an outside, and the two insides lie within the inside of the whole and together fill it. So a diagonal cuts a polygon into two polygons. The ends of the diagonal are the only corners the two share, and since those ends are not neighbours, each chain holds at least one corner between them. Each of the two polygons therefore lacks a corner that the other has, and so has fewer corners than the whole. Cut the pieces again in the same way, and the pieces of the

pieces. A piece with more than three corners always has a diagonal and can be cut again, and the number of corners falls at every cut, so the cutting ends only at triangles, and every polygon can be cut into them. With that the rule above holds for every polygon.

each piece: width times the average of its two gaps



the outer gaps are 0, so the whole is $(x_2 - x_1)g / 2$

Figure 31.1. A triangle with its corners ordered from left to right and the vertical line through the middle one. Each piece lies between two straight lines, so each has width times the average of the vertical gaps at its two ends; one gap is 0 at each outer corner, so the two areas are $(x_2 - x_1)g/2$ and $(x_3 - x_2)g/2$ and the whole is $(x_3 - x_1)g/2$. Assumes: that the area between two straight lines over a stretch is the difference of the two areas beneath them, which needs the figure to lie above the horizontal, arranged here by sliding it up; and that areas of regions set side by side add

* * *

Let a *grid point* be a point both of whose coordinates are whole numbers, and take polygons whose corners are grid points. Call such a triangle *bare* when it holds no grid point but its three corners, inside or on its sides.

A bare triangle has area 1/2.

Let its corners be P , Q and R , let u be the arrow from P to Q and v the arrow from P to R , and complete the parallelogram on u and v , whose fourth corner is $P + u + v$. The half turn about the midpoint of the side from Q to R sends a point w to $Q + R - w$, so it carries grid points to grid points; it sends P to $Q + R - P$, which is $P + u + v$, and it carries the triangle onto the other half of the parallelogram. The triangle is bare, and so is its image: the same half turn would carry a grid point of the image, other than a corner, back to a grid point of the triangle, other than a corner, and there is none. So the parallelogram holds no grid point but its four corners.

Take any grid point w at all. The aim is to show that the arrow from P to w is a whole-number combination of u and v , that is, a whole number times u plus a whole number times v . The arrows u and v do not lie along one line, so $T(u, v)$ is not 0. Write x for the arrow from P to w , and put $s = T(x, v)/T(u, v)$ and $t = T(u, x)/T(u, v)$. Multiplying out, $u_1T(x, v) + v_1T(u, x)$ is $x_1T(u, v)$ and $u_2T(x, v) + v_2T(u, x)$ is $x_2T(u, v)$, so $su + tv$ is x : every arrow of the page is a combination of u and v , and the arrow from P to w is $su + tv$.

Let a be the largest whole number not above s and b the largest not above t , and put $z = w - au - bv$. Then z is a grid point, since w is one and u and v have whole coordinates. The arrow from P to z is $(s - a)u + (t - b)v$, with both coefficients at least 0 and below 1, so z lies in the parallelogram. The corners Q, R and $P + u + v$ have the coefficients 1 and 0, 0 and 1, and 1 and 1, so z is none of them. The parallelogram holds no other grid point, so z is P . Then w is $P + au + bv$, and the arrow from P to w is $au + bv$, a whole-number combination of u and v .

In particular the two arrows $(1, 0)$ and $(0, 1)$ are whole-number combinations: $(1, 0) = au + bv$ and $(0, 1) = cu + dv$ with a, b, c and d whole. Twist those two. By the rules for the twist, $T(au + bv, cu + dv)$ is $acT(u, u) + adT(u, v) + bcT(v, u) + bdT(v, v)$. The first and last terms are 0, and $T(v, u)$ is $-T(u, v)$, so

$$1 = T((1, 0), (0, 1)) = (ad - bc) \cdot T(u, v)$$

and both factors are whole numbers: $ad - bc$ because a, b, c and d are whole, and $T(u, v)$ because u and v have whole coordinates. Two whole numbers whose product is 1 are both 1 or both -1 . So $T(u, v)$ is 1 or -1 , and the triangle's area is half its size, which is $1/2$.

* * *

Now the count. For a polygon with grid points for corners, let I be the number of grid points strictly inside it and B the number on its boundary, the corners included.

Its area is $I + B/2 - 1$.

Cut the polygon into triangles by diagonals. If one of those triangles is not bare it holds a grid point other than its corners: when the point is inside, join it to the three corners and three triangles stand where one did; when it lies on a side, join it to the opposite corner and two stand where one did. Each step raises the number of triangles, and it makes a corner of one grid point that was not a corner before. The grid points in or on the polygon are finite in number, so the steps run out, and every triangle is bare when they do.

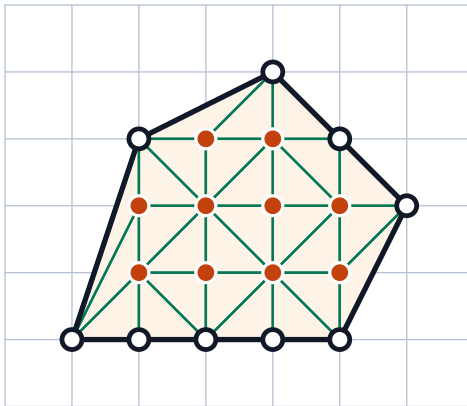
What is left is a network drawn on the page. Its dots are the grid points in or on the polygon, all $I + B$ of them, since a grid point that were not a dot would sit inside some triangle or on its side and that triangle would not be bare. Its edges are the sides of the triangles, and no dot lies partway along an edge, for the same reason. Its faces are the triangles and one more, the region outside the polygon. The boundary separates that region from everything within, as a closed polygon does. The formula for networks on the page applies: dots less edges plus faces is 2.

Count the sides of the triangles in two ways. If T is the number of triangles they have $3T$ sides between them, counted with repeats. Now let E be the number of edges of the network. A side inside the polygon belongs to two triangles and a side on the boundary to one.

The boundary carries B grid points, which cut it into B edges, so the other $E - B$ edges lie inside. Counting each inside edge twice and each boundary edge once gives $3T = 2(E - B) + B$, which is $2E - B$, and so $E = (3T + B)/2$. With $I + B$ dots and $T + 1$ faces,

$$(I + B) - (3T + B)/2 + (T + 1) = 2$$

Multiplying by 2 and gathering, $2I + 2B - 3T - B + 2T + 2 = 4$, so $T = 2I + B - 2$. Every triangle has area $1/2$ and areas add, so the polygon has area $T/2$, which is $I + B/2 - 1$. The formula is Pick's, from 1899.



10 inside

9 on the boundary

$$10 + 9/2 - 1 = 27/2$$

and 27 bare triangles,
each of area one half

Figure 31.2. A polygon with grid points for corners, cut into triangles each of which holds no grid point but its own three. Filled dots lie inside, ringed dots on the boundary. The number of triangles is twice the inside count plus the boundary count less 2, and each has area one half, so the area is the inside count plus half the boundary count less 1. *Assumes:* that the cutting can always be pushed until every triangle is bare, which the text argues; and that the formula for networks on the page applies to the network of dots and sides drawn here, the outside of the polygon counting as one face

The two rules can be set against each other. The triangle with corners $(0, 0)$, $(4, 0)$ and $(0, 3)$ has area 6 by half the base times the height; the two legs and the long side carry 8 grid points between them and 3 lie inside, and $3 + 4 - 1 = 6$. The field with corners

$(0, 0)$, $(7, 2)$, $(9, 7)$, $(3, 9)$ and $(-1, 4)$ carries 6 grid points on its boundary, the five corners and the midpoint of the side from $(9, 7)$ to $(3, 9)$, and 54 inside: $54 + 3 - 1 = 56$, the number the sum of twists gave.

Two things follow that a drawing would not suggest. The first is that no equilateral triangle has all three of its corners at grid points. If one did, the square of its side would be a whole number, being a sum of two whole squares. Call that whole number k , and let u and v be the arrows from one corner to the other two. Then $u_1^2 + u_2^2$, $v_1^2 + v_2^2$ and $(u_1 - v_1)^2 + (u_2 - v_2)^2$ are all k , and the third less the first two is $-2(u_1v_1 + u_2v_2)$, so $u_1v_1 + u_2v_2$ is $k/2$. Multiplying out shows that $T(u, v)^2 + (u_1v_1 + u_2v_2)^2$ is $(u_1^2 + u_2^2)(v_1^2 + v_2^2)$, so $T(u, v)^2$ is $k^2 - k^2/4$, and the square of the area, a quarter of that, is $3k^2/16$. If the area were a fraction p/q , then 3 would be the square of the fraction $4p/(qk)$, and it is not. But the count makes the area of every grid polygon a whole number of halves, which is a fraction. The second is that a bare triangle can be as long and thin as wanted: the corners $(0, 0)$, $(1, 0)$ and $(n, 1)$ give area $1/2$ for every whole n , so by the count $I + B/2 - 1$ is $1/2$ with B at least 3, which leaves $B = 3$ and $I = 0$ and makes the triangle bare, while the longest side grows without bound.

The count also points the other way, from area to number. Take any region with an area and lay the grid over it more and more finely, or what is the same, blow the region up. The number of grid points it swallows is close to its area, measured in squares of the grid. The two can differ only on account of the points near its edge, and how small that difference is depends on how the edge behaves. Counting grid points in a growing disc, and asking how far the count strays from the area, is a question that arithmetic and geometry share and neither has finished.

A field has corners $(0, 0)$, $(6, 1)$, $(8, 5)$, $(4, 8)$ and $(-1, 5)$, in order. What is its area, and how many grid points lie inside it?

A grid polygon has area 30 and 10 grid points on its boundary. How many lie inside?

Cut the square with corners $(0, 0)$, $(2, 0)$, $(2, 2)$ and $(0, 2)$ into

bare triangles in two different ways, and count the triangles each time.

Take the square with corners $(0, 0)$, $(5, 0)$, $(5, 5)$ and $(0, 5)$ and cut out of it the square with corners $(1, 1)$, $(3, 1)$, $(3, 3)$ and $(1, 3)$. Count the grid points inside the figure that remains and on its two boundaries, and set $I + B/2 - 1$ against the area. What has to be added, and what would a figure with two holes need?

Can a grid polygon have area $1/3$?

Show that if the twist of two arrows with whole-number coordinates is 1, then every grid point is a whole-number combination of them.

BRACKETS THAT DO NOT CLOSE

Some regions have no area in that sense. Let f be 1 at every fraction and 0 at every other number, on the interval from 0 to 1. Cut the interval into pieces, however short. Every piece contains fractions: once $1/n$ is shorter than the piece, the multiples of $1/n$ advance in steps too short to step over it, so one of them lands inside. Every piece also contains numbers that are not fractions. Take a fraction inside the piece and add to it the diagonal of the unit square divided by a whole number, chosen large enough that the sum is still inside the piece. The sum is not a fraction, since if it were, subtracting the fraction and multiplying by the whole number would make the diagonal a fraction, which it is not. So on every piece f takes the value 1 somewhere and 0 somewhere. A staircase lying under the graph has every step at height at most 0, and a staircase lying over it has every step at height at least 1. The interval has length 1, so every lower staircase has area at most 0 and every upper staircase has area at least 1. Every number between 0 and 1 then lies between all the lower staircase areas and all the upper ones. A region has an area only when exactly one number lies between them, so the region under f has no area.

The parabola's staircases differed in one respect: the gap be-

tween the upper and lower areas was $1/n$, which falls below every tolerance. For the parabola that was enough, because a number lying between all the brackets was already in hand. For a curve whose brackets shrink but for which no candidate is known, does some number always lie between them? That is a question about numbers, not about area.

* * *

The phrase “falls below every tolerance” deserves an exact form. Someone claims that a sequence approaches the value L . An opponent names a tolerance, as small as they like. The claimant answers with a stage, and wins the round if every term beyond that stage is within the tolerance of L . The claim is true when the claimant can answer every tolerance, and the answer is allowed to depend on the tolerance named. The sequence $1/n$ approaches 0: against the tolerance $1/1000$ the stage 1000 wins, since every later term is at most $1/1001$, which is below $1/1000$, while the stage 999 loses, since the next term is $1/1000$, which is not below $1/1000$. A sequence approaches at most one value. Suppose two different values were claimed for the same sequence, and let the opponent name half the distance between them as the tolerance. A term within that tolerance of both values would put the two values closer together than they are, so no term is within the tolerance of both. But if both claims were true, every term beyond the later of the two answering stages would be within the tolerance of both.

Limits of sums and products follow. If a_n approaches L and b_n approaches M , then $a_n + b_n$ approaches $L + M$: against a tolerance ε , play $\varepsilon/2$ against each sequence, obtain a stage for each, and answer with the later one; beyond it the two errors are each below $\varepsilon/2$, so the error of the sum is below ε . Products need one more step. Write $a_nb_n - LM$ as $a_n(b_n - M) + M(a_n - L)$. Beyond the stage that answers the tolerance 1 for a_n , every a_n is within 1 of L , so the size of a_n is at most $|L| + 1$. The first piece is then at most $(|L| + 1) \cdot |b_n - M|$ and the second at most $|M| \cdot |a_n - L|$. Play the tolerance

$\varepsilon/2$ divided by $|L| + 1$ against b_n , and beyond the stage that answers it the first piece is below $\varepsilon/2$. Play $\varepsilon/2$ divided by $|M| + 1$ against a_n , and beyond the stage that answers it the second piece is below $\varepsilon/2$. Beyond the latest of the three stages the two pieces together are below ε . So $a_n b_n$ approaches LM .

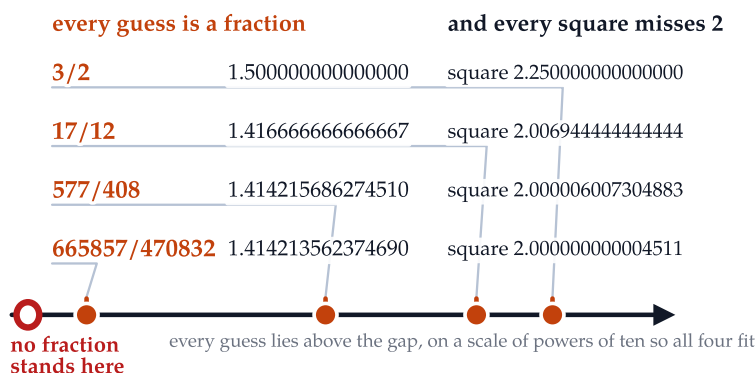
* * *

A procedure that anyone can run shows a sequence that approaches something no fraction is. To improve a guess x at the number whose square is 2, replace x by the average of x and $2/x$. If x is too small then $2/x$ is too large, and the other way round, so the two bracket the target and their average lands between them. Working out the square of that average and subtracting 2 gives the square of $(x^2 - 2)/(2x)$. A square is never negative, so every guess after the first has square at least 2. And from a guess whose square is above 2 the next guess is smaller: $2/x$ is then below x , so the average of the two is below x . So whatever the first guess does, from the second guess on the guesses only come down. Starting from 1, four steps give

3/2	1.500000000000000	square	2.250000000000000
17/12	1.416666666666667	square	2.006944444444444
577/408	1.414215686274510	square	2.000006007304883
665857/470832	1.414213562374690	square	2.0000000000004511

Every entry in the first column is a fraction, since averages and quotients of fractions are fractions. The squares close in on 2: they exceed it by $1/4$, $1/144$, $1/166464$ and $1/221682772224$, each excess about an eighth of the square of the one before. No entry, at any step, has square exactly 2, since no fraction does. Suppose the terms approached a fraction r . Their squares would then approach r^2 , by the limit of a product found above, with both sequences taken to be the terms. But the squares approach 2, since from $3/2$ on each excess is less than a thirtieth of the one before, as is worked out below; and a sequence approaches at most one value, so r^2 would be

2. No fraction has square 2, so the terms, which come down step after step, have nothing among the fractions to approach.



each step lands vastly closer than the last, and the place they close on is not a fraction

Figure 32.1. Each step replaces a guess by the average of the guess and two divided by it, and the squares fall toward 2 without ever reaching it, so the fractions close in on a position on the line where no fraction is available to meet them. *Assumes:* that the sequence really does close in on one definite place rather than wandering, which needs the argument in the text; a picture of four points is compatible with many different fates

The missing number is not in a gap between two fractions, because there are no such gaps: between any two fractions lies their average, which is another fraction, and so infinitely many fractions lie between any two. So the fractions show no hole at any place, no pair of neighbours with room between them where a missing point could be patched in. What is absent has to be supplied by a new assumption about numbers, an axiom.

* * *

The axiom concerns nested intervals:

a sequence of intervals, each containing its two ends and lying inside the one before, whose lengths fall below every tolerance, has exactly one number lying in all of them.

Uniqueness needs nothing more: two different numbers are a definite distance apart, and an interval shorter than that distance cannot hold both. Existence is the axiom. The axiom is asserted of a larger system of numbers, called the real numbers. That system contains the fractions, and it adds, multiplies and compares its members by the rules the fractions obey.

For the fractions alone the axiom fails, and the averaging sequence shows how. If x' is the average of x and $2/x$, then $x'^2 - 2 = (x^2 - 2)^2/(4x^2)$. As before, the right side is a square, and not 0 since no fraction has square 2, so from $3/2$ on every term x has square above 2. Three things follow for each term. First, $2/x$ is below x , and the next term x' , their average, lies between them. Second, x' is below x , so $2/x'$ is above $2/x$; and x'^2 is above 2, so $2/x'$ is below x' . The interval from $2/x'$ to x' therefore lies inside the interval from $2/x$ to x , and the intervals from $2/x$ to x are nested.

Third, the length of the interval from $2/x$ to x is $x - 2/x$, which is $(x^2 - 2)/x$. Suppose $x^2 - 2$ is at most $1/4$, as it is for $3/2$. Since x^2 is above 2, $4x^2$ is above 8, so the quotient $(x^2 - 2)/(4x^2)$ is below $1/32$, and the next excess $(x^2 - 2)^2/(4x^2)$ is less than a thirtieth of $x^2 - 2$; in particular it is again at most $1/4$. An excess cut to less than a thirtieth at every step falls below every tolerance. Each length $(x^2 - 2)/x$ is smaller than the excess $x^2 - 2$, since x is above 1, so the lengths fall below every tolerance.

A number c in all of them lies between $2/x$ and x , so $c^2 - 2$ lies between $(2/x)^2 - 2$ and $x^2 - 2$. The second of these is the excess, which approaches 0. The first works out to $-2(x^2 - 2)/x^2$, and since x^2 is above 2 its size is smaller than the excess, so it approaches 0 as well. So the fixed number $c^2 - 2$ has size below every tolerance, and such a number is 0, since otherwise the tolerance equal to its size would defeat it. So $c^2 = 2$. No fraction has that property, so c is not a fraction: these nested intervals, whose ends are all fractions, have no fraction lying in all of them, and for the fractions alone the axiom fails.

The real numbers have been described by the axiom and not built. A fraction is a pair of whole numbers with a rule for same-

ness, and nothing comparable has yet been said about what a real number is; whether some system satisfies the description is a separate question, answered later by building one out of sequences of fractions. Two further facts go with the axiom. The first is that enough copies of any length, however small, exceed any given length. It was used when tenths of a side were laid along the diagonal, and it is what makes $1/n$ fall below every tolerance. It is assumed too, and it becomes a theorem when the real numbers are built. The second is another form of the axiom, available once the first fact is granted: every collection of numbers that has something in it and an upper bound has a least upper bound. That the two forms say the same thing is proved when the real numbers are built.

* * *

The first consequence is used everywhere later. An increasing sequence that stays below a fixed bound approaches a number.

Take such a sequence and a bound B above all its terms, and start from the interval from the first term to B . That interval contains a term, and its right end is at least every term. Halve it. If the upper half contains a term, keep the upper half: it contains a term, and its right end is the old right end. Otherwise no term reaches the midpoint, so the midpoint is at least every term; keep the lower half, which contains the term the whole interval contained. So every kept interval contains a term and has a right end at least every term, and the kept intervals are nested with lengths halving each time. The axiom gives a number L in all of them. Given a tolerance, some kept interval is shorter than the tolerance. It contains some term. Every later term is at least that term, because the sequence increases, and no term exceeds the interval's right end. So from that term on, every term lies in that interval. So does L , and two numbers in an interval shorter than the tolerance are within the tolerance of each other: from that term on, every term is within the tolerance of L . A decreasing sequence that stays above a fixed

bound approaches a number by the same argument turned upside down.

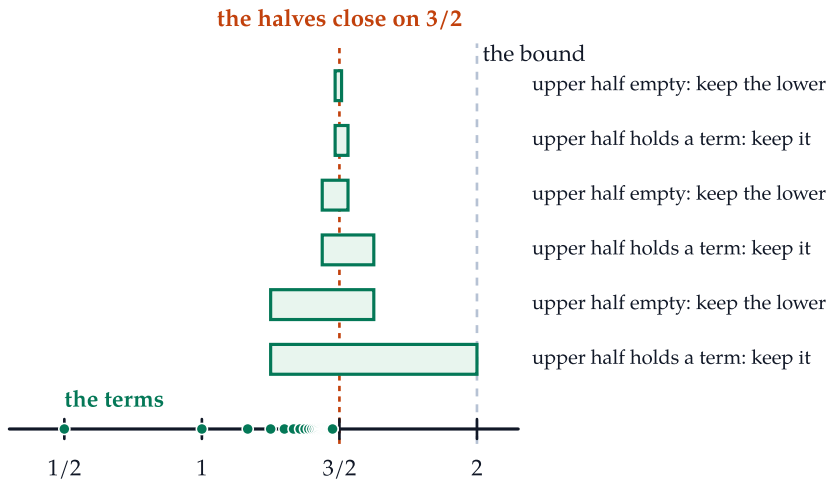


Figure 32.2. An increasing sequence, here $3/2$ minus $1/n$, with the bound 2 above it. The interval from the first term to the bound is halved; the upper half is kept if it contains a term, and otherwise the lower half is kept. The kept halves nest and close on $3/2$, which no term of the sequence ever reaches. *Assumes:* that the nested halves close on exactly one number, which is the axiom and nothing more; that the number they close on is where the sequence is going is what the argument in the text proves, and a picture of forty dots cannot show it

Increase was used in one sentence: once a term lies in a kept interval, every later term does. A sequence that only stays within bounds, such as $0, 1, 0, 1, \dots$, approaches nothing: whatever value L is claimed for it, the tolerance $1/2$ defeats every stage, since beyond any stage both 0 and 1 occur, and they cannot both be within $1/2$ of L .

The same pattern of argument recurs. Start with an interval that passes some test, the test being chosen so that whenever an interval passes it, at least one of its halves passes it too. Keep a half that passes, and repeat. The kept intervals are nested and their lengths halve, so the axiom gives one point lying in all of them. What the

test guaranteed at every stage is then turned into a statement about that point, as it was here about L .

A decimal expansion is an instance. Writing $1.414\dots$ does not exhibit a number; it names the sequence of truncations $1, 1.4, 1.41, 1.414, \dots$, each a fraction whose denominator is a power of ten, each at least the one before, and none above 2. That is an increasing sequence below a bound, and the axiom is what makes the decimal name a number.

Does the sequence whose n -th term is $1 + 1/4 + 1/9 + \dots + 1/n^2$, the sum of the reciprocals of the first n squares, approach a number?

* * *

Areas follow for a whole family of curves. Let f increase on an interval of length ℓ , rising by R from its left end to its right. Cut the interval into n equal pieces, each of width ℓ/n . Because f increases, the lower staircase takes the left-hand height on each piece and the upper staircase takes the right-hand height. The right-hand height on each piece is the left-hand height on the next, so when the upper heights are added up and the lower heights taken away, everything cancels except the last height less the first, which is R . The upper staircase therefore exceeds the lower one by R times ℓ/n , which is $R\ell/n$.

Cut into 2, 4, 8, ... equal pieces, each cutting refining the one before: refining a cutting raises the lower staircase or leaves it alone, since the lowest value on a smaller piece is at least the lowest value on the piece containing it, and lowers the upper staircase or leaves it alone. And every lower staircase, for any cutting, has area at most that of every upper staircase, for any cutting. To see it, take a lower staircase on one cutting and an upper staircase on another, and cut at every point where either cutting cuts. On that common refinement the lower area is at least the area of the given lower staircase, the upper area is at most the area of the given upper staircase, and the lower area is at most the upper one. So along the cuttings into

2^k pieces the lower areas increase, the upper areas decrease, each lower area is below each upper area, and the gaps $R\ell/2^k$ fall below every tolerance. By the axiom exactly one number c lies in all the intervals they form.

Take the area s of any lower staircase, on any cutting. Compare it with the cutting into 2^k pieces. By the last remark, s is at most the upper area of that cutting. And c is at least its lower area, which is its upper area less $R\ell/2^k$. So $s - c$ is at most $R\ell/2^k$, and this holds for every k . Since $R\ell/2^k$ falls below every tolerance, $s - c$ is not a positive amount, and s is at most c ; in the same way the area of every upper staircase is at least c . So c is the one number between all the lower and all the upper staircases: every increasing function with no negative values has an area under it, and so does every decreasing one. The parabola is one case, with $R = 1$, $\ell = 1$ and gap $1/n$, and there the number was known in advance.

Areas defined this way obey the rule they were modelled on: regions side by side add. First a remark. If exactly one number P lies between all the lower and all the upper staircase areas of a region, then for every positive δ some lower staircase has area above $P - \delta$; otherwise every lower area would be at most $P - \delta$, and $P - \delta/2$, which is above $P - \delta$ and below P , would be a second number between them. Upper staircases come within δ of P in the same way.

Now let f have no negative values on the interval from a to c , let b lie between, and suppose the region over the stretch from a to b has area P and the region over the stretch from b to c has area Q . A staircase lying inside the whole region can have b added to its cutting without changing its area, and it then splits into a staircase inside each part. The first has area at most P and the second at most Q , so the area of the whole staircase is at most $P + Q$; likewise a staircase containing the whole region has area at least $P + Q$.

So $P + Q$ lies between all the lower and all the upper areas. And lower staircases within $\delta/2$ of P and of Q , set side by side, make a lower staircase of the whole within δ of $P + Q$, and the same for upper ones. A number below $P + Q$ falls short of it by some positive

δ , and then some lower staircase of the whole has area above that number; a number above $P + Q$ is beaten by an upper staircase in the same way. So no other number lies between. The region over the whole interval has area $P + Q$.

Three things follow. A function that rises on one stretch and falls on the next has an area under it, the sum of the two. A region that lies inside another has at most its area, since every staircase inside the first lies inside the second. And a rectangle with sides a and b is its own lower and upper staircase, so it has area ab under the definition. Two of the three rules about area, the second in the form of regions side by side over neighbouring stretches, are now theorems about the numbers the staircases give, and the third takes one line: for a function with no negative values the flat staircase at height 0 lies inside the region, so no area these definitions give is negative.

Let t be $1/q$ at each fraction p/q in lowest terms and 0 at every number that is not a fraction. Does the region under t from 0 to 1 have an area?

SEQUENCES AT WORK

A clay tablet from Babylon, now at Yale and catalogued as YBC 7289, shows a square with its diagonals, and along one diagonal the number 1;24,51,10 in the sixties of the scribes: $1 + 24/60 + 51/3600 + 10/216000$, which is 1.41421296. The diagonal of the unit square is 1.41421356. The scribe's value is right in every place it gives, an error of 4 parts in ten million, and it was written down about 1700 BC. How was it found?

Nobody knows, but one procedure gives it in two steps. Take a guess x at the square root of 2. If x is too large then $2/x$ is too small, and if x is too small then $2/x$ is too large, so the root lies between x and $2/x$, and their average, halfway between the two, is the natural next guess. Starting from $3/2$, the first average is $17/12$, which the scribes would write 1;25, and the second is $577/408$, which is 1;24,51,10,35. The procedure is among the oldest recursions on record, written down by Heron about AD 60, and with the axiom of nested intervals in hand it can be understood completely: that it works, for any number and any start, and how fast.

Let a be positive, start from any positive x , and replace x by $x' = (x + a/x)/2$. A line of algebra gives

$$x'^2 - a = (x^2 - a)^2 / (4x^2)$$

and the right side is a square divided by a positive number, so it

is never negative. After the first step, then, every term has square at least a , whatever the start. From then on x^2 is at least a , and dividing by x shows that a/x is at most x . The average of x and a number no larger than x is at most x , so x' is at most x : the terms decrease. They stay above 0. A decreasing sequence that stays above a bound approaches a number c . What is c ? The rule says $2x'x = x^2 + a$. Along the sequence x' and x both approach c , so by the law for products the left side approaches $2c^2$ and the right side approaches $c^2 + a$. But the two sides are equal term by term, so they are one sequence, and a sequence approaches one value only. So $2c^2 = c^2 + a$, which is $c^2 = a$. And c is not negative. If it were, take the tolerance equal to the size of c : every number within that tolerance of c is negative, and no term of the sequence is negative, so no term would ever come within the tolerance. Every positive number has a square root, and the averaging finds it.

How fast? Write d for the excess $x^2 - a$. The identity says $d' = d^2/(4x^2)$, which is less than $d^2/(4a)$ once x^2 is above a , and that is so from the first step on. For $a = 2$ each excess is less than an eighth of the square of the one before: $1/4, 1/144, 1/166464, 1/221682772224$. Squaring a small number roughly doubles the number of zeros after its decimal point, so at each step the excess has about twice as many zeros as it had before. The distance from x to the root has about as many zeros as the excess, because $x^2 - a$ is that distance multiplied by $x + \sqrt{a}$, a number near 3 when a is 2. So the number of correct places roughly doubles at each step: from $3/2$ the distances to the root carry 1, 2, 5, 11, 24 and 48 zeros after the point. Compare the halving of intervals. Starting from the interval from 1 to 2 and keeping, each time, the half in which the square crosses 2, the interval has length $1/2^n$ after n steps, and since 2^{10} is about 10^3 , ten halvings buy three decimal places. After 40 halvings the root is known to twelve places. The averaging has eleven places at its third step and twenty-four at its fourth.

The two procedures differ in kind. Halving uses only the sign of the function $x^2 - 2$, and it gains a fixed amount each step. It would work as well for any function that changes sign across the interval

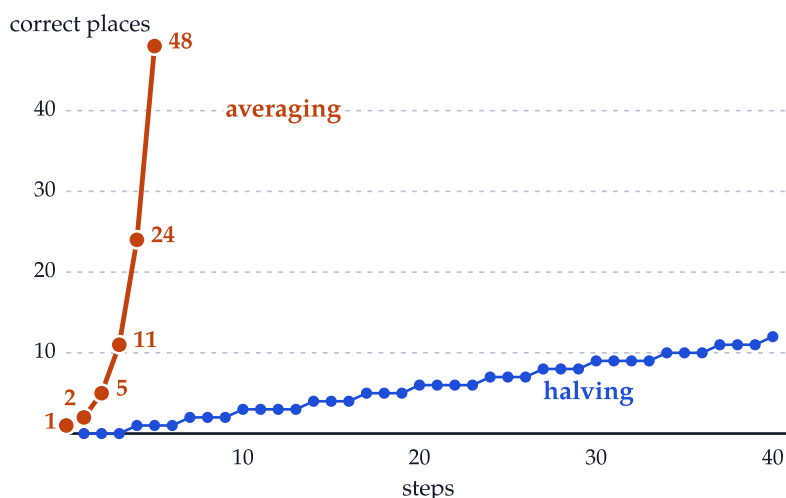


Figure 33.1. The number of correct decimal places in the square root of 2 after each step, for the averaging of x and $2/x$ started at $3/2$ and for the halving of the interval from 1 to 2. Halving gains three places in ten steps. Averaging doubles its places at every step. *Assumes:* places counted as the whole number of zeros after the decimal point in the error

and makes no jumps, a condition that will be made exact when curves are replaced by straight lines. Averaging uses the shape of the function, and each step squares the error; the reason for the squaring, and the general form of the procedure, will appear at that same point.

* * *

Two small facts about limits and order are used constantly. If $a_n \leq b_n$ for every n , and the sequences approach L and M , then $L \leq M$: suppose L were above M , and take the tolerance $(L - M)/2$. Late enough, every a_n is within that tolerance of L and so above the midpoint $(L + M)/2$, while every b_n is within it of M and so below the midpoint. That puts a_n above b_n , against what was given. Strict inequality can be lost, since $1/n$ is above 0 for every n and

approaches 0. And if $a_n \leq b_n \leq c_n$, where a_n and c_n approach the same L , then b_n approaches L : as soon as a_n and c_n are both within the tolerance of L , so is b_n , which lies between them.

Quotients follow limits as sums and products do. If b_n approaches a number M other than 0, then $1/b_n$ approaches $1/M$: once b_n is within $|M|/2$ of M its size is at least $|M|/2$, and then $|1/b_n - 1/M| = |b_n - M|/(|b_n| \cdot |M|)$ is at most $2|b_n - M|/M^2$, which falls below every tolerance. A quotient a_n/b_n is the product of a_n and $1/b_n$, so the law for products does the rest.

With these, a sequence defined by a recursion can often be settled in two moves: show that it approaches something, and then let the recursion say what. Take $x_1 = \sqrt{2}$ and $x_{n+1} = \sqrt{2 + x_n}$, so that the terms are

$$\sqrt{2}, \sqrt{2 + \sqrt{2}}, \sqrt{2 + \sqrt{2 + \sqrt{2}}}, \dots$$

which are 1.4142, 1.8478, 1.9616, 1.9904, 1.9976. They increase. The recursion says $x_{n+1}^2 = 2 + x_n$ and $x_n^2 = 2 + x_{n-1}$, and subtracting gives $x_{n+1}^2 - x_n^2 = x_n - x_{n-1}$. So if x_n is above x_{n-1} , then x_{n+1}^2 is above x_n^2 , and since the terms are positive, x_{n+1} is above x_n . The first step, from 1.4142 to 1.8478, is up, and each step up forces the next. They stay below 2: if x_n is below 2 then x_{n+1}^2 is below 4. So they approach a number L , at most 2. Squaring the recursion, $x_{n+1}^2 = 2 + x_n$, and letting n grow gives $L^2 = 2 + L$, whose only positive solution is 2. The speed can be read off as well. Subtracting both sides of $x_{n+1}^2 = 2 + x_n$ from 4 gives $4 - x_{n+1}^2 = 2 - x_n$, and the left side factors: $(2 - x_{n+1})(2 + x_{n+1}) = 2 - x_n$. So each distance to 2 is the one before divided by $2 + x_{n+1}$, which is close to 4 because the terms are close to 2, and the successive distances 0.5858, 0.1522, 0.0384, 0.0096, 0.0024 show it.

The order of the two moves matters. The recursion $x_{n+1} = x_n^2 - 2$, started at 3, gives 3, 7, 47, 2207, If those approached a number L , then L would be $L^2 - 2$, so L would be 2 or -1 . They approach nothing, since they pass every bound. The algebra began by supposing a limit L , there is none, and so its answer means nothing. The same mistake in a famous disguise: if S stands for $1 - 1 + 1 - 1$

+ ..., then $S = 1 - S$, so S is $1/2$. The algebra is faultless and S stands for nothing, the partial sums being 1, 0, 1, 0,

A recursion need not rise or fall to be settled. Let $x_1 = 1$ and $x_{n+1} = 1 + 1/x_n$. The terms are 1, 2, $3/2$, $5/3$, $8/5$, $13/8$, ..., which are the convergents of the continued fraction with every quotient 1. They are alternately below and above one another. What was proved about convergents says that each differs from the one before by one over the product of their denominators, with the sign alternating. The alternating sign means each step goes in the opposite direction to the last, and because the denominators increase, each step is shorter than the last. A step up followed by a shorter step down ends above where it began, and a step down followed by a shorter step up ends below where it began. So the odd-numbered terms increase, the even-numbered terms decrease, every odd one is below every even one, and neighbours differ by an amount that falls below every tolerance. So each odd term with the even term after it makes an interval, the intervals are nested, and the axiom gives one number in all of them, which both the odd and the even terms approach. Then the recursion, with the law for quotients, says that the limit x satisfies $x = 1 + 1/x$: it is the proportion of the diagonal of a regular pentagon to its side, and the endless expansion $[1; 1, 1, 1, \dots]$, which was a record of a subtraction, is now a number.

* * *

A recursion can also refuse to settle, and in a manner far wilder than 0, 1, 0, 1. Take a population of insects that breed once a year, measured as a fraction x of the largest number their food can support. In a thin year they multiply; in a crowded year they starve. The simplest rule with both features is

$$x' = c \cdot x(1 - x)$$

with a constant c between 0 and 4 for the rate of breeding, and it keeps x between 0 and 1. Start at $x = 0.3$.

With $c = 2$ the terms are 0.42, 0.4872, 0.49967, 0.4999998, and they approach $1/2$ at the pace of the root-finder: since $1/2 - 2x(1 - x) = 2(1/2 - x)^2$, each distance from $1/2$ is twice the square of the one before.

With $c = 3.2$ the terms never settle on one value. After a while they alternate between 0.5130 and 0.7995, a good year and a lean year in turn. The rule exchanges those two values exactly: they are $(4.2 - \sqrt{0.84})/6.4$ and $(4.2 + \sqrt{0.84})/6.4$. The equation saying that two applications of the rule bring x back has degree four. The two alternating values are two of its roots. The other two are 0 and $11/16$, the values the rule leaves where they are: they come back after two steps because they never left. In the computation the even-numbered terms close in on one of the two alternating values, 0.5130 and 0.7995, and the odd-numbered terms on the other, and that they do so from this start is relied on here and not proved. With $c = 3.5$ the pattern has four values in turn.

With $c = 4$ there is no pattern at all. The terms wander up and down the whole stretch from 0 to 1 and show no sign of settling. Worse, start a second population at 0.3000001. The two agree to five places after 5 years, to four after 10, to two after 15, and at the twenty-first year they stand at 0.22 and 0.43. Each step roughly doubles the difference between them, and twenty-one doublings turn a ten-millionth into a fifth. The rule is as simple as a rule can be, nothing in it is random, and its outcome twenty-five steps on cannot be predicted from any measurement that a biologist could make. Robert May, who was one, drew the attention of scientists to this in 1976, and the study of such rules has been a subject of its own since.

* * *

The terms for $c = 3.2$ do not approach anything, but every second one does. A *subsequence* of a sequence is what remains when some terms are struck out and infinitely many are kept, in their original order. A sequence that approaches L has all its subse-

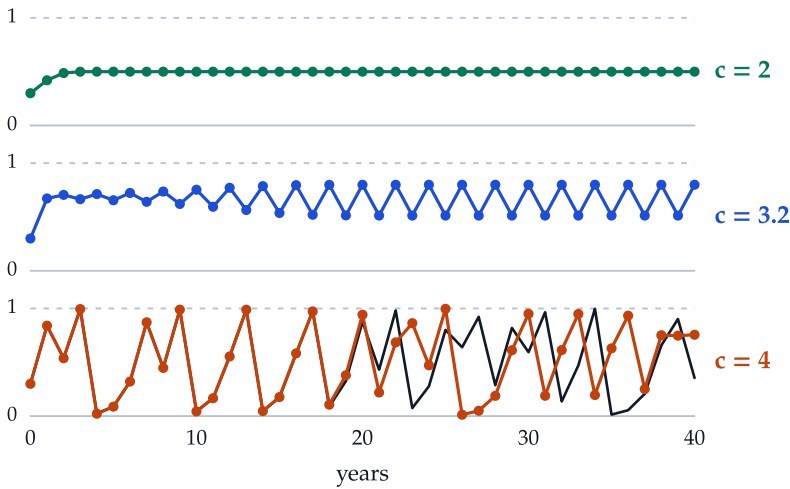


Figure 33.2. Forty years of the rule $x' = c \cdot x(1 - x)$, started at 0.3. With $c = 2$ the population settles at once. With $c = 3.2$ it alternates between two values. With $c = 4$ it wanders, and the second curve, started at 0.3000001, parts company with the first after about twenty years. *Assumes: arithmetic to sixteen figures; for $c = 4$ the computed terms past the fiftieth year or so are themselves an instance of the same sensitivity, and are not drawn*

quences approaching L . A sequence that approaches nothing may have subsequences that do, and if it is bounded it must have one.

Every sequence whose terms all lie in one interval, ends included, has a subsequence that approaches a number of that interval.

The proof is the halving pattern. Halve the interval. At least one half contains infinitely many terms of the sequence, since if each half held finitely many, the whole would. Keep such a half, and halve again. The kept intervals are nested, their lengths fall below every tolerance, and each contains infinitely many terms; the axiom gives a number L in all of them. Now choose the subsequence: a term in the first kept interval; then a later term in the second, which exists because infinitely many terms lie there; then a later one in the third; and so on. The k -th chosen term and L lie together in the k -

th kept interval, so they differ by at most its length, and the chosen terms approach L . The theorem is due to Bolzano, who found it in 1817, and to Weierstrass, who made it a tool.

It says that a bounded sequence cannot avoid coming back, again and again, as near as anyone likes to some number. For $c = 4$ it says that some subsequence of that wandering population approaches a number, though it does not say which. It is known that for nearly every starting value each number from 0 to 1 is approached by some subsequence.

* * *

Every test for convergence so far has needed either a candidate for the limit, as in the game of tolerances, or a sequence that rises or falls. There is a test that needs neither. Say that a sequence *huddles* when for every tolerance there is a stage past which any two of its terms are within the tolerance of each other: any two, and not only neighbours. The sentence mentions no limit, only the terms.

A sequence of real numbers approaches a number exactly when it huddles.

If the terms approach L , then past the stage for half the tolerance every term is within half the tolerance of L , so any two are within the tolerance of each other. Conversely, let the sequence huddle. It is bounded. Take the stage for the tolerance 1 and fix one term past it. Every term past that stage is within 1 of the fixed term, and the terms up to that stage are finitely many, so one interval holds all the terms. By the theorem of Bolzano and Weierstrass, then, some subsequence approaches a number L . And then the whole sequence does. Given a tolerance, take the stage past which any two terms are within half of it. Some chosen term of the subsequence lies past that stage and within half the tolerance of L . Every term past the stage is within half the tolerance of that one, and so within the tolerance of L .

Neighbours are not enough. The sums $1 + 1/2 + \dots + 1/n$ have neighbouring terms differing by $1/n$, which falls below every tolerance. Yet the $2n$ -th sum is above the n -th by $1/(n+1) + \dots + 1/(2n)$, which is n fractions each at least $1/(2n)$, and so by at least $1/2$, however large n is. Past every stage there are two terms at least $1/2$ apart, so the sums do not huddle. In the same way the square roots of the whole numbers have neighbours differing by less than $1/(2\sqrt{n})$, and they pass every bound.

Among the fractions the test fails in one direction: the averaging sequence for the square root of 2 consists of fractions, it huddles, and no fraction is its limit. That a huddling sequence always has a limit is the axiom of nested intervals in another form. The axiom gave it, and it gives the axiom back. Take nested intervals whose lengths shrink below every tolerance. Their left ends huddle, because any two left ends past the k -th lie in the k -th interval and so differ by at most its length. So the left ends have a limit, and by the facts about limits and order that limit is at least every left end and at most every right end: it lies in every one of the intervals. Huddling, then, is the form in which the axiom does most of its later work, because it certifies a limit without anyone having to name it.

Here is that work done once in general. Suppose f carries an interval, ends included, into itself, and shrinks distances: there is a number k below 1 with $|f(x) - f(y)|$ at most $k|x - y|$ for all x and y in the interval. Start anywhere in the interval and iterate, $x_{n+1} = f(x_n)$. Neighbouring terms satisfy $|x_{n+2} - x_{n+1}| \leq k|x_{n+1} - x_n|$, so $|x_{n+1} - x_n|$ is at most k^{n-1} times the first step $s = |x_2 - x_1|$. Two terms far apart in the sequence are joined by a chain of neighbours, so for m above n

$$|x_m - x_n| \leq s(k^{n-1} + k^n + \dots + k^{m-2}) < s \cdot k^{n-1}/(1 - k)$$

by the subtraction that summed the powers of $5/6$. And k^{n-1} falls below every tolerance as n grows, since the powers of $1/k$ pass every bound. So the sequence huddles, and approaches a number c . Every term lies between the two ends of the interval, and limits respect order, so c lies between them as well. Then $f(x_n)$ ap-

proaches $f(c)$, being within $k|x_n - c|$ of it; but $f(x_n)$ is x_{n+1} , which approaches c . So $f(c) = c$. There is no second such point: two of them would satisfy $|c - c'| = |f(c) - f(c')| \leq k|c - c'|$, which forces $c = c'$. And letting m grow in the display gives a guarantee that can be computed before the iteration is run: the n -th term is within $s \cdot k^{n-1}/(1 - k)$ of c .

For $f(x) = 1 + 1/x$ on the interval from $3/2$ to 2 : the values lie between $3/2$ and $5/3$, inside the interval, and $|f(x) - f(y)| = |x - y|/(xy)$ is at most $(4/9)|x - y|$. From $x_1 = 2$ the first step is $1/2$, so the n -th term is within $(9/10)(4/9)^{n-1}$ of the limit: the tenth term, $144/89$, is guaranteed within 0.0007 of it, and is in fact within 0.0001 .

Replacing x by $(2x + a/x^2)/3$ is meant to find the cube root of a . Does it, from every positive start, and how does the error at one step compare with the error at the step before?

For which whole numbers a is $\sqrt{(a + \sqrt{(a + \sqrt{(a + \dots)})})}$ a whole number?

A sequence has neighbouring terms whose differences fall below every tolerance, and all its terms lie between 0 and 1 . Must it approach a number?

Does every sequence have a subsequence that never decreases or a subsequence that never increases?

With $c = 3.5$ in the rule for the insects, what are the four values to four decimal places, and is there a value of x at which the population would stay the same from year to year for that c ?

STRAIGHT LINES INSTEAD OF CURVES

The area under the parabola from 0 to b is $b^3/3$. The map sending (x, y) to (bx, b^2y) carries the region under the parabola from 0 to 1 onto the region under it from 0 to b , since it sends (x, x^2) to $(bx, (bx)^2)$; it carries staircases to staircases, and it multiplies the area of every rectangle by $b \cdot b^2 = b^3$. So the brackets for the second region are b^3 times those for the first, and the one number between them is $b^3/3$. That gives a function, the accumulated area $A(b)$, which grows as b moves right. How fast does it grow? At b the region is being fed by a strip of height b^2 , so a guess is that it grows at the rate b^2 . But “rate” has a meaning so far only for straight lines.

Curves are hard and straight lines are easy, and the response is to replace a curve near a point by a straight line and keep account of the error. A function f has *derivative* m at a point a when

$$f(a + h) = f(a) + m \cdot h + r(h)$$

where the error $r(h)$ is small compared with h , meaning that $r(h)/h$ approaches 0 as h approaches 0. The value a step h away is the value at a , plus a correction proportional to the step, plus a leftover that vanishes even measured against the step. The statement that $r(h)/h$ approaches 0 is the game of tolerances with a window in

place of a stage: for every tolerance there is a window around 0 such that $r(h)/h$ is within the tolerance for every h other than 0 in the window. Sums, products, quotients and squeezes pass to such a limit by the arguments already made for sequences. Nothing in those arguments used the whole numbers, and wherever a stage answered a tolerance there, a window answers it here.

Subtracting $f(a)$ from the defining form and dividing by h turns it into $(f(a+h) - f(a))/h = m + r(h)/h$, so f has derivative m at a exactly when that quotient, the *difference quotient*, comes within every tolerance of m for every small enough h other than 0. The same game defines continuity: f is *continuous* at c when for every tolerance there is a window around c , one-sided if c is an end of an interval, throughout which f is within the tolerance of $f(c)$; and f is continuous on an interval when it is continuous at every point of it.

Back to the derivative: at most one number m can serve at a given point a . If m and m' both serve, with leftovers r and r' , then the two forms describe the same value $f(a+h)$, so $m \cdot h + r(h) = m' \cdot h + r'(h)$, and dividing by h gives $m - m' = (r'(h) - r(h))/h$ for every h other than 0 near 0, and the right side comes within every tolerance of 0 as h approaches 0, since $r'(h)/h$ and $r(h)/h$ each come within half the tolerance. A fixed number that is within every tolerance of 0 is 0: if it were not 0, its own size would be a tolerance, and the number is not within that tolerance of 0, its distance from 0 being exactly its size. So $m = m'$.

For $f(x) = x^2$, expanding gives $(a+h)^2 = a^2 + 2a \cdot h + h^2$. The correction is $2a \cdot h$ and the leftover h^2 satisfies $h^2/h = h$, which approaches 0, so the derivative of x^2 at a is $2a$. For x^3 , $(a+h)^3 = a^3 + 3a^2h + 3ah^2 + h^3$, so the derivative is $3a^2$, with leftover $3ah^2 + h^3$; at $a = 1$ and $h = 0.01$ that leftover is 0.000301. For every power the same happens. Multiplied out, $(a+h)^{k+1}$ is a sum of products of $k+1$ factors, each factor an a or an h . The product with no h is a^{k+1} ; the $k+1$ products with exactly one h add up to $(k+1)a^k h$; and every other product has at least two factors h . Divide such a product by h and set one more factor h aside: $k-1$ factors remain, each an

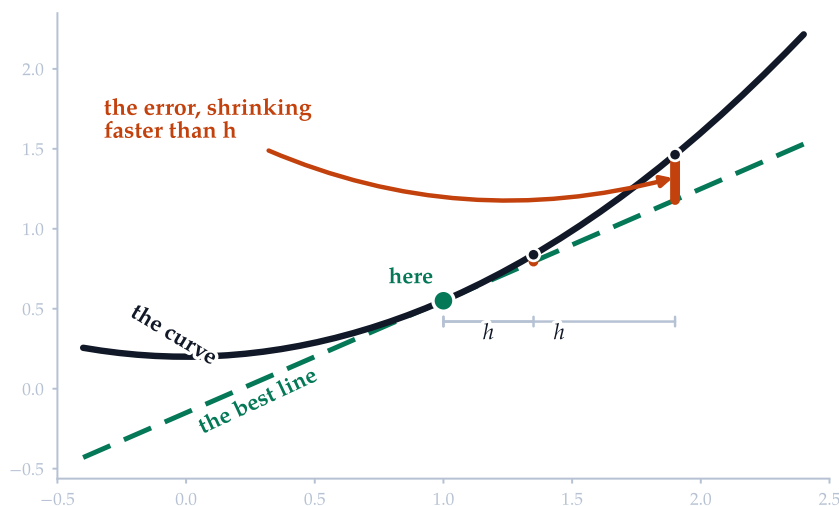


Figure 34.1. Near a point, the curve and its best straight line agree closely, and the gap between them, the error, closes faster than the step h does. That is the whole definition: the leftover vanishes even when measured against the step it came from. *Assumes:* that a single best line exists at this point; where it does not, as at a corner, there is no derivative and the picture cannot be drawn

a or an h , and once $|h|$ is at most 1 each of them has size at most $|a| + 1$. So the product divided by h has size at most $|h|(|a| + 1)^{k-1}$. There are 2^{k+1} products in all, two choices for each of the $k + 1$ factors, hence fewer than 2^{k+1} of this kind, so the leftover divided by h has size at most $2^{k+1}(|a| + 1)^{k-1}|h|$, which approaches 0. So the derivative of x^{k+1} is $(k + 1)x^k$.

Derivatives of sums and multiples are sums and multiples of derivatives. If f and g have derivatives m and n at a , with leftovers r and s , then adding the two defining forms gives $(f + g)(a + h) = (f + g)(a) + (m + n) \cdot h + r(h) + s(h)$, and $(r(h) + s(h))/h$ approaches 0 by the law for sums; multiplying a defining form by a constant multiplies m and r by it; and a line $cx + d$ has derivative c , with leftover 0. So $A(b) = b^3/3$, a third of b^3 , has as derivative a third of $3b^2$, which is b^2 , the height of the parabola at b , as guessed.

For a curve that rises or falls, the rate of the accumulated area

can be computed exactly. Let f be continuous and falling, with no negative values, on an interval, and let $A(x)$ be the area under f from the left end of the interval to x . For a positive step h , $A(x+h) - A(x)$ is the area of the region under f from x to $x+h$, since regions side by side add. That region has width h and heights between $f(x+h)$ and $f(x)$, since the curve falls, so it contains a rectangle of area $h \cdot f(x+h)$ and lies inside one of area $h \cdot f(x)$. The quotient $(A(x+h) - A(x))/h$ is therefore between $f(x+h)$ and $f(x)$, both of which tend to $f(x)$ as h shrinks, since f is continuous; a negative step is handled the same way, and for a rising curve the two rectangles exchange places.

One remark about the two ends, where a step can go only one way: between two points A changes by at most the largest height of f times the distance between them, so A is unbroken there as well as inside. By the squeeze, A has derivative $f(x)$ at every point inside the interval: the area has, as its rate, the height of the curve at the moving end. That is the fundamental theorem of the calculus, for curves that rise or fall.

The same squeeze serves every continuous curve, once every continuous curve has an area under it. In place of the heights at the two ends of the short strip it uses the smallest and largest heights on the strip, and those are reached because a continuous function on an interval that contains its two ends reaches both, as is proved further on. So far the staircases have given areas only to curves that rise or fall, or do so on finitely many stretches. Area, a global quantity, is found by reversing rate, a local one.

* * *

Not every function has a derivative. At 0 the absolute value function has difference quotient $|h|/h$, which is 1 for every positive h and -1 for every negative h , so the quotient approaches no single number, and there is no derivative at 0. The function is continuous there, its graph drawn without lifting the pen, and it has a corner.

Corners need not be isolated. There are functions continuous

everywhere with a derivative nowhere, curves without breaks and without a tangent at any point, which look as rough under every magnification as they do at first sight; one is built from a sum of tents once infinite sums of functions have a meaning.

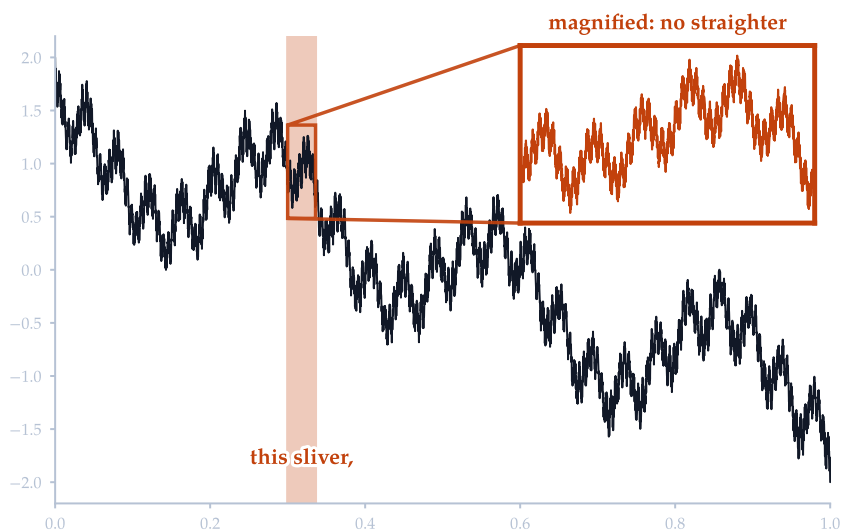


Figure 34.2. A continuous curve and, inset, a narrow slice of it enlarged. The slice is no straighter than the whole, and pushing the magnification further would change nothing: the curve keeps the same roughness at every scale, so no point of it has a tangent. *Assumes:* that the roughness continues below the resolution drawn; a partial sum is smooth when looked at closely enough, and only the infinite sum is genuinely nowhere-differentiable

* * *

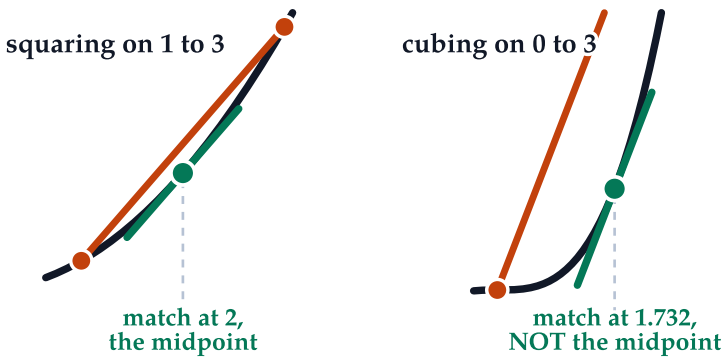
Reversing a rate finds a function whose rate is the given curve. That it finds the only one, apart from an added constant, rests on the statement that a function whose derivative is 0 throughout an interval does not change, and that statement rests on a theorem joining an average to an instant. Across an interval a function has an average slope, the change in value divided by the change in po-

sition. At each interior point it has a tangent slope, its derivative there, which knows nothing of the ends. Call an interval that contains its two ends a *closed* interval. The *average-slope theorem* says: if a function is continuous on a closed interval and has a derivative at every interior point, then at some interior point the tangent slope equals the average slope.

The function x^2 from 1 to 3 has average slope 4, and its tangent slope $2x$ equals 4 at $x = 2$, the midpoint. The function x^3 from 0 to 3 has average slope 9, and its tangent slope $3x^2$ equals 9 at $x = \sqrt{3}$, which is not the midpoint. The theorem asserts that such a point exists, not where it is.

Where does the tangent slope of x^3 equal its average slope across the interval from -1 to 2 ?

the chord's slope is achieved by some tangent, always



the first case invites a false rule; the second kills it

Figure 34.3. The chord across an interval and the tangent that matches its slope: for the squaring function on 1 to 3 the match happens at the midpoint 2, and for the cubing function on 0 to 3 it happens at the square root of 3, which is not the midpoint. *Assumes:* that the curve has no break and no corner anywhere between the endpoints; one break can leave the chord unmatched by any tangent, and the picture cannot show the absence of a break

The proof needs another theorem first: a function continuous

on a closed interval reaches a largest value. Suppose f is continuous on the interval from a to b . If f were not bounded above, it would not be bounded above on one of the two halves; keep that half, and repeat. The halves close on a point c of the interval, and by continuity f is below $f(c) + 1$ throughout some window around c ; but the halves eventually lie inside that window, and on them f was not bounded above. So f is bounded above, by some number B .

Now halve along values, starting from the interval from $f(a)$ to B . At each step look at the midpoint of the current interval: if some value of f is at or above that midpoint, keep the upper half, and otherwise keep the lower half. The kept intervals close on a number M , which no value of f exceeds. Every right end is at least every value: B is, and a right end moves only when the lower half is kept, and then it moves to a midpoint that every value of f is below. If some value $f(x)$ exceeded M , a kept interval shorter than $f(x) - M$ would contain M and so have its right end below $f(x)$, which no right end is.

And values of f come within every tolerance of M . Every kept interval has a value of f at or above its left end: the first has $f(a)$, and a left end moves only to a midpoint that some value of f is at or above. A kept interval shorter than the tolerance contains M , and the value at or above its left end is at most M , so that value is within the tolerance of M .

Finally halve the domain, keeping a half on which values of f come within every tolerance of M ; one of the two halves always has that property, since if each half had a tolerance such that none of its values came within that tolerance of M , then no value on the interval being halved would come within the smaller of the two tolerances of M , and that interval has the property. The kept halves close on a point c . If $f(c)$ were below M , say by 2δ , then by continuity f would be below $M - \delta$ throughout a window around c , and a kept half inside that window would contain no value within δ of M . So $f(c) = M$.

Both continuity and the closed interval are needed: $1/x$ on the

interval from 0 to 1 with 0 left out has no largest value, and neither does the function equal to $1/x$ for x above 0 and to 0 at 0, on the closed interval, which is not continuous at 0.

Now the average-slope theorem. Subtract from f the straight line through the two ends of its graph. The result g is continuous: f is, the line changes by less than any tolerance over a narrow enough window, and splitting the tolerance between the two works as it did for sums. At both ends g has the same value, 0, since the line meets the graph there. At each interior point the derivative of g is the derivative of f minus the slope of the line, which is the average slope. So it is enough to find an interior point where the derivative of g is 0. If the largest and smallest values of g are both taken at the ends, they are equal, g is constant, and its derivative is 0 everywhere inside. Otherwise g takes its largest value, or its smallest value, which is the largest value of $-g$, at some interior point c . If the derivative m of g at c were positive, then in $g(c+h) = g(c) + m \cdot h + r(h)$ the leftover would be smaller in size than $m \cdot h$ for every small enough h other than 0, since $r(h)/h$ comes within the tolerance m of 0. So $m \cdot h + r(h)$ would have the sign of h , and $g(c+h)$ would exceed $g(c)$ for small positive h and fall below it for small negative h , so $g(c)$ would be neither the largest nor the smallest value; and a negative m fails in the same way. So the derivative of g at c is 0.

A function with a derivative at a point is continuous there, since its change over a step h is $m \cdot h + r(h)$, which approaches 0. So a function whose derivative is 0 at every point of an interval is continuous on it, and if two of its values differed, the average slope between them would not be 0, and the theorem would give an interior point with that nonzero derivative. So the function is constant. In the same way, a function whose derivative is never negative on an interval never decreases there. Two functions with the same derivative differ by a function with derivative 0, which is constant: a function whose rate is a given curve is determined up to an added number. So for a continuous curve f that rises or falls, with no negative values, the area under it from a to b is $F(b) - F(a)$ for any

function F with derivative f inside and unbroken at the ends. Write $A(x)$ for the accumulated area from a to x . Then A and F have the same derivative inside, so $A - F$ is a constant there, and since both are unbroken at the ends, $A - F$ is that same constant at a and at b . So $A(b) - F(b) = A(a) - F(a)$, and $A(a)$, the area from a to a , is 0, which leaves $A(b) = F(b) - F(a)$. Under x^3 from 1 to 2, take $F(x) = x^4/4$, whose derivative is x^3 : the area is $(16 - 1)/4 = 15/4$, with no staircase.

For every k and n , does the sum $1^k + 2^k + \dots + n^k$ lie between $n^{k+1}/(k+1)$ and $(n+1)^{k+1}/(k+1)$?

The hypotheses matter. Let a function take one value on the left part of an interval and a larger value on the right part, jumping at a point p inside, with the larger value at p itself. Its average slope across the interval is positive, and wherever it has a derivative that derivative is 0, so no interior point matches the average. The proof fails at a definite step. Subtracting the line through the ends gives a function decreasing on each side of p , whose largest value is taken at p , an interior point; but the function has no derivative at p , and the step "there the derivative of g is 0" has nothing to act on. For a second example the proof fails at a different step. The function equal to x from 0 up to but not including 1, and equal to 0 at 1, has derivative 1 at every interior point and average slope 0. The line through the ends of its graph is 0, so g is the function itself, which has no largest value, and the proof stops at its first step, because continuity fails at one end. Since a derivative at a point forces continuity there, the two ends are the only places where continuity is a separate demand.

Is there a function whose derivative is -1 at every negative number and 1 at every other number?

EVERY CONTINUOUS CURVE HAS AN AREA

The staircases gave an area to every curve that rises, to every curve that falls, and to every curve made of finitely many rising and falling stretches. The theorem that the area has the height of the curve as its rate was proved for those, with the remark that the same proof would serve every continuous curve, once every continuous curve has an area. Does it?

Here is a continuous curve that the earlier theorems do not reach. Cut the interval from 0 to 1 at the points $1/2, 1/3, 1/4, \dots$, and over each stretch from $1/(n+1)$ to $1/n$ raise a tent: let $z(x)$ be the distance from x to the nearer end of the stretch it lies in, and let $z(0) = 0$. The tents grow narrower and lower toward 0. At any point above 0 the function z is continuous, being made of straight pieces that meet. At 0 it is continuous as well. The value $z(x)$ is at most half the length of the stretch that holds x . The stretch from $1/(n+1)$ to $1/n$ has length $1/(n(n+1))$, and half of that is less than the left end $1/(n+1)$, which is at most x . So $z(x)$ is less than x , and as x approaches 0 the values $z(x)$ approach 0, which is $z(0)$. But z rises and falls infinitely often between 0 and 1, and no cutting into finitely many stretches makes it rise or fall on each.

What did the staircases need? On each piece of a cutting, the

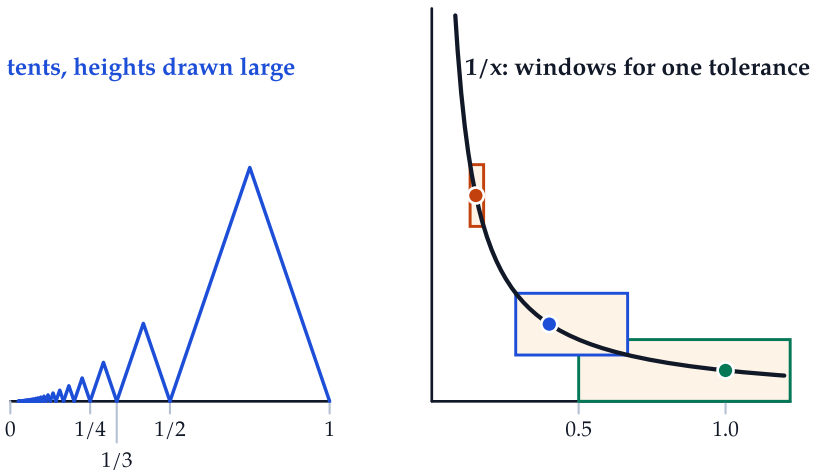


Figure 35.1. On the left, the tents over the stretches from $1/(n+1)$ to $1/n$, with the heights drawn larger than the widths so that the small ones can be seen: a continuous curve that rises and falls infinitely often before it reaches 0. On the right, the curve $1/x$ with windows that keep it within one unit of its value at three points; the windows narrow without limit as the point moves toward 0, and no one width serves them all. *Assumes:* the tents drawn only down to the stretch that begins at $1/40$, below which they are finer than the ink; and a tolerance of 1, not $1/10$, for the windows, so that they can be seen

upper staircase stands at the largest value of the function on that piece and the lower at the smallest, and a continuous function has both on a piece that contains its ends, as was proved. Call their difference the *swing* of the function on the piece. The gap between the upper and the lower staircase is the sum of swing times width over the pieces. For a rising curve the swings added up to the total rise, and the widths were made small.

For a curve that wanders, the total of the swings need have no bound. Make the tent over the stretch from $1/(n+1)$ to $1/n$ as tall as $1/n$. The curve is still continuous at 0, because its height at x is at most $2x$. But its rises add up to $1 + 1/2 + 1/3 + \dots$, and those sums pass every number: the terms from $1/n$ to $1/(2n)$ add at least $1/2$

however large n is, so there is always another $1/2$ to come. So the swings themselves have to be made small, all of them at once. If every swing is at most some small number, the gap, being the sum of swing times width, is at most that number times the sum of the widths, and the widths add up to the length of the interval. The area exists, then, if for every tolerance the interval can be cut into pieces on each of which the swing is within the tolerance.

Continuity promises something that sounds like that and is weaker. It says that for every point and every tolerance there is a window around the point throughout which the function stays within the tolerance of its value there. The window is allowed to depend on the point, and it can depend on it badly. Take $1/x$ on the stretch from 0 to 1 with 0 left out, and the tolerance $1/10$. At $x = 1$ a window of width 0.09 on either side will do. At $x = 1/10$, where the curve is a hundred times steeper, the value changes by a tenth when x moves by a thousandth, so the window there has to be narrower than a thousandth. At $x = 1/100$ a move of a hundred-thousandth already changes the value by a tenth, and the window has to be narrower than that. Every point has its window, and no one width serves all the points, because the windows shrink to nothing as x approaches 0. On that stretch the swings cannot all be made small with finitely many pieces, since the piece that reaches down to 0 has no largest value at all.

A function is *uniformly continuous* on an interval when for every tolerance there is one width such that any two points of the interval closer together than that width have values within the tolerance of each other. The definition of continuity said: for every point and every tolerance there is a width. This one says: for every tolerance there is a width that serves every point. The phrase “for every point” has moved from before the choice of the width to after it. The question is which choice is allowed to depend on what. It is the question on which the game of tolerances and stages turned, and it will come back more than once.

* * *

Uniform continuity asks for more than continuity does. On an interval that contains its ends it asks for nothing more: every function continuous on such an interval meets it already.

A function continuous on an interval that contains its two ends is uniformly continuous on it.

Suppose not. Then for some tolerance ε no width serves: for each whole number n the width $1/n$ fails, so there are points x_n and y_n of the interval, closer together than $1/n$, whose values differ by at least ε . The x_n all lie in the interval, so they are bounded, and by the theorem about bounded sequences some subsequence of them approaches a number c . It is here that the ends are needed. Because the interval contains its ends, c belongs to it; without them the subsequence could approach a missing end, a point where f is not defined. Along that subsequence the y_n approach c too, being within $1/n$ of the x_n . Now use continuity at the one point c : there is a window around c in which every value is within $\varepsilon/2$ of $f(c)$. Far enough along the subsequence both x_n and y_n lie in that window, so their values are within $\varepsilon/2$ of $f(c)$, and within ε of each other. That contradicts the way they were chosen.

The proof names no width. It shows that a width exists by showing that, if there were none, continuity would fail at some point c , and it finds c by the theorem that a bounded sequence has a subsequence approaching some number. For $1/x$ on the stretch with 0 left out, the same steps run until the point c is called for, and c is 0, where there is no function to be continuous. Heine published the theorem in 1872.

Now the area. Let f be continuous on the interval from a to b , with no negative values. Given a tolerance ε , take the width that serves the tolerance $\varepsilon/(b-a)$, and cut the interval into equal pieces narrower than that width. On each piece the largest and smallest values are taken at two points of the piece, and those are closer together than the width, so the swing is at most $\varepsilon/(b-a)$. The gap between the upper and lower staircases is the sum of swing times width. With every swing at most $\varepsilon/(b-a)$ and the widths adding

up to $b - a$, the gap is at most $\epsilon / (b - a)$ times $b - a$, which is ϵ . From here the argument that gave areas to rising curves goes through as it stands, for it used the rise only to make the gap small. Along the cuttings into 2, 4, 8, ... equal pieces the lower areas increase, the upper areas decrease, each lower area is below each upper one, and the gaps fall below every tolerance; one number lies in all the intervals they form, and every lower staircase on any cutting is at most that number, and every upper staircase at least.

Every continuous curve over an interval that contains its ends has an area under it.

The shrinking tents have an area, then, although no one has computed it yet. And the rule that regions side by side add was proved from the definition of area alone, so it holds here.

For a function that takes negative values as well, count the area below the axis as negative. It is convenient to do this by lifting: add a constant K large enough that $f + K$ has no negative values, take the area under $f + K$, and subtract the rectangle $K(b - a)$; the answer does not depend on K , since raising K by d adds the rectangle $d(b - a)$ to the area and to what is subtracted. The number obtained is called the *integral* of f from a to b . For a function with no negative values it is the area under the curve, and in general it is the area above the axis less the area below it. The integral from a to b plus the integral from b to c is the integral from a to c , and an integral taken backwards, from b to a , is counted as the negative of the integral from a to b , so that the rule holds in whatever order the three points lie.

* * *

The theorem on rates can now be proved without the word rising. Let f be continuous on an interval, fix a point a of it, and let $A(x)$ be the integral of f from a to x . For a positive step h , $A(x + h) - A(x)$ is the integral of f from x to $x + h$. On that short stretch f lies between its smallest value m and its largest value M there, so

the integral lies between $h \cdot m$ and $h \cdot M$: taking the whole stretch as a single piece, the lower staircase is $h \cdot m$ and the upper staircase is $h \cdot M$, and the integral lies between any lower staircase and any upper one. If f takes negative values there, lifting by K adds the same rectangle $h \cdot K$ to all three and changes nothing. The quotient $(A(x+h) - A(x))/h$ therefore lies between m and M . As h shrinks, m and M both approach $f(x)$: they are values of f at points within h of x , and f is continuous at x . A negative step goes the same way. So

A has derivative $f(x)$ at every point x inside the interval: the integral of a continuous function, taken up to a moving end, has the function as its rate.

That is the fundamental theorem of the calculus, in full. Every continuous function is the derivative of something, namely of its own integral. Finding a formula for that something is another matter. The rule for powers, read backwards, makes x^k the derivative of $x^{k+1}/(k+1)$ for $k = 0, 1, 2, \dots$, and it cannot make $1/x$ in that way. In the list $\dots, 1/x, 1, x, x^2, \dots$, each entry is x times the one before, and the rule takes each power to a multiple of the entry before it. So $1/x$ could only come from the entry after it, the constant 1, and the derivative of a constant is 0. But a function with derivative $1/x$ exists, by the theorem. So do functions with derivatives $1/(1+x^2)$ and $\sqrt{1-x^2}$, whatever formulas there may or may not be for them. These functions are defined by areas. Their values can be computed to any accuracy from staircases, and their properties can be worked out from the curves they measure.

It was already proved that two functions with the same derivative on an interval differ by a constant. With that, the rule for computing holds for every continuous f : if F is any function with derivative f , the integral of f from a to b is $F(b) - F(a)$. For F and A have the same derivative f at every point inside the interval from a to b , and both are unbroken at a and at b : F because a function with a derivative at a point is continuous there, and A because over a step h it changes by at most h times the largest size of f . So F is A

plus a constant, and the constant cancels in the difference: $F(b) - F(a) = A(b) - A(a)$. Since $A(a)$ is 0, that is $A(b)$, the integral of f from a to b .

* * *

Staircases are not the only sums that find an integral. Cut the interval into pieces, choose any point at all in each piece, and add up the value of f at the chosen point times the width of the piece. Such a sum lies between the lower and the upper staircase on the same cutting, since each chosen value lies between the smallest and the largest on its piece. As the pieces are made narrow, the lower and the upper staircase both approach the integral, and the sum, held between the two, approaches it as well, whatever points are chosen. Riemann took these sums as the definition in 1854, and they carry his name.

This turns some limits of sums into areas at sight. The sum

$$1/(n+1) + 1/(n+2) + \dots + 1/(2n)$$

has n terms, none of which is large, and it is not plain what it does as n grows. Divide top and bottom of each term by n : the k -th term is $(1/n) \cdot 1/(1+k/n)$. So the sum is a sum for the function $1/(1+x)$ on the interval from 0 to 1, cut into n equal pieces, with the right-hand end of each piece chosen. It approaches the integral of $1/(1+x)$ from 0 to 1. The curve $1/(1+x)$ over the interval from 0 to 1 is the curve $1/x$ over the interval from 1 to 2 moved one unit to the left, so that integral is the area under the curve $1/x$ from 1 to 2. For $n = 10, 100$ and 1000 the sums are 0.66877, 0.69065 and 0.69290.

The choice of points does matter to the speed. For the curve $1/(1+x^2)$ from 0 to 1, the staircases on ten pieces give 0.7600 and 0.8100, and on a thousand pieces 0.78515 and 0.78565: the gap is the total fall of the curve, $1/2$, times the width of a piece, so a thousand pieces buy three decimal places. Choose the midpoint of each piece instead, and ten pieces give 0.78561, a hundred give 0.7854002, and

the integral is 0.7853982. With midpoints the error falls a hundred-fold when the pieces are made ten times narrower. The reason is that on a narrow piece a curve is nearly a straight line, and for a straight line the value at the midpoint gives the area exactly; what is left over comes from the bending, and measuring it takes the derivative of the derivative.

A function with a jump is not continuous, and it can still have an area. One change is needed first, because a function with a jump need not have a largest value on the piece that holds the jump. Let the upper staircase on a piece stand at any height that no value on the piece exceeds, and the lower at any height that no value falls below. For a bounded function with finitely many jumps, each jump can be enclosed in a piece of its own. On that piece the upper and lower staircase heights stay some distance apart, but the piece can be made so narrow that this distance times the width of the piece, which is what the piece adds to the gap, is as small as desired. Elsewhere the function is continuous on stretches that contain their ends, and there the gap can be made small as before.

The function that is 1 at every fraction and 0 elsewhere has no area: every piece, however narrow, holds a fraction and a number that is not a fraction, so every upper staircase stands at 1 or higher and every lower staircase at 0 or lower, and the gap never shrinks. That function is discontinuous at every point. Between those cases the boundary was found by Lebesgue in 1902: a bounded function has an area exactly when, for any given tolerance, the points where it fails to be continuous can be covered by a sequence of intervals whose lengths, however many of them are added up, stay below the tolerance. The proof is not given here, and nothing later depends on it.

Is x^2 uniformly continuous on the whole line? Is $\sqrt{x}$, on the numbers from 0 upwards?

Which width serves the tolerance $1/100$ for the function x^3 on the interval from 0 to 2?

What do the sums $(1/n)(\sqrt{1/n} + \sqrt{2/n} + \dots + \sqrt{n/n})$ approach as n grows?

Every continuous curve has an area

If f is continuous on an interval with no negative values, and the area under it is 0, must f be 0 at every point?

What is the area under the shrinking tents, as a sum over the tents, and does that sum approach a number?

THE CALCULUS AS A MACHINE

So far every derivative has been found from the definition, by expanding $f(a + h)$ and watching what is left over. That is how the rule for powers was found, and it is slow work. The subject became a calculus, a procedure that could be carried out without thought, when Leibniz and Newton found rules that build the derivative of a complicated function from the derivatives of its parts. There are four, and each comes from the defining form

$$f(a + h) = f(a) + m \cdot h + r(h)$$

in which m is the derivative at a and the error $r(h)$ is small compared with h . Sums and constant multiples have been dealt with.

Products. Let $g(a + h) = g(a) + n \cdot h + s(h)$ likewise, and multiply the two forms. The result is $f(a)g(a)$, plus $(f(a)n + g(a)m) \cdot h$, plus the six terms $mn \cdot h^2$, $f(a)s(h)$, $g(a)r(h)$, $m \cdot h \cdot s(h)$, $n \cdot h \cdot r(h)$ and $r(h)s(h)$, each of which is h times something that approaches 0. So the derivative of the product fg at a is $f(a)n + g(a)m$: each factor in turn is differentiated while the other stands still. A rectangle whose sides both grow shows the rule. Its area grows by a strip along one side, a strip along the other, and a corner where the

two strips cross; the strips are in proportion to h , and the corner, in proportion to h^2 , is the part that is small compared with h .

A function of a function. Let g be applied after f , and let g have derivative n at the point $f(a)$, with error s . Write $k = m \cdot h + r(h)$ for the step that f takes when its input takes the step h . Then

$$\begin{aligned} g(f(a+h)) &= g(f(a) + k) = g(f(a)) + n \cdot k + s(k) \\ &= g(f(a)) + nm \cdot h + [n \cdot r(h) + s(k)] \end{aligned}$$

The bracket is small compared with h , and its two parts can be taken one at a time. The part $n \cdot r(h)$ is small compared with h because $r(h)$ is. For $s(k)$, notice first that $k = m \cdot h + r(h)$ shrinks to 0 as h does. At an h where k is not 0, $s(k)/h$ is $s(k)/k$ times k/h . The first factor approaches 0, because s is small compared with its own step k and k is shrinking; the second factor is $m + r(h)/h$, which approaches m . At an h where k is 0, the input of g has taken no step at all, so $s(k)$ is 0 and there is nothing to bound. So the derivative of g after f , at a , is the derivative of g at $f(a)$ times the derivative of f at a . Rates multiply along a chain: if g changes n times as fast as its input, and its input changes m times as fast as x , then g changes nm times as fast as x . For $(x^2 + 1)^{10}$ the rule gives $10(x^2 + 1)^9 \cdot 2x$ at once, where expanding would produce eleven terms first.

Reciprocals and quotients. The function $1/x$ has derivative $-1/x^2$ at every x other than 0, since $(1/(x+h) - 1/x)/h$ is $-1/(x(x+h))$, which approaches $-1/x^2$ by the law for quotients of limits. Now $1/g$ is the function $1/x$ applied after g . By the rule for a function of a function its derivative at a is the derivative of $1/x$ at $g(a)$, which is $-1/g(a)^2$, times the derivative n of g : that is $-n/g(a)^2$, wherever g is not 0. And f/g is the product of f and $1/g$, so by the rule for products its derivative is $m/g(a) - f(a)n/g(a)^2$, which over the common denominator is $(g(a)m - f(a)n)/g(a)^2$. For $1/(1+x^2)$ that is $-2x/(1+x^2)^2$, and for $x/(1+x^2)$ it is $(1-x^2)/(1+x^2)^2$.

* * *

The fourth rule is for a function run backwards, and first it has to be known that functions can be run backwards.

A continuous function that is below a value y at one point of an interval and above it at another takes the value y somewhere between.

Suppose the function is below y at the left point and above it at the right one; the other case is the same with the roles exchanged. Halve the interval between the two points and look at the function at the midpoint. If it equals y there, stop. If it is below y , keep the right half; if it is above y , keep the left half. Either way the function is below y at the left end of the half that is kept and above y at its right end, so the step can be repeated on that half, and on the half kept from it, and so on, either until it stops or without end.

The halves close on a point c . Could the value at c be below y ? Then, by continuity, the function would be below y throughout some window around c . But the right-hand ends of the halves close on c , so eventually they lie in that window, and at every one of them the function is above y . That is a contradiction. The value at c cannot be above y either, for the mirrored reason, with the left-hand ends in place of the right. So it equals y .

Bolzano proved this in 1817, in the paper in which he found that bounded sequences accumulate, and he was the first to see that it needed proof: a curve that is below a line at one place and above it at another must cross it, as anyone can see, but what anyone can see is the pencil, and the theorem is about numbers. Among the fractions it is false: $x^2 - 2$ is below 0 at 1 and above 0 at 2, and is 0 at no fraction.

Now let f be continuous and increasing on an interval, in the strict sense that a larger input always gives a larger value. Each value between two of its values is taken, by the theorem, and taken once only, since f increases. So every y in the range of f comes from exactly one x , and the rule sending y to that x is the *inverse* of f , written here as g . It increases: if a larger y came from a smaller x , or from the same x , then f would not be increasing. It is continuous. Take a tolerance around $x = g(y)$; it marks out an interval around

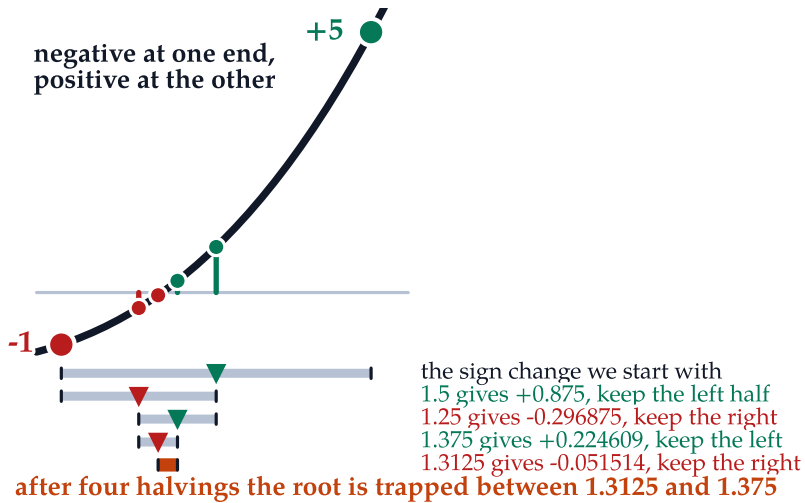


Figure 36.1. The curve of x cubed minus x minus 1 runs from minus one at the input 1 to plus five at the input 2, and four halvings keep a sign change every time, trapping the crossing between 1.3125 and 1.375. Assumes: that the curve has no break between the sampled points, which is exactly the hypothesis of continuity; among the fractions the same halving can be run on x squared minus 2, and there it traps no crossing at all

x . Since f is increasing and takes every value in between, f carries that interval to an interval around y , and this is the window. Every point of the window is the value of f at some point of the first interval, so g sends it back to that point, inside the tolerance. And if f has derivative m at x , not 0, then g has derivative $1/m$ at $y = f(x)$. For take a step k from y , and let $x + h = g(y + k)$. Then $f(x + h) = y + k$, and since $f(x) = y$, the step k is $f(x + h) - f(x)$, while the step taken by g is h . Here h is not 0 when k is not, since g sends different numbers to different numbers, and h shrinks as k does, g being continuous. The quotient for g is

$$(g(y + k) - g(y))/k = h/(f(x + h) - f(x))$$

which is the reciprocal of a quotient that approaches m . Reflected in the diagonal line $y = x$, the graph of f becomes the graph of g ,

and a line of slope m becomes a line of slope $1/m$.

The first use is for roots. For a whole number q above 1, the function x^q is continuous and increasing on the numbers from 0 upwards and passes every bound, so every number at least 0 is the q -th power of exactly one such number, its q -th root, written $x^{1/q}$. Its derivative at $y = x^q$ is $1/(q \cdot x^{q-1})$, by the rule for inverses. Now x is $y^{1/q}$, so x^{q-1} is $y^{(q-1)/q}$, which is $y^{1-1/q}$, and dividing by it is multiplying by $y^{1/q-1}$. In terms of y , then, the derivative is $(1/q) \cdot y^{1/q-1}$. For a fraction p/q , the power $x^{p/q}$ is the q -th root raised to the power p , and the rule for a function of a function gives the derivative $p \cdot (x^{1/q})^{p-1} \cdot (1/q) \cdot x^{1/q-1}$, which comes to $(p/q) \cdot x^{p/q-1}$. To see it, write t for $x^{1/q}$, so that x is t^q : the second factor is $1/q$ divided by t^{q-1} , as found for roots, so the product is $(p/q) \cdot t^{p-1}/t^{q-1}$, which is $(p/q) \cdot t^{p-q}$, the q -th root raised to the power $p - q$, and that is $x^{p/q-1}$. So for positive x

the derivative of x^r is $r \cdot x^{r-1}$ for every fraction r

the rule that was proved for whole powers, and negative fractions follow by the rule for reciprocals. The square root has derivative $1/(2\sqrt{x})$, and $\sqrt{1+x^2}$ has derivative $x/\sqrt{1+x^2}$.

* * *

With the rules in hand, questions about largest and smallest become routine, by a fact that was proved on the way to the average-slope theorem: at a point inside an interval where a function takes its largest or its smallest value, its derivative, if it has one, is 0. Now a continuous function on an interval containing its ends certainly has a largest value. By that fact, the largest value is taken in one of three kinds of place: at an end of the interval, at an interior point where the derivative is 0, or at an interior point where there is no derivative. Such places are usually few, and the largest value is found by comparing the values at them.

From a square sheet of side a , equal squares of side x are cut from the four corners and the flaps folded up to make an open

box. Its volume is $x(a - 2x)^2$, which is 0 at $x = 0$ and at $x = a/2$. The derivative, by the rules for products and chains, is $(a - 2x)^2 - 4x(a - 2x)$, which is $(a - 2x)(a - 6x)$. Inside the interval it is 0 only at $x = a/6$. The volume is 0 at both ends of the interval and positive between them, so its largest value is not at an end; it is at an interior point where the derivative is 0, and there is only one. So the largest box has depth a sixth of the side of the sheet, and its volume, $(a/6)(2a/3)^2$, is $2a^3/27$.

A lifeguard on a beach sees a swimmer in trouble, out to sea and some way along the shore. She runs faster than she swims. Where should she enter the water? Let her stand at distance p from the waterline and the swimmer at distance q beyond it, with a distance d along the shore between them, and let her enter the water at distance x along the waterline, measured from the point of the waterline nearest to her towards the swimmer. Her time is

$$T(x) = \sqrt{(p^2 + x^2)}/v + \sqrt{(q^2 + (d - x)^2)}/w$$

where v is her speed on sand and w her speed in water. Its derivative is $x/(v\sqrt{(p^2 + x^2)}) - (d - x)/(w\sqrt{(q^2 + (d - x)^2)})$. The first fraction, without the v , is the sine of the angle α between her path on the sand and the perpendicular to the waterline, and the second, without the w , is the sine of the corresponding angle β in the water. The derivative is negative at $x = 0$ and positive at $x = d$. As the entry point slides along the shore the angle α opens and the angle β closes, so $\sin \alpha$ grows and $\sin \beta$ shrinks, and the derivative, which is $\sin \alpha$ over v less $\sin \beta$ over w , increases throughout. The derivative is continuous, negative at one end and positive at the other, so it is 0 somewhere between; and since it increases, it is 0 only once. Before that point the derivative is negative and T is falling, after it the derivative is positive and T is rising, so there T is smallest:

$$\sin \alpha / v = \sin \beta / w$$

Light crossing from air into water obeys the same equation, with its two speeds. The law of refraction was found by experiment by Snell about 1621, and Fermat showed in 1662 that it is

what light would do if it took the path of least time. Heron of Alexandria had explained the law of reflection by a shortest path; Fermat's was the first such explanation that needed a derivative. The speeds it implied for light, slower in water than in air, were confirmed by measurement in 1850, and principles of this kind, that nature does the least of something, have stood near the centre of physics ever since.

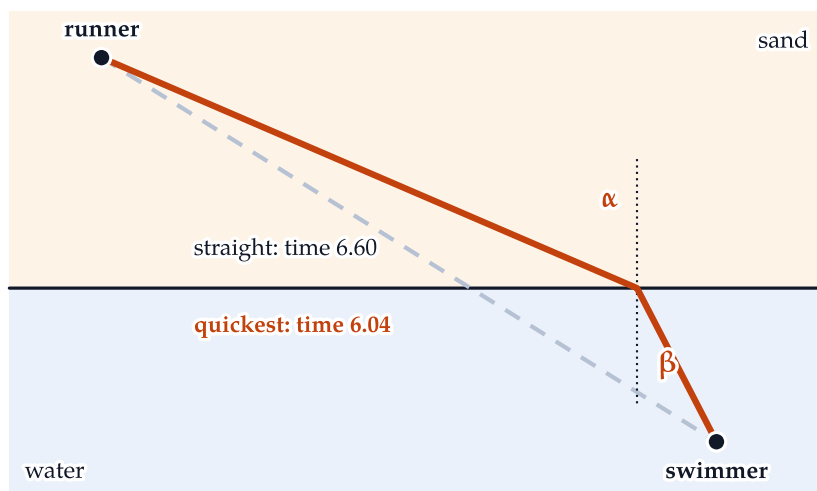


Figure 36.2. A runner on the sand and a swimmer in the water, with running twice as fast as swimming. The straight path is the shortest and not the quickest. The quickest path spends more of its length on the sand, and at the waterline the sines of the two angles with the perpendicular stand in the proportion of the two speeds, 2 to 1. *Assumes:* a straight waterline and one speed on each side of it; the crossing point was found by halving the interval on which the derivative of the time changes sign

* * *

The derivative is itself a function, and its derivative, the *second derivative*, says how the slope is changing. If the second derivative is never negative on an interval, the slope never decreases there, and the curve bends upwards like a cup. Such a function lies above

each of its tangents. For by the average-slope theorem $f(x) - f(a)$ is $(x - a)$ times the slope at some point c between a and x , so

$$f(x) - f(a) - f'(a)(x - a) = (f'(c) - f'(a))(x - a)$$

where f' stands for the derivative. The left side is the height of the curve at x less the height there of the tangent at a . On the right, if x is to the right of a then so is c , and since the slope never decreases, both factors are at least 0; if x is to the left of a , both are at most 0. Either way the product is not negative, so the curve is nowhere below the tangent.

It also lies below each of its chords. Take a point of the curve between the two ends of a chord, and join it by one straight line to the left end and by another to the right end. By the average-slope theorem, the slope of the first line is the slope of the curve at some place to the left of the point, and the slope of the second is the slope of the curve at some place to the right of it, which is no smaller. So the line on to the right end is at least as steep as the line from the left end. For a point above the chord it would be the other way round: the line up to it from the left end would be steeper than the chord, and the line on from it to the right end less steep than the chord. So the point lies on the chord or below it.

For x^2 , whose second derivative is 2, lying below the chord at its midpoint says that the square of an average is at most the average of the squares, which is the statement that a variance is not negative. At a point where the derivative is 0 and the second derivative is positive, the slope is negative just before the point and positive just after, so the function has there its smallest value among nearby points; where the second derivative is negative, its largest. For the box, the derivative of the volume multiplies out to $a^2 - 8ax + 12x^2$, so the second derivative is $-8a + 24x$, which at $x = a/6$ is $-4a$: negative, in agreement with the largest volume found there.

The tangent line was the best straight line. The second derivative gives the best parabola, and says how good it is.

If f has a second derivative throughout an interval containing a and

$a + h$, then $f(a + h) = f(a) + f'(a)h + f''(c)h^2/2$ for some c between a and $a + h$.

Here f'' is the second derivative. Choose the number K that makes $f(a + h)$ equal to $f(a) + f'(a)h + K \cdot h^2$, and consider the function $G(t) = f(t) - f(a) - f'(a)(t - a) - K(t - a)^2$. It is 0 at $t = a$, and 0 at $t = a + h$ by the choice of K . So its derivative is 0 at some point c_1 between them, by the average-slope theorem. But its derivative, $f'(t) - f'(a) - 2K(t - a)$, is also 0 at $t = a$. So the derivative of that, which is $f''(t) - 2K$, is 0 at some point c between a and c_1 . That says $K = f''(c)/2$. Repeating the step gives the same kind of statement with more terms: with third derivative f''' ,

$$f(a + h) = f(a) + f'(a)h + f''(a)h^2/2 + f'''(c)h^3/6$$

and so on, each new term a derivative at a , times a power of h , divided by a factorial, and the last term taken at an unknown point c . Taylor published the expansion in 1715 without the last term; Lagrange supplied it in 1797, and it is what turns an approximation into a statement.

For $\sqrt{1+x}$ at $a = 0$ the first and second derivatives at 0 are $1/2$ and $-1/4$, and the third derivative at a point c is $(3/8)(1+c)^{-5/2}$. So $\sqrt{1+h}$ is $1 + h/2 - h^2/8$, with an error of exactly $(3/8)(1+c)^{-5/2} \cdot h^3/6$, which is $(1+c)^{-5/2} \cdot h^3/16$. For positive h the point c is positive, so the power of $1+c$ is below 1, and the error is at most $h^3/16$. At $h = 1/10$ that gives 1.04875 with an error of at most 0.0000625, and $\sqrt{1.1}$ is 1.0488088.

The expansion settles a question left open about sums that choose the midpoint of each piece. Around the midpoint m of a piece of width w , $f(x)$ is $f(m) + f'(m)(x - m) + f''(c)(x - m)^2/2$. Integrate over the piece, one term at a time. The first term gives $f(m) \cdot w$, which is the term of the sum. The second gives 0, its positive and negative halves cancelling. The third is all of the error. It is at most $M(x - m)^2/2$ in size, where M bounds the second derivative, and the integral of $(x - m)^2$ over the piece, which reaches $w/2$ to each side of m , is $w^3/12$. So the error on a piece is

at most $M \cdot w^3/24$, and over an interval of length ℓ , which has ℓ/w pieces, at most $M \cdot \ell \cdot w^2/24$. It goes as the square of the width: ten times narrower, a hundred times better. For $1/(1 + x^2)$, whose second derivative is at most 2 in size, ten pieces are guaranteed an error of at most 0.00083, and the error found was 0.00021.

* * *

The same idea finds roots. To solve $f(x) = 0$, start from a guess x , replace the curve by its tangent line at x , and solve that instead: the line $f(x) + f'(x)(t - x)$ is 0 at

$$x_1 = x - f(x)/f'(x)$$

and x_1 is the next guess, written with a subscript rather than a prime, the prime being already spoken for by the derivative. For $f(x) = x^2 - a$ it says $x_1 = x - (x^2 - a)/(2x)$, which is $(x + a/x)/2$: the averaging of the Babylonian scribes is the tangent line in disguise. Newton described the method in 1669 on the equation $x^3 - 2x - 5 = 0$. From $x = 2$, where the value is -1 and the slope 10, the guesses run

$$2.1, 2.0945681, 2.0945514817, 2.0945514815$$

and the last is right in every place shown.

The expansion with remainder says why the correct places double. Let ρ be the root. Expand around the guess x , with the step $\rho - x$: $0 = f(\rho) = f(x) + f'(x)(\rho - x) + f''(c)(\rho - x)^2/2$. Divide by $f'(x)$. By the rule for the next guess, $f(x)/f'(x)$ is $x - x_1$, so the equation reads $0 = (x - x_1) + (\rho - x) + f''(c)(\rho - x)^2/(2f'(x))$. The x and the $-x$ cancel, and moving $x_1 - \rho$ to the other side leaves

$$x_1 - \rho = (f''(c)/(2f'(x))) \cdot (x - \rho)^2$$

The new error is the square of the old one, times a factor that stays bounded as long as the slope stays away from 0 and the bending stays bounded. Squaring is what doubles the places: an error

of about one part in a million, six correct places, becomes an error of about one part in a million million, twelve correct places, apart from the factor. For $x^2 - a$ the factor is $1/(2x)$. The excess $x^2 - a$ is the error $x - \rho$ times $x + \rho$, which is close to $2x$ times the error. So the old error is about the old excess divided by $2x$, and the new excess is about $2x$ times the new error. The new error is the old error squared, divided by $2x$; multiplied by $2x$, that is the old error squared, which is the old excess squared divided by $4x^2$. In terms of the excess, then, the factor becomes $1/(4x^2)$, and for $a = 2$, where x^2 is close to 2, that is the eighth that was seen before.

The method is not safe. It needs a start near enough to the root, and away from places where the curve is flat. For $x^3 - 2x + 2$, starting from 0, the tangent at 0 sends the guess to 1, and the tangent at 1 sends it back to 0, forever, though the equation has a root near -1.77 . A second failure needs the cube root on the whole line. The cube x^3 increases on the whole line, so its inverse, the cube root, is defined on the whole line, the cube root of $-y$ being the negative of the cube root of y . Apply the tangent method to the cube root, which is 0 only at 0. By the rule for inverses, its derivative at x is 1 over 3 times the square of the cube root of x , so $f(x)/f'(x)$ is 3 times the cube of the cube root of x , which is $3x$. The next guess from any x other than 0 is therefore $x_1 = x - 3x = -2x$, on either side of 0 alike: twice as far from the root, and on the other side of it, at every step. Halving an interval on which the function changes sign cannot fail and gains one binary place at a time; the tangent method doubles its places when it works. The practice is to halve until the root is cornered and then switch.

Where does $x/(1 + x^2)$ take its largest value, and what is it?

A fence 8 feet high runs parallel to the wall of a house, 27 feet from it. How long is the shortest ladder that reaches from the ground outside the fence, over it, to the wall?

A ray of light goes from one point to another by way of a flat mirror. If it takes the shortest path, what can be said about the two angles it makes with the mirror?

For which fractions r is $(1 + x)^r$ at least $1 + rx$ for every x above

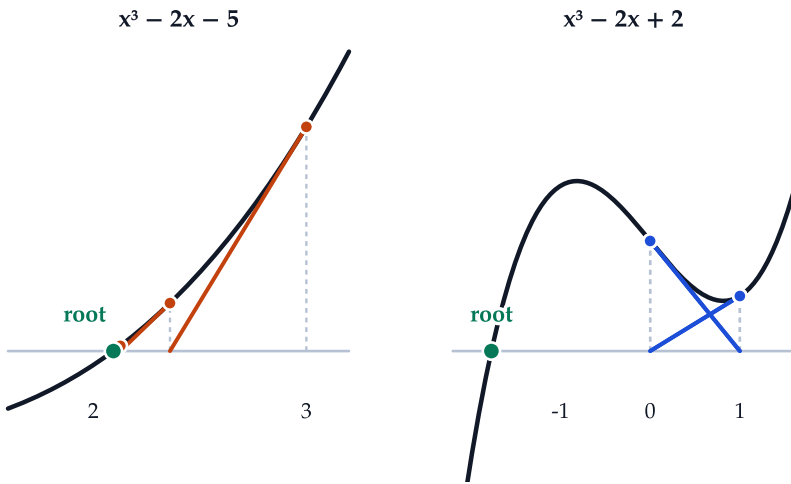


Figure 36.3. The tangent method at work and at fault. On the left, for $x^3 - 2x - 5$ from the guess 3, each tangent line meets the axis nearer the root than the last. On the right, for $x^3 - 2x + 2$ from the guess 0, the tangent at 0 points to 1 and the tangent at 1 points back to 0, and the guesses never approach the root near -1.77 . Assumes: two particular cubics; the vertical scales of the two panels differ

-1?

The equation $x^2 + 1 = 0$ has no solution among the real numbers. What does the tangent method do with it, from the start $1/\sqrt{3}$, and from the start 1?

REVERSING RATES

The integral of a continuous function from a to b is $F(b) - F(a)$ for any function F whose derivative is the given one. Finding derivatives is a procedure; finding a function with a given derivative is a search, and the search is helped by reading the rules for derivatives backwards. Each rule about derivatives is a rule about integrals, and two of them do most of the work.

The rule for a function of a function says that $G(u(x))$ has derivative $g(u(x)) \cdot u'(x)$, where g is the derivative of G . Read backwards: $G(u(x))$ is a function whose derivative is $g(u(x)) \cdot u'(x)$, so the integral of $g(u(x)) \cdot u'(x)$ from a to b is its change, $G(u(b)) - G(u(a))$. But G is also a function whose derivative is g , so the same difference is the integral of g from $u(a)$ to $u(b)$. So

where u has an unbroken derivative and g is continuous, the integral of $g(u(x)) \cdot u'(x)$ over x from a to b is the integral of $g(u)$ over u from $u(a)$ to $u(b)$.

This is *substitution*: a factor that is the derivative of an inner function is absorbed, and the inner function becomes the variable. For $x\sqrt{1-x^2}$ from 0 to 1, take $u = 1 - x^2$, whose derivative is $-2x$; the integrand is $-(1/2)\sqrt{u} \cdot u'$. As x goes from 0 to 1, u goes from 1 to 0, so the integral is $-1/2$ times the integral of $\sqrt{u}$ from 1 to 0. An integral taken backwards is the negative of the integral taken

forwards, so that is $1/2$ times the integral of $\sqrt{u}$ from 0 to 1. Since $u^{3/2}$ has derivative $(3/2)\sqrt{u}$, that is $(1/2)(2/3) = 1/3$. For $x/(1+x^2)^2$ from 0 to b , take $u = 1 + x^2$, whose derivative is $2x$, so that the integrand is $(1/2)(1/u^2) \cdot u'$. As x goes from 0 to b , u goes from 1 to $1 + b^2$, so the integral is $1/2$ times the integral of $1/u^2$ from 1 to $1 + b^2$. Since $-1/u$ has derivative $1/u^2$, that is $(1/2)(1 - 1/(1 + b^2))$.

The rule for products says that fg has derivative $f'g + fg'$. Integrate both sides from a to b . The left side gives $f(b)g(b) - f(a)g(a)$, since fg is a function whose derivative is being integrated. The right side gives the integral of $f' \cdot g$ plus the integral of $f \cdot g'$. Move the first of these to the other side:

*where f and g have unbroken derivatives, the integral of $f \cdot g'$ is
 $f(b)g(b) - f(a)g(a)$ less the integral of $f' \cdot g$.*

This is *integration by parts*. It moves the derivative from one factor to the other, at the price of a term at the ends, and it pays when the moved derivative makes things simpler. Take the integral of $x^m(1-x)^n$ from 0 to 1, for whole numbers m and n . Let f be $(1-x)^n$ and g' be x^m , so that g is $x^{m+1}/(m+1)$ and f' is $-n(1-x)^{n-1}$. The term at the ends is 0, since g is 0 at 0 and f is 0 at 1. So the integral of $f \cdot g'$ is the negative of the integral of $f' \cdot g$, and $f' \cdot g$ is $-n(1-x)^{n-1} \cdot x^{m+1}/(m+1)$. The two minus signs cancel, and the integral is $n/(m+1)$ times the integral of $x^{m+1}(1-x)^{n-1}$: one power has moved across. Doing it n times moves them all, and leaves $n(n-1) \cdots 1$ over $(m+1)(m+2) \cdots (m+n)$, times the integral of x^{m+n} , which is $1/(m+n+1)$. The top is $n!$. The bottom is now $(m+1)(m+2) \cdots (m+n+1)$, which is $(m+n+1)!$ with its first m factors missing, so it is $(m+n+1)!/m!$. So

the integral of $x^m(1-x)^n$ from 0 to 1 is $m! \cdot n!/(m+n+1)!$

For $m = 3$ and $n = 2$ that is $6 \cdot 2/720 = 1/60$.

The same number can be found with no calculus at all, and the second way explains the first. A ball is rolled onto a table and stops at a random distance x across it, as a fraction of the width. Then N more balls are rolled, and each stops to the left of the first

with chance x . The chance that exactly k of them do is $(N \text{ choose } k)x^k(1-x)^{N-k}$. But x itself is random, and random here means this: however the width is cut into j equal pieces, the first ball is equally likely to stop in each of them, with chance $1/j$ for each.

Suppose it stops in a given piece. Then the chance that exactly k of the others stop to its left is a value between the smallest and the largest that $(N \text{ choose } k)x^k(1-x)^{N-k}$ takes on that piece. So the chance overall is a sum over the pieces of $1/j$ times such a value. Now $1/j$ is also the width of a piece, so that sum lies between the lower and the upper staircase on the cutting. This holds for every j , and as j grows both staircases approach the integral. So the chance overall is the integral of $(N \text{ choose } k)x^k(1-x)^{N-k}$ from 0 to 1.

Now look at the same experiment differently. Roll all $N + 1$ balls first and only then decide which one was the first. By symmetry it is equally likely to be the leftmost, the second from the left, and so on: $N + 1$ cases, each with chance $1/(N + 1)$. And it has exactly k balls to its left when it is in place $k + 1$. So

the integral of $(N \text{ choose } k)x^k(1-x)^{N-k}$ from 0 to 1

is $1/(N + 1)$, for every k

which is the formula above with $m = k$ and $n = N - k$. This is the table on which Bayes, in the paper published in 1763, two years after his death, first reasoned from observed outcomes back to an unknown chance: having seen k of N balls stop to the left, what should be believed about x ?

* * *

Substitution is how the physics of motion is done. Newton's law says that a force F on a body of mass m changes its velocity v at the rate F/m . Let the body move along a line, with position x at time t , so that v is the rate of change of x . The quantity $mv^2/2$ has rate of change $mv \cdot v'$, by the rule for a function of a function, and mv' is the force, so the rate is $F \cdot v$. If the force depends only on

the position, then $F \cdot v$ is $F(x(t)) \cdot x'(t)$, and substitution turns its integral over time into the integral of F over the positions passed through:

the change in $mv^2/2$ is the integral of the force over the positions passed through.

The quantity $mv^2/2$ is called the kinetic energy and the integral of the force the work, so the change in kinetic energy is the work done. That statement answers questions that look hard. How fast must a stone be thrown upwards never to come back, the air being ignored? Gravity pulls a mass m at distance r from the centre of the earth with the force GMm/r^2 , where M is the mass of the earth and G a constant. Climbing from the surface, at $r = R$, to the distance $r = b$ costs the work

$$GMm(1/R - 1/b)$$

since $-1/r$ has derivative $1/r^2$. As b grows that approaches GMm/R and never exceeds it. So a stone that starts with kinetic energy above GMm/R can never be stopped. Gravity pulls against the climb, so on the way out to the distance b the stone loses the kinetic energy $GMm(1/R - 1/b)$. That is less than GMm/R , so some kinetic energy is left at every distance, and the stone is still moving. A stone that starts with less kinetic energy than GMm/R does come back: far enough out, $GMm(1/R - 1/b)$ is more than it started with, and since $mv^2/2$ is never negative the stone cannot reach that distance, so it stops short and gravity pulls it back. The condition is $mv^2/2 \geq GMm/R$. The pull at the surface, GMm/R^2 , is the familiar mg , so GM/R is gR , and the condition says $v^2 \geq 2gR$. With $g = 9.81$ metres per second per second and $R = 6371$ kilometres, the speed is 11.2 kilometres a second. It does not depend on the mass of the stone, and the argument from units would have found $\sqrt{(gR)}$ and missed the $\sqrt{2}$.

* * *

The climb to infinity used an integral over a stretch with no end. Such an integral is a limit of ordinary ones: the integral of f from a onwards is the limit, as b grows, of the integral from a to b , when that limit exists. For $1/x^2$ from 1 onwards it is the limit of $1 - 1/b$, which is 1: a region that runs on forever, and whose area is 1.

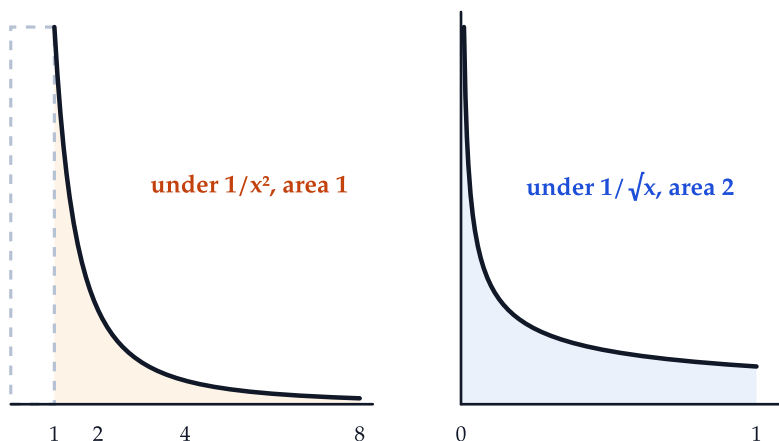


Figure 37.1. Two regions without end. The one under $1/x^2$ runs on forever to the right and has area 1, the same as the dashed unit square beside it, which the two scales of the drawing stretch into a tall rectangle. The one under $1/\sqrt{x}$ runs up forever along the vertical axis and has area 2. The curve $1/x$ is the boundary case between them, and is not drawn here. Assumes: the regions drawn as far as the page allows; what is cut off has area $1/8$ on the left and $1/10$ on the right

Whether a region that runs on forever has an area is a matter of how fast the curve falls. For $1/x^p$ with a fraction p other than 1, the integral from 1 to b is $(b^{1-p} - 1)/(1 - p)$. When p is above 1 the power b^{1-p} approaches 0 and the area is $1/(p - 1)$. When p is below 1 the power passes every bound, and there is no area. At $p = 1$ the rule for powers gives no formula, but the matter can be settled by rectangles: under $1/x$, between 2^k and 2^{k+1} , lies a rectangle of width 2^k and height $1/2^{k+1}$, of area $1/2$, so the area from 1 to 2^n is at least $n/2$ and passes every bound. The curve $1/x$ is the boundary, and it

falls on the side of no area.

The same happens at a place where a curve shoots up. The integral of $1/\sqrt{x}$ from c to 1 is $2 - 2\sqrt{c}$, which approaches 2 as c approaches 0, so the region under $1/\sqrt{x}$ over the stretch from 0 to 1, which is unbounded upwards, has area 2. For $1/x^p$ at 0 the roles are reversed: there is an area when p is below 1 and none when p is 1 or more.

These areas measure sums. Give each term $1/k^p$ a step of width 1 and height $1/k^p$. Drawn over the stretch from $k - 1$ to k , to the left of k , the step lies under the curve $1/x^p$, because the curve falls and comes down to the height $1/k^p$ only at the right end of the stretch. Drawn over the stretch from k to $k + 1$, to the right of k , the step lies over the curve, which starts at that height and falls. So for p above 1

$$1/2^p + 1/3^p + \dots + 1/n^p \text{ is less than the area from 1 to } n, \text{ which is less than } 1/(p - 1)$$

and the sums $1 + 1/2^p + \dots + 1/n^p$, which increase, stay below $1 + 1/(p - 1)$ and approach a number. For $p = 3/2$ the bound is 3, and the sums approach 2.612. For p above 0 and at most 1 the staircase over the curve shows that the sums pass every bound: the sum as far as $1/n^p$ is more than the area from 1 to $n + 1$, and that area passes every bound. For p at most 0 no term is less than 1, and the sums pass every bound as well.

* * *

What is the length of a curve? Straight pieces have lengths, by the theorem on squares, and a curve can be approached by chains of them. Take the graph of a function f over the interval from a to b , mark the points of the curve over a cutting of the interval, and join each to the next. The chain is inscribed in the curve, and its length is the sum of the hypotenuses $\sqrt{(w^2 + k^2)}$, where w is the width of a piece and k the rise of the curve across it. Suppose f has a continuous derivative. By the average-slope theorem the

rise across a piece is $k = f'(c) \cdot w$ for some c in the piece. So each hypotenuse is $\sqrt{(w^2 + f'(c)^2 \cdot w^2)}$, which is $\sqrt{(1 + f'(c)^2)} \cdot w$, and the length of the chain is

$$\text{the sum over the pieces of } \sqrt{(1 + f'(c)^2)} \cdot w$$

and that is one of the sums met before: a point c is chosen in each piece, and the value at c of the continuous function $\sqrt{(1 + f'(x)^2)}$ is multiplied by the width of the piece. As the pieces narrow, such sums approach the integral, whatever points are chosen. So the inscribed chains have lengths that approach one number, and that number is taken as the *length* of the curve:

the length of the graph of f from a to b is the integral of $\sqrt{(1 + f'(x)^2)}$.

For a straight line of slope s it gives $\sqrt{(1 + s^2)}$ times the width, as it should. Torricelli had found the exact length of a spiral about 1645; the first curve with an equation in powers of x and y whose length was found exactly was the graph of $x^{3/2}$, by Neile in 1657, when he was nineteen. There f' is $(3/2)\sqrt{x}$, so the length from 0 to 4 is the integral of $\sqrt{(1 + 9x/4)}$, and with $u = 1 + 9x/4$ that is $4/9$ times the integral of $\sqrt{u}$ from 1 to 10:

$$(8/27)(10\sqrt{10} - 1) = 9.0734$$

The chord from one end to the other is $\sqrt{(16 + 64)}$, which is 8.944. Twenty years earlier Descartes had written, with the authority of Aristotle behind him, that the proportion between straight lines and curved ones was not known and could not be. For most curves the integral is one of those with no formula; the parabola itself leads to the area under the curve $1/x$.

The circle is one that matters. On the upper half of the unit circle $y = \sqrt{(1 - x^2)}$, and y' is $-x/y$, so $1 + y'^2$ is $(y^2 + x^2)/y^2$, which is $1/y^2$. Fix a point of the upper half circle, over x_1 , and let a second point, over x , move to the left of it. The length of the arc between them is $A(x)$, the integral of $1/y$ from x to x_1 , and its rate of change

as x moves is $-1/y$. The sign is a minus because x is the lower end of the integral: moving x to the right shortens the arc.

Compare the sector that the two points cut off with the centre. Its area is found from a simpler sector, the one between the ray to $(1, 0)$ and the ray to the point (x, y) . That sector is the region under the circle from x to 1, together with the triangle whose corners are the centre, $(x, 0)$ and (x, y) , or less that triangle when x is negative. The region has area the integral of y from x to 1, and $xy/2$ is the area of the triangle when x is positive and the negative of it when x is negative, so in both cases the sector has area $xy/2$ plus the integral of y from x to 1. Its rate of change is $(y + x \cdot y')/2 - y$: the first part by the rule for products, and the $-y$ because x is the lower end of the integral. With $y' = -x/y$ that is $(y - x^2/y)/2 - y$, and with $x^2 + y^2 = 1$ that is $-1/(2y)$. The sector between the two points is that sector less the fixed one for x_1 , so its area $S(x)$ has the same rate, $-1/(2y)$. So $A - 2S$ has rate of change 0 and is constant, and it is 0 when the two points coincide:

on a circle of radius 1, the length of an arc is twice the area of its sector.

The ends of the half circle, where $1/y$ is unbounded, are reached by letting the points approach them, the arc being a limit of arcs and the sector a limit of sectors.

The two ways of measuring how far a ray has turned, by the arc it sweeps and by the sector it sweeps, are one measure. For the whole circle of radius 1 it says that the circumference is twice the area of the disc. For radius r lengths scale by r and areas by r^2 , so there the circumference times the radius is twice the area. That is the statement of Archimedes that a disc has the area of a triangle whose base is the circumference and whose height is the radius.

* * *

Volumes are found by slices, as areas were by strips. If the slice of a solid at height t has area $a(t)$, continuous in t , the volume of the solid is *taken* to be the integral of $a(t)$ over the heights. That

is a definition and not a theorem. Nothing so far has given a solid a volume. And the staircases do not carry over as they stand: a rectangle could be fitted under a curve, but a slab whose base is the slice at height t need not lie inside the solid at all. It is the right definition because it gives a box its length times its width times its height, and gives a solid contained in another the smaller number. Two solids with equal slices at every height have equal volumes, whatever their shapes; Cavalieri made a method of that in 1635, and it is how a stack of coins pushed askew keeps its volume.

Take a pyramid of height h on any base, of area B . The slice at distance t below the apex is the base shrunk by the factor t/h , and shrinking a figure by a factor multiplies its area by the square of the factor, so its area is $B(t/h)^2$, and the volume is B/h^2 times the integral of t^2 from 0 to h :

the volume of a pyramid, or a cone, on any base is a third of the base times the height.

The $1/3$ is the area under the parabola, the first number that the staircases found. Democritus knew the rule for the pyramid and Eudoxus proved it, by a bracketing argument, about 370 BC. No way of cutting a pyramid into finitely many pieces and reassembling them into a box is possible in general, as Dehn proved in 1900, so here the limit cannot be avoided.

* * *

Integrals that have no formula have to be computed, and a little theory makes the computing fast. The sum that takes the mid-point of each piece has an error that goes as the square of the width. So does the sum of trapezoids, which joins the points of the curve over the cutting by straight lines. Where the curve bends one way the two sums err on opposite sides, and the trapezoid sum errs by about twice as much as the midpoint sum, each error being a multiple of the total change of slope across the piece. The reason is that the midpoint sum is a sum of trapezoids too. The rectangle

whose height is the value at the midpoint has the same area as the trapezoid under the tangent at the midpoint, since what the tilted top gains on one side of the midpoint it loses on the other. So one sum replaces the curve by chords and the other by tangents, and a curve that bends one way lies between its chords and its tangents.

For the parabola x^2 over a piece of width w the chord cuts off an area of $w^3/6$, by the theorem of Archimedes, and the tangent at the midpoint, which is parallel to the chord, cuts off $w^3/12$, exactly half; and a narrow piece of any curve with a second derivative is nearly a piece of a parabola. So take two parts of the midpoint sum and one of the trapezoid sum, and divide by three. Twice the smaller error, on one side, all but cancels the larger error on the other. For $1/(1+x^2)$ from 0 to 1, on ten pieces, the midpoint sum is 0.7856064963, too large by 0.0002083; the trapezoid sum is 0.7849814972, too small by 0.0004167; and

$$(2 \cdot 0.7856064963 + 0.7849814972)/3 = 0.785398163$$

which is the true value to all nine places, the difference being in the tenth. The rule is named after Simpson, who published it in 1743; Kepler had used it in 1615 to gauge wine barrels. It is exact for every polynomial of degree three or less. On a piece with midpoint m such a polynomial is a sum of multiples of 1 , $x - m$, $(x - m)^2$ and $(x - m)^3$. For the constant both sums are exact; for the two odd powers the integral is 0, the value at the midpoint is 0, and the values at the two ends cancel; and for $(x - m)^2$ the integral is $w^3/12$, the midpoint sum is 0 and the trapezoid sum is $w \cdot w^2/4$, of which a third is $w^3/12$.

What does the area under $x/(1+x^2)^2$ from 0 to b approach as b grows, and what does the area under $x/(1+x^2)$ do?

The graph of $x^3/6 + 1/(2x)$ is one of the few curves whose length comes out in closed form. How long is it between $x = 1$ and $x = 2$?

Two pipes of radius r cross at right angles, axis through axis. What is the volume of the region inside both? Its slices parallel to both axes are squares.

Is there a fraction p for which the region under $1/x^p$ over the whole stretch from 0 onwards has an area? Does the region under $1/(\sqrt{x} \cdot (1+x))$ over that stretch have one?

Of the $N + 1$ balls on the table, the first has k to its left. What is the average, over all such tables, of the distance x at which the first ball stopped?

TWO LIMITS, TWO ANSWERS

In row m and column n of an endless square array stands the number

$$m/(m+n)$$

What happens far out in the array, where m and n are both large? There are two ways to go far out, and they disagree.

Fix the row, so that m stays put while n grows. The numerator is fixed and the denominator grows without bound, so the entries tend to 0; every row tends to 0, and the limit of the row limits is 0. Fix the column instead, so that n stays put while m grows. The entry $m/(m+n)$ is $1 - n/(m+n)$, and the piece taken away now has a fixed numerator and a denominator that grows without bound, so it tends to 0 and the entries tend to 1; every column tends to 1, and the limit of the column limits is 1. The entry at $m = 100, n = 1$ is $100/101$ and the entry at $m = 1, n = 100$ is $1/101$, and deep in the array entries near 1 and entries near 0 sit side by side.

Neither computation is wrong. What fails is a sentence nobody wrote down, that the order in which two limits are taken cannot change the answer. Many theorems of analysis allow two limiting processes to be exchanged under some hypothesis, and many counterexamples show an exchange failing without it. Is the rate of change of an infinite sum of functions the sum of their rates? Is

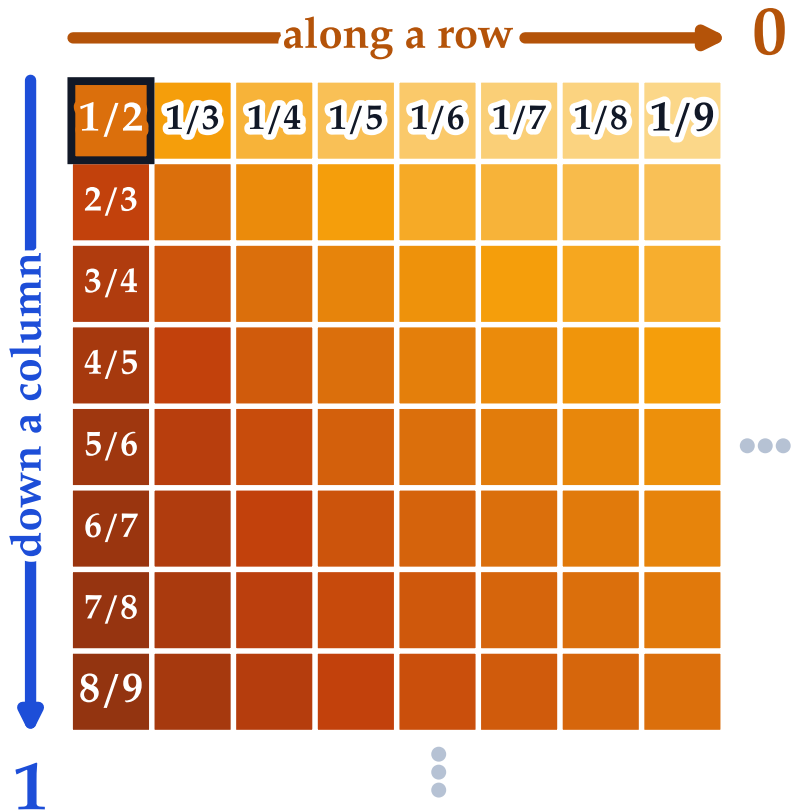


Figure 38.1. The entry in row m , column n is $m/(m+n)$, shaded from pale to deep. Reading along any row toward the right the values fade to zero; reading down any column they deepen to one, so the two ways of going to infinity reach opposite answers. *Assumes:* that the pattern of the drawn corner continues forever; the disagreement is a statement about the infinite array, and no finite block of it can settle the two limits

the area under a limit of functions the limit of their areas? Is a limit of continuous functions continuous? Each question asks whether two limits may be exchanged.

* * *

In 1821 Cauchy published a proof that a convergent sum of continuous functions is continuous. The statement is false. Abel observed five years later that a certain sum of sine waves converges to a function with a jump, and a simpler example needs no sines.

Let $f_n(x) = x^n$ on the interval from 0 to 1. For x strictly below 1 the powers decrease and never go below 0, so they converge to some number L . Now multiply every power by x . The products $x \cdot x^n$ are the same powers with the first one dropped, so they still tend to L ; they are also x times a sequence that tends to L , so they tend to xL . A sequence has only one limit, so $L = xL$, which says $(1 - x)L = 0$, and since $1 - x$ is not 0, $L = 0$. At $x = 1$ every f_n is 1. So the sequence converges at every point, to the function that is 0 everywhere except for the value 1 at the right-hand end. Every f_n is continuous and the limit is not.

To see which step of the proof breaks, write down the estimate that continuity of the limit f at a point a would need. The distance from $f(x)$ to $f(a)$ is at most the distance from $f(x)$ to $f_n(x)$, plus that from $f_n(x)$ to $f_n(a)$, plus that from $f_n(a)$ to $f(a)$. The middle piece is small when x is close to a , since f_n is continuous, and the last is small when n is large. The first is small when n is large, but how large n must be depends on x , while continuity at a concerns all x near a at once. What is needed is one n that makes the first piece small for every x near a , and convergence at each point separately does not supply one. The powers of x show the gap is real. For x^n to fall below $1/2$, $x = 0.9$ needs $n = 7$, $x = 0.99$ needs $n = 69$, and $x = 0.999$ needs $n = 693$; as x approaches 1 the required stage runs off, and no single stage serves every point.

So a stronger kind of convergence is needed. A sequence of functions converges *uniformly* to f on an interval when for every tol-

erance there is a stage beyond which every function of the sequence is within the tolerance of f at every point of the interval. Convergence at each point says: for every point and every tolerance there is a stage. Uniform convergence says: for every tolerance there is a stage that serves every point. As with uniform continuity, “for every point” has moved from before the choice of the stage to after it, so the stage may no longer depend on the point.

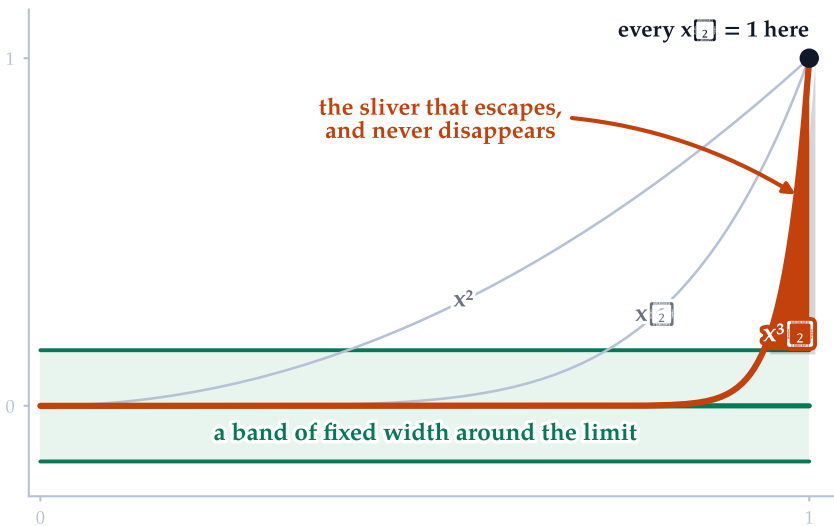


Figure 38.2. A band of fixed width drawn around the value 0, which the limit takes at every point of the interval but the right-hand end, where every power of x is 1. A sequence converges uniformly when, for every band however narrow, past some stage every function lies wholly inside it; the powers of x fail because near the right-hand end they escape it, and the escaping sliver never quite goes. Assumes: that the band has the same width everywhere; letting the width shrink near the endpoint would hide exactly the failure the picture is meant to show

With that hypothesis the theorem is true. Given a tolerance ε , uniform convergence supplies a stage beyond which every f_n is within $\varepsilon/3$ of f at every point. Fix one such n . The continuity of f_n at a supplies a window around a inside which $f_n(x)$ is within $\varepsilon/3$ of $f_n(a)$. For x in that window the three pieces are each below $\varepsilon/3$, so $f(x)$ is within ε of $f(a)$. The powers of x fail the hypothesis.

Take the tolerance $1/2$. For a fixed n , x^n is continuous and equal to 1 at $x = 1$, so it exceeds $1/2$ at every x below 1 close enough to 1. At those points the limit is 0, so x^n is more than $1/2$ away from its limit there. This happens for every n , so no stage brings the functions within $1/2$ of the limit at every point. Each fixed x below 1 does stop being such a point once n passes the stage at which x^n falls below $1/2$, the stages 7, 69 and 693 found above for 0.9, 0.99 and 0.999. So as n grows the points where x^n exceeds $1/2$ crowd toward 1, they are never absent, and they lie over the jump in the limit.

The array at the start was the same failure. Let $f_n(x) = 1/(1 + nx)$ on the interval from 0 to 1; the entry $m/(m + n)$ is $f_n(1/m)$. For each x above 0, $f_n(x)$ tends to 0 as n grows, and $f_n(0)$ is 1 for every n , so the limit function is 0 except for the value 1 at 0: a jump at 0, as the powers had one at 1. Fixing the row and letting n grow takes the limit function at the points $1/m$, and those values are all 0. Fixing the column and letting m grow follows f_n along the points $1/m$ toward 0, which by the continuity of f_n gives $f_n(0) = 1$. The two orders of limit are the two sides of the jump. The diagonal of the array shows that the convergence is not uniform: $f_n(1/n) = 1/2$ for every n , at a point where the limit is 0, so when the tolerance is below $1/2$ there is no stage beyond which every f_n is within the tolerance of the limit at every point. Away from a jump there is no trouble. For the powers, on the interval from 0 to 0.9 the single stage 66 brings x^n within $1/1000$ of 0 at every point, since 0.9^{66} is 0.00096.

Areas can fail to follow a limit as well. Let $h_n(x)$ be n for x above 0 and up to $1/n$, and 0 elsewhere on the interval from 0 to 1, so that $h_n(0)$ is 0. Each h_n is a staircase with a single step of height n and width $1/n$, so its area is 1. At each x above 0, $h_n(x)$ is 0 once n exceeds $1/x$, and $h_n(0)$ is 0 always, so the limit function is 0 everywhere, with area 0, while the areas are all 1. If functions with no negative values each have an area on an interval and converge uniformly there, does the limit have an area, and is it the limit of the areas?

* * *

The phrase “infinite sum” has been used for decimal expansions and for sums of functions without being defined. Nobody performs infinitely many additions. The finite sums can be formed, the first term, the first two, the first three, and the only definition available watches where they go:

an infinite sum is the limit of its partial sums, and where that limit does not exist, the sum does not exist.

An infinite sum is a limit written with plus signs, and most errors with series come from treating those plus signs as the familiar ones.

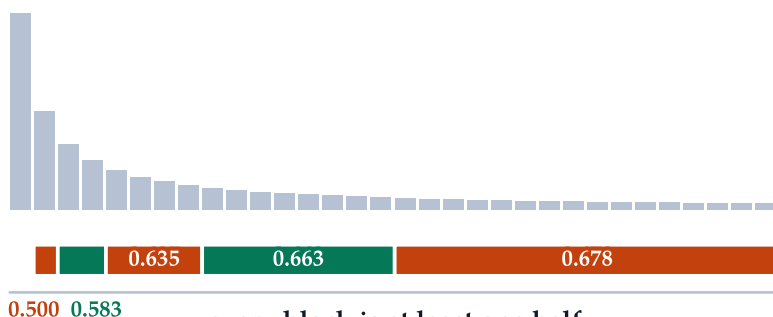
For the terms $1, 1/2, 1/4, 1/8, \dots$ the partial sums are $1, 3/2, 7/4, 15/8, \dots$, and the sum of the first n terms is $2 - 1/2^{n-1}$. The sum is 2 because the gap $1/2^{n-1}$ falls below every tolerance; no partial sum equals 2. The terms shrink too, and it is tempting to guess that a series converges whenever its terms shrink to 0.

The guess is false, and the counterexample is the sum $1 + 1/2 + 1/3 + \dots + 1/N$ that appeared in the expected waiting time for every face of a die, where a table of its values could not decide whether it grows without bound. After the first two terms, $1 + 1/2$, group the rest into blocks. The block $1/3 + 1/4$ has two terms, each at least $1/4$, so it is at least $1/2$. The block from $1/5$ to $1/8$ has four terms, each at least $1/8$, so it is at least $1/2$, and the block from $1/9$ to $1/16$ has eight terms, each at least $1/16$. In general the block from $1/(2^k + 1)$ to $1/2^{k+1}$ has 2^k terms, each at least $1/2^{k+1}$, so it is at least $1/2$. Each doubling of the number of terms therefore adds at least another $1/2$: the partial sums are at least 2 after four terms, $5/2$ after eight and 3 after sixteen. Adding $1/2$ often enough passes every bound, so the series diverges.

The divergence is slow. The sum of the first 64 terms is about 4.7439, and the first partial sum to pass 10 is the 12,367th. A series cannot converge unless its terms shrink to 0, since otherwise the

shrinking terms are necessary and are not enough

the terms shrink to nothing



so the total passes 4 by the 64th term, and never stops climbing

Figure 38.3. The reciprocals grouped into blocks of doubling length: each block sums to at least one half however far out it is taken, so the running total climbs by at least a half forever, although the individual terms are heading to zero. *Assumes:* that the blocks continue forever with the same bound, which the general argument establishes and these five blocks only illustrate; five agreeing cases are not a proof

partial sums keep jumping by at least a fixed amount. But terms that shrink to 0 are not enough to make a series converge.

* * *

Attach alternating signs to the same terms: $1 - 1/2 + 1/3 - 1/4 + \dots$. The partial sums with an even number of terms, $1/2, 7/12, 37/60, \dots$, increase, since each new pair adds a positive term and subtracts a smaller one. The partial sums with an odd number of terms, $1, 5/6, 47/60, \dots$, decrease, since each new pair subtracts a term and adds back a smaller one. Every even sum lies below every odd sum. To compare an even sum with an odd sum, go further along the series than both: by then the even sums have only risen and the odd sums have only fallen, and out there an even sum is still below the odd sum that follows it, which adds one positive term.

The n -th partial sum and the next differ by $1/(n+1)$, which falls below every tolerance. So each odd sum and the even sum after it are the ends of an interval, the intervals are nested with lengths shrinking to 0, and they close on exactly one number: the series converges. The first 1000 terms add up to 0.692647 and the first 1001 to 0.693646, and the sum lies between them.

Change nothing except the order in which the terms are used. Take two positive terms for every negative one: $1 + 1/3 - 1/2$, then $1/5 + 1/7 - 1/4$, then $1/9 + 1/11 - 1/6$, and so on. Every term of the alternating series appears exactly once. The partial sums of this arrangement tend to 1.039721, one and a half times the original sum, and the factor can be proved.

Call the original sum S . The series with every term halved, $1/2 - 1/4 + 1/6 - 1/8 + \dots$, has partial sums half the original ones, so its sum is $S/2$. Put a 0 before each of its terms, $0 + 1/2 + 0 - 1/4 + 0 + 1/6 + \dots$; this repeats each partial sum once and so leaves the limit at $S/2$. Add this series to the original term by term. The odd positions give $1 + 0, 1/3 + 0, 1/5 + 0, \dots$, and the even positions give $-1/2 + 1/2 = 0, -1/4 - 1/4 = -1/2, -1/6 + 1/6 = 0, -1/8 - 1/8 = -1/4, \dots$. Limits add, so this series has sum $S + S/2$; deleting its zero terms only deletes repeated partial sums, so the limit stays the same; and with the zeros deleted it is $1 + 1/3 - 1/2 + 1/5 + 1/7 - 1/4 + \dots$, the rearranged series. And S is not 0, since it lies between $1/2$ and 1, so $S + S/2$ is a different number from S . The same numbers, added in a different order, have a different sum.

What is the sum of $1 - 1/2 - 1/4 + 1/3 - 1/6 - 1/8 + 1/5 - 1/10 - 1/12 + \dots$, which takes one positive term and then two negative ones?

Nothing is wrong in either computation, as nothing was wrong in either pass through the array. The rearranged series has a different sequence of partial sums, and two sequences need not share a limit. So a line has to be drawn between series that tolerate reordering and series that do not. Replace every term by its size and ask whether the new series converges; if it does, the original is called *absolutely convergent*.

A convergent series of positive terms converges to the same sum

however its terms are reordered. Every partial sum of the rearrangement is a sum of finitely many of the original terms, so it lies below the original sum. The rearranged partial sums increase and stay below that bound, so they converge, to something at most the original sum. The original series is in turn a rearrangement of the new one, so the same argument with the two series exchanged gives the reverse inequality, and the two sums are equal.

Take a series with signs that converges absolutely, and split it into the series of its positive terms and the series of the sizes of its negative terms. Each of the two has increasing partial sums, and these are bounded by the sum of all the sizes, so each converges. Every partial sum of the original is a partial sum of the first minus a partial sum of the second, so the original converges to the difference of the two totals. A rearrangement reorders the positive terms among themselves and the negative terms among themselves, neither total changes, and neither does the difference. Absolutely convergent series behave like finite sums. The alternating series is not absolutely convergent, since without its signs it is the series that passes every bound, and Riemann proved in 1854 that a convergent series that is not absolutely convergent can be reordered to converge to any number c at all.

First, the positive terms of such a series have partial sums that pass every bound, and so do the sizes of its negative terms. If the positive terms had bounded partial sums they would converge; the partial sums of the sizes of the negative terms are differences between partial sums of the positive terms and of the whole series, so they would converge too, and the series would converge absolutely. The same holds with the two kinds exchanged. Second, the terms tend to 0, since the series converges.

Now take positive terms, in their original order, until the running sum first exceeds c ; then negative terms until it first drops below c ; then positive terms again, and so on. A term equal to 0 changes no running sum, and each one is placed when its turn comes in the original order. Every run ends, because the positive terms have partial sums that pass every bound and so do the sizes

of the negative terms. Each run takes at least one term of its kind, in the original order, and the runs alternate without end, so the turn of each term comes: every term is used, once.

When a run of positive terms ends, the sum exceeds c by at most the last term added, because before that term it had not yet exceeded c . Through the run of negative terms that follows, the sum only falls, so it stays below that level, and it ends below c by at most the size of the last negative term added. So from the first switch on, the running sum differs from c by no more than the size of a term that ended one of the last two runs, and those terms tend to 0: the rearranged series converges to c . For the alternating series and the target 2, the terms $1 + 1/3 + \dots + 1/15$ reach 2.021800, the first sum above 2; then $-1/2$ brings it to 1.521800; the terms from $1/17$ to $1/41$ bring it to 2.004063; and $-1/4$ brings it to 1.754063.

* * *

A series whose terms are functions is a way of building new functions, and two facts make it safe. The first is a test for uniform convergence. Suppose each function g_n is at most a number M_n in size at every point, and the series of the M_n converges. At every point x , the partial sums of the sizes $|g_n(x)|$ are bounded by the sum of the M_n , so the series of the $g_n(x)$ converges absolutely, to a value $g(x)$; and the distance from $g(x)$ to the n -th partial sum is at most the sum of the M_k past n , which does not depend on x and falls below every tolerance. So the convergence is uniform. The second fact is the theorem above: a uniform limit of continuous functions is continuous.

For a number y , write $d(y)$ for the distance from y to the nearest whole number. It is 0 at whole numbers and $1/2$ halfway between them, and in between it runs up and down in straight pieces of slope 1 and -1 , a row of tents, continuous everywhere. Let $g_n(x) = d(2^n x)/2^n$, the same row of tents shrunk by 2^n in both directions, and let

$$T(x) = g_0(x) + g_1(x) + g_2(x) + \dots$$

Every g_n is continuous and at most $1/2^{n+1}$ in size, and those bounds add up to 1, so the series converges uniformly and T is continuous everywhere. At $1/2$ the first tent gives $1/2$ and every later one sits at a whole number and gives 0, so $T(1/2) = 1/2$. At $1/4$ the first two give $1/4$ each and the rest 0, so $T(1/4) = 1/2$. At $3/8$ they give $3/8, 1/8, 1/8$ and then 0, so $T(3/8) = 5/8$. At $1/3$, no power of 2 is a multiple of 3, so every $2^n/3$ is a whole number plus $1/3$ or plus $2/3$ and lies at distance $1/3$ from the nearest whole number, so $T(1/3)$ is $1/3$ times $1 + 1/2 + 1/4 + \dots$, which is $2/3$.

What is the largest value of T ?

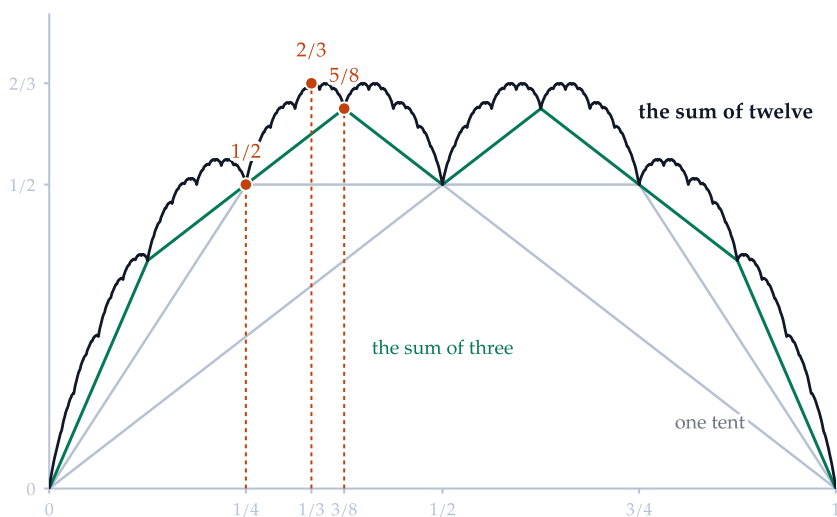


Figure 38.4. The first tent, the sum of two, the sum of three, and the sum of the first twelve: each stage adds a row of tents half the size, the stages converge uniformly, and the curve they close on is continuous everywhere and has a tangent nowhere. The marked heights are $T(1/4) = 1/2$, $T(3/8) = 5/8$ and $T(1/3) = 2/3$. Assumes: that twelve stages stand in for infinitely many; at any drawn resolution the partial sum is a polygon with corners, and the absence of a tangent at every point is a statement the text proves and no drawing can show

T has a derivative at no point. T repeats with period 1, since $2^n(x+1)$ and $2^n x$ differ by a whole number, so it is enough to take x with $0 \leq x < 1$; for each m let $a_m \leq x < b_m$ be consecutive multiples

of $1/2^m$. For n at least m , the numbers $2^n a_m$ and $2^n b_m$ are whole, so g_n is 0 at both ends. For n below m , g_n is a straight piece of slope 1 or -1 between consecutive multiples of $1/2^{n+1}$, and the interval from a_m to b_m lies inside one such piece: since $n + 1$ is at most m , every multiple of $1/2^{n+1}$ is also a multiple of $1/2^m$, so no corner of g_n falls strictly between the consecutive multiples a_m and b_m . So the change in T from a_m to b_m is the change in the first m tents alone, and the quotient

$$(T(b_m) - T(a_m)) / (b_m - a_m)$$

is a sum of m terms, each 1 or -1 . At $x = 1/3$ the first eight quotients are 1, 0, 1, 0, 1, 0, 1, 0; the first three come from the values of T above. For every x , if all m terms of the m -th quotient are 1 the sum is m , and switching a term from 1 to -1 lowers the sum by 2, so the quotient is even when m is even and odd when m is odd. Consecutive quotients are then different whole numbers, so they differ by at least 1, and the quotients do not settle.

If T had a derivative at x , they would settle. When x lies strictly inside the interval from a_m to b_m , the change in T across the interval is the change from a_m to x plus the change from x to b_m , and each change is its quotient times its length. Dividing by the whole length, the quotient across the interval is a weighted average of the quotient from a_m to x and the quotient from x to b_m , with weights the two lengths divided by the whole. Both ends are within $1/2^m$ of x , so they close in on x , and each of those two quotients tends to the derivative at x . A weighted average of two quantities that tend to one number tends to that number. When x is the left end of the interval, the quotient across it is already one of the two. Either way the quotients would settle, and they do not, so T has no derivative at x , and x was arbitrary. The graph of T is an unbroken curve with a finite slope at no point. At some points the chords do not wander: at $1/7$ the quotient across the interval from a_m to b_m is at least $m/3$, which runs off past every bound, and a chord standing ever steeper is no tangent either.

Weierstrass described a function of this kind in 1872, built from cosines, and the tents give one with less machinery, found by Takagi in 1903. It had been widely believed that a continuous function has a derivative except at isolated points, and Ampère published an attempted proof of that in 1806; the belief fails at every point of a single definite curve. The question about rates asked at the start has its answer here too. The functions $d(n^2x)/n$ are at most $1/(2n)$ in size, so they tend to 0 uniformly, and the limit, the function 0, has derivative 0 everywhere. Yet each of them is made of straight pieces of slope n or $-n$, and those slopes grow past every bound and do not tend to 0 or to any other number. The rate of change of the limit is not the limit of the rates of change. Uniform convergence delivers continuity, and areas with it, but not a derivative, and the sum of tents separates continuity from having a derivative.

THE FUNCTION THAT MAKES ITSELF

The functions met so far were given by recipes and then examined: squaring, cubing, the row of tents. Here a property of a derivative is demanded first, and the question is whether anything has it. Which function is its own derivative and takes the value 1 at 0? Any quantity that grows at a rate equal to itself obeys that demand, as a population does whose births are proportional to its size, or a debt whose interest is proportional to the debt.

No polynomial meets the demand. Differentiating a polynomial of degree d gives one of degree $d - 1$, or the zero polynomial when d is 0, so a polynomial and its derivative never have the same degree. The only polynomial equal to its own derivative is therefore the zero polynomial, and that one does not take the value 1. Whatever meets it has to be built.

At most one function meets it. If two did, their difference g would be its own derivative, with $g(0) = 0$. Take a positive a below 1. On the interval from 0 to a the size of g , which is continuous, reaches a largest value M at some point c . If $c = 0$, then $M = |g(0)| = 0$. Otherwise, by the average-slope theorem, there is a point s between 0 and c at which the derivative of g equals $(g(c) - g(0))/c$, which is $g(c)/c$ because $g(0) = 0$. The derivative of g is g , so $g(s)$

$= g(c)/c$, that is, $|g(c)| = c \cdot |g(s)|$. Now $|g(c)|$ is M , the point c is at most a , and $|g(s)|$ is at most M , since M is the largest size anywhere on the interval. So $M \leq aM$. With a below 1 and M not negative, that can hold only if $M = 0$. Hence g vanishes from 0 to a . In particular $g(a) = 0$, and that was all the argument used about the starting point; so started at a it shows that g vanishes from a to $2a$, then from $2a$ to $3a$, and so on, and mirrored it does the same to the left of 0. The vanishing is carried along the whole line in steps of length a , and the two functions were one function.

* * *

The function arrives from area. The curve $1/t$ decreases for t above 0, so the region under it between any two positive points has an area. Let $L(x)$ be the area under $1/t$ from 1 to x when x is at least 1, and minus the area from x to 1 when x lies between 0 and 1. Then $L(1) = 0$ and L increases.

L turns multiplication into addition. Stretch the region under $1/t$ from 1 to b horizontally by a factor a , so that the point t goes to at , and squash it vertically by the same factor, so that a height h goes to h/a . The point $(t, 1/t)$ goes to $(at, 1/(at))$, which is on the same curve, so the region goes onto the region under $1/t$ from a to ab . A rectangle of width w and height h goes to a rectangle of width aw and height h/a , of the same area, so every staircase inside the first region goes to a staircase of the same area inside the second, and every staircase containing the first goes to one of the same area containing the second. The brackets are the same, and the two regions have the same area. The region from 1 to ab is the region from 1 to a with the region from a to ab beside it, so

$$L(ab) = L(a) + L(b)$$

for a and b at least 1. The same stretch by a carries the region from $1/a$ to 1 onto the region from 1 to a , and $L(1/a)$ is by definition minus the area of the first, so $L(1/a) = -L(a)$. Writing any factor below 1 as the reciprocal of one above 1 then extends the rule to all positive

a and b . Take a at least 1, b below 1 and ab at least 1, for instance: a is the product of ab and $1/b$, both at least 1, so $L(a) = L(ab) + L(1/b) = L(ab) - L(b)$, which is the rule again. The other cases go the same way. Taking $a = b = 2$ again and again gives $L(2^n) = n \cdot L(2)$. The region under $1/t$ from 1 to 2 contains the rectangle of width 1 and height $1/2$ and lies inside the rectangle of width 1 and height 1, so $L(2)$ is between $1/2$ and 1; with four strips the brackets are 0.6345 and 0.7595. So $L(2^n)$ is at least $n/2$, L passes every bound, and by $L(1/a) = -L(a)$ it goes below every bound as well.

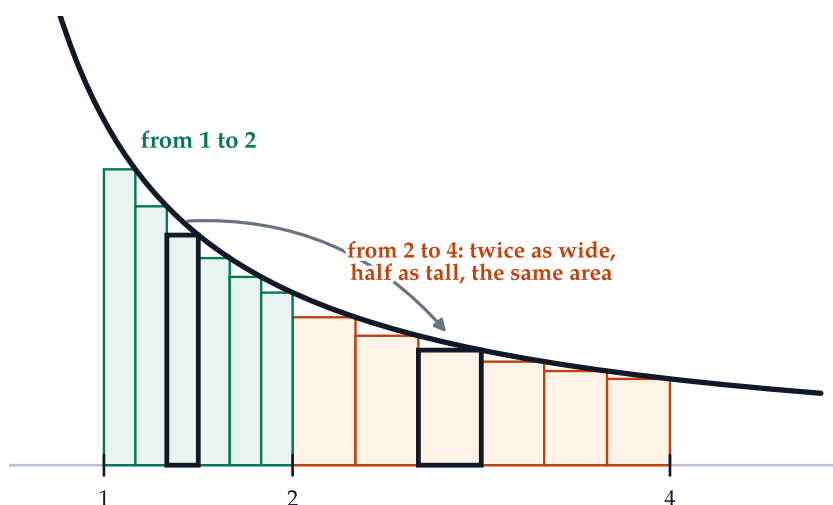


Figure 39.1. The region under the curve $1/t$ from 1 to 2, and the region from 2 to 4. Stretching the first horizontally by 2 and squashing it vertically by 2 carries it exactly onto the second, and carries each inscribed rectangle to a rectangle of the same area, so the two regions have equal area and the area from 1 to 4 is twice the area from 1 to 2. *Assumes:* that a region's area is fixed by its brackets, which the nested-interval axiom guarantees; and that stretching by a and squashing by a sends the curve onto itself, which is the one fact about $1/t$ the argument uses

L has a derivative that can be computed exactly. On every interval from c to d with c above 0 the curve $1/t$ falls, and $L(x)$ is $L(c)$ plus the area under $1/t$ from c to x . For a falling curve, the rate at which the area under it grows is the height of the curve. So L has

derivative $1/x$ at every positive x . In particular L is continuous, as every function with a derivative is.

L is the logarithm.

* * *

L can be run backwards. Take any number y . Since L passes every bound in both directions, some value of L is below y and some value is above it; L is continuous, and a continuous function takes every value between two of its values, so $L(x) = y$ for some positive x . Since L increases, no second x gives the same value. So for every number y there is exactly one positive x with $L(x) = y$. Call it $E(y)$. Then $E(0) = 1$, E increases, and the rule for L read backwards gives

$$E(y + z) = E(y) \cdot E(z)$$

since $L(E(y) \cdot E(z)) = L(E(y)) + L(E(z)) = y + z$, which makes $E(y) \cdot E(z)$ the one positive number whose L is $y + z$, and that number is $E(y + z)$. E is the inverse of L , so it is continuous, and the rule for a function run backwards gives its derivative: at $y = L(x)$ it is the reciprocal of the derivative of L at x . That derivative is $1/x$. So the derivative of E at y is x , which is $E(y)$. The demand is met, by a function built from an area, and by uniqueness nothing else meets it. The number $E(1)$, the one number whose logarithm is 1, is called e . It is above 2: $L(2)$ is below 1, which is $L(e)$, and L increases.

Is $E(x)$ at least $1 + x$ for every number x ?

* * *

A formula for E follows from the demand, by a lemma. Suppose $g(0) = 0$ and the derivative of g has size at most $M \cdot t^k / k!$ at every point t between 0 and x . Then $|g(x)|$ is at most $M \cdot x^{k+1} / (k + 1)!$. To see it, let $D(t) = M \cdot t^{k+1} / (k + 1)! - g(t)$. By the rules for powers, sums and multiples, the derivative of D is $M \cdot t^k / k!$ minus the

derivative of g , and by the supposition that difference is never negative between 0 and x . So by the average-slope theorem D never decreases there. Since $D(0) = 0$, $D(x)$ is at least 0, which says that $g(x)$ is at most $M \cdot x^{k+1}/(k+1)!$. The same with $-g$ in place of g shows that $-g(x)$ is at most that as well, and the two together bound $|g(x)|$.

Let $R_n(x)$ be $E(x)$ minus the whole sum $1 + x + x^2/2! + \dots + x^n/n!$, for x at least 0. Differentiating turns each term of the sum into the term before it, $x^2/2!$ into x , $x^3/3!$ into $x^2/2!$, and the 1 into 0, while E is its own derivative. So the derivative of R_n is R_{n-1} , and $R_n(0) = 0$. The derivative of R_0 is $E(t)$, which is at most $E(x)$ between 0 and x because E increases, so the lemma, with $M = E(x)$ and $k = 0$, gives $|R_0(x)| \leq E(x) \cdot x$. The same reasoning stopped at any t between 0 and x gives $|R_0(t)| \leq E(x) \cdot t$. Then the derivative of R_1 is R_0 , of size at most $E(x) \cdot t$, so the lemma with $k = 1$ gives $|R_1(x)| \leq E(x) \cdot x^2/2!$; and step by step, $|R_n(x)| \leq E(x) \cdot x^{n+1}/(n+1)!$. That bound tends to 0 as n grows, for every x , since each $x^{n+1}/(n+1)!$ is the one before times $x/(n+1)$, and once n exceeds $2x$ the factor is at most a half. So

$$E(x) = 1 + x + x^2/2! + x^3/3! + \dots$$

for every x at least 0. To the left of 0, first $E(y) \cdot E(-y) = E(0) = 1$, so $E(-y) = 1/E(y)$. The lemma holds to the left of 0 too, with $|t|$ and $|x|$ in place of t and x : the function sending t to $g(-t)$ is 0 at 0 and its derivative at t has the same size as the derivative of g at $-t$, so the lemma applied to it is the mirrored statement. Between a negative x and 0 the values of E are at most 1, since E increases and $E(0) = 1$. So the same steps, with 1 in place of $E(x)$, give $|R_n(x)|$ at most $|x|^{n+1}/(n+1)!$, which tends to 0, and the series holds for negative x as well. At $x = -1$ the terms through $1/9!$ add up to 0.367879, the reciprocal of 2.718282. At $x = 1$ the partial sums are 1, 2, $5/2$, $8/3$, $65/24$, $163/60$, ..., eight terms give 2.718254, and eleven give 2.718282, correct to six places; the remainder after n terms is at most $e/n!$. And e is below 3, since $n!$ is at least 2^{n-1} , so the series for e is at most $1 + 1 + 1/2 + 1/4 + \dots$, which is 3.

Multiply the partial sums of the series for $E(y)$ and $E(z)$ and collect the terms of equal total degree. Which series do the collected terms form, and why is its sum the product of the two sums?

The same lemma identifies the logarithm of 2. Let $R_n(x)$ be $L(1+x)$ minus the whole sum $x - x^2/2 + x^3/3 - \dots \pm x^n/n$, for x between 0 and 1. Its derivative is $1/(1+x)$ minus the finite sum $1 - x + x^2 - \dots \pm x^{n-1}$. Multiplying that finite sum by $1+x$ cancels each term against its neighbour and leaves $1 \pm x^n$, so the finite sum is $(1 \pm x^n)/(1+x)$. Taking it from $1/(1+x)$ leaves $\pm x^n/(1+x)$, and since $1+t$ is at least 1 that has size at most t^n at each t . The lemma, with $k = n$ and $M = n!$, so that $M \cdot t^n/n!$ is t^n , gives $|R_n(x)| \leq n! \cdot x^{n+1}/(n+1)!$, which is $x^{n+1}/(n+1)$; at $x = 1$ that is $1/(n+1)$ and tends to 0. So $L(2)$ is the sum of the alternating series $1 - 1/2 + 1/3 - \dots$

That series is a poor way to compute $L(2)$: its error after n terms is at most $1/(n+1)$, and after n or after $n+1$ terms it is at least half of that, since the sum lies between those two partial sums, which differ by $1/(n+1)$. The error settles at about $1/(2n)$, so six decimal places would need a million terms. A faster one comes from the other side of 1. For x between 0 and $1/2$ let $R_n(x)$ be $L(1-x)$ plus $x + x^2/2 + \dots + x^n/n$. The derivative of $L(1-x)$ is $-1/(1-x)$: when x moves by h the point $1-x$ moves by $-h$, so the difference quotient of $L(1-x)$ is minus the difference quotient of L at $1-x$, and that one tends to $1/(1-x)$. The derivative of the finite sum is $1 + x + \dots + x^{n-1}$, which times $1-x$ is $1 - x^n$, so it equals $(1 - x^n)/(1-x)$. Adding the two, the derivative of R_n is $-x^n/(1-x)$. For t up to $1/2$ the divisor $1-t$ is at least $1/2$, so the size is at most $2t^n$, and the lemma, with $k = n$ and $M = 2 \cdot n!$, gives $|R_n(x)| \leq 2x^{n+1}/(n+1)$. At $x = 1/2$, where $L(1/2) = -L(2)$,

$$L(2) = 1/2 + 1/(2 \cdot 2^2) + 1/(3 \cdot 2^3) + 1/(4 \cdot 2^4) + \dots$$

with an error after n terms of at most $1/((n+1)2^n)$. Nineteen terms give 0.6931471, the error being below 1 in the seventh place. At $x = 1/5$ the same series gives $L(5/4) = 1/5 + 1/(2 \cdot 5^2) + 1/(3 \cdot$

$5^3) + \dots$, and eleven terms give 0.2231436. Since $10 = 2^3 \cdot 5/4$, the rule for products gives $L(10) = 3L(2) + L(5/4) = 2.302585$.

The logarithm also measures the series that passes every bound. Write H_n for $1 + 1/2 + \dots + 1/n$. On the stretch from k to $k + 1$ the curve $1/t$ lies between $1/(k + 1)$ and $1/k$, so $1/(k + 1) \leq L(k + 1) - L(k) \leq 1/k$. Add the right-hand inequalities for k from 1 to n : the differences of L add to $L(n + 1) - L(1)$, which is $L(n + 1)$, and the right sides add to H_n . Add the left-hand ones for k from 1 to $n - 1$: the left sides add to $H_n - 1$ and the differences add to $L(n)$. Together,

$$L(n + 1) \leq H_n \leq 1 + L(n)$$

so the harmonic sums are staircases under and over the curve $1/t$, and they grow as the logarithm does. More is true. The difference $H_n - L(n)$ decreases as n grows, since from n to $n + 1$ it changes by $1/(n + 1) - (L(n + 1) - L(n))$, which is at most 0; the difference $H_n - L(n + 1)$ increases, since from n to $n + 1$ it changes by $1/(n + 1) - (L(n + 2) - L(n + 1))$, which is at least 0; and the second is below the first, by $L(n + 1) - L(n)$, which is at most $1/n$ and so falls below every tolerance. So both approach one number, written γ , which lies between them at every stage. At $n = 64$, where H_n is 4.743891 and $L(64) = 6L(2) = 4.158883$, the first difference is 0.585008, and the second is at least the first less $1/64$, which is 0.569383; so γ lies between 0.569383 and 0.585008.

How many terms does the series need to pass 100? Since H_n lies between $L(n) + \gamma$ and $L(n + 1) + \gamma$, the first n with H_n above 100 is within 1 of $E(100 - \gamma)$. With $L(10) = 2.302585$, the number $100 - \gamma$ is between 43.175 and 43.183 times $L(10)$. Now $E(43 \cdot L(10))$ is 10^{43} , since $L(10^{43}) = 43 \cdot L(10)$. So by the rule for E , $E(100 - \gamma)$ is 10^{43} times $E(s \cdot L(10))$ for some s between 0.175 and 0.183, and that second factor lies between 1.49 and 1.53. The harmonic sums pass 100 only after about 1.5×10^{43} terms, a number forty-four digits long, which no table could have reached.

Rearrange the alternating series to take p positive terms for every q negative ones. What is its sum, and which rearrangement of

this kind has sum 0?

* * *

A second route reaches e from lending money. A sum lent at a hundred percent interest for a year doubles. Paid in two instalments of fifty percent, it grows by a factor $3/2$ twice, to 2.25. In ten instalments the factor is 2.593742, in a hundred 2.704814, in a thousand 2.716924, and in a million 2.718280. The factors climb slowly toward the number the series reached quickly.

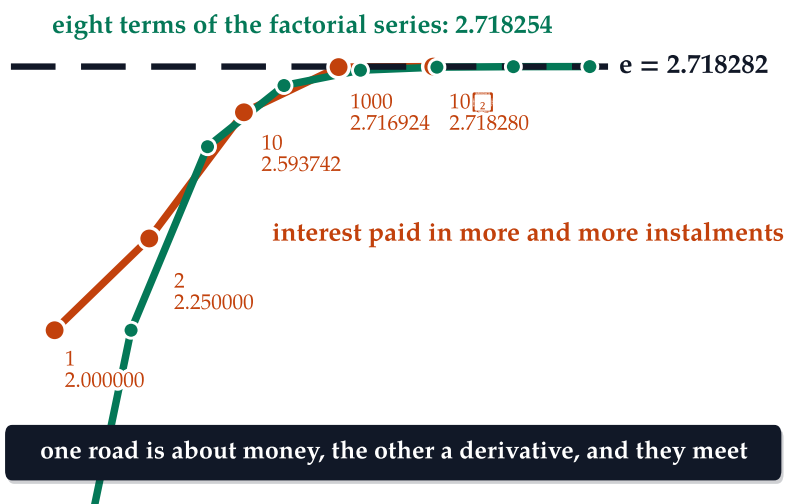


Figure 39.2. Compounding more often climbs slowly toward the same number the factorial series reaches almost at once: at a thousand instalments the interest route has 2.716924 while eight terms of the series already give 2.718254. Assumes: that both routes converge to one number rather than to two very close ones, which the argument in the text establishes and no finite table can

The logarithm of $(1 + 1/n)^n$ is $n \cdot L(1 + 1/n)$, by the rule for L used n times. And $L(1 + h)/h$ is $(L(1 + h) - L(1))/h$, which tends to the derivative of L at 1, which is 1. With $h = 1/n$ the quotient $L(1 + h)/h$ is $n \cdot L(1 + 1/n)$, and h tends to 0 as n grows. So $n \cdot L(1 + 1/n)$ tends to 1, and since E is continuous, $(1 + 1/n)^n =$

$E(n \cdot L(1 + 1/n))$ tends to $E(1)$, which is e .

Is every $(1 + 1/n)^{n+1}$ above e , and how far below e can a million instalments leave the factor?

* * *

Is e a fraction? Suppose $e = p/q$ with whole numbers p and q . Multiply the series by $q!$. Every term up to $1/q!$ becomes a whole number, since $q!$ is a multiple of each of those denominators, and $q! \cdot e = q! \cdot p/q$ is a whole number too. Now $q! \cdot e$ is $q!$ times the terms up to $1/q!$ plus $q!$ times the tail $1/(q+1)! + 1/(q+2)! + \dots$. So $q!$ times the tail is the difference of two whole numbers, and is a whole number. It is positive. And it is small. Each term of the tail is at most the one before divided by $q+2$, so the tail is at most $1/(q+1)!$ times the geometric series $1 + 1/(q+2) + 1/(q+2)^2 + \dots$. That series has sum $(q+2)/(q+1)$, and $q!/(q+1)!$ is $1/(q+1)$. So $q!$ times the tail is at most $(q+2)/(q+1)^2$, which is $3/4$ when $q=1$ and smaller for larger q . No whole number lies strictly between 0 and $3/4$, so e is not a fraction. For $q=3$, three factorial times the first four terms $1 + 1 + 1/2 + 1/6$ is exactly 16, and three factorial times the rest is about 0.3097.

The argument turns on how fast the tail shrinks, and factorial denominators are what make it shrink fast enough. For the series $1 + 1/2 + 1/4 + \dots$, whose sum is the whole number 2, the matching step multiplies by 2^q , which turns the terms up to $1/2^q$ into whole numbers; but 2^q times the tail $1/2^{q+1} + 1/2^{q+2} + \dots$ is exactly 1, not below 1, and the argument proves nothing.

* * *

The expansion of a function around a point, with its remainder at an unknown place, was proved for finitely many terms. Letting the terms run on forever raises two separate questions, and one function shows how separate they are. The *series* of a function g at 0, when g has derivatives of every order there, is the series whose

n -th coefficient is the n -th derivative of g at 0 divided by $n!$. For a polynomial it returns the polynomial, since the n -th derivative of x^k at 0 is $n!$ when $k = n$ and 0 otherwise; for E every derivative at 0 is 1, and the series is $1 + x + x^2/2! + \dots$, the one found above.

A function can have derivatives of every order at a point and still differ from its series there. Let $F(x) = E(-1/x^2)$ when x is not 0, and $F(0) = 0$. Away from 0, F is positive: $F(1/2) = E(-4)$, about 0.0183. Near 0 it is flatter than any polynomial: $F(1/10) = E(-100)$, about 3.7×10^{-44} , and $F(1/20) = E(-400)$ is below 10^{-173} .

At 0 the difference quotient is $(F(h) - F(0))/h$, which is $F(h)/h$. Write $y = 1/h^2$. Then $F(h) = E(-y) = 1/E(y)$, and $1/|h| = \sqrt{y}$, so the size of the quotient is $\sqrt{y}/E(y)$. The series shows $E(y) \geq y^2/2$ for positive y , all its terms being positive, so the quotient is at most $\sqrt{y}$ divided by $y^2/2$, which is $2/y^{3/2}$. As h tends to 0, y passes every bound, and $2/y^{3/2}$ tends to 0. So F has derivative 0 at 0. Away from 0, the rules for products and for functions of functions give the derivative $2/x^3 \cdot E(-1/x^2)$ and the second derivative $(4/x^6 - 6/x^4) \cdot E(-1/x^2)$, and each further derivative is again a polynomial in $1/x$ times $E(-1/x^2)$. The quotient argument, with the bound $E(y) \geq y^m/m!$ for an m large enough to beat the power of y in the numerator, shows that each of those derivatives has derivative 0 at 0 as well. So every derivative of F at 0 is 0.

The series that F 's derivatives at 0 produce therefore has every coefficient 0. It converges everywhere, to the function 0, and F is 0 only at the origin.

For E the lemma showed that the series converges to E , with an explicit remainder. Replacing a function by its series needs two things: that the series converge, which depends on the coefficients, and that the remainder after n terms tend to 0, which is what makes it converge to the function. The logarithm fails the first once x is past 1. The n -th derivative of $L(1+x)$ is $(n-1)/(1+x)^n$ with alternating signs, so its series at 0 is $x - x^2/2 + x^3/3 - \dots$, which the lemma showed equal to $L(1+x)$ from 0 to 1. At $x = 2$ each term $2^n/n$ is the one before times $2(n-1)/n$, at least 1 from the second term on, so the terms do not shrink and the series diverges: its

The function that makes itself

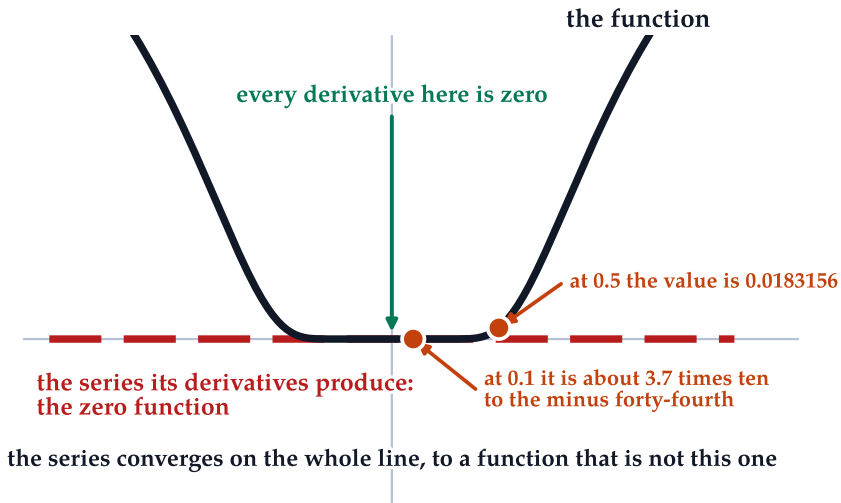


Figure 39.3. The function that is zero at the origin and the exponential of minus one over x squared elsewhere, drawn against the series its own derivatives produce: the series is the zero function, it converges everywhere, and it is wrong everywhere but one point. *Assumes:* that the curve really is flat to infinite order at the origin rather than only very flat; the drawing cannot distinguish those, and the whole content of the example is which one is true

partial sums are 2, 0, 2.667, -1.333 , 5.067, while $L(3) = 1.098612$ exists, with derivatives of every order there. The lemma's bound $x^{n+1}/(n+1)$ shrinks only when x is at most 1. F meets the first and fails the second at every point but one. Its derivatives at 0 describe what happens at 0 and nothing else. The tents had no derivative anywhere; F has derivatives of every order everywhere and is still not what they say.

THE CIRCLE AND THE EXPONENTIAL

Two laws have the same shape. On the line, $E(y) \cdot E(z) = E(y + z)$. On the unit circle, the point at angle α times the point at angle β is the point at angle $\alpha + \beta$, which followed from the addition formulas for the sine and cosine. To compare them, the angle has to be a number, and not a number in a unit chosen at pleasure: degrees, three hundred and sixty to a turn, would carry that arbitrary choice into every formula. An angle, being a proportion of a turn, has a measure of its own.

The measure comes from area. The unit disc has an area. Its upper right quarter is the region under the curve that gives the height of the circle above each point of the base. That curve decreases from 1 to 0 as the base runs from 0 to 1, and every decreasing curve has an area under it. With a thousand strips the brackets trap the quarter between 0.78488 and 0.78589, so the disc's area lies between 3.1395 and 3.1436. That area is the number π . An angle at the centre cuts a sector out of the disc, and the *radian measure* of the angle is twice the area of its sector. The same angle can sit anywhere round the centre, opening either way, and its measure must not depend on where it sits. So the definition uses the fact that a sector turned about the centre, or mirrored in a line through

it, keeps its area. The staircase definition of area, built from upright strips, does not show that; for that step area is taken in its everyday sense. A full turn cuts the whole disc and has measure 2π , a half turn has measure π , and a right angle $\pi/2$. Two angles set side by side cut two sectors that do not overlap, so the measure of the combined angle is the sum of the two measures.

Every number from 0 to 2π is the measure of some angle. For a point (x, y) on the upper half of the circle, the sector from $(1, 0)$ to (x, y) is the region under the circle from x to 1 together with the triangle with corners O , $(x, 0)$ and (x, y) when x is at least 0, and that region with the triangle removed when x is negative; either way its area is the area under the circle from x to 1 plus $xy/2$.

The area of the sector changes continuously as x moves, and its two parts can be taken one at a time. The area under the circle changes by at most the distance x moves, since the curve has height at most 1. For the part $xy/2$, one inequality about square roots is needed: for a and b at least 0, $|\sqrt{a} - \sqrt{b}|$ is at most $\sqrt{|a - b|}$. The reason is that $|\sqrt{a} - \sqrt{b}|$ is at most $\sqrt{a} + \sqrt{b}$, so $(\sqrt{a} - \sqrt{b})^2$ is at most $|\sqrt{a} - \sqrt{b}| \cdot (\sqrt{a} + \sqrt{b})$, which is $|a - b|$.

Now let x move by k . A difference of two squares is the difference of the two numbers times their sum, and here the sum is at most 2 in size, so x^2 moves by at most $2k$. Then $1 - x^2$ moves by at most $2k$, and by the inequality the height $y = \sqrt{1 - x^2}$ moves by at most $\sqrt{(2k)}$. In the product xy each factor is at most 1 in size, so xy moves by at most the movement of x plus the movement of y , which is $k + \sqrt{(2k)}$, and that is as small as desired once k is small enough. So the area of the sector is a continuous function of x . It is 0 at $x = 1$ and $\pi/2$ at $x = -1$, where the sector is the half disc, so it takes every value between, and the measure takes every value from 0 to π . Mirroring in the horizontal axis carries each such sector to a sector below the axis, of the same area, turned from $(1, 0)$ the other way. The rest of the disc is then the sector turned from $(1, 0)$ in the first direction round to the mirrored point, and its measure is 2π minus a value from 0 to π , so the values from π to 2π occur as well.

And no number is the measure of two different angles. Take two different rays, both turned from the ray through $(1, 0)$ in the same direction. The sector of the one turned further is the sector of the other with the sector between the two rays set beside it, so it is enough that the sector between the rays has positive area. When the turn between the rays is less than a half turn, take the triangle with one corner at O and the other two where the rays meet the circle. Its three corners are not in one line, because the rays are different and do not point in opposite directions, so the triangle has positive area; and it lies inside the sector, since the chord between the two points on the circle lies inside the disc. When the turn is a half turn or more, the sector between the rays contains a half disc, of area $\pi/2$. So the two measures differ. A ray turned through k whole turns and then through an angle of measure s has measure $2\pi k + s$, and a ray turned the other way has the negative measure; measures of angles set side by side still add, since whole turns add.

* * *

The sine and the cosine have derivatives, and with angles in radian measure those derivatives carry no stray factor. Both facts come from one squeeze. Let h be a small positive angle, and take three regions sharing the corner O and the point $A = (1, 0)$. The triangle with third corner $P = (\cos h, \sin h)$ lies inside the sector of angle h , since the chord from A to P lies inside the disc. The sector lies inside the triangle with third corner $T = (1, \tan h)$. Here $\tan h$, the sine of h over its cosine, is the height at which the ray at angle h meets the vertical line through A . To see the containment, take any point of the sector. It lies on a ray at some angle t at most h , at distance at most 1 from O . That ray does not reach the vertical line through A until distance $1/\cos t$, which is at least 1. So the point comes at or before the line, on the stretch of its ray that lies in the triangle. So

$$(1/2) \sin h \leq h/2 \leq (1/2) \tan h$$

the first triangle having base 1 and height $\sin h$, the sector having area $h/2$ by the definition of the measure, and the second triangle having base 1 and height $\tan h$. Dividing by $(1/2) \sin h$, which is positive, gives $1 \leq h/(\sin h) \leq 1/(\cos h)$. Taking one over each of three positive numbers reverses their order, so $\cos h \leq (\sin h)/h \leq 1$. And $\cos h$ tends to 1. The addition formula for the cosine with both angles equal to $h/2$ gives $\cos h = \cos^2(h/2) - \sin^2(h/2)$, which is $1 - 2 \sin^2(h/2)$, so $1 - \cos h = 2 \sin^2(h/2)$. The left-hand inequality of the squeeze says $\sin h \leq h$; applied to $h/2$ it makes $2 \sin^2(h/2)$ at most $2(h/2)^2 = h^2/2$, which tends to 0. So $(\sin h)/h$, caught between $\cos h$ and 1, tends to 1. And $(1 - \cos h)/h$ is at most $(h^2/2)/h = h/2$ in size, so it tends to 0. A negative h changes neither conclusion, since $\sin(-h) = -\sin h$ and $\cos(-h) = \cos h$. At $h = 0.1$ the quotient $(\sin h)/h$ is 0.99833 and $\cos h$ is 0.99500; at $h = 0.01$ the quotient is 0.999983. Had angles been measured in degrees, the quotient would tend to $\pi/180$, and that factor would appear in every formula that follows.

The squeeze holds for every angle h between 0 and a right angle, and it computes π . Put $h = \pi/n$ and multiply through by $2n$: $n \sin(\pi/n) \leq \pi \leq n \tan(\pi/n)$. The left side, s_n , is half the perimeter of the regular polygon with n corners on the unit circle, and the right side, t_n , is half the perimeter of the regular polygon whose n sides touch it. Going from n to $2n$ needs no new sine to be found, only the addition formulas, which turn into two rules of arithmetic. Write $\theta = \pi/(2n)$, so that $s_n = n \sin 2\theta$ and $t_n = n \tan 2\theta$, while $s_{2n} = 2n \sin \theta$ and $t_{2n} = 2n \tan \theta$. With both angles equal to θ , the addition formulas give $\sin 2\theta = 2 \sin \theta \cos \theta$ and $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$, which is $2\cos^2 \theta - 1$. For the first rule, $1/s_n + 1/t_n$ is $(1 + \cos 2\theta)/(n \sin 2\theta)$, which is $2\cos^2 \theta/(2n \sin \theta \cos \theta)$, that is, $\cos \theta/(n \sin \theta)$; and $\cos \theta/(n \sin \theta)$ is $2/t_{2n}$. So $2/t_{2n} = 1/s_n + 1/t_n$. For the second rule, $s_n \cdot t_{2n}$ is $2n \sin \theta \cos \theta$ times $2n \sin \theta/\cos \theta$, which is $4n^2 \sin^2 \theta$, the square of s_{2n} . So $s_{2n}^2 = s_n \cdot t_{2n}$. From the hexagon, where $s_6 = 3$ and $t_6 = 2\sqrt{3}$ because $\sin 30^\circ = 1/2$ and $\cos 30^\circ = \sqrt{3}/2$, four doublings give

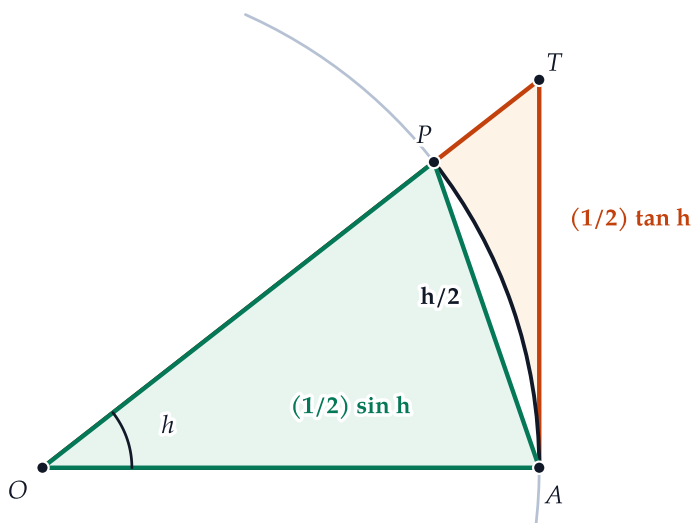


Figure 40.1. A sector of the unit disc with angle h , drawn large so that the three regions can be told apart: the triangle with corners O, A, P lies inside it and has area half the sine of h ; the sector itself has area $h/2$ by the definition of the radian; the triangle with corners O, A, T contains it and has area half the tangent of h . Three regions each inside the next, so $(\sin h)/h$ is squeezed between $\cos h$ and 1. *Assumes:* that a region inside another has at most its area, which is the first two sentences about area; and that the containments hold for every angle below a right angle, which the text argues and one drawn angle cannot show

12	3.1058285	3.2153903
24	3.1326286	3.1596599
48	3.1393502	3.1460862
96	3.1410320	3.1427146

so π lies between 3.1410 and 3.1428, with no strips under a curve. Archimedes stopped at the same polygons of 96 sides and stated the bounds $3 \frac{10}{71}$ and $\frac{22}{7}$, which are 3.1408 and 3.1429.

By the addition formula for the sine,

$$(\sin(x + h) - \sin x)/h = \sin x \cdot (\cos h - 1)/h + \cos x \cdot (\sin h)/h$$

and as h shrinks the first term tends to 0 and the second to $\cos x$, so the derivative of the sine is the cosine. By the addition formula for the cosine, $(\cos(x+h) - \cos x)/h$ is $\cos x \cdot (\cos h - 1)/h - \sin x \cdot (\sin h)/h$, which tends to $-\sin x$, so the derivative of the cosine is minus the sine.

A derivative at every point need not behave well. Let $f(x) = x^2 \sin(1/x)$ when x is not 0, and $f(0) = 0$. Away from 0 the rules for products and for functions of functions give the derivative $2x \sin(1/x) - \cos(1/x)$. At 0 the difference quotient is $h \sin(1/h)$, which lies between $-h$ and h and so tends to 0; the derivative at 0 is 0. As x approaches 0 the term $2x \sin(1/x)$ tends to 0, so everything depends on the term $-\cos(1/x)$. At the points $1/(k\pi)$, which crowd toward 0, $\sin(1/x)$ is $\sin(k\pi) = 0$, so the derivative is $-\cos(k\pi)$, which is 1 and -1 in turn. So f has a derivative at every point, and the derivative does not settle as x approaches 0. The average-slope theorem applies to f on every interval, since it asks for a derivative at every interior point and not for a continuous one.

The series for both follow from the lemma that produced the series for E , since every derivative of either function is one of $\sin$, $\cos$, $-\sin$ and $-\cos$, all of size at most 1. For x at least 0, $\sin x$ has derivative $\cos x$, of size at most 1, so $|\sin x| \leq x$; then $1 - \cos x$ has derivative $\sin x$, of size at most x , so $|1 - \cos x| \leq x^2/2$; then $x - \sin x$ has derivative $1 - \cos x$, so $|x - \sin x| \leq x^3/6$; then $\cos x - 1 + x^2/2$ has derivative $x - \sin x$, so its size is at most $x^4/24$; and $\sin x - x + x^3/6$ has derivative $\cos x - 1 + x^2/2$, so its size is at most $x^5/120$. At $x = 1$, $x - x^3/6$ is 0.83333 against $\sin 1 = 0.84147$, a shortfall of 0.00814, inside $1/120$. Continuing,

$$\sin x = x - x^3/3! + x^5/5! - \dots \quad \cos x = 1 - x^2/2! + x^4/4! - \dots$$

for every x , the remainder after the term in x^n being at most $|x|^{n+1}/(n+1)!$ in size. To the left of 0 nothing has to be redone. On one side, $\sin(-x)$ is $-\sin x$ and $\cos(-x)$ is $\cos x$. On the other, the first series has only odd powers of x , so every partial sum changes sign when x does, and the second has only even powers,

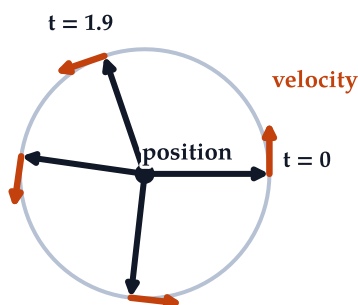
so every partial sum stays the same. So each remainder at $-x$ is the remainder at x with at most a change of sign.

Using the series for the cosine and its remainder, between which two consecutive thousandths does π lie?

* * *

Let a point move, with position at time t the point of the unit circle at angle t , which is $\cos t + i \sin t$. Its velocity is the rate of change of each coordinate, $-\sin t + i \cos t$, and that is i times the position, since $i \cdot (\cos t + i \sin t) = -\sin t + i \cos t$. Multiplying by i is a quarter turn, so at every instant the velocity is the position turned through a right angle: it is perpendicular to the position and has the same length, 1. The point's distance from the centre stays the same, and the derivative of $\cos^2 t + \sin^2 t$, which is $2\cos t \cdot (-\sin t) + 2\sin t \cdot \cos t = 0$, says so.

velocity = position turned a right angle



perpendicular, so the length never changes: speed 1, radius 1, angle = time
 at $t = 1.9$ the sector has area 0.95 and the angle is 1.9

Figure 40.2. At four instants, the position arrow from the origin and the velocity arrow at its tip: the velocity is always the position rotated a quarter turn, so it can never lengthen or shorten the position, and the point runs the unit circle at unit speed. *Assumes:* that the perpendicularity holds at every instant, not only the four drawn; the whole derivation is a claim about all of them at once, and four arrows can only ever illustrate it

Compare the demand that produced E . By the rule for a function of a function, $E(ct)$, for a fixed real number c , has derivative $c \cdot E(ct)$ and value 1 at 0. Nothing else does, and the uniqueness argument used for E shows it, with the size of c carried along. The difference g of two functions that meet the demand has derivative $c \cdot g$ and $g(0) = 0$, so it is enough to show that such a g is 0 everywhere.

Take a positive a with $a|c|$ below 1 and run the argument used for E . The size of g , which is continuous, reaches a largest value M on the interval from 0 to a , at some point p ; if p is 0 then $M = |g(0)| = 0$, and otherwise the average-slope theorem gives a point q between 0 and p with $g(p) = p \cdot c \cdot g(q)$, so $M = p|c| \cdot |g(q)|$, which is at most $a|c|M$, and M is 0 again because $a|c|$ is below 1. So g is 0 from 0 to a , and, started again at a , from a to $2a$, and so the vanishing is carried along the line in steps of a . Applied to the function sending t to $g(-t)$, whose derivative is $-c$ times itself and whose multiple $-c$ has the same size as c , the same argument carries it to the left of 0. The moving point has velocity i times its position and starts at 1. So it meets the same demand with $c = i$, and if E of an imaginary number is to mean anything consistent with the demand that defines E , $E(it)$ has to be the moving point.

Only one motion meets that demand. The argument for a real c cannot be reused as it stands, because the average-slope theorem concerns real-valued functions and the moving point is not one; a different argument does the work. Suppose two motions have velocity i times position and both start at 1, and let $d = u + iv$ be their difference, which has velocity i times d and starts at 0. Since $i \cdot (u + iv) = -v + iu$, the velocity condition says that the rate of u is $-v$ and the rate of v is u , so the rate of $u^2 + v^2$ is $2u \cdot (-v) + 2v \cdot u = 0$. The size of d stays constant, it starts at 0, and so d is 0 at every time. The motion with velocity i times position starting at 1 is the circular one.

So $E(it) = \cos t + i \sin t$ is forced by the demand, and the series for E gives the same answer independently. Put the imaginary number $i \cdot t$ into the series. The powers of i cycle through 1, i , -1 , $-i$, so the terms with even powers of t are real with alternating

signs, and the terms with odd powers carry a factor i and alternating signs. The real terms are $1 - t^2/2! + t^4/4! - \dots$, the series for the cosine, and the imaginary terms are i times $t - t^3/3! + \dots$, the series for the sine.

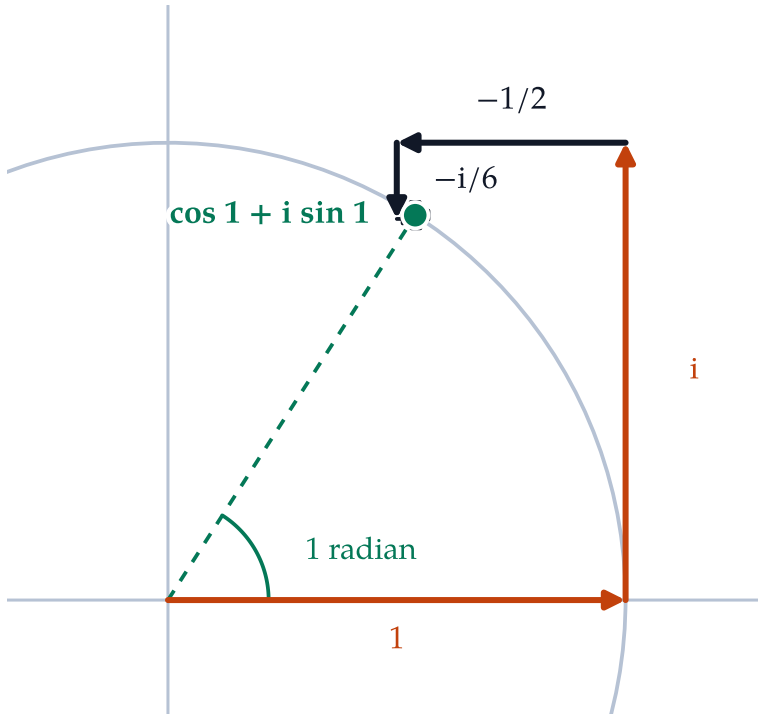


Figure 40.3. The terms of the series for E at the input i , laid head to tail: 1, then i , then $-1/2$, then $-i/6$, then $1/24$, each a quarter turn from the one before and one over the next factorial in length. The partial sums spiral in on the point of the unit circle at angle 1 radian, whose coordinates are $\cos 1 = 0.5403$ and $\sin 1 = 0.8415$. Assumes: that the spiral continues to close on that point past the eight arrows drawn, which the convergence of the series guarantees and the drawing only suggests

At $t = 1$ the terms are $1, i, -1/2, -i/6, 1/24, i/120, \dots$, each a quarter turn from the one before, and the partial sums spiral in: four terms give $0.5 + 0.8333i$, at distance 0.972 from the centre; eight give $0.54028 + 0.84147i$, at distance 0.99998; and $\cos 1$ and $\sin 1$ are

0.54030 and 0.84147. At $t = \pi$ the point on the circle is -1 , and forty terms of the series at $i\pi$ give -1 to fifteen decimal places. The equation $E(i\pi) = -1$ says that a point moving at unit speed perpendicular to its radius, released at 1 and running for a time π , the measure of a half turn, arrives at -1 .

* * *

The two laws are one. Fix s and let t be the time. Then $E(is) \cdot E(it)$ and $E(i(s+t))$ are two motions. Both have velocity i times position: the first is the moving point multiplied by the fixed number $E(is)$, and the second is the moving point with its clock set forward by s . Both are at $E(is)$ when $t = 0$. The uniqueness argument worked with the difference of two motions, and needed only that the difference starts at 0, that is, that the two starts agree. So $E(is) \cdot E(it) = E(i(s+t))$. In coordinates, $(\cos s + i \sin s)(\cos t + i \sin t) = \cos(s+t) + i \sin(s+t)$, and multiplying out and matching real and imaginary parts gives the addition formulas $\cos(s+t) = \cos s \cos t - \sin s \sin t$ and $\sin(s+t) = \sin s \cos t + \cos s \sin t$. That is not a second proof of them, since the derivatives of the sine and cosine that made the motion circular were computed from those formulas; it shows that the addition formulas and the rule $E(y+z) = E(y) \cdot E(z)$ are one law. At $s = \pi/12$ and $t = \pi/6$, the angles of 15° and 30° , $\cos(\pi/12) \cos(\pi/6) - \sin(\pi/12) \sin(\pi/6)$ is 0.70711, the cosine of $\pi/4$.

Facts also pass the other way, from the circle to E . A full turn brings the moving point back to its start, so $E(i(t+2\pi)) = E(it)$ for every t : E of an imaginary number repeats. E of a real number never repeats a value, since it increases, and it passes every bound.

Which motion has velocity $-0.1 + i$ times its position and starts at 1, and how far from the centre is it after one full turn?

The number π , which arrived as the area of a disc, and the number e , which arrived as the point where the area under $1/t$ reaches 1, sit together in $E(i\pi) = -1$. Like e , π is not a fraction. Lambert proved it in 1761 from an expansion of the tangent as a continued

fraction, which shows that $\tan x$ is not a fraction when x is a fraction other than 0; since $\tan(\pi/4) = 1$, $\pi/4$ is not a fraction, and neither is π . That expansion does not come from the series built so far. Growth along the line and rotation round the circle are one mechanism: the rate equals a fixed multiple of the position. The multiple decides what appears. A real multiple gives pure growth or decay, an imaginary one pure turning, and a multiple with both parts gives both at once.

ENDLESS POLYNOMIALS

The exponential, the sine and the cosine each turned out to be the sum of a series of powers of x , and the logarithm was one on a limited stretch. A series of that shape, $a_0 + a_1x + a_2x^2 + \dots$, is a *power series*. It looks like a polynomial that does not stop, and the question is how far it may be treated as one: added, multiplied, differentiated and integrated term by term. Within limits that can be stated exactly, it may.

The simplest is $1 + x + x^2 + x^3 + \dots$. Its partial sums are $(1 - x^{n+1})/(1 - x)$, so for x of size below 1 it converges to $1/(1 - x)$, and for x of size 1 or more its terms do not shrink and it diverges. That pattern, convergence inside a symmetric stretch and divergence outside it, is general.

If a power series converges at a point c , it converges absolutely at every point x closer to 0 than c is.

The terms $a_n c^n$ of a convergent series approach 0, so they are bounded, by B say. Then $a_n x^n$, which is $a_n c^n$ multiplied by $(x/c)^n$, has size at most $B \cdot k^n$, where $k = |x|/|c|$ is below 1. The series $B(1 + k + k^2 + \dots)$ converges, so by comparison with it the series of the sizes of the $a_n x^n$ converges too. More is true: on any stretch from $-r$ to r with r below $|c|$, the same bound with $k = r/|c|$ serves every x at once: every term $a_n x^n$ has size at most $B \cdot k^n$ on the whole stretch. Taking

M_n to be $B \cdot k^n$, the test with bounds M_n says that the convergence is uniform there, and the sum is a continuous function.

So for every power series there is a number R , possibly 0 and possibly endless, such that the series converges absolutely for $|x|$ below R and diverges for $|x|$ above R . Take R to be the least upper bound of the sizes of the points where the series converges, or endless if those sizes have no bound. If $|x|$ is below R , then some point c of size above $|x|$ is a point where the series converges, since otherwise $|x|$ would be a smaller upper bound; and the result above gives absolute convergence at x . If $|x|$ is above R , the series cannot converge at x , because R bounds the sizes of all the points where it converges.

R is called the *radius* of the series. For the exponential series the radius is endless. For $1 + x + x^2 + \dots$ it is 1. For $1 + x + 2!x^2 + 3!x^3 + \dots$ it is 0, the terms failing to shrink at any x but 0. At the two ends of the stretch anything can happen: the series for the logarithm of $1 + x$ converges at 1 and diverges at -1 .

* * *

Two theorems make the endless polynomial usable. The first is about integrals, and it is the answer to a question asked when uniform convergence was defined.

If continuous functions f_n converge uniformly to f on the interval from a to b , then the integrals of the f_n converge to the integral of f .

The limit f is continuous, so it has an integral. Past the stage at which every f_n is within ε of f at every point, f_n lies between $f - \varepsilon$ and $f + \varepsilon$, so its integral is within $\varepsilon(b - a)$ of the integral of f . Recall the narrow spikes of area 1. They converged to 0 at every point, yet their areas stayed at 1 and did not converge to the area under the limit, which is 0. Those spikes did not converge uniformly, and ruling out such behaviour is what uniformity buys.

A power series converges uniformly on any stretch inside its radius, and its partial sums are polynomials, which are continuous

and are integrated one power at a time. By the theorem, the integrals of the partial sums converge to the integral of the sum. So the series can be integrated term by term: the integral of $a_0 + a_1t + a_2t^2 + \dots$ from 0 to x is $a_0x + a_1x^2/2 + a_2x^3/3 + \dots$, for $|x|$ below R .

The second theorem is about derivatives, where uniform convergence of the functions themselves is known to be useless: the tents $d(n^2x)/n$ approached 0 uniformly while their slopes grew. What helps is uniform convergence of the derivatives. Differentiate a power series term by term, to get $a_1 + 2a_2x + 3a_3x^2 + \dots$. This new series has the same radius. It cannot have a larger one, since $a_n x^n$ has size at most $|x|$ times the size of $n \cdot a_n x^{n-1}$, so wherever the new series converges absolutely the old one does.

Nor can it have a smaller one. At $x = 0$ every power series converges, so take $|x|$ below R with x not 0, and choose c with $|x|$ below $|c|$ below R . The original series converges at c , so its terms $a_n c^n$ are bounded in size, by B say, as before. Now $n \cdot a_n x^{n-1}$ is $(n/x) \cdot a_n c^n \cdot (x/c)^n$, so it has size at most $(B/|x|) \cdot n k^n$ with $k = |x|/|c|$ below 1, and the series of $n k^n$ converges, since the ratio of each term to the one before is $k(n+1)/n$, which is below a fixed number short of 1 once n is large, and from that stage on the terms are at most those of a geometric series with that fixed number as its ratio, which converges. So the new series converges for every $|x|$ below R . It is itself a power series, so by the first result it converges uniformly on stretches inside the radius, to a continuous function g . Integrate g term by term from 0 to x : the result is the original series less a_0 . So the original sum f is a_0 plus the integral of g , and by the theorem on the rate of an integral, f has derivative g .

Inside its radius a power series can be differentiated term by term, as often as desired.

Differentiate n times and put $x = 0$. The terms below $a_n x^n$ have been differentiated away to 0. The terms above it still carry a power of x , so they vanish at 0. The term $a_n x^n$ itself has become $n! \cdot a_n$. So the coefficients are determined by the function: a_n is the n -th derivative at 0 divided by $n!$. Two power series that agree on a

stretch around 0, however short, are the same function there, with the same derivatives at 0, so they have the same coefficients.

* * *

The series $1 - t^2 + t^4 - t^6 + \dots$ is the one for $1/(1 + t^2)$, with radius 1. Integrated term by term from 0 to x it gives

$$x - x^3/3 + x^5/5 - x^7/7 + \dots$$

as the integral of $1/(1 + t^2)$ from 0 to x . That integral is a function already promised, the one whose derivative is $1/(1 + x^2)$, and it has a meaning. The tangent of an angle, the sine over the cosine, has derivative $(\cos^2 + \sin^2)/\cos^2$, which is $1 + \tan^2$, by the rule for quotients. Its derivative is positive, so the tangent increases as the angle runs from minus a right angle to a right angle, and it passes every bound in both directions near the ends. So it has an inverse, the *arctangent*, which sends a number x to the angle between minus a right angle and a right angle whose tangent is x . By the rule for inverses, the derivative of the arctangent at x is 1 over $1 + \tan^2$ of that angle, which is $1/(1 + x^2)$. The arctangent is 0 at 0, and so is the integral from 0 to x . Both have derivative $1/(1 + x^2)$, and two functions with the same derivative and the same value at 0 are the same function. So the arctangent is the integral, and

$$\arctan x = x - x^3/3 + x^5/5 - x^7/7 + \dots \text{ for } |x| \text{ below } 1.$$

The end $x = 1$ is of special interest, since the angle whose tangent is 1 is half a right angle, $\pi/4$. A series need not converge to anything at the end of its stretch, and this one needs an argument. The finite sum $1 - t^2 + \dots \pm t^{2n-2}$ is exactly $(1 \pm t^{2n})/(1 + t^2)$, where the sign is that of the last term of the sum. So it differs from $1/(1 + t^2)$ by $t^{2n}/(1 + t^2)$, which is at most t^{2n} . Now integrate from 0 to 1. The finite sum gives the first n terms of the series at $x = 1$, the function $1/(1 + t^2)$ gives the arctangent of 1, which is $\pi/4$, and t^{2n} gives $1/(2n + 1)$. So the first n terms of the series at $x = 1$ differ from $\pi/4$ by at most $1/(2n + 1)$, which approaches 0. So

$$\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \dots$$

The circle, which entered as an area, is here tied to the odd numbers and nothing else. Madhava knew the series in Kerala about 1400, and Gregory and Leibniz found it again in the 1670s. It is of no use for computing: a thousand terms give 3.1406 and the next one 3.1426.

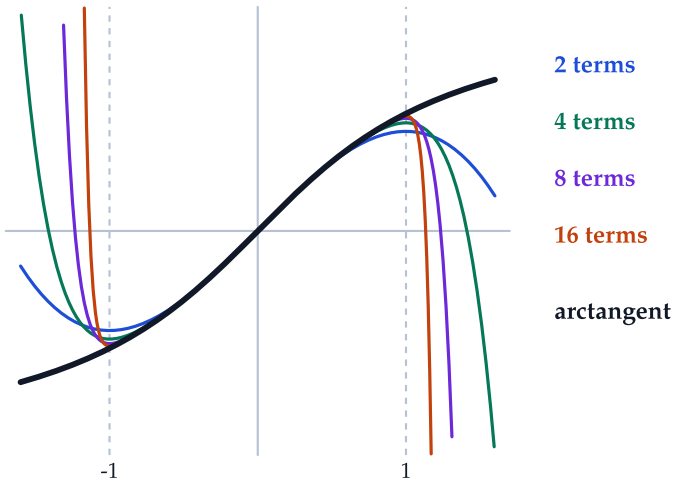


Figure 41.1. The arctangent, in black, and the partial sums of its series with 2, 4, 8 and 16 terms. Between -1 and 1 the polynomials settle onto the curve, slowly at the two ends. Beyond 1 they fly off, the sooner the more terms they have, although the arctangent itself continues without incident. Assumes: real x only; the reason the series stops at 1 , where nothing happens to the function, lies off the line, at the points i and $-i$ where $1 + x^2$ is 0

Powers of a small x shrink fast, so the series for the arctangent is quick when x is small, and slow at $x = 1$. The multiplication of the numbers of the plane gives a way to write $\pi/4$ using arctangents of small numbers only. Multiplying two such numbers adds their angles. The number $5 + i$ has an angle whose tangent is $1/5$. Its fourth power is

$$(5 + i)^2 = 24 + 10i, (24 + 10i)^2 = 476 + 480i$$

which has an angle just over half a right angle, its two coordinates being nearly equal. Multiply by $239 - i$, which takes away the angle whose tangent is $1/239$:

$$(476 + 480i)(239 - i) = 114244 + 114244i$$

and that lies exactly on the diagonal. So four times the angle of $5 + i$, less the angle of $239 + i$, is half a right angle:

$$\pi/4 = 4 \arctan(1/5) - \arctan(1/239)$$

Machin found this in 1706 and used it to compute π to a hundred places. Eight terms of the first series and four of the second give 3.14159265359, right in every place. The powers of $1/5$ are convenient in decimals, and each further term of that series gains more than a place.

* * *

Newton's first great discovery, made at Cambridge in the winter of 1664 to 1665, some months before the plague sent him home, was that the expansion of a power of $1 + x$ does not need a whole-number exponent. For a whole number r the expansion of $(1 + x)^r$ has coefficients $r(r - 1) \cdots (r - n + 1)/n!$, the number of ways of choosing n things from r , and it stops, because at $n = r + 1$ a factor 0 appears. Put for r a fraction that is not a whole number and no factor is ever 0. The series goes on forever:

$$(1 + x)^r = 1 + rx + r(r - 1)x^2/2! + r(r - 1)(r - 2)x^3/3! + \dots \text{ for } |x| \text{ below } 1.$$

Write c_n for the coefficients. The ratio of each to the one before is $(r - n)/(n + 1)$, whose size approaches 1. So the terms $c_n x^n$ have ratios whose sizes approach $|x|$. For $|x|$ below 1 those sizes are in the end below a fixed number short of 1, and the series converges absolutely. For $|x|$ above 1 they are in the end above 1, so the terms grow in size. The radius is 1. Inside it call the sum $f(x)$.

The ratio says that $(n+1)c_{n+1} = (r-n)c_n$, which is $(n+1)c_{n+1} + n \cdot c_n = r \cdot c_n$. Now the coefficient of x^n in $f'(x)$ is $(n+1)c_{n+1}$, and in $x \cdot f'(x)$ it is $n \cdot c_n$. So the left side is the coefficient of x^n in $(1+x)f'(x)$, and the right side is the coefficient of x^n in $r \cdot f(x)$. Since that holds for every n , $(1+x)f'(x) = r \cdot f(x)$. Now differentiate the quotient $f(x)/(1+x)^r$, using the rule for fraction powers: the result is $(f'(x)(1+x) - r \cdot f(x))/(1+x)^{r+1}$, which is 0. So the quotient is constant, and at $x=0$ it is 1. It is therefore 1 for every $|x|$ below 1, which says that $f(x)$ is $(1+x)^r$.

With $r = 1/2$ it gives $\sqrt{1+x} = 1 + x/2 - x^2/8 + x^3/16 - 5x^4/128 + \dots$, whose first terms were found before with a bound on the error. With $r = -1/2$ and $-4x$ in place of x , the coefficients simplify in a surprising way:

$$1/\sqrt{1-4x} = 1 + 2x + 6x^2 + 20x^3 + 70x^4 + 252x^5 + \dots$$

The coefficients are the middle entries of the even rows of the triangle of choices, the numbers $(2n \text{ choose } n)$ that were used to count primes. The reason is that the two lists grow by the same rule. Each coefficient of the series is the one before multiplied by $-4(-1/2 - n)/(n+1)$, which is $(4n+2)/(n+1)$. And $(2n+2 \text{ choose } n+1)$ is $(2n \text{ choose } n)$ multiplied by $(2n+2)(2n+1)/(n+1)^2$, which is $(4n+2)/(n+1)$ again, once $n+1$ is cancelled from $2n+2$. Both lists start at 1, so they agree all the way along.

* * *

Such a series, whose coefficients are a sequence of interest, is called the *generating function* of the sequence, and the principle that a function has only one expansion turns facts about functions into facts about sequences.

Take the list 1, 1, 2, 3, 5, 8, ..., in which each number is the sum of the two before. Write F_n for the number in place n and let $F(x) = F_1x + F_2x^2 + F_3x^3 + \dots$. Since F_n is at most 2^n , the series converges for $|x|$ below $1/2$. Multiply it by x and by x^2 , which shifts the coeffi-

cients one and two places, and subtract: in $F(x) - xF(x) - x^2F(x)$ the coefficient of x^n is $F_n - F_{n-1} - F_{n-2}$, which is 0 from $n = 3$ on. The coefficient of x is F_1 , which is 1, and the coefficient of x^2 is $F_2 - F_1$, which is 0. So what survives is x . So

$$F(x) = x/(1 - x - x^2)$$

The denominator is $(1 - \varphi x)(1 - \psi x)$, where $\varphi = (1 + \sqrt{5})/2$ and $\psi = (1 - \sqrt{5})/2$ are the two numbers with $\varphi + \psi = 1$ and $\varphi\psi = -1$. Splitting the fraction,

$$x/(1 - x - x^2) = (1/\sqrt{5}) \cdot (1/(1 - \varphi x) - 1/(1 - \psi x))$$

as can be checked by putting the right side over a common denominator, and each piece is a series of the simplest kind, with coefficients φ^n and ψ^n . Comparing coefficients,

$$F_n = (\varphi^n - \psi^n)/\sqrt{5}$$

Since ψ is -0.618 , its powers shrink, and $\psi^n/\sqrt{5}$ always has size below $1/2$. So F_n is the whole number nearest to $\varphi^n/\sqrt{5}$. The list grows by the factor φ , the proportion of the pentagon, at each step, and by φ^5 , which is 11.09 , in five steps. That was the fact behind the bound on the length of Euclid's algorithm.

A second example needs products of series.

If two series converge absolutely, to A and B , then the series whose n -th term collects all the products $a_k b_l$ with $k + l = n$ converges absolutely to AB .

Suppose first that no term is negative. The partial sum C_N of the collected series contains the products with $k + l$ at most N ; the product $A_N B_N$ of the partial sums contains those with k and l both at most N ; and C_{2N} contains all of those again. So $C_N \leq A_N B_N \leq C_{2N}$. The left half shows that the increasing sums C_N are bounded by AB , so they have a limit. The middle, $A_N B_N$, approaches AB , while C_N and C_{2N} both approach that limit. So the limit is at most

AB and at least AB , and it is AB . For series with signs, $A_N B_N - C_N$ is a sum of products $a_k b_l$ with k and l at most N and $k + l$ above N , so its size is at most the same expression for the series of sizes, which approaches 0 by the first case. So C_N , like $A_N B_N$, approaches AB . And each collected term has size at most the matching collected term for the series of sizes, whose sum converges by the first case, so the collected series converges absolutely. The question asked about the series for $E(y)$ and $E(z)$, collected by total degree, is answered by this: the collected terms are $(y + z)^n/n!$, by the binomial theorem.

Now let c_n be the number of ways of bracketing a product of $n + 1$ letters, so that $c_0 = 1, c_1 = 1, c_2 = 2, c_3 = 5$. The last multiplication performed splits the letters into a left block of $k + 1$ letters and a right block of $n - k$, each bracketed in its own way: c_k ways for the left block and c_{n-k-1} for the right, where k can be anything from 0 to $n - 1$. So

$$c_n = c_0 c_{n-1} + c_1 c_{n-2} + \dots + c_{n-1} c_0$$

which is the rule for the coefficients of a product: the right side is the coefficient of x^{n-1} in $c(x)^2$, and so the coefficient of x^n in $x \cdot c(x)^2$. That holds for every n from 1 on, and only the constant term $c_0 = 1$ is left out. So the generating function $c(x)$ satisfies $c(x) = 1 + x \cdot c(x)^2$, a quadratic equation for $c(x)$, whose solution with $c(0) = 1$ is $(1 - \sqrt{1 - 4x})/(2x)$. The coefficients of the square root come from the series just found: $\sqrt{1 - 4x}$ has derivative $-2/\sqrt{1 - 4x}$, which is -2 times the series with coefficients $(2n \text{ choose } n)$, and integrating term by term, $\sqrt{1 - 4x}$ is 1 less twice the sum of $(2n \text{ choose } n)x^{n+1}/(n + 1)$. So $1 - \sqrt{1 - 4x}$ is twice that sum, and dividing by $2x$ removes the 2 and lowers each power by one. So the coefficients of $c(x)$ are

$$c_n = (2n \text{ choose } n)/(n + 1)$$

the count that was found before by reflecting routes across a line. The argument as given assumed that the series $c(x)$ converges somewhere, which was not shown. It can be run the other way.

Start from the series with coefficients $(2n \text{ choose } n)/(n+1)$. It has radius $1/4$ by the ratio of its terms, and its sum is the closed form, which satisfies the quadratic equation. So its coefficients satisfy the same recursion as the c_n with the same start, and the two sequences are one.

What is the sum of $1/2 + 2/4 + 3/8 + 4/16 + \dots$, and of which wait is it the average, as 6 was the average number of throws of a die until a six?

Why is $\pi/4 = \arctan(1/2) + \arctan(1/3)$, and how many terms of the two series are needed for six places of π ?

The numerators 1, 3, 7, 17, 41, ... of the fractions closing in on the diagonal of the square satisfy $a_n = 2a_{n-1} + a_{n-2}$. What is their generating function, and what formula does it give?

In how many ways can a dollar be changed into coins of 1, 5, 10, 25 and 50 cents, and which product of series has that number as the coefficient of x^{100} ?

What is the radius of the series $1 + 2x + 6x^2 + 20x^3 + \dots$, and what does the series say when x is $-1/4$, if it may be trusted there?

SUMS THAT LAND ON π

The number π entered as the area of a disc. It has since turned up in the series $1 - 1/3 + 1/5 - \dots$, which has no circle in it, and it is about to turn up in the tossing of coins, in the size of $n!$, and in the sum of the reciprocals of the squares. Each time the route is the same: some integral of powers of the sine or cosine, handled by parts. One family of integrals carries all of it.

Let W_n be the integral of $\sin^n x$ from 0 to $\pi/2$. Then $W_0 = \pi/2$ and $W_1 = 1$. For n at least 2, write $\sin^n x$ as $\sin^{n-1} x \cdot \sin x$ and integrate by parts, moving the derivative off the second factor, which is the derivative of $-\cos x$. The term at the ends is 0, and what remains is $(n-1)$ times the integral of $\sin^{n-2} x \cdot \cos^2 x$. Since $\cos^2 x$ is $1 - \sin^2 x$, that integral is $W_{n-2} - W_n$, so $W_n = (n-1)(W_{n-2} - W_n)$. Here W_n stands on both sides; collecting it on the left gives $n \cdot W_n = (n-1) \cdot W_{n-2}$, that is,

$$W_n = ((n-1)/n) \cdot W_{n-2}$$

Each step down by 2 multiplies by one such fraction. Starting from an even n the steps end at $W_0 = \pi/2$, and starting from an odd n they end at $W_1 = 1$:

$$W_{2n} = (1/2)(3/4)(5/6) \cdots ((2n-1)/(2n)) \cdot \pi/2, \quad W_{2n+1} = (2/3)(4/5)(6/7) \cdots (2n/(2n+1))$$

One family has π in it and the other has not. But the two are close. Since $\sin x$ lies between 0 and 1 on the stretch, each higher power of it is smaller, so $W_{2n+1} \leq W_{2n} \leq W_{2n-1}$. By the rule, the outer two are related by $W_{2n+1} = (2n/(2n+1)) \cdot W_{2n-1}$, a factor which approaches 1. Dividing through by W_{2n+1} , the ratio W_{2n}/W_{2n+1} lies between 1 and $(2n+1)/(2n)$, so it approaches 1. Written out from the two products, that ratio is $\pi/2$ times $(1/2)(3/2)(3/4)(5/4) \cdots ((2n-1)/(2n))((2n+1)/(2n))$. So this product of fractions approaches $2/\pi$, and turning every fraction upside down,

$$\pi/2 = (2/1)(2/3)(4/3)(4/5)(6/5)(6/7) \cdots$$

Wallis found the product in 1655, by an audacious interpolation and without the calculus. It converges slowly: twice the product, which should be π , is 3.0677 after 10 pairs of factors, 3.1338 after 100 pairs and 3.1408 after 1000. Its value lies in what it says about the middle entry of the triangle of choices. The chance that $2n$ tosses of a fair coin show exactly n heads is $u = (2n \text{ choose } n)/4^n$, which is $(2n)!$ divided by the square of $2^n \cdot n!$. Now $2^n \cdot n!$ is the product $2 \cdot 4 \cdot 6 \cdots 2n$ of the even numbers up to $2n$, and $(2n)!$ is that product times the product $1 \cdot 3 \cdot 5 \cdots (2n-1)$ of the odd ones. So u is the odd numbers over the even ones, the product $(1/2)(3/4) \cdots ((2n-1)/(2n))$, which is W_{2n} divided by $\pi/2$. Next, multiply the products for W_{2n} and W_{2n+1} together. Interleaved, the fractions are $(1/2)(2/3)(3/4) \cdots (2n/(2n+1))$, where each denominator cancels the next numerator and only $1/(2n+1)$ is left, so $W_{2n} \cdot W_{2n+1} = \pi/(2(2n+1))$. The ratio of the two factors approaches 1, so W_{2n}^2 is close to $\pi/(2(2n+1))$, which is close to $\pi/(4n)$. Then W_{2n} is close to $\sqrt{\pi/(2\sqrt{n})}$, and dividing by $\pi/2$,

$(2n \text{ choose } n)/4^n$ is close to $1/\sqrt{(\pi n)}$, the ratio of the two approaching 1.

For $2n = 20$ the chance of ten heads is 0.1762 and the formula gives 0.1784; for $2n = 100$ they are 0.07959 and 0.07979. The count of level scores, which was found by adding up entries of the triangle, is governed by the circle.

* * *

How large is $n!$? It is below n^n and above $(n/2)^{n/2}$, and between those there is a great deal of room. The logarithm turns the product into the sum $L(1) + L(2) + \dots + L(n)$, which is a staircase for the integral of L . Since $x \cdot L(x) - x$ has derivative $L(x)$, the integral of L from 1 to n is $n \cdot L(n) - n + 1$. So $L(n!)$ should be near $n \cdot L(n) - n$, and $E(n \cdot L(n) - n)$ is $(n/e)^n$: this suggests comparing $n!$ with $(n/e)^n$. The comparison is made exact by watching the ratio

$$a_n = n! / (\sqrt{n} \cdot (n/e)^n)$$

from each n to the next. In the ratio a_n/a_{n+1} the factorials leave $1/(n+1)$, the square roots leave $\sqrt{(1+1/n)}$, and the powers leave $(n+1)^{n+1}/(n^n \cdot e)$. Together that is $(1+1/n)$ raised to the power $n+1/2$, divided by e . So $L(a_n/a_{n+1}) = (n+1/2) \cdot L(1+1/n) - 1$. Put $t = 1/(2n+1)$, so that $1+1/n$ is $(1+t)/(1-t)$ and $n+1/2$ is $1/(2t)$. The series for $L(1+t)$ and for $L(1-t)$, subtracted, give $L((1+t)/(1-t)) = 2(t + t^3/3 + t^5/5 + \dots)$. Multiplied by $1/(2t)$ that is $1 + t^2/3 + t^4/5 + \dots$, and the leading 1 is cancelled by the -1 , so

$$L(a_n/a_{n+1}) = t^2/3 + t^4/5 + t^6/7 + \dots$$

That is positive, so the a_n decrease. And it is less than $(t^2/3)(1 + t^2 + t^4 + \dots)$, which is $t^2/(3(1-t^2))$. With $t = 1/(2n+1)$ that is $1/(12n(n+1))$, which is exactly $1/(12n) - 1/(12(n+1))$. So $L(a_n) - L(a_{n+1})$ is less than $1/(12n) - 1/(12(n+1))$, and moving terms across, $L(a_n) - 1/(12n)$ is less than $L(a_{n+1}) - 1/(12(n+1))$: the sequence $L(a_n) - 1/(12n)$ increases. Now $L(a_n)$ is always above $L(a_n) - 1/(12n)$, which is never below its first term $L(a_1) - 1/12$. So the a_n decrease and stay above the positive number $E(L(a_1) - 1/12)$, and a decreasing sequence with a floor under it approaches a limit: the a_n approach a positive number c . They approach it from above, so $c < a_n$. The increasing sequence $L(a_n) - 1/(12n)$ approaches the same $L(c)$ from below, so $L(a_n) < L(c) + 1/(12n)$. At every stage, then,

$$c < a_n < c \cdot E(1/(12n))$$

What is c ? The circle again. The chance of n heads in $2n$ tosses is $(2n)!/(n!)^2/4^n$. Put $a_{2n} \cdot \sqrt{(2n)} \cdot (2n/e)^{2n}$ for $(2n)!$ and $a_n \cdot \sqrt{n} \cdot (n/e)^n$ for each $n!$. The powers of e cancel, the powers of n cancel, and the 2^{2n} that is left on top cancels the 4^n . What remains is $a_{2n} \cdot \sqrt{(2n)} / (a_n^2 \cdot n)$, which is $(a_{2n}/a_n^2) \cdot \sqrt{(2/n)}$. As n grows, a_{2n} and a_n both approach c , so a_{2n}/a_n^2 approaches c/c^2 , which is $1/c$, and the chance is close to $(1/c) \sqrt{(2/n)}$. It is also close to $1/\sqrt{(\pi n)}$. Two quantities close to the same chance have a ratio approaching 1; the ratio of these two is the fixed number $\sqrt{(2\pi)}/c$, so that number is 1. So $c = \sqrt{(2\pi)}$:

$n!$ is close to $\sqrt{(2\pi n)} \cdot (n/e)^n$, the ratio of the two lying between 1 and $E(1/(12n))$.

The formula carries the name of Stirling, who found the constant in 1730; de Moivre had the rest. For $n = 10$ it gives 3598696 against 3628800, a ratio of 1.0083654, while $E(1/120)$ is 1.0083682: the upper bound is nearly attained. For $n = 100$, where $n!$ has 158 digits, the ratio is 1.00083.

* * *

In $2n$ tosses the chance of $n + k$ heads, compared with the chance of n heads, is

$$\begin{aligned} & (2n \text{ choose } n + k) / (2n \text{ choose } n) \\ &= n/(n+1) \cdot (n-1)/(n+2) \cdots (n-k+1)/(n+k) \end{aligned}$$

since passing from $n + j - 1$ heads to $n + j$ multiplies the entry by $(n - j + 1)/(n + j)$. Divide the top and bottom of the j -th factor by n : its logarithm is $L(1 - (j - 1)/n) - L(1 + j/n)$. When t is small, $L(1 + t)$ is close to t , the first term of its series. So when j is small compared with n , the two logarithms are close to $-(j - 1)/n$ and j/n , and the factor contributes about $-(j - 1)/n - j/n$, which is

$-(2j-1)/n$. Add these for j from 1 to k : the odd numbers up to $2k-1$ add up to k^2 , so the logarithm of the ratio is close to $-k^2/n$. So the ratio is close to $E(-k^2/n)$, and with the value in the middle already known,

the chance of $n+k$ heads in $2n$ tosses is close to $(1/\sqrt{(\pi n)}) \cdot E(-k^2/n)$

as long as k is small compared with n . Carrying the logarithms two terms further, the next contributions add up to $k^2/(2n^2) - k^4/(6n^3)$ and smaller things, so the ratio of the true chance to the formula stays close to 1 as long as k^4 is small compared with n^3 . For a hundred tosses the chances of 50, 55, 60 and 65 heads are 0.07959, 0.04847, 0.01084 and 0.00086, and the formula gives 0.07979, 0.04839, 0.01080 and 0.00089. De Moivre found this in 1733. The bars of the chances, which the triangle of choices supplies one at a time, lie along a single curve: the curve $E(-x^2)$ stretched sideways by $\sqrt{n}$. That stretch is $\sqrt{2}$ times the deviation of the count of heads. The same curve appears, as Laplace showed, whenever many small independent effects are added, which is why measurements of almost anything are distributed along it.

The chances add up to 1. Write x for $k/\sqrt{n}$: the chance of $n+k$ heads is close to $(1/\sqrt{\pi}) \cdot E(-x^2) \cdot (1/\sqrt{n})$, and when k goes up by 1, x goes up by $1/\sqrt{n}$. So each chance is $1/\sqrt{\pi}$ times the area of a bar of height $E(-x^2)$ and width $1/\sqrt{n}$, and the sum of the chances is $1/\sqrt{\pi}$ times a staircase for the curve. A staircase with ever narrower steps approaches the area, so that requires the whole area under $E(-x^2)$, from far left to far right, to be $\sqrt{\pi}$. The argument is suggestive and not a proof, since the approximation was good only for k small compared with n . But the conclusion is true, and the integrals W_n prove it. The exponential lies above its tangent at 0, so $E(y) \geq 1+y$ for every y . With $y = -x^2$ this gives $1-x^2 \leq E(-x^2)$, and with $y = x^2$ it gives $E(-x^2) \leq 1/(1+x^2)$. Raise to the power n , which keeps the order between positive numbers and turns $E(-x^2)$ into $E(-nx^2)$. In the first inequality $1-x^2$ is positive only for x below 1, so integrate it from 0 to 1; the middle integrand is positive beyond 1 as well, so carrying its integral onwards only

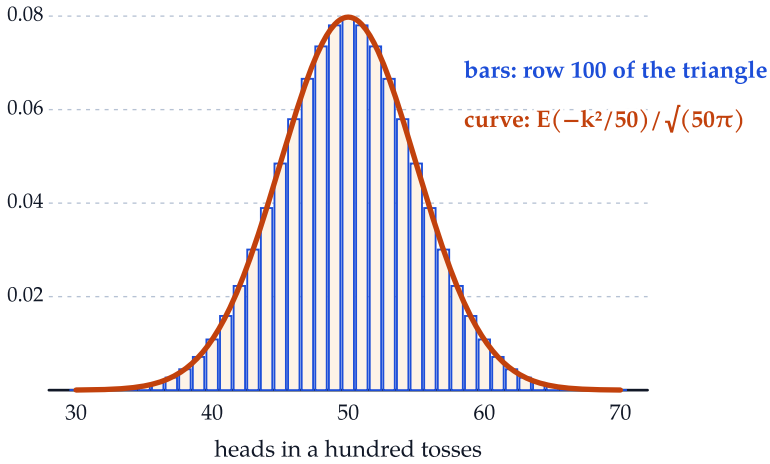


Figure 42.1. The chance of each number of heads in a hundred tosses of a fair coin, as bars, and the curve that the analysis predicts, whose height at $50 + k$ heads is $E(-k^2/50)$ divided by the square root of 50π . The bars come from row 100 of the triangle of choices, one entry at a time, and the curve from the area of a circle. *Assumes:* a fair coin and independent tosses; the agreement is closest near the middle and is off by three parts in a hundred at 65 heads

enlarges it. The second inequality holds for every x , so integrate it from 0 onwards:

the integral of $(1 - x^2)^n$ from 0 to 1 $\leq$ the integral of $E(-nx^2)$ from 0 onwards $\leq$ the integral of $1/(1 + x^2)^n$ from 0 onwards

In the first, substitute $x = \cos \theta$, with θ running from $\pi/2$ down to 0. Then $1 - x^2$ is $\sin^2 \theta$ and the derivative of x is $-\sin \theta$, the minus sign turning the ends round, so the integrand becomes $\sin^{2n} \theta \cdot \sin \theta$ and the integral is W_{2n+1} . In the last, substitute $x = \cos \theta / \sin \theta$, the cotangent, which runs from 0 onwards as θ runs from $\pi/2$ down to 0 and whose derivative is $-1/\sin^2 \theta$. Since $1 + x^2$ is $1/\sin^2 \theta$, the integrand $1/(1 + x^2)^n$ is $\sin^{2n} \theta$, the derivative takes two powers of the sine away, and the integral is W_{2n-2} . In the middle, substitute $x = s/\sqrt{n}$: it becomes $G/\sqrt{n}$, where G is the integral of $E(-s^2)$ from 0

onwards. So $\sqrt{n} \cdot W_{2n+1} \leq G \leq \sqrt{n} \cdot W_{2n-2}$ for every n . Neighbouring members of the family have a ratio approaching 1, so W_{2n+1}^2 and W_{2n-2}^2 are both close to $W_{2n} \cdot W_{2n+1}$, which is $\pi/(2(2n+1))$. Multiplied by n , that approaches $\pi/4$. So $n \cdot W_{2n+1}^2$ and $n \cdot W_{2n-2}^2$ both approach $\pi/4$, and the two ends of the bracket, which are their square roots, approach $\sqrt{\pi/2}$. G is caught between them, so $G = \sqrt{\pi/2}$. That is the area from 0 onwards, and since $E(-x^2)$ takes the same value at $-x$ as at x , the other half of the line gives the same again:

the area under $E(-x^2)$, over the whole line, is $\sqrt{\pi}$.

Add the entries of row 100 from 45 to 55 and divide by 2^{100} : the chance that a hundred tosses give between 45 and 55 heads is 0.7287, and between 40 and 60 it is 0.9648. These are areas under this curve. Give each count a bar of width 1. The bars for 45 to 55 then reach from 44.5 to 55.5, so the first chance is the area within 5.5 heads of the middle. For a hundred tosses the deviation is 5, so that is 1.1 deviations; the area under the curve within 1.1 deviations of the middle is 0.7287 of the whole, and within 2.1 deviations it is 0.9643.

* * *

In 1650 Mengoli asked for the sum of the reciprocals of the squares,

$$1 + 1/4 + 1/9 + 1/16 + 1/25 + \dots$$

It converges, slowly: a thousand terms give 1.6439, and the rest of the series is about $1/1000$. The Bernoullis tried and failed, and the problem took its name from their city, Basel. In 1735 Euler, then twenty-eight, found the sum: it is $\pi^2/6$, which is 1.644934. His argument treated $\sin x/x$ as a polynomial of endless degree, with roots at $\pm\pi, \pm2\pi, \dots$, and therefore equal to the product of the factors $1 - x^2/(k^2\pi^2)$; comparing the coefficient of x^2 in that product with the $-1/6$ of the series gave the sum. It was a daring argument

and he knew that it was not a proof; the product was justified only a century later. There is a proof within reach.

Let A_n be the integral of $\cos^{2n} x$ from 0 to $\pi/2$, which by the symmetry of the sine and cosine is W_{2n} , so that $A_n = ((2n-1)/(2n)) \cdot A_{n-1}$; and let B_n be the integral of $x^2 \cos^{2n} x$ over the same stretch. Integrate A_n by parts twice. First against the factor 1, whose integral is x : the ends give 0, and $A_n = 2n$ times the integral of $x \cdot \sin x \cdot \cos^{2n-1} x$. Then against x , whose integral is $x^2/2$: the ends give 0 again, and the derivative of $\sin x \cdot \cos^{2n-1} x$ is $\cos^{2n} x - (2n-1) \sin^2 x \cdot \cos^{2n-2} x$. Replacing $\sin^2$ by $1 - \cos^2$, that derivative is $2n \cdot \cos^{2n} x - (2n-1) \cdot \cos^{2n-2} x$. The second integration by parts gives A_n as $-2n$ times the integral of $x^2/2$ times this derivative, which is $-n \cdot (2n \cdot B_n - (2n-1) \cdot B_{n-1})$. So

$$A_n = n(2n-1) \cdot B_{n-1} - 2n^2 \cdot B_n$$

Divide by $n^2 \cdot A_n$, and use the rule for A_n on the first term: $(2n-1)/(n \cdot A_n)$ is $2/A_{n-1}$. So

$$1/n^2 = 2 \cdot (B_{n-1}/A_{n-1} - B_n/A_n)$$

and the sum telescopes: $1 + 1/4 + \dots + 1/N^2 = 2B_0/A_0 - 2B_N/A_N$. Here $A_0 = \pi/2$ and B_0 is the integral of x^2 , which is $\pi^3/24$, so $2B_0/A_0 = \pi^2/6$. It remains to see that B_N/A_N approaches 0. On the stretch from 0 to $\pi/2$ the sine lies above its chord, its second derivative being negative there, so $\sin x \geq 2x/\pi$, that is, $x \leq (\pi/2) \sin x$. So $x^2 \leq (\pi^2/4) \sin^2 x$, and B_N is at most $\pi^2/4$ times the integral of $\sin^2 x \cdot (\cos x)^{2N}$. Writing $\sin^2$ as $1 - \cos^2$, that integral is $A_N - A_{N+1}$, and since $A_{N+1} = ((2N+1)/(2N+2)) \cdot A_N$, it is $A_N/(2N+2)$. So B_N/A_N is at most $\pi^2/(8(N+1))$, and

$$\pi^2/6 - (1 + 1/4 + \dots + 1/N^2) = 2B_N/A_N, \text{ which is between 0 and } \pi^2/(4(N+1)).$$

The sum is $\pi^2/6$, with a bound on the error at every stage.

The number answers a question that was left open. The chance that two whole numbers chosen at random have no common divisor was the product of the factors $1 - 1/p^2$ over all the primes p ,

which settled numerically at 0.6079. Each factor has the reciprocal $1 + 1/p^2 + 1/p^4 + \dots$. Multiply these series together for the primes up to some bound P , by the rule for the product of two absolutely convergent series, used once for each prime. Each term of the product, multiplied out, is 1 over the square of a product of powers of those primes. Every n whose prime divisors are all at most P is such a product, and in one way only. So the product is the sum of $1/n^2$ over exactly those n , each once. That sum includes every n up to P and is part of the whole series, so it lies between the partial sum up to P and $\pi^2/6$. As P grows,

the product of $1/(1 - 1/p^2)$ over all primes is the sum of $1/n^2$ over all whole numbers, which is $\pi^2/6$

and the chance is $6/\pi^2$, which is 0.6079. The uniqueness of factorisation, written as an equation between a product and a sum, is Euler's as well, from 1737, and it is where the study of the primes by the calculus begins.

How many digits has 1000!, and how many has (2000 choose 1000)?

In ten thousand tosses of a fair coin, what, to two figures, is the chance of exactly 5000 heads, and of exactly 5100?

No combination of powers, roots, E , L , sines and cosines has $E(-x^2)$ for its derivative; Liouville proved that in 1835, and it is relied on here and not proved. But the integral of $x \cdot E(-x^2)$ from 0 onwards can be found. What is it, and what is the integral of $x^2 \cdot E(-x^2)$ over the whole line?

What is the sum of the reciprocals of the odd squares, $1 + 1/9 + 1/25 + \dots$, and of the alternating series $1 - 1/4 + 1/9 - 1/16 + \dots$?

What is the chance that a whole number chosen at random is divisible by no square other than 1?

GROWTH, OSCILLATION, ORBIT

The exponential was found by asking for a function equal to its own rate of change, and the circular motion $E(it)$ by asking for a motion whose velocity is its position turned through a right angle. In both cases a law was written as a statement about a rate, and the function came out of the law. Most laws of nature are written that way, because what can be observed and reasoned about is how a thing changes from one instant to the next, and what is wanted is how it behaves over a span of time. An equation that prescribes the rate of an unknown function in terms of the function is called a *differential equation*. Three of them account for a great deal of the world.

The first says that a quantity changes at a rate proportional to itself: $y' = ky$. The function $C \cdot E(kt)$ satisfies it, with C its value at time 0, and nothing else does. Suppose y obeys the law, and look at the product $y \cdot E(-kt)$. By the product rule its derivative is $y' \cdot E(-kt)$ less $ky \cdot E(-kt)$, and since y' is ky , that is $ky \cdot E(-kt)$ less $ky \cdot E(-kt)$, which is 0. A function whose derivative is 0 is constant, so the product keeps the value it had at time 0, which is C . So $y \cdot E(-kt) = C$ at every time, and multiplying both sides by $E(kt)$ gives $y = C \cdot E(kt)$.

If k is positive the quantity grows. Moving on by a time $L(2)/k$ multiplies $E(kt)$ by $E(L(2))$, which is 2, so the quantity doubles in every stretch of time of length $L(2)/k$, whatever its size at the start: money at three per cent a year doubles in about $0.693/0.03$ years, which is 23, and a population growing at 1.1 per cent a year, as the world's was in 2015, doubles in 63. If k is negative the quantity decays, halving in every stretch of $L(2)/|k|$, its *half-life*.

Radioactive atoms decay at a rate proportional to the number remaining, because each atom has the same chance of decaying in the next second however old it is. Carbon of the kind with atomic weight 14 is made in the upper air at a steady rate and taken up by everything living; when a plant or animal dies, the uptake stops and the carbon 14 in it decays with a half-life of 5730 years. Charcoal from the painted cave at Lascaux was found to have 14.5 per cent of the carbon 14 of living wood. After n half-lives the fraction left is $1/2$ to the power n , so the number of half-lives elapsed is the power of $1/2$ that gives 0.145. Taking logarithms, $n \cdot L(2) = L(1/0.145)$, so n is $L(1/0.145)/L(2) = 2.79$. That many half-lives of 5730 years each come to about 16000 years, and the fires were lit that long ago. Libby devised the method in 1949.

A cup of coffee at 90 degrees stands in a room at 20, and cools at a rate proportional to its excess over the room, which is Newton's law of cooling. The excess obeys $y' = -ky$. If the coffee is at 70 after five minutes, the excess has fallen from 70 to 50 in that time, so $E(-5k) = 5/7$. It will be at 40, with an excess of 20, when $E(-kt) = 20/70 = 2/7$. Taking logarithms of both statements gives $5k = L(7/5)$ and $kt = L(7/2)$. Dividing the second by the first removes the unknown k : $t/5 = L(7/2)/L(7/5)$, so $t = 5 \cdot L(7/2)/L(7/5)$, which is 18.6 minutes.

Growth in proportion to size cannot go on. A population whose growth is checked by crowding obeys, in the simplest case, $y' = ky(1 - y)$, where y is the population as a fraction of the most that the land will carry. This is the law of the insects with time running smoothly in place of year by year. The function

$$y = 1/(1 + C \cdot E(-kt))$$

satisfies it, as the rules for derivatives confirm: writing u for $C \cdot E(-kt)$, the value y is $1/(1 + u)$, its derivative is $ku/(1 + u)^2$, and $1 - y$ is $u/(1 + u)$, so that $y' = ky(1 - y)$. Nothing else strictly between 0 and 1 obeys the law. For such a y put $u = (1 - y)/y$, which is the same as $1/y - 1$. Its derivative is $-y'/y^2$. Putting $ky(1 - y)$ for y' turns that into $-k(1 - y)/y$, which is $-ku$. So $u' = -ku$, and u is $C \cdot E(-kt)$ by the rule settled for the first equation. Solving $u = 1/y - 1$ for y gives $y = 1/(1 + u)$, the function displayed. So every such solution rises along an S-shaped curve, at first like the exponential and then levelling off at 1, and none of them overshoots, oscillates or turns chaotic, however large k is. The wild behaviour of the yearly rule came from taking steps of finite size. Verhulst proposed the smooth law in 1838 and fitted it to the populations of France and Belgium.

* * *

The second equation is $y'' = -y$: the acceleration is the negative of the position. The sine and the cosine satisfy it, each being minus its own second derivative, and so does $a \cos t + b \sin t$ for any numbers a and b . Those are all. The reason has three steps. First, for any solution the quantity $y^2 + y'^2$ has derivative $2y'(y + y'')$, which is 0 because $y'' = -y$, so that quantity is constant. Second, a solution that starts with $y = 0$ and $y' = 0$ therefore has $y^2 + y'^2 = 0$ forever; two squares that add up to 0 are both 0, so y is 0 at every time, and the solution is the zero function. Third, take any solution y and subtract from it the solution $y(0) \cos t + y'(0) \sin t$. The difference of two solutions is a solution, since the second derivative of a difference is the difference of the second derivatives. At time 0 the difference $y - y(0) \cos t - y'(0) \sin t$ has the value $y(0) - y(0)$, which is 0, and the rate $y'(0) - y'(0)$, which is 0. By the second step it is the zero function. So

every solution of $y'' = -y$ is $y(0) \cos t + y'(0) \sin t$.

A position and a velocity at one instant determine the motion for all time, and every such motion repeats after the time 2π . If a constant ω^2 multiplies the right-hand side, the equation $y'' = -\omega^2 y$ has the solutions $a \cos \omega t + b \sin \omega t$, since each differentiation of $\cos \omega t$ or $\sin \omega t$ brings out a factor ω . These repeat after the time $2\pi/\omega$.

A weight of mass m on a spring is pulled back towards its resting place with a force proportional to its displacement, so $m \cdot x'' = -cx$, where c is the stiffness of the spring. So ω^2 is c/m , and the weight bobs with period $2\pi\sqrt{(m/c)}$, whatever the size of the swing. The same equation governs anything that is pushed back towards a resting state in proportion to its displacement from it. A smooth restoring force is nearly proportional to the displacement as long as the displacement is small, so the equation governs nearly everything that is disturbed a little from rest: a plucked string, the air in an organ pipe, the current in a tuned circuit, a building in an earthquake.

A pendulum of length ℓ displaced by an angle θ from the vertical moves along an arc of length $\ell\theta$, and gravity pulls it back along the arc with the force $mg \sin \theta$. The distance along the arc is $\ell\theta$, so the acceleration along it is $\ell\theta''$, and mass times acceleration equals force: $m\ell\theta'' = -mg \sin \theta$, with the minus sign because the pull is back towards the vertical. Dividing by $m\ell$ gives

$$\theta'' = -(g/\ell) \sin \theta$$

which is not the equation above. But for small angles $\sin \theta$ is close to θ , within one part in six hundred up to a tenth of a radian, and then the equation is $\theta'' = -(g/\ell)\theta$, with period

$$T = 2\pi\sqrt{(\ell/g)}$$

The argument from units gave $\sqrt{(\ell/g)}$ times a pure number that it could not supply. The number is 2π , and it comes from the circle: an oscillation is a circular motion seen from the side. For $\ell = 1$ metre and $g = 9.81$ the period is 2.006 seconds; a pendulum that beats seconds, one second each way, is 0.994 metres long, which is why

the pendulums of tall clocks are all of a size, and why the metre was nearly defined as the length of one.

Galileo believed that the period is the same for swings of every size. It is not. Multiply the pendulum's equation by θ' : the quantity $\theta'^2/2 - (g/\ell) \cos \theta$ has derivative $\theta'(\theta'' + (g/\ell) \sin \theta)$, which is 0, so it is constant. It is the energy again. Suppose the swing reaches the angle α . There the bob stops to turn round, so θ' is 0, and the constant is $-(g/\ell) \cos \alpha$. At every other angle, then, $\theta'^2/2 - (g/\ell) \cos \theta = -(g/\ell) \cos \alpha$, which says that $\theta'^2 = (2g/\ell)(\cos \theta - \cos \alpha)$. That gives the rate θ' at each angle, and a rate turns into a time in the usual way: while the angle advances by a small amount, the time that passes is that amount divided by θ' , so the time taken is the integral of $1/\theta'$ over the angle traversed. Here $1/\theta'$ is $\sqrt{(\ell/(2g))}$ divided by $\sqrt{(\cos \theta - \cos \alpha)}$. A whole swing is made of four equal quarters, each between the bottom and the angle α , so the period is

$$T = 4 \cdot \sqrt{(\ell/(2g))} \cdot \text{the integral of } 1/\sqrt{(\cos \theta - \cos \alpha)} \text{ from } 0 \text{ to } \alpha$$

No combination of powers, roots, E , L , sines and cosines gives that integral, which is Liouville's theory again, relied on here and not proved. Computed, it gives the period as a multiple of $2\pi\sqrt{(\ell/g)}$: 1.0019 for swings of 10 degrees either side, 1.0174 for 30 degrees, 1.0732 for 60, 1.1803 for 90, and 2.44 for 170, where the bob lingers near the top. A clock whose pendulum swings 3 degrees either side loses about thirteen seconds a day compared with one swinging 1 degree. Huygens, who built the first pendulum clock in 1656, found the curve along which the period does not depend on the swing at all, the cycloid traced by a point on a rolling wheel, and built clocks whose pendulums were constrained to follow it.

* * *

The third equation governs the planets. Kepler, from the observations of Tycho Brahe, extracted three laws between 1609 and 1619. Each planet moves on an ellipse with the sun at one focus;

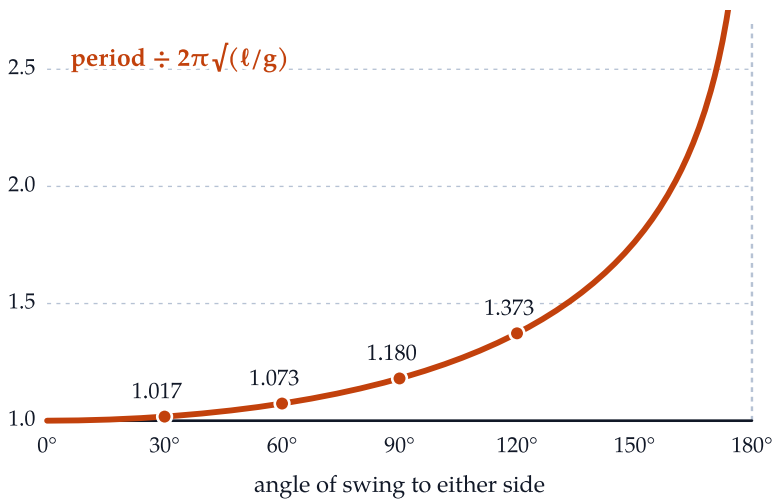


Figure 43.1. The period of a pendulum, as a multiple of $2\pi\sqrt{\ell/g}$, against the angle it swings to either side of the vertical. For small swings the multiple is 1, within two parts in a thousand up to 10 degrees. It reaches 1.18 at a right angle and grows without bound as the swing approaches the upright position, where the bob would balance. *Assumes:* a rigid rod without weight and no friction; the curve was computed from the energy integral by the method of arithmetic and geometric means, and checked by direct summation

an *ellipse* is the curve of all points whose distances from two fixed points, its *foci*, add up to a fixed length, the oval that a gardener draws with two pegs and a loop of string. The line from the sun to the planet sweeps out equal areas in equal times. And the squares of the times of revolution are as the cubes of the distances: the square of the time divided by the cube of the distance is the same number for every planet. That law was checked earlier against the numbers for five planets, and an argument from units arrived at it. But the argument took each planet to keep one distance from the sun, and it had nothing to say about flattened orbits. In 1684 Halley asked Newton what path a planet would follow if it were drawn towards the sun by a force varying as the inverse square of the distance, and Newton answered at once: an ellipse, he had cal-

culated it. The calculation grew into the Principia of 1687. With the numbers of the plane it takes a page.

Put the sun at the origin, and let the planet be at z , a number of the plane depending on the time. Write $z = r \cdot u$, where r is the distance and $u = E(i\theta)$ is the point of the unit circle in the planet's direction. The law of force says that the acceleration points from the planet to the sun and has size μ/r^2 :

$$z'' = -(\mu/r^2) \cdot u$$

where μ is a constant belonging to the sun. First, areas. For two numbers $z = x + iy$ and w of the plane, write $z^* = x - iy$ for the mirror image of z . It has the size of z and the opposite angle. Multiplying two numbers of the plane multiplies their sizes and adds their angles, so $z^* \cdot w$ has the size $|z||w|$, and its angle is the angle of w less the angle of z , which is the angle from z round to w . The imaginary part of a number is its size times the sine of its angle, so the imaginary part of $z^* \cdot w$ is $|z||w|$ times the sine of the angle from z round to w . Now look at the triangle with corners $0, z$ and w . Take the side from 0 to z as its base, of length $|z|$; its height is then $|w|$ times the sine of the same angle, and its area is half the base times the height. So the imaginary part of $z^* \cdot w$ is twice the area of that triangle, counted with a sign: positive when w lies anticlockwise from z , negative when it lies clockwise.

In a short time s the line from the sun sweeps the triangle with corners $0, z(t)$ and $z(t + s)$, apart from a sliver between the chord and the path, and the sliver is at most the chord times the path's departure from it. The chord is of the order of s , and over so short a time the path departs from its chord by an amount of the order of s^2 , so the sliver is of the order of s^3 . The triangle has one side of length r and a height of the order of s , so its area is of the order of s , and once everything is divided by s the sliver's share shrinks to nothing.

By the rule for triangles, twice the triangle's area is the imaginary part of $z^* \cdot z(t + s)$, where z stands for $z(t)$. Subtracting $z^* \cdot z$, which is real, leaves an imaginary part as it was, so twice the area

is also the imaginary part of $z^* \cdot (z(t+s) - z(t))$. Divide by s and let s shrink: the quotient $(z(t+s) - z(t))/s$ becomes the velocity z' . The rate at which the line from the sun sweeps out area is therefore half of h , where h is the imaginary part of $z^* \cdot z'$. Its derivative is the imaginary part of $z^{*'} \cdot z' + z^* \cdot z''$. The first product is the size of z' squared, which is real. The second is $-(\mu/r^2) z^* \cdot u$, which is $-\mu/r$, real again, since $z^* \cdot u$ is $r \cdot u^* \cdot u = r$. So h is constant:

under any force directed towards a fixed centre, equal areas are swept in equal times.

Nothing about the inverse square was used. The step runs backwards as well. Whatever the force, the derivative of h is the imaginary part of $z^* \cdot z''$, since the other product was real. If equal areas are swept in equal times, h is constant, so that imaginary part is 0 and $z^* \cdot z''$ is a real number; call it q . Multiply both sides by z : since $z \cdot z^*$ is r^2 , this gives $r^2 \cdot z'' = q \cdot z$. So z'' is the real number q/r^2 times z , and it points along the line through the planet and the sun. Kepler's second law says exactly that the force on a planet points at the sun, and nothing more.

Since $z = r \cdot E(i\theta)$, the product rule gives $z' = (r' + ir\theta') \cdot E(i\theta)$, while z^* is $r \cdot E(-i\theta)$. The two exponentials multiply to 1, so the product $z^* \cdot z'$ is $r(r' + ir\theta')$, whose imaginary part is $r^2\theta'$. So $h = r^2\theta'$. If h is 0 then θ' is 0, the direction never changes and the planet falls along a straight line into the sun; set that case aside. Otherwise $h = r^2\theta'$ says that $1/r^2$ is θ'/h , so μ/r^2 can be written $(\mu/h)\theta'$. Now the inverse square does its work. The derivative of $u = E(i\theta)$ is $i\theta'u$, so the derivative of iu is $-\theta'u$, and the law of force reads

$$z'' = -(\mu/h)\theta'u = (\mu/h) \cdot (iu)'$$

Both sides are derivatives. Two functions with the same derivative differ by a constant, so $z' = (\mu/h)(iu + c)$ for some constant number c of the plane. Turning the axes turns z , u and c through the same angle and leaves the equation as it stands, so the axes can be chosen to put c on the upper half of the imaginary axis: $c = ie$ with e real and not negative. The velocity z' is (μ/h) times the point

$iu + ie$. As the planet goes round, u runs round the unit circle, and so does iu ; adding ie shifts that circle so that its centre is ie , and multiplying by the fixed number μ/h scales a circle to a circle. So the velocity, drawn as an arrow from a fixed point, has its tip on a circle, with its centre at $(\mu/h) \cdot ie$. Hamilton noticed that in 1846.

Now compute h once more from its definition, with this velocity: $z^* \cdot z'$ is $r \cdot u^* \cdot (\mu/h)(iu + ie)$, which is $(\mu r/h) \cdot i(1 + e \cdot u^*)$, since $u^* \cdot u$ is 1. The imaginary part of i times a number is that number's real part, and the real part of $u^* = E(-i\theta)$ is $\cos \theta$, so the imaginary part of $z^* \cdot z'$ is $(\mu r/h)(1 + e \cos \theta)$. That is h . So $h^2 = \mu r(1 + e \cos \theta)$, and solving for r gives

$$r = p/(1 + e \cos \theta), \text{ where } p = h^2/\mu$$

This is the equation of the path. Write it as $r = p - e \cdot x$, since $r \cos \theta$ is x , and square: $x^2 + y^2 = (p - ex)^2$. When e is 0 that is a circle. When e is between 0 and 1 it is $(1 - e^2)x^2 + 2pex + y^2 = p^2$, a closed oval, and it is an ellipse with the origin at one focus.

Put $a = p/(1 - e^2)$ and take the point $-2ea$ on the axis as the second peg; the first peg is the origin, where the sun is. The square of the distance from the second peg to (x, y) is $(x + 2ea)^2 + y^2$, which is $r^2 + 4eax + 4e^2a^2$, since $x^2 + y^2$ is r^2 . Compare it with the square of $2a - r$, which is $r^2 - 4ar + 4a^2$. Because $r = p - ex$ and $p = a(1 - e^2)$, the product ar is $a^2(1 - e^2) - eax$, so $-4ar + 4a^2$ is $4eax + 4e^2a^2$, and the two squares agree. Since r is never more than $p/(1 - e)$, which is $a(1 + e)$ and less than $2a$, the number $2a - r$ is positive, and it is the distance from the second peg. The distance from the first peg is r . So the two distances add up to $2a$ at every point of the path.

When e is 1 the equation is $y^2 = p^2 - 2px$, a parabola lying on its side. When e is more than 1 the x^2 and y^2 terms have opposite signs and the curve is open; the same computation goes through with $a = p/(1 - e^2)$ now negative, the distance from the second peg coming out as $r - 2a$, so the two distances differ by the fixed length $-2a$ instead of adding up. Such a curve is called a *hyperbola*, and the path is one of its two branches. These are the paths of bodies

that pass the sun once and never return. That is Kepler's first law, and more.

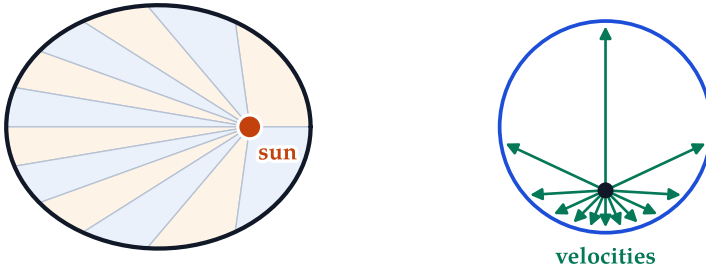


Figure 43.2. On the left, an orbit computed step by step from the law that the acceleration points at the sun and varies as the inverse square of the distance. It closes into an ellipse with the sun at a focus, and the twelve sectors, swept in equal times, have equal areas. On the right, the velocity at the same twelve instants, drawn from a fixed point: the tips of the arrows lie on a circle. *Assumes:* one attracting body that does not move; the orbit was integrated in twelve thousand small steps and checked against the ellipse the text derives, and the circle drawn is the one the text derives, not fitted

For the third law, let the ellipse have half-axes a and b , the longer one a . The nearest and farthest distances $p/(1+e)$ and $p/(1-e)$ add up to the long axis, which agrees with $a = p/(1-e^2)$, and completing the square in the equation of the path turns it into $(1-e^2)(x+ea)^2 + y^2 = pa$, because $p^2 + (1-e^2)e^2a^2$ is $p^2 + pe^2a$, which is $p(p+e^2a)$, and $p+e^2a$ is $a(1-e^2) + e^2a = a$. The curve is highest where $x = -ea$, and there $y^2 = pa$. So $b^2 = pa$, which is the same as $p^2/(1-e^2)$. An ellipse with half-axes a and b is the circle of radius a squashed in one direction by the factor b/a . The squashing multiplies the area of every rectangle, and so every area, by b/a , so the area of the ellipse is πa^2 times b/a , which is πab . Area is swept at the rate $h/2$, so the time T of one revolution is πab divided by $h/2$, which is $2\pi ab/h$. Square it, and use $b^2 = pa$ together with $h^2 = p\mu$, which is the definition $p = h^2/\mu$ read the other way:

$$T^2 = 4\pi^2 a^2 b^2 / h^2 = 4\pi^2 a^2 (pa) / (p\mu) = (4\pi^2 / \mu) \cdot a^3$$

The squares of the times are as the cubes of the distances, for flattened orbits as for round ones, provided the distance is taken to be half the long axis. And the constant, which the argument from units could not reach, is $4\pi^2/\mu$, the same for every body that circles the same sun. Halley applied it to the comet of 1682. It returns every 76 years. The earth goes round the same sun in 1 year, and half its long axis is 1 earth's distance, so for the comet the cube of half the long axis, measured in earth's distances, is 76 squared, and half its long axis is 76 to the power $2/3$ times the earth's distance, which is 17.9. The comet passes within 0.59 of the earth's distance from the sun. The nearest distance on an ellipse is $p/(1+e)$, which is $a(1-e)$, so $1-e$ is $0.59/17.9$ and $e = 0.97$. The farthest distance, $a(1+e)$, is about 35 times the earth's distance, which reaches out beyond Neptune. He predicted its return for 1758, and it came, sixteen years after his death.

What is the half-life of a substance of which one part in a thousand decays in a year, and how long until a tenth of it is left?

A population following the smooth law of crowded growth is at a tenth of its ceiling now and at a fifth of it one year from now. When will it reach a half, and when nine tenths?

A tunnel is bored straight through the earth from one city to another, and a train runs through it with no engine and no friction, pulled by gravity alone. Inside the earth the pull towards the centre is proportional to the distance from the centre. How long does the journey take, and how does the time depend on which two cities are joined?

How much does a pendulum clock lose in a day if its swing grows from 2 degrees either side to 4?

A satellite circles the earth just above the ground. The pull there is g , so μ for the earth is gR^2 , with R the radius of the earth. How long does it take to go round, and how does that compare with the train?

THE SPHERE MEASURED

Archimedes asked that his tomb carry a sphere inside a cylinder, and the proportion between them. When Cicero was an official in Sicily in 75 BC he went looking for the grave, which the people of Syracuse had forgotten, and found it by the carving, overgrown with brambles. The proportion was 2 to 3, and it holds twice over: the sphere has two thirds of the volume of the cylinder that encloses it, and two thirds of its surface. Archimedes thought these his finest results. With the integral both take a few lines, and his own arguments, which had no integral, are still the ones that show why.

The volume first. Let the sphere have radius r and centre at the origin, and slice it by horizontal planes. The slice at height t is a disc whose radius ρ satisfies $\rho^2 + t^2 = r^2$, so its area is $\pi(r^2 - t^2)$, and the volume is the integral of that from $-r$ to r . The constant r^2 gives r^2 times the length $2r$ of the interval, which is $2r^3$, and the integral of t^2 over the same interval, $2r^3/3$, is taken away, so the volume is

$$\pi(2r^3 - 2r^3/3) = (4/3)\pi r^3$$

The enclosing cylinder has base πr^2 and height $2r$, so its volume is $2\pi r^3$, and the sphere is two thirds of it.

Now look at the areas of the slices again, without integrating. At height t the cylinder's slice has area πr^2 . The sphere's has area $\pi(r^2 - t^2)$. The difference is πt^2 , which is the area of a disc of radius t . Discs whose radius equals their distance from the centre stack up into a cone: its point is at the centre, its side slopes at half a right angle, and it widens to radius r where it meets the top of the cylinder. Below the centre the same discs make a second cone, point to point with the first and widening downwards, and the two together are the double cone. So at every height

$$\text{slice of cylinder} = \text{slice of sphere} + \text{slice of double cone}$$

and solids with equal slices at every height have equal volumes. The double cone is two cones of base πr^2 and height r , with volume $(2/3)\pi r^3$ together, by the rule for cones. That is a third of the cylinder's $2\pi r^3$. So the sphere is the cylinder less a third of itself. Archimedes found the result by a comparison of the same kind, weighing the slices of a sphere and a cone against those of a cylinder on an imagined balance, with unequal arms; he then proved it by brackets, in the manner of the parabola. The comparison in the form given here, slice against slice with no balance needed, is the later one of Cavalieri. A cone, a half sphere and a cylinder on the same base and with the same height have volumes as 1, 2 and 3.

* * *

Surface area is more delicate than volume, because a curved surface cannot be approached by flat tiles as carelessly as a curve by chords. Schwarz gave the example in 1880. Fold a concertina of triangles round a plain cylinder, with all their corners on the cylinder. Now make the triangles smaller, but refine the fold in one direction faster than in the other. The corners crowd the surface as closely as wanted, and yet the total area of the triangles can be sent to any number above the cylinder's own area, or made to grow without bound.

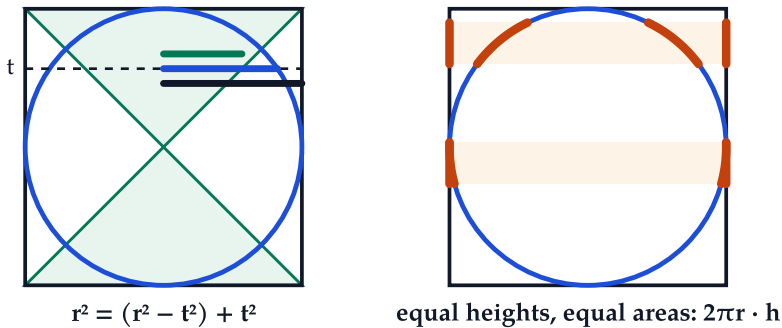


Figure 44.1. A sphere, its cylinder and the double cone, seen from the side. On the left, at any height t the three slices are discs of radius r , $\sqrt{r^2 - t^2}$ and t , and the first has the area of the other two together. On the right, a zone of the sphere between two horizontal planes and the band of the cylinder between the same planes, which have equal areas wherever the planes are put. *Assumes:* a side view, in which discs appear as horizontal segments; the two statements are about the areas of the discs and of the curved bands, which the drawing cannot show

Chords of a curve never misbehave so. As a chord shortens, its direction approaches the direction of the curve, and so its length and the length of the piece of curve it spans come to agree, the ratio of the two approaching 1. A triangle is under no such constraint: however small it is, with all three corners on the surface, it can still stand steeply tilted to the surface, and a tilted triangle has more area than the patch of surface beneath it.

For a surface made by turning a curve about an axis there is a safe way: replace the curve by an inscribed chain of chords, turn the chain, and take the limit of the areas of the bands it sweeps out. Each band is a slice of a cone. A whole cone, of slant length s and base radius ρ , unrolls flat into a sector of a circle of radius s whose arc is the rim, $2\pi\rho$; a sector is to its circle as its arc is to the circumference, so its area is πs^2 times $2\pi\rho/(2\pi s)$, which is $\pi\rho s$.

A band between radii ρ_1 and ρ_2 is a large cone less a small one. Say ρ_2 is the larger radius, let the band have slant length s , and let the small cone that was cut away have slant length c . The large

cone then has slant length $c + s$, and the band has area $\pi\rho_2(c + s) - \pi\rho_1c$. The two cones have the same shape, so $\rho_1/c = \rho_2/(c + s)$, which gives $\rho_2c - \rho_1c = \rho_1s$. Putting that in, the band has area $\pi\rho_2s + \pi\rho_1s$, which is

$$\pi(\rho_1 + \rho_2) \cdot s$$

that is, the mean of the two circumferences, times the slant. When the two radii are equal the band is instead a piece of a cylinder, which unrolls into a rectangle of sides $2\pi\rho_1$ and s , and the same formula gives its area.

For a sphere, take a chord of the generating half circle, of length s , running from a point at distance ρ_1 from the axis to a point at distance ρ_2 , and let w be the length of its shadow on the axis. Let M be its midpoint, at distance $(\rho_1 + \rho_2)/2$ from the axis and at distance d from the centre. The line from the centre to M is perpendicular to the chord, because the two ends of the chord are equally far from the centre. Now compare two right triangles. The first has the line from the centre to M as its hypotenuse, of length d , with one leg along the axis; its other leg is the distance of M from the axis, $(\rho_1 + \rho_2)/2$. The second has the chord as its hypotenuse, of length s , and one of its legs is the chord's shadow, of length w . The hypotenuse of the first is perpendicular to the hypotenuse of the second, and the leg of the first that runs straight out from the axis is perpendicular to the shadow, which runs along it. So the first triangle is the second turned through a right angle and changed in size, the angles of the two agree, and those two legs are the same fraction of their hypotenuses: $((\rho_1 + \rho_2)/2)/d = w/s$. Multiplying out, $(\rho_1 + \rho_2) \cdot s = 2d \cdot w$. So the band swept by the chord has area

$$\pi(\rho_1 + \rho_2) \cdot s = 2\pi \cdot d \cdot w$$

exactly. Now take the arc of the half circle that generates the zone wanted, the whole half circle or the part of it between two heights a distance h apart, and inscribe in it a chain of equal chords. Equal chords of a circle are equally far from its centre, so they all lie at the same distance d , and their shadows, laid end to end along the

axis, add up to h , which for the whole sphere is the diameter. Each chord sweeps a band of area $2\pi d$ times its own shadow, so the area swept by the whole chain is $2\pi d \cdot h$. As the chords shorten they draw in towards the circle, and d approaches r .

The part of a sphere of radius r between two parallel planes a distance h apart has area $2\pi r \cdot h$.

It does not matter where the planes cut. A band around the equator and a cap at the pole have equal areas if they have equal heights, the cap making up in slope what it lacks in girth. And $2\pi r \cdot h$ is the area of the band of the enclosing cylinder between the same two planes. So the sphere and the side of its cylinder have equal areas between any two horizontal planes, and the whole sphere has the area of the whole side:

the surface of a sphere is $4\pi r^2$, four times its great circle

and with the two ends of the cylinder, each πr^2 , the cylinder's surface is $6\pi r^2$, of which the sphere's is two thirds.

The calculus agrees. A curve $y = f(x)$ turned about the axis sweeps out a surface whose area is the integral of $2\pi y \cdot \sqrt{1 + y'^2}$. The reason is the one that gave the length of a curve. There, a short chord over a short stretch of the axis had length close to $\sqrt{1 + y'^2}$ times that stretch, and the chain's lengths were sums for the integral of $\sqrt{1 + y'^2}$. Here the band swept by that chord has area close to $2\pi y$ times the chord's length, so the areas of the chain's bands are sums for the integral of $2\pi y \cdot \sqrt{1 + y'^2}$. On the half circle $y = \sqrt{r^2 - x^2}$, and y' is $-x/y$, as it was on the unit circle. Then $1 + y'^2$ is $(y^2 + x^2)/y^2$, which is r^2/y^2 , and $\sqrt{1 + y'^2}$ is r/y , so the integrand is $2\pi y \cdot r/y$, which is $2\pi r$, a constant. Archimedes' exact identity for chords is what stands behind the cancellation. There is also a check through the volume. A ball of radius $r + k$ contains the ball of radius r and a shell around it, whose volume is close to the surface times the thickness k ; so the surface should be the rate of change of the volume with the radius, and the derivative of $(4/3)\pi r^3$ is $4\pi r^2$.

* * *

The theorem on zones says more than it seems to. Press the sphere outwards onto its cylinder, each point moving horizontally away from the axis. A zone goes to the band of equal area, and since both are unchanged by turning about the axis, the part of the zone between two meridians goes to the same share of the band. So pieces bounded by parallels and meridians keep their areas, and any region can be approached by such pieces: the pressing keeps areas. Lambert made a map of the world on that principle in 1772: latitude is plotted by its height on the cylinder, which is the sine of the latitude, and the cylinder is unrolled. Shapes near the poles are squashed beyond recognition, and every country has its true share of the paper. On that map the share of the earth's surface between two latitudes can be read off a ruler. Half the earth's surface lies between 30 degrees north and 30 degrees south: the sine of 30 degrees is $1/2$, so that band of the map reaches half way up and half way down from the equator, and takes half the height of the paper. The two caps beyond 60 degrees, which look so large on the maps of the schoolroom, are together 13.4 per cent: each cap takes the top $1 - \sin 60^\circ$, which is 0.134, of its own half of the map, so the two together take 0.134 of the whole.

It also gives a meaning to a point chosen at random on a sphere, and a way of choosing one. If "at random" means that equal areas get equal chances, then the height of the point above the equatorial plane is evenly spread between $-r$ and r , and its longitude is evenly spread around the circle: choose those two numbers, and the point is chosen. Choosing the latitude evenly instead, which seems as natural, gives a degree of latitude near the pole the same chance as a degree at the equator, though the band it covers has far less height and so, by the theorem on zones, far less area; that crowds the points at the poles, and the two meanings differ as the three meanings of a random chord did.

* * *

Slices work in any number of dimensions, and the results run against intuition. Call the set of points $(x_1, x_2, \dots, x_n)$ with $x_1^2 + \dots + x_n^2$ at most 1 the unit ball in n dimensions, and write V_n for its volume, defined by slicing. The slice of the n -dimensional ball at height t is a ball of one dimension fewer, of radius $\sqrt{1-t^2}$. Its volume is V_{n-1} times that radius to the power $n-1$, because enlarging every length by a factor k multiplies a volume of dimension $n-1$ by k to the power $n-1$.

The slicing says why. In two dimensions the rule is the rectangle rule, where both sides grow by k and the area by k^2 . Now go up one dimension at a time. The slices of an enlarged body are the slices of the old one enlarged by k . So if the rule holds one dimension down, each slice of a body of dimension $n-1$ grows by k to the power $n-2$; the stack of slices is also k times as long; and the volume grows by k to the power $n-1$, so the rule holds here too.

Put $t = \sin \theta$. Then $1 - t^2$ is $\cos^2 \theta$, so the radius to the power $n-1$ is $\cos \theta$ to the power $n-1$; the substitution supplies one more factor $\cos \theta$, the derivative of $\sin \theta$; and θ runs from $-\pi/2$ to $\pi/2$, which by symmetry gives twice the integral from 0 to $\pi/2$. So

$$V_n = V_{n-1} \cdot \text{the integral of } (1-t^2)^{(n-1)/2} \text{ from } -1 \text{ to } 1 = V_{n-1} \cdot 2W_n$$

where W_n is the integral of $\cos^n \theta$ from 0 to $\pi/2$, which is the same as the integral of $\sin^n \theta$ and obeys $W_n = ((n-1)/n)W_{n-2}$. Starting from the segment, $V_1 = 2$:

$$V_2 = \pi = 3.14, V_3 = 4\pi/3 = 4.19, V_4 = \pi^2/2 = 4.93, V_5 = 8\pi^2/15 = 5.26, V_6 = \pi^3/6 = 5.17, V_7 = 16\pi^3/105 = 4.72$$

The volume of the unit ball rises to a largest value in five dimensions and then falls, and since the factors $2W_n$ shrink towards 0, it falls towards 0. In ten dimensions the ball has volume 2.55, inside a cube of side 2 and volume 1024: it fills a quarter of one per cent of its box. In twenty dimensions it fills one part in forty million. A cube in many dimensions is nearly all corners. And a ball in many dimensions is nearly all skin: the smaller ball that reaches 0.9 of the way out from the centre is the whole ball shrunk by the factor

0.9, so by the rule for enlargement it has 0.9^n of the volume, which in fifty dimensions is half of one per cent. All the rest lies in the thin outer layer. These are practical facts. A table of data with fifty columns is a cloud of points in fifty dimensions, and methods that rely on points having near neighbours, which work in two or three dimensions, fail there for the reason these numbers give.

A round hole is drilled through the centre of a ball, leaving a ring like a napkin ring, 6 centimetres high. What is the volume of the ring, and what does it depend on?

A bowl in the shape of a half sphere of radius r holds water to a depth d at its centre. What is the volume of the water?

What share of the earth's surface lies north of latitude 45 degrees, and north of which latitude does a tenth of it lie?

A sphere is cut by parallel planes into slices of equal thickness, like a loaf. Which slice has the most crust?

The volumes of the unit balls of even dimensions 2, 4, 6, ... are π , $\pi^2/2$, $\pi^3/6$, What is the pattern, and what is their sum?

EVERY POLYNOMIAL HAS A ROOT IN THE PLANE

The numbers of the plane were introduced to give $x^2 + 1 = 0$ a root, and they did more than was asked. Every number but 0 has its n n -th roots. So every quadratic has its two roots there, because completing the square asks for nothing beyond a square root. And the cubic formula, which passes through square roots of negative numbers on its way to real answers, makes sense at every step. Does the process have to be repeated? Is there some equation with coefficients in the plane whose roots lie outside it, in a still larger system that would have to be built in turn? There is not.

Every polynomial of degree at least 1, with coefficients in the plane, has a root in the plane.

Girard asserted something like it in 1629. D'Alembert attempted a proof in 1746, and Gauss, in his doctoral thesis of 1799, criticised every earlier attempt and gave his own, which by later standards has a gap as well. The proof below is due to Argand, of whom little is known beyond that he kept accounts in Paris, and who published it in 1806 and again in 1814. It is an argument about largest and smallest values, and the one piece it took for granted, that a

continuous function on a disc reaches its smallest value, was not provable until the real numbers were understood.

On the real line the situation is mixed. A polynomial of odd degree with real coefficients has a real root, since for large positive x it has the sign of its leading term, for large negative x the opposite sign, and a continuous function that changes sign is 0 somewhere between. A polynomial of even degree need not: $x^2 + 1$ stays above 0. In the plane there are no exceptions, and the reason is that the plane has room to go round.

* * *

Let $p(z) = z^n + a_{n-1}z^{n-1} + \dots + a_1z + a_0$, with the leading coefficient made 1 by division. The proof studies the size $|p(z)|$, a real function on the plane that is never negative, and shows three things: far out it is large; somewhere it is smallest; and where it is smallest it is 0.

Far out. Sizes multiply, and the size of a sum is at most the sum of the sizes. Turned round, the same fact gives a bound from below. The largest term is the whole sum less the other terms, so its size is at most the size of the sum plus the sizes of the others; and that says the size of the sum is at least the largest size less the others. Let S be the sum of the sizes of the coefficients $a_0, \dots, a_{n-1}$. For $|z|$ at least 1 a higher power of $|z|$ is no smaller than a lower one, and k is at most $n - 1$, so each $|a_k z^k|$ is at most $|a_k| \cdot |z|^{n-1}$. The terms after z^n therefore have sizes adding up to at most $S \cdot |z|^{n-1}$, and setting the size of z^n against them gives

$$|p(z)| \geq |z|^n - S \cdot |z|^{n-1} = |z|^{n-1}(|z| - S)$$

Outside the disc of radius $R = S + 1 + |a_0|$ the factor $|z| - S$ is above $1 + |a_0|$ and the factor $|z|^{n-1}$ is at least 1, so this exceeds $|a_0|$, which is the value of $|p|$ at 0. So whatever the smallest value of $|p|$ is, it is not to be found outside that disc, and the search is confined to the disc, edge included. Nor is it found on the edge: there $|z| - S$ is $1 + |a_0|$ and $|z|$ is at least 1, so $|p|$ is at least $1 + |a_0|$, again above

the value at 0. So the smallest value of $|p|$ on the disc, which the next step shows to exist, is taken at a point inside the edge and not on it. Such a point has room around it: a short step from it in any direction stays inside the disc. The last step of the proof takes exactly such a short step.

Somewhere smallest. A polynomial is continuous on the plane. The difference $p(z) - p(w)$ is a sum of terms $a_k(z^k - w^k)$, and $z^k - w^k$ is $z - w$ times $z^{k-1} + z^{k-2}w + \dots + w^{k-1}$. For z and w in the disc that second factor is a sum of k products, each of size at most R^{k-1} , so its size stays below a fixed bound. Hence the size of $p(z) - p(w)$ is at most a fixed multiple of $|z - w|$ for z and w in the disc, and $p(z)$ is within any tolerance of $p(w)$ once z is close enough to w . The same holds for the sizes, since $|p(z)|$ and $|p(w)|$ differ by at most $|p(z) - p(w)|$.

And a continuous real function on a closed disc reaches a smallest value. For an interval this was proved by halving; here the theorem on bounded sequences, used once for each coordinate, does the same work. The values of $|p|$ on the disc are bounded below by 0, so their negatives are bounded above by 0 and have a least upper bound, whose negative is the greatest lower bound m . For each k the number $m + 1/k$ is above m and so is not a lower bound, which means some point z_k of the disc has $|p(z_k)|$ below $m + 1/k$. That gives points $z_1, z_2, \dots$ of the disc with $|p(z_k)|$ approaching m .

Their real coordinates are a bounded sequence, so some subsequence of the points has real coordinates approaching a number; of that subsequence, some further subsequence has imaginary coordinates approaching a number as well. So a subsequence of the points approaches a point z_0 . It is a point of the disc because the edge is included: points of size at most R cannot approach a point of size above R . Along that subsequence the values $|p(z_k)|$ approach $|p(z_0)|$ by continuity, and they approach m by the way the points were chosen, so $|p(z_0)| = m$.

Where it is smallest it is 0. This is Argand's step, and it is where the plane differs from the line. Suppose $p(z_0)$ is not 0. Write $z = z_0 + h$, so that h is the step away from z_0 . Put $z_0 + h$ for z in p , multiply

out every power of $z_0 + h$, and collect the result by powers of h . This is a finite piece of algebra, and the part with no h in it is what is left when $h = 0$, namely $p(z_0)$:

$$p(z_0 + h) = p(z_0) + c \cdot h^k + (\text{terms with higher powers of } h)$$

where $c \cdot h^k$ is the first term after the constant whose coefficient is not 0. There is one, since p is not constant. Now choose the direction of h . Numbers of the plane have k -th roots, and c is not 0, so the number $-p(z_0)/c$ has a k -th root d , which is a number with $c \cdot d^k = -p(z_0)$: moving from z_0 in the direction d makes the term $c \cdot h^k$ point exactly against $p(z_0)$. Put $h = td$, with t a small positive real number. Then $c \cdot h^k = -t^k \cdot p(z_0)$, and

$$p(z_0 + td) = p(z_0)(1 - t^k) + (\text{terms with } t^{k+1} \text{ and higher})$$

Once t is below 1 the factor $1 - t^k$ is positive, so the size of the first part is $|p(z_0)|(1 - t^k)$. The rest is a finite sum of terms, each a fixed number times a power of t that is $k + 1$ or higher, and for t below 1 such a power is at most t^{k+1} . So the rest has size at most $K \cdot t^{k+1}$, where the constant K is the sum of the sizes of those fixed numbers. So $|p(z_0 + td)|$ is at most the two sizes added, $|p(z_0)|(1 - t^k) + K \cdot t^{k+1}$, and regrouping turns this into $|p(z_0)| - t^k(|p(z_0)| - Kt)$, which is less than $|p(z_0)|$ as soon as $K \cdot t$ is below $|p(z_0)|$. A smaller value than the smallest has been found. So $p(z_0)$ was 0 after all.

On the real line the argument stops at the choice of direction. At the smallest value of $x^2 + 1$, which is at 0, the first term after the constant is x^2 , and to make it point against the constant 1 requires a square root of -1 . The line has two directions, and both go uphill.

* * *

A root z_1 gives a factor, as was shown when the roots of unity were counted. Since $p(z_1)$ is 0, $p(z)$ is the same as $p(z) - p(z_1)$, which is a sum of terms $a_k(z^k - z_1^k)$, and each $z^k - z_1^k$ has the

factor $z - z_1$. So $p(z) = (z - z_1)q(z)$, where q has degree $n - 1$ and leading coefficient 1. If that degree is at least 1, q has a root z_2 in turn and gives up a factor $z - z_2$, and the step repeats until the degree has run down to 0 and only the constant 1 is left. So

$$p(z) = (z - z_1)(z - z_2) \cdots (z - z_n)$$

Every polynomial of degree n splits completely, with exactly n roots when they are counted as often as they occur. The fractions, the real numbers and the fields of remainders all lack this property.

When the coefficients are real, the mirror image of a root is a root. Taking mirror images respects sums and products and leaves real numbers alone, so the mirror image of $p(z_1)$ is $p(z_1^*)$: every z_1 in the expression turns into z_1^* and the coefficients stay as they are. If $p(z_1)$ is 0, its mirror image is 0 as well, and that says $p(z_1^*) = 0$. So the roots off the real line come in mirror pairs, and a pair z_1 and z_1^* gives the factor $(z - z_1)(z - z_1^*) = z^2 - 2xz + (x^2 + y^2)$, where $z_1 = x + iy$, a quadratic with real coefficients. So every real polynomial is a product of real factors of degree 1 and 2. Leibniz, in 1702, believed that $x^4 + 1$ was an exception. Its roots are the four odd eighth roots of 1, at $(\pm 1 \pm i)/\sqrt{2}$, and pairing each with its mirror image gives

$$x^4 + 1 = (x^2 + \sqrt{2}x + 1)(x^2 - \sqrt{2}x + 1)$$

The cubic $x^3 - 2x - 5$ on which Newton tried his method has the real root 2.0946 and the mirror pair $-1.0473 \pm 1.1359i$, of size 1.5450; the three add up to 0 and multiply to 5, as the coefficients require.

The proof shows that roots exist and gives no formula for them. For degrees up to four there are formulas, built from roots of numbers. For degree five there is none, and the reason is the subject of the end of this development. But the roots can always be computed. The tangent method works in the plane exactly as on the line, $z' = z - p(z)/p'(z)$, the rules for derivatives of polynomials being pure algebra. Take $x^2 + 1$, whose roots no real start can reach. From 1 the next guess is 0, where the tangent is flat, and from $1/\sqrt{3}$ the

guesses swing between $1/\sqrt{3}$ and $-1/\sqrt{3}$ for ever. The start $1 + i$, by contrast, leads to i in a few steps. For Newton's cubic, the start i gives

$$\begin{aligned} & -1 + 0.4i, -0.563 + 1.772i, -0.778 + 1.186i, \\ & -1.042 + 1.090i, -1.0469 + 1.1372i, -1.04727 + 1.13594i \end{aligned}$$

and the seventh guess is right to twelve places, the eighth to twenty-four. Which root a start leads to is another matter. Cayley asked in 1879 how the plane is divided among the roots, settled it for quadratics, where the dividing line is the perpendicular bisector of the two roots, and reported that the cubic appeared to present considerable difficulty. It does: for $z^3 - 1$ the three regions are so interleaved that every point on the boundary of one is on the boundary of all three. That was settled only around 1918, when Fatou and Julia built a theory of a map of the plane applied over and over. The fact is stated here as they proved it.

* * *

There is a second way to see the theorem, and it explains where the roots are. It is close kin to the idea of Gauss's first proof, which is that on a large circle $p(z)$ behaves as z^n does. Let z go once round a circle of radius r about the origin, and watch the path of $p(z)$. For very small r , $p(z)$ stays close to a_0 , and if a_0 is not 0 its path is a small loop that does not go round the origin at all. For very large r , $p(z)$ is z^n times a factor close to 1, and z^n goes round the origin n times as z goes round once, so the path of $p(z)$ winds round the origin n times. As r grows from small to large the loop changes continuously, and a loop cannot change the number of times it winds round a point without passing through the point. At each radius where that happens, some z on the circle has $p(z) = 0$. With roots counted as often as they occur, the winding number of the loop at radius r is the number of roots inside the circle. Making this precise takes a definition of the count that does not appeal to the eye, and that is built next; the proof above does not depend on it.

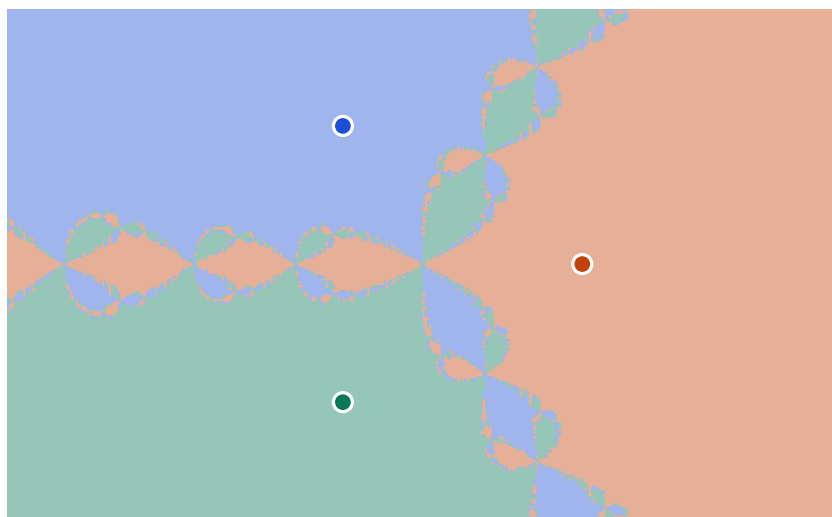


Figure 45.1. The plane shaded by the root of $z^3 = 1$ that the tangent method reaches from each starting point, for starts up to 2.6 to either side of the origin and 1.6 above and below it. The dots are the three roots. Near each root the method goes to that root. Between them the three regions are interleaved at every scale, and every boundary point of one region is a boundary point of all three. *Assumes:* a grid of 360 by 220 starting points and at most forty steps from each; the few starts that have not settled by then are left white

For Newton's cubic the loop winds 0 times for r below 1.545, twice between 1.545 and 2.0946, where the mirror pair is inside and the real root is not, and three times beyond.

Factorise $x^4 + 4$ into two quadratics with whole-number coefficients, and $x^6 + 1$ into real quadratics.

All the roots of $z^n + a_{n-1}z^{n-1} + \dots + a_0$ lie in the disc whose radius is 1 plus the largest of the sizes of the coefficients. Why?

The exponential $E(z) = 1 + z + z^2/2! + \dots$, for z in the plane, is never 0. At which step does the proof above fail for it?

If a polynomial with real coefficients has odd degree, the pairing of roots gives a second proof that it has a real root. What is it, and which properties of the real numbers does each of the two proofs use?

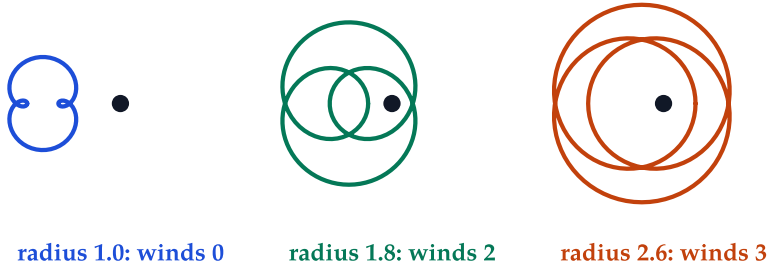


Figure 45.2. The path of $z^3 - 2z - 5$ as z goes once round a circle about the origin, for three radii. The dot is the origin. At radius 1 the path does not go round it. At radius 1.8, with the mirror pair of roots inside the circle, it goes round twice, and at radius 2.6, with all three roots inside, three times. *Assumes:* three radii chosen away from 1.545 and 2.0946, the sizes of the roots, at which the path passes through the origin; each panel has its own scale

How many times does the path of $z^5 + z + 1$ wind round the origin as z goes round the circle of radius 0.9, of radius 1.1, and of radius 2? The polynomial is $(z^2 + z + 1)(z^3 - z^2 + 1)$.

WHAT A LOOP COUNTS

The second proof that every polynomial has a root watched a loop and counted how often it went round the origin, and the counting was left as a picture. The picture is right, and what it needs is a definition of the count that does not appeal to the eye, together with two facts: that the count is a whole number, and that it cannot change while the loop is moved continuously without meeting the origin. Both come from the tools already in hand.

A *loop* is a continuous map z from an interval of times, say 0 to 1, into the plane, with $z(1) = z(0)$. Take a loop that never passes through the origin. At each time it has a direction, the point $z(t)/|z(t)|$ of the unit circle, and the question is how many times that direction goes round while t runs from 0 to 1. The direction alone does not answer it: the point of the circle at the end is the point at the start, and the circle does not record how many turns were taken to return. What records the turns is an angle that is allowed to grow past a full turn and keep growing.

For a loop avoiding the origin there is a continuous function $\theta(t)$ with $z(t) = |z(t)| \cdot E(i\theta(t))$, and any two such differ by a constant, which is a whole multiple of 2π .

Here is the construction. The function z is continuous on an interval containing its ends, so it is uniformly continuous there, and

its size, being continuous and never 0, has a smallest value m above 0. Uniform continuity supplies a width such that any two times closer together than that width have their points closer together than half of m . Cut the interval from 0 to 1 into pieces narrower than that width, at times $t_0 = 0, t_1, \dots, t_k = 1$. Take one piece, beginning at t_j . Throughout it the point $z(t)$ stays within half of m of $z(t_j)$, while $z(t_j)$ itself is at least m from the origin. Seen from the origin, then, the point stays inside a disc whose radius is less than half the distance to its centre. The directions to such a disc lie within a third of a right angle of the direction to its centre, that being the angle whose sine is a half; in particular the point never crosses the ray from the origin opposite to $z(t_j)$. So the directions on the piece fill an arc of less than two thirds of a right angle, which is shorter than a right angle.

Such an arc lies inside one of the four half planes x above 0, x below 0, y above 0, y below 0. The reason is that the four axis directions, right, up, left and down, are a quarter turn apart, so an arc shorter than a quarter turn contains at most one of them. If it contains the direction right, say, it reaches less than a quarter turn to either side of it, and so stays where x is above 0; the other three go the same way. If it contains none of the four, it lies within a single quarter, where two of the half planes serve. On each of the four a formula gives the angle continuously: $\arctan(y/x)$ where x is above 0, $\arctan(y/x) + \pi$ where x is below 0, $\pi/2 - \arctan(x/y)$ where y is above 0, and $-\pi/2 - \arctan(x/y)$ where y is below 0. So on each piece there is a continuous angle. At a shared end the angle of one piece and the angle of the next describe the same point, so they differ by a whole multiple of 2π , and shifting the later one by that multiple makes the two agree there. Fitting them together gives θ on the whole interval.

Suppose two such functions are given. At each time both are angles of the same point $z(t)$, so their difference is a whole multiple of 2π , and the difference is continuous. A continuous function that took two different whole multiples of 2π would take every value between them, and not all of those are whole multiples of 2π . So

the difference is constant.

Since $z(1) = z(0)$, the two angles $\theta(1)$ and $\theta(0)$ describe one point, so they differ by a multiple of 2π , and that multiple does not depend on which θ was chosen, since two choices differ by a constant, and the constant cancels in $\theta(1) - \theta(0)$. Define

the winding number of the loop about the origin is $(\theta(1) - \theta(0))/2\pi$

a whole number. For the loop $t \mapsto E(2\pi int)$ it is n , the angle being $2\pi nt$. For a loop that stays on one side of a line through the origin it is 0: with the axes turned so that one of them lies along that line, the formulas of the construction make the angle a continuous function of the point on that side, so $\theta(1)$ and $\theta(0)$, being that function at the same point, are equal. The winding number about any other point w is the winding number of the loop $z(t) - w$ about the origin, defined whenever the loop avoids w .

* * *

If two loops avoiding the origin can be carried into one another by a continuous family of loops, none of which passes through the origin, they have the same winding number.

Let the family be $z_s(t)$, continuous in the pair (s, t) for s from 0 to 1, with every z_s a loop avoiding the origin. The whole family is a continuous map on a square containing its edges. A continuous function on an interval with its ends has a smallest value, and so does one on a closed disc. The argument for both runs here word for word: a sequence of points of the square has a subsequence whose first coordinates settle, and a further subsequence whose second coordinates settle too, and the limit point lies in the square. So $|z_s(t)|$ has a smallest value m above 0, and the map is uniformly continuous on the square by the same run of the argument. Run the construction above with the width chosen for that m : it gives, for each s , an angle θ_s built from the same cutting of the time interval. The shifts by multiples of 2π do not alter how much the angle

changes across a piece, so $\theta_s(1) - \theta_s(0)$ is the sum, over the pieces, of the change of angle across each piece.

Fix one piece and one s . For every u close enough to s , the points $z_u(t)$ on that piece still stay within half of m of $z_s(t_j)$, so one half plane, and the one arctan formula on it, serves for all those u together. The change of angle across the piece is that formula at the point $z_u(t_{j+1})$ minus the same formula at $z_u(t_j)$, and both points move continuously with u . So each change is a continuous function of s , and so is their sum. The winding number, which is $(\theta_s(1) - \theta_s(0))/2\pi$, is therefore a continuous function of s taking whole-number values, and by the intermediate value theorem it is constant.

Two consequences follow at once. A loop that can be shrunk to a point without meeting the origin winds 0 times, since a loop that stays at one point has a constant angle, and so winding 0. And if a loop avoids the origin and lies inside a disc that does not contain the origin, it winds 0 times, since it can be shrunk inside that disc.

Now the second proof of the theorem on roots can be given in full, with nothing resting on the picture. Let p have degree n and leading coefficient 1, and suppose p has no root. Then for every r the loop L_r given by $t \mapsto p(r \cdot E(2\pi it))$ avoids the origin, and the family is continuous in the pair (r, t) , so all the L_r have the same winding number. At $r = 0$ the loop is the constant $p(0)$, of winding 0. For large r write $p(z) = z^n(1 + q(z))$ with q a sum of negative powers of z , whose size is below 1 once r is large enough. Then $p(r \cdot E(2\pi it))$ is the product of two loops. The first, $r^n E(2\pi int)$, winds n times. The second, $1 + q$, stays closer than 1 to the number 1, hence in the half plane of numbers with positive real part, and so winds 0 times. When two loops are multiplied their angles add, since $E(i\alpha) \cdot E(i\beta) = E(i(\alpha + \beta))$; so the sum of the two angles is a continuous angle for the product, and the winding numbers add. So the winding is $n + 0 = n$. If n is at least 1 the two counts disagree, and the supposition fails.

* * *

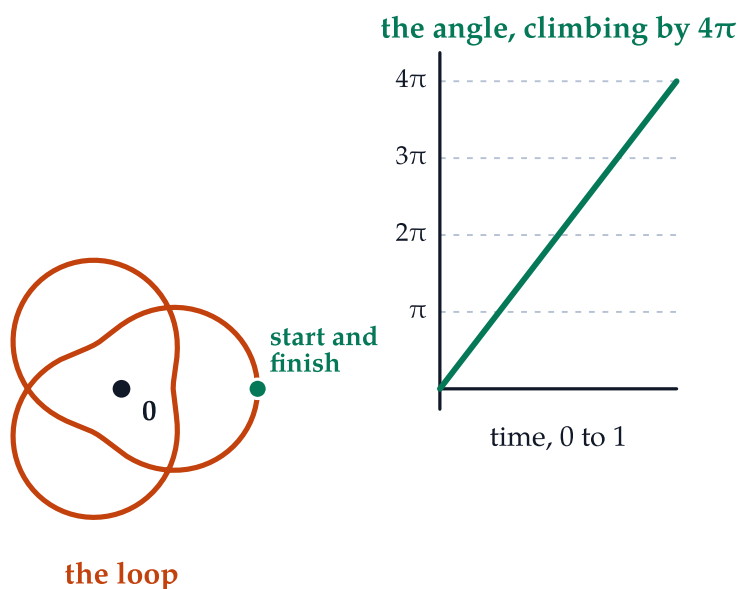


Figure 46.1. A loop that goes twice round the origin, and the angle that records it. On the left the path returns to where it started, and the direction it points returns with it, so the start and finish alone cannot say how many turns were taken. On the right the angle is allowed to keep climbing: it ends 4π above where it began, and the winding number is that rise divided by 2π . Assumes: the loop drawn is $t \mapsto (1 + 0.45 \cos 6\pi t) \cdot E(4\pi i t)$, chosen so the two turns are visible as separate strands; the angle is built piece by piece, as the text builds it

Its first use beyond that proof is a theorem about maps of a disc that looks unrelated.

A continuous map of a disc into itself leaves some point where it was.

Let D be the disc of points with $|z|$ at most 1, and suppose, aiming at a contradiction, that f carries D into D continuously with $f(z)$ never equal to z . Then $h(z) = z - f(z)$ is continuous on D and never 0. Consider the loops H_r , given by $t \mapsto h(r \cdot E(2\pi i t))$, for r from 0 to 1: a continuous family avoiding the origin, so all have one winding number, and at $r = 0$ the loop is constant, so the number is 0.

Now compute it at $r = 1$. For $|z| = 1$, look at the real part of $z^* \cdot h(z)$, which is $|z|^2$ — the real part of $z^* \cdot f(z)$, that is 1 — the real part of $z^* f(z)$. Since $|z^* \cdot f(z)| = |f(z)|$ is at most 1 , the real part of $z^* \cdot f(z)$ is at most 1 , so the real part of $z^* \cdot h(z)$ is at least 0 . It is 0 only when $z^* \cdot f(z)$ has real part 1 , and a number of size at most 1 with real part 1 is the number 1 itself; multiplying $z^* \cdot f(z) = 1$ by z , with $z \cdot z^* = 1$, turns it into $f(z) = z$, excluded. So the real part of $z^* h(z)$ is positive for every z on the circle. Now join z to $h(z)$ by the family $(1 - s) \cdot z + s \cdot h(z)$, for s from 0 to 1 . It never vanishes. At $s = 0$ it is z , which is not 0 . At any other s , if it vanished then $h(z)$ would equal $-((1 - s)/s) \cdot z$, which is z multiplied by a number that is negative or 0 , and the real part of $z^* \cdot h(z)$ would then be negative or 0 , which it is not. In particular $h(z)$ and z are never opposite in direction. That family carries the loop $t \mapsto E(2\pi it)$, of winding 1 , to the loop H_1 , so H_1 winds once. One loop cannot wind both 0 and 1 times, so f has a fixed point.

Brouwer proved this in 1910 for every dimension. In one dimension it is the intermediate value theorem applied to $f(x) - x$, which is at least 0 at one end of the interval and at most 0 at the other. In the plane it is the winding number. A cup of coffee stirred and allowed to settle has some particle of liquid where it began; two copies of the same map, one laid crumpled on the other, have some point of the country lying over itself.

A second use is a statement about the disc and its rim.

No continuous map of the disc onto its rim leaves every point of the rim where it was.

Suppose g were such a map. Every value of g lies on the rim, so the loops $t \mapsto g(r \cdot E(2\pi it))$, for r from 0 to 1 , form a continuous family avoiding the origin, and all have the same winding number. At $r = 0$ the loop is constant, of winding 0 . At $r = 1$ the points $E(2\pi it)$ are on the rim, where g moves nothing, so the loop is $t \mapsto E(2\pi it)$, of winding 1 . The two counts disagree, so there is no such g . That fact is the usual route to the fixed-point theorem in higher

dimensions, where no angle is available and the count has to be built another way.

A third use goes back to the polynomials and makes the counting exact.

If p has no root on the circle of radius r about the origin, the winding number of the loop $t \mapsto p(r \cdot E(2\pi it))$ is the number of roots inside that circle, counted with repetition.

Write p as its leading coefficient times the product of the factors $z - r_j$ over its n roots, which the theorem on roots supplies. The angle of a product is the sum of the angles, so the winding number of the loop is the sum, over the roots, of the winding numbers of the loops $t \mapsto r \cdot E(2\pi it) - r_j$. Each of those is the circle of radius r shifted by $-r_j$, and its winding about the origin is the winding of the circle about the point r_j . Suppose first that r_j is inside the circle. As s falls from 1 to 0 the family $t \mapsto r \cdot E(2\pi it) - s \cdot r_j$ never meets the origin, because $|s \cdot r_j|$ is smaller than the radius r ; and the family ends at the circle about the origin, which winds once. So the loop winds once. Suppose instead that r_j is outside. Then the circle lies inside a disc missing r_j , so the shifted loop lies inside a disc missing the origin, and it winds 0 times. Adding gives the count.

The same statement holds for functions built from power series rather than polynomials, and it is how roots are located in practice. It allows a comparison: if at every point of a circle the difference of two functions is smaller in size than one of the two, then the two have the same number of roots inside. That is Rouché's theorem of 1862.

For the loop $t \mapsto E(2\pi it) + 2$, what is the winding number about 0, about 2 and about 4, and why is there no answer at all about 3?

A loop crosses the positive horizontal axis four times, twice upward and twice downward. What is its winding number about the origin, and why does the answer not depend on anything else about the loop?

Show that $z^5 + 3z + 1$ has exactly one root inside the circle of radius 1, by comparing it on that circle with $3z$.

Does the map of the circle into itself that sends z to z^4 leave some point where it was? Decide, and say what the winding numbers of $z \mapsto z^4 - z$ on the circles of radius 0.9 and 1.1 tell, and why that loop has no winding number on the circle of radius 1.

Why does the argument for a fixed point break down for a map of the plane into itself, and what map shows that it must?

THE REALS, BUILT

Nested intervals whose lengths shrink past every tolerance close on exactly one number. That sentence was laid down as an axiom about a system called the real numbers, and the limit laws, the area under the parabola, the derivative, the logarithm, e and π , and the moving point on the circle were all computed inside that system. Nobody exhibited the system. A description with no object behind it may describe nothing, and then every computation made under it is a computation about nothing; Russell said that postulating what is wanted has “the advantages of theft over honest toil”. What has to be built is a system that contains the fractions, carries addition, multiplication and order obeying the familiar rules, and satisfies the axiom as a theorem, and it has to be built from materials that exist: fractions, which are pairs of whole numbers with a rule for sameness, and sequences of fractions, which are rules assigning a fraction to each whole number.

* * *

The averaging sequence for the square root of 2 shows what is missing. Each term is the average of the one before and 2 divided by it, so the terms are $3/2$, $17/12$, $577/408$, $665857/470832$, ..., and their squares exceed 2 by $1/4$, $1/144$, $1/166464$ and $1/221682772224$;

$17^2 = 289$ is one more than $2 \cdot 144$, for instance. As was shown when the axiom failed for the fractions, each term x has square above 2, the next term lies between $2/x$ and x , and these intervals are nested, so every later term lies in the interval from $2/x$ to x ; that interval is shorter than the excess $x^2 - 2$, since x is above 1, and each excess is less than a thirtieth of the one before. The distance from any term to any later one is therefore less than that term's excess.

So the terms huddle, in the sense already given that name: for every tolerance there is a stage past which any two of them, and not only neighbours, are within the tolerance of each other. That is a statement about the sequence alone, and it mentions no destination.

Every sequence that converges huddles. The converse fails among the fractions. The averaging sequence huddles and converges to no fraction, since a fraction it converged to would have square 2, and no fraction does. The terms gather at a place that is not in the system they belong to.

So let the huddle be the number. The obstacle is that many sequences gather at the same place. The truncations huddle too: each repeats the digits of the one before and adds one more, so any two from the one with k places onward lie within $1/10^k$ of each other. The rule that makes them names no destination: the k -th is the largest decimal with k places whose square falls below 2. The squares 1, 1.96, 1.9881, 1.999396, 1.99996164 therefore climb. Adding one unit of the last place to the k -th truncation gives a decimal whose square is above 2, so 2 lies between the two squares. The two squares differ by the difference of the two decimals times their sum, and the decimals differ by $1/10^k$ while their sum is below 4. That puts the k -th square within $4/10^k$ of 2. So the squares climb to 2, and the terms are different fractions from the averaging sequence at every stage.

Two huddling sequences *head for the same place* when their difference tends to 0, and for these two that can be checked without naming a destination: the difference of their squares tends to 0, since both squares tend to 2, and the difference of the terms is the

difference of the squares divided by the sum of the terms. That sum is at least 2, since every term of either sequence is at least 1, so the difference of the terms is at most half the difference of the squares, and tends to 0 with it.

Heading for the same place behaves as a rule for sameness must. Every huddling sequence heads for the same place as itself, since its difference from itself is 0, 0, 0, If a heads for the same place as b , then b heads for the same place as a , since the two differences have the same size. And if a heads for the same place as b , and b as c , then a heads for the same place as c , by splitting the tolerance: past the stage at which the terms of a are within half the tolerance of those of b , and those of b within half of those of c , the terms of a are within the tolerance of those of c . So it sorts the huddling sequences of fractions into classes, and

a real number is the class of all huddling sequences of fractions that head for the same place as a given one.

The square root of 2 is the class of all huddling sequences of fractions whose terms are positive from some stage on and whose squares tend to 2; the averaging sequence is one member and the truncations are another. Each fraction q is a real number too, the class of the constant sequence $q, q, q, \dots$, so the fractions sit inside the new system, and a real number is a fraction exactly when its class contains a constant sequence.

The same act made the fractions out of pairs of whole numbers, the clocks out of the integers and a collapsed group out of a group: sort into classes by a rule for sameness, define the arithmetic on chosen members, and show that the choice does not matter.

* * *

Add two real numbers by adding member sequences term by term, and multiply them term by term. Three checks are needed before either sentence means anything.

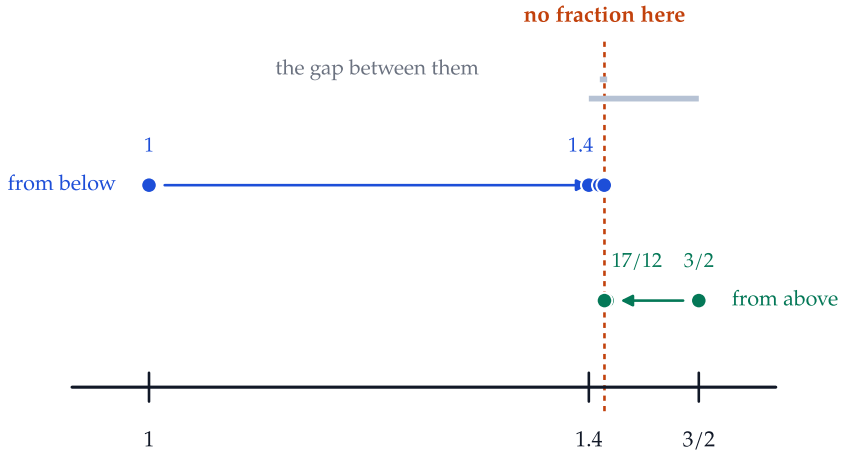


Figure 47.1. The averaging sequence $3/2, 17/12, 577/408$ comes down from above and the truncations $1, 1.4, 1.41, 1.414$ climb from below; no term of either is a term of the other, and the distance between the two sequences shrinks past every tolerance, so they head for one place. That shared destination, which no fraction occupies, is what the class of all such sequences is made to be. *Assumes:* that the two sequences go on huddling past the terms drawn, which the argument in the text establishes; and that a number line can be drawn at all before the numbers on it have been built, which is the picture's own small theft

First, the termwise sum and product of huddling sequences huddle. For the sum, split the tolerance: past the later of the two stages for half the tolerance, the sum of two terms differs from the sum of two later terms by at most the two separate differences. For the product, write $ab - a'b'$ as $a(b - b') + (a - a')b'$, where the primes mark later terms. A huddling sequence is bounded, because past its stage for the tolerance 1 every term is within 1 of one fixed term, and there are finitely many terms before that stage; so there is a positive fraction M that no term of either sequence exceeds in size. The first piece, $a(b - b')$, then has size at most M times the size of $b - b'$, and the second at most M times the size of $a - a'$. Past the later of the stages at which the two sequences

huddle to within the tolerance divided by $2M$, each piece is below half the tolerance, and $ab - a'b'$ is below the whole of it.

Second, the choice of members. If a and a' head for the same place, and b and b' do, then $a + b$ and $a' + b'$ head for the same place, since the difference of the sums is the sum of two differences that tend to 0; and $ab - a'b' = a(b - b') + (a - a')b'$ tends to 0, since a and b' are bounded. So sums and products of classes do not depend on the members used. Boundedness is where huddling is needed: the sequences $1/n$ and $0, 0, 0, \dots$ head for the same place, but multiplied term by term by $1, 2, 3, \dots$, a sequence that is not bounded and does not huddle, they give $1, 1, 1, \dots$ and $0, 0, 0, \dots$, which do not.

Third, the rules. Addition and multiplication of classes are commutative, associative and distributive, because at every term they are fractions obeying those rules; the class of $0, 0, 0, \dots$ is a zero, the class of $1, 1, 1, \dots$ is a one, and the negative of a class is the class of the negated sequence.

Reciprocals need a lemma. Suppose a class is not the class of 0, so that a member sequence a does not tend to 0. Tending to 0 would mean that, for every tolerance, the terms are all smaller than it from some stage on. So there is a tolerance ε for which that never happens: however late the stage, some term beyond it has size at least ε . Let N be the huddling stage for $\varepsilon/2$, and pick one term of size at least ε that comes after N . Every term past N is within $\varepsilon/2$ of the picked term, so every term from that stage on has size at least $\varepsilon/2$ and the same sign as the picked term. A sequence that does not tend to 0 settles on one side of 0, at a definite distance.

Earlier terms may be 0 and have no reciprocal, so build a new sequence from the reciprocals of the terms from that stage on, with anything written in the earlier places. It huddles. The difference of two reciprocals is the difference of the originals divided by their product, and the product has size at least $\varepsilon^2/4$, so two reciprocals differ by at most $4/\varepsilon^2$ times what the originals differ by, which falls below every tolerance. And the class of the reciprocals times the class of a is the class of a sequence that is 1 from some stage on,

which is the class of 1. So every real number other than 0 has a reciprocal, and the real numbers form a field.

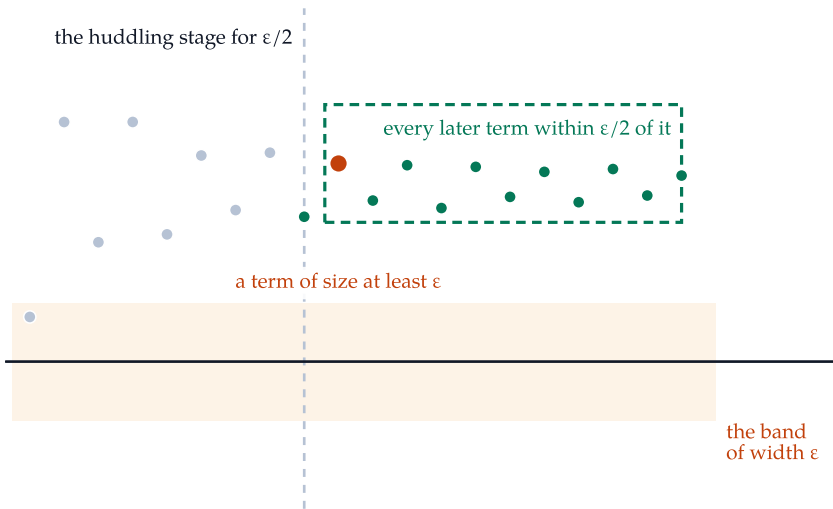


Figure 47.2. A huddling sequence whose terms do not tend to 0. Some term past the huddling stage has size at least the tolerance, every later term lies within half the tolerance of it, so every later term stays on the same side of 0 and outside the shaded band. That is what makes the reciprocals huddle, and what makes every real number positive, zero or negative. *Assumes:* that the drawn tail stands in for an endless one; the lemma is about every term past the stage, and twenty dots illustrate it without proving it

Order comes from the same lemma. Call a class positive when one of its member sequences, and hence every one, is from some stage on above a fixed positive fraction; members differ by sequences tending to 0, so if one member is from some stage on above a positive fraction, any other member is from some later stage on within half that fraction of it, and so above half that fraction. By the lemma every class is exactly one of positive, zero and negative, and sums and products of positive classes are positive, since past the later of the two stages one sequence stays above a positive fraction r and the other above a positive fraction s , and then their termwise sum stays above $r + s$ and their termwise

product above rs . One real number is below another when the difference is positive. This is an order, it agrees with the order of the fractions on constant sequences, and it obeys the rules that order obeyed. And for every positive real number there is a whole number n with $1/n$ below it, because the number is from some stage on above some fraction p/q , and $1/(q+1)$ is below p/q . It is no longer an assumption that enough copies of any positive quantity exceed 1; the construction inherited it from the whole numbers.

Take a huddling sequence of fractions, positive from some stage on, whose squares tend to 3. Is the square of its class equal to the class of 3? And is its class a fraction?

* * *

Now the axiom, after two lemmas. Every real number is approached by fractions: given a class, a member sequence a , and a tolerance, let N be the huddling stage of a for that tolerance; the constant sequence $a_N, a_N, \dots$ differs from a by a sequence whose terms past N are within the tolerance, so the fraction a_N is within the tolerance of the class.

And a huddling sequence of real numbers converges to a real number. Let $r_1, r_2, \dots$ huddle, and for each k let q_k be found as follows: the first lemma says some fraction lies within $1/k$ of r_k ; among all such fractions take those whose denominator in lowest terms is smallest, and of these, which are finitely many because fractions with one denominator d stand at least $1/d$ apart, take the least. That is a rule, not a run of choices made one after another. The fractions q_k huddle. To get from q_k to q_j , go from q_k to r_k , then from r_k to r_j , then from r_j to q_j : the first step is at most $1/k$, the last is at most $1/j$, and the middle one is the distance between two of the r 's. Given a tolerance, all three steps are below a third of it once k and j are past some stage, the outer two because $1/k$ and $1/j$ fall below every positive number, and the middle one because the r 's huddle. So $q_1, q_2, \dots$ is a huddling sequence of fractions, and its

class is a real number c .

The distance from r_k to c is at most $1/k$ plus the distance from q_k to c . Now $c - q_k$ is the class of the sequence whose j -th term is $q_j - q_k$. Given a tolerance, let k be past the huddling stage of the q 's for it; then every term $q_j - q_k$ with j past that stage is within the tolerance of 0, and so the distance from q_k to c is at most the tolerance. Both parts, $1/k$ and the distance from q_k to c , fall below every tolerance as k grows. So r_k converges to c . Among the real numbers, huddling and converging are the same.

The axiom follows. Take nested intervals with left ends $x_1 \leq x_2 \leq \dots$ and right ends $y_1 \geq y_2 \geq \dots$, with lengths shrinking past every tolerance. Any two left ends from the k -th on lie in the k -th interval, so they are within its length of each other, and the lengths shrink past every tolerance. So the left ends huddle, and by the second lemma they converge to a real number c . Each x_k is at most c , since all later left ends are at least x_k ; and c is at most each y_k , since every left end is at most every right end. So c lies in every interval, and no second number does, since two numbers in every interval differ by less than every length. The sentence that was laid down as an axiom is a theorem about an object made of fractions.

The axiom has a second form: every collection of real numbers that has something in it and an upper bound has a least upper bound. The two forms are equivalent. Take a collection S with an upper bound B and a member s , and halve the interval from s to B . If some member of S lies at or beyond the midpoint, keep the upper half; otherwise keep the lower half. Two things stay true of every kept interval. Its right end is an upper bound of S : when the upper half is kept the right end is unchanged, and when the lower half is kept the new right end is the midpoint, which no member reaches. And it contains a member of S : when the upper half is kept, the member found at or beyond the midpoint lies in it, since no member passes the right end; when the lower half is kept, the member held by the previous interval lies below the midpoint and so in the lower half.

The kept lengths are the first length halved again and again, so

the k -th is the first length divided by 2^k , which is at most the first length divided by k , since 2^k is at least k . It was proved above that some $1/n$ falls below any positive number, and that puts the kept lengths below every tolerance. The intervals close on a number c . No member of S exceeds c . Suppose a member m did. Some kept interval is shorter than $m - c$, and it contains c , so its right end is below m ; yet that right end is an upper bound of S , which m contradicts. No number d below c is an upper bound either. Some kept interval is shorter than $c - d$, and it contains c , so it lies entirely above d ; and it contains a member of S , which is therefore above d . So c is the least upper bound.

In the other direction, given nested intervals, the left ends are bounded by y_1 and have a least upper bound c . Every right end is an upper bound of the left ends, since $x_j \leq x_k \leq y_k$ when $j \leq k$ and $x_j \leq y_j \leq y_k$ when $j > k$; so c , being least, is at most every right end, and at least every left end, and it lies in every interval. The shrinking lengths are needed only for uniqueness.

* * *

Nothing new was assumed: fractions, rules assigning them to whole numbers, and collections of such rules were the only materials, so the description describes something.

The plane has an object too. The pairs $a + bi$ were built with fractions in both places, while the sine and cosine of an arbitrary angle, the moving point on the circle and the exponential of an imaginary number all needed real numbers there. Every check that made the pairs of fractions a field used only rules that the real numbers also obey, together with one fact about order: $a^2 + b^2$ is 0 only when a and b both are, which gives each nonzero pair the reciprocal $(a - bi)/(a^2 + b^2)$. Squares of nonzero real numbers are positive, so the same checks go through, and the pairs of real numbers form a field containing the pairs of fractions.

And the diagonal of the unit square has a length. Square the averaging sequence term by term: the squares $9/4$, $289/144$,

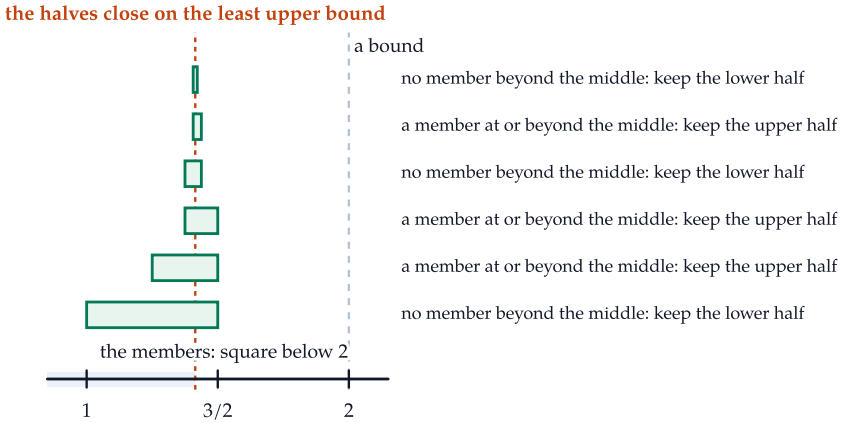


Figure 47.3. The collection of fractions whose square is below 2, and its bound 2. The interval from the member 1 to the bound is halved six times: a half is kept if it holds a member at or beyond its midpoint, and otherwise the lower half is kept. Every kept half has an upper bound for its right end and a member inside it, and the halves close on the least upper bound, which is the square root of 2. *Assumes:* that the halves close on a single number, which is the nested-interval sentence, now a theorem; and that a shaded ray can stand for a collection with no largest member, which no drawing can show at its right-hand end

332929/166464, ... differ from the constant sequence 2, 2, 2, ... by $1/4, 1/144, 1/166464, \dots$, which tends to 0. So the square of the class of the averaging sequence is the class of 2. The number that cannot be a fraction is built out of fractions, and its square is 2 exactly.

IS THERE ONLY ONE LINE?

The real numbers were built as classes of huddling sequences of fractions, and the axiom of nested intervals came out as a theorem about them. But that was one way of building, and there are others. Would another construction give other numbers, with other theorems? The question matters, because nearly everything since the parabola has been proved about “the real numbers” with no mention of how they were made.

The best known of the other constructions is older than the one by sequences in its idea, though not in its publication. Dedekind found it in 1858, when he first had to teach the calculus and felt keenly, he wrote, that he had no proof that an increasing bounded sequence approaches a limit; he published it in 1872. A fraction divides the fractions into those below it and those above. A gap among the fractions does the same. Where the square root of 2 ought to be, the fractions fall into two sides: below the gap, the negative fractions together with the fractions whose squares are below 2; above it, the positive fractions whose squares are above 2. Dedekind’s idea was to let the division be the number. A *cut* is a collection of fractions that has something in it, does not contain everything, contains every fraction below any of its members, and has no largest member. The cut of a fraction q is the collection of fractions below q . The cut for the square root of 2 is the collection

of fractions that are negative or have square below 2, and it is the cut of no fraction.

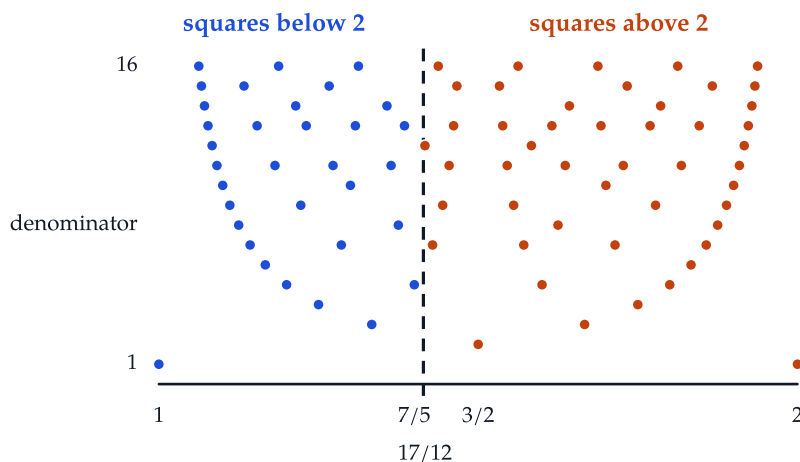


Figure 48.1. The fractions between 1 and 2 with denominators up to 16, each drawn at the height of its denominator. Those whose squares are below 2 are on one side, those whose squares are above 2 on the other, and none is on the line between. The division is the number: nothing else about the square root of 2 is needed to say where it is. *Assumes:* denominators up to 16; with larger denominators the two sides close in on the line without limit and never reach it

One cut is below another when it is contained in it and is not the whole of it. Two cuts are added by adding their members in all possible ways. Multiplying takes more care, because of the signs, and is a page of cases. But the property for which the whole construction exists takes only a few lines. Let a collection of cuts have something in it and an upper bound. Their union, the collection of all fractions that belong to any of them, is again a cut. It has something in it. It does not contain everything, because it lies inside the upper bound, and the upper bound leaves some fraction out. A fraction below one of its members lies in the same cut as that member. And a largest member of the union would be the largest member of the cut it came from, which has none. Now or-

der among cuts is containment. The union contains each of the cuts, so it is an upper bound of them; and every cut that contains them all contains their union, so no upper bound is smaller. So it is their least upper bound.

The two constructions have opposite strengths. With sequences the arithmetic is immediate, term by term, and completeness needs an argument. With cuts completeness is immediate and the arithmetic needs an argument. And the cut is older than Dedekind by twenty-two centuries. When the diagonal of the square refused a common measure with its side, the proportion of two lengths was given a meaning by comparing whole-number multiples: two proportions are the same when every comparison of m of the first length against n of the second comes out the same way for both. That definition is Eudoxus's. For two lengths a and b , the fraction n/m falls below the proportion of a to b exactly when n copies of b come out shorter than m copies of a , so the comparisons pick out a collection of fractions: the fractions n/m that fall below the proportion. Since m and n count copies, they are whole numbers and these fractions are all positive. The collection assigned to the pair of lengths is these fractions together with 0 and every negative fraction. It is a cut. Dedekind said as much, and added that what Euclid lacks is the statement that every cut is filled.

A third construction is the one everybody meets first. A real number is an integer followed by an endless string of decimal digits. Decimal reckoning came into general use through Stevin, whose pamphlet of 1585 taught it and urged decimal coinage and measures. Letting his digits run on without end gives a good definition of a real number, with two blemishes. Arithmetic on endless decimals is awkward, since addition starts from the right and these have no right-hand end. And two strings can name one number:

$$0.999\ldots = 1$$

The truncations 0.9, 0.99, 0.999, ... differ from 1 by $1/10$, $1/100$, $1/1000$, ..., which fall below every tolerance, so the number they approach is 1. A number is what the truncations approach, not the

string of symbols. This is the only kind of coincidence. Suppose two different strings first differ at some place, and call the one with the larger digit there the larger string. At that place the larger is ahead by at least 1 in the digit, which is one unit of that place. The later digits of the smaller string can make up some of this, but at most one unit of that place, and a whole unit only when every one of them is 9. The later digits of the larger string add to its lead unless every one of them is 0. So the truncations of the two strings approach the same number only when the smaller string runs on in 9s and the larger in 0s, which is how $0.999\dots$ and $1.000\dots$ coincide.

* * *

Three constructions, then, and more are known. Do they produce the same thing? They cannot produce the same objects: a class of sequences is not a collection of fractions, and neither is a string of digits. What can be asked is whether they produce the same structure, so that every statement about sums, products and order that is true in one is true in the others. They do.

Call a system of numbers a *complete ordered field* when it has three things. First, addition and multiplication obeying the rules of a field. Second, an order that respects them: every number is exactly one of positive, 0, and the negative of a positive; sums and products of positive numbers are positive; and one number is below another exactly when the second minus the first is positive. Third, completeness: every collection with something in it and an upper bound has a least upper bound.

Any two complete ordered fields can be matched, number for number, so that sums go to sums, products to products, and larger to larger; and there is only one way to do it.

Let R and S be two of them. Each contains a 1, and so $1 + 1$, and so on: copies of the whole numbers, all different because each is larger than the one before. That needs 1 to be positive, and it is: were 1 the negative of a positive p , it would equal $(-p) \cdot (-p)$,

which is $p \cdot p$ and positive. Then come negatives and quotients: a copy of the fractions. The match has no choice on these. It sends 0 to 0: since $x + 0$ is x , the image of x is the image of x with the image of 0 added, and the only number that can be added without changing anything is 0. And 1 is its own square, so it goes to a number that is its own square. A number x with $x \cdot x = x$ has $x \cdot (x - 1) = 0$, so it is 0 or 1; and 0 is already the image of 0, which leaves 1. So $1 + 1$ goes to $1 + 1$, and the fraction p/q of R to the fraction p/q of S .

Next, in a complete ordered field the whole numbers have no upper bound. If they had one they would have a least one, b . Then $b - 1$ would not be an upper bound, so some whole number n would exceed it, and $n + 1$ would exceed b . But $n + 1$ is a whole number and b was an upper bound of the whole numbers, so there is no such b . This is the axiom of Archimedes, now a theorem. It follows that between any two members of the field lies a fraction: if $y - x$ is positive, some whole number n exceeds $1/(y - x)$, since the whole numbers have no upper bound, and then $1/n$ is below $y - x$. The multiples of $1/n$ pass every bound in both directions, and each step is shorter than the stretch from x to y , so they cannot step over it: the first multiple above x is still below y .

So every member x of R is the least upper bound of the fractions below it. It is an upper bound of them. And any smaller number has a fraction between itself and x , which is a fraction below x that the smaller number fails to bound; so no smaller number is an upper bound. Send x to the least upper bound, in S , of the same fractions. There is one, since those fractions are bounded above in S by any fraction above x .

The rule respects order. If x is below y , some fraction c lies between them, and then some other fraction d between c and y . Every fraction below x is below c , so c is an upper bound for them in S and the image of x is at most c . And d is one of the fractions below y , so the image of y is at least d . Since c is below d , the image of x is below the image of y . It can be undone, by the same rule run from S to R . It respects sums. The fractions below $x + y$ are

the sums of a fraction below x and a fraction below y , by one more use of the fractions lying between. The same statement holds in S for the images of x and y , and because the rule respects order, the fractions below the image of x are the fractions below x , and likewise for y . So the image of $x + y$ and the sum of the two images are both the least upper bound, in S , of one and the same collection of fractions, and they are equal. For positive numbers it respects products in the same way, and signs take care of the rest. And no other match is possible, since a match must send fractions to the same fractions and respect order, which pins down where every least upper bound goes.

The real numbers are therefore determined by their description. Anything proved from the rules of a field, the rules of order and completeness is true of the classes of sequences, of the cuts, and of the decimals. That takes for granted that the cuts and the decimals are complete ordered fields, which has to be checked. For the cuts, the sum of two is again a cut: it has no largest member, and it misses the sum of any two fractions the two miss. For the decimals, order two strings by the first place at which they differ, with a string ending in all 9s counted as the one it agrees with; sums and products are then read off the truncations, which huddle. With that checked, nobody needs to remember which construction was used, and the axiom of nested intervals could be laid down in advance and the building left for later.

Not every description is like this. The rules of a group describe a great many different groups, and the rules of geometry without the postulate of parallels will turn out to describe more than one plane. A description satisfied by only one structure is called *categorical*, and it is the best that a description can be. It is also fragile. Drop completeness, and the fractions are an ordered field, and so are the numbers $a + b\sqrt{2}$ with a and b fractions, and they cannot be matched. A match sends $1 + 1$ to $1 + 1$ and products to products, so it would send a number whose square is 2 to a number whose square is 2, and one of these fields has such a number and the other has not.

Those two fields still obey the theorem of Archimedes. Without completeness even that can fail, and there are ordered fields with members that are positive and smaller than every $1/n$. Take the quotients of polynomials with real coefficients, and call such a quotient positive when its values are positive for all large enough t . A quotient of polynomials other than 0 keeps one sign for all large t , since numerator and denominator have finitely many roots between them and past the largest of those neither can change sign without taking the value 0. So exactly one of a quotient and its negative is called positive, and that is an order which respects sums and products. In this order $1/t$ is positive. It is also below $1/n$ for every n , since $1/n - 1/t$ is positive once t exceeds n . There, infinitely small quantities exist, as the inventors of the calculus believed they did on the line. On the line they do not, and the theorem of Archimedes is the proof: if a positive x were below every $1/n$, then $1/x$ would be above every whole number n , and the whole numbers have no upper bound.

What is the cut of the cube root of 2, and what, written out as a collection of fractions, is the sum of the cut of the square root of 2 and the cut of 1?

Which real numbers have two decimal expansions, and which have two expansions in base two?

Can the real numbers be matched with themselves, sums going to sums and products to products, in any way other than by leaving every number where it is? Can the numbers of the plane?

In the ordered field of quotients of polynomials, which is larger, t or 1000, and t^2 or 1000 t ? Does the collection of whole numbers have a least upper bound there?

Is the square of 0.333... equal to 0.111...? What does the question have to do with how decimals are multiplied?

HOW MANY ARE THERE?

The construction of the real numbers added to the fractions every class of huddling sequences that holds no constant sequence. How much did it add? Counting gives an answer only when it stops, and nothing endless can be counted to the end. So the question first needs a meaning for the word “size” that does not depend on counting.

The meaning was in use long before. Choices of two things from five were shown to be as many as choices of three from five by a correspondence exact in both directions, each choice of two determining the choice of the other three and each choice of three determining the two left over, with nothing unmatched on either side and no arithmetic at all. Counting a finite pile is the same act with the numbers 1, 2, 3, ... as the second pile. So keep the pairing and drop the numbers: two collections have the same size when their members can be paired off perfectly, each with exactly one partner and none left over. A collection that can be paired off with the whole numbers 1, 2, 3, ... can be *listed*: the partner of 1 is its first entry, the partner of 2 its second, and every member appears somewhere on the list.

The whole numbers pair off with the even numbers by doubling, 1 with 2, 2 with 4, 3 with 6, and so on without end. Every whole number has exactly one even partner and every even number is the

partner of exactly one whole number, so the two collections have the same size, although the second is a part of the first with half of the first removed. That is no contradiction.

A finite collection can never be paired off with a part of itself that leaves something out. For the collection $1, 2, \dots, n$ the claim is that a pairing of it with some of its own members uses all of them, and the claim climbs. It holds for $n = 1$. Suppose it holds for n , and take a pairing of $1, \dots, n + 1$ with some of those numbers. If no member had $n + 1$ for its partner, the partners of $1, \dots, n$ would lie among $1, \dots, n$ and so, by the case before, would be all of them. The partner of $n + 1$ is not $n + 1$, by the supposition, and every one of $1, \dots, n$ is already taken, so $n + 1$ would have no partner. So some member j has the partner $n + 1$. If j is not $n + 1$ itself, exchange the partners of j and $n + 1$, which does not change which numbers are used. Now $n + 1$ is paired with itself, the rest of the pairing pairs $1, \dots, n$ with some of $1, \dots, n$, hence with all of them, and every number is used.

An endless collection, on the other hand, can always be paired off with a part of itself: choose members $a_1, a_2, a_3, \dots$ one after another, all different, which never fails because the collection is endless, and pair each a_k with a_{k+1} and every other member with itself; a_1 is left out. Making infinitely many choices in succession is an assumption, the axiom of choice, and it is named here because it is used. So the collections that pair off with a part of themselves are exactly the endless ones.

* * *

The fractions are the first real test, because they are packed in a way the whole numbers are not. Between any two of them lies another, so no fraction has a next one, and a walk through them in increasing order cannot take a second step.

Lay the positive fractions out in a grid, with the numerator across and the denominator down, so that $3/5$ sits in column 3 of row 5 and every positive fraction as written sits in exactly one

cell. Walking along the first row never ends, so the second row is never reached and a fraction such as $1/2$ gets no place on the list. Walking the diagonals works. The cells whose two numbers add to a fixed total form a diagonal with finitely many cells: the total 2 gives $1/1$, the total 3 gives $1/2$ and $2/1$, the total 4 gives $1/3$, $2/2$ and $3/1$. Cross each diagonal from its lower left end to its upper right end, so that the numerator rises, and finish it before starting the next. Every fraction a/b is reached, since it lies on the diagonal with total $a + b$, which comes after finitely many earlier diagonals of finite length; and no cell is visited twice. The fraction $5/3$ lies on the diagonal with total 8, after the twenty-one cells of the six earlier diagonals and after $1/7$, $2/6$, $3/5$ and $4/4$ on its own, so it is the twenty-sixth cell visited.

The grid holds $1/2$ and $2/4$ in different cells, so the walk names some numbers more than once. Striking out every cell whose value has already appeared leaves a list in which each positive fraction appears exactly once; on it $5/3$ is the twentieth entry. Zero and the negative fractions are added by weaving, 0 first and then each positive fraction followed by its negative. So the fractions can be listed. Density is not size: a collection can have a member between any two of its members and still pair off with 1, 2, 3, ...

Can the points of the plane with fractions for both coordinates be listed? Can the finite sequences of 0s and 1s?

* * *

If every endless collection could be listed, the new notion of size would have a single infinite value and would say nothing. The collections that cannot be listed include one simpler than the real numbers: the endless sequences of 0s and 1s.

Four rows show the argument at a size where nothing is hidden. Take sequences beginning 0110, 1010, 1100 and 0001, and read the diagonal: the first digit of the first, the second of the second, the third of the third, the fourth of the fourth, which gives 0, 0, 0, 1. Flip every digit of the diagonal, 0 to 1 and 1 to 0, and the result begins

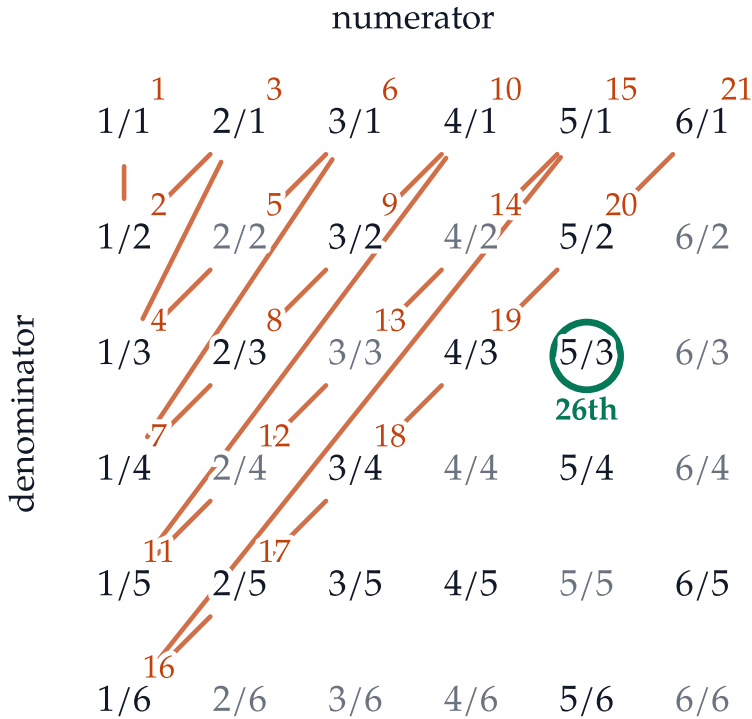


Figure 49.1. The positive fractions laid out with the numerator across and the denominator down, and the walk that crosses each diagonal of constant total from bottom to top before starting the next. Every cell is reached after finitely many steps; 5/3, on the diagonal with total 8, is the twenty-sixth cell visited, the steps being numbered here through the sixth diagonal only. Cells whose value has already appeared are greyed: the walk lists fractions as written, and striking those out leaves a list of numbers. Assumes: that the grid continues in both directions forever and the walk with it; six diagonals are drawn and the argument is about all of them

How many are there?

1110. It differs from the first row in the first place, from the second in the second, from the third in the third and from the fourth in the fourth, so it is none of the four.

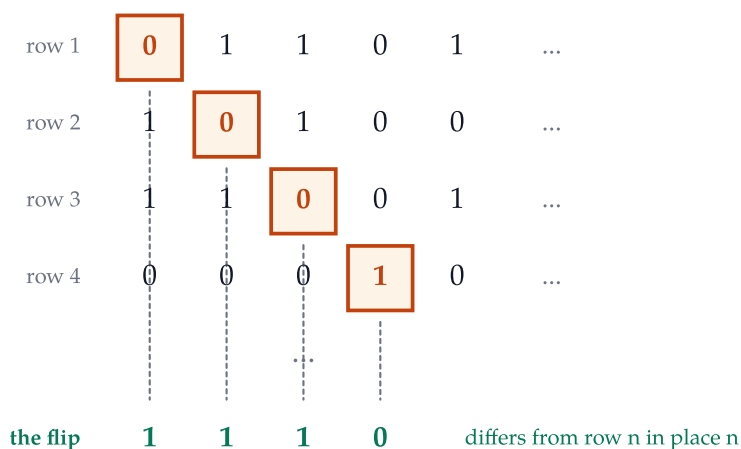


Figure 49.2. Four sequences of 0s and 1s, with their diagonal marked: 0, 0, 0, 1. Flipping each diagonal digit gives 1, 1, 1, 0, a sequence that disagrees with the first row in the first place, the second in the second, and so on. Nothing used the number four, and the same flip escapes any list however long. Assumes: that the rows continue to the right and the list continues downward without end; the argument is about the endless case and the picture is its first four rows

Nothing used the number four. Suppose every endless sequence of 0s and 1s had been listed, as a first sequence, a second, a third, and so on. Build a sequence whose n -th digit is the flip of the n -th digit of the n -th listed sequence. It differs from the n -th entry in the n -th place, for every n , so it is nowhere on the list, and the list was any list at all. Adding the escaped sequence to the list as a new first line does not help, since the flip, run on the longer list, escapes again. No listing of the 0/1 sequences is complete, and there is a second infinite size.

The argument does not need the whole numbers. A sequence of 0s and 1s is a collection of whole numbers, the collection of those

n whose digit is 1, and the flipped diagonal is the collection D of those n that do not belong to the n -th listed collection. Read to the fourth place, the four rows above are the collections $\{2, 3\}$, $\{1, 3\}$, $\{1, 2\}$ and $\{4\}$; the numbers not in their own row's collection are 1, 2 and 3, and $D = \{1, 2, 3\}$ is the flip 1110.

Now let X be any collection, and suppose each member x of X is paired with a part P_x of X . Let D be the part made of those x that do not belong to P_x . Then D is not the part paired with any member. Suppose it were the part P_x of some particular x , and ask whether that x belongs to D . If x belongs to D , then by the rule that made D it does not belong to P_x ; but P_x is D , so x does not belong to D . If x does not belong to D , then it does not belong to P_x , and the rule puts it in D . Either answer contradicts itself, so D is no P_x . The pairing was any pairing at all, and it missed the part D . So no collection can be paired off with the collection of all its parts.

Say that X is *smaller* than Y when X pairs off with a part of Y and not with the whole of Y . Every X pairs off with a part of the collection of its parts, x going to the part whose only member is x . So the collection of the parts of X is larger than X , whatever X is, and the sizes never stop. Russell ran the argument on the collection of all collections and found that there is no such thing: the collection of those collections that are not members of themselves would be a member of itself exactly when it is not.

* * *

The real numbers now. Read each endless 0/1 sequence as instructions for cutting intervals in thirds. Start with the interval from 0 to 1. For a first digit 0 keep the left third, from 0 to $1/3$; for a 1 keep the right third, from $2/3$ to 1; the middle third is discarded either way. Read the next digit and keep the left or right third of what was kept, and so on. The kept intervals nest, and their lengths $1/3, 1/9, 1/27, \dots$ shrink past every tolerance, so they close on exactly one real number. The sequence 0101... keeps 0 to $1/3$, then $2/9$ to $3/9$, then $6/27$ to $7/27$, then $20/81$ to $21/81$, and

every one of these contains $1/4$, which is the number that sequence names; 0111... closes on $1/3$, and 1000... on $2/3$.

Thirds rather than halves make different sequences name different numbers. Take two different sequences and the first digit where they differ. Up to that digit they have kept the same interval. At that digit one keeps its left third and the other its right third, and everything either keeps later lies inside its own third. So the two numbers they close on lie in different thirds, with the whole middle third between them. With halves the two kept pieces would share an endpoint, and 0111... and 1000... would both close on $1/2$.

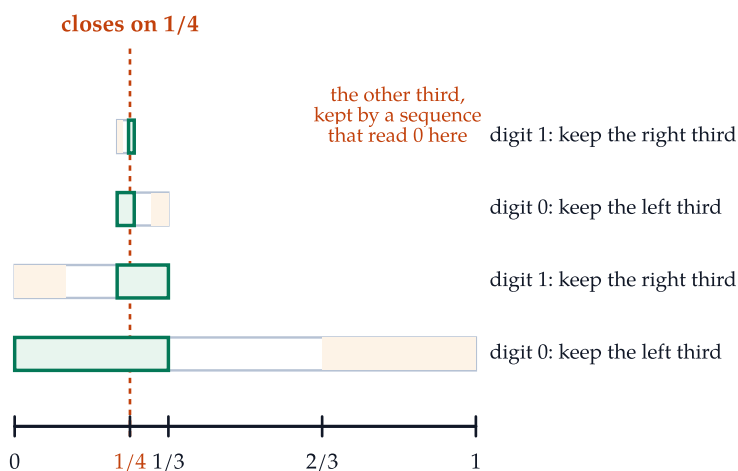


Figure 49.3. The sequence 0101... read by the thirds machine: keep the left third, then the right third of that, then the left, then the right. The kept intervals are 0 to $1/3$, $2/9$ to $3/9$, $6/27$ to $7/27$, $20/81$ to $21/81$, and they close on $1/4$. A sequence that reads 0 where this one reads 1 keeps the other third at that step, and the discarded middle separates the two numbers forever. Assumes: that the nested thirds close on exactly one number, which is the nested-interval theorem; four steps are drawn and the machine runs without end

So the 0/1 sequences are paired off with a part of the real numbers, different numbers for different sequences. If the real numbers could be listed, the numbers in that part could be read off the

list in order, skipping the rest, and that would list the sequences, which cannot be listed. The real numbers cannot be listed. Nor can the new numbers alone, the ones added to the fractions to make the real numbers: a list of those, woven with a list of the fractions, would list the real numbers. The added numbers, far from being occasional repairs between fractions, outnumber them in the sense of pairing. Length and size disagree in both directions. In one direction, the numbers the thirds machine names take up no length and still cannot be listed: after n steps they all lie in 2^n intervals of total length $(2/3)^n$, which falls below every tolerance. In the other direction, the fractions can be listed and still turn up everywhere. List them as $q_1, q_2, q_3, \dots$ and place an interval of length $\varepsilon/2$ round q_1 , one of length $\varepsilon/4$ round q_2 , one of length $\varepsilon/8$ round q_3 , and so on. These intervals cover all the fractions with total length at most ε , however small ε is, and yet every interval of the line, however short, contains fractions.

Which of $1/10$, $1/2$ and $7/9$ does the thirds machine name, and by which sequences?

* * *

Call a number *algebraic* when it satisfies a polynomial equation with integer coefficients, not all 0. Every fraction is algebraic, by an equation of the first degree; so is the square root of 2, by $x^2 - 2 = 0$, and i , by $x^2 + 1 = 0$; and every root of every such polynomial of every degree is algebraic. The algebraic numbers can be listed, by the grid argument in different clothes. Give each polynomial a *height*, its degree plus the sizes of its coefficients added together, so that $x^2 - 2$ has height 5. Each height allows only finitely many polynomials, since the degree and every coefficient are bounded by it: height 1 allows 1 and -1 , height 2 allows ± 2 and $\pm x$, and height 3 allows the ten polynomials ± 3 , $\pm 2x$, $\pm x \pm 1$ and $\pm x^2$.

A polynomial of degree d has at most d roots. That was proved over any field when the squares modulo a prime were sorted, and the proof bears repeating. If α is a root, the polynomial equals itself

less its value at α , which is a sum of terms $c(x^k - \alpha^k)$, each of which has the factor $x - \alpha$; so the polynomial is $x - \alpha$ times a quotient of degree one lower: a root α splits off a factor $x - \alpha$. A second root β of the polynomial is a root of the quotient. The value of the polynomial at β is $\beta - \alpha$ times the value of the quotient at β , and it is 0; $\beta - \alpha$ is not 0; and in a field a product is 0 only when a factor is. (Where that fails, as with remainders on division by 8, $x^2 - 1$ has the four roots 1, 3, 5 and 7.) So each root split off lowers the degree by one, and a polynomial of degree d runs out of degree after d roots. So the algebraic numbers come in finite batches, one batch for each height, and a sequence of finite batches, with repeats struck out, is a list.

The real numbers cannot be listed, so there are real numbers that are not algebraic, satisfying no polynomial equation with integer coefficients of any degree, and most real numbers are of that kind. The argument names none of them, and yet, run against a particular listing $r_1, r_2, r_3, \dots$ of the algebraic numbers, such as the one by heights, it is a procedure. Start with the interval from 0 to 1 and at step n keep the left third or the right third of the interval kept so far, one that does not contain r_n ; the two thirds share no point, so r_n cannot lie in both, and at least one of them does not contain it. The kept intervals close on exactly one real number, and it lies in the interval kept at step n , so it is not r_n , for any n . The comparison of sizes alone is like the average over all colourings, which showed that the pairs among 1024 people can be coloured so that no twenty of them are all joined by one colour without building any colouring.

A number that is not algebraic can also be exhibited, by a formula, because an algebraic number that is not a fraction keeps fractions at a distance. Let P be a polynomial of degree d at least 1 with integer coefficients, let α be a real root of it, and let M be a bound for the size of the derivative of P between $\alpha - 1$ and $\alpha + 1$; there is one, the derivative being continuous on that closed interval. If p/q is a fraction within 1 of α that is not itself a root of P , then $q^d \cdot P(p/q)$ is an integer, since each term of $P(p/q)$ is an integer coef-

ficient times p^k/q^k with k at most d , and multiplying by q^d clears the denominator. It is not 0, since p/q is not a root, and an integer other than 0 has size at least 1. So $|P(p/q)|$ is at least $1/q^d$; and by the average-slope theorem $|P(p/q)|$, which is $|P(p/q) - P(\alpha)|$, is at most M times $|p/q - \alpha|$. So

$$|\alpha - p/q| \geq 1/(M \cdot q^d)$$

Now let $\lambda = 1/10 + 1/10^2 + 1/10^6 + 1/10^{24} + \dots$, the sum of the numbers $1/10^{k!}$. As a decimal it is

$$\lambda = 0.110001000000000000000001\dots$$

with a 1 in the places 1, 2, 6, 24, 120, ... and 0 elsewhere. The sum of its first n terms is a fraction p_n/q_n with $q_n = 10^{n!}$. These fractions increase, so they are all different, and each falls short of λ by the rest of the series, which is at most $(10/9)/10^{(n+1)!}$: its first term is $1/10^{(n+1)!}$, and each later term is less than a tenth of the one before, so the whole is at most $1 + 1/10 + 1/100 + \dots = 10/9$ times the first term. Since $(n+1)!$ is $n!$ times $n+1$, the number $10^{(n+1)!}$ is q_n^{n+1} , and the bound reads $(10/9)/q_n^{n+1}$.

Suppose λ were a root of a polynomial P as above. At most d of the fractions p_n/q_n would be roots of P . For all the others, which lie within 1 of λ since the rest of the series is less than 1, the inequality would give $1/(M \cdot q_n^d) \leq |\lambda - p_n/q_n| \leq (10/9)/q_n^{n+1}$, that is, $q_n^{n+1-d} \leq (10/9) \cdot M$. The right side is a fixed number. On the left the exponent is at least 1 as soon as n is d or more, so the left side is at least q_n , which grows past every bound. The inequality fails once n is large. So λ satisfies no polynomial equation with integer coefficients, and with $d = 1$ the same argument shows that it is not a fraction.

Liouville proved in 1844 that numbers of this kind exist, and gave this decimal in 1851. The argument from listing came some twenty years later and is Cantor's. So are the pairing definition of size, the listing of the fractions, the flipped diagonal of 1891 and the theorem on parts.

Does the same argument show that the sum of the numbers $1/10^{k^2}$ is not algebraic?

The 0/1 sequences pair off with a part of the real numbers, by the thirds machine. The real numbers pair off with a part of the 0/1 sequences too, in two steps. First send a real number y to $(1 + y/(1 + |y|))/2$, which lies strictly between 0 and 1 and is different for different y . Then name a number strictly between 0 and 1 by halving. Start with the interval from 0 to 1; at each step write 0 when the number lies in the left half of the interval kept so far, not counting the midpoint, and 1 otherwise, and keep the half it lies in. Two different numbers get different names, since they fall in different halves once the halves are shorter than the distance between them.

Two collections each of which pairs off with a part of the other pair off with each other, and that is a theorem. Let f pair X with a part of Y , and g pair Y with a part of X . Let C_0 be the members of X that g never reaches, those that are not $g(y)$ for any y in Y , let C_1 be the members $g(f(x))$ for x in C_0 , let C_2 be made from C_1 in the same way, and so on, and let C be everything that lies in any of these. Send each x in C to $f(x)$, and each x outside C to the y with $g(y) = x$, which exists because x is not in C_0 .

Different members of X go to different members of Y . Two members of C are kept apart by f , which sends different members to different members. Two members outside C that went to the same y would both be $g(y)$, so they would be one member. And if x in C and x' outside C both went to y , then $y = f(x)$ and $g(y) = x'$, so $x' = g(f(x))$; but x lies in one of $C_0, C_1, C_2, \dots$, and then $g(f(x))$ lies in the next, so x' would be in C . And every y is reached: if $g(y)$ is outside C , it is sent to y ; if $g(y)$ is in C , it is not in C_0 , so it is $g(f(x))$ for some x in C , and then $y = f(x)$, since g sends different members to different members.

So the 0/1 sequences pair off with all of the real numbers, and the size of the collections that can be listed and the size of the real numbers are the first two sizes of the ladder that never stops. Does some collection sit strictly between, too large to be listed and too small to pair with the real numbers? The usual rules of mathe-

matics settle that question in neither direction: Gödel showed in 1938 that the assertion that no such collection exists cannot be refuted from those rules, and Cohen showed in 1963 that it cannot be proved from them, both on the condition that the rules are free of contradiction.

NUMBERS NO EQUATION REACHES

Nearly every real number is a root of no polynomial with whole-number coefficients, and one such number has been written down, a decimal made for the purpose. That leaves the numbers anyone cares about. Is π algebraic? Is e ? Numbers that are not algebraic are called *transcendental*, and proving that a number met in the wild is one of them is a different task from manufacturing an example, because the number arrives with its own definition and the proof has to work with that. The proofs that exist share one design. Assume the number is a fraction, or satisfies an equation. Build, from that assumption, a quantity that must be a whole number and cannot be 0. Then show by an estimate that the same quantity is smaller than 1. The proof that e is not a fraction was of this design: $q!$ times the tail of the series was a whole number strictly between 0 and $3/4$.

Here is the design applied to π , in a proof found by Niven in 1947. Suppose π were a fraction a/b . For a whole number n , to be chosen large at the end, let

$$f(x) = x^n(a - bx)^n/n!$$

On the interval from 0 to π , which on the supposition is the

interval from 0 to a/b , both x and $a - bx$ are positive, so f is positive inside the interval; and x is at most π while $a - bx$ is at most a , so f is at most $(\pi a)^n/n!$. Two facts about f do the work.

First, f and all its derivatives take whole-number values at 0 and at π . Multiplied out, $n! \cdot f(x)$ is a polynomial with whole-number coefficients in which every power of x is at least the n -th. Differentiate a polynomial j times and put $x = 0$: the powers below x^j have been differentiated away, the powers above it still carry a factor x and vanish, and the term in x^j has become $j!$ times its coefficient. So the j -th derivative of f at 0 is $j!$ times the coefficient of x^j in $n! \cdot f(x)$, divided by $n!$. For j below n that coefficient is 0, and for j at least n the quotient $j!/n!$ is a whole number. So every derivative of f is a whole number at 0. And $f(a/b - x) = f(x)$, since $f(x)$ is $b^n/n!$ times $x^n(a/b - x)^n$, which is unchanged when x and $a/b - x$ change places; and differentiating both sides of $f(a/b - x) = f(x)$ brings out a factor -1 on the left each time, so after j differentiations the j -th derivative of f at $a/b - x$ is the j -th derivative at x , apart from sign. Put $x = 0$: the derivatives at a/b , which is π , are those at 0, apart from signs.

Second, let F be f , less its second derivative, plus its fourth, less its sixth, and so on: the alternating sum of the derivatives of even order, which stops because the derivatives of a polynomial are 0 from some order on. Differentiating twice moves every term two places along, so F'' is the second derivative of f , less its fourth, plus its sixth, and so on. Add F'' to F . Each term of F'' meets the same term of F with the opposite sign, and only the leading f has no partner. So $F'' + F = f$. The function $F'(x) \sin x - F(x) \cos x$ has derivative $F''(x) \sin x + F(x) \sin x$, by the rule for products, and that is $f(x) \sin x$. So

the integral of $f(x) \sin x$ from 0 to π is $F(\pi) + F(0)$

since the integral of a derivative is the change in the function, here the change in $F'(x) \sin x - F(x) \cos x$ from 0 to π . At both ends $\sin$ is 0, so the term with F' drops out. At π the cosine is -1 , so the function is $F(\pi)$ there; at 0 the cosine is 1, so the function is

$-F(0)$ there; and the first less the second is $F(\pi) + F(0)$. Now $F(\pi)$ and $F(0)$ are sums and differences of the values of f and its derivatives at π and at 0 , so by the first fact the right side is a whole number. But the left side is the integral of a function that is positive inside the interval and at most $(\pi a)^n/n!$, so it lies strictly between 0 and $\pi(\pi a)^n/n!$. Powers lose to factorials: each step from n to $n + 1$ multiplies that bound by $\pi a/(n + 1)$, which is below a half once n exceeds $2\pi a$. From there on the bound is at least halved at every step, and a number halved often enough falls below 1 . For such an n , a whole number lies strictly between 0 and 1 . So π is not a fraction.

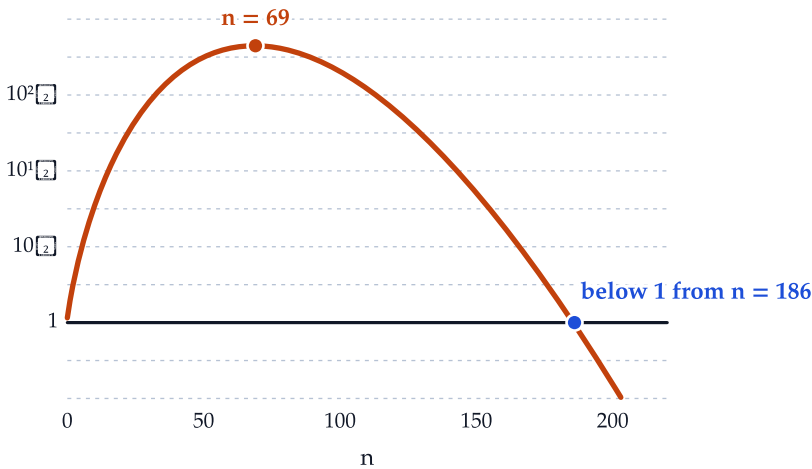


Figure 50.1. The bound $\pi(\pi a)^n/n!$ for the integral in the proof, with $a = 22$, on a scale where each line is a factor of ten thousand. It climbs for the first 69 values of n , to about 10^{29} , and then falls without end, passing below 1 at $n = 186$. The proof needs only that it falls below 1 at last, which is where a whole number would be trapped. Assumes: $a = 22$, as if π were $22/7$; a larger numerator moves the peak further out and higher, and changes nothing else

Lambert's proof of 1761 was the first, by a continued fraction for the tangent, and Legendre showed in 1794 that π^2 is not a fraction either, which is more: it says that π is not the square root of a

fraction.

* * *

For e the target is higher. In 1873 Hermite proved that e is transcendental. It was the first time this had been shown for a number that was not constructed for the purpose. His proof was long, and Hilbert in 1893 reduced it to the form that follows. It has the same design, with one new idea: a prime number p , to be chosen large at the end, is used to show that the whole number cannot be 0.

Suppose e satisfied an equation

$$a_0 + a_1 e + a_2 e^2 + \dots + a_n e^n = 0$$

with whole-number coefficients and a_0 not 0. The powers of e are the values $E(1), E(2), \dots, E(n)$ of the exponential. Let p be a prime, and let

$$f(x) = x^{p-1} \cdot ((x-1)(x-2) \dots (x-n))^p / (p-1)!$$

This time let $F = f + f' + f'' + \dots$, the sum of all the derivatives, which again stops. Differentiating moves every term one place along, so $F' = f' + f'' + \dots$, the same sum with its first term missing. Then $F - F' = f$, and the function $-E(-x)F(x)$ has derivative $E(-x)(F(x) - F'(x))$, which is $E(-x)f(x)$. Integrate from 0 to k . The integral of $E(-x)f(x)$ is the change in $-E(-x)F(x)$, which is $F(0) - E(-k)F(k)$, since $E(0) = 1$. Multiply through by $E(k)$, remembering that $E(k) \cdot E(-k) = 1$:

$$E(k) \cdot F(0) - F(k) = E(k) \cdot \text{the integral of } E(-x)f(x) \text{ from } 0 \text{ to } k$$

for $k = 0, 1, \dots, n$. Multiply the k -th of these by a_k and add them up. On the left the terms with $F(0)$ add up to $F(0)$ times $a_0 + a_1 e + \dots + a_n e^n$, which is 0 by the supposition. So

$$-(a_0 F(0) + a_1 F(1) + \dots + a_n F(n)) = \text{the sum of } a_k \cdot E(k) \cdot \text{the integral of } E(-x)f(x) \text{ from } 0 \text{ to } k$$

The plan is to show that the left side is a whole number other than 0 and that the right side is small.

The left side. Expand f in powers of x . The lowest power is x^{p-1} , and its coefficient comes from taking the constant term of every bracket: the product $(-1)(-2) \cdots (-n)$ is $(-1)^n n!$, so the coefficient is $((-1)^n n!)^p / (p-1)!$, and every coefficient is a whole number divided by $(p-1)!$. Differentiating j times and putting $x = 0$ leaves $j!$ times the coefficient of x^j . So the derivatives of order below $p-1$ are 0 at 0; the derivative of order $p-1$ is $((-1)^n n!)^p$; and each derivative of order j at least p is a whole number times $j! / (p-1)!$, which is the product $p(p+1) \cdots j$ and so a multiple of p . Now $F(0)$ is the sum of all these derivatives at 0. So $F(0)$ is a whole number, and every term in it is a multiple of p except the one of order $p-1$, which is $\pm(n!)^p$. On division by p , then, $F(0)$ leaves what $\pm(n!)^p$ leaves.

At the point k , for k from 1 to n , expand f in powers of $x - k$ instead. The coefficients are still whole numbers divided by $(p-1)!$, since substituting $x = k + y$ in a polynomial with whole-number coefficients gives one in y ; and now the lowest power is $(x - k)^p$, because of the factor $(x - k)^p$ in f . So the derivatives of order below p are 0 at k , and each of order j at least p is a whole number times $j! / (p-1)!$, a multiple of p . So $F(1), \dots, F(n)$ are whole numbers divisible by p . The left side is therefore a whole number, and on division by p it leaves what $\pm a_0(n!)^p$ leaves.

Choose the prime p larger than n and larger than the size of a_0 ; there are primes beyond every bound. Then p divides neither a_0 nor any of the numbers $1, 2, \dots, n$, each being smaller in size than p and not 0. But $a_0(n!)^p$ is a product of a_0 and copies of $1, 2, \dots, n$, and a prime that divides a product divides one of the factors. So p does not divide $a_0(n!)^p$, with either sign, and the left side, which differs from $\pm a_0(n!)^p$ by a multiple of p , is not divisible by p . In particular it is not 0, and being a whole number, its size is at least 1.

The right side. On the interval from 0 to n the factor x^{p-1} is at most n^{p-1} and each factor $|x - j|^p$ is at most n^p , so $|f(x)|$ is at most $n^{np+p-1} / (p-1)!$, which is n^n times $c^{p-1} / (p-1)!$, where $c = n^{n+1}$.

In each integral $E(-x)$ is at most 1, so the function being integrated has size at most that bound, and the stretch from 0 to k has length at most n . So each integral has size at most n times the bound. So the right side has size at most

$$(|a_0| + |a_1| + \dots + |a_n|) \cdot E(n) \cdot n \cdot n^n \cdot c^{p-1}/(p-1)!$$

in which only the last factor depends on p , and powers lose to factorials: it approaches 0 as p grows. For a large enough prime p the right side has size below 1, while the left side, which equals it, has size at least 1. So there is no such equation, and

e satisfies no polynomial equation with whole-number coefficients.

An equation whose constant term is 0 reduces to one whose constant term is not: divide by the largest power of x dividing the polynomial, which throws away no root but 0, and e is not 0. So nothing is lost by supposing a_0 is not 0.

Several things follow at once. No power of e with a whole-number exponent other than 0 is a fraction: $e^m = u/v$ with m positive would make e a root of $vx^m - u$, and a negative m is the same with u and v exchanged. So $E(r)$ is not a fraction for any fraction r other than 0, since $E(m/k) = u/v$ gives $e^m = (u/v)^k$, a fraction. Read backwards, the logarithm of a positive fraction other than 1 is never a fraction: $L(2) = m/k$ would make e^m equal to 2^k .

* * *

The same method, carried further, reaches π . In 1882 Lindemann proved that $E(\alpha)$ is transcendental for every algebraic number α other than 0, real or not. Now $E(i\pi) = -1$, and -1 is as algebraic as a number can be: it is the root of $x + 1$. If $i\pi$ were algebraic, Lindemann's theorem would make $E(i\pi)$ transcendental. So $i\pi$ is not algebraic. And then π is not algebraic either. For suppose π were a root of a polynomial P with whole-number coefficients.

Then $i\pi$ would be a root of $P(ix) \cdot P(-ix)$, because putting $x = i\pi$ turns the second factor into $P(\pi)$, which is 0. That product, multiplied out, has coefficients of the form $u + vi$ with u and v whole numbers. Exchanging i and $-i$ swaps the two factors and so leaves the product unchanged, which forces every v to be 0. So the product has whole-number coefficients, and $i\pi$ would be algebraic after all. The proof follows Hermite's plan, but the assumed equation now involves all the roots of the polynomial that α satisfies, and to show that the quantity built from them is a whole number takes the theory of expressions that are unchanged when the roots are rearranged. That theory comes later, and the proof for π is given in full as soon as it does. Lindemann's theorem settled the oldest open problem in mathematics, in a way that will be plain once it is known which lengths a ruler and compasses can construct.

The subject remains hard, and its open questions are embarrassing in their simplicity. It is known that $E(\pi)$ is transcendental, and that 2 to the power $\sqrt{2}$, which can only mean $E(\sqrt{2} \cdot L(2))$, is; both follow from a theorem of Gelfond and Schneider of 1934. It is not known whether $e + \pi$ is a fraction. It is not known whether $e \cdot \pi$ is. At least one of the two is transcendental. If both were algebraic, then e and π would be the roots of $x^2 - (e + \pi)x + e\pi$, an equation whose coefficients are algebraic, and a root of an equation with algebraic coefficients is itself algebraic. So e would be algebraic, and it has been proved that e is not. That fact about roots is relied on here and proved later, when the dimensions of number systems are counted. Nobody can say which of $e + \pi$ and $e \cdot \pi$ it is. The constant γ , the limit of $1 + 1/2 + \dots + 1/n - L(n)$, has been computed to billions of decimal places, and it is not known whether it is a fraction. Apéry astonished a conference in 1978 by proving that the sum of the reciprocals of the cubes, $1 + 1/8 + 1/27 + \dots$, is not a fraction; for the fifth powers the question is open, though it is known that at least one of the sums for the fifth, seventh, ninth and eleventh powers is not a fraction.

Which of these are algebraic, and by which equations: $\sqrt{2} + \sqrt{3}$, the cosine of 20 degrees, $E(L(2)/2)$, and $E(1) + E(-1)$?

In the proof for π , where exactly was it used that a and b are whole numbers, and where that n may be taken as large as desired?

Suppose π^2 were a fraction a/b . Let $f(x) = x^n(1-x)^n/n!$ and let F be b^n times the combination of π^{2n} times f , less π^{2n-2} times its second derivative, plus π^{2n-4} times its fourth, and so on. What is the derivative of $F'(x) \sin \pi x - \pi F(x) \cos \pi x$, and how does the proof end?

Why can a number whose decimal expansion eventually repeats not be transcendental, and can a transcendental number have a continued fraction with all its quotients equal to 1 or 2?

If x and y are transcendental, must $x + y$ be? Must xy be? Must at least one of $x + y$ and xy be?

MONSTERS

When the real numbers had been built and the words continuous, length and curve had been given exact meanings, objects began to appear that obeyed the definitions and defied every expectation the words had carried. Hermite wrote to a friend that he turned away “with fright and horror” from functions with no derivatives, and Poincaré wrote that logic sometimes makes monsters, and that the new functions were invented to put the reasoning of our fathers in the wrong. They were made for that purpose, and they did more. Each one separates two ideas that had always been met together.

The first has been met. The machine that names numbers by keeping the left or the right third of an interval, discarding the middle third at every step, names a collection of numbers between 0 and 1. Call it the *thirds set*. Build it from the other side: from the interval from 0 to 1 remove the open middle third, from $1/3$ to $2/3$; from each of the two pieces left remove the middle third; from each of the four pieces left, the middle third; and so on forever. The thirds set is what is never removed. The ends of the pieces are never removed, so 0, 1, $1/3$, $2/3$, $1/9$, $2/9$ and the rest are in it; so are numbers that are the end of nothing, such as $1/4$, which is named by the sequence 0, 1, 0, 1, ...: keep the left third, then the right third of that, then the left third again, and so on, and $1/4$ lies in every piece kept.

How long is the thirds set? The lengths removed are $1/3$, then $2/9$, then $4/27$, each step removing a third of what is left, and they add up to

$$1/3 + 2/9 + 4/27 + \dots = (1/3)/(1 - 2/3) = 1$$

The whole length of the interval is removed. What remains can be covered, at stage n , by 2^n intervals of total length $(2/3)^n$, which falls below every tolerance. It contains no interval, however short. An interval lying in the set would have to lie inside a single piece at every stage, since everything between two pieces has been removed; but the pieces of stage n are only $1/3^n$ long, and for large n that is shorter than the interval. And it has as many members as the whole line, since its members correspond to the endless sequences of 0s and 1s, and those were paired off with all of the real numbers. Size and length have nothing to do with each other: here is a set as numerous as the line, with no length at all.

The thirds set is large in another way. Every number from 0 to 2 is the sum of two of its members. Take the square of all points (x, y) with x and y from 0 to 1, and ask whether the line $x + y = t$ meets a point with both coordinates in the set. At stage 1 the points with both coordinates in kept pieces form four corner squares of side $1/3$. Over the lower left corner square $x + y$ runs from 0 to $2/3$; over the upper left and over the lower right it runs from $2/3$ to $4/3$; over the upper right it runs from $4/3$ to 2. Between them those stretches cover every t from 0 to 2, so a line $x + y = t$ that meets the big square meets at least one of the four. Inside that corner square the picture repeats at a third of the size, so the line meets one of its four corner squares; and so on. The squares met are nested and shrink to a point, which lies on the line and has both coordinates in the thirds set. Those two coordinates are two members of the set, and because the point is on the line their sum is t . Dust, added to dust, fills an interval.

* * *

The second monster is a function built on the first. Let s be $1/2$ on the removed middle third, from $1/3$ to $2/3$. Let it be $1/4$ on the first of the two thirds removed next and $3/4$ on the second; then $1/8, 3/8, 5/8$ and $7/8$ on the four removed after that; and so on. The rule is that the value on each removed piece is halfway between the values on the nearest pieces already removed to its left and to its right. Where there is no such piece on the left, the value 0 at the left end of the whole interval is used, and where there is none on the right, the value 1 at the right end. That defines s on every removed interval, and s never decreases. On the thirds set itself, define it by the kept-third instructions: a member named by a sequence of 0s and 1s is sent to the number whose digits in base two are that sequence, so that $1/4$, named by 0, 1, 0, 1, ..., goes to 0.010101... in base two, which is $1/3$.

The two definitions fit together. The first removed third runs from $1/3$, named by 0, 1, 1, 1, ..., to $2/3$, named by 1, 0, 0, 0, ..., and in base two 0.0111... and 0.1000... are both $1/2$, the value on that third. Every removed piece is the middle third of the piece kept after some finite string of 0s and 1s, and its ends are named by that string followed by 0, 1, 1, 1, ... and by the string followed by 1, 0, 0, 0, ...; in base two these give the same number, the string followed by a single 1, and the halfway rule gives that same number to the piece between them. And the function is continuous: two numbers closer together than $1/3^n$ have values within $1/2^n$ of each other. The reason is that s rises by $1/2^n$ across each kept piece of stage n and not at all across a removed one, and two numbers that close lie in the same piece of stage n or in neighbouring pieces, removed or kept. They cannot lie in two different kept pieces, since any two of those are separated by a removed piece at least $1/3^n$ long. So all of the rise between them happens inside at most one kept piece, and is at most $1/2^n$.

So s climbs continuously from 0 at 0 to 1 at 1. But on every removed interval it is constant, with derivative 0, and the removed intervals have total length 1. The function rises from 0 to 1 while having slope 0 everywhere except on a set of no length. All of its

rising is done on the thirds set.

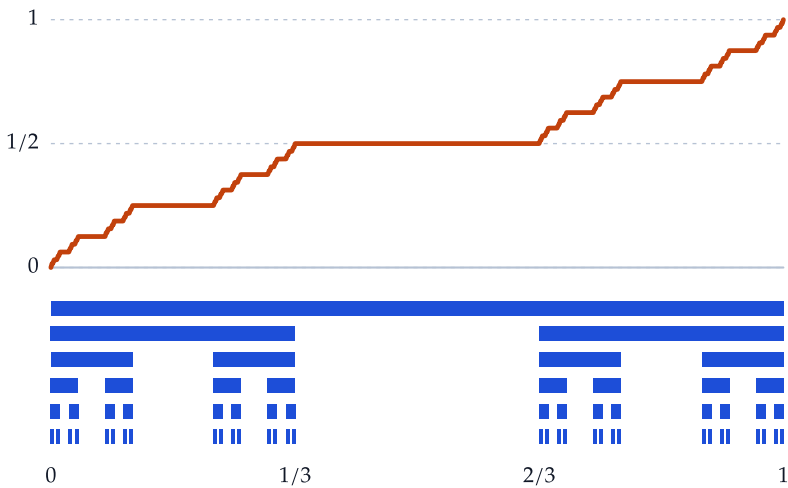


Figure 51.1. Below, the first six stages of the thirds set, each stage keeping the outer thirds of every piece of the stage before. Above, the staircase: flat over every removed interval, at heights $1/2$, then $1/4$ and $3/4$, then the odd eighths, and yet continuous and rising from 0 to 1. Assumes: six stages of a process that does not end; at the scale of the page the seventh stage could not be told from the sixth

A function with derivative 0 throughout an interval is constant. This one is not constant, and the theorem is not contradicted, because at the points of the thirds set s has no derivative. Take such a point. The kept piece of stage n that holds it is $1/3^n$ wide and s rises by $1/2^n$ across it, an average slope of $(3/2)^n$ over the whole piece. Measured from the point to the two ends of the piece, the slopes cannot both be below that average, so from the point to one end or the other the slope is at least $(3/2)^n$. As n grows that end closes in on the point and these slopes pass every bound. A derivative would be the number such slopes approach, and slopes that pass every bound approach no number.

The theorem asked for a derivative at every interior point. The hypothesis that looked like pedantry was load-bearing: “derivative 0 except on a set of length 0” is not enough. Nor does integrating

the rate of s recover its change: that rate is 0 wherever it exists, so its integral is 0, while s rises by 1.

The graph has one more surprise. A chain inscribed in it is at most 2 long, since each step is at most as long as its width and its rise together, and the widths add to 1 and the rises add to 1. Put the corners of the chain at the ends of the pieces of stage n : the steps across the removed pieces are flat and add to $1 - (2/3)^n$. The steps across the kept pieces rise by 1 between them, and each of them is longer than its rise, so together they are longer than 1. The chain is therefore longer than $2 - (2/3)^n$. The lengths of the chains approach 2.

Length was given a meaning for a curve with a continuous derivative: it is the one number its inscribed chains approach. For any curve at all, length can be taken to be the smallest number that no inscribed chain exceeds. Where there is a continuous derivative the two meanings agree. Adding a corner never shortens a chain, since two sides of a triangle are together at least as long as the third; so any chain is at most as long as the finer chains that keep its corners, and those approach the one number. No chain exceeds it, then, and chains come as close to it as desired. By that measure the graph of s , a continuous rising curve from one corner of the unit square to the opposite corner, has length 2, as long as going round by the edges. Cantor described the function in 1884.

* * *

The third monster is a curve. Take a segment, remove its middle third, and replace it by the other two sides of an equilateral triangle built on it, pointing outwards. The result is a chain of four segments, each a third as long as the original. Do the same to each of the four, then to each of the sixteen, and so on. Von Koch proposed this in 1904. He wanted a curve with no tangent at any point, built in a way that could be seen. Three of them, built outwards on the sides of an equilateral triangle, make a closed curve, the snowflake.

The corners made at one step are never moved again, so they lie

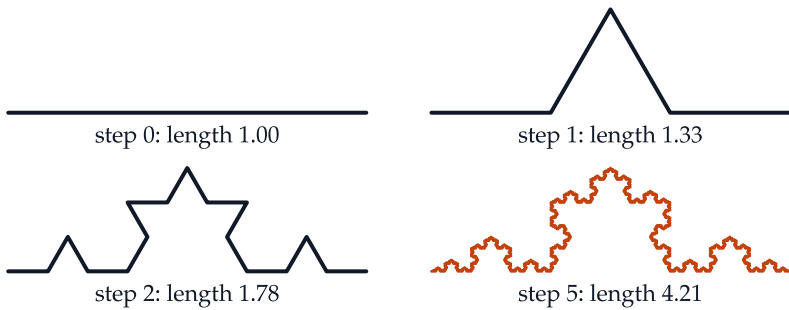


Figure 51.2. Von Koch's curve after 0, 1, 2 and 5 steps. Each step replaces every segment by four segments a third as long, so the length is multiplied by $4/3$ each time: 1, 1.33, 1.78 and, at the fifth step, 4.21. In the limit the curve has no finite length, no tangent anywhere, and is made of four copies of itself at a third of the size. *Assumes:* the limit curve cannot be drawn; the fifth stage, with 1024 segments, stands in for it

on the finished curve and the chain through the corners of stage n is inscribed in it. At each step the length is multiplied by $4/3$, so those inscribed chains have lengths $(4/3)^n$, which pass every bound: the curve has no finite length, and neither has any piece of it, however small, since every piece contains a small copy of the whole. The area enclosed is finite. If the first triangle has area a , the first step adds 3 triangles of area $a/9$, and each later step adds four times as many triangles, each a ninth of the size, so the area is

$$a(1 + 1/3 + (1/3)(4/9) + (1/3)(4/9)^2 + \dots) = a(1 + (1/3)/(1 - 4/9)) = (8/5)a$$

An endless boundary encloses eight fifths of a triangle.

Such curves turned out to describe something. Richardson had asked why Spain and Portugal gave different figures for the length of their common border. In a paper printed in 1961, eight years after his death, he reported that the measured length of a coast or a frontier depends on the length of the ruler. A ruler half as long follows more of the wiggles and gives a larger total, and each further halving multiplies the total again by the same factor. For

the west coast of Britain the measured length grows as the ruler length to the power -0.25 . For a smooth curve the power is 0, the chains settling to a limit; for the snowflake curve, a ruler a third as long finds $4/3$ of the length, which is a power of -0.26 , since $1/3$ raised to the power -0.26 is very nearly $4/3$.

That power measures how much rougher than a line the curve is, and it can be read as a dimension. A segment is made of 3 copies of itself at a third of the size, a square of 9, a cube of 27: a figure of dimension d is made of 3^d copies of itself at a third of the size. Von Koch's curve on one segment is made of 4, and the snowflake, three of them joined, is no rougher. So its dimension should be the d with $3^d = 4$, which is $L(4)/L(3) = 1.26$. That agrees with the ruler, and the agreement is no accident: for a figure made of 3^d copies of itself, a ruler a third as long needs 3^d times as many steps, each a third as long, so it finds $3^d/3$ of the length, which is $1/3$ raised to the power $1 - d$. With $d = 1.26$ that power is the -0.26 the ruler found. The thirds set, made of 2 copies at a third of the size, has dimension $L(2)/L(3) = 0.63$: more than a point, less than a line. Hausdorff made the definition exact in 1918, and Mandelbrot, who read Richardson's forgotten paper, made a subject of it from 1967 on, and in 1975 gave such shapes the name fractals.

* * *

The fourth monster is the one that upset most. In 1877 Cantor found that the points of a square can be paired off with the points of one of its sides: from a point (x, y) of the square form the number whose decimal digits are those of x and y taken alternately. A number named by two strings of digits, such as $0.5000\dots$ and $0.4999\dots$, is always written as the one that does not end in nines, except that 1 is written $0.999\dots$, so that every coordinate is named by digits after the point alone; then different points give different numbers. That pairs the square with a part of the side. The side, in turn, is a part of the square. And when each of two collections pairs off with a part of the other, the two collections pair off with each other. He

wrote to Dedekind, “*I see it, but I do not believe it.*” A square and a segment have the same size. The pairing is wildly discontinuous, and it was hoped that continuity would rescue the idea of dimension: surely a continuous curve, the path of a moving point, cannot cover a square.

It can. Peano showed it in 1890, and Hilbert gave the construction the following year in a form that can be drawn. Cut the square into four quarter squares and the time interval from 0 to 1 into four quarters, and decide that during the first quarter of the time the moving point is somewhere in the lower left square, during the second in the upper left, during the third in the upper right, and during the fourth in the lower right. Now cut each quarter square into four, and each quarter of the time into four, and assign the sixteen small squares to the sixteen short times. Two things are asked of the order. Squares visited at consecutive times must share a side, and the four small squares of each quarter square must be visited during its quarter of the time. That can be done, by turning and reflecting the pattern of the first step, and it can be continued forever.

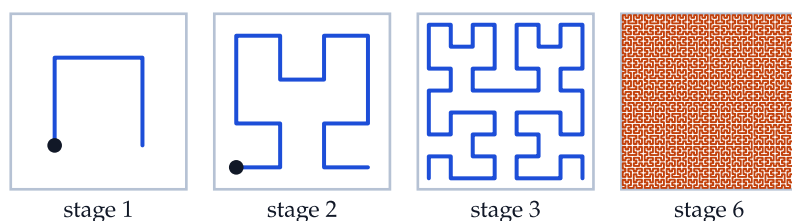


Figure 51.3. The order in which Hilbert’s curve visits the squares at stages 1, 2, 3 and 6, drawn by joining the centres of the squares in the order of their times. Squares visited at consecutive times always share a side, and each quarter of the time is spent in one quarter of the square. The curve itself is the limit, and passes through every point. *Assumes:* the chains drawn are not the curve and do not converge to it as drawings; they show the order of the visits, which is what defines the curve

Where is the point at time t ? At stage n the time t lies in one of 4^n short intervals, to which a square of side $1/2^n$ has been assigned,

and these squares are nested as the intervals are. Each square covers a stretch of x and a stretch of y . The stretches of x are nested intervals, which close on one number x by the axiom of nested intervals, and the stretches of y close on one number y in the same way. So the squares close on the single point (x, y) . That point is the position $h(t)$. When t lies at the end of a short interval it belongs to two of them at every later stage, and there are two chains of squares; but at each stage the two squares share a side. Together they make a rectangle $1/2^n$ by $2/2^n$, whose diagonal is $\sqrt{5}/2^n$, and both of the points closed on lie in it. So the two points are within $\sqrt{5}/2^n$ of each other for every n , and are one point.

The motion is continuous. Two times closer together than $1/4^n$ lie in the same short interval of stage n or in neighbouring ones, so the two positions lie in the same square of side $1/2^n$ or in two that share a side, and are closer together than $\sqrt{5}/2^n$.

And it passes through every point of the square. Let P be any point. It lies in one of the four quarter squares, in one of the four quarters of that, and so on: a nested chain of squares containing P . The times assigned to them are nested intervals, which close on a time t , and $h(t)$ is the point on which those squares close, which is P . A point on the border between squares lies in more than one chain and is visited more than once: the centre of the square is passed at the times $1/6$, $1/2$ and $5/6$.

A continuous motion lasting one second visits every point of a square. It cannot do so without visiting some points twice. Suppose a motion visited each point once. It would pair the times of the segment with the points of the square continuously, and the pairing read backwards, from points to times, would be continuous as well. For suppose points P_k approached P while their times stayed away from the time of P . Those times all lie in the interval from 0 to 1, ends included, so some subsequence of them would approach a time u of that interval, and u is not the time of P , since the times stayed away from it. The motion is continuous, so its positions along that subsequence, which approach P , also approach its position at u . The motion would be at P at the time u too, and

P would be visited twice. Netto proved in 1879 that no continuous pairing of a segment with a square, continuous in both directions, exists, and Brouwer in 1911 that none exists between spaces of different dimensions of any kind, so dimension does survive. But it survives as a theorem, and a hard one. Size does not protect it and continuity alone does not protect it.

Every one of the monsters is a limit: of finite sets, of staircases, of chains, of nested squares. Each stage is tame, and nothing visible at any stage warns of what the limit will be. The theorems about limits say exactly what passes to a limit and what need not, and that is why they carry the hypotheses they do.

What is the value of the staircase s at $1/10$ and at $3/4$, and what is its integral from 0 to 1?

Remove from the interval from 0 to 1 a middle piece of length $1/4$, then from each of the two pieces left a middle piece of length $1/16$, then from each of the four left a middle piece of length $1/64$, and so on. How much length is removed in all, and does what remains contain an interval?

Shade the odd entries in the first sixty-four rows of the triangle of choices. What figure appears, of how many copies of itself at half the size is it made, and what is its dimension?

In the snowflake construction, replace the middle third of each segment by the other three sides of a square. What are the length, the enclosed area if the sides of a square are treated so, and the dimension?

At which times does Hilbert's curve pass through the point $(1/2, 1/4)$?

THE AXIOM NOBODY STATED

The sine of an angle was defined as a ratio of two sides of a right triangle containing the angle, and the definition means something only if two right triangles with the same acute angle have proportional sides, so that the ratio belongs to the angle and not to the triangle it was measured in. That belief about the page had been used before: when rectangles were taken to tile the plane with area equal to the product of their sides, and when the radian was measured by the area of a sector. Is it a theorem, or an assumption?

In working form: given any triangle and any segment, there is a triangle with the same three angles that has the segment as one of its sides, in the corresponding place. Call this the scaled-copy principle, that shapes come in every size. Every drawing agrees with it, and that is no evidence that it follows from anything.

A construction Wallis gave in a lecture at Oxford in 1663, printed only thirty years later in his collected works, shows what the principle amounts to. Take a line l and a point P not on it. Drop the perpendicular from P to l , meeting it at Q , and let m be any line through P other than the one perpendicular to PQ at P . The claim is that m meets l , and the scaled-copy principle is what proves it.

Since m is not perpendicular to PQ , it makes an acute angle θ with the ray from P toward Q , on one side of that ray. Pick a point X on m on that side, close enough to P that the perpendicular from

X to the line PQ meets it at a point Y between P and Q . The triangle PYX has a right angle at Y and the angle θ at P . Apply the scaled-copy principle to that triangle and the segment PQ : there is a triangle PQZ , on the same side, with a right angle at Q , the angle θ at P , and PQ as the side between them.

Where is the corner Z ? The side PZ leaves P at the angle θ from PQ , on the chosen side, and m is the line that does that, so Z lies on m . The side QZ is perpendicular to PQ at Q , and l is the line that does that, so Z lies on l . The point Z is on both lines, so m meets l . Every line through P meets l , with one possible exception: the perpendicular to PQ at P . That line does miss l , by the theorem on alternate angles below. It is therefore the only line through P that misses l .

That last statement is the parallel postulate, in the form Playfair gave it. Euclid's own version is the fifth of the assumptions he laid down: if two lines are cut by a third, and the inside angles on one side together make less than two right angles, then the two lines meet on that side. Given the other assumptions, the two versions are equivalent. Each form gives the other by an argument of Proclus, relied on here and not set out.

The construction shows that the scaled-copy principle implies it. The reverse is also true. With the postulate, the copy is built as follows. Let the triangle be ABC , and suppose the given segment is to take the place of the side AB . Lay the segment along AB from the corner A , extending AB if need be, and through its far end draw the parallel to the opposite side BC . The parallel meets the line AC , since otherwise AC and BC would be two lines through C both missing it, and it makes the same angles with the lines AB and AC as BC did, so the triangle it cuts off at A has the same three angles as ABC and has the segment as its side. Euclid's theory of similar triangles, in the sixth book of the *Elements*, rests on that construction. So the principle and the postulate are one assumption in two forms.

Each of Euclid's other assumptions is about a finite drawing: two points determine a line, a line can be extended, a circle can

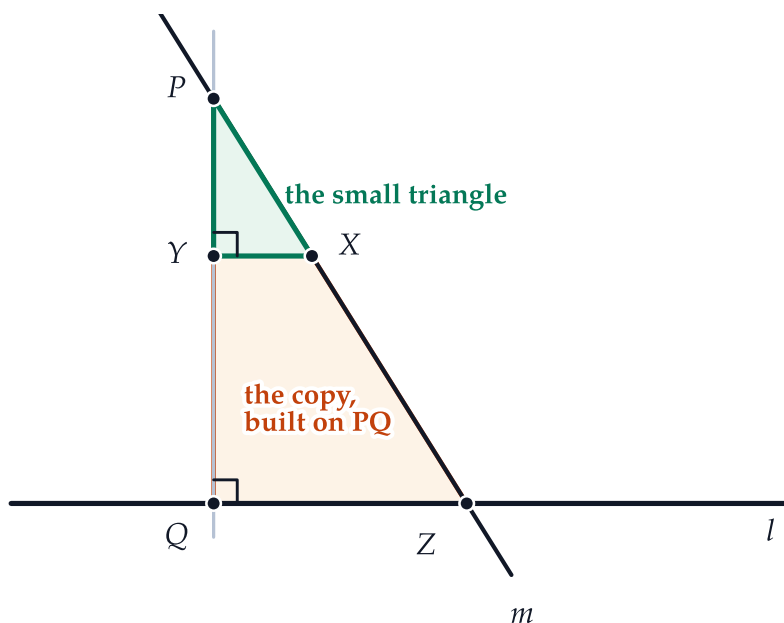


Figure 52.1. A line l , a point P off it, the perpendicular PQ , and a line m through P making an acute angle with PQ . A small right triangle PYX is cut off by dropping a perpendicular from a point of m ; if a copy of it with the same angles can be built on the side PQ , the copy's third corner Z lies on m by its angle at P and on l by its right angle at Q , so m meets l . Shapes in every size force the parallel through P to be unique. *Assumes:* that the scaled copy exists, which is the whole hypothesis under examination; the drawing shows the copy already built and cannot show whether building it was allowed

be drawn about any centre, all right angles are equal. The fifth, in either form, speaks of what two lines do however far they run, whether they meet at last or never meet at all, and no ruler can check that.

* * *

Where does the postulate enter the theorem that the angles of a triangle add up to two right angles? Separating the proof into two

halves locates the step.

The first half needs no postulate: if a line crosses two others and makes equal alternate angles with them, the two lines do not meet. It rests on an older theorem, that an exterior angle of a triangle exceeds each of the two interior angles not next to it. In the triangle ABC , extend BC beyond C to D , take the midpoint M of AC , and extend BM beyond M to E with ME equal to BM . Compare the triangles ABM and CEM . The sides AM and CM are equal because M is the midpoint of AC , the sides BM and EM are equal by construction, and the angles between them at M are vertically opposite. Two sides and the angle between them agree, so the triangles agree, and the angle ECM equals the angle BAC ; and the angle ECM lies inside the exterior angle ACD , so the exterior angle exceeds the angle at A . The same construction on the side BC , using the midpoint of BC and extending AC beyond C , shows that the exterior angle on that side of C exceeds the angle at B . That exterior angle is vertically opposite ACD and so equal to it, so ACD exceeds the angle at B as well. Nothing was used beyond extending lines, halving a segment, and the agreement of two triangles with two sides and the included angle equal.

Now suppose two lines making equal alternate angles with a transversal met. The two lines and the transversal would form a triangle. Of the two equal alternate angles, the one on the side where the lines meet is an interior angle of that triangle, and the other is an exterior angle at another corner, so not next to it. The theorem forbids an exterior angle equal to such an interior angle, so the lines do not meet.

The second half is where the postulate enters. In a triangle with angles A , B and C at the corners of the same names, draw through C the line making with CA an alternate angle equal to A ; by the first half it does not meet the line AB . Draw also through C the line making with CB an alternate angle equal to B ; it does not meet AB either. If the two lines are one line, then the angles A , C and B lie side by side at C along it and fill a straight angle, which is two right angles. That they are one line is exactly the postulate: through C

there is only one line missing AB . Without the postulate the two lines may differ, the three angles at C may fail to fill the straight angle, and the angle sum may fall short.

Fall short, and never exceed: that much survives without the postulate. One more assumption is needed, the axiom of Archimedes in the form it takes for angles: halving an angle often enough makes it smaller than any given angle. From that and Euclid's other assumptions, no triangle has an angle sum above two right angles. Two facts carry the proof. First, any two angles of a triangle add up to less than two right angles, since the exterior angle next to one of them exceeds the other, and an angle with its exterior angle makes two right angles. Second, any triangle can be traded for one with the same angle sum in which one angle is at most half of a chosen angle of the first.

In the triangle ABC with angles α , β and γ , let M be the midpoint of BC and extend AM beyond M to E with ME equal to AM , as in the proof of the exterior-angle theorem. The triangles AMB and EMC agree in two sides and the angle between them, so the angle MEC equals the angle MAB , and the angle MCE equals β . The triangle AEC therefore has the angle MAC at A , the angle MAB at E , and the angle $\gamma + \beta$ at C , the ray CM lying between CA and CE . The angles MAC and MAB are the two parts into which the ray AM cuts α , so the angles at A and E add up to α , and one of them is at most $\alpha/2$. The angle sum of AEC is then $\alpha + \beta + \gamma$, the same as that of ABC .

Now suppose some triangle had angle sum two right angles plus δ , with δ positive. Trade it n times, each time halving the angle the last trade made small: the result is a triangle with the same angle sum and an angle of at most $\alpha/2^n$, which for n large is below δ . Its angle sum is still two right angles plus δ , and one angle is less than δ , so its other two angles add up to more than two right angles, against the first fact. Saccheri and Legendre both found this theorem.

For any triangle, call its angle sum minus two right angles its *excess*. Under the postulate every excess is 0. The converse holds too, by a ladder of Legendre's: if every triangle has excess 0, then

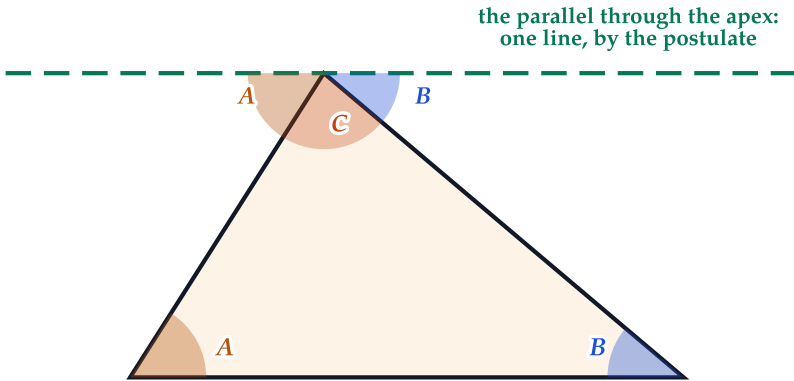


Figure 52.2. A triangle with angles A , B , C . Through the apex, one line makes with CA the alternate angle equal to A , and one makes with CB the alternate angle equal to B ; each is parallel to the base by the exterior-angle theorem, which needs no postulate. If the two are one line, the angles A , C , B fill a straight angle at the apex. That they are one line is the postulate, and nothing else in the proof is. *Assumes:* that the two lines drawn through the apex coincide, as the picture makes them; without the postulate they may be two lines with a gap between them at the apex, and the drawing cannot show the possibility it was made to deny

through a point P only one line misses a line l . Let Q be the foot of the perpendicular from P to l , and let m be a line through P making with PQ an angle θ less than a right angle. Mark Q_1 on l , on the side of PQ where m makes its angle θ , with QQ_1 equal to PQ . The triangle PQQ_1 has a right angle at Q between equal sides, so, its excess being 0, its other two angles are half a right angle each. Mark Q_2 beyond Q_1 with Q_1Q_2 equal to PQ_1 . The triangle PQ_1Q_2 has equal sides PQ_1 and Q_1Q_2 , so its angles at P and Q_2 are equal. Its angle at Q_1 sits on l next to the angle PQ_1Q , which is half a right angle, so it is two right angles less half of one. Its excess being 0, the angles at P and Q_2 share the remaining half of a right angle, a quarter each. So the angle QPQ_2 is a half and a quarter of a right

angle.

Continuing, the angle from PQ to PQ_n is a right angle less a 2^n -th part of one, 45° , 67.5° , 78.75° , ..., and by the axiom of Archimedes for angles it exceeds θ for some n . Then m enters the triangle with corners P , Q and Q_n at the corner P , between its two sides there, and a line that does so leaves through the opposite side, which lies on l . So m meets l .

That last step grants the page one more thing that no list of Euclid's records: a line drawn into a triangle at one corner leaves through the side facing it. It follows from an assumption Pasch stated in 1882, that a line entering a triangle through one side, away from the corners, leaves through one of the other two, and the argument about the outside angle above leaned on it too. Turned round: if the postulate fails, some triangle has an excess other than 0. By the theorem above the excess cannot be positive, so that triangle's angle sum falls short of two right angles, and a protractor could in principle detect the shortfall. The ladder cannot do without the axiom of Archimedes. In 1900 Dehn built a plane in which that axiom fails: there every triangle has angle sum two right angles, and yet many lines through a point miss a given line.

Gauss's survey of Hanover in the early 1820s measured the angles of a triangle whose corners were the summits of the Hohenhagen, the Brocken and the Inselsberg, with sides of about 69, 85 and 107 kilometres. On a sphere the size of the earth, where the angle sum of every triangle exceeds two right angles, a triangle with those sides has an excess of about fifteen seconds of arc, some four thousandths of a degree, and on a larger sphere the excess is smaller still. The excess, where there is one, shrinks with the triangle, and a small triangle on any smooth surface looks flat. No measurement of a finite figure can confirm the postulate, and failing to refute it is no evidence for it.

* * *

On a sphere the excess is not 0, and its consequences can be

watched. The part of straight lines is played by great circles, the circles whose centre is the centre of the sphere, since a thread stretched tight between two points on the surface runs along one. Take the triangle whose corners are the north pole and two points on the equator a quarter of the way round from each other. Its sides are arcs of great circles, and its three angles are right angles: the two on the equator because meridians cross the equator at right angles, and the one at the pole because the two meridians are a quarter turn apart. Its angle sum is three right angles, its excess is a right angle, and it covers one eighth of the sphere.

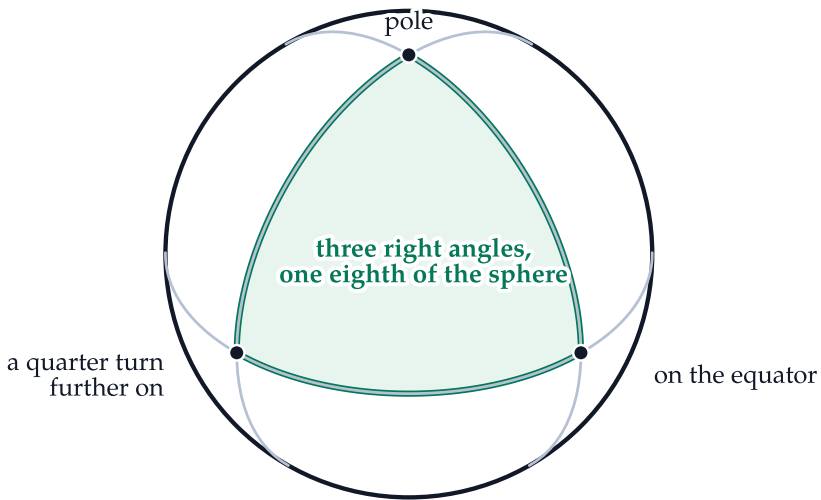


Figure 52.3. The triangle on a sphere whose corners are the pole and two points on the equator a quarter turn apart. Its three sides are arcs of great circles, each of its three angles is a right angle, its angle sum is three right angles, and it covers one eighth of the surface. A copy of it at half the size would have to have a smaller angle sum, so no such copy exists. *Assumes:* that great circles play the part of straight lines, being the paths of a thread stretched tight on the surface; the drawing is a flat projection and its right angles do not all look right

The excess of a spherical triangle is proportional to its area. Har-

riot found this in 1603, and Girard published it in 1629. The proof counts one covering in two ways. Extend the three sides of a triangle T to full great circles. The two great circles through the corner with angle α cut the sphere into four wedges, and the wedge containing T together with the opposite wedge makes up the fraction α/π of the surface, α being measured in radians, since the pair of wedges takes up 2α of a full turn of 2π .

Area on the sphere is taken here in its everyday sense, as the area of a sector was. Turning the sphere about the axis through the corner changes no area, and it carries a wedge at that corner onto any other wedge there of the same angle, so wedges of equal angle have equal area. If n equal wedges fill the sphere, each therefore has one n -th of the sphere's area, and k of them side by side have k times as much. Any wedge is squeezed between wedges of that kind.

Do the same at the other two corners, with angles β and γ , and add up the areas of the three pairs of wedges. The points of T lie in all three pairs, and so do the points of the opposite triangle, whose corners are the points diametrically opposite those of T ; every other point of the sphere lies in exactly one pair. So the three pairs cover the whole surface once, T twice more, and the opposite triangle twice more, and the opposite triangle has the same area as T , being its mirror image through the centre. With S for the area of the sphere and $A(T)$ for that of the triangle,

$$(\alpha + \beta + \gamma) \cdot S/\pi = S + 4 \cdot A(T)$$

so $A(T) = S \cdot (\alpha + \beta + \gamma - \pi)/(4\pi)$. The area is the excess multiplied by the fixed number $S/(4\pi)$. For the triangle with three right angles the excess is $\pi/2$ and the formula gives $S/8$. With the area $4\pi R^2$ of a sphere of radius R , which Archimedes found and which the turning of a half circle about its axis gave earlier, the excess in radians is the area of the triangle divided by R^2 . That leads to the fifteen seconds of Gauss's triangle. A flat triangle with sides 69, 85 and 107 kilometres has area 2929 square kilometres, by the law of cosines and half $ab \sin C$, and a spherical triangle so small has

very nearly that area; divided by the square of 6371 kilometres it is 0.0000722 radians, which is 14.9 seconds of arc. Doubling the radius divides the excess by four.

What fraction of a sphere does the triangle cover whose corners are the north pole and two points on the equator a sixth of the way round from each other?

Two consequences follow. Triangles with the same angles and different sizes do not exist on a sphere, because the angles fix the area; the scaled-copy principle is false there, and a sine defined as a ratio of sides depending only on the angle does not exist. Both can be watched. Take a triangle with right angles at both ends of one side. Its other two sides run along great circles perpendicular to that side's great circle, and all such circles pass through the two poles of that great circle, a quarter circle away from it. The other two sides meet at the third corner, so the third corner is one of those poles, and each of the two sides is a quarter circle long. In a triangle with three right angles this applies to every side, so all three sides are a quarter circle each, and the triangle comes in one size.

Now cut that triangle in half by the meridian from the pole to the middle of its equator side. Each half has angles of 45° , 90° and 90° . The side facing its angle of 45° is half of the equator side, an eighth of a circle, and each side facing a right angle is a quarter circle, so the first is half as long as the second. On the flat page the side facing an angle of 45° in a right triangle is 0.7071 of the side facing the right angle, that number being the sine of 45° .

And there is a length that the geometry picks out by itself. On the flat plane no length is distinguished from any other, and the unit had to be chosen; on the sphere the side of the triangle with three right angles is a definite length that nobody chose.

A convex solid with flat faces has a point inside it from which every ray meets its surface exactly once. Project the surface from that point onto a sphere centred there. What does the area formula for spherical triangles say about corners minus edges plus faces?

* * *

The sphere does not show that the postulate is independent of Euclid's other assumptions, because it breaks some of those as well. Two great circles always meet, twice, so two points diametrically opposite lie on many lines, and a great circle is finite, so a line cannot be extended beyond every length. The exterior-angle argument used that the segment BM can be doubled without its far end coming back round, and on the sphere it can come back round, which is where that argument fails. A surface on which every one of Euclid's other assumptions holds and the postulate alone fails would settle the question.

Meanwhile the earlier uses of the postulate can be listed. The unit square whose diagonal was measured, with four right angles and four equal sides, needs the postulate. Under Euclid's other assumptions together with the axiom of Archimedes, a quadrilateral with four right angles exists only if the postulate holds, as Saccheri and Legendre showed. The sphere has none: there the angles of every quadrilateral add up to more than four right angles, since a diagonal splits it into two triangles each with a positive excess. The rectangle with area ab , the tiling of the plane by rectangles, and the sector whose area measured an angle are objects of the flat plane, and the calculus built on them is the calculus of a flat plane. None of it is wrong: it is a body of theorems about a geometry with one more assumption than had been stated.

THE SPHERE'S OWN TRIGONOMETRY

On a sphere there are no triangles of the same shape and different sizes, so the sine and cosine of the page, which are ratios of sides that depend on the shape alone, have nothing to say directly about a triangle drawn there. Yet the oldest trigonometry is spherical. It was made for the sky, which looks like the inside of a sphere, and for the earth, which is the outside of one, by astronomers who needed the distance between two stars or the direction of one city from another. What takes the place of the theorem on squares, and of the rules that solve a flat triangle from some of its parts?

Take a sphere of radius 1. A point of it is a direction from the centre, and the distance between two points, along the great circle through them, is the angle between the two directions, in radians, because on a circle of radius 1 an arc is as long as the angle it spans. So the sides of a spherical triangle are angles too. Write a , b and c for the sides and α , β and γ for the angles facing them, the angle at a corner being the angle between the two great circles that meet there, which is the angle between their planes.

Two points of the unit sphere whose directions are an angle c apart are joined, through the inside of the sphere, by a chord, and in the flat triangle formed by the chord and the two radii the law

of cosines gives

$$\text{chord}^2 = 1 + 1 - 2 \cos c$$

So $\cos c$ can be found from the chord, and the chord can be found from coordinates, by the theorem on squares used twice. Let one point be d_1 east, d_2 north and d_3 up from another. In the floor, the point directly below it is at distance $\sqrt{(d_1^2 + d_2^2)}$ from the other. Up the wall, that floor distance and the height d_3 are the legs of a second right triangle, whose hypotenuse joins the two points. So the distance between them is $\sqrt{(d_1^2 + d_2^2 + d_3^2)}$.

Place the corner C of the triangle at the north pole. Let the axis through the pole be the third coordinate. A point at angular distance b from the pole has its direction tilted by the angle b away from the axis, so its height is $\cos b$ and its distance from the axis is $\sin b$. On the meridian of longitude 0 that distance from the axis lies wholly along the first coordinate, so the point is $(\sin b, 0, \cos b)$: that is the corner A . A point at distance a from the pole has height $\cos a$ and distance $\sin a$ from the axis, and on the meridian of longitude γ that distance is turned through the angle γ , which splits it into $\sin a \cos \gamma$ along the first coordinate and $\sin a \sin \gamma$ along the second. So the point is $(\sin a \cos \gamma, \sin a \sin \gamma, \cos a)$: that is B . The angle at the pole between the two meridians is γ . The square of the chord from A to B is the sum of the squares of the three differences, and multiplied out, with $\sin^2 + \cos^2 = 1$ used three times, it is $2 - 2(\sin a \sin b \cos \gamma + \cos a \cos b)$. The chord formula gives the same square as $2 - 2 \cos c$. Setting the two equal,

$$\cos c = \cos a \cos b + \sin a \sin b \cos \gamma$$

This is the law of cosines of the sphere. It gives the third side from two sides and the angle between them, and any angle from the three sides. When γ is a right angle it reads

$$\cos c = \cos a \cdot \cos b$$

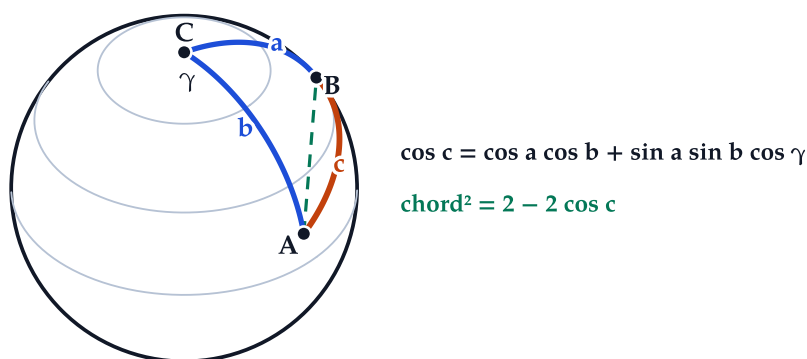


Figure 53.1. A spherical triangle placed with its corner C at the north pole. The sides b and a run down two meridians to the corners A and B, the angle γ at the pole is the difference of their longitudes, and the third side c is the arc of the great circle from A to B. The straight chord from A to B, through the inside of the ball, gives $\cos c$. *Assumes:* a view from a point above latitude 38° , seen from far away; the latitude circles are drawn only where they face the viewer, while the triangle is drawn whole, one of its corners lying a little past the edge of the visible half

which is what the theorem on squares becomes on a sphere: the cosine of the hypotenuse is the product of the cosines of the legs.

* * *

Where did the theorem on squares go? It is still there, for small triangles. The series for the cosine begins $1 - x^2/2 + x^4/24$, and the series for the sine begins $x - x^3/6$. Put these into the law of cosines and keep the terms up to the second power of the sides: the left side is $1 - c^2/2$, and the right side is $(1 - a^2/2)(1 - b^2/2) + ab \cos \gamma$, which to that accuracy is $1 - a^2/2 - b^2/2 + ab \cos \gamma$. So

$$c^2 = a^2 + b^2 - 2ab \cos \gamma, \text{ apart from terms of the fourth power of the sides}$$

and that is the law of cosines of the page. Take a triangle whose sides are small fractions of the radius. Flat trigonometry governs

it with an error that, as a proportion of the sides, is of the size of the square of those fractions, which is smaller still. This is the exact sense in which a sphere is flat in the small, and the reason nobody standing on the earth notices that it is round.

The size of the error can be seen in a right triangle on the earth, of radius 6371 kilometres, with legs of 1000 kilometres each. The legs are $1000/6371 = 0.15696$ of a radian. With equal legs the rule $\cos c = \cos a \cos b$ reads $\cos c = \cos^2 a$, which gives $c = 0.22152$, and times 6371 that is 1411.3 kilometres. The theorem on squares gives 1414.2. The difference is 2.9 kilometres in fourteen hundred, two parts in a thousand, for a triangle the size of western Europe.

Surveyors use a more refined statement, found by Legendre in 1787. A small spherical triangle, with sides measured along the ground, may be solved as a flat triangle with the same sides, provided each of its angles is first reduced by one third of its excess. For the triangle of Hohenhagen, the Brocken and the Inselsberg, with sides of 69, 85 and 107 kilometres and an excess of 14.89 seconds of arc, each angle can be computed twice from the three sides, once by the law of cosines of the sphere and once by the law of cosines of the page. The spherical value is the larger each time, and the three differences are 4.9623, 4.9623 and 4.9623 seconds, where a third of the excess is 4.9623. The excess is shared equally among the corners to an accuracy far beyond any theodolite, whatever the shape of the triangle. The proof carries the expansion above one order further; it is relied on here and not given.

* * *

Keep the corner C at the pole and let the other two corners be two places on the earth. The distance of a place from the pole is a right angle less its latitude, so a and b are a right angle less each of the two latitudes, and γ , the angle between the two meridians, is the difference of the two longitudes. The cosine of a right angle less φ is $\sin \varphi$, and its sine is $\cos \varphi$, so in the law of cosines the sines and cosines of the sides change places. For two places with latitudes φ_1

and φ_2 whose longitudes differ by λ , the angle d between them satisfies

$$\cos d = \sin \varphi_1 \sin \varphi_2 + \cos \varphi_1 \cos \varphi_2 \cos \lambda$$

and the distance over the ground is d times the radius of the earth. London lies at 51.5 degrees north and 0.1 west, New York at 40.7 north and 74.0 west. The formula gives $\cos d = 0.6412$, so d is 50.12 degrees, 0.8747 of a radian, and the distance is 5573 kilometres.

The route of that length is not the one a flat map suggests. A course held at a fixed compass bearing crosses every meridian at the same angle. On the map of Mercator the meridians are parallel upright lines, and the scale of latitude is stretched by the amount that keeps angles true; the stretching is relied on here and not derived. So the course drawn on that map crosses the parallel meridians always at the same angle, and a path that does that is a straight line. Such a course from London to New York is 5797 kilometres long, a number that comes out of the same map. The great circle is 224 kilometres shorter, and it sets out from London heading 18 degrees north of west, although New York lies eleven degrees of latitude to the south; it climbs to latitude 53.8 before turning south. The direction of departure comes from the second rule of the sphere.

That rule is the law of sines. From the law of cosines, $\cos \gamma = (\cos c - \cos a \cos b) / (\sin a \sin b)$. Then $\sin^2 \gamma$ is 1 less the square of that. Written as a single fraction, it has the denominator $\sin^2 a \sin^2 b$ and the numerator $\sin^2 a \sin^2 b - (\cos c - \cos a \cos b)^2$. Replace each $\sin^2$ by 1 less $\cos^2$: the numerator is $(1 - \cos^2 a)(1 - \cos^2 b) - (\cos c - \cos a \cos b)^2$, which multiplies out to

$$1 - \cos^2 a - \cos^2 b - \cos^2 c + 2 \cos a \cos b \cos c$$

an expression that stays the same when a , b and c change places. So $\sin^2 \gamma / \sin^2 c$, which is that expression divided by $\sin^2 a \sin^2 b \sin^2 c$, also stays the same when the corners are renamed. But renaming the corners turns $\sin^2 \gamma / \sin^2 c$ into $\sin^2 \alpha / \sin^2 a$ or into $\sin^2 \beta / \sin^2 b$, so the three ratios are equal. The sides and angles of a triangle



Figure 53.2. Two routes from London to New York on a map of Mercator's kind, where a course held at one compass bearing is a straight line. The great circle, the route along the sphere's own straight lines, bulges north to latitude 53.8 and is 5573 kilometres long; the straight line on the map is 5797. Assumes: a spherical earth of radius 6371 kilometres; the flattening of the real earth changes these distances by a few kilometres

lie between 0 and π , where the sine is positive, so square roots may be taken:

$$\sin \alpha / \sin a = \sin \beta / \sin b = \sin \gamma / \sin c$$

On the page the law of sines has the sides themselves in the denominators, where this one has their sines. The flat law follows from this one as the sides shrink, because the sine of a small side is very nearly the side itself. For the route, take the triangle with corners at the pole, London and New York. The side facing London's corner is New York's distance from the pole, 49.3 degrees, the side facing the pole is d , and the angle at the pole is the difference of longitudes, 73.9 degrees. The law of sines says that the sine of the angle at London divided by $\sin 49.3^\circ$ equals $\sin 73.9^\circ$ divided by $\sin 50.12^\circ$. So the sine of the angle at London is $\sin 73.9^\circ \cdot$

$\sin 49.3^\circ / \sin 50.12^\circ$, which is 0.9492. Two angles of a triangle have that sine, 71.7 degrees and 108.3, and the law of sines cannot choose between them. The law of cosines can: with London 38.5 degrees from the pole, the cosine of the same angle is $(\cos 49.3^\circ - \cos 38.5^\circ \cos 50.12^\circ) / (\sin 38.5^\circ \sin 50.12^\circ)$, which is 0.3146. It is positive, so the angle is the smaller one, 71.7 degrees from north towards the west: 18.3 degrees north of west. For a destination far enough to the south the cosine comes out negative, and the larger angle is the one to take.

* * *

There is a third rule, which has no counterpart on the page. Each side of a spherical triangle lies on a great circle, and a great circle has two poles, the ends of the axis perpendicular to its plane. For each side take the pole that lies on the same side of it as the opposite corner. The three poles A' , B' and C' are the corners of the *polar triangle*.

The relation is mutual. A' is a quarter circle from every point of the great circle BC , in particular from B and from C . Likewise B' is a quarter circle from C and from A . So C is a quarter circle from both A' and B' . The direction of C is then perpendicular to the directions of A' and of B' , and so to the whole plane of the great circle through them, which makes C a pole of the side $A'B'$. Of the two poles it is the one on the side of C' . For C' was chosen on the same side of AB as C , which means less than a quarter circle from C ; then C is less than a quarter circle from C' , which puts C on the same side of $A'B'$ as C' . So the original triangle is the polar triangle of its polar triangle.

And the sides of the polar triangle are the supplements of the angles of the original. Both A' and B' are a quarter circle from C , so they lie on the great circle whose pole is C , which for an observer standing at C is the horizon. A point of the horizon is fixed by the direction in which it is seen from C , and the arc between two points of the horizon is the difference of their directions.

Measure directions at C so that the side CA leaves C in the direction 0 and the side CB in the direction γ . A' is a pole of the great circle through C and B , so from C it is seen at a right angle to the direction γ , to one side or the other: in the direction γ plus a right angle, or γ less a right angle. It was chosen on the side where A lies, and A is seen in the direction 0 , which is on the lesser side of γ , so A' is in the direction $\gamma - \pi/2$. In the same way B' is a right angle from the direction 0 on the side where B lies, which is the direction $\pi/2$. The arc between them is the difference, $\pi/2 - (\gamma - \pi/2)$, so $c' = \pi - \gamma$. Since the original is the polar triangle of the polar triangle, its sides are in turn the supplements of the polar triangle's angles.

Apply the law of cosines to the polar triangle: $\cos c' = \cos a' \cos b' + \sin a' \sin b' \cos \gamma'$. Now translate. The sides a' , b' and c' are $\pi - \alpha$, $\pi - \beta$ and $\pi - \gamma$, and the angle γ' is $\pi - c$. The cosine of a supplement is the cosine with its sign changed, and the sine of a supplement is the same sine. So the left side becomes $-\cos \gamma$, the first product becomes $\cos \alpha \cos \beta$ with its two changes of sign cancelling, and the last term becomes $-\sin \alpha \sin \beta \cos c$. With every sign reversed, the formula reads

$$\cos \gamma = -\cos \alpha \cos \beta + \sin \alpha \sin \beta \cos c$$

The three angles determine the sides: solved for $\cos c$, the formula reads $\cos c = (\cos \gamma + \cos \alpha \cos \beta) / (\sin \alpha \sin \beta)$, and the other two sides come in the same way. On the page three angles determine a shape and leave the size free; on the sphere, as the area already showed, they determine everything, and this formula says how. The triangle with three right angles has $\cos c = 0$, a quarter circle for each side. The triangle with three angles of 72 degrees, five of which fit round a point, has

$$\cos c = (\cos 72^\circ + \cos^2 72^\circ) / \sin^2 72^\circ = \cos 72^\circ / (1 - \cos 72^\circ) = 0.4472$$

so c is 63.43 degrees. The excess of such a triangle is $216 - 180 = 36$ degrees, which is $\pi/5$, and that is its area; the sphere has area 4π ,

so it takes twenty such triangles to cover the sphere, five at every corner. They have sixty corners between them, and each point is shared by five, so there are twelve points, the twelve corners of the solid with twenty triangular faces. Its neighbouring corners are therefore 63.43 degrees apart as seen from the centre, an angle whose tangent is exactly 2, and from that one number the solid can be built.

Menelaus of Alexandria wrote the first treatise on spherical triangles, about the year 100. The law of sines was found in the tenth century in Baghdad and Persia, by Abu al-Wafa and his contemporaries, who needed the direction of Mecca from every city of Islam, which is the angle at London in the computation above with Mecca in place of New York. The law of cosines, as a statement about every spherical triangle, is in Regiomontanus, in a work finished in 1464 and printed in 1533; it is stated there in versed sines. The versed sine of an angle is 1 less its cosine, and the statement is that the versed sine of c is the versed sine of $a - b$, plus $\sin a \sin b$ times the versed sine of γ . The form written above came later.

In which direction does Mecca, at 21.4 degrees north and 39.8 east, lie from London, and how far is it?

On a sphere of radius 1, how long are the sides of the triangle with three angles of 120 degrees, and of the triangle with three angles of 90 degrees and 60 degrees and 60 degrees? What solids do they belong to?

How large is the angle sum of the right triangle on the earth with legs of 1000 kilometres, first from the rules above and then from its area?

What does the rule $\cos c = \cos a \cos b$ say about a right triangle with one leg a quarter circle, and why is that what a globe shows?

Along a parallel of latitude, a place keeps a fixed distance from the pole, but the parallel is not a great circle. How much longer is the journey from London due west along its parallel to longitude 74 west than the great circle between the same two points?

CIRCLES THAT CROSS AT RIGHT ANGLES

The flat plane, with the postulate of parallels and the angle sum of two right angles, has a theory of circles that the sphere and every other surface lack, and the world in which the postulate fails is going to be built out of that theory. It is the third book of Euclid. Three theorems carry it, and each is a consequence of the angle sum of a triangle.

An angle at the circumference is half the angle at the centre. Let A , B and P be points of a circle with centre O , and look at the chord AB from P and from O . Draw the diameter from P through O to the far side, at Q . The triangle OPA has two equal sides, the radii, so its angles at P and A are equal. The angle AOQ is an exterior angle of that triangle: it is two right angles less the triangle's angle at O . The three angles of the triangle add up to two right angles, so AOQ is equal to the sum of the other two, the angles at P and at A , and that sum is twice the angle OPA . Likewise BOQ is twice OPB . Adding, or subtracting when A and B lie on the same side of the diameter, the angle AOB is twice the angle APB . So

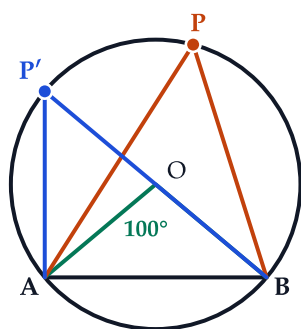
every point of an arc of a circle sees the chord of the arc under the same angle, half the angle at the centre.

When the chord is a diameter, the angle at the centre is two right angles and the angle at the circumference is a right angle, which is the theorem of Thales. The chord cuts the circle into two arcs, and a point on either arc sees the chord under half the angle at the centre that opens towards the other arc. Those two angles at the centre together make a full turn, which is four right angles, so the two angles at the circumference add up to two right angles. Now take a four-sided figure with its corners on a circle. A diagonal of it is a chord, and the two corners that the diagonal does not join lie on opposite arcs of it. So a four-sided figure with its corners on a circle has opposite angles that add up to two right angles. The converse holds too: if the opposite angles of a four-sided figure add up to two right angles, its corners lie on a circle. Draw the circle through three of the corners, and look at the diagonal that joins the two corners next to the fourth. A fourth corner inside that circle would see the diagonal under a larger angle than a point of the arc does, and one outside under a smaller, by the exterior angle of a triangle. Either way the sum of the opposite angles would not be two right angles, so the fourth corner is on the circle.

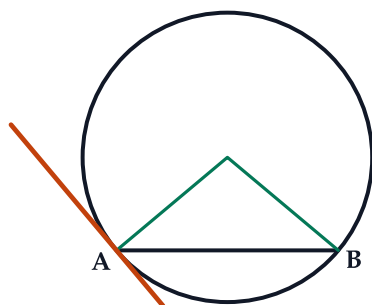
The tangent and the chord. Let P slide along the circle towards A . The line PA turns, and in the limit it is the tangent at A , while the line PB becomes the chord AB . The angle between them stays what it was. So the angle between the tangent at A and a chord AB is half the angle under which the chord is seen from the centre. It can be seen directly as well. The tangent is perpendicular to the radius OA : the triangle OAP has two equal sides, the radii, so its angle at A is a right angle less half its angle at O , and as P slides to A its angle at O shrinks to nothing, so the line PA comes to stand at right angles to OA . And the triangle OAB has base angles of a right angle less half the angle at O , and the angle between tangent and chord is what is left of the right angle between tangent and radius when that base angle is taken away, which is half the angle at O .

The power of a point. Let X be a point inside a circle, and let two chords AB and CD pass through it. The triangles XAC and XDB have equal angles at X , and the angles at A and at D are equal, since

Circles that cross at right angles



50° from every point of the arc



tangent and chord: 50° again

Figure 54.1. On the left, the chord AB seen from the centre O and from two points P and P' of the same arc: the angle at the centre is 100° , and the angle at each point of the arc is 50° . On the right, the same chord with the tangent at A : the angle between tangent and chord is again 50° . *Assumes:* the flat plane, where the angles of a triangle add up to two right angles; on a sphere the angle at the circumference is not the same from every point of the arc

both see the chord CB from the circumference. Triangles with the same angles have proportional sides, so $XA/XD = XC/XB$, that is,

$$XA \cdot XB = XC \cdot XD$$

The product of the two pieces into which X cuts a chord is the same for every chord through X . For X outside the circle, with one line through it cutting the circle at A and then B and another at C and then D , the triangles XAC and XDB share the angle at X , and the angles at A and at D are equal as well. For A and D see the chord CB from opposite arcs, so the angles CAB and CDB add up to two right angles; and the angles XAC and CAB lie side by side along the straight line XB , so they add up to two right angles too. Hence the angle XAC equals the angle CDB . The triangles again have the same angles, and the same equation follows, the distances now being measured from X to the near and the far crossing. When one of the lines turns until its two crossings merge, it becomes a tan-

gent, and the product becomes the square of the tangent's length. That common product is the *power* of the point X . Taking the chord through the centre, it is $(d - r)(d + r)$, or $d^2 - r^2$, for a point outside at distance d from the centre of a circle of radius r , and $r^2 - d^2$ for a point inside.

An observer whose eye is at height h above a smooth sea looks along a tangent to the earth. The line from the eye through the centre of the earth cuts the surface at distances h and $h + 2R$ from the eye, so the square of the distance to the horizon, which is the length of a tangent from the eye, is the power $h(h + 2R)$, and since h is tiny beside R the distance is very nearly $\sqrt{(2Rh)}$. With $R = 6371$ kilometres, an eye at 1.7 metres sees 4.7 kilometres; from a cliff of 100 metres the horizon is at 36 kilometres; and from an aircraft at 10 kilometres it is at 357. An archaeologist holding a fragment of the rim of a plate lays a ruler across it. Suppose the chord is 10 centimetres and the rim stands 1 centimetre above the middle of the ruler. The middle of the ruler is a point inside the full circle of the plate. It cuts the chord along the ruler into pieces of 5 and 5, and it cuts the diameter through it into a piece of 1 and a piece of $2r - 1$. The two products are equal, so $5 \cdot 5 = 1 \cdot (2r - 1)$, and the plate had a radius of 13 centimetres.

* * *

The power of a point explains when two circles cross at right angles. Let a circle K , of centre c and radius r , cross a circle Γ , of centre O and radius 1, at a point T . The circles cross at right angles when their tangents at T do. Each tangent is perpendicular to its own radius, so the tangents are at right angles exactly when the radii OT and cT are. And the line through T at right angles to the radius cT is the tangent to K at T , so that happens when the radius OT of Γ is tangent to K . Then the tangent from O to K has length 1, and the power of O with respect to K is 1:

$$|Oc|^2 - r^2 = 1$$

and conversely. Now take any line through O that cuts K , at P and at a second point P^* . The product $|OP| \cdot |OP^*|$ is the power of O , which is 1. So a circle that crosses Γ at right angles is carried onto itself by the map that sends each point P to the point P^* on the same ray from O with

$$|OP| \cdot |OP^*| = 1$$

This map is *inversion* in Γ . For a circle of radius ρ the product is to be ρ^2 . It exchanges the inside of the circle with the outside, leaves each point of the circle itself in place, and undoes itself. The centre has no image; points near the centre go far away, and it is convenient to say that the centre goes to a single point at infinity and to count straight lines as circles through that point. Inversion behaves like a reflection in a curved mirror, and it has two properties that make it a tool.

It sends circles to circles, straight lines counting as circles. For inversion in the circle of radius 1 about the origin, a circle or line is the set of points with $A(x^2 + y^2) + Bx + Cy + D = 0$ for some constants, and the inversion sends (x, y) to $(x, y)/(x^2 + y^2)$. Since the inversion undoes itself, the image of the curve is the set of points whose own images lie on the curve. So substitute the image of (x, y) into the equation. The image has $x/(x^2 + y^2)$ and $y/(x^2 + y^2)$ for coordinates, and the sum of their squares is $1/(x^2 + y^2)$, so the equation reads $A/(x^2 + y^2) + Bx/(x^2 + y^2) + Cy/(x^2 + y^2) + D = 0$. Multiplying through by $x^2 + y^2$ gives $A + Bx + Cy + D(x^2 + y^2) = 0$, which is again a circle or a line. The constants A and D have changed places: a line missing the origin, which has $A = 0$ and D not 0, goes to a circle through the origin, which has $D = 0$ and A not 0, and such a circle goes back to such a line. A line through the origin has both constants 0 and goes to itself. Inversion in any other circle is this one with the plane shifted and scaled.

It preserves the size of every angle, and reverses its sense. Inversion in the unit circle is the map $z \mapsto 1/z$ on the numbers of the plane followed by taking the mirror image in the horizontal axis. For distances from 0 multiply and angles add, and z times $1/z$ is 1, so $1/z$

lies at distance 1 over the distance of z , and its angle is the angle of z taken the other way round; its mirror image in the horizontal axis therefore lies on the ray from 0 through z , at the distance that inversion asks for. The map $z \mapsto 1/z$ sends a small displacement h from z to $1/(z+h) - 1/z$, which is h times $-1/(z(z+h))$, that is, h times $-1/z^2$ plus an error small compared with h . Multiplying every displacement from z by the same number $-1/z^2$ turns them all through one angle and stretches them all by one factor, so the angle between two curves through z is unchanged; taking the mirror image then reverses the sense of every angle and keeps its size.

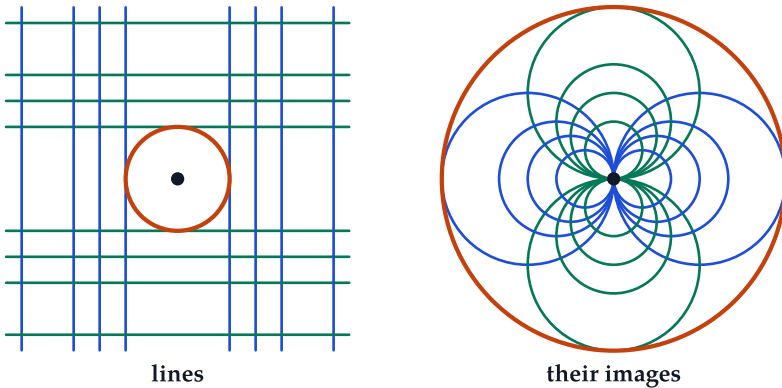


Figure 54.2. On the left, a grid of straight lines, none through the origin, and the unit circle. On the right, the image of the grid under inversion in that circle. Each line has become a circle through the origin, and the circles still cross at right angles, as the lines did; squares far from the origin have become small curved squares near it. *Assumes:* the lines $x = \pm 1, \pm 3/2, \pm 2, \pm 3$ and the same for y ; a line through the origin would go to itself

A machine shows what the first property is good for. The steam engines of the eighteenth century needed to guide a piston rod in a straight line by means of rods turning on pivots, which by themselves trace only circles. Watt's linkage of 1784, of which he was prouder than of the engine, traces a long thin figure of eight whose middle stretch is very nearly straight, and is not straight. The problem of exact straight-line motion by rods moving in a plane was

thought to be impossible until Peaucellier, a French army officer, solved it in 1864.

Take a rhombus of four equal rods, with corners P , A , Q and B in order, and join the opposite corners A and B to a fixed pivot O by two equal longer rods. Each of O , P and Q is equally far from A and from B , so all three lie on the line that cuts AB in half at right angles: O , P and Q are in line. Let M be the centre of the rhombus. The diagonals of a rhombus cut each other in half at right angles, so M is the middle of PQ and the triangles OMA and PMA have right angles at M . Then $|OP| \cdot |OQ| = (|OM| - |PM|)(|OM| + |PM|) = |OM|^2 - |PM|^2$. By the theorem on squares $|OM|^2 = |OA|^2 - |AM|^2$ and $|PM|^2 = |PA|^2 - |AM|^2$, so the product is $|OA|^2 - |PA|^2$, a constant of the machine. So Q is the image of P under inversion in a circle about O , whatever position the linkage is in. Now add one more rod, from a second fixed pivot to P , of such a length that P is forced to move on a circle through O . The inversion of a circle through O is a straight line, and Q moves along it.

* * *

The oldest use of the second property is on the sky. To map a sphere onto a plane, place the plane through the equator and the eye at the north pole N , and send each point of the sphere to the point where the line from N through it meets the plane. The southern half of the sphere lands inside the equator, the northern half outside, and the point of the unit sphere with coordinates (x, y, z) goes to

$$(x, y)/(1 - z)$$

This is *stereographic projection*. It is an inversion of space, in the sphere about N whose radius has square 2. A point of the sphere and its image lie on one ray from N , and their distances from N multiply to 2. For the square of the first distance is $x^2 + y^2 + (1 - z)^2$, which is $2 - 2z$ because $x^2 + y^2 + z^2 = 1$. And along the ray the point lies $1 - z$ below the level of N while the image lies 1 below it,

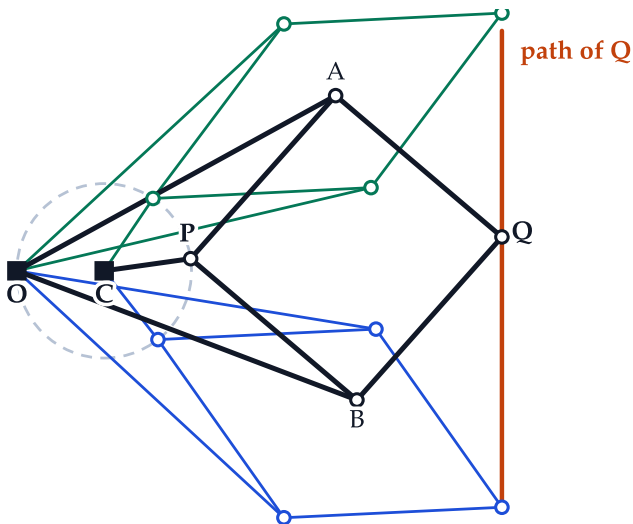


Figure 54.3. Peaucellier's linkage in three positions. The pivots O and C are fixed. The long rods OA and OB have length 5, the four rods of the rhombus length 3, and the rod CP , of length 1.2, keeps P on a circle through O . In every position O , P and Q are in line and the distances OP and OQ multiply to 16, so Q is the inverse of P and moves along a straight line. Assumes: rigid rods and perfect pivots; the three positions were computed from the rod lengths alone

so the second distance is the first divided by $1 - z$. The product is $(2 - 2z)/(1 - z)$, which is 2.

Inversion in a sphere sends spheres and planes to spheres and planes, by the same algebra with $x^2 + y^2 + z^2$ in place of $x^2 + y^2$. It preserves angles as well. Every small displacement from a point P lies in a plane through N and P , and in that plane the map is the inversion of the plane already studied. There, take a point on the horizontal axis at distance s to the right of the centre, to which any point can be brought by turning the plane about the centre. As a number of the plane that point is s . Multiplying a displacement by $-1/s^2$ reverses it and divides its length by s^2 , and the mirror image in that axis then turns its vertical part back again. So the part of the displacement along the ray from the centre is reversed, the part

across the ray is kept, and the length is divided by the square of the distance from the centre, or multiplied by ρ^2 over that square when the circle has radius ρ .

Here the product is 2, so the map stretches the displacement by the factor $2/|NP|^2$, the same in every such plane, keeps the part of it perpendicular to NP in its direction and reverses the part along NP . So all displacements from P are reflected in one plane, the plane through P at right angles to NP , and stretched by one factor, and the angle between any two of them is kept.

A circle on the sphere is where the sphere meets a plane. The inversion sends the sphere to the plane of the equator and sends the cutting plane to a sphere or a plane, so the image of the circle is where the plane of the equator meets a sphere or a plane: a circle, or a line if the circle passed through N .

Stereographic projection sends circles to circles and keeps every angle.

So on such a map of the sky every circle of the heavens, the equator, the tropics, the path of the sun, the horizon of the observer, is a true circle that can be drawn with compasses, and the angle between two of them is the angle in the sky. That is what made the astrolabe possible, the brass computer of astronomers from late antiquity to the seventeenth century. The projection was known to Hipparchus about 150 BC, and the fact about circles is used throughout the Planisphere of Ptolemy; the oldest general proof that survives is al-Farghani's, from the ninth century. In one dimension less it has been met already. Projecting a circle from one of its points onto a line is the construction that found every point of the circle with fraction coordinates: the slope t of the chord through $(-1, 0)$ was the image, and $((1 - t^2)/(1 + t^2), 2t/(1 + t^2))$ the point.

* * *

Inversion can simplify a figure before anything is proved about it, and a theorem of Steiner's shows how far that goes. Take two circles, one inside the other, not concentric. In the ring between

them draw a circle touching both, then a second touching both and the first, a third touching both and the second, and so on round the ring. Usually the chain does not close: the last circle overlaps the first. For some pairs of circles it closes exactly, after some number of circles. Steiner proved in 1826 that if it closes from one starting circle, it closes from every starting circle, with the same number of links.

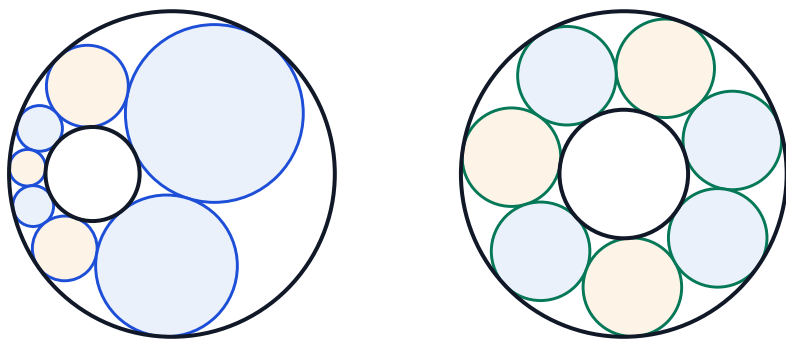


Figure 54.4. On the right, seven equal circles in a round ring, each touching its neighbours and both walls; turning the ring shows that the chain closes from any start. On the left, the image of that figure under an inversion: the two walls are no longer concentric, the seven circles have seven sizes, and every touching has survived, so this chain too closes from any start. *Assumes:* a ring whose radii stand in the proportion that seven circles require, $(1 + \sin(\pi/7))/(1 - \sin(\pi/7))$

If the two circles were concentric there would be nothing to prove, since the whole figure could be turned about the common centre and any starting circle carried to any other. And every pair of circles that do not meet can be made concentric by an inversion. For a point (x, y) outside the circle $x^2 + y^2 + Bx + Cy + D = 0$, the left side is the power of the point, since it is the square of the distance to the centre less the square of the radius.

Now take two circles, each with an equation of this form. A point outside both has equal powers with respect to them when the two left sides are equal there, that is, when their difference is

zero. In the difference the terms $x^2 + y^2$ cancel and an equation of the first degree is left, so the points of equal power lie on a straight line. When the circles do not meet, this line lies outside both. For suppose it crossed one of the circles. At the crossing point one left side is zero, and on the line the two left sides are equal, so the other is zero too, and the point would lie on both circles, which have no point in common. From a point X of that line the tangents to the two circles have the same length, since the square of each is the power of X . A circle centred at X with that length for radius crosses both circles at right angles: the square of its radius is the power of its centre with respect to each of them, and that was the test for crossing at right angles.

All such circles pass through the same two points L and L' of the line joining the two centres. To see it, take the line of centres as the horizontal axis. Both centres lie on it, so neither equation has a term in y , and neither does their difference; the line of equal powers, where that difference is zero, therefore stands at right angles to the line of centres, and it can be taken as the vertical axis. Each circle then has its centre at $(a, 0)$ say, and its equation reads $x^2 + y^2 - 2ax + k = 0$. The difference of the two equations has to vanish on the vertical axis, where $x = 0$, and nowhere else, so it is a multiple of x , and the two circles have different values of a and the same k . On the vertical axis the equation of either circle becomes $y^2 + k = 0$. The circles do not meet that axis, so this has no solution, and k is positive. The point $(0, t)$ has power $t^2 + k$, and the circle about it with that for the square of its radius, $x^2 + (y - t)^2 = t^2 + k$, meets the horizontal axis where $x^2 = k$, whatever t is. So L and L' are the points at distance $\sqrt{k}$ on either side.

Invert the whole figure in a circle centred at L . The circles through L and L' become straight lines through the image of L' . The two original circles become two circles that cross all those lines at right angles, angles being preserved, and they are circles and not lines, since neither passes through L : putting $x = \sqrt{k}$ and $y = 0$ into the left side gives $2k - 2a\sqrt{k}$, which is zero only when $a = \sqrt{k}$, and then the square of the radius, $a^2 - k$, would be zero. A

line that crosses a circle at right angles passes through its centre, so a circle that crosses at right angles every line through a point is centred at that point. The two circles have become concentric, touching circles have become touching circles, and the chain has become a chain in a round ring, which closes from every start if it closes from one.

How high must an observer be to see a lighthouse lamp 30 metres above the sea from 40 kilometres away, and how far apart can two ships with masts of 30 metres be and still see each other's mastheads?

What does inversion in the unit circle do to the line $x = 1/2$, to the circle of radius $1/2$ about the point $(1/2, 0)$, and to the circle of radius 1 about the point $(2, 0)$?

For four points A, B, C and D in order on a circle, invert in a circle centred at A . What does the fact that the images of B, C and D lie on a line, with the image of C between the other two, say about the six distances between the four points?

Where do the meridians and the parallels of latitude go under stereographic projection, and what is the image of a great circle that does not pass through the pole?

In the rod machine of Peaucellier the long rods are 5 units, the short ones 3, and P moves on a circle of radius 2 through O . Along which line does Q move, and how far from O is it?

WHAT A MODEL PROVES

For two thousand years the fifth postulate was attacked by people who believed it to be a theorem. In the eighteenth century Saccheri and Lambert assumed it false and derived consequence after consequence in the hope of reaching an absurdity. The angle sum of every triangle came out less than two right angles, and decreasing as the triangle grows, and no two triangles of different sizes came out with the same angles. No absurdity came. Is the denial of the postulate coherent, or only unrefuted?

A question of that shape is settled by a model. The construction of the real numbers showed that the completeness axiom describes something by building an object that satisfies it. To show that the postulate cannot be proved from Euclid's other assumptions, build an object that satisfies all of those and violates the postulate. A proof is a chain of steps each of which holds in every world where the assumptions hold. If some world obeys the other assumptions and disobeys the fifth, no chain of steps from the others can reach the fifth, since it would have to reach it in that world too. One world where the conclusion fails defeats every possible proof at once, as one tunnelled solid defeated every possible proof of the polyhedron formula for all solids.

* * *

The world is a disc. Its points are the points strictly inside a fixed circle Γ of radius 1 about a centre O , and its lines are of two kinds: the diameters of Γ , and the arcs inside Γ of circles that cross Γ at right angles. Everything about these lines is checked with the ordinary theorems of Euclid's plane, postulate included, since the disc is drawn in that plane. Nothing is circular in that: the question is only whether Euclid's assumptions without the fifth force the fifth, and a world where they hold and it fails answers it wherever it is built.

A circle with centre c and radius r crosses Γ at right angles exactly when the radii of the two circles to a crossing point are perpendicular, and then the theorem of Pythagoras, in the right triangle those radii form with the segment joining the centres, reads $|c|^2 = r^2 + 1$. Call such a circle orthogonal.

For a point P inside Γ other than O , let P^* be the point on the ray from O through P at distance $1/|OP|$ from O , so that $|OP| \cdot |OP^*| = 1$. Every orthogonal circle through P passes through P^* : O lies outside it, since the power of O with respect to it is $|Oc|^2 - r^2 = 1$, and so the line OP crosses it a second time at distance $1/|OP|$, which is P^* . Conversely, take the circle through P, P^* and any third point. The line OP cuts it at P and at P^* , so the product for O is $|OP| \cdot |OP^*| = 1$. Hence $|Oc|^2 - r^2 = 1$, and the circle is orthogonal.

So two points P and Q of the disc lie on exactly one line of the disc. If O, P and Q are in a row, that line is the diameter through them. Otherwise it is the arc of the circle through P, Q and P^* . Three points not in a row lie on exactly one circle, and that circle is orthogonal by the converse, so P and Q lie on at least one line of the disc. They lie on no other: every orthogonal circle through P and Q passes through P^* as well, and so it is the one circle through those three points. For P at 0.5 on the horizontal radius and Q at (0.3, 0.4), the point P^* is at 2 on the same radius, and the circle through the three has centre (1.25, 0.625) and radius squared 0.953125, which is 1 less than $|c|^2 = 1.953125$, as the converse requires.

* * *

Now the postulate. Let l be the horizontal diameter, and let P be the point half way up the vertical radius, so that P^* is at height 2 on the vertical axis. Every line of the disc through P other than the vertical diameter is an arc of a circle through P and P^* , whose centre is equally far from P and P^* and so lies on the horizontal line at height $5/4$, half way between them. Write the centre as $(t, 5/4)$. The point $P = (0, 1/2)$ is on the circle, at t across and $3/4$ down from the centre, so the radius satisfies $r^2 = t^2 + 9/16$.

The circle meets the horizontal axis at the points x with $(x - t)^2 + 25/16 = t^2 + 9/16$, that is, $(x - t)^2 = t^2 - 1$. When t is between -1 and 1 there is no such x , and the circle misses l . When t exceeds 1 the equation has the two solutions $t - \sqrt{(t^2 - 1)}$ and $t + \sqrt{(t^2 - 1)}$. Their product is $t^2 - (t^2 - 1) = 1$, and the larger exceeds 1 and so lies outside Γ . The smaller therefore lies strictly between 0 and 1 , and the circle meets l there; when t is below -1 it meets the axis at $t + \sqrt{(t^2 - 1)}$, strictly between -1 and 0 ; for $t = 2$ the crossing is at $2 - \sqrt{3}$, about 0.268 . At $t = 1$ and $t = -1$ the circle touches the axis at 1 and at -1 , which lie on Γ and not in the disc.

Through P pass infinitely many lines of the disc that never meet l , one for each t from -1 to 1 . The fifth postulate is false in the disc, and it fails in the way the eighteenth-century derivations had described: not by having no parallel, as on the sphere, but by having too many.

The lines through P that meet l are divided from those that miss it by a definite angle. The circle with centre $(t, 5/4)$ passes through $P = (0, 1/2)$, and its radius to P runs along $(-t, -3/4)$. The tangent at P is perpendicular to that radius, so it runs along $(3/4, -t)$, which is $3/4$ across for every t down. The line PO runs straight down, so the tangent makes with PO an angle whose tangent is $3/(4|t|)$, and a right angle when t is 0 . The lines that meet l are those with $|t|$ above 1 , and for them $3/(4|t|)$ is below $3/4$. So they leave P within a fixed angle on either side of PO , the angle whose tangent is $3/4$, which is the angle of 36.87° in the triangle with sides 3 , 4 and 5 . The lines that miss l fill the rest. In Euclid's plane the dividing angle is a right angle wherever P is; in the disc it depends

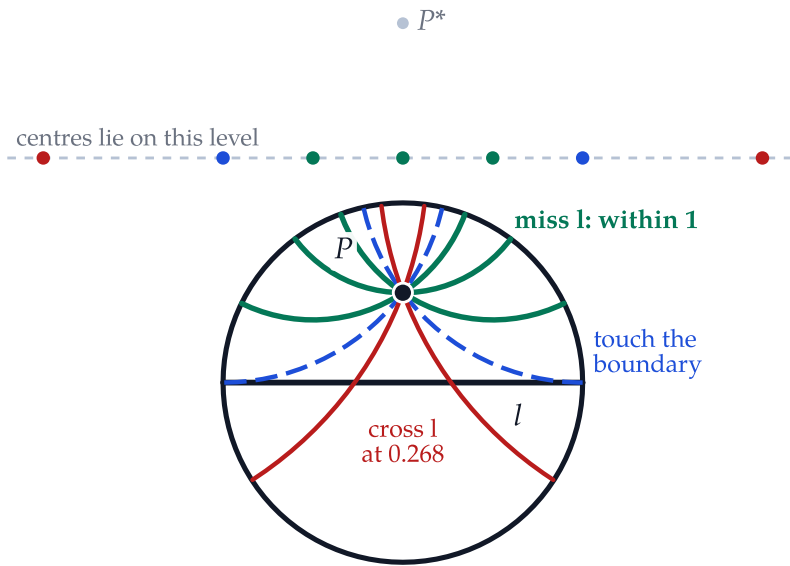


Figure 55.1. The disc model. The horizontal diameter is a line l , and P is the point half way up the vertical radius; the lines through P are arcs of circles through P and its inverse P^* , with centres on the dashed level. The arcs whose centres lie within 1 of the vertical axis never meet l ; the two with centres exactly 1 away touch the boundary at the ends of l ; and only the arcs with centres further out cross l inside the disc. The parallel through P is not unique, and the postulate fails. *Assumes:* that the arcs drawn stand for the whole family; the computation in the text covers every centre and the drawing shows seven

on P . The angle between two lines of the disc is taken to be the ordinary angle between their tangents, and the motions introduced next keep it.

For P at height s on the vertical radius, what is the dividing angle?

* * *

The other assumptions remain to be checked, and most of them

concern congruence, when two segments or two angles or two triangles are the same size, which so far means nothing in the disc. It is given a meaning by saying what the rigid motions are. The motions of the disc are the chains of its reflections, and the reflection in a line of the disc is ordinary mirror reflection when the line is a diameter, and inversion in the circle when the line is an arc: the point at distance d from the circle's centre, along a ray from that centre, goes to the point at distance r^2/d along the same ray. Inversion exchanges the inside and outside of its circle and leaves the circle itself in place, as a mirror should. Four facts make it a motion of the disc.

Two of them are the properties that made inversion a tool: it sends circles to circles, counting straight lines as circles, and it preserves the size of every angle while reversing its sense.

Inversion in an orthogonal circle C sends Γ to itself. The image of Γ is a circle, since Γ does not pass through the centre c of C , which is outside Γ . It passes through the two points where Γ meets C , which stay in place, and it crosses C at right angles there, since angles are preserved. Only one circle through those two points crosses C at right angles. Its centre must lie on the perpendicular bisector of the two points, being equally far from both. It must also lie on the tangent to C at either of them: at a right-angled crossing its radius to the crossing point is perpendicular to the radius of C there, and the line through that point perpendicular to the radius of C is the tangent to C . Those two lines meet in one point. So the image is Γ . Since Γ goes to itself, the inside of Γ goes either all to the inside or all to the outside, and a single point decides which. It goes to the inside: O goes to the point on the ray from c through O at distance $r^2/|c|$ from c , which is at distance $(|c|^2 - r^2)/|c| = 1/|c|$ from O , and $1/|c|$ is below 1 because c lies outside Γ .

It follows that inversion in an orthogonal circle sends lines of the disc to lines of the disc: a line of the disc lies on a circle or straight line crossing Γ at right angles, and its image lies on a circle or straight line crossing the image of Γ , which is Γ , at right angles.

So every reflection, and every chain of reflections, carries the

disc onto itself, lines to lines, and angles to angles of the same size. Two figures in the disc are *congruent* when some chain of reflections carries one onto the other. Euclid's assumptions about congruence become statements about what reflections can do, and the decisive one is that they can do enough. Any point P can be carried to O . The circle centred at P^* with radius $\sqrt{(|OP^*|^2 - 1)}$ is orthogonal, since $|OP^*|$ exceeds 1, and inversion in it sends P to O , because $|P^*P| \cdot |P^*O| = (|OP^*| - |OP|) \cdot |OP^*| = |OP^*|^2 - 1$, the square of its radius; for P at 0.5 on a radius, P^* is at 2 and the radius is $\sqrt{3}$. Any direction at O can be turned to any other by a rotation, which is two reflections in diameters. So any point with a direction through it can be carried to any other point with any direction, which is the freedom that laying off segments along rays and comparing triangles requires. One further fact is needed: two segments laid off along one ray from one point, and carried onto each other by a chain of reflections that leaves the point in place, coincide. A chain that carries one such segment onto the other can be made to leave the point in place. Carry the point to O first; if the chain then sends O to the far end A of the other segment, follow it by the inversion that carries A to O , which carries O back to A , being its own undoing, and the two together carry the one segment onto the other with O left in place. It is then enough to show that a chain of reflections leaving O in place is an ordinary rotation or reflection about O , since those keep ordinary distances from O .

Such a chain f sends lines of the disc through O , which are diameters, to diameters, and keeps the sizes of angles at O . Follow f by the rotation about O that brings the right-hand half of the horizontal diameter back onto itself. The upper half of the vertical diameter makes a right angle with it, so it now lies along the vertical diameter; if it points down, follow with the reflection in the horizontal diameter as well. The result is a chain h that carries each of those two radii onto itself. Then h carries every radius onto itself, since the angles a radius makes with those two are kept and they settle its direction.

What remains open is whether h slides points along their radii.

Let P lie at distance s from O on the right-hand horizontal radius, and $h(P)$ at distance s' on the same radius. The line of the disc through P perpendicular to that radius is the arc of the circle through P and P^* , whose centre is on the horizontal axis at $d = (1 + s^2)/(2s)$. The radius at angle φ from the right-hand horizontal radius meets that arc exactly when $d \cos \varphi$ exceeds 1. The points $t(\cos \varphi, \sin \varphi)$ of the radius that lie on the circle satisfy $t^2 - 2td \cos \varphi + 1 = 0$. That equation has positive roots exactly when $d \cos \varphi$ is at least 1. The two roots multiply to 1, so when they differ exactly one of them is below 1 and lies inside the disc; when $d \cos \varphi$ equals 1 the only root is $t = 1$, a point of Γ and not of the disc.

The chain h carries the perpendicular at P to the perpendicular at $h(P)$, and carries each radius to itself, so the two perpendiculars meet the same radii. The perpendicular at P meets the radii with $\cos \varphi$ above $1/d = 2s/(1 + s^2)$, and the perpendicular at $h(P)$ meets those with $\cos \varphi$ above $2s'/(1 + s'^2)$. The two sets of radii are the same, so $2s/(1 + s^2) = 2s'/(1 + s'^2)$, and since that expression increases as s runs from 0 to 1, $s' = s$. The same argument applies along any other radius, after a rotation that turns it onto the horizontal one, so h leaves every point in place. Now h is f followed by a rotation and perhaps a reflection about O ; undoing those, f is itself a rotation or a reflection about O .

With that fact the congruence assumptions hold. Take two triangles that agree in two sides and the angle between them. In each, carry the corner that holds that angle to O , and turn one triangle so that one of its two sides lies along the ray of the matching side of the other. If its second side then lies on the wrong side of that ray, reflect in the diameter along the ray. The equal angles now put the second pair of sides along one ray as well. The equal sides laid off along the two rays then coincide, and so do the third corners.

The points P for which QP is congruent to a given QA are carried, by the reflection that sends Q to O , onto an ordinary circle about O , since the motions leaving O in place are rotations and reflections; so a circle can be drawn about any centre. The remaining congruence assumptions go the same way. Two angles of the disc

are congruent exactly when they have the same size: reflections keep sizes, and conversely, if each vertex is carried to O , one side of each angle turned onto the right-hand horizontal radius, and the angle reflected in the horizontal diameter if need be, two angles of the same size end with their second sides on the same radius. So all right angles are congruent, which is Euclid's fourth assumption, and an angle of a given size can be laid off on a given side of a given ray in one way only.

A chain of reflections can be undone, each reflection undoing itself, and two chains can be run one after the other, so congruence passes along a chain of figures and back. A segment is congruent to itself with its ends exchanged: carry one end to O , and the inversion that carries the other end A to O carries O to A , being its own undoing, and carries the diameter through A to itself; for A at 0.5 it is the inversion in the circle with centre 2 and radius $\sqrt{3}$.

And congruent segments added to congruent segments give congruent wholes. If B lies between A and C , and B' between A' and C' , with AB congruent to $A'B'$ and BC to $B'C'$, carry B and B' to O and both lines onto the horizontal diameter with A and A' on the same side. Some chain carries the segment OA onto OA' . If it sends O to A' and A to O , follow it by the motion that exchanges the ends of OA' ; so the chain can be taken to leave O in place. It is then a rotation or reflection about O , which keeps ordinary distances from O , and since A and A' lie on the same radius, A is A' ; likewise C is C' .

A line of the disc can be extended without end in the disc's own terms. Given a segment from A to B , the reflection in the line through B perpendicular to AB carries A to a point A' beyond B , and BA' is congruent to AB . Repeating from A' produces a row of congruent segments marching toward Γ . The reflection in the line of the disc perpendicular to the horizontal diameter at the point b is the inversion in the circle centred on the axis at $d = (1 + b^2)/(2b)$ with radius squared $d^2 - 1$, and it sends a point a of the diameter to $d - (d^2 - 1)/(d - a)$. Starting from the segment from O to $1/2$, the reflection at $1/2$, where d is $5/4$, sends O to $4/5$, and the reflection

at $4/5$ sends $1/2$ to $13/14$: the congruent segments from 0 to $1/2$, from $1/2$ to $4/5$ and from $4/5$ to $13/14$ have lengths $1/2$, $3/10$ and $9/70$ on the page. On the page they look shorter and shorter and never arrive, since Γ is not in the disc and no reflection reaches it. Measured by the disc's congruence they all have the same length: every segment can be extended beyond either end by a congruent segment, which is what the assumption asks.

Where is the n -th point of the row that starts with O and $1/2$?

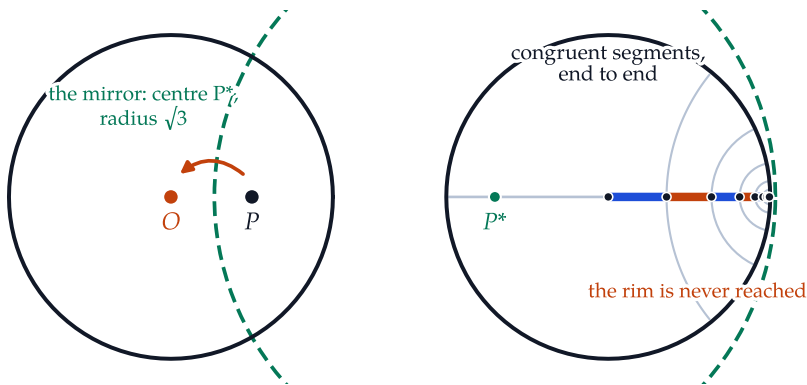


Figure 55.2. Left: the orthogonal circle centred at P^* , the inverse of P , has radius $\sqrt{3}$ when P sits half way along a radius; inversion in it swaps the inside and outside of that circle and sends P exactly to the centre O , while carrying the disc onto itself. Right: a segment reflected in the perpendicular line at its far end, again and again; every segment in the row is congruent to the first, and the row never reaches the rim. Assumes: that the drawn reflections are the disc's motions, which the four facts in the text establish; the eye reads the right-hand row as shrinking, and that reading is the one the model was built to correct

The tiling of the disc by triangles shows the same thing on a larger scale. One triangle, with a right angle at the centre, an angle of 45° at its corner on the horizontal radius and an angle of 36° at its corner on the vertical radius, is reflected in its own sides again and again. Its angle sum is 171° , and no triangle with those angles exists in Euclid's plane. That the copies so made cover the disc once over, meeting edge to edge and never overlapping, is a fact about such triangles taken here from other hands; what is shown above

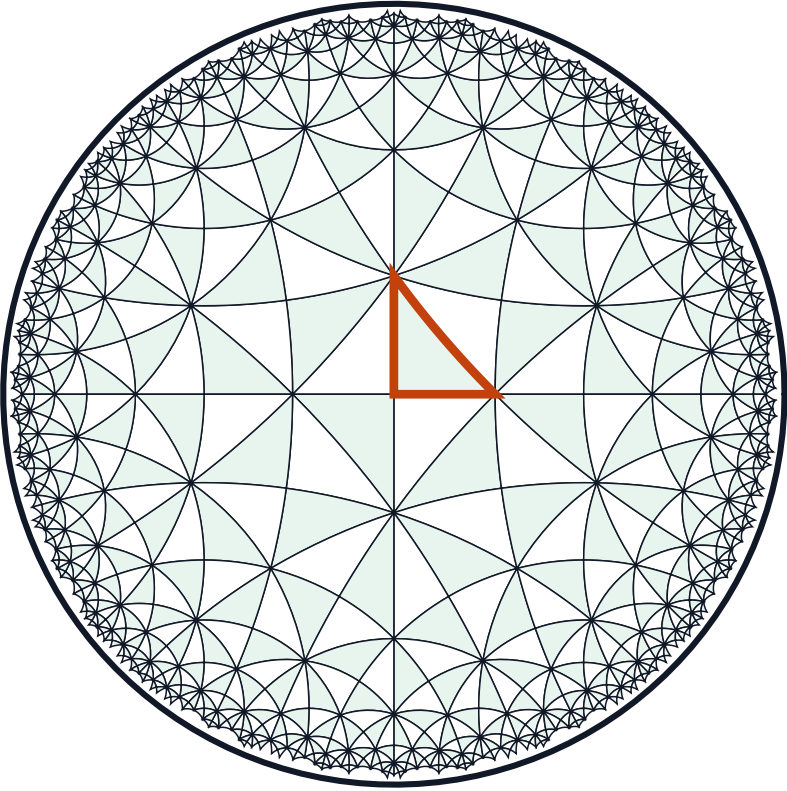


Figure 55.3. A triangle of the disc with angles 90° , 45° and 36° , reflected in its own sides again and again. Every triangle in the picture is congruent to the first, being its image under a chain of reflections; the ones near the rim are the same shape and the same size in the disc's own measure, and the rim, which they crowd toward, is not part of the disc and is never reached. *Assumes:* that the reflections continue past the depth drawn, and that no chain of reflections reaches the rim, which the text argues and the eye refuses

is only that each copy, being the image of the first under a chain of reflections, is congruent to it. The triangles near the rim look smaller, and they are the same size.

* * *

A model proves two things. The first is relative consistency. Every statement about the disc is a statement about points and circles of an ordinary plane, so a contradiction derived from Euclid's other assumptions together with the denial of the fifth would be a contradiction in ordinary plane geometry; if Euclid's geometry is free of contradiction, so is the geometry in which the fifth postulate is denied, and the eighteenth-century consequences are theorems about the disc. The second is independence. No proof of the fifth postulate from the other assumptions exists, because the disc satisfies the others and not the fifth. The postulate can be added to the others, giving Euclid's plane, or its denial can, giving the disc, and the rest cannot force the choice. An earlier sentence said that the usual rules of mathematics settle the size question between the lists and the real numbers in neither direction. It means the same kind of thing: provided those rules are free of contradiction, there is a model of them in which a collection of intermediate size exists, and a model in which none does.

A model does not say which geometry physical space has. Every measurement of a finite figure is consistent with both, and the plane and the disc are two descriptions, each of somewhere.

The angle sum of a particular triangle in the disc can be computed. Take the triangle with corners at O , at the point half way along the horizontal radius and at the point half way up the vertical radius. Two of its sides lie along diameters, so the angle at O is a right angle. The third side is the arc of the circle through the other two corners and through the point 2 on the horizontal axis; its centre is $(5/4, 5/4)$ and its radius squared is $17/8$, and $25/8$ exceeds $17/8$ by exactly 1, so it is orthogonal. The radius from $(5/4, 5/4)$ to $(1/2, 0)$ runs $3/4$ across and $5/4$ down, so the tangent to the circle

there, perpendicular to that radius, runs $5/4$ across for every $3/4$ up. It meets the horizontal diameter at the angle whose tangent is $3/5$, which is 31.0° , and by symmetry the arc meets the vertical diameter at the same angle. The angle sum is 151.9° , short of two right angles by 28.1° . Push the two outer corners along their radii toward Γ and the arc joining them meets the diameters at smaller and smaller angles: with the corners at 0.9 the angles are 6.0° , and at 0.99 they are 0.6° . The angle sums of these triangles approach a single right angle.

Legendre's ladder can be watched failing here. With P half way up the vertical radius and Q at O , the first rung Q_1 is the point 0.5 on the horizontal radius, and the angle QPQ_1 is the 31.0° computed above; the next rungs lie at 0.8831 and 0.9954 , crowding toward Γ , and the angles QPQ_2 and QPQ_3 , 36.66° and 36.8696° , never pass 36.87° , the dividing angle for that P .

Every triangle of the disc falls short, by an amount that can be read off the page. Carry one corner of the triangle to O . Two of its sides then lie along radii, out to corners B and C , and the third side is an arc of an orthogonal circle K , with centre c . Let Δ be the angle BcC that the chord BC subtends at c . Then the angle sum of the triangle is two right angles less Δ .

The proof is carried out in Euclid's plane. A point p of K has $|p - c|^2 = |c|^2 - 1$, which says $|p|^2 = 2p \cdot c - 1$, where $p \cdot c$ is the product of $|c|$ and the length of p measured along the direction of c ; so p lies inside Γ exactly when $p \cdot c$ is below 1 . The line of points with $p \cdot c = 1$ passes between O and c , since $p \cdot c$ is 0 at O and is $|c|^2$, which is above 1 , at c . So the part of K inside the disc lies on the far side of that line from c , and it is less than half of K , because a line cuts off less than half of a circle on the side away from the circle's centre. The arc BC therefore lies on the far side of the chord BC from c , and Δ is less than two right angles.

At B the tangent to K makes with the chord BC the angle $\Delta/2$, half the angle at the centre, by the theorem on the tangent and the chord. The line OB meets K at B and again at B^* , further out, and O lies outside K , so the segment BO lies outside K , on the other side of

the tangent line from the chord. So the tangent line at B separates BO from the chord, and one half of it, starting from B , lies inside the ordinary angle OBC . The two halves make with the chord the angles $\Delta/2$ and two right angles less $\Delta/2$, and the half at $\Delta/2$ is the one on C 's side, along which the arc sets out from B . That is the half inside the angle OBC . Were it the other half, the angle OBC would exceed two right angles less $\Delta/2$, leaving less than $\Delta/2$ for the angle OCB . But the angle OCB contains half of the tangent line at C in the same way, and so it is more than $\Delta/2$.

So the angle of the disc triangle at B , between BO and the arc, is the ordinary angle OBC less $\Delta/2$, and the same holds at C , while the angle at O is the same in both senses. The ordinary triangle OBC has angle sum two right angles, so the triangle of the disc has two right angles less Δ .

Any triangle of the disc can be carried to one with a corner at O , and the carrying keeps its angles, so every triangle of the disc falls short of two right angles: the first of the consequences that Saccheri and Lambert reached and could not refute is a theorem about the disc. For the triangles above, with a right angle at O and corners at s on both radii, the tangent of $\Delta/2$ is s^2 , and the shortfall is 28.07° at $s = 1/2$, 78.01° at 0.9 and 88.85° at 0.99 . Shortfalls add. Cut a triangle by a line of the disc from a corner to the opposite side. The angles of the two pieces are the angles of the whole together with the two angles made at the cut, which fill a straight angle. So the two angle sums add up to the angle sum of the whole plus two right angles, and the shortfalls of the two pieces add up to the shortfall of the whole. Each piece therefore falls short by less than the whole does. So the angle sum decreases as a triangle grows, which is the second consequence.

The third consequence follows: two triangles of the disc with the same three angles are congruent. Carry the corner A' of the second onto the corner A of the first, with the side $A'B'$ along AB and C' on the same side of it as C ; the equal angles at A put C' on the ray AC . If B' is B , the equal angles at B lay off the same line, so C' is C . Write β for the angle both triangles have at B and B' , and γ

for the angle both have at C and C' .

If B' lies between A and B and C' between A and C , look at the four-sided figure $B'BCC'$. The triangle $AB'C'$ has the angle β at B' , so the figure's angle at B' , which lies beside it along the line AB , is two right angles less β ; likewise its angle at C' is two right angles less γ . With the angles β and γ at B and C , that makes four right angles in all. But a diagonal cuts the figure into two triangles, each of them short of two right angles, so its four angles make less than four right angles. If B' lies between A and B while C lies between A and C' , the sides BC and $B'C'$ cross at a point X , and the triangle $BB'X$ has the angle β at B and the angle two right angles less β at B' , which leaves nothing for its third angle. The remaining cases are these with the two triangles exchanged. So the triangles coincide. As on the sphere, the angles of a triangle fix its size, and the disc has a length that nobody chose.

Find the triangle with a corner at O , its other two corners on radii 45° apart, and all three angles 45° . Is there a triangle of the disc with three angles of 60° ?

MEASURING IN THE DISC

The disc has lines, angles and congruence, and a row of congruent segments marching towards the rim showed that its own idea of length is not the page's. What is its idea of length? Nothing has yet assigned a number to a segment of the disc. Congruence says when two segments are equally long, and a length is a number that respects that: equal for congruent segments, and adding up when segments are laid end to end along a line.

The row of segments says what the number must be. Starting from O , the points of the row along a radius were at $1/2$, $4/5$ and $13/14$ on the page. Compute $(1+x)/(1-x)$ for each: the values are 3, 9 and 27. Each congruent step multiplies that quantity by 3. The step is made by sliding the disc along the horizontal diameter so that O goes to the point a . That motion is made of two reflections. The first is the reflection in the vertical diameter, which sends x to $-x$. The second is the reflection in a line of the disc perpendicular to the horizontal diameter: the one whose circle has its centre on that diameter at $d = 1/a$. The square of that circle's radius is $d^2 - 1$, and the inversion in it sends x to $d - (d^2 - 1)/(d - x)$. Together they send the point x of the diameter to

$$(x + a)/(1 + ax)$$

and this motion multiplies $(1+x)/(1-x)$ by the fixed number

$(1 + a)/(1 - a)$. To see it, put $(x + a)/(1 + ax)$ in place of x and clear the fractions: the top becomes $1 + ax + x + a$, which is $(1 + a)(1 + x)$, and the bottom becomes $1 + ax - x - a$, which is $(1 - a)(1 - x)$. For the row of segments a is $1/2$, and the fixed number is 3. Take the logarithm and multiplying turns into adding: each congruent step adds the same fixed amount to $L((1 + x)/(1 - x))$. That is how a length has to behave. A length is positive, grows as the segment grows, and adds when segments are laid end to end, and $L((1 + x)/(1 - x))$, measured from O along the radius, does all three.

It is also the only candidate, up to a fixed multiple. Suppose two quantities both grow and both add, and on one segment the first is some multiple of the second. Then it is the same multiple on that segment laid end to end any number of times, since both add, and the same multiple on half the segment, since two congruent halves make the whole. Segments built by repeating and halving come as close as wanted to any segment, and both quantities grow, so the multiple is the same on every segment. So the length of the segment from O to a point at distance s from O on the page has to be a fixed multiple of

$$\rho = L((1 + s)/(1 - s))$$

and the multiple is a choice of unit, which is taken to be 1. The steps of the row have length $L(3)$, which is 1.0986. As s approaches 1 the length passes every bound: the rim is infinitely far away, which is how a disc that fits on a page can serve as a plane that extends without end. Solving for s , the point at distance ρ from O in the disc is at $s = (E(\rho) - 1)/(E(\rho) + 1)$ on the page: 0.4621 for $\rho = 1$, then 0.7616, 0.9051 and 0.9640 for 2, 3 and 4.

For two points neither of which is O , carry one of them to O by a motion and measure there. The result can be written down without moving anything. The line of the disc through P and Q runs out to two points of the rim, A beyond P and B beyond Q , and

the distance from P to Q is L of $(|PB| \cdot |QA|)/(|PA| \cdot |QB|)$

where the four lengths are those of straight segments on the page. When P is O and Q is at distance s from it along a diameter, the four lengths are $|PB| = 1$, $|QA| = 1 + s$, $|PA| = 1$ and $|QB| = 1 - s$, and the expression is $(1 + s)/(1 - s)$, in agreement with the length found from O . And it is unchanged by every motion. A motion is made of reflections, and it carries the rim points A and B to the rim points of the new line, so it is enough to look at one reflection acting on the four points. A reflection in a diameter keeps all lengths on the page. A reflection in any other line of the disc is an inversion in a circle, say of centre c and radius k . It sends X and Y to points X' and Y' with $|cX| \cdot |cX'| = |cY| \cdot |cY'| = k^2$. The triangles cXY and $cY'X'$ share the angle at c , and those equal products make the sides about it proportional, so the triangles are similar and $|X'Y'| = k^2|XY|/(|cX| \cdot |cY|)$. Apply this to the four lengths in the expression above. Each of the points P , Q , A and B appears once in the top, in $|PB| \cdot |QA|$, and once in the bottom, in $|PA| \cdot |QB|$. So the top and the bottom are both multiplied by $k^4/(|cP| \cdot |cQ| \cdot |cA| \cdot |cB|)$, and the expression does not change.

* * *

Small things in the disc look on the page as they are, except for size. Near O the length $L((1 + s)/(1 - s))$ is $L(1 + s) - L(1 - s)$, and for small s the first is very nearly s and the second very nearly $-s$. So the length is very nearly $2s$, and there the disc's lengths are twice the page's. A small figure near a point P at distance s from O can be carried to O by the inversion that exchanges P and O . Its circle has its centre at P^* , at distance $1/s$ from O , and radius squared $1/s^2 - 1$, and P is at distance $1/s - s$ from that centre. By the rule $|X'Y'| = k^2|XY|/(|cX| \cdot |cY|)$, with X and Y close to P , the inversion enlarges small figures at P by the factor $(1/s^2 - 1)/(1/s - s)^2$, which is $1/(1 - s^2)$. Motions keep the disc's lengths and keep angles. So a small figure at P has, in the disc's own terms, the shape it shows on the page, and its size comes in two steps: carrying it to O enlarges it on the page by $1/(1 - s^2)$, and at O the disc's lengths are twice

the page's. Together they make

every small length at distance s from O equal to $2/(1 - s^2)$ times its length on the page

and every small area $(2/(1 - s^2))^2$ times its area on the page. The factor grows without bound towards the rim, which is why the congruent tiles of a tiling look smaller and smaller there.

Now circles. The points of the disc at distance ρ from O form an ordinary circle about O , of radius s on the page, and its circumference in the disc is $2\pi s$ times the factor:

$$\text{circumference} = 2\pi \cdot 2s/(1 - s^2) = 2\pi \cdot (E(\rho) - E(-\rho))/2$$

the second form coming from putting $s = (w - 1)/(w + 1)$, where w stands for $E(\rho)$. Then $2s/(1 - s^2)$ is $2(w - 1)(w + 1)/((w + 1)^2 - (w - 1)^2)$, which is $(w^2 - 1)/(2w)$, that is $(w - 1/w)/2$, and $1/w$ is $E(-\rho)$. The combination $(E(\rho) - E(-\rho))/2$ is called the *hyperbolic sine* of ρ , written $\sinh \rho$. For small ρ it is close to ρ , and the circumference is close to the $2\pi\rho$ of the flat plane. But it grows as the exponential does. A circle of radius 1 has circumference 7.4 where Euclid's has 6.3; radius 2 gives 22.8 against 12.6; radius 4 gives 171.5 against 25.1; and a circle of radius 10 has a circumference of 69198, more than a thousand times the flat one. The area inside, found by adding up rings, is $2\pi(\cosh \rho - 1)$, where $\cosh \rho$ is $(E(\rho) + E(-\rho))/2$, and it grows in the same way.

There is a great deal of room in this plane. In the flat plane the number of things of a fixed size that fit within distance ρ grows as ρ^2 , and in the disc it grows as $E(\rho)$. A tree that branches in two at every step needs 2^n places at depth n . The flat plane cannot supply them without crowding, and the disc supplies them easily. That is why, since 2017, systems that learn from data have stored hierarchies as points of this disc, where each level can be given as much room as it needs; the tree of all the species and the pages of a web site are examples.

The dividing angle can be read in the same terms. For a point at height s on the vertical radius the computation made for $s = 1/2$

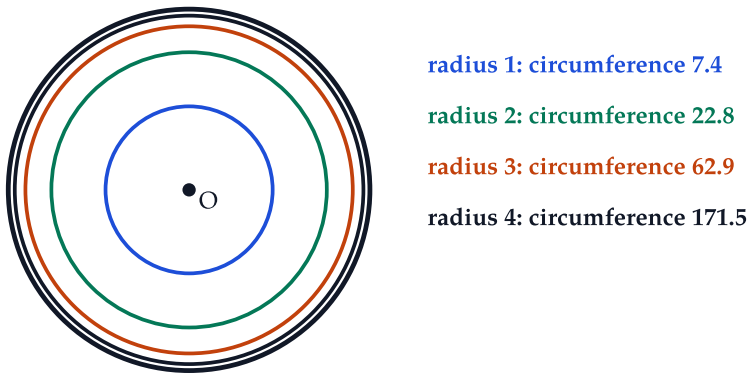


Figure 56.1. Circles about O whose radii, measured in the disc, are 1, 2, 3 and 4. On the page their radii are 0.462, 0.762, 0.905 and 0.964, crowding towards the rim. Their circumferences in the disc are 7.4, 22.8, 62.9 and 171.5, where circles of those radii in the flat plane would have 6.3, 12.6, 18.8 and 25.1. *Assumes:* the natural unit of length of the disc, in which a small length at O is twice its length on the page

goes through unchanged. The circles through the point and its partner at height $1/s$ have their centres at $(t, (s + 1/s)/2)$. The equation for the crossing with the horizontal diameter comes out as $(x - t)^2 = t^2 - 1$ again, so they meet the diameter exactly when $|t|$ exceeds 1. And the tangent at the point runs along $((1/s - s)/2, -t)$. The border cases are $t = 1$ and $t = -1$. There the tangent drops 1 towards the diameter for every $(1/s - s)/2$ it moves sideways, to the right for $t = 1$ and to the left for $t = -1$, and $(1/s - s)/2$ is $(1 - s^2)/(2s)$. A larger $|t|$ makes the tangent steeper. So the lines through the point that meet the diameter are those within an angle Π of the perpendicular, where $\tan \Pi = (1 - s^2)/(2s)$. That is $1/\sinh \rho$, where ρ is the distance of the point from the line, because the circumference computation found $\sinh \rho$ to be $2s/(1 - s^2)$.

Now put u for $\tan(\Pi/2)$. The addition formulas give $\sin \Pi = 2 \sin(\Pi/2) \cos(\Pi/2)$ and $\cos \Pi = \cos^2(\Pi/2) - \sin^2(\Pi/2)$; dividing the first by the second, and then the top and the bottom by $\cos^2(\Pi/2)$, gives $\tan \Pi = 2u/(1 - u^2)$. Putting $u = E(-\rho)$ makes

that $2E(-\rho)/(1 - E(-2\rho))$, which is $2/(E(\rho) - E(-\rho))$, that is $1/\sinh \rho$. Since Π is at most a right angle, u lies between 0 and 1, and only one such u can do this, since $2u/(1 - u^2)$ grows with u . So the dividing angle takes the form found by Bolyai and Lobachevsky, each working alone, around 1830:

$$\tan(\Pi/2) = E(-\rho)$$

At distance 0 the dividing angle is a right angle, as in Euclid. At distance 1 it is 40.4 degrees, and at distance 5 it is 0.77 degrees: from five units away, all the lines that ever reach a given line leave within less than a degree of the perpendicular. Had the length been defined as k times $L((1 + s)/(1 - s))$ for some other multiple k , the formula would read $\tan(\Pi/2) = E(-\rho/k)$. So the multiple taken to be 1 at the start was not only a convenience: it gives the disc a natural unit of length, one that nobody chose, the distance at which the dividing angle is 40.4 degrees.

* * *

The shortfall of a triangle, two right angles less its angle sum, was additive: cut a triangle in two and the shortfalls of the pieces add up to the shortfall of the whole. It is kept by motions. Those are the defining properties of an area, and the shortfall is the area. With angles in radians and lengths in the natural unit, the area computed from the factor $(2/(1 - s^2))^2$ agrees with it exactly. Take the triangle with corners at O , $(1/2, 0)$ and $(0, 1/2)$. Its shortfall was 28.07 degrees, which is 0.48996 radians. Adding up the factor $(2/(1 - s^2))^2$ over the region, by machine, gives 0.48996 as well.

A path of the disc that leaves the line through its two ends is longer. To see it, carry one end to O by a motion, so that the line through the two ends becomes a diameter and the other end lies on a radius. Break any path from O to the other end into small steps. On the page each step has a part directed along the radius through it, away from O or towards it, and a part across, and the step is at least as long as its part along the radius. The factor $2/(1 - s^2)$

depends only on the distance s from O , so it stretches a step and its part along the radius alike. Each step therefore counts, in the disc, at least as much as the change it makes in the distance from O , measured along a radius. Added up, the path is at least as long as the radius to its far end, and longer if any step goes across. So no path from O beats the radius. The lines of the disc are its shortest paths.

Gauss proved in 1827 a theorem about triangles drawn with shortest paths on any curved surface. Some surfaces are the same at every one of their points, as the sphere and the disc are. For such a surface the theorem says that the area of a triangle is a fixed multiple of the amount by which its angle sum departs from two right angles. The sphere's excess and the disc's shortfall are two cases of that one theorem. The general proof is relied on here and not given.

It has a consequence without any parallel in Euclid. The angle sum of a triangle is more than 0, so its shortfall is less than π :

no triangle of the disc has area as large as π , however long its sides.

The triangles with all three corners on the rim, which are not triangles of the disc but limits of them, have sides of infinite length, three angles of 0, and area exactly π . A polygon of n sides is cut by diagonals into $n - 2$ triangles, so its area is $(n - 2)\pi$ less its angle sum. A pentagon can have five right angles, which no Euclidean pentagon can, and then its area is $3\pi - 5\pi/2$, which is $\pi/2$. Four of them fit round each corner, as squares do in the flat plane.

Which regular tilings are there? Let regular polygons of p sides meet q at each corner, so that each angle of the polygon is a q -th part of four right angles. In the flat plane the angle of a regular polygon of p sides is fixed, at two right angles less a p -th part of four, and setting the two equal gives $2\pi/q = \pi - 2\pi/p$ in radians, and dividing by 2π turns that into $1/p + 1/q = 1/2$, which has the three solutions 4 and 4, 3 and 6, 6 and 3: squares, triangles and hexagons. On a sphere polygons have an excess, their angles are larger than in the plane, and the condition becomes $1/p + 1/q >$

$1/2$, which for p and q both at least 3 has the five solutions that are the five regular solids, blown up onto the sphere.

In the disc polygons have a shortfall, their angles are smaller than in the plane, and as a regular polygon about O grows from a speck to the full disc its angle falls continuously from the flat value to 0, passing through every value between. So whenever $1/p + 1/q$ is below $1/2$ there is a regular polygon of p sides whose angle is the q -th part of four right angles, and q copies of it fit round a corner. That the copies made by reflecting in the sides again and again go on fitting, covering the disc once and never overlapping, was proved by Poincaré in 1882; it is relied on here and not proved. So there is a tiling whenever

$$1/p + 1/q < 1/2$$

and that leaves out only the eight cases above and the pairs with a 2 in them. The corners of the central tile are at the distance s from O on the page, where $s^2 = \cos(\pi/p + \pi/q)/\cos(\pi/p - \pi/q)$. Two requirements give this. The side is a line of the disc, so its circle, with centre c and radius r , crosses the rim at right angles, and $|Oc|^2 = r^2 + 1$. And the side meets the radius to its end V at the angle π/q , half the angle of the polygon, so the radius cV , which is perpendicular to the side, makes with VO the angle $\pi/2 + \pi/q$. In the triangle OVc the angle at O is π/p , half the angle $2\pi/p$ between the radii to the two ends of the side, and so the angle at c is $\pi/2 - \pi/p - \pi/q$. The law of sines gives $r/\sin(\pi/p) = |Oc|/\cos(\pi/q) = s/\cos(\pi/p + \pi/q)$. Putting the first of these into $|Oc|^2 - r^2 = 1$, and using $\cos^2 x - \sin^2 y = \cos(x + y)\cos(x - y)$, gives $|Oc|^2 = \cos^2(\pi/q)/(\cos(\pi/q + \pi/p)\cos(\pi/q - \pi/p))$, and the second then gives the formula for s^2 . The right-hand side is a number strictly between 0 and 1 exactly when $1/p + 1/q$ is below $1/2$. For heptagons meeting three at a corner s is 0.3007. The flat plane has three regular tilings, the sphere five, and the disc infinitely many. Escher, who learned of the disc from a figure in a paper by the geometer Coxeter that reached him in 1958, made four woodcuts of it, and wrote that the figure had given him quite a shock.

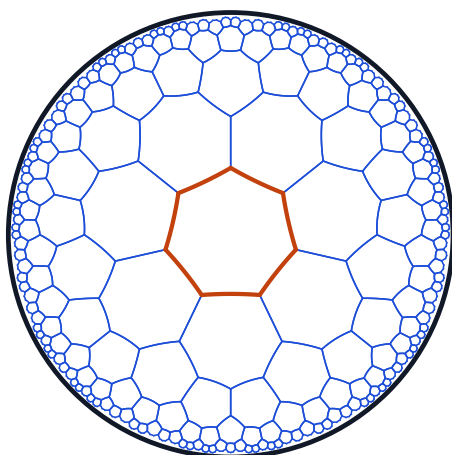


Figure 56.2. Regular heptagons of the disc, three at every corner, made by reflecting the central one in its sides four times over. Every tile is congruent to the central one. The corners of the central tile lie 0.3007 from O on the page, the value the formula gives for $p = 7$ and $q = 3$. Assumes: four generations of reflections; the tiling continues to the rim, where the tiles are too small for the page

How far apart, in the disc, are the points $(1/2, 0)$ and $(0, 1/2)$?

What is the radius of a circle of the disc whose circumference is 1000, and what would it be in the flat plane?

A regular polygon of the disc has four sides and four angles of 60 degrees. What is its area, how many of them meet at a corner, and how far from the centre are its corners on the page?

For which of the pairs p and q from 3 to 7 does the regular tiling lie on the sphere, in the flat plane, and in the disc?

A limit of triangles of the disc has two of its corners on the rim, where its angles are 0, and a right angle at its third corner. What is its area, and what is the dividing angle at the third corner for the side opposite?

TOWERS

With a straightedge and a compass, double a cube: build the side of a cube that holds twice the volume of a given one. Trisect an angle: cut any given angle into three equal parts. Square a circle: build a square with the area of a given circle. The three problems were posed in Greece and stayed open for two thousand years. A thousand failed trisections prove nothing about the next attempt, because “no construction trisects” is a claim about every construction that will ever be devised, and inspecting constructions settles them one at a time. The way through is to describe everything that can be built, completely, so that what cannot be built falls out with no construction examined, as no fraction was examined when the diagonal of the square was shown to have no fraction for its length.

A straightedge draws the line through two known points; a compass draws the circle through one known point centred at another; and the plane is Euclid’s, with the parallel postulate in force, since similar triangles will be needed. Start from a segment of length 1. Sums and differences of lengths come from laying segments end to end. That a length can be carried from one place to another by these two instruments is the second proposition of Euclid’s first book, and its construction is relied on here. Products and quotients come from similar triangles. On one ray from a point O mark U at distance 1 and B at distance b , and on another ray mark A at dis-

tance a . The line through B parallel to UA meets the second ray at a point P . The triangles OUA and OBP are similar, so OP is to OA as OB is to OU , that is, $OP/a = b/1$, and OP is ab . The line through U parallel to BA meets the second ray at a point Q . Now the triangles OUQ and OBA are similar, so OQ is to OA as OU is to OB , that is, $OQ/a = 1/b$, and OQ is a/b .

Square roots come from a semicircle. Lay segments of lengths 1 and w end to end, draw the semicircle with the whole as diameter, and raise the perpendicular at the join. Put the join at the origin, so that the diameter runs from -1 to w . Its length is $w + 1$, so the radius is $(w + 1)/2$, and the centre sits at $(w - 1)/2$, so its distance from the join is $(w - 1)/2$ when w is at least 1 and $(1 - w)/2$ when w is less, with square $((w - 1)/2)^2$ either way. Let the perpendicular at the join meet the semicircle at height h . The join, the centre and that meeting point are the corners of a right triangle: its legs are h and that distance, and its long side is a radius, $(w + 1)/2$. The theorem of Pythagoras gives $h^2 + ((w - 1)/2)^2 = ((w + 1)/2)^2$. So $h^2 = ((w + 1)^2 - (w - 1)^2)/4$, and since $(w + 1)^2 - (w - 1)^2 = 4w$, this is $h^2 = w$. So h is $\sqrt{w}$, and with $w = 3$ the instruments produce $\sqrt{3}$.

Put coordinates on the plane with the unit segment along an axis from the origin, and call a number constructible when it is a coordinate of a point that the instruments can build from the two ends of that segment. Every construction is a finite sequence of moves, and each move produces a new point where two lines meet, or a line and a circle, or two circles. A line through two known points has an equation of the first degree whose coefficients come from the known coordinates by the four operations, and a circle has an equation of the second degree in which x^2 and y^2 both carry the coefficient 1. Where two lines meet, two equations of the first degree are solved, and the four operations suffice. Where two circles meet, subtracting one equation from the other cancels x^2 and y^2 and leaves an equation of the first degree, the equation of a line. A point on both circles satisfies both equations, so it satisfies their difference: it lies on that line and on the first circle. So the case is

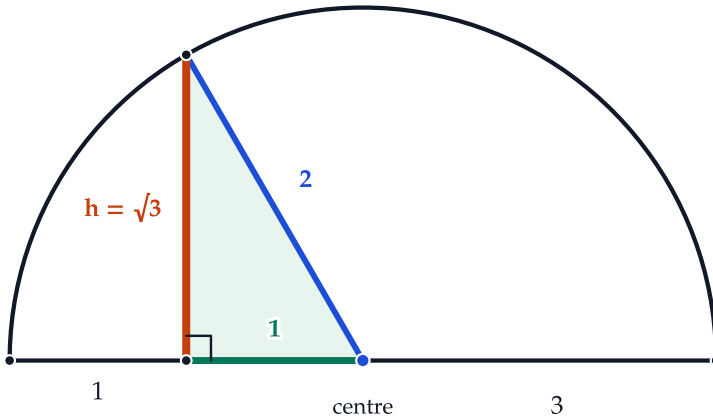


Figure 57.1. Segments of lengths 1 and 3 laid end to end, the semicircle on the whole as diameter, and the perpendicular raised at the join. The centre lies at distance 1 from the join and the radius is 2, so the altitude h satisfies $h^2 + 1^2 = 2^2$, which makes h the square root of 3. Assumes: the theorem of Pythagoras, a theorem of Euclid's plane with the parallel postulate in force

that of a line and a circle. Where a line meets a circle, substituting the line into the circle gives a quadratic, and when the two meet its roots need the four operations and one square root of a positive number. So each move adds at most one square root to what was known, and reading a whole construction at once:

a number is constructible exactly when it lies at the top of a tower of fields $Q = F_0 \subset F_1 \subset \dots \subset F_k$ in which each field is the one before with the square root of one of its positive members adjoined.

Each F_{i+1} is a field. Its members are the numbers $a + b\sqrt{w}$ with a and b in F_i , and sums, differences and products of such numbers are again of that form, because $\sqrt{w}$ times $\sqrt{w}$ is w , a member of F_i . The reciprocal of a nonzero member is $(a - b\sqrt{w})/(a^2 - wb^2)$, as multiplying it by $a + b\sqrt{w}$ confirms. Its denominator could vanish only if $a^2 = wb^2$. With $b = 0$ that forces $a = 0$, and the member was

0. With b not 0 it would put $\sqrt{w} = \pm a/b$ in F_i already, and then nothing was adjoined.

The reading of the moves gave one half of “exactly”: every constructible number lies at the top of a tower. The other half says that every number at the top of a tower can be built. For that the instruments must carry out, on coordinates, each thing a tower does: add, subtract, multiply, divide, and take the square root of a positive number. The circle about the origin through $(1, 0)$ meets the horizontal axis again at $(-1, 0)$. The circle about $(-1, 0)$ through $(1, 0)$ and the circle about $(1, 0)$ through $(-1, 0)$ cross at two points of the vertical axis, so that axis can be drawn, and in the same way the perpendicular bisector of any segment. A circle about the origin through $(c, 0)$ gives $(-c, 0)$, $(0, c)$ and $(0, -c)$, which supplies negatives and carries lengths from one axis to the other. For the perpendicular from a point P to a line, the circle about P through a known point X of the line meets the line again at Y , and the perpendicular bisector of XY passes through P ; if the circle only touches the line at X , the line PX is the perpendicular. A parallel is a perpendicular to a perpendicular.

With parallels in hand the product and the quotient above are built on the two axes, the semicircle on the segment from $(-1, 0)$ to $(w, 0)$ meets the vertical axis at $(0, \sqrt{w})$, and the point (a, b) is where the perpendiculars at $(a, 0)$ and $(0, b)$ cross. So every number at the top of a tower is a coordinate of a point that can be built.

* * *

A cubic equation stands at the centre of two of the problems, and one argument disposes of both. It is a descent: a root of the cubic at one level of a tower forces a root one level down, and so on down to the fractions.

Suppose a cubic equation with fraction coefficients has no fraction root, and suppose some constructible number satisfies it. That number lies in some tower. Call the fractions the ground level of

a tower, the field after one adjoined square root the first level, and so on up. Among all towers and all constructible roots of the cubic, choose a tower and a root so that the root appears at the lowest level possible: no constructible root of the cubic appears at a lower level of any tower. That level is not the ground level, where the numbers are the fractions and the cubic was supposed to have no fraction root. At the level chosen the root has the form $p + q\sqrt{w}$, with p, q and w from the level below and $\sqrt{w}$ not in the level below, since if it were, the chosen level would be the level below itself and the root would lie one level down; and q is not 0, since otherwise the root would lie one level down.

Substitute $p + q\sqrt{w}$ into the cubic and expand. The only powers of $\sqrt{w}$ that appear are the first, the second and the third; the second is w , a number from the level below, and the third is $w\sqrt{w}$. So every term is either free of $\sqrt{w}$ or is a number from the level below times $\sqrt{w}$, and the result reads $A + B\sqrt{w} = 0$ with A and B from the level below. Then $B = 0$, since otherwise $\sqrt{w} = -A/B$ would lie at the level below, and so $A = 0$ as well. Now substitute $p - q\sqrt{w}$ instead. The expansion is the same as before with $-\sqrt{w}$ written for $\sqrt{w}$ throughout. The even powers of $\sqrt{w}$ are unchanged and the odd powers change sign, so the part free of $\sqrt{w}$ is still A and the part carrying it is now $-B\sqrt{w}$. The result is $A - B\sqrt{w}$, and since A and B are both 0, it is 0 too.

So the cubic has two roots at this level, $p + q\sqrt{w}$ and $p - q\sqrt{w}$, and they are different because q is not 0. Divide the cubic by x minus the first, and the quotient by x minus the second. What is left is a factor of the first degree, and a factor of the first degree gives a third root. Multiplying out $a \cdot (x - r_1)(x - r_2)(x - r_3)$ shows that the three roots add up to the negative of the coefficient of x^2 divided by the coefficient a of x^3 , which is a fraction. The two known roots add up to $2p$, since $q\sqrt{w}$ and $-q\sqrt{w}$ cancel. So the third root is that fraction minus $2p$, and since p and every fraction lie at the level below, so does the third root. The cubic has a constructible root one level down, contrary to the choice of the lowest level. So no constructible number satisfies the cubic.

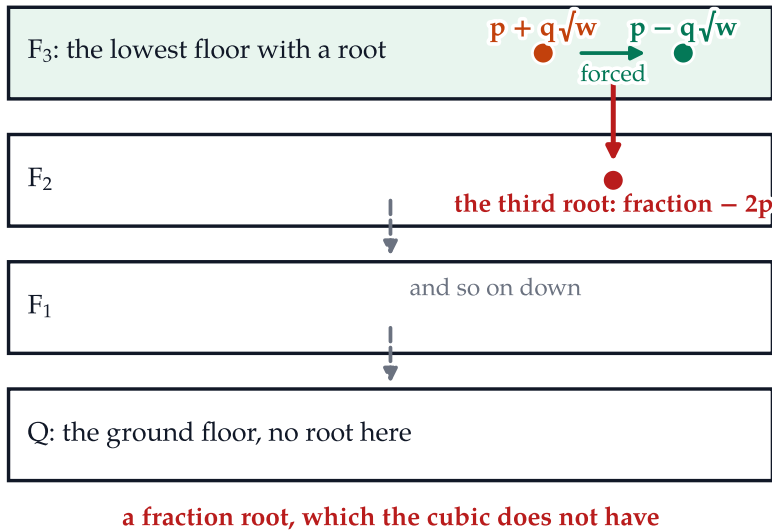


Figure 57.2. A tower of square-root steps drawn as floors. A root of a cubic sitting at the lowest floor that holds one has the form $p + q\sqrt{w}$ with p, q, w from the floor below; the conjugate $p - q\sqrt{w}$ is then a root on the same floor, and the third root, a fraction minus $2p$, lives one floor down, contradicting the choice of floor. The descent ends on the ground floor, where the cubic was assumed to have no root. *Assumes:* that the cubic has three roots whose sum is the negative of the coefficient of x^2 divided by that of x^3 , which follows from dividing out the two known ones; and that $\sqrt{w}$ is absent from the floor below, which is what makes the step a step

Doubling the cube asks for $\sqrt[3]{2}$, a root of $x^3 - 2$. A fraction root p/q in lowest terms, with q positive, would give $p^3 = 2q^3$. So q divides p^3 . But q shares no prime factor with p , so it shares none with p^3 , and q is 1. That leaves $p \cdot p^2 = 2$, so p divides 2. The candidates 1, -1 , 2 and -2 give $x^3 - 2$ the values -1 , -3 , 6 and -10 , none of them 0. The cubic has no fraction root, so $\sqrt[3]{2}$ is not constructible, and the cube cannot be doubled.

A method that trisects every angle would trisect 60° , an angle the instruments do build, as the corner of an equilateral triangle. The result would be an angle of 20° , and a built angle gives its co-

sine as a length: mark the point at distance 1 from the vertex along one side and drop the perpendicular to the other side. So trisecting 60° means constructing the cosine of 20° . Two uses of the addition formula for the cosine give $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, and with $\theta = 20^\circ$ the left side is $1/2$, so $c = \cos 20^\circ$ satisfies $8c^3 - 6c - 1 = 0$.

A fraction root p/q in lowest terms gives, after multiplying through by q^3 , $8p^3 - 6pq^2 - q^3 = 0$. Every term except q^3 carries a factor p , so p divides q^3 ; since p shares no prime factor with q , it is ± 1 . Every term except $8p^3$ carries a factor q , so q divides $8p^3$, and for the same reason q divides 8. The eight candidates $1, -1, 1/2, -1/2, 1/4, -1/4, 1/8$ and $-1/8$ give the values $1, -3, -3, 1, -19/8, 3/8, -111/64$ and $-17/64$, none of them 0. So the cosine of 20° is not constructible, the angle of 60° cannot be trisected, and no method trisects every angle.

The refutation depends on the tools. Put two marks on the straightedge, and allow it to be slid until the marks rest on two given curves. Then any angle can be trisected, by a construction known to Archimedes. Take an angle θ less than a right angle, which is enough, since a right angle can be trisected and every angle is some right angles and an acute one. Draw a circle of radius r about the vertex O , meeting the sides of the angle at A and B , and put two marks on the straightedge a distance r apart. Lay the straightedge through B and slide it until one mark, D , rests on the line AO extended beyond O , and the other, C , rests on the circle between D and B . Then DC is r , the distance between the marks, and CO is r , a radius. So the triangle DCO has two equal sides, and its angles at D and at O are equal; call each of them φ .

The points D, C and B lie along the straightedge, so the angle BCO is an exterior angle of that triangle, and it is the sum of those two angles, 2φ . Next, OC and OB are both radii, so the triangle OCB has equal angles at C and at B , and the angle OBC is 2φ as well. Last, D, O and A lie on one line, so θ , the angle AOB , is the exterior angle at O of the triangle DOB . It is the sum of the angles at D and at B , which is $\varphi + 2\varphi = 3\varphi$. The angle at D is a third of θ .

The perpendicular from C to the line DO cuts the triangle DCO

into two equal right triangles, each with long side r and angle φ , so each has a side $r \cos \varphi$ along DO . The distance OD is therefore $2r \cos \varphi$, and for $\theta = 60^\circ$ and $r = 1$ it is $2\cos 20^\circ$, a root of $z^3 - 3z - 1$, which is the cubic $8c^3 - 6c - 1$ with $z = 2c$: the marked straight-edge reaches exactly the number that the argument on the cubic showed straightedge and compass cannot reach.

The same argument settles two regular polygons. Place a regular n -sided polygon with its centre at the origin and one corner at $(1, 0)$. The next corner has for coordinates the cosine and the sine of $2\pi/n$, and the sine is the square root of one minus the square of the cosine. So if the polygon can be built, the cosine of $2\pi/n$ is a constructible number; and if the cosine can be built, a square root gives the sine, the corner can be built, and the compass steps the side from $(1, 0)$ to that corner around the circle to give the rest. A regular n -sided polygon can be built exactly when the cosine of $2\pi/n$ can.

For $\theta = 2\pi/7$ the angles 4θ and 3θ add up to a full turn. Two angles that add up to a full turn sit at mirror-image points of the unit circle, above and below the horizontal axis, so their cosines are equal: $\cos 4\theta = \cos 3\theta$. By the addition formula this reads $8y^4 - 8y^2 + 1 = 4y^3 - 3y$ for $y = \cos \theta$. The angle 0 satisfies $\cos 4\theta = \cos 3\theta$ as well, so $y = 1$ is a root of this equation and $y - 1$ is a factor of it. The cosine of $2\pi/7$ is not 1, so it is a root of what is left when that factor is divided out, which is $8y^3 + 4y^2 - 4y - 1 = 0$. For $\theta = 2\pi/9$, which is 40° , the angle 3θ has cosine $-1/2$, so $\cos 40^\circ$ satisfies $8x^3 - 6x + 1 = 0$.

Doubling the unknown clears the 8s. Putting $z = 2y$ turns the first into $z^3 + z^2 - 2z - 1$, and putting $z = 2x$ turns the second into $z^3 - 3z + 1$. Both have leading coefficient 1 and whole-number coefficients, and since twice a fraction and half a fraction are fractions, a cubic before the change has a fraction root exactly when the cubic after it has one. By the argument used for $x^3 - 2$, a fraction root of such a cubic must be a whole number, positive or negative, that divides the constant term. The constant terms here are -1 and 1 , so only 1 and -1 need trying. They give -1 and 1 for the first and -1

and 3 for the second, so neither has a fraction root. So the regular polygons with seven and with nine sides cannot be built.

The pentagon can. Three times 72° and twice 72° add to a full turn, so $y = \cos 72^\circ$ satisfies $4y^3 - 3y = 2y^2 - 1$; dividing out $y - 1$ leaves $4y^2 + 2y - 1 = 0$, and y is its positive root, $(\sqrt{5} - 1)/4$, one square root away from the fractions.

Which angles of a whole number of degrees can be built with straightedge and compass, and which of those can be trisected?

WHAT A DIMENSION COUNTS

The descent refuses one cubic at a time, and an equation of degree five would need a descent of its own. A single number attached to a field, measuring how large it is over the fractions, refuses at once every number whose field has the wrong size, and it comes from treating a field as a collection of things that can be added and scaled.

When a field K contains a field F , the elements of K can be added to one another and multiplied by members of F , and K is called a *space over F* . A list of elements of K *spans* K over F when every element of K is a combination of the list with coefficients in F . The plane of pairs is spanned over the real numbers by 1 and i ; the pairs with fraction entries, written $Q(i)$, are spanned over the fractions by 1 and i ; and $Q(\sqrt{2})$, the field of all $a + b\sqrt{2}$ with a and b fractions, is spanned by 1 and $\sqrt{2}$. The collection of all $a + b\sqrt[3]{2} + c\sqrt[3]{4}$ with fractions a , b and c is spanned by 1, $\sqrt[3]{2}$ and $\sqrt[3]{4}$, and it is closed under multiplication, because $\sqrt[3]{2} \cdot \sqrt[3]{4} = 2$ and $\sqrt[3]{4} \cdot \sqrt[3]{4} = 2\sqrt[3]{2}$ keep every product inside; whether each of its nonzero members has a reciprocal inside it is not yet clear.

Maps that respect the two operations of a space have been in use for a long time without a name. Multiplication by the pair $2 + i$ sends 1 to $2 + i$ and i to $-1 + 2i$, it sends a sum to the sum of the images, and it sends a scaled element to the scaled image.

The correction term in the derivative, which sends a step h to $m \cdot h$, does the same. A map with those two properties, sending a sum to the sum of the images and a scaled element to the scaled image, is called *linear*. A linear map is settled by where it sends a spanning list: multiplication by $2 + i$ sends $a + bi$ to $a(2 + i) + b(-1 + 2i)$.

Not every natural map is linear. The change of x^2 at 3, which sends a step h to $(3 + h)^2 - 9 = 6h + h^2$, sends 0.01 to 0.0601. A linear map would send the doubled step 0.02 to the doubled value 0.1202, and this one sends it to 0.1204. Its linear part, sending h to $6h$, is what the derivative keeps.

Not every linear map of the plane is a multiplication. Reflection in the horizontal axis, which sends $a + bi$ to $a - bi$, is linear; but a multiplication that leaves 1 in place is multiplication by 1, which moves nothing, while the reflection moves i . Among the maps that are linear over the real numbers, the multiplications are exactly those that commute with multiplication by i , meaning that multiplying by i and then applying the map gives the same result as applying the map and then multiplying by i . A multiplication by z does commute with it, since $z(iu) = i(zu)$. In the other direction, a linear map T that commutes with it has $T(i) = iT(1)$, so it sends $a + bi$ to $aT(1) + bT(i) = aT(1) + biT(1) = (a + bi)T(1)$, and T is multiplication by the number $T(1)$.

And a multiplication by z respects addition but not, in general, multiplication. To respect multiplication it would have to send a product uv to the product of the images, which is $(zu)(zv) = z^2uv$, and it sends uv to zuv . The two agree for all u and v only when $z^2 = z$, that is, when z is 0 or 1.

* * *

Is the number of elements in a spanning list a fact about the space or about the list? Other lists span $Q(\sqrt{2})$. The elements $1 + \sqrt{2}$ and $1 - \sqrt{2}$ do, since $a + b\sqrt{2}$ is $((a + b)/2)(1 + \sqrt{2}) + ((a - b)/2)(1 - \sqrt{2})$; and $1, \sqrt{2}$ and $1 + \sqrt{2}$ do, although the third is the sum of the first two and can be dropped. No single element

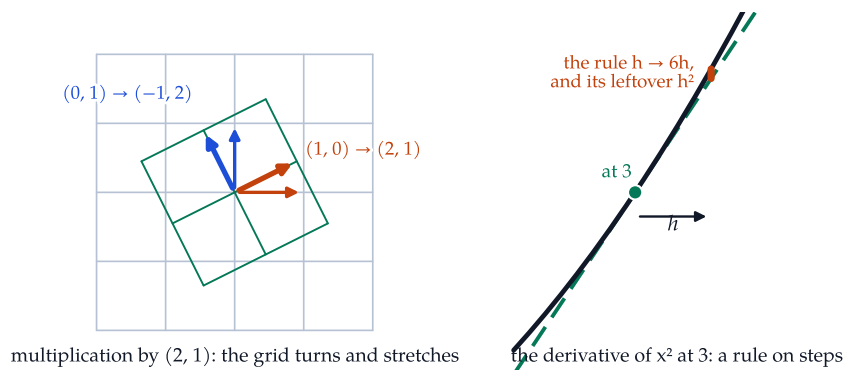


Figure 58.1. Left: multiplication by the pair $(2, 1)$ carries the square grid to a turned and stretched grid; $(1, 0)$ goes to $(2, 1)$ and $(0, 1)$ to $(-1, 2)$, and every other image is forced from those two. Right: near the input 3 the squaring function is replaced by 9 plus the rule that sends a step h to $6h$; at $h = 0.01$ that prediction is 9.06 and the truth is 9.0601. Both maps send sums to sums and scaled steps to scaled images, and that pair of properties is all the word linear means. *Assumes:* that the grid on the left extends and the rule on the right applies to every step, not only the ones drawn; four numbers determine the left map and one number the right

spans $Q(\sqrt{2})$, because the nonzero fraction multiples of one element are either all fractions or all irrational, and so never include both 1 and $\sqrt{2}$.

A list is *independent* over F when the only combination of it with coefficients in F that equals 0 is the one with every coefficient 0. Equivalently, no member of the list is a combination of the others. In one direction, a combination equal to 0 with some coefficient other than 0 can be solved for the member carrying that coefficient, which makes that member a combination of the others. In the other direction, a member that is a combination of the others gives, when it is moved to the other side, a combination equal to 0 in which it carries the coefficient -1 . A list that spans and is independent is a *basis*; $1, \sqrt{2}$ is one, and so is $1 + \sqrt{2}, 1 - \sqrt{2}$.

In a space over F , an independent list is never longer than a spanning list.

The proof exchanges one list for the other. Let $u_1, \dots, u_m$ be inde-

pendent and $w_1, \dots, w_n$ span. Put u_1 in front of the w 's. The new list still spans. Because the w 's span, u_1 is a combination of them, and moving u_1 to the other side gives a combination of the new list that equals 0 in which u_1 carries the coefficient -1 , so not all of its coefficients are 0. Read that combination from left to right and stop at the last member with a coefficient other than 0. The combination can be solved for that member, so it is a combination of the members before it. That member is not u_1 , which has nothing before it and is not 0, so it is one of the w 's. Remove it. The list still spans, because wherever the removed w was used it can be replaced by its combination of the members before it. The list has length n again, and it holds u_1 and $n - 1$ of the w 's.

Now put u_2 in front and repeat, and go on in the same way with u_3 and the rest. At each round the list spans and a vanishing combination exists, and its last member with a coefficient other than 0 cannot be one of the u 's standing at the front, since then the combination would involve u 's alone and contradict their independence; so it is again one of the w 's, and it is removed. Each round brings in one u and removes one w . If m were larger than n , then after n rounds no w would remain, the list $u_n, \dots, u_1$ would span, and u_{n+1} would be a combination of $u_1, \dots, u_n$, which independence forbids. So m is at most n .

In $Q(\sqrt{2})$, take the spanning list $1, \sqrt{2}, 1 + \sqrt{2}$ and the independent list $1 + 2\sqrt{2}, 3 - \sqrt{2}$. Bringing in $1 + 2\sqrt{2}$ gives the list $1 + 2\sqrt{2}, 1, \sqrt{2}, 1 + \sqrt{2}$, in which $\sqrt{2}$ is half of $(1 + 2\sqrt{2}) - 1$, so $\sqrt{2}$ is removed. Bringing in $3 - \sqrt{2}$ gives $3 - \sqrt{2}, 1 + 2\sqrt{2}, 1, 1 + \sqrt{2}$, in which $1 = (2/7)(3 - \sqrt{2}) + (1/7)(1 + 2\sqrt{2})$, so 1 is removed. Three members remain, both independent elements among them, and 2 is at most 3.

Two consequences follow. Two bases of one space have the same length: the first is independent and the second spans, so the first is no longer than the second, and with the roles exchanged the second is no longer than the first; that common length is the *dimension* of the space over F , a number that belongs to the space and not to any list. And a finite spanning list contains a basis, ob-

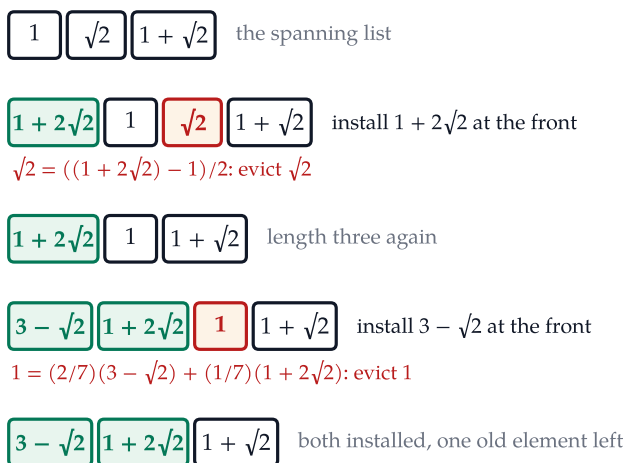


Figure 58.2. The procedure run on the spanning list $1, \sqrt{2}, 1 + \sqrt{2}$ of the field $Q(\sqrt{2})$ and the independent list $1 + 2\sqrt{2}, 3 - \sqrt{2}$. Each round puts the next independent element at the front; the list then carries a passenger, which is always one of the old elements, and it is evicted. The length stays three, and after two rounds one old element survives, so the independent list was no longer than the spanning one. *Assumes:* that the evicted element really was a combination of the ones before it, which is the identity written beside each arrow

tained by removing, one at a time, members that are combinations of the others. So $Q(\sqrt{2})$ and $Q(i)$ have dimension 2 over Q , the plane of pairs has dimension 2 over the real numbers, and Z/p has dimension 1 over itself.

* * *

The fields that the instruments build are stacked in towers, and along a tower dimensions multiply. Suppose K has a basis $a_1, \dots, a_m$ over F , and L contains K and has a basis $b_1, \dots, b_n$ over K . Then the mn products $a_i b_j$ form a basis of L over F . They span: an element of L is a combination of the b 's with coefficients in K , each of those coefficients is a combination of the a 's with coefficients in F , and

substituting writes the element as a combination of the products. They are independent: a combination of the products that equals 0 can be regrouped as a combination of the b 's whose coefficients are combinations of the a 's; those coefficients lie in K , so each is 0 by the independence of the b 's over K ; and then every coefficient of every a is 0 by the independence of the a 's over F .

The tower $Q \subset Q(\sqrt{2}) \subset Q(\sqrt{2}, \sqrt{3})$ shows the law at work. The number $\sqrt{3}$ is not in $Q(\sqrt{2})$. If $\sqrt{3} = a + b\sqrt{2}$ with fractions a and b , squaring gives $3 = a^2 + 2b^2 + 2ab\sqrt{2}$. If ab were not 0, this could be solved for $\sqrt{2}$ and would make $\sqrt{2}$ a fraction; so $ab = 0$, and one of a and b is 0. If $a = 0$, then $3 = 2b^2$, which makes $3/2$ the square of a fraction, and then $6 = (2b)^2$ is the square of the fraction $2b$. If $b = 0$, then $3 = a^2$ is the square of a fraction. Neither can happen, because the argument that refused a fraction square root to 2 refuses one to 3 and to 6.

The members of the top field are the numbers $x + y\sqrt{3}$ with x and y in $Q(\sqrt{2})$, so 1 and $\sqrt{3}$ span it over $Q(\sqrt{2})$; and they are independent, because $x + y\sqrt{3} = 0$ with y not 0 would give $\sqrt{3} = -x/y$, a member of $Q(\sqrt{2})$, while with $y = 0$ it says that $x = 0$ as well. So the second step has basis 1, $\sqrt{3}$ over $Q(\sqrt{2})$, and the top has dimension $2 \cdot 2 = 4$ over Q , with basis 1, $\sqrt{2}$, $\sqrt{3}$, $\sqrt{6}$.

A single number generates the top: the smallest field containing $\sqrt{2} + \sqrt{3}$ is already all of $Q(\sqrt{2}, \sqrt{3})$. The cube of $\sqrt{2} + \sqrt{3}$ is $11\sqrt{2} + 9\sqrt{3}$, so the cube minus nine times the number is $2\sqrt{2}$. A field containing $\sqrt{2} + \sqrt{3}$ contains that cube and that multiple, since it is closed under multiplication and subtraction, so it contains $2\sqrt{2}$ and with it $\sqrt{2}$. Then it contains $\sqrt{3}$, which is $(\sqrt{2} + \sqrt{3}) - \sqrt{2}$, and $\sqrt{6}$, which is $\sqrt{2} \cdot \sqrt{3}$.

For constructions the consequence is immediate. Every constructible number lies at the top of a tower in which each field is the one before with the square root of one of its members adjoined. Such a step has dimension 2 over the field before, with basis 1 and the new square root, or dimension 1 when the square root was there already; and along the tower the dimensions multiply. So every constructible number lies in a field whose dimension over Q is a

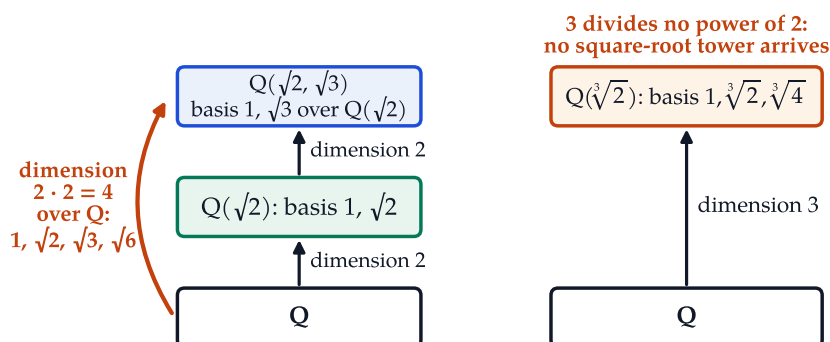


Figure 58.3. Left: the tower from the fractions to $Q(\sqrt{2})$ to $Q(\sqrt{2}, \sqrt{3})$, a step of dimension 2 on a step of dimension 2, with the products of the two bases, $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$, as a basis of the top over the bottom. Right: the field generated by the cube root of 2 has dimension 3 over the fractions, and 3 divides no power of 2, so no square-root tower reaches it. *Assumes:* that dimensions multiply along a tower, which is the regrouping argument in the text; and that the cube root's field has dimension exactly 3, which the descent supplies

product of 1s and 2s, which is a power of 2.

If α lies in such a field F , of dimension 2^k , then $Q(\alpha)$, the smallest field containing α , lies between Q and F . It has finite dimension over Q . An independent list inside $Q(\alpha)$ is also an independent list inside F , which is spanned by 2^k elements, so it has at most 2^k members; take one that is as long as possible. That list spans $Q(\alpha)$. For if some element of $Q(\alpha)$ lay outside its span, putting the element on the end would give a longer list, and the longer list would still be independent: in a combination of it that equals 0, the new element must carry the coefficient 0, since otherwise the combination could be solved for it and it would lie in the span after all; and once that coefficient is 0, what is left is a vanishing combination of the old list, whose coefficients are all 0.

A longer independent list contradicts the choice of the longest, so the longest list is a basis of $Q(\alpha)$. And F has finite dimension over $Q(\alpha)$, because the basis of F over Q still spans F when the

coefficients are allowed to come from the larger field $Q(\alpha)$. So 2^k is the dimension of F over $Q(\alpha)$ times the dimension of $Q(\alpha)$ over Q , and the dimension of $Q(\alpha)$ over Q divides a power of 2.

The cube root of 2 now has a dimension. The elements $1, \sqrt[3]{2}$ and $\sqrt[3]{4}$ are independent over Q . Suppose $a + b\sqrt[3]{2} + c\sqrt[3]{4} = 0$ with fractions a, b and c not all 0. If $c = 0$, then b is not 0, and $\sqrt[3]{2} = -a/b$ is a fraction, which it is not. If c is not 0, then $\sqrt[3]{2}$ satisfies the quadratic $cx^2 + bx + a = 0$. Completing the square writes the roots of that quadratic as $p + q\sqrt{w}$ with p, q and w fractions. Here w is positive: a negative w would make the roots not real, and $w = 0$ would make $\sqrt[3]{2}$ the fraction p . So $\sqrt[3]{2}$ would lie in the field reached from the fractions by adjoining the one square root $\sqrt{w}$, which would make it constructible in a single step; and the descent showed that it is not constructible at all.

So the collection of all $a + b\sqrt[3]{2} + c\sqrt[3]{4}$ has dimension 3 over Q , and it is a field. Multiplication by a nonzero member u is a linear map of the collection to itself, and it sends different elements to different elements, since $ux = uy$ forces $u(x - y) = 0$ and a product of real numbers is 0 only when a factor is. So it sends the basis $1, \sqrt[3]{2}, \sqrt[3]{4}$ to three independent elements. A combination of the images $u, u\sqrt[3]{2}$ and $u\sqrt[3]{4}$ is u times the same combination of $1, \sqrt[3]{2}$ and $\sqrt[3]{4}$; if it equals 0 then, since u is not 0, the combination of the originals equals 0, and so all its coefficients are 0. Three independent elements in a space of dimension 3 span it, since otherwise an element outside their span could be added to make four independent elements in a space spanned by three. So 1 is a combination of the images, which says $1 = u \cdot x$ for some x in the collection, and x is the reciprocal of u .

Nothing about $\sqrt[3]{2}$ was used except that the collection contains the fractions, is closed under addition and multiplication, has finite dimension over Q , and sits inside a field, where a product is 0 only when a factor is; every collection of that kind is a field. This one is $Q(\sqrt[3]{2})$, of dimension 3, and 3 divides no power of 2. The count reaches beyond cubics: a number whose field has dimension 5, 6 or 7 over Q is out of reach of the instruments, with no descent written

for it.

What is the dimension of the smallest field containing both $\sqrt{2}$ and $\sqrt[3]{2}$ over Q ?

* * *

A dimension count also forces an equation. If α lies in a field K of dimension n over F , the $n + 1$ elements $1, \alpha, \alpha^2, \dots, \alpha^n$ cannot be independent, since an independent list is never longer than a spanning list. So some combination $c_0 + c_1\alpha + \dots + c_n\alpha^n$ with coefficients in F , not all 0, is 0, and α satisfies a polynomial equation over F of degree at most n . Nothing about α was used except that it lives in a field of finite dimension. The number $\sqrt{2} + \sqrt{3}$ lives in $Q(\sqrt{2}, \sqrt{3})$, of dimension 4, so it satisfies an equation of degree at most 4 before anyone writes one down. And every constructible number satisfies a polynomial equation with fraction coefficients, and so, after clearing denominators, one with integer coefficients: every constructible number is algebraic.

That disposes of the third problem, on one condition: that π is not algebraic. A square with the area of a circle of radius 1 has side $\sqrt{\pi}$, and if $\sqrt{\pi}$ were constructible then so would be its square π , which would then be algebraic. Lindemann's theorem of 1882, met among the numbers no equation reaches, shows that π is not algebraic, and the impossibility of squaring the circle rests on that theorem, whose proof is given once it has been shown that an expression in the roots of an equation that is unchanged when the roots are rearranged can be computed from the coefficients.

MANY EQUATIONS AT ONCE

The eighth of the Nine Chapters on the Mathematical Art, compiled in China some two thousand years ago, opens with a problem about grain. Three bundles of the best grain, two of middling and one of poor yield 39 measures. Two of the best, three of middling and one of poor yield 34. One of the best, two of middling and three of poor yield 26. How much does one bundle of each yield? The text lays the numbers out in columns on a counting board and gives a procedure: multiply one column through by the top number of another and subtract until the top number vanishes; do the same again one row down; and read the answers off from the bottom upwards. In modern order,

$$3x + 2y + z = 39$$

$$2x + 3y + z = 34$$

$$x + 2y + 3z = 26$$

Subtract twice the first equation from three times the second, and the first from three times the third: what is left is $5y + z = 24$ and $4y + 8z = 39$, with no x . Subtract four times the first of these from five times the second: $36z = 99$. So z is $2 \frac{3}{4}$, then y is $4 \frac{1}{4}$, then x is $9 \frac{1}{4}$. The method is called Gaussian elimination, after a mathematician who used it eighteen centuries later on the orbit of an asteroid.

Its three moves are to exchange two equations, to multiply an equation by a number other than 0, and to add a multiple of one equation to another. Each can be undone by a move of the same kind, so none of them changes the collection of solutions. By these moves any system, over any field, can be brought to a staircase form, in which each equation that still has an unknown in it has its first unknown further to the right than the first unknown of the equation above, and the equations with no unknown left stand at the bottom. The first unknown of each equation that still has one is called a *pivot*; the pivots are where the steps of the staircase begin. In that form everything about the system can be read off. If one of the bottom equations reads $0 = c$ with c not 0, there is no solution. Otherwise the unknowns that are not pivots can be given any values at all. The lowest equation with an unknown in it then fixes its pivot, since every other unknown in it has a value already. The equation above it fixes the next pivot in the same way, and so on, from the bottom upwards.

One consequence is worth setting apart, because counts of the symmetries of a field will later depend on it. A system whose right-hand sides are all 0 always has the solution in which every unknown is 0. If it has more unknowns than equations, it has another. The staircase has at most one pivot for each equation, so some unknown is not a pivot. The right-hand sides stay 0 under all three moves, so no equation reads $0 = c$ with c not 0, and that unknown can be given the value 1. The solution that results is not the one in which every unknown is 0.

* * *

Elimination is a procedure. What it computes is better seen through linear maps. A system of d linear equations in n unknowns, with coefficients in a field F , is a linear map T from the space of lists of length n to the space of lists of length d : the list of unknowns goes to the list of left-hand sides. Lists of one length with entries in F can be added entry by entry, and multiplied by a

member of F entry by entry. Adding, and multiplying by members of F , is all that the word *space over F* asked of a field holding F . It is also all that was used in proving that an independent list is never longer than a spanning one, where one list was exchanged for the other. So from here on the word covers any collection with those two operations, field or not, and span, independence, basis and dimension mean there what they meant before. In the space of lists of length n , the n lists with a single 1 and 0 elsewhere are a basis: the list $a_1, \dots, a_n$ is a_1 times the first of them plus a_2 times the second and so on, and it is a combination of them in no other way. So the dimension of that space is n . Solving the system with right-hand side b means finding the lists that T sends to b .

Two collections belong to any linear map T from a space V to a space W . The *kernel* is everything in V that T sends to 0. The *image* is everything in W that T reaches. Both are closed under addition and scaling, so they are spaces themselves. Both have dimensions when V is spanned by a finite list. The image is spanned by what T does to that list. An independent list inside the kernel is independent in V , so it is no longer than that list; take one as long as possible, and it spans the kernel, since an element of the kernel outside its span could be put on the end to give a longer independent list. And the two dimensions are tied together:

the dimension of V is the dimension of the kernel plus the dimension of the image.

Take a basis $k_1, \dots, k_r$ of the kernel. It is an independent list in V , and it can be extended to a basis of V . While some element of V lies outside the span of the list, add one such element; the list stays independent. The adding has to stop, since V is spanned by a finite list and an independent list is never longer than a spanning one. It stops only when nothing is outside the span, and then the list is a basis of V . Call the added elements $v_1, \dots, v_s$. Then $T(v_1), \dots, T(v_s)$ is a basis of the image. It spans the image: every member of V is a combination of the whole basis, and T sends it to the same combination of the $T(k_i)$ and the $T(v_j)$. The $T(k_i)$ are all 0, so what

T reaches is a combination of the $T(v_j)$ alone. It is independent: if some combination of the $T(v_j)$ is 0, then T sends the same combination of the v_j to 0, so that combination of the v_j lies in the kernel and equals some combination of the k_i . Subtracting one from the other gives a combination of the whole basis that is 0, and the independence of the whole basis makes every coefficient 0, those of the v_j among them. So the dimension of V is $r + s$, where r is the dimension of the kernel and s is the dimension of the image.

The theorem has been met before in another form. A linear map respects addition, so it is a homomorphism of groups under addition, and for a homomorphism the elements sent to any one target form a coset of the kernel. That says what the solutions of a system look like: if one list x_0 is sent to b , the lists sent to b are exactly x_0 plus the members of the kernel. Over the field $\mathbb{Z}/p$ a space of dimension m has p^m members, since each member is exactly one combination of a basis and each of the m coefficients can be chosen in p ways; there the group has p^n members, the kernel p^r and the image p^s . The count for groups says that the size of the group is the size of the kernel times the size of the image, so $p^n = p^r \cdot p^s$. Comparing exponents, $n = r + s$, which is the theorem on dimensions once more.

The dimension of the image is called the *rank* of the map. For a system it is the number of pivots, for two reasons. First, a move of elimination does one thing to the whole list of d equations, and that thing is a linear map of the space of lists of length d that can be undone. Such a map carries a basis of the old image to a basis of the new one, so the moves do not change the dimension of the image. Second, in staircase form the equations below the last pivot have left-hand side 0, so the image lies among the lists that have 0 in those places. Every such list is reached, since a staircase system with no equation $0 = c$, c not 0, has a solution. Those lists form a space with one dimension for each equation that has a pivot, which is one for each pivot. The kernel then has dimension n less the rank, one dimension for each unknown that is free. A system with as many equations as unknowns has exactly one solution for every

right-hand side when the rank is n , because then the kernel is 0 alone and the image is everything; and when the rank is lower, the image is not everything and the kernel is more than 0 alone, so some right-hand sides have no solution and each of the others has more than one.

* * *

A toy shows all of this at once, over the field with two elements. Twenty-five lamps stand in a square of five rows by five, each with a button. Pressing a button changes the state of its own lamp and of the lamps next to it in its row and column, on to off and off to on. All the lamps are on. Which buttons put them all out?

Pressing a button twice does nothing, and the order of pressing does not matter, so a plan is a list of 25 entries from $\mathbb{Z}/2$, saying for each button whether to press it. Each lamp gives one equation: the presses that affect it must add up to 1. That is a system of 25 equations in 25 unknowns over $\mathbb{Z}/2$, and elimination solves it. The rank turns out to be 23, not 25. So the kernel has dimension 2: there are four plans, the empty plan among them, that change nothing at all. The image has dimension 23: of the 2^{25} patterns of lamps, only 2^{23} , a quarter, can be reached from darkness or returned to it. The pattern with every lamp on is one of them. Its solutions are one plan plus each of the four members of the kernel, so there are exactly four, and the one that elimination produces presses 15 buttons. On a board of three by three the rank is 9, and every pattern can be put out in exactly one way; on a board of four by four the rank is 12 out of 16.

A different use reads the kernel directly. When propane burns, some number of molecules of C_3H_8 combine with some number of O_2 to give some number of CO_2 and of H_2O , and each kind of atom must be as plentiful after as before. For the four numbers a , b , c and d that is three equations, $3a = c$ for carbon, $8a = 2d$ for hydrogen and $2b = 2c + d$ for oxygen, with right-hand sides 0. Four unknowns and three equations: the kernel is not 0. It has dimension 1,

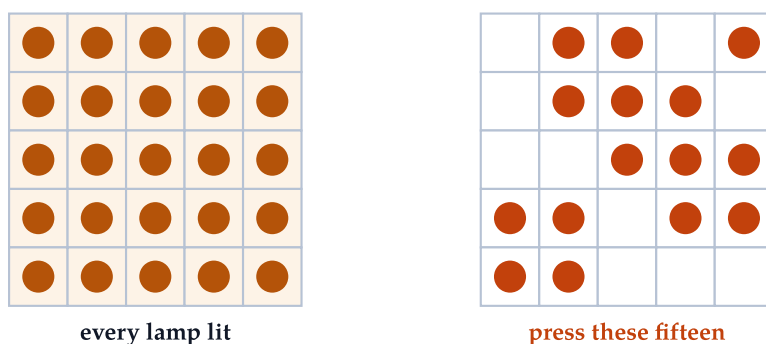


Figure 59.1. On the left, the board with every lamp lit. On the right, a plan that puts them all out: press the fifteen marked buttons, in any order. It was found by elimination over the field with two elements, and there are exactly three other plans, which differ from it by the plans that change nothing. *Assumes:* the rule that a button changes its own lamp and the lamps beside it in its row and column; with diagonal neighbours included the ranks are different

spanned by the list 1, 5, 3, 4, and that is the balanced equation of the chemists: one propane and five oxygen give three carbon dioxide and four water.

* * *

The theorem on dimensions converts uniqueness into existence. Given n different numbers $x_1, \dots, x_n$ in a field, and any n values $y_1, \dots, y_n$, is there a polynomial of degree below n that takes the value y_i at each x_i ? Consider the map that sends a polynomial of degree below n , which is a list of n coefficients, to its list of values at the n points. It is linear. Its kernel consists of the polynomials of degree below n that take the value 0 at all n points, which is to say that have n different roots. A polynomial other than 0 has no more roots than its degree, and here the degree is below n . So the kernel is 0 alone, which says that no two polynomials of degree below n take the same values at the n points. By the theorem on dimensions the

image then has dimension n . It lies inside the space of lists of length n , which also has dimension n . If some list lay outside the image, adding it to a basis of the image would give $n + 1$ independent lists in a space of dimension n , one too many. So the image is everything, which says that every list of values is taken by some polynomial:

through n points with different first coordinates there passes exactly one polynomial of degree below n .

There is a formula for the polynomial, published by Waring in 1779 and now named after Lagrange, but the proof did not need one.

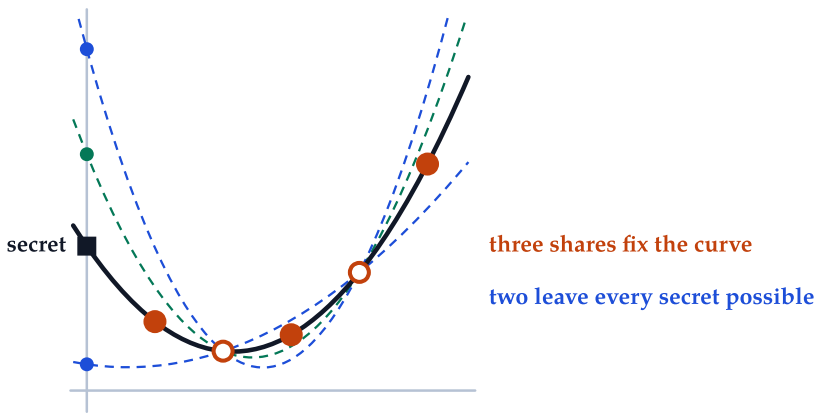


Figure 59.2. Five shares of a secret are five points of one parabola, and the secret is the height at which the parabola crosses the vertical axis. Any three of the points fix the parabola. Two of them do not: through the two outlined points pass parabolas crossing the axis at every height, three of which are drawn. Assumes: real numbers, for the sake of a picture; the scheme itself works with remainders modulo a prime, where there are no small and large values to give anything away

In 1979 Shamir built a method of sharing a secret on this theorem, over a field $\mathbb{Z}/p$. A secret number s is to be divided among five people so that any three of them together can recover it and

no two of them learn anything at all. Choose two more numbers a and b at random from $\mathbb{Z}/p$ and form the polynomial $f(x) = s + ax + bx^2$, then give person k the pair $(k, f(k))$. Three people hold three points, which determine the polynomial of degree below 3, and so its value at 0, which is s . Two people hold two points, and for every candidate value of s there is exactly one polynomial of degree below 3 through their two points and $(0, s)$: every secret is equally consistent with what they know. With $p = 31$, the secret 11 and the polynomial $11 + 8x + 7x^2$, the five shares are 26, 24, 5, 0 and 9.

How much does each kind of grain yield if three bundles of the best, two of middling and one of poor yield 39 measures, as before, but the other two totals are 35 and 25?

On the board of three by three, with every lamp off, which buttons light the middle lamp alone?

A magic square of three rows has the same sum along every row, every column and both diagonals. What is the dimension of the space of all such squares with entries in the fractions, and why is the middle entry always a third of the common sum?

Ethane, C_2H_6 , burns to carbon dioxide and water. What is the smallest balanced equation in whole numbers?

Three of the five holders of the shares above, numbers 2, 3 and 5, meet. How do they recover the secret from 24, 5 and 9, working modulo 31?

VOLUME, AND WHAT A MAP STRETCHES

A linear map of the plane sends the unit square, with corners at $(0, 0)$, $(1, 0)$, $(1, 1)$ and $(0, 1)$, to a parallelogram, and sends the grid of all such squares to a grid of copies of that parallelogram. A square of a finer grid is a scaled copy of the unit square, and the map sends it to the same scaled copy of the parallelogram, so every grid square, coarse or fine, has its area multiplied by one and the same number, the area of that one parallelogram. Any figure can be bracketed by grid squares, fine ones if need be. So the map multiplies the area of every figure by that number. What is that number, in terms of the map?

Let the map send $(1, 0)$ to (a, c) and $(0, 1)$ to (b, d) . Those two lists are the sides of the parallelogram, and they are recorded in an array with a and b in the first row and c and d in the second, a *matrix*. Shearing a parallelogram along one of its sides slides the opposite side parallel to itself, which keeps the base and the height and so does not change the area. In terms of the lists, a shear adds a multiple of one side to the other. If a is 0, the side (a, c) already lies along the upright axis, and subtracting d/c times it from (b, d) leaves $(b, 0)$, a rectangle of area bc , which is $ad - bc$ up to sign; if c is 0 as well, that side is 0, the parallelogram is flat, and $ad - bc$ is

0. Suppose now that a is not 0. Subtract b/a times the side (a, c) from the side (b, d) : what is left is $(0, d - bc/a)$, which lies along the upright axis. If that side is 0 the parallelogram is flat, its area is 0, and $ad - bc$ is 0 as well. Otherwise subtract from (a, c) the multiple of that upright side that removes the c : what is left is $(a, 0)$. Two shears have brought the parallelogram to a rectangle with sides a and $d - bc/a$. So its area is

$$ad - bc$$

up to sign. The sign has a meaning. It is positive when the turn from the first side to the second is anticlockwise, as it is from $(1, 0)$ to $(0, 1)$, and negative when the map has turned the plane over. The number $ad - bc$, with its sign, is the *determinant* of the map.

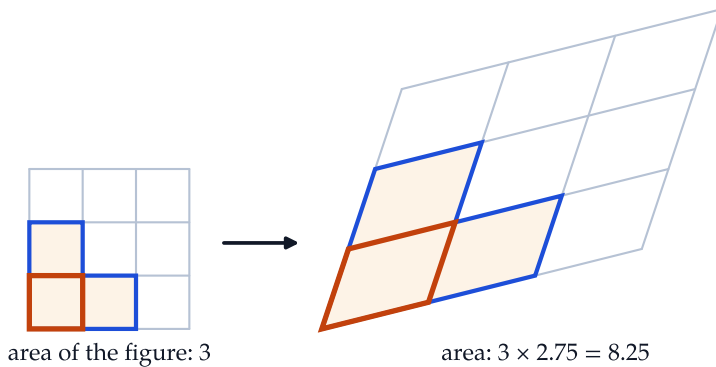


Figure 60.1. The map that sends $(1, 0)$ to $(2, 0.5)$ and $(0, 1)$ to $(0.5, 1.5)$, acting on a grid of unit squares and on a figure drawn in it. Every square goes to a parallelogram of area $2 \cdot 1.5 - 0.5 \cdot 0.5 = 2.75$, so the area of the figure, and of anything else, is multiplied by 2.75. Assumes: a map with positive determinant; a negative one would also turn the figure over, as a mirror does

What should the corresponding number be in three dimensions, or in n ? The signed volume D of the box with sides $v_1, \dots, v_n$, as a function of the sides, ought to have three properties. It is linear in each side separately, the others being held fixed: doubling one side doubles the volume, and a box whose first side is $u + w$ has

the volume of the boxes with first sides u and w together, which is the principle of the shear. It is 0 when two sides are equal, the box being flat. And it is 1 for the unit cube. From the first two it follows that exchanging two sides changes the sign. Put $u + w$ in both of two places and leave the other sides alone. That box has two equal sides, so its D is 0. Expanding by linearity in each of the two places gives four terms, $D(u, u) + D(u, w) + D(w, u) + D(w, w)$. The first and the last are 0, again because two sides are equal. What survives is $D(u, w) + D(w, u) = 0$, which says that $D(w, u)$ is $-D(u, w)$.

Those properties leave no freedom at all. Write each side as a combination of the unit lists $e_1, \dots, e_n$, and expand D by linearity in every place. Each term is a product of n coefficients, one from each side, times D of some list of unit lists. If a unit list appears twice the term is 0. Otherwise the unit lists appear in some rearranged order, and sorting them back by exchanges produces the sign of the rearrangement, +1 for an even one and -1 for an odd one. So

the determinant is the sum, over all rearrangements π of $1, \dots, n$, of the sign of π times the product of the entries in row $\pi(j)$ of column j

and conversely that formula has the three properties. Each term takes exactly one entry from each column, so the formula is linear in each column. Suppose next that columns j and k agree. Pair each rearrangement π with the one that is the same except that its values at j and k are exchanged. The term of π takes the entry in row $\pi(j)$ of column j and the entry in row $\pi(k)$ of column k ; the term of its partner takes row $\pi(k)$ of column j and row $\pi(j)$ of column k . Since the two columns hold the same entries, the two products are equal. The two rearrangements differ by one exchange, so their signs are opposite. The terms cancel in pairs and the sum is 0. Last, on the unit lists every entry off the diagonal is 0, so only the term of the identity survives, and it is 1.

The formula also reads the same with rows and columns exchanged. A term is the product of n entries, the one in row $\pi(j)$ of column j for each j . List the same n entries by their rows instead:

the entry taken from row i sits in column $\sigma(i)$, where σ is the rearrangement that undoes π . So the same term is the product of the entries in column $\sigma(i)$ of row i . The sign is the same too, because σ has the same parity as π , the two together making the identity, which is even. And as π runs through all the rearrangements, so does σ . So the sum is the same formula with the words row and column exchanged, and whatever the three properties give for the sides, which are the columns, they give for the rows as well. For $n = 2$ the two rearrangements give $ad - bc$. For $n = 3$ there are six terms, three with each sign. It was for this formula that the evenness and oddness of rearrangements was first needed, a century before anyone spoke of groups.

The formula has $n!$ terms and is useless for computing: for the 25 lamps it has more than 10^{25} . The three properties compute quickly. Adding a multiple of one row to another does not change D : by linearity in that row the change is the multiple times a D with a repeated row, which is 0. Exchanging two rows changes its sign. Use those two moves alone, with no row multiplied by a number, to bring the array to staircase form. Each row then begins further to the right than the one above it, so every entry below the diagonal is 0. Then only one term of the formula survives, the product of the entries on the diagonal, because every rearrangement other than the identity takes some entry from below the diagonal. That product is D of the original array, with one change of sign for each exchange made. For the array of the three kinds of grain, those moves give the diagonal entries 3, $5/3$ and $12/5$, whose product is 12.

* * *

Let T be a linear map of the space of lists of length n to itself, and consider the function that sends n lists $v_1, \dots, v_n$ to $D(Tv_1, \dots, Tv_n)$. It is linear in each place, because T is linear, and it is 0 when two places agree, because then two of the lists Tv agree. The expansion above used only those two properties until its last step, and what it

shows is that any function with those two properties is the formula times the function's own value on the unit lists. So this function is a fixed multiple of D , and the multiple is its value on the unit lists, $D(\text{Te}_1, \dots, \text{Te}_n)$. The lists $\text{Te}_1, \dots, \text{Te}_n$ are the columns of the array of T , so that value is the determinant of T . So

$$D(Tv_1, \dots, Tv_n) = \det T \cdot D(v_1, \dots, v_n)$$

which says that T multiplies every signed volume by $\det T$. Apply this to a second map S after T , and the volumes are multiplied first by $\det T$ and then by $\det S$:

$$\det(ST) = \det S \cdot \det T$$

Multiplied out for arrays of three rows, the identity is a page of algebra with no visible reason to come out; here the reason is the one just given.

A map with determinant 0 flattens every box, so its image is not the whole space: if the unit lists were the images $Tv_1, \dots, Tv_n$ of some lists, the identity would give $1 = \det T \cdot D(v_1, \dots, v_n) = 0$. So its kernel is not 0, by the theorem on dimensions. A map with determinant other than 0 sends the unit lists to independent lists. For suppose one of the images were a combination of the others. Expanding D of the images in that place by linearity would leave terms each of which repeats a side, and every one of those is 0, so the determinant would be 0. Since the images of the unit lists are independent, the map has kernel 0, and by the theorem on dimensions its image is then the whole space, so it can be undone. So a system of n equations in n unknowns has exactly one solution for every right-hand side exactly when its determinant is not 0. Leibniz in 1693 and Seki in Japan ten years earlier formed these sums to decide that question, and Cramer in 1750 gave the solution itself as a quotient of two determinants. The name came later and from elsewhere: Gauss in 1801 called the number $b^2 - ac$ the determinant of the expression $ax^2 + 2bxy + cy^2$, because its behaviour depends on that number, and Cauchy in 1815 carried the word over to the sums.

An identity met long ago is a special case. Multiplying by the number $a + b\sqrt{2}$ is a linear map of the space of numbers $x + y\sqrt{2}$ over the fractions: it sends 1 to $a + b\sqrt{2}$ and $\sqrt{2}$ to $2b + a\sqrt{2}$, so its array has rows $a, 2b$ and b, a , and its determinant is $a^2 - 2b^2$. Multiplying by one such number and then by another is multiplying by the product, and $(a + b\sqrt{2})(c + d\sqrt{2})$ is $(ac + 2bd) + (ad + bc)\sqrt{2}$. The determinant of the combined map is the product of the two determinants. So

$$(a^2 - 2b^2)(c^2 - 2d^2) = (ac + 2bd)^2 - 2(ad + bc)^2$$

which is the rule by which solutions of $x^2 - 2y^2 = 1$ were multiplied together, and the rule that the sizes of numbers in the plane multiply is the same statement for $a + bi$. Every such identity is the multiplication of determinants in a field of finite dimension over the fractions.

* * *

A linear map usually turns directions as well as stretching them. A list v , not 0, is an *eigenvector* of T with *eigenvalue* λ when $Tv = \lambda v$: along that line the map is multiplication by a number. That happens exactly when the map $T - \lambda$, which sends v to $Tv - \lambda v$, has a kernel that is not 0, which is when

$$\det(T - \lambda) = 0$$

For a space of dimension n that is a polynomial equation of degree n for λ , and its roots are the eigenvalues.

Take the map of the plane that sends (x, y) to $(y, x + y)$. Starting from $(0, 1)$ it produces $(1, 1), (1, 2), (2, 3), (3, 5), (5, 8)$: each application steps the pair one place along the list in which every number is the sum of the two before. Its array has rows 0, 1 and 1, 1, so $\det(T - \lambda)$ is $(-\lambda)(1 - \lambda) - 1$, which is $\lambda^2 - \lambda - 1$. The eigenvalues are $\phi = 1.618$ and $\psi = -0.618$, the two roots of $x^2 = x + 1$, with eigenvectors $(1, \phi)$ and $(1, \psi)$. Along the first line the

map stretches by φ , and along the second it shrinks by 0.618 and flips.

Any starting pair is a combination of the two eigenvectors; after n steps the part along the first has grown by φ^n and the part along the second has all but vanished. So the pairs of neighbouring numbers of the list line up along the direction $(1, \varphi)$, which says that their ratios approach φ , and the exact formula comes the same way. Since $\varphi - \psi = \sqrt{5}$, the starting pair $(0, 1)$ is $((1, \varphi) - (1, \psi))/\sqrt{5}$. Applying the map n times multiplies the first part by φ^n and the second by ψ^n , and the first entry of the result, which is the n -th number of the list, is $(\varphi^n - \psi^n)/\sqrt{5}$. It was found before from a generating function. Here it is geometry: the list is what the larger eigenvalue does to a vector.

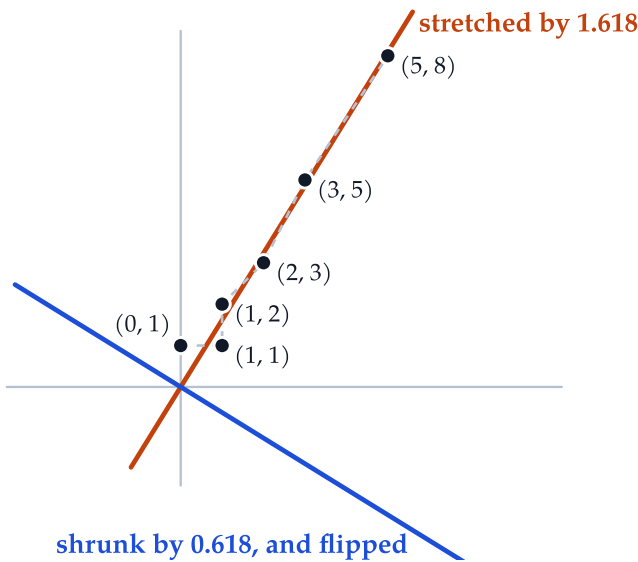


Figure 60.2. The two lines that the map from (x, y) to $(y, x + y)$ carries to themselves. Along the line of slope 1.618 it stretches by 1.618; along the line of slope -0.618 it shrinks by 0.618 and flips. The dots are $(0, 1)$, $(1, 1)$, $(1, 2)$, $(2, 3)$, $(3, 5)$ and $(5, 8)$: they hop from side to side of the first line, closing in on it. Assumes: five steps from $(0, 1)$, so six dots; the distance of the n -th dot from the first line is 0.618^n times a constant

The same reasoning ranks the pages of the web. Imagine a reader who, once a second, follows a link chosen at random from the page being read. Let the web have four pages: A links to B and C , B links to C , C links to A , and D links to A and C . If the chances of finding the reader at the four pages are p_A, p_B, p_C and p_D now, then a second later they are

$$p_C + p_D/2, p_A/2, p_A/2 + p_B + p_D/2, 0$$

which is a linear map applied to the list of chances. Start the reader anywhere, say with chance $1/4$ at each page, and apply the map again and again: the chances go to $0.375, 0.125, 0.5, 0$, then to $0.5, 0.1875, 0.3125, 0$, and they settle down to

$$0.4, 0.2, 0.4, 0$$

That limit is a list which the map leaves unchanged, an eigenvector with eigenvalue 1, and it can be found at once by solving $Tp = p$: $p_D = 0, p_B = p_A/2, p_C = p_A$, and the chances add up to 1. The map has the eigenvalue 1 because it keeps the total of the chances. The list Tp has the same total as p , so $Tp - p$ has total 0: the map $T - 1$ sends every list to a list of total 0. Those lists form a space of dimension 3 only, being the kernel of the map that takes a list of four numbers to their total. So the image of $T - 1$ falls short of the whole space, which has dimension 4, and by the theorem on dimensions its kernel is not 0. The other roots of $\det(T - \lambda)$, which is $\lambda(\lambda - 1)(\lambda^2 + \lambda + 1/2)$, are 0 and $(-1 \pm i)/2$, of size 0.707.

Eigenvectors with different eigenvalues are independent. Suppose not, and among all the dependences, combinations of them equal to 0 with no coefficient 0, take one with the fewest terms. It has at least two terms, since an eigenvector is not 0. Let λ be the eigenvalue of one of its terms and apply $T - \lambda$. Each term is multiplied by its own eigenvalue minus λ , which removes that one term and leaves every other with a coefficient that is still not 0. That is a shorter dependence, which cannot be. So the four eigenvectors, two of them with entries in the plane, are independent and span

the lists with entries in the plane. Write the starting list as a combination of the four. Each application of T multiplies each part by its eigenvalue: the part with eigenvalue 1 stays as it is, the part with eigenvalue 0 goes at once, and the other two shrink by the factor 0.707 each time, so that every part but the first dies away as the ψ part did.

The limit is the long-run share of the reader's time spent on each page, and it is a measure of importance that a page cannot award to itself: C matters because A and B send readers to it, and D , to which nobody links, does not matter at all, however many links it sends out. In 1998 Brin and Page ranked the pages of the real web by this eigenvector, for a map on a space of some tens of millions of dimensions, and founded a company on it.

* * *

Eigenvalues give a mechanical answer to a question that was settled before only in principle. The number $\sqrt{2} + \sqrt{3}$ lies in $Q(\sqrt{2}, \sqrt{3})$, a field of dimension 4 over the fractions, so a count of dimensions says that its powers from the 0-th to the fourth cannot all be independent, and it satisfies some equation of degree at most 4 with fractions as coefficients. But the count never writes that equation down, and eigenvalues do. Multiplication by $\gamma = \sqrt{2} + \sqrt{3}$ is a linear map of the field $Q(\sqrt{2}, \sqrt{3})$, whose basis is $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$. It sends 1 to $\sqrt{2} + \sqrt{3}$, it sends $\sqrt{2}$ to $2 + \sqrt{6}$, it sends $\sqrt{3}$ to $3 + \sqrt{6}$, and it sends $\sqrt{6}$ to $3\sqrt{2} + 2\sqrt{3}$. Write down the array and compute $\det(T - \lambda)$: it is

$$\lambda^4 - 10\lambda^2 + 1$$

The number γ is an eigenvalue of that map, taken on lists with real entries. Read a list x_1, x_2, x_3, x_4 as the real number $x_1 + x_2\sqrt{2} + x_3\sqrt{3} + x_4\sqrt{6}$. Reading is linear, and since column j of the array records γ times the j -th basis element, reading Tx gives γ times reading x . So every list that $T - \gamma$ produces reads as 0: reading $Tx - \gamma x$ gives γ times reading x , less γ times reading x . The reading

map sends the lists, a space of dimension 4, onto the real numbers, a space of dimension 1, so by the theorem on dimensions its kernel, the lists that read as 0, has dimension 3. The image of $T - \gamma$ lies inside that kernel, so it is not the whole space. By the same theorem the kernel of $T - \gamma$ is not 0, and so $\det(T - \gamma) = 0$. So γ satisfies the equation, which is the equation that squaring reaches directly: $\gamma^2 = 5 + 2\sqrt{6}$, so $(\gamma^2 - 5)^2 = 24$, so $\gamma^4 - 10\gamma^2 + 1 = 0$. The procedure is the same for any sum or product of algebraic numbers, and needs no ingenuity. If α satisfies an equation of degree m and β one of degree n , the mn products $\alpha^i \beta^j$, with i below m and j below n , can stand where the basis stood: multiplying one of them by $\alpha + \beta$, or by $\alpha\beta$, and using the two equations to bring the powers of α below m and those of β below n , gives a combination of them with fractions as coefficients. The reading argument used nothing more, with entries in the plane when the numbers are not real; it never needed the products to be independent.

What is the volume of the box with sides (1, 2, 3), (4, 5, 6) and (7, 8, 10), and what happens to it if the last entry is 9?

A quarter turn of the plane sends (x, y) to $(-y, x)$. What are its eigenvalues, and what does the answer say about lines that the quarter turn does not move?

What is the array of multiplication by $\sqrt[3]{2}$ on the field $Q(\sqrt[3]{2})$, with basis 1, $\sqrt[3]{2}$ and $\sqrt[3]{4}$, and which equation does the determinant give for the number $1 + \sqrt[3]{2} + \sqrt[3]{4}$?

A web has three pages: A links to B, B links to A and to C, and C links to A. What share of a random reader's time does each page get?

The eigenvalues of the map sending (x, y) to $(y, x + y)$ multiply to -1 and add to 1. What, in terms of the array of a map of the plane, are the product and the sum of its two eigenvalues, and why?

THE SMALLEST EQUATION

A number in a field of finite dimension satisfies many equations, since any multiple of one is another. Which of them belongs to the number, and what fixes its degree? Answering needs polynomials to be objects that can be divided.

A polynomial over F is a finite list of coefficients from F . Two lists are added slot by slot and multiplied by the rule that the coefficient of x^k in the product is the sum of the products of coefficients whose indices add to k , and the *degree* is the index of the last coefficient that is not 0. Polynomials can be divided with remainder, as whole numbers can. Given g and a nonzero m , there are q and r with $g = qm + r$ and r either 0 or of degree below that of m . Long division finds them: it subtracts a multiple of m that removes the leading term of g and repeats on a polynomial of smaller degree, and degrees cannot decrease forever. The pair is unique, since $qm + r = q'm + r'$ gives $(q - q')m = r' - r$. The right side is 0 or has degree below that of m , while a nonzero multiple of m has degree at least that of m . So both sides are 0, which gives $q = q'$ and then $r = r'$. Dividing $x^3 - 2$ by $x + 1$ gives the quotient $x^2 - x + 1$ and the remainder -3 , since $(x + 1)(x^2 - x + 1) = x^3 + 1$.

With division in hand, Euclid's algorithm runs on polynomials as it ran on whole numbers. Replace the pair g, m by the pair m, r , where r is the remainder of g on division by m , and repeat. The

degrees descend, so the process stops at a remainder of 0. The last remainder that is not 0 divides both g and m . Substituting back through the steps writes it as a combination $ug + vm$ with polynomial coefficients u and v . Every common divisor of g and m divides that combination. Made monic, with leading coefficient 1, it is the *greatest common divisor* of g and m .

A polynomial is not the function it computes. Over $\mathbb{Z}/2$ the polynomial $x^2 + x$ takes the value 0 at both of the two inputs, since $0 + 0 = 0$ and $1 + 1 = 0$, and yet its list of coefficients is not the zero list. So when a number is said to satisfy a polynomial equation, the polynomial meant is a list of coefficients, and the list must not be the zero list; the zero list would say nothing, since every number satisfies it. The count of dimensions supplies such a list: for a number α in a field of dimension n over F , the $n + 1$ powers $1, \alpha, \dots, \alpha^n$ are dependent, so some combination of them with coefficients not all 0 equals 0.

* * *

Among the nonzero polynomials over F that vanish at α , take one of smallest degree and divide it by its leading coefficient. Call the result m .

It divides every polynomial g that vanishes at α . Divide g by m : the remainder r satisfies $r(\alpha) = g(\alpha) - q(\alpha)m(\alpha) = 0$. If r were not the zero polynomial, it would be a nonzero polynomial vanishing at α of degree below that of m , and m was chosen with the smallest such degree. So r is the zero polynomial, and $g = qm$. In particular m is unique. Two such polynomials would divide each other, so they would have the same degree and each would be a constant multiple of the other; since both are monic, the constant is 1 and they are equal. It is the *minimal polynomial* of α over F .

It is irreducible over F : it is not a product of two polynomials over F of smaller degree. If it were, the product would vanish at α . A product of elements of a field is 0 only when a factor is, so one of the two factors would vanish at α . That factor would be a

nonzero polynomial of smaller degree than m vanishing at α , and m was chosen so that there is none.

If m has degree d , the powers $1, \alpha, \dots, \alpha^{d-1}$ form a basis over F of the smallest field containing F and α , which is written $F(\alpha)$. They are independent: a combination of them that equals 0, with coefficients not all 0, would be a nonzero polynomial of degree below d vanishing at α , and d is the smallest degree such a polynomial can have. Every polynomial in α is a combination of them, since dividing by m replaces a polynomial by one of degree below d with the same value at α . And the polynomials in α already form a field, so they are the whole of $F(\alpha)$. Sums, differences and products of polynomials in α are again polynomials in α , so only reciprocals need an argument. If $g(\alpha)$ is not 0, then m does not divide g . The greatest common divisor of g and m is a monic divisor of m , and because m is irreducible its only monic divisors are 1 and m itself; since m does not divide g , the greatest common divisor is 1. So $ug + vm = 1$ for some polynomials u and v , and at α , where $m(\alpha) = 0$, this reads $u(\alpha)g(\alpha) = 1$. The reciprocal of $g(\alpha)$ is $u(\alpha)$, again a polynomial in α . For $x^3 - 2$ and $x + 1$ the division above gives $3 = (x + 1)(x^2 - x + 1) - (x^3 - 2)$, and at $x = \sqrt[3]{2}$ the second term vanishes, so

$$1/(1 + \sqrt[3]{2}) = (1 - \sqrt[3]{2} + \sqrt[3]{4})/3$$

which is Euclid's algorithm computing a reciprocal that the count of dimensions only proved to exist.

What are the fractions a , b and c with $1/(2 + \sqrt[3]{4}) = a + b\sqrt[3]{2} + c\sqrt[3]{4}$?

So the dimension of $F(\alpha)$ over F is the degree of the minimal polynomial of α . The polynomial $x^3 - 2$ is irreducible over $\mathbb{Q}$, because a factorisation of a cubic has a factor of degree 1 and so a fraction root, which $x^3 - 2$ does not have; so it is the minimal polynomial of $\sqrt[3]{2}$, and the dimension 3 found by the descent is the degree 3 of the equation. The number $\gamma = \sqrt{2} + \sqrt{3}$ generates $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, of dimension 4, so its minimal polynomial has degree 4; and $\gamma^2 = 5 + 2\sqrt{6}$ gives $(\gamma^2 - 5)^2 = 24$, so $\gamma^4 - 10\gamma^2 + 1 = 0$. The minimal polynomial of γ divides that quartic, since the quartic vanishes

at γ , and both are monic of degree 4; so the quartic is the minimal polynomial, and so it is irreducible over Q .

	1	$\sqrt{2}$	$\sqrt{3}$	$\sqrt{6}$	
1	1	0	0	0	
γ	0	1	1	0	
γ^2	5	0	0	2	row 5 – 10·row 3 + row 1
γ^3	0	11	9	0	
γ^4	49	0	0	20	
=	0	0	0	0	so $\gamma^4 - 10\gamma^2 + 1 = 0$

Figure 61.1. The powers of $\gamma = \sqrt{2} + \sqrt{3}$ written in the basis $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$ of the four-dimensional field $Q(\sqrt{2}, \sqrt{3})$. Five rows in a space of four coordinates cannot be independent, and the dependence is visible: the fifth row minus ten times the third plus the first is the zero row, which says $\gamma^4 - 10\gamma^2 + 1 = 0$. A count forced the equation before any algebra found it. *Assumes:* that the four listed elements really are a basis, which is the tower law; each row comes from multiplying out

If α and β both satisfy polynomial equations over F , so do $\alpha + \beta$, $\alpha\beta$, $\alpha - \beta$ and, when β is not 0, α/β , and their equations never have to be found. The minimal polynomial of β over F still vanishes at β when its coefficients are read in the larger field $F(\alpha)$, so the minimal polynomial of β over $F(\alpha)$ divides it, and the degree of β over $F(\alpha)$ is at most its degree over F . That degree is the dimension of $F(\alpha, \beta)$ over $F(\alpha)$. Dimensions multiply along a tower, so $F(\alpha, \beta)$ has finite dimension over F . The sum, product, difference and quotient of α and β all lie in $F(\alpha, \beta)$, and by the count of dimensions every element of a field of finite dimension over F satisfies an equation over F . The algebraic numbers, which could be listed one by one, are closed under the four operations.

What is the minimal polynomial of $\sqrt{2} + \sqrt[3]{2}$ over $\mathbb{Q}$?

The polynomial $x^4 - 10x^2 + 1$ carries a warning about irreducibility. Over $\mathbb{Q}$ it cannot be factored. Over $\mathbb{Q}(\sqrt{2})$ it is $(x^2 - 2\sqrt{2}x - 1)(x^2 + 2\sqrt{2}x - 1)$, over $\mathbb{Q}(\sqrt{3})$ it is $(x^2 - 2\sqrt{3}x + 1)(x^2 + 2\sqrt{3}x + 1)$, and over $\mathbb{Q}(\sqrt{6})$ it is $(x^2 - 5 - 2\sqrt{6})(x^2 - 5 + 2\sqrt{6})$; each factorisation comes from writing the quartic as a difference of two squares, which can be done in three ways: $(x^2 - 1)^2 - 8x^2$, $(x^2 + 1)^2 - 12x^2$ and $(x^2 - 5)^2 - 24$, in that order.

Modulo every prime it can be factored. Modulo 2 it is $x^4 + 1$, which is $(x + 1)^4$. Modulo an odd prime p , at least one of 2, 3 and 6 is a square. The rule met for squares modulo an odd prime, that a non-square times a non-square is a square, settles it: if neither 2 nor 3 is a square modulo p , their product 6 is.

The prime $p = 3$ stands apart, because there 3 is 0 and is not among the nonzero remainders; but 0 is the square of 0, so 3 is a square in that case too.

Whichever of the three has a square root s modulo p supplies a factorisation from the corresponding difference of two squares. If $s^2 = 2$, for instance, then $8x^2 = (2sx)^2$, so $(x^2 - 1)^2 - 8x^2$ is $(x^2 - 1 - 2sx)(x^2 - 1 + 2sx)$, and the other two cases go the same way. Modulo 5, where 6 is 1 and $s = 1$, the third reads $(x^2 - 5 - 2)(x^2 - 5 + 2)$, which is $(x^2 + 3)(x^2 + 2)$. A polynomial can therefore be irreducible over the fractions while every reduction of it modulo a prime factors, and irreducibility over $\mathbb{Q}$ cannot always be detected by searching for a prime modulo which the polynomial stays irreducible.

Modulo 11, which of 2, 3 and 6 has a square root, and into which two quadratics does $x^4 - 10x^2 + 1$ factor there?

A prime can still prove irreducibility over the fractions, by another route. The first step is a fact about factorisations: when a monic polynomial with whole-number coefficients is a product of two monic polynomials with fraction coefficients, the two factors have whole-number coefficients after all. Suppose a monic polynomial f with whole-number coefficients is gh , where g and h are monic, nonconstant, and have fraction coefficients. Let d be the

smallest positive whole number for which dg has whole-number coefficients, and e the same for h . Then $(de)f = (dg)(eh)$, with whole-number coefficients on both sides. Suppose de exceeds 1, take a prime p that divides it, and reduce every coefficient modulo p . The left side becomes zero. The right side is a product of two polynomials over $\mathbb{Z}/p$. Since $\mathbb{Z}/p$ is a field, if neither factor were zero their leading coefficients would multiply to something nonzero, which would be the leading coefficient of the product, and the product would not be zero. So one factor reduces to zero, say dg , and then p divides every coefficient of dg , including its leading coefficient d , because g is monic. Then $(d/p)g$ has whole-number coefficients, contradicting the choice of d . So $de = 1$, and g and h have whole-number coefficients.

Now suppose that a prime p divides every coefficient of f except the leading 1, and that p^2 does not divide the constant term. A factorisation of f into two polynomials of smaller degree can be taken with both factors monic, since their leading coefficients multiply to the leading 1 of f and dividing each factor by its own leaves the product unchanged. If $f = gh$ with g and h monic and non-constant, then g and h have whole-number coefficients, by the fact proved above. Write n for the degree of f . Reducing every coefficient modulo p turns f into x^n , since p divides every coefficient but the leading 1, so the reductions of g and h multiply to x^n .

Two facts about polynomials over the field $\mathbb{Z}/p$ are needed. The first is that a monic polynomial is a product of monic irreducible ones: one that is not irreducible is a product of two of smaller degree, each made monic by dividing through, and degrees cannot fall for ever. The second is that an irreducible polynomial q that divides a product gh divides g or h . If q does not divide g , their greatest common divisor is 1, because q has no divisors but constants and constant multiples of itself. So $uq + vg = 1$ for some polynomials u and v , and $h = uqh + vgh$ is divisible by q , since q divides each of the two terms.

So every monic irreducible factor of the reduction of g divides x^n , which is a product of copies of x . By the second fact, applied

to one copy of x after another, it divides x ; being monic and not constant, it is x . The reduction of g is therefore a power of x , and its degree is at least 1, because g is monic and keeps its degree when reduced. The same holds for h . A power of x of degree at least 1 has constant term 0, so p divides the constant terms of g and of h , and p^2 divides their product, which is the constant term of f . That contradicts the hypothesis, so f cannot be factored.

This is Eisenstein's criterion. For $x^5 - 2$ and the prime 2, every coefficient below the leading one is 0 or -2 , and 4 does not divide -2 ; so $x^5 - 2$ is irreducible over $\mathbb{Q}$, and the fifth root of 2 generates a field of dimension 5. A number that can be built with straightedge and compass generates a field whose dimension over $\mathbb{Q}$ divides a power of 2. Since 5 divides no power of 2, the fifth root of 2 cannot be built, a case the descent on cubics never reached.

For which whole numbers n can the real n -th root of 2 be built?

* * *

The minimal polynomial belongs to a number that already lives in some field. The plane of pairs was built to supply a root of $x^2 + 1$ where none existed. Can a root be manufactured in the same way for any nonconstant irreducible polynomial p over any field F ?

It can, by a collapse of the kind that made the clocks out of the integers. Call two polynomials over F congruent when p divides their difference. Addition and multiplication of classes are defined on chosen members, and the choice does not matter: if g and g' differ by a multiple of p , and h and h' do, then $g + h$ and $g' + h'$ differ by a multiple of p , and so do gh and $g'h'$, since $gh - g'h' = g(h - h') + (g - g')h'$. Every class contains exactly one polynomial that is 0 or of degree below that of p , namely the remainder that any of its members leaves on division by p . So the classes can be taken to be the remainders themselves: two remainders are added or multiplied as polynomials, and the result is then replaced by its own remainder on division by p . If p has degree d , every remainder is a combination of $1, x, \dots, x^{d-1}$ in exactly one way, so these form

a basis over F . The class of x , call it θ , is a root of p . Sums and products of classes are computed on members, so putting θ into p , which is nothing but sums and products of θ and elements of F , gives the class of the polynomial p itself. And p differs from 0 by a multiple of p , so the class of p is the class of 0.

The rules of a field other than reciprocals hold for the classes because they hold for polynomials and the operations on classes are computed on members. Reciprocals are where the irreducibility of p is used, as primality was used for the clocks. If p does not divide g , then, as for reciprocals in $F(\alpha)$, irreducibility makes the greatest common divisor of g and p equal to 1, so $ug + vp = 1$ for some u and v , and the class of u is the reciprocal of the class of g . The classes form a field, written $F[x]/(p)$, of dimension d over F , in which p has a root.

polynomials over $\mathbb{Z}/2$		the classes multiplied, then reduced			
x^2	$\longrightarrow$	$x + 1$			
$x^2 + 1$	$\longrightarrow$	x			
x^3	$\longrightarrow$	1			
$x^3 + x$	$\longrightarrow$	$x + 1$			
$x^4 + x$	$\longrightarrow$	0			
remainder mod $x^2 + x + 1$					

Figure 61.2. The polynomials over $\mathbb{Z}/2$ sorted into classes by their remainder on division by $x^2 + x + 1$, which has no root and is irreducible. Four remainders, so four classes; the class of x satisfies the polynomial by construction, since x^2 reduces to $x + 1$; and the multiplication table, with every product reduced, has no zero in it except where a factor is zero, so every nonzero class has a reciprocal. This is a field with four elements, although the clock with four hours is not a field. *Assumes:* that the class operations do not depend on the chosen member, which is the check made in the text; the table is computed on the remainders and the check is what licenses that

Over $\mathbb{Z}/2$ the polynomial $x^2 + x + 1$ has no root, since it takes the value 1 at 0 and at 1, and a quadratic with no root has no factor of degree 1, so it is irreducible. Its classes are the four remainders 0, 1, x and $x + 1$, with x^2 reduced to $x + 1$. Writing w for the class of x , the products are $w \cdot w = w + 1$, $w \cdot (w + 1) = 1$ and $(w + 1) \cdot (w + 1) = w$, and every nonzero element has a reciprocal. So there is a field with four elements, written F_4 . It is not the clock with four hours, which is not a field, since $2 \cdot 2 = 0$ on it. Over $\mathbb{Z}/3$ the polynomial $x^2 + 1$ has no root, since the squares there are 0, 1 and 1, and its classes form a field with nine elements, which is not the clock with nine hours, where $3 \cdot 3 = 0$. Over $\mathbb{Z}/2$ the cubic $x^3 + x + 1$ has no root, since it takes the value 1 at 0 and at 1, so it has no factor of degree 1, and none of degree 2 either, since a factor of degree 2 of a cubic leaves a factor of degree 1 beside it. It is irreducible, and it gives a field with eight elements.

The sizes 4, 9 and 8 are powers of primes, and every finite field has a power of a prime for its size. In a finite field the sums $1, 1 + 1, 1 + 1 + 1, \dots$ cannot all be different, so some sum of n ones is 0; take the smallest such n . It is prime, since if $n = ab$ with a and b smaller than n , the sum of a ones times the sum of b ones is the sum of n ones, which is 0, and one of the two factors would have to be 0, although each is a sum of fewer than n ones and n was the smallest. Write p for this prime n . The sums of fewer than p ones then form a copy of $\mathbb{Z}/p$ inside the field, so the field is a space over $\mathbb{Z}/p$, of some finite dimension d , and a space of dimension d over $\mathbb{Z}/p$ has p^d elements, one choice among p for each coordinate. No field has six elements, since 6 is not a power of a prime.

Over the reals the polynomial $x^2 + 1$ has no root, and the classes of remainders $a + bx$ multiply by the rule $(a + bx)(c + dx) = ac + (ad + bc)x + bdx^2$, which reduces to $(ac - bd) + (ad + bc)x$. That is the rule for multiplying pairs: the plane of pairs was $\mathbb{R}[x]/(x^2 + 1)$. Over the fractions $x^3 - 2$ is irreducible, and $\mathbb{Q}[x]/(x^3 - 2)$ holds a root of it, built with no real numbers at all.

WHAT THE COEFFICIENTS KNOW

The roots of $x^3 - 2x - 5$ are one real number, 2.0946, and a mirror pair in the plane, and Cardano's formula reaches them only through cube roots of numbers that have to be dug out of the coefficients first. Yet the sum of the three is exactly 0, their product is exactly 5, and the sum of their squares is exactly 4, and these can be read off the equation without finding a single root. Which questions about the roots can the coefficients answer?

Multiply out a polynomial written by its roots. For three roots,

$$(x - r_1)(x - r_2)(x - r_3) = x^3 - (r_1 + r_2 + r_3)x^2 + (r_1r_2 + r_1r_3 + r_2r_3)x - r_1r_2r_3$$

and for n roots the coefficient of x^{n-k} is, apart from the sign $(-1)^k$, the sum of all products of k different roots. Call that sum e_k : e_1 is the sum of the roots, e_2 the sum of the products of pairs, and e_n the product of them all. These are the *elementary symmetric polynomials*. When the leading coefficient is 1, the coefficients of an equation are, up to sign, the elementary symmetric polynomials of its roots. For any other leading coefficient they are those polynomials multiplied by it. Viète knew this for positive roots in 1591, and Girard stated

it in general in 1629. For $x^3 - 2x - 5$ they are $e_1 = 0$, $e_2 = -2$ and $e_3 = 5$.

An expression in the roots is *symmetric* when it does not change however the roots are rearranged. The sum of the squares of the roots is symmetric, and it can be written in terms of the e 's: squaring e_1 gives every square once and every product of a pair twice, so

$$r_1^2 + r_2^2 + \dots + r_n^2 = e_1^2 - 2e_2$$

which for the cubic above is $0 + 4$. The coefficients cannot tell one root from another, since they do not change when the roots are rearranged, so they can answer only symmetric questions. The theorem is that they can answer all of those.

Every symmetric polynomial in $r_1, \dots, r_n$, with whole-number coefficients, is a polynomial in $e_1, \dots, e_n$ with whole-number coefficients.

The proof is a descent, and it is also a procedure. Order the products of powers of the roots as words are ordered in a dictionary: compare the powers of r_1 first, the higher power coming first; if they agree compare the powers of r_2 ; and so on. In a symmetric polynomial the first term in this order, say with powers $a_1, a_2, \dots, a_n$, has $a_1 \geq a_2 \geq \dots \geq a_n$. If some a_i were smaller than a_{i+1} , exchanging r_i and r_{i+1} would turn that term into one with the larger power in the earlier place; that term is also in the polynomial, because the polynomial is symmetric, and it comes earlier in the dictionary, which is impossible for the first term.

Now take e_1 to the power $a_1 - a_2$, times e_2 to the power $a_2 - a_3$, and so on, ending with e_n to the power a_n . That product has the same first term. The first term of a product is the product of the first terms, and the first term of e_k is $r_1 r_2 \dots r_k$. The root r_1 appears in every one of $e_1, e_2, \dots, e_n$, so its power in the product of first terms is $(a_1 - a_2) + (a_2 - a_3) + \dots + a_n = a_1$; the root r_2 appears in all but e_1 , so its power is a_2 ; and so on down to r_n , which appears only in e_n and gets power a_n . Multiply that product by the coefficient of the first term of the symmetric polynomial and subtract it. What

is left is symmetric, and its first term comes later in the dictionary. The subtracted product has the same total degree as the term it cancels, so no term of higher degree than the polynomial's ever appears, and only finitely many terms of at most that degree exist; so after finitely many subtractions nothing is left, and the subtracted products, added up, are the polynomial.

Take the sum of the cubes of three roots. Its first term is r_1^3 , with powers 3, 0, 0, so subtract e_1^3 . What is left is -3 times the sum of all $r_i^2 r_j$, less $6r_1 r_2 r_3$. Its first term is $r_1^2 r_2$, with powers 2, 1, 0, so add $3e_1 e_2$. The product $e_1 e_2$ contains each $r_i^2 r_j$ once and $r_1 r_2 r_3$ three times, so adding $3e_1 e_2$ removes the first part and turns $-6r_1 r_2 r_3$ into $3r_1 r_2 r_3$, which is $3e_3$. So

$$r_1^3 + r_2^3 + r_3^3 = e_1^3 - 3e_1 e_2 + 3e_3$$

and for the roots of $x^3 - 2x - 5$ the sum of the cubes is 15.

* * *

The sums of powers, $p_k = r_1^k + \dots + r_n^k$, can be reached faster, by a chain of identities that Newton found in 1666:

$$p_k - e_1 p_{k-1} + e_2 p_{k-2} - \dots \pm e_{k-1} p_1 \mp k \cdot e_k = 0$$

where e_j is taken as 0 when j exceeds n . When k is n or more, this is the equation itself in disguise. Each root satisfies $r^n - e_1 r^{n-1} + e_2 r^{n-2} - \dots = 0$; multiply by r^{k-n} and add up over the roots. For $k = n$ the last term, e_n times the sum of the 0-th powers, is $n \cdot e_n$. When k is below n , write the left side out in the roots $r_1, \dots, r_n$; the task is to show that every term cancels. Every term has total degree k , so it involves at most k of the roots. Pick any term, and put equal to 0 all the roots except k of them, keeping those the term uses. The p 's and e 's become the p 's and e 's of the k roots that remain, so the left side becomes the left side of the identity for k roots and the k -th powers, where the power equals the number of roots; that is the case just proved, and it is 0. The substitution deletes only terms

that contain a root set to 0, so the chosen term keeps its coefficient, and that coefficient must therefore be 0. Every term can be treated this way, so the identity holds.

For $x^3 - 2x - 5$ the chain gives $p_1 = 0$, then $p_2 = 0 \cdot 0 - 2(-2) = 4$, then $p_3 = 0 - 0 + 3 \cdot 5 = 15$, then $p_4 = e_1 p_3 - e_2 p_2 + e_3 p_1 = 8$, and so on forever, each from the three before. Every sum of powers of the roots is a whole number, though two of the roots are not even real.

That explains a curiosity. The roots of $x^2 - x - 1$ are $\varphi = 1.618$ and $\psi = -0.618$, with $e_1 = 1$ and $e_2 = -1$, so the sums of their powers obey $p_k = p_{k-1} + p_{k-2}$ and run 1, 3, 4, 7, 11, 18, 29, 47, 76, 123, They are whole numbers, and the powers of ψ shrink to nothing. Since $\varphi^k = p_k - \psi^k$, the powers of φ come closer and closer to whole numbers: φ^{10} is 122.9919 and φ^{20} is 15126.99993. The same happens for $1 + \sqrt{2}$, whose partner $1 - \sqrt{2}$ is -0.414 , and for any root of a monic equation with whole-number coefficients whose other roots all have size below 1: the leading coefficient 1 is what makes the e 's, and with them every p_k , whole numbers.

* * *

One symmetric expression matters more than the rest. For an equation with roots $r_1, \dots, r_n$, the product of the squares of all the differences of two roots,

$$\Delta = \text{the product of } (r_i - r_j)^2 \text{ over all pairs}$$

is the *discriminant*. It is symmetric, so it is a polynomial in the coefficients, and it is 0 exactly when two roots coincide. For $x^2 + bx + c$ the difference of the roots is $\sqrt{(b^2 - 4c)}$, and Δ is $b^2 - 4c$. For the cubic $x^3 + px + q$ there is a shorter route through the derivative. Write the cubic as $(x - r_1)(x - r_2)(x - r_3)$. By the rule for products its derivative is $(x - r_2)(x - r_3) + (x - r_1)(x - r_3) + (x - r_1)(x - r_2)$, one term for each factor left out. At $x = r_1$ the last two terms contain the factor $x - r_1$ and are 0, so the derivative there is $(r_1 - r_2)(r_1 - r_3)$; likewise at r_2 and r_3 . So the product of the three

values of the derivative at the roots is $(r_1 - r_2)(r_1 - r_3)(r_2 - r_1)(r_2 - r_3)(r_3 - r_1)(r_3 - r_2)$. Each pair of roots appears twice, as $(r_i - r_j)(r_j - r_i)$, which is $-(r_i - r_j)^2$. There are three pairs, so the product is $(-1)^3$ times the product of the three squared differences: it is $-\Delta$.

But the derivative is also $3x^2 + p$, so the same product is the product of the three numbers $p + 3r_i^2$, which is $p^3 + 3p^2(r_1^2 + r_2^2 + r_3^2) + 9p(r_1^2r_2^2 + r_1^2r_3^2 + r_2^2r_3^2) + 27r_1^2r_2^2r_3^2$. Here $e_1 = 0$, $e_2 = p$ and $e_3 = -q$. The first sum is $e_1^2 - 2e_2 = -2p$. Squaring e_2 gives every product $r_i^2r_j^2$ once and every product $r_1r_2r_3 \cdot r_k$ twice. The second kind add up to $2e_3e_1$, so the second sum is $e_2^2 - 2e_1e_3$, and since $e_1 = 0$ it is e_2^2 . The third is $e_3^2 = q^2$. So the product is $p^3 - 6p^3 + 9p^3 + 27q^2$, which is $4p^3 + 27q^2$. That product is $-\Delta$, and

$$\Delta = -4p^3 - 27q^2$$

For a cubic with real coefficients the sign of Δ says how the roots lie. If all three are real and different, every difference is real and not 0, and Δ , a product of their squares, is positive. If one is real, say r , and two are a mirror pair $a \pm bi$, the difference of the pair is $2bi$, whose square is $-4b^2$, a negative number. The other two differences, $r - (a + bi)$ and $r - (a - bi)$, are mirror images of each other, and a number other than 0 times its mirror image is positive, the square of its size. So the product of those two differences is positive, and its square is positive too. Δ is a negative number times a positive one: Δ is negative. For $x^3 - 2x - 5$, Δ is $32 - 675 = -643$, and there is one real root. For $x^3 - 3x + 1$, which is satisfied by twice the cosine of 40 degrees, Δ is $108 - 27 = 81$, and there are three.

The square root of the discriminant, δ = the product of the differences $r_i - r_j$ with i below j , is not symmetric. Exchanging two roots changes its sign. Exchange r_a and r_b , with a below b , and sort the factors of δ by what happens to them. Factors that involve neither r_a nor r_b do not change. The factor $r_a - r_b$ becomes $r_b - r_a$, reversed. For each root r_c with c between a and b , the factors $r_a - r_c$ and $r_c - r_b$ become $r_b - r_c$ and $r_c - r_a$, both reversed. Every

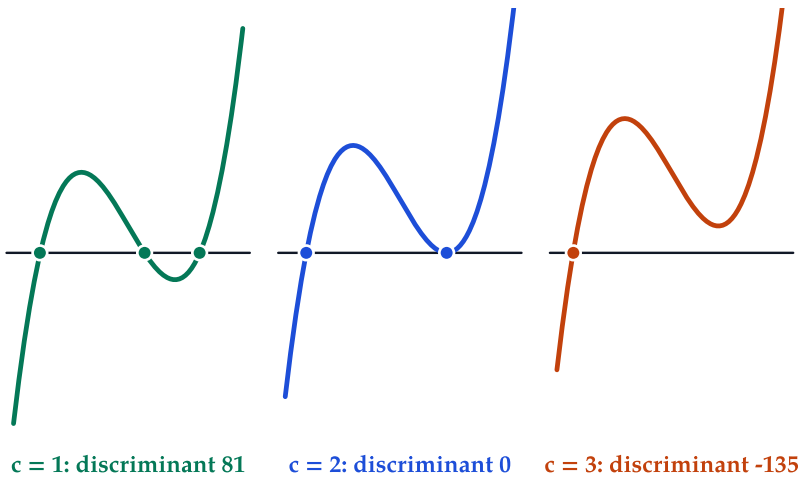


Figure 62.1. The cubic $x^3 - 3x + c$ for $c = 1, 2$ and 3 , whose discriminant $108 - 27c^2$ is $81, 0$ and -135 . With a positive discriminant the curve crosses the axis three times. At 0 two of the crossings have merged into a touching. With a negative discriminant one crossing is left, and the other two roots are a mirror pair off the line. *Assumes:* cubics with no square term and real coefficients; for equations of higher degree the sign of the discriminant says less

other factor involving r_a or r_b becomes another factor of δ with its subscripts still in order. So the number of reversed factors is 1 plus twice the number of roots between, an odd number, and δ changes sign. So every even rearrangement of the roots leaves δ alone and every odd one reverses it: δ is the simplest expression that the even rearrangements keep and the others do not. For $x^3 - 3x + 1$ it is 9 or -9 , depending on how the roots are numbered, a whole number either way, and that will turn out to say that the symmetries of this equation are the three rotations of its roots and not all six rearrangements. For $x^3 - 2x - 5$ it is $\sqrt{-643}$, which is not even real, and the equation has all six.

The theorem also gives equations for new numbers. If α is a root of f and β a root of g , both with fraction coefficients, consider the product of all the factors $x - \alpha_i - \beta_j$, over all roots α_i of f and all

roots β_j of g . Its coefficients are symmetric in the α_i and symmetric in the β_j . Viewed as polynomials in the α_i , their coefficients are polynomials in the β_j rather than whole numbers.

The descent that proved the theorem used nothing about coefficients except that they can be added, subtracted and multiplied, so it runs unchanged. At each step it subtracts a product of e 's times the coefficient of the first term, and that coefficient is symmetric in the β_j : rearranging the β_j leaves the whole polynomial as it was, and so leaves each of its coefficients as it was. So each coefficient of the product is a polynomial in the e 's of the α_i whose coefficients are symmetric polynomials in the β_j with whole-number coefficients, and the theorem applied a second time makes those polynomials in the e 's of the β_j . The coefficients of the product are therefore polynomials, with whole-number coefficients, in the e 's of the α_i and of the β_j , and those are, up to sign, the coefficients of f and g divided by their leading coefficients: fractions.

So $\alpha + \beta$ is a root of a polynomial with fraction coefficients, of degree the product of the two degrees, which can be computed without knowing any root. The same goes for $\alpha\beta$. This is the third proof that sums and products of algebraic numbers are algebraic; the first counted dimensions, and the second took a determinant. In the same way, the polynomial whose roots are the squares of the roots of $x^3 - 2x - 5$ is $y^3 - 4y^2 + 4y - 25$: its coefficients are, up to sign, the e 's of the squares: their sum is $p_2 = 4$, the sum of their products in pairs is $e_2^2 - 2e_1e_3 = 4$, and their product is $e_3^2 = 25$, with e_1, e_2, e_3 those of the original roots.

For the roots of $x^3 - 2x - 5$, what are the sum of the reciprocals, the sum of the fifth powers, and the sum of the squares of the three differences?

Write $r_1^4 + r_2^4 + r_3^4$ in terms of e_1, e_2 and e_3 , by the procedure and then by the chain of identities.

Which of the cubics $x^3 - 4x + 1$, $x^3 + x + 1$ and $x^3 - 7x + 7$ have three real roots, and for which is δ a whole number?

To which whole number is $(2 + \sqrt{3})^{12}$ close, and how close?

What is the discriminant of $x^4 + 1$, found from its four roots, and

what is the polynomial whose roots are the squares of the roots of $x^3 + x + 1$?

THE CIRCLE CANNOT BE SQUARED

A square with the area of a circle of radius 1 has side $\sqrt{\pi}$, and the circle can be squared with ruler and compasses only if $\sqrt{\pi}$ is constructible. Every constructible number is algebraic, as the count of dimensions showed, and if $\sqrt{\pi}$ were algebraic so would be its square. So the oldest of the three problems comes down to one question: does π satisfy a polynomial equation with whole-number coefficients? Hermite's proof that e satisfies no such equation was carried through in full. The same method was said to reach π , but the argument stopped at one step, and that step needed the theory of expressions unchanged by rearranging roots. That theory is now in hand, and the proof can be finished. It is Lindemann's, from 1882, in the shape Hilbert gave it in 1893, with one change: where that proof integrated, the tail of the series for E does the work, so no integrals appear.

First, E in the plane. For a number z of the plane let $E(z)$ be the sum of the series $1 + z + z^2/2! + z^3/3! + \dots$, which converges absolutely, the sizes of its terms being the terms of the series for $E(|z|)$. The rule for multiplying two absolutely convergent series was proved using sizes alone, so it holds in the plane as it did on the line. It gives $E(z) \cdot E(w) = E(z + w)$. Collecting the products

of degree n gives the sum of $z^k w^{n-k} / (k!(n-k)!)$ over k , and by the binomial theorem this is $(z + w)^n / n!$, the term of degree n in $E(z + w)$. For $z = it$ the series was shown to give $\cos t + i \sin t$, so $E(i\pi) = -1$. And the tail of the series has a bound that will be needed: $E(z)$ less its terms up to $z^t / t!$ has size at most the sum of $|z|^j / j!$ over j above t . That difference is the infinite sum of the terms beyond $z^t / t!$. For any finite stretch of those terms, the size of the sum is at most the sum of the sizes, which is at most the whole sum of $|z|^j / j!$ over j above t . The finite stretches approach the difference, so the same bound holds for it in the limit.

* * *

Suppose π satisfied an equation with whole-number coefficients. Then so would $i\pi$. If P is a polynomial with whole-number coefficients and $P(\pi) = 0$, the product $P(ix) \cdot P(-ix)$, written down when the transcendence of e was proved, has whole-number coefficients, and at $x = i\pi$ it equals $P(-\pi) \cdot P(\pi)$, which is 0. Take such an equation for $i\pi$, of degree d , with leading coefficient ℓ , a positive whole number. It has d roots in the plane, because every polynomial with coefficients in the plane splits there into linear factors; a root is counted as often as its factor occurs. Call the roots $\theta_1 = i\pi, \theta_2, \dots, \theta_d$. The numbers $\lambda_i = \ell\theta_i$ are the roots of the equation obtained by writing x/ℓ for x and multiplying through by the $(d-1)$ -th power of ℓ . That equation is monic with whole-number coefficients: the leading term $\ell(x/\ell)^d$ becomes x^d , and a lower term $a_k x^k$ becomes $a_k \ell^{d-1-k} x^k$, a whole number times x^k because k is at most $d-1$. The coefficients of a monic polynomial are, up to sign, the elementary symmetric polynomials of its roots. So the elementary symmetric polynomials of $\lambda_1, \dots, \lambda_d$ are whole numbers, and by the theorem on symmetric expressions,

every symmetric polynomial in $\lambda_1, \dots, \lambda_d$ with whole-number coefficients is a whole number.

Since $E(\theta_1) = -1$, the product $(1 + E(\theta_1))(1 + E(\theta_2)) \dots (1 +$

$E(\theta_d)$ is 0. Multiply it out. There are 2^d terms, one for each subset S of the roots, and the term for S is the product of $E(\theta_i)$ over the roots in S , which is E of their sum, by the law for E in the plane; the empty subset gives $E(0) = 1$. Some of these sums may be 0. The empty sum is, and others can be: for the equation $x^3 - 2 = 0$ the sum of all three roots is 0, since the equation has no square term. Let q be the number of subsets whose sum is 0; q is at least 1. Let $\beta_1, \dots, \beta_m$ be the sums that are not 0, one for each of the remaining subsets, with a value listed as often as it occurs. Not every sum is 0, since θ_1 alone is not, so m is at least 1. Each of the q subsets with sum 0 contributes $E(0) = 1$ to the multiplied-out product, and the other subsets contribute $E(\beta_1), \dots, E(\beta_m)$, so

$$q + E(\beta_1) + E(\beta_2) + \dots + E(\beta_m) = 0$$

with q a positive whole number.

The β 's need not be whole numbers, or even real, but they are the roots of an equation with whole-number coefficients that can be written down. Consider the polynomial whose roots are the 2^d sums of subsets of $\lambda_1, \dots, \lambda_d$, the product of x less each sum. A rearrangement of the λ 's rearranges the subsets among themselves, so it rearranges the factors and leaves the product alone: the coefficients of this polynomial are symmetric polynomials in the λ 's, with whole-number coefficients, and so they are whole numbers. The sums that are 0 contribute a factor x^q , and what is left is

$$H(x) = (x - \gamma_1)(x - \gamma_2) \cdots (x - \gamma_m), \text{ where } \gamma_k = \ell\beta_k$$

a monic polynomial of degree m whose coefficients are whole numbers, namely the coefficients of the full product shifted down q places, since the full product is x^q times $H(x)$, and whose constant term $b = H(0)$ is not 0, being $\pm$ the product of the γ 's. For $x^3 - 2$ the six nonzero sums are the three roots and their negatives, q is 2, and H is $x^6 - 4$. Put $\psi(x) = H(\ell x)$. It has whole-number coefficients. Its roots are the β 's, since $\psi(\beta_k) = H(\ell\beta_k) = H(\gamma_k) = 0$. It is ℓ^m times the product of the factors $x - \beta_k$, and $\psi(0) = b$.

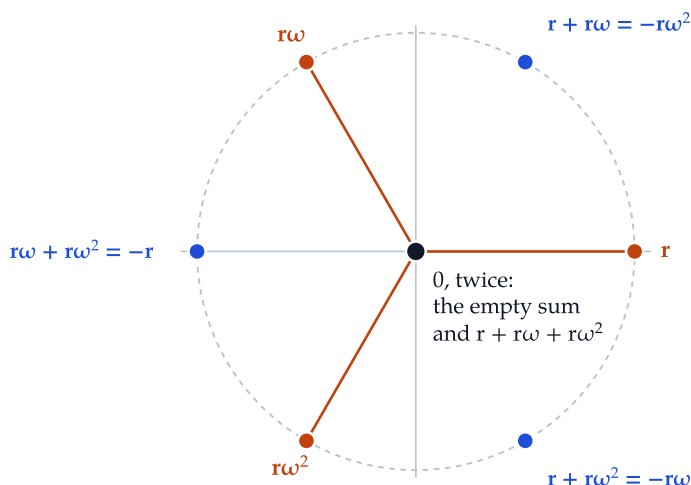


Figure 63.1. The three roots of $x^3 - 2$, at radius $\sqrt[3]{2}$ in the plane, and the eight sums of subsets of them. The empty subset and the whole set both add up to 0, so q is 2; the six other sums are the roots and their negatives, the roots of $x^6 - 4$. Were the equation of $i\pi$ this cubic, the proof would work with these six numbers. *Assumes:* the roots drawn to scale; the same picture for the equation of $i\pi$, whatever it is, has 2^d sums for d roots, and the proof needs only that the nonzero ones, multiplied by the leading coefficient ℓ , are the roots of a monic polynomial with whole-number coefficients

* * *

Now Hermite's design. Let p be a prime, to be chosen large at the end, and let

$$f(x) = x^{p-1} \cdot \psi(x)^p / (p-1)!$$

a polynomial of degree $p - 1 + mp$, and let F be f plus all its derivatives, $f + f' + f'' + \dots$, a sum that ends, because every derivative of a polynomial beyond its degree is 0. In the proof for e , the identity $E(k)F(0) - F(k) = E(k)$ times an integral supplied the quantity to be estimated. Here is the same identity without the integral. Write $f(x)$ as the sum of its terms $c_t x^t$. The derivatives of x^t are $t \cdot x^{t-1}$,

$t(t-1) \cdot x^{t-2}$, and so on down to $t!$; the one of order s is $t!/(t-s)!$ times x^{t-s} . So x^t and all its derivatives, added up, give $t!$ times $1 + x + x^2/2! + \dots + x^t/t!$, so $F(\alpha)$ is the sum over t of $c_t \cdot t!$ times the series for $E(\alpha)$ cut off after the term in α^t , and $F(0)$ is the sum over t of $c_t \cdot t!$. Therefore

$$E(\alpha)F(0) - F(\alpha) = \text{the sum over } t \text{ of } c_t \cdot t! \cdot (\text{the tail of the series for } E(\alpha) \text{ beyond the term in } \alpha^t)$$

for every α in the plane. The tail has size at most the sum over j above t of $|\alpha|^j/j!$. Multiply by $t!$ and write $j = t + s$, with s at least 1. Then $j!$ is $t!$ times $(t+1)(t+2) \dots (t+s)$, and each of those s factors is at least the matching factor of $1 \cdot 2 \dots s$, so $t!/j!$ is at most $1/s!$; and $|\alpha|^j$ is $|\alpha|^t$ times $|\alpha|^s$. So $t!$ times the tail has size at most $|\alpha|^t$ times the sum of $|\alpha|^s/s!$ over s at least 1, which is below $|\alpha|^t \cdot E(|\alpha|)$. Adding up,

$$E(\alpha)F(0) - F(\alpha) \text{ has size at most } E(|\alpha|) \text{ times the value at } |\alpha| \text{ of } f \text{ with every coefficient replaced by its size.}$$

Take this for $\alpha = \beta_k$ and add over k , using the relation $q + E(\beta_1) + \dots + E(\beta_m) = 0$ to replace the sum of the $E(\beta_k)$ by $-q$:

$$-q \cdot F(0) - (F(\beta_1) + \dots + F(\beta_m)) = \text{the sum over } k \text{ of } (E(\beta_k)F(0) - F(\beta_k))$$

The left side will be a whole number that is not 0, and the right side will be smaller than 1.

* * *

The left side. First $F(0)$. The polynomial $(p-1)! \cdot f(x) = x^{p-1}\psi(x)^p$ has whole-number coefficients, its lowest term is $b^p x^{p-1}$, and the derivative of order j at 0 is $j!$ times the coefficient of x^j . So the derivatives of f at 0 of order below $p-1$ are 0; the derivative of order $p-1$ is b^p ; and each derivative of order j at least p is a whole number times $j!/(p-1)!$, which is $p(p+1) \dots j$, a multiple of p . So

$F(0)$ is a whole number, and on division by p it leaves the same remainder as b^p .

Next the sum of the $F(\beta_k)$, the step that needs the symmetric expressions. Look at f near β_k by writing $x = \beta_k + y$, so that $\ell x = \gamma_k + \ell y$. Then x^{p-1} is $(\gamma_k + \ell y)^{p-1}$ divided by the $(p-1)$ -th power of ℓ , and $\psi(x)$, which is the product of the factors $\ell x - \gamma_i$, becomes the product of the factors $\gamma_k - \gamma_i + \ell y$, of which the one with $i = k$ is ℓy . Raising to the p -th power turns that factor into $\ell^p y^p$, and ℓ^p divided by the $(p-1)$ -th power of ℓ leaves a single ℓ . So

$$(p-1)! \cdot f(\beta_k + y) = \ell \cdot y^p \cdot (\gamma_k + \ell y)^{p-1}$$

times the product over i other than k of $(\gamma_k - \gamma_i + \ell y)^p$

and the right side is a polynomial in y whose coefficients are polynomials in ℓ and $\gamma_1, \dots, \gamma_m$ with whole-number coefficients. The derivative of f of order j at β_k is $j!$ times the coefficient of y^j , divided by $(p-1)!$. Every coefficient below y^p is 0, so the derivatives of order below p vanish at each β_k ; that is the factor $(x - \beta_k)^p$ in f at work. For j at least p , $j!/(p-1)!$ is again a multiple of p , and what multiplies it is the coefficient of y^j , a polynomial in the γ 's with whole-number coefficients. Now add over k . Rearranging $\gamma_1, \dots, \gamma_m$ rearranges the m versions of the display above, one for each k , among themselves, and leaves their sum alone. So the sum over k of the coefficients of y^j is a symmetric polynomial in the γ 's with whole-number coefficients. By the theorem it is a polynomial with whole-number coefficients in the elementary symmetric polynomials of the γ 's. Those are, up to sign, the coefficients of H , which are whole numbers. So for each j at least p the sum over k of the j -th derivatives of f at the β_k is a whole number divisible by p , and for j below p it is 0. Adding over j ,

$$F(\beta_1) + \dots + F(\beta_m) \text{ is a whole number divisible by } p.$$

No single $F(\beta_k)$ need be a whole number, or real; only the sum is. The left side, $-qF(0)$ less that sum, is therefore a whole number which leaves on division by p the remainder of $-q \cdot b^p$. Choose p larger than q and larger than the size of b . A prime larger than q

cannot divide the positive whole number q , and a prime larger than the size of b cannot divide b , which is not 0. A prime that divides a product divides one of its factors, so p divides neither b^p nor $q \cdot b^p$, and the left side, whose remainder is that of $-q \cdot b^p$, is not divisible by p . In particular it is not 0, and being a whole number it has size at least 1.

The right side. Let B be the largest of the sizes $|\beta_k|$, and let C be the value at B of ψ with each coefficient replaced by its size. Replace every coefficient of f by its size. Each coefficient of the result is at most the matching coefficient of x^{p-1} times the p -th power of ψ with its coefficients so replaced, divided by $(p-1)!$. The reason is that each coefficient of a product is a sum of products of coefficients, and the size of such a sum is at most the same sum taken with sizes. At $x = B$ every power of x is at least 0, so the same comparison holds for the values there, and the second value is $B^{p-1}C^p / (p-1)!$. Each of the m terms on the right has size at most $E(B)$ times that, since $|\beta_k|$ is at most B , and for x at least 0 neither E nor a polynomial whose coefficients are at least 0 decreases as x increases. So the whole right side has size at most

$$m \cdot E(B) \cdot B^{p-1} \cdot C^p / (p-1)!$$

in which B , C , m and $E(B)$ are fixed by the supposed equation and only p moves. Powers lose to factorials: each step from p to $p+1$ multiplies the bound by BC/p , which is below a half once p exceeds $2BC$. From then on each step at least halves the bound, so it falls below 1 and stays there. So for a large enough prime p , and there are primes beyond every bound, the right side has size below 1 while the left side, which equals it, has size at least 1. The supposition fails:

π satisfies no polynomial equation with whole-number coefficients.

So $\sqrt{\pi}$ does not either, its square being π , and the circle cannot be squared. The problem is at least as old as Anaxagoras, who is said to have worked on it in an Athenian prison about 450 BC,

and Hippocrates of Chios squared several regions bounded by circular arcs not long after, which kept the hope alive for two thousand years. Lindemann settled it in 1882, a decade after Hermite. The proof above used one fact about $i\pi$ and nothing else: that its exponential is -1 .

Everything else was Hermite's design, with the symmetric expressions keeping the roots of the equation from ever having to be known.

In the same paper Lindemann proved the general theorem that $E(\alpha)$ is transcendental for every algebraic α other than 0, and he announced more. In 1885 Weierstrass supplied the proof that when $\alpha_1, \dots, \alpha_n$ are different algebraic numbers, no sum of algebraic multiples of $E(\alpha_1), \dots, E(\alpha_n)$ is 0 unless every multiple is. When moreover no whole-number combination of the α 's is 0 except the one with every coefficient 0, the values $E(\alpha_1), \dots, E(\alpha_n)$ satisfy no polynomial relation with algebraic coefficients at all. Both are the same design carried across more roots at once, and the general statement is relied on wherever it is used and not proved here.

The first of them already makes $\cos \alpha$ and $\sin \alpha$ transcendental for every algebraic α other than 0. If $\cos \alpha$ were algebraic, $E(i\alpha)$ would be a root of $x^2 - 2(\cos \alpha)x + 1$, whose coefficients lie in a field of finite dimension over the fractions; then $E(i\alpha)$ would lie in one of finite dimension too and be algebraic, although $i\alpha$ is algebraic and not 0. The same theorem makes $L(\alpha)$ transcendental for every algebraic α other than 0 and 1, since an algebraic value of $L(\alpha)$ would be an algebraic number other than 0 whose exponential, α , is algebraic. So $\cos 1$, $L(2)$ and $E(\sqrt{2})$ are all transcendental.

Suppose, against the theorem, that $i\pi$ satisfied $x^2 + 2 = 0$. What are q , H and b , and for which primes p does the bound on the right side fall below 1?

For the equation $x^4 - 2 = 0$ in the same role, list the sixteen sums, find q , H and b , and say which nonzero sums occur more than once.

The proof used only that $E(i\pi)$ is -1 . What does it say about the equation $E(\alpha) = -1$ among algebraic numbers α , and which

numbers does it therefore show to be transcendental besides π ?

Given that π is transcendental, which of π^2 , $\sqrt{\pi}$, $\pi + 1$, $1/\pi$ and $\pi + e$ follow, and by what argument?

Where in the proof was it used that p is a prime and not any large whole number?

WHAT A SYMMETRY OF A FIELD CAN MOVE

A matching of a field K with itself that respects both operations and leaves every element of a smaller field F where it is will be called an *automorphism of K over F* . It is a linear map of K over F . It respects addition, and it respects scaling by a member c of F : writing σ for the automorphism, $\sigma(cx) = \sigma(c)\sigma(x) = c\sigma(x)$, since σ fixes c . It also respects multiplication, which a linear map need not do. And it can be undone. Composing two automorphisms gives an automorphism, and so the automorphisms of K over F form a group under composition, as the ways of matching any system with itself do. How much freedom does such a map have?

Less than it seems, for two reasons. An automorphism σ fixes more than it was asked to. It fixes 1. First, $\sigma(0) = \sigma(0 + 0) = \sigma(0) + \sigma(0)$, and taking $\sigma(0)$ from both sides gives $\sigma(0) = 0$. Since σ is one-to-one and already sends 0 to 0, it cannot also send 1 to 0, so $\sigma(1)$ is not 0. Next, $\sigma(1) = \sigma(1 \cdot 1) = \sigma(1)\sigma(1)$, so $\sigma(1)$ is a nonzero element equal to its own square, and the only such element is 1. With 1 fixed, $1 + 1$ is fixed, since $\sigma(1 + 1) = \sigma(1) + \sigma(1)$, and in the same way every whole number is fixed. Finally, a fraction a/b times b is a , so $\sigma(a/b)\sigma(b) = \sigma(a)$; with a and b fixed this reads $\sigma(a/b) \cdot b = a$, and so $\sigma(a/b) = a/b$. So σ fixes every fraction, and for

the same reason an automorphism of a field containing Z/p fixes every element of Z/p .

And an automorphism sends roots to roots. If p is a polynomial with coefficients in F and α is a root of p in K , then $\sigma(p(\alpha)) = p(\sigma(\alpha))$, because σ respects every sum and product in the expression and leaves each coefficient alone; so $\sigma(\alpha)$ is also a root of p . In particular σ sends α to a root of its minimal polynomial that lies in K . If K is $F(\alpha)$, then σ is settled by $\sigma(\alpha)$, since every element of K is a polynomial in α and $\sigma(g(\alpha)) = g(\sigma(\alpha))$.

* * *

One kind of automorphism can be written down at once. Let w be an element of F whose square root is not in F . Every element of $F(\sqrt{w})$ is $a + b\sqrt{w}$ with a and b in F , and the map sending $a + b\sqrt{w}$ to $a - b\sqrt{w}$ respects addition. It respects multiplication too. Multiplying first gives $(a + b\sqrt{w})(c + d\sqrt{w}) = (ac + bdw) + (ad + bc)\sqrt{w}$, which the map sends to $(ac + bdw) - (ad + bc)\sqrt{w}$; applying the map first and then multiplying gives $(a - b\sqrt{w})(c - d\sqrt{w}) = (ac + bdw) - (ad + bc)\sqrt{w}$, the same element. Applying the map twice gives back $a + b\sqrt{w}$, so the map undoes itself. So it is an automorphism of $F(\sqrt{w})$ over F .

$Q(\sqrt{2})$ over Q has two automorphisms, since $\sqrt{2}$ can go only to $\sqrt{2}$ or $-\sqrt{2}$ and both choices occur, the second being the map $a + b\sqrt{2} \mapsto a - b\sqrt{2}$. $Q(i)$ over Q has two, the identity and conjugation. The plane of pairs over the reals has the same two. $Q(\sqrt{2}, \sqrt{3})$ over Q has at most four, since $\sqrt{2}$ goes to $\pm\sqrt{2}$ and $\sqrt{3}$ to $\pm\sqrt{3}$. All four occur. One is the identity. The map that changes the sign of $\sqrt{3}$ and fixes $Q(\sqrt{2})$ is the automorphism of $Q(\sqrt{2})(\sqrt{3})$ over $Q(\sqrt{2})$ just described. The map that changes the sign of $\sqrt{2}$ and fixes $Q(\sqrt{3})$ is the same construction on $Q(\sqrt{3})(\sqrt{2})$; it applies because the squaring argument that kept $\sqrt{3}$ out of $Q(\sqrt{2})$ also keeps $\sqrt{2}$ out of $Q(\sqrt{3})$. Composing these two gives the fourth, which changes both signs.

F_4 over $Z/2$ has two. Its generator w , a root of $x^2 + x + 1$, can go to w or to the other root $w + 1$. The second choice is the map

$x \mapsto x^2$. This map does send w to $w + 1$: from $w^2 + w + 1 = 0$ and $1 + 1 = 0$ we get $w^2 = w + 1$. It respects multiplication in any field, since $(xy)^2 = x^2y^2$. It respects addition here, since $(a + b)^2 = a^2 + 2ab + b^2$ and 2 is 0 . It is one-to-one: if $x^2 = y^2$, then $(x - y)^2 = x^2 - y^2 = 0$, again because 2 is 0 , and a square is 0 only when the element squared is 0 , so $x = y$. A one-to-one map of a finite field to itself misses no element, so it can be undone.

In five cases the number of automorphisms equals the dimension. Is that always so? $Q(\sqrt[3]{2})$ over Q says no. Its generator must go to a root of $x^3 - 2$ that lies in $Q(\sqrt[3]{2})$, every element of $Q(\sqrt[3]{2})$ is a real number, and $x^3 - 2$ has only one real root, $\sqrt[3]{2}$ itself. So the only automorphism is the identity, one against a dimension of 3 . The field holds one root of its generator's equation and not the other two, and the generator has nowhere else to go.

A field with eight elements was built over $Z/2$ from $x^3 + x + 1$, and it is generated by a root θ of that cubic. How many automorphisms does it have over $Z/2$, and where do they send θ ?

Where are the other two roots of $x^3 - 2$? They can be manufactured. Take any polynomial p over F . Split it into irreducible factors; descent on the degree shows this can always be done. Use the quotient construction to adjoin a root of one of the factors of degree above 1 . Over the field obtained, that factor has a factor of degree 1 . Split what remains into irreducible factors over the new field, and repeat. The degree of the part not yet split into factors of degree 1 falls at every round, so the process stops at a field over which p is a product of factors of degree 1 and which is generated over F by the roots of p . Such a field is called a *splitting field* of p .

For $x^3 - 2$ over Q the process runs in two steps. Over $Q(\sqrt[3]{2})$ the polynomial is $(x - \sqrt[3]{2})(x^2 + \sqrt[3]{2}x + \sqrt[3]{4})$, and the quadratic has no root in $Q(\sqrt[3]{2})$, since its discriminant is $-3\sqrt[3]{4}$, which is negative, and $Q(\sqrt[3]{2})$ consists of real numbers. Adjoining a root β of the quadratic gives a field of dimension $3 \cdot 2 = 6$ over Q . Write ω for $\beta/\sqrt[3]{2}$. Putting $\beta = \omega\sqrt[3]{2}$ into $\beta^2 + \sqrt[3]{2}\beta + \sqrt[3]{4} = 0$ gives $\sqrt[3]{4}(\omega^2 + \omega + 1) = 0$, so ω is a root of $x^2 + x + 1$. Multiplying $\omega^2 + \omega + 1 = 0$ by $\omega - 1$ gives $\omega^3 = 1$, so $(\omega\sqrt[3]{2})^3 = 2$ and $(\omega^2\sqrt[3]{2})^3 = 2$, and the

field contains the three roots $\sqrt[3]{2}$, $\omega\sqrt[3]{2}$ and $\omega^2\sqrt[3]{2}$ of $x^3 - 2$. In the plane of pairs the three roots lie at 120° from each other on a circle of radius $\sqrt[3]{2}$. An automorphism of this field over Q is settled by where it sends $\sqrt[3]{2}$, to one of three roots, and ω , to one of two, so there are at most six.

Can a field of odd dimension over Q contain two different roots of $x^3 - 2$?

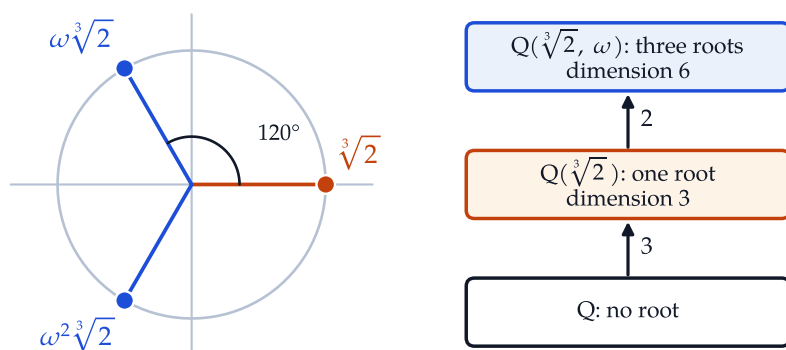


Figure 64.1. The three roots of $x^3 - 2$ in the plane of pairs, at 120° from each other on a circle of radius $\sqrt[3]{2}$; only one is real. Adjoining the real one gives a field of dimension 3 over the fractions holding a single root; adjoining a root of $x^2 + x + 1$ to that gives a field of dimension $3 \cdot 2 = 6$ holding all three, the splitting field of $x^3 - 2$. Assumes: that the quadratic factor left after dividing out the real root has no root in the three-dimensional field, which its discriminant shows since its roots are not real

The real numbers have only one automorphism. An automorphism σ of the reals fixes every fraction. It also preserves order, for a reason with no analysis in it: a real number is at least 0 exactly when it is a square, and σ sends squares to squares, so it sends numbers at least 0 to numbers at least 0. Now if $x < y$, then $y - x$ is positive, so $\sigma(y) - \sigma(x) = \sigma(y - x)$ is at least 0, and it is not 0 because σ is one-to-one and sends only 0 to 0. So $\sigma(x) < \sigma(y)$. If $\sigma(x)$ differed from x , some fraction q would lie strictly between them. Say $x < q < \sigma(x)$. Since σ preserves order, $x < q$ gives $\sigma(x) < \sigma(q)$, and $\sigma(q) = q$ because q is a fraction; so $\sigma(x) < q$, which contradicts

$q < \sigma(x)$. If instead $\sigma(x) < q < x$, the same steps give $q < \sigma(x)$, again a contradiction. So $\sigma(x) = x$ for every x .

The plane of pairs gives no such conclusion. An automorphism of the plane over the fractions sends i to a root of $x^2 + 1$, that is, to i or to $-i$, so on the numbers $a + bi$ with a and b fractions it acts as the identity or as conjugation; and these numbers come arbitrarily close to every point of the plane. It is tempting to conclude that every automorphism of the plane is the identity or conjugation, but that step assumes automorphisms are continuous, sending nearby points to nearby points. For the reals nothing of the kind was assumed: what did the work was the order, and the order came from squares, since the numbers at least 0 are exactly the squares. In the plane every element is a square, and squares single out nothing. Automorphisms of the plane over the fractions other than the identity and conjugation do exist; their existence follows from the axiom of choice and cannot be proved without some form of it.

the field	dimension	automorphisms	
$\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$	2	2	
$\mathbb{Q}(i)$ over $\mathbb{Q}$	2	2	
$\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}$	4	4	
$\mathbb{F}_4$ over $\mathbb{Z}/2$	2	2	
$\mathbb{C}$ over $\mathbb{R}$	2	2	
$\mathbb{Q}(\sqrt[3]{2})$ over $\mathbb{Q}$	3	1	the conjecture dies here
$\mathbb{R}$ over $\mathbb{Q}$	no finite dimension	1	

never more than the dimension

Figure 64.2. For each field built so far, its dimension over the base and the number of automorphisms over the base. The rows that agree invite the guess that they always do; the cube root has dimension 3 and a single automorphism, because two of the three roots of its equation are not in the field to be sent to; the reals have one automorphism over the fractions and no finite dimension at all. *Assumes:* that the counts are exhaustive, which the root-to-root argument in the text guarantees, since an automorphism can move a generator only to a root of its own equation inside the field

* * *

The count of automorphisms never exceeded the dimension in any of these cases, and that inequality is a theorem. Its proof has three parts. The first is a fact about distinct automorphisms. The other two compare the dimension of a field with the size of a group of its automorphisms, one showing that the dimension is at most that size and the other that it is at least that size.

Distinct automorphisms $\sigma_1, \dots, \sigma_m$ of a field K are *independent over K* : no combination $c_1\sigma_1 + \dots + c_m\sigma_m$ with coefficients in K , not all 0, sends every element of K to 0. Suppose one did, and take such a vanishing combination with the fewest coefficients other than 0, say $c_1, \dots, c_r$, all nonzero. One coefficient alone is impossible: the combination would then send 1 to $c_1\sigma_1(1) = c_1$, which is not 0. So r is at least 2, and since σ_1 and σ_2 differ, some y in K has $\sigma_1(y) \neq \sigma_2(y)$. The combination vanishes at xy for every x , which says $c_1\sigma_1(x)\sigma_1(y) + \dots + c_r\sigma_r(x)\sigma_r(y) = 0$; and it vanishes at x , which after multiplying through by $\sigma_1(y)$ says $c_1\sigma_1(x)\sigma_1(y) + \dots + c_r\sigma_r(x)\sigma_1(y) = 0$. Subtracting the second from the first removes the first term and leaves a vanishing combination of $\sigma_2, \dots, \sigma_r$ with coefficients $c_i(\sigma_i(y) - \sigma_1(y))$. The coefficient of σ_2 is $c_2(\sigma_2(y) - \sigma_1(y))$, which is not 0, because c_2 is not 0 and y was chosen with $\sigma_2(y) \neq \sigma_1(y)$. So this is a vanishing combination with coefficients not all 0, and it has fewer coefficients other than 0, contrary to the choice.

Now let G be a finite group of automorphisms of K , with n members, and let F be the collection of elements of K that every member of G leaves where they are. F is a field, since sums, products and reciprocals of fixed elements are fixed, and it is called the *fixed field* of G . The dimension of K over F is exactly n . This is the *fixed-field theorem*.

The linear algebra needed was set apart, with this use in view, when systems of equations were solved by elimination: a system of linear equations over any field whose right-hand sides are all 0 and which has more unknowns than equations has a solution

other than zero, because its staircase has at most one pivot for each equation and an unknown that is not a pivot can be given the value 1.

The dimension is at most n . Suppose $a_1, \dots, a_{n+1}$ in K were independent over F . The n equations

$$\sigma_j(a_1)y_1 + \sigma_j(a_2)y_2 + \dots + \sigma_j(a_{n+1})y_{n+1} = 0, \text{ one for each } \sigma_j \text{ in } G,$$

in $n + 1$ unknowns have a solution other than zero. Among all such solutions take one with the fewest entries other than 0, and scale it so that one of them, say y_1 , is 1. If every y_i were in F , the equation for the identity, which belongs to G , would read $a_1y_1 + \dots + a_{n+1}y_{n+1} = 0$, contradicting independence over F . So some y_k lies outside F , and some τ in G moves it. Apply τ to every equation. The equation for σ_j becomes $\tau\sigma_j(a_1)\tau(y_1) + \dots + \tau\sigma_j(a_{n+1})\tau(y_{n+1}) = 0$, which is the equation for the member $\tau\sigma_j$ of G , with the $\tau(y_i)$ in place of the unknowns. As σ_j runs through G , so does $\tau\sigma_j$, since G is a group, so the new equations are the old ones in a different order. So the $\tau(y_i)$ form another solution of the same system. Subtract it from the first. The difference is again a solution. Its k -th entry $y_k - \tau(y_k)$ is not 0, so it is a solution other than zero. Wherever y_i is 0, $\tau(y_i)$ is 0 too, so the difference is 0 there as well; and its first entry is $1 - 1 = 0$, although y_1 was not 0. So it has fewer entries other than 0 than the solution chosen, which is impossible. So no $n + 1$ elements of K are independent over F , and K has a basis over F with at most n members.

The dimension is at least n . Suppose K had a basis $b_1, \dots, b_d$ over F with d below n . The d equations

$$x_1\sigma_1(b_i) + x_2\sigma_2(b_i) + \dots + x_n\sigma_n(b_i) = 0, \text{ one for each } b_i,$$

in the n unknowns x_j have a solution other than zero. Every z in K is a combination of the b_i with coefficients f_i in F , and each σ_j leaves each f_i alone, so $x_1\sigma_1(z) + \dots + x_n\sigma_n(z)$ is the sum over i of $f_i(x_1\sigma_1(b_i) + \dots + x_n\sigma_n(b_i))$, which is 0. So the combination $x_1\sigma_1 + \dots + x_n\sigma_n$ vanishes on all of K without all its coefficients being 0, contradicting independence.

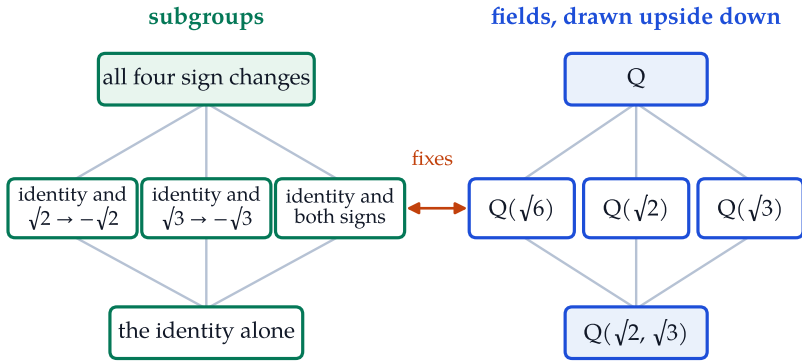
The second count used nothing about G except that its members are distinct and fix F . So any n' distinct automorphisms of K that fix a field F force the dimension of K over F to be at least n' . Two consequences follow. When F is the fixed field of G , every automorphism of K over F already belongs to G . If one did not, it and the n members of G would be $n + 1$ distinct automorphisms fixing F , forcing the dimension of K over F to be at least $n + 1$; but that dimension is exactly n . And for any field K of finite dimension over any field F , the automorphisms of K over F are at most as many as that dimension. The plane of pairs has no finite dimension over the fractions, since every member of a field of finite dimension over Q satisfies an equation with fraction coefficients, and there are real numbers that satisfy none; so the many automorphisms of the plane that the axiom of choice supplies lie outside the bound.

A field K of finite dimension over F is *Galois over F* when its automorphisms over F are exactly as many as the dimension. By the theorem, this says that F is the fixed field of the whole group. The fixed field of the whole group contains F , and the theorem gives K a dimension over it equal to the number of automorphisms. If that number is also the dimension of K over F , the tower law leaves dimension 1 between F and the fixed field, so they are the same field. Conversely, if F is the fixed field, the theorem gives the count and the dimension as equal directly. $Q(\sqrt{2}, \sqrt{3})$ is Galois over Q , since $a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}$ is left alone by all four sign changes only when $b = c = d = 0$. $Q(\sqrt[3]{2})$ is not Galois over Q : the fixed field of its one automorphism is all of $Q(\sqrt[3]{2})$, of dimension 3 over Q , which is the factor by which the count fell short.

A count can also fall strictly between. The fourth root of 2 is not in $Q(\sqrt{2})$, since $(a + b\sqrt{2})^2 = \sqrt{2}$ would need $a^2 + 2b^2 = 0$ and $2ab = 1$. So $Q(\sqrt[4]{2})$, which is $Q(\sqrt{2})$ with the square root of $\sqrt{2}$ adjoined, has dimension 4 over Q . An automorphism sends $\sqrt[4]{2}$ to a member x of the field with $x^4 = 2$. The field lies inside the real numbers, where $x^4 = 2$ forces $x^2 = \sqrt{2}$ and then $x = \sqrt[4]{2}$ or $x = -\sqrt[4]{2}$, and both occur, the second because sending $a + b\sqrt{w}$ to $a - b\sqrt{w}$ is an automorphism of $F(\sqrt{w})$ over F , here with $F = Q(\sqrt{2})$

and $w = \sqrt{2}$. So $Q(\sqrt[4]{2})$ has two automorphisms over Q , strictly between 1 and 4, and their fixed field is $Q(\sqrt{2})$. The field $Q(\sqrt[4]{2})$ is Galois over $Q(\sqrt{2})$, and $Q(\sqrt{2})$ is Galois over Q , and yet $Q(\sqrt[4]{2})$ is not Galois over Q : being Galois does not pass up a tower.

The subgroups of the group of $Q(\sqrt{2}, \sqrt{3})$ have fixed fields too. The subgroup made of the identity and the sign change of $\sqrt{2}$ alone fixes exactly the elements with $b = d = 0$, which form $Q(\sqrt{3})$; by the theorem $Q(\sqrt{2}, \sqrt{3})$ has dimension 2 over $Q(\sqrt{3})$, as the tower law says. The subgroup containing the sign change of $\sqrt{3}$ alone fixes $Q(\sqrt{2})$, and the subgroup containing the change of both signs fixes $Q(\sqrt{6})$, since it sends $\sqrt{6} = \sqrt{2} \cdot \sqrt{3}$ to $(-\sqrt{2})(-\sqrt{3}) = \sqrt{6}$. Three subgroups of order 2 and three fields of dimension 2 over Q , matched one to one, with larger subgroups belonging to smaller fields.



dimension of the top over each middle field: $4/2 = 2$, the order of its subgroup

Figure 64.3. The four automorphisms of $Q(\sqrt{2}, \sqrt{3})$ over Q are the four choices of sign for $\sqrt{2}$ and $\sqrt{3}$. Each subgroup of order two fixes exactly one of the three fields: the one changing the sign of $\sqrt{2}$ fixes $Q(\sqrt{3})$, the one changing the sign of $\sqrt{3}$ fixes $Q(\sqrt{2})$, and the one changing both fixes $Q(\sqrt{6})$, whose dimension over Q is 4 divided by 2; the whole group fixes only Q , and the identity alone fixes everything. Two lists, the same length, matched one to one, with inclusion running the opposite way on the two sides. Assumes: that the fixed field of each subgroup is exactly the field named, read off on $a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}$ from which coefficients must vanish

WHEN IS A FIELD GALOIS

Is the matching between subgroups and fields in $Q(\sqrt{2}, \sqrt{3})$ a theorem about every field, and what would a proof need?

Let G be the group of automorphisms of K over F , and take the two maps the matching suggests. A subgroup H goes to its fixed field K^H , which lies between F and K . A field L between F and K goes to $\text{Aut}(K/L)$, the automorphisms of K that leave every element of L alone, which form a subgroup of G . For the matching to be a theorem the two maps must undo each other. One direction holds already: the automorphisms of K that fix K^H are exactly the members of H , by the fixed-field theorem. The other direction is the difficulty. Give the fixed field of $\text{Aut}(K/L)$ the name M . Every member of $\text{Aut}(K/L)$ leaves L alone, so M contains L , and the dimension of K over L is the dimension of K over M times the dimension of M over L . So M equals L exactly when K has the same dimension over M as over L . By the fixed-field theorem the dimension of K over M is the number of members of $\text{Aut}(K/L)$. So M equals L exactly when K has as many automorphisms over L as its dimension over L , which is to say exactly when K is Galois over L .

So everything turns on one question. If K is Galois over F , is K Galois over every field between?

The definition does not answer it. Enlarging the base from F to L shrinks the dimension and shrinks the group, and nothing in the

definition says the two shrink by the same factor. What is needed is a second description of the same property, one that visibly survives enlarging the base.

* * *

In $Q(\sqrt{2}, \sqrt{3})$ apply the four automorphisms to $\alpha = \sqrt{2} + \sqrt{3}$. They produce four different values, $\sqrt{2} + \sqrt{3}$, $-\sqrt{2} + \sqrt{3}$, $\sqrt{2} - \sqrt{3}$ and $-\sqrt{2} - \sqrt{3}$. Pair the first value with the second and the third with the fourth. The first pair has sum $2\sqrt{3}$ and product 1, and the second pair has sum $-2\sqrt{3}$ and product 1. Two factors $x - a$ and $x - b$ multiply to $x^2 - (a + b)x + ab$, so the product of the four factors x minus each value is

$$(x^2 - 2\sqrt{3}x + 1)(x^2 + 2\sqrt{3}x + 1) = x^4 - 10x^2 + 1$$

the minimal polynomial of $\sqrt{2} + \sqrt{3}$. The irrational parts cancel for a reason. Each automorphism rearranges the four values, so applying it to the product only changes the order of the four factors, and the product comes out as the same polynomial as before. So the automorphism leaves every coefficient of the product where it is, and the only elements that every automorphism leaves where they are form Q . Applied to $\sqrt{2}$ the four automorphisms produce only two values, $\sqrt{2}$ and $-\sqrt{2}$, and the product of the two factors is $x^2 - 2$, again the minimal polynomial; applied to $\sqrt{6}$ they produce $\sqrt{6}$ and $-\sqrt{6}$, and $x^2 - 6$.

That is a general fact. Let K be Galois over F , take α in K , and call the distinct values $\sigma(\alpha)$, as σ runs through G , the *orbit* of α . Form

$f(x) = \text{the product of } x - \beta \text{ over the distinct } \beta \text{ in the orbit.}$

Any σ in G rearranges the orbit, since it sends $\sigma'(\alpha)$ to $\sigma\sigma'(\alpha)$, and distinct values stay distinct because σ can be undone. So σ rearranges the factors of f and leaves each coefficient of f alone; each coefficient lies in the fixed field of G , which is F , because K is Galois over F . So f is a polynomial over F . The identity is a member

of G , so α is in its own orbit, $x - \alpha$ is one of the factors of f , and $f(\alpha) = 0$. The minimal polynomial m of α divides every polynomial over F that has α as a root, so m divides f . In the other direction every β in the orbit is a root of m , since automorphisms send roots to roots. Each root β gives a factor $x - \beta$ of m , and different values of β give different factors, so the product of all of them divides m . That product is f , so f divides m . Two monic polynomials that divide each other are equal, so $m = f$. When K is not Galois over F the product need not have its coefficients in F : in $Q(\sqrt[4]{2})$, with its two automorphisms, the images of $\sqrt[4]{2}$ are $\sqrt[4]{2}$ and $-\sqrt[4]{2}$, and the product $(x - \sqrt[4]{2})(x + \sqrt[4]{2}) = x^2 - \sqrt{2}$ does not have fraction coefficients, while the minimal polynomial of $\sqrt[4]{2}$ over Q is $x^4 - 2$.

In a Galois extension, then, the minimal polynomial of every element is a product of factors of degree 1, all different, and all its roots lie in K . From this a second description follows. Take a basis $a_1, \dots, a_n$ of K over F , and let p be the product of the distinct minimal polynomials of the a_i . Each of them has distinct roots in K , and two distinct monic irreducible polynomials share no root, since both would be the minimal polynomial of a shared root. So p has no repeated root. All the roots of p lie in K , so p is a product of factors of degree 1 over K , and the field its roots generate over F lies inside K . That field contains every a_i , and so every combination of the a_i with coefficients in F , which is every element of K . So it is K :

a field Galois over F is a splitting field over F of a polynomial with no repeated root.

That description survives enlarging the base. If L lies between F and K , then K is still generated over L by the roots of p , and p still has no repeated root. Whether the description forces the number of automorphisms to equal the dimension is the question that remains.

* * *

When can a polynomial have a repeated root? A repeated root

of p is a root α with $(x - \alpha)^2$ dividing p . Detecting one without knowing α needs an instrument, and the instrument is the derivative, with every trace of a limit removed. For a polynomial over any field,

$$\text{the derivative of } c_0 + c_1x + c_2x^2 + \dots + c_nx^n \text{ is } c_1 + 2c_2x + \dots + nc_nx^{n-1}$$

where nc_n means c_n added to itself n times. This is a rule for making one list of coefficients from another. It obeys the sum rule, coefficient by coefficient. It also obeys the product rule. For the monomials x^j and x^k the product rule asks whether $(j+k)x^{j+k-1}$ equals $jx^{j-1} \cdot x^k + x^j \cdot kx^{k-1}$, and it does, since each term on the right is a multiple of x^{j+k-1} and the multiples add to $j+k$. Every product of polynomials is a sum of constants times products of monomials, so by the sum rule the product rule holds for all of them.

If $p = (x - \alpha)^2g$, the product rule gives $p' = 2(x - \alpha)g + (x - \alpha)^2g'$, so α is a root of p' as well. Conversely, let α be a root of both p and p' . Write $p = (x - \alpha)h$; then $p' = h + (x - \alpha)h'$, and putting α for x gives $h(\alpha) = p'(\alpha) = 0$. So $x - \alpha$ divides h , and a second factor $x - \alpha$ divides p . So the repeated roots of p are exactly the common roots of p and p' , and a common root of two polynomials is a root of their greatest common divisor, which Euclid's algorithm computes without knowing any root.

For which primes r does $x^3 + x + 1$ have a repeated root in some field containing $\mathbb{Z}/r$?

Let p be irreducible. Its greatest common divisor with p' divides p , so it is 1 or p itself up to a constant. It can be p only if p divides p' , and p' has smaller degree than p , so only if p' is the zero polynomial. A repeated root of p is a root of that greatest common divisor, and the constant 1 has no roots; while if p' is the zero polynomial the greatest common divisor is p itself and every root of p is repeated. So an irreducible polynomial has a repeated root exactly when its derivative is the zero polynomial.

Over the fractions that never happens. The derivative of a polynomial of degree n contains the term nc_n with c_n not 0, and adding

a nonzero fraction to itself n times does not give 0. So every irreducible polynomial over $\mathbb{Q}$ has distinct roots, and so does every irreducible polynomial over any field in which no sum $1 + 1 + \dots + 1$ is 0, however many ones it has.

In other fields it can happen. Start from $\mathbb{Z}/2$, where $1 + 1 = 0$, and take all ratios of polynomials in a letter t with coefficients in $\mathbb{Z}/2$, two ratios counting as the same when cross-multiplying gives the same polynomial; the construction that made the fractions from the whole numbers makes this a field. Over it consider

$$x^2 + t$$

It is irreducible. A polynomial of degree 2 or 3 that factors has a factor of degree 1, and a factor of degree 1 gives a root in the field, which is a ratio g/h of polynomials in t . Then $(g/h)^2 = -t$, and $-t$ is t because $-1 = 1$ in $\mathbb{Z}/2$, so $g^2 = t \cdot h^2$. Comparing degrees in t gives $2 \cdot (\text{degree of } g) = 1 + 2 \cdot (\text{degree of } h)$, an even number equal to an odd one. Its derivative is $2x$, which is the zero polynomial, since 2 is 0. Adjoin a root θ by the quotient construction, so that $\theta^2 = t$; then $(x + \theta)^2 = x^2 + 2\theta x + \theta^2 = x^2 + t$, since $2\theta x$ is 0. So $x^2 + t$, irreducible, has the single root θ , repeated. A polynomial with no repeated root is called *separable*, and $x^2 + t$ shows that the word rules something out: an irreducible polynomial need not be separable. The phenomenon is not special to $1 + 1$ being 0. Over the ratios of polynomials in t with coefficients in $\mathbb{Z}/3$, where $1 + 1$ is 2 but $1 + 1 + 1$ is 0, the polynomial $x^3 - t$ is irreducible by the same comparison of degrees, $3 \cdot (\text{degree of } g) = 1 + 3 \cdot (\text{degree of } h)$, its derivative $3x^2$ is the zero polynomial, and a root θ gives $(x - \theta)^3 = x^3 - 3\theta x^2 + 3\theta^2 x - \theta^3 = x^3 - t$: one root, three times over.

Can an irreducible polynomial over a finite field have a repeated root?

* * *

The converse, that a splitting field of a separable polynomial has as many automorphisms as its dimension, is awkward to attack di-

rectly, since an automorphism has to be defined on the whole field at once. A stronger statement is easier, because it can be proved one root at a time: count the ways an *isomorphism* between two smaller fields, a matching of one field with another that respects both operations, extends upward. Two fields have to be allowed even though the aim is a single one, because once one root has been sent to another, the map built so far goes from the field generated by the first root to the field generated by the second, and these need not be the same field.

Let φ be an isomorphism from a field F to a field F' , let p be a polynomial over F with no repeated root, and write φp for the polynomial over F' obtained by applying φ to every coefficient. Let K be a splitting field of p over F and K' a splitting field of φp over F' . The claim is that the number of isomorphisms from K to K' that agree with φ on F is exactly the dimension of K over F . The descent is on that dimension.

If the dimension is 1, then $K = F$, so p is already a product of factors of degree 1 over F . Applying φ to that factorisation shows that φp is a product of factors of degree 1 over F' . So the roots of φp already lie in F' , and K' , which they generate over F' , is F' . An isomorphism from $K = F$ to $K' = F'$ that agrees with φ on F is φ itself, so the count is 1, which is the dimension.

If the dimension is larger, some irreducible factor q of p over F has degree d greater than 1, since otherwise p would already be a product of factors of degree 1 over F . Let α in K be a root of q . An isomorphism ψ from K to K' that extends φ sends α to a root of φq , since applying ψ to $q(\alpha) = 0$ gives $(\varphi q)(\psi(\alpha)) = 0$. So the extensions are counted in two stages: where α goes, and then the extensions from there.

The first stage has exactly d choices. The polynomial φq is irreducible over F' , since a factorisation of it would come back through φ^{-1} to a factorisation of q . It divides φp , since q divides p . What has to be shown is that φq has d different roots in K' , and for that it is enough that φp has no repeated root. The hypothesis says this of p and not of φp , so it has to be carried across by φ , and what φ can

carry is the greatest common divisor with the derivative.

Over K the polynomial p is a constant times a product of factors $x - \gamma$, one for each root γ , and an irreducible polynomial that divides a product divides one of the factors, by the argument that worked for primes; so every divisor of p over K is a constant times a product of some of those factors. If the greatest common divisor of p and its derivative were not constant, some factor $x - \gamma$ would divide both, and γ would be a repeated root of p , which p does not have. So that divisor is 1 over K .

Euclid's algorithm, run over F , returns the same answer. What it produces is a combination of p and its derivative that divides both of them. It still divides both over K , so it divides their greatest common divisor over K ; and every common divisor over K divides any combination of p and its derivative, so it divides what the algorithm produces. The two answers divide each other, so the algorithm run over F returns 1 as well. Every step of that computation adds, subtracts, multiplies or divides coefficients in F , and φ respects all four; φ also turns the derivative of p into the derivative of φp , since $\varphi(nc)$ is $\varphi(c)$ added to itself n times. So applying φ to every step gives Euclid's algorithm for φp and its derivative over F' , and the greatest common divisor of φp and its derivative is 1 as well. So φp has no repeated root. Over K' , then, φp is a product of distinct factors of degree 1, and by the same description of divisors φq , which divides it and has degree d , is a constant times d of them. So φq has exactly d distinct roots in K' .

For each root β of φq there is exactly one isomorphism from $F(\alpha)$ to $F'(\beta)$ that agrees with φ on F and sends α to β . Every element of $F(\alpha)$ is written in exactly one way as $c_0 + c_1\alpha + \dots + c_{d-1}\alpha^{d-1}$ with the c_i in F , and a map that respects both operations, agrees with φ on F and sends α to β has to send that element to $\varphi(c_0) + \varphi(c_1)\beta + \dots + \varphi(c_{d-1})\beta^{d-1}$. So there is at most one. That rule is an isomorphism, because both fields are copies of the remainders, $F(\alpha)$ of the remainders modulo q and $F'(\beta)$ of the remainders modulo φq , and φ carries sums, products and remainders modulo q to sums, products and remainders modulo φq .

The second stage is the descent. Fix one choice, an isomorphism ψ_0 from $F(\alpha)$ to $F'(\beta)$. The coefficients of p lie in F , where ψ_0 agrees with φ , so applying ψ_0 to p gives φp again. K is a splitting field of p over $F(\alpha)$, K' is a splitting field of φp over $F'(\beta)$, and p still has no repeated root. So ψ_0 , $F(\alpha)$ and $F'(\beta)$ can take the places of φ , F and F' in the claim. The dimension of K over $F(\alpha)$ is the dimension of K over F divided by d , which is smaller. The claim is already established at that smaller dimension, so the number of isomorphisms from K to K' that extend ψ_0 is exactly the dimension of K over $F(\alpha)$.

Every extension of φ arises exactly once, from one choice in the first stage followed by one extension in the second, so the total is d times the dimension of K over $F(\alpha)$, which is the dimension of K over F .

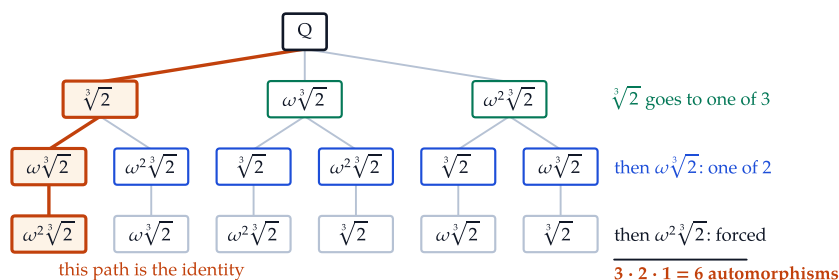


Figure 65.1. The count run on $x^3 - 2$ over $\mathbb{Q}$, drawn as a tree. The first row is where the root $\sqrt[3]{2}$ may go: to any of the three roots. The second row is where $\omega\sqrt[3]{2}$ may then go: to either root not already used. The last row is forced. Each path from the top to the bottom is one automorphism, and there are $3 \cdot 2 \cdot 1 = 6$ of them, which is the dimension of the splitting field over $\mathbb{Q}$. Assumes: that the quadratic left after removing one root is irreducible over the field just built, which the text checks; if it factored, the second row would offer one choice and not two, and the product would be smaller

Two statements follow. Taking $F' = F$, $K' = K$ and φ the identity, the isomorphisms counted are the automorphisms of K over F : a splitting field of a separable polynomial is Galois. Taking K and K'

to be two splitting fields of the same polynomial over the same F , with ϕ the identity, the count is at least 1: any two splitting fields of a separable polynomial are matched by an isomorphism that leaves F alone, whatever choices were made in building them.

For which whole numbers m and n , neither of them a square, can $Q(\sqrt{m})$ and $Q(\sqrt{n})$ be matched by a map that respects both operations?

For $x^3 - 2$ over Q the count runs as follows. The irreducible factor is $x^3 - 2$ itself, so $\sqrt[3]{2}$ has three places to go. Over $Q(\sqrt[3]{2})$ what remains is $x^2 + \sqrt[3]{2}x + \sqrt[3]{4}$, irreducible there, so the next root has two places to go, and the last has one. The count is $3 \cdot 2 \cdot 1 = 6$, the dimension of the splitting field, and all six of the possible automorphisms occur.

* * *

So if K is Galois over F , it is Galois over every field L between. K is a splitting field over F of a separable polynomial p , so it is a splitting field of the same p over L , p is still separable, and a splitting field of a separable polynomial is Galois.

So for every L between F and K the fixed field of $\text{Aut}(K/L)$ is L , because K is Galois over L , and the automorphisms fixing K^H are exactly H . The two maps undo each other:

if K is Galois over F with group G , the subgroups of G and the fields between F and K correspond one to one, by $H \mapsto K^H$ and $L \mapsto \text{Aut}(K/L)$.

The correspondence reverses inclusion, since a larger subgroup fixes fewer elements and a larger field is fixed by fewer automorphisms. It carries orders to dimensions. K has dimension over K^H equal to the order of H , by the fixed-field theorem, and dimension over F equal to the order of G , because K is Galois over F . Dimensions multiply along the chain F, K^H, K , so the dimension of K^H over F is the order of G divided by the order of H . And it turns a question about fields into a question about a finite group. Finding every field between F and K is, over the fractions, a search among

infinitely many elements; finding every subgroup of a finite group is a finite check.

How many automorphisms does $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ have over $\mathbb{Q}$, and which fields between $\mathbb{Q}$ and it have dimension 4 over $\mathbb{Q}$?

EVERY FINITE FIELD

Fields with 4, 8 and 9 elements were made by dividing polynomials by an irreducible one, and every finite field was shown to have a power of a prime for its size. Two questions were left. Is there a field of every such size, of 16 elements, of 125, of 2^{256} ? And can two fields of the same size differ? With splitting fields in hand both have short answers, and the answers turn out to matter to anyone who has played a scratched disc.

Let p be a prime, d a whole number at least 1, and $q = p^d$. Consider the polynomial $x^q - x$ over $\mathbb{Z}/p$, and let K be a splitting field for it. Inside K , look at the roots of that polynomial, the elements a with $a^q = a$. There are exactly q of them. In K the polynomial splits into q factors of degree 1, so it has q roots if none is repeated, and none is. The derivative of $x^q - x$ is $q \cdot x^{q-1} - 1$, and the term $q \cdot x^{q-1}$ vanishes because q is a sum of p 's, so the derivative is -1 . A polynomial whose derivative is -1 has no root in common with its derivative, and so has no repeated root. And the roots form a field. Products and reciprocals of roots are roots. For sums, the binomial theorem gives

$$(a + b)^p = a^p + b^p$$

in any field where a sum of p ones is 0, because every entry of row p of the triangle of choices, apart from the two ends, is divisible

by p . Take k strictly between 0 and p . Then $k!$ times p choose k is $p(p-1)\cdots(p-k+1)$, a multiple of p . A prime that divides a product divides one of the factors, and p divides none of $1, 2, \dots, k$, all of them smaller than p , so p does not divide $k!$. Since p divides $k!$ times p choose k and does not divide $k!$, it divides p choose k . Raising to the power p is therefore a map that respects both addition and multiplication. Doing it d times is raising to the power q , which respects them too, so if $a^q = a$ and $b^q = b$ then $(a+b)^q = a+b$. So the q roots are a field, and since K is generated by them they are all of K :

for every prime p and every d , there is a field with p^d elements.

Now let K be any field with q elements. Its nonzero elements are a group of $q-1$ members under multiplication, so, since the order of an element divides the size of the group, every one of them satisfies $a^{q-1} = 1$, and multiplying by a gives $a^q = a$. Zero satisfies $a^q = a$ as well, so every element of K is a root of $x^q - x$. That polynomial therefore splits into factors of degree 1 over K , and K is generated by its roots, since every element of K is one of them. So K is a splitting field of $x^q - x$ over $\mathbb{Z}/p$, and $x^q - x$ has no repeated root, its derivative being -1 , so two splitting fields of it can be matched by a map that respects both operations:

any two fields with p^d elements are the same field under different names.

It is written F_q . Galois described these fields in 1830, at eighteen, and Moore proved in 1893 that there are no others. The three fields made earlier are F_4 , F_8 and F_9 , and any other irreducible polynomial of the right degree would have made the same fields. Over $\mathbb{Z}/2$ the cubics $x^3 + x + 1$ and $x^3 + x^2 + 1$ are both irreducible, and the two fields of eight remainders they define are matched by sending the class of x in the first to the class of $x + 1$ in the second.

* * *

The nonzero elements of F_q form a group under multiplication, and it has the plainest shape a group can have: it consists of the powers of one element. The proof given for the remainders modulo a prime used nothing but the fact that an equation of degree m has at most m roots in a field. Every element of the group has an order that divides $q - 1$. Take a divisor m of $q - 1$ and suppose some element has order m . Its m powers are m different roots of $x^m = 1$, so they are all the roots, and every element of order m is among them. Exactly $\phi(m)$ of those powers have order exactly m . So for each divisor m of $q - 1$ there are either $\phi(m)$ elements of order m or none. Every element has exactly one order, so these counts, taken over all the divisors of $q - 1$, add up to $q - 1$, the size of the group. The values of ϕ at the divisors of $q - 1$ also add up to $q - 1$. If any count were 0 in place of $\phi(m)$, the total would fall short of $q - 1$, so none of the counts can be 0. In particular there are $\phi(q - 1)$ elements of order $q - 1$. In F_9 , made from $Z/3$ by adjoining i with $i^2 = -1$, the element $1 + i$ has the powers

$$1 + i, 2i, 1 + 2i, 2, 2 + 2i, i, 2 + i, 1$$

eight different elements, and it is one of $\phi(8) = 4$ generators.

A generator g of the nonzero elements of F_q generates F_q as a field over Z/p , so F_q is Z/p with the single element g adjoined. The minimal polynomial of g is irreducible, and its degree is the dimension of the field g generates over Z/p . That field is all of F_q , of dimension d , so the degree is d . So over Z/p there are irreducible polynomials of every degree, which no search through polynomials would have shown. And the map that raises every element to the power p , which respects both operations, is an automorphism of F_q over Z/p . It sends different elements to different ones, since $a^p = b^p$ gives $(a - b)^p = 0$ and so $a = b$. A map from a finite set to itself that never sends two elements to one must reach every element, so it is onto. It is named after Frobenius. Applied k times it raises to the power p^k . Since g generates F_q , this is the identity exactly when it fixes g , that is when $g^{p^k - 1} = 1$, which needs $p^k - 1$ to be a multiple of $p^d - 1$, the order of g . For k from 1 to $d - 1$ the number $p^k -$

1 is positive and smaller than $p^d - 1$, so the map first becomes the identity when $k = d$. So it has order d , and since d is the dimension of F_q over Z/p , its powers are all the automorphisms:

*F_q is Galois over Z/p , with a group of d members,
the powers of the map $a \mapsto a^p$.*

The field with eight elements has three automorphisms, which send any generator θ of its nonzero elements to θ , θ^2 and θ^4 . This group of automorphisms is a clock with d hours: the map applied k times plays the part of the hour k , and applying two of them one after the other adds their hours. A clock with d hours has one subgroup for each divisor s of d , made of the multiples of s , and no others. A subgroup that contains a nonzero hour contains a smallest positive hour s . Any hour h in it, and the hour d , which is 0, can be written as a multiple of s plus a remainder below s ; the remainder is the hour less a multiple of s , so it is an hour of the subgroup, and being below s it is 0. So every hour of the subgroup is a multiple of s , s divides d , and the subgroup is the d/s multiples of s ; the subgroup with the single hour 0 belongs to the divisor d itself, and different divisors give subgroups of different sizes.

The fields between Z/p and F_q match the subgroups of this group one to one, so they correspond to the divisors of d as well. The subgroup of the multiples of m has d/m members, so the field it fixes has dimension d divided by d/m , which is m , over Z/p . So F_q contains a field with p^m elements exactly when m divides d , and then contains only one. The field of 64 elements contains the fields of 2, 4 and 8 elements and no field of 16 or 32.

How many irreducible polynomials are there? Every element of F_q , with $q = p^n$ as before but the exponent now called n , is a root of exactly one monic irreducible polynomial over Z/p , its minimal polynomial, whose degree is the dimension of the field the element generates, a divisor of n . Conversely, take a monic irreducible polynomial of degree m , with m dividing n . It has all its m roots in F_q , and they are different. In a splitting field of the polynomial, a root generates a field with p^m elements, and F_q contains a copy of that

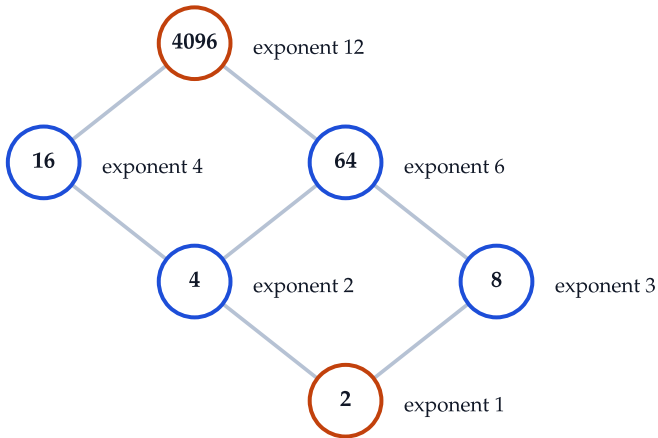


Figure 66.1. The fields contained in the field with $2^{12} = 4096$ elements. There is one for each divisor of 12, and one contains another exactly when the exponent of the first is a multiple of the exponent of the second. The diagram is the diagram of the divisors of 12, and also, upside down, of the subgroups of a clock with twelve hours. *Assumes:* the exponent 12; for an exponent that is a power of a prime the fields form a single chain

field, so the polynomial has a root a in F_q . Then $a^q = a$, and the polynomial, being the minimal polynomial of a , divides $x^q - x$. Over F_q that is a product of q different factors of degree 1, so the polynomial is a product of m of them. So $x^q - x$ is the product of all the monic irreducible polynomials whose degree divides n , each once, and counting degrees,

$$p^n = \text{the sum, over the divisors } m \text{ of } n, \text{ of } m \text{ times } N_m$$

where N_m is the number of monic irreducible polynomials of degree m . That determines the numbers one after another. Over $\mathbb{Z}/2$ they are 2, 1, 2, 3, 6, 9, 18 and 30 for the degrees 1 to 8. For a prime degree n the formula reads $p^n = p + n \cdot N_n$, which shows once more that n divides $p^n - p$. In general N_n is close to p^n/n . The formula with m in place of n gives $m \cdot N_m \leq p^m$, since $m \cdot N_m$ is one of the

terms of its sum. Every divisor m of n other than n itself is at most $n/2$, so the terms of the sum other than $n \cdot N_n$ add up to at most the sum of p^m over the whole numbers m from 1 to $n/2$; since p is at least 2, each of these powers is at most half the next, so that sum is less than twice its largest term, and less than $2p^{n/2}$. So $n \cdot N_n$ lies between $p^n - 2p^{n/2}$ and p^n . There are p^n monic polynomials of degree n , so about one in every n of them is irreducible. Among the whole numbers of d digits about one in $2.3d$ is prime, a fact that took a century to prove. For polynomials over a finite field the corresponding fact is a count.

* * *

In 1947 Hamming, working at Bell Laboratories on a computer that ran unattended over weekends and stopped dead whenever it read a wrong bit, asked why a machine that could detect an error could not find it and put it right. His answer can be read off the field with eight elements.

Let θ generate F_8 , with $\theta^3 = \theta + 1$. Its seven powers, written as triples of coefficients of θ^2, θ and 1, are

$$\theta^0 = 001, \theta^1 = 010, \theta^2 = 100, \theta^3 = 011, \theta^4 = 110, \theta^5 = 111, \theta^6 = 101$$

the seven nonzero triples of bits. Call a word of seven bits $c_0c_1\dots c_6$ a *code word* when the sum of the powers θ^i over the places i where c_i is 1 comes to 0. The word 1101000 is one, since $\theta^0 + \theta^1 + \theta^3$ is $001 + 010 + 011$, which is 000. The rule that sends a word to that sum is a linear map from the space of words, of dimension 7 over $\mathbb{Z}/2$, to F_8 , of dimension 3. It reaches every element of F_8 , since each nonzero one is some θ^i , so its kernel has dimension 7 less 3, which is 4: there are 16 code words, enough to carry four bits of message in seven bits of signal.

Suppose a code word is sent and one bit is changed on the way, in place j . The receiver computes the sum for the word received. The code word contributed 0, and the changed bit contributes θ^j , so the sum is θ^j , and it names the place of the error. The receiver flips

that bit back. If 1101010 arrives, the sum is $001 + 010 + 011 + 111 = 111$, which is θ^5 , so place 5 is wrong and the word sent was 1101000. Every single error is corrected, at a cost of three extra bits in seven. Two code words never differ in fewer than three places. Adding two code words place by place gives a word with 1s exactly where they differ, and its sum is $0 + 0 = 0$. So if they differed in one or two places, the sum of one or two different powers of θ would be 0, which it is not: a power of θ is never 0, and two different elements of F_8 never add to 0, since each element of F_8 is its own negative.

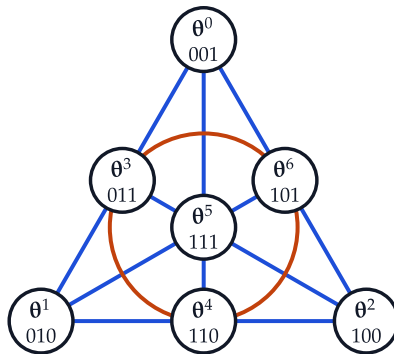


Figure 66.2. The seven nonzero elements of the field with eight elements, as powers of θ and as triples of bits, and the seven sets of three that add up to zero: six straight lines and the circle. Any two points lie on exactly one of them. Each line is a code word with three 1s, and the picture is the smallest projective plane. Assumes: the generator θ with $\theta^3 = \theta + 1$; the other cubic gives the same picture with the labels moved

The seven code words with exactly three 1s are the triples of powers of θ that add up to 0. Any two powers of θ belong to exactly one such triple, the third member being their sum. Seven points, seven triples, any two points in exactly one triple: that is the smallest projective plane, whose points are the seven triples of bits other than 000 and whose lines are the triples of points adding to 000.

The codes that protect recorded music and the pictures sent back by spacecraft are built on F_{256} , whose elements are bytes, and on the theorem that a polynomial has no more roots than its de-

gree. Reed and Solomon proposed them in 1960. Take a message of k bytes as the coefficients of a polynomial of degree below k over F_{256} , and send its values at n different elements of the field, with n larger than k . Two different messages give polynomials whose difference is not zero and has degree below k , so it has fewer than k roots. The two signals agree exactly at the elements where that difference is 0, so they agree in fewer than k places and differ in more than $n - k$.

Now suppose a signal arrives with up to $(n - k)/2$ wrong bytes. It differs from the right signal in at most $(n - k)/2$ places. Any other signal differs from the right one in more than $n - k$ places, and the wrong bytes can close at most $(n - k)/2$ of those gaps, so the received signal still differs from that other one in more than $(n - k)/2$ places. It is therefore closer to the right signal than to any other, and can be corrected. It is the scheme that shared a secret among five people, with the extra values used for repair in place of concealment.

On a compact disc the code turns every 24 bytes of music into 32, in two interleaved layers. The player reconstructs the sound across a scratch two and a half millimetres long. The square patterns that telephones read carry the same code, which is why they still work with a corner torn off.

Which elements of F_9 generate its nonzero elements, and which of them are squares?

How many monic irreducible polynomials of degree 2 are there over $\mathbb{Z}/3$, and of degree 12 over $\mathbb{Z}/2$? What are the three of degree 4 over $\mathbb{Z}/2$?

The word 0110101 arrives. Which bit is wrong, and what was sent?

What is the product of all the nonzero elements of a finite field, and what is the sum of all its elements?

A message of 3 elements of $\mathbb{Z}/7$ is sent as the 7 values of a polynomial of degree below 3. How many wrong values can be corrected, and how many can be detected without being corrected?

TWO LATTICES, ONE UPSIDE DOWN

Inside the six-dimensional splitting field of $x^3 - 2$ sit two fields that look equally modest. $Q(\omega)$ has dimension 2 over Q and two automorphisms. $Q(\sqrt[3]{2})$ has dimension 3 over Q and a single automorphism, because it is a field of real numbers and holds only one of the three roots of $x^3 - 2$. The first is Galois over Q and the second is not. The correspondence matches the six subgroups of the group with the six fields between Q and the top, and it treats these two alike, each being the fixed field of one subgroup. Where, in the lattice of subgroups, does the difference show?

It shows in how a subgroup sits among the subgroups got from it by relabelling, and one computation says so. Let K be Galois over F with group G , take σ in G and a subgroup H , and ask what σ does to the fixed field K^H . An element x lies in $\sigma(K^H)$ exactly when $\sigma^{-1}x$ lies in K^H , which says that $h(\sigma^{-1}x) = \sigma^{-1}x$ for every h in H . Apply σ to both sides: the left side becomes $\sigma h \sigma^{-1}(x)$ and the right side becomes x , and applying σ^{-1} undoes the step. So x lies in $\sigma(K^H)$ exactly when $\sigma h \sigma^{-1}(x) = x$ for every h in H , that is, when x is fixed by every member of $\sigma H \sigma^{-1}$. So

$$\sigma(K^H) = K^{\sigma H \sigma^{-1}}$$

Read the formula for one σ at a time. The automorphism σ carries the field K^H onto itself exactly when $K^{\sigma H \sigma^{-1}} = K^H$, and since different subgroups have different fixed fields, that happens exactly when $\sigma H \sigma^{-1} = H$. So K^H is carried onto itself by every automorphism of K over F exactly when $\sigma H \sigma^{-1} = H$ for every σ in G . A subgroup with that property is a normal subgroup, the kind a group can be collapsed by. Passing from h to $\sigma h \sigma^{-1}$, which undoes σ , performs h and performs σ again, is relabelling h by σ , and it is called *conjugation*. It first appeared as the condition for collapsing a group into blocks. Here it is a condition about fields, and the computation that produces it never mentions blocks.

Two steps join normality to being Galois. Suppose first that H is normal, and write L for K^H , so that $\sigma(L) = L$ for every σ in G . Restricting each σ to L gives an automorphism of L over F , and restriction respects composition, so restriction is a homomorphism from G to $\text{Aut}(L/F)$. Its kernel consists of the automorphisms that leave every element of L alone, which is $\text{Aut}(K/L)$, which is H . The image of a homomorphism has as many members as the group has, divided by the size of the kernel, so the image has the order of G divided by the order of H . The order of G is the dimension of K over F and the order of H is the dimension of K over L , so that number is the dimension of L over F . The image therefore supplies L with as many automorphisms over F as its dimension. An extension never has more automorphisms than its dimension, so these are all of them: L is Galois over F , and its group, being the whole image, is the collapse of G by H .

Suppose next that L is Galois over F . In a Galois extension the minimal polynomial of an element is the product of x minus the members of its orbit, so the minimal polynomial over F of every element of L splits inside L . Any σ in G sends an element of L to a root of that minimal polynomial, and all those roots lie in L , so $\sigma(L)$ is contained in L . But σ carries a basis of L over F to a basis of $\sigma(L)$, so $\sigma(L)$ has the same dimension over F as L , and a field inside L of the same dimension as L is all of L . So $\sigma(L) = L$. Then $K^{\sigma H \sigma^{-1}} = \sigma(K^H) = K^H$, and since the correspondence matches subgroups

with fields one to one, $\sigma H \sigma^{-1} = H$.

So a field between F and K is Galois over F exactly when its subgroup is normal, and then its group is the collapse of G by that subgroup.

* * *

Where every element commutes with every other, normality cannot fail: $\sigma h \sigma^{-1} = h \sigma \sigma^{-1} = h$, so conjugation moves no element at all, and every subgroup is carried onto itself. In $K = Q(\sqrt[3]{2}, \sqrt[3]{3})$, with G the four sign changes, $\sigma H \sigma^{-1} = H$ for every subgroup, so every field between Q and K is Galois over Q . The group has five subgroups: all of it, three of order 2, and the identity alone. So there are exactly five fields between Q and K , namely Q , $Q(\sqrt[3]{2})$, $Q(\sqrt[3]{3})$, $Q(\sqrt[3]{6})$ and K , and the proof that there are no others never examines a field. It counts the subgroups of a group of order 4.

In the splitting field of $x^3 - 2$, normality does fail. An automorphism is settled by what it does to the roots $\sqrt[3]{2}$, $\omega\sqrt[3]{2}$ and $\omega^2\sqrt[3]{2}$, so different automorphisms rearrange the roots differently. The field is the splitting field of a polynomial with no repeated root and has dimension 6, so it has six automorphisms, and three things have only six rearrangements. So all six rearrangements occur, and the group is the six rearrangements of three things. Its subgroups are the whole group; the three rotations of the roots, of order 3; three subgroups of order 2, each generated by one swap of two roots; and the identity. The list is complete. By Lagrange's theorem a subgroup has 1, 2, 3 or 6 members. A subgroup with three members consists of the powers of an element of order 3; the only such elements are the two rotations other than the identity, and both give the same subgroup. A subgroup with two members consists of the identity and an element of order 2, and the elements of order 2 are the three swaps.

Conjugating the swap of the first two roots by a rotation gives the swap of a different pair, so no subgroup of order 2 is normal, and their fields, $Q(\sqrt[3]{2})$, $Q(\omega\sqrt[3]{2})$ and $Q(\omega^2\sqrt[3]{2})$, one for each root,

are the three fields between Q and K that fail to be Galois over Q . The rotations form a normal subgroup, having only two cosets; its field is $Q(\omega)$, of dimension 2 over Q and Galois as the splitting field of $x^2 + x + 1$, and its group is the collapse by the rotations, a group of two members whose non-identity member sends ω to ω^2 .

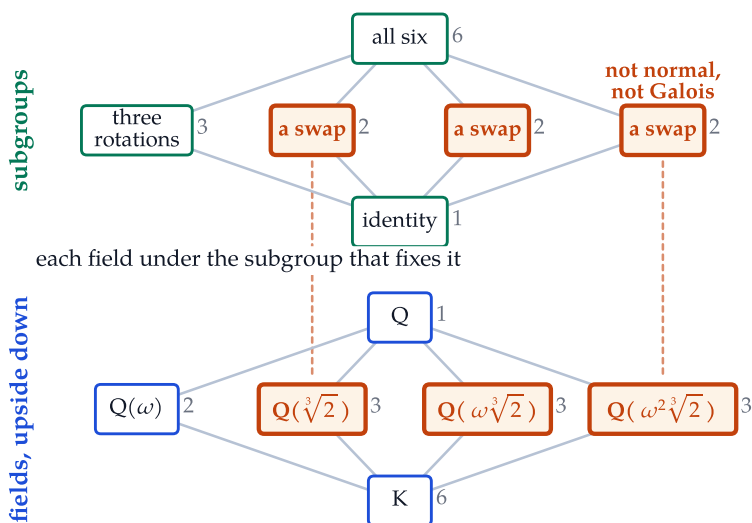


Figure 67.1. The subgroups of the group of the splitting field of $x^3 - 2$, above, and the fields between Q and that splitting field, below, on the same geometry with the field side upside down, so that each field sits under the subgroup that fixes it. The small numbers are orders above and dimensions over Q below, and each pair multiplies to 6. The three subgroups of order 2 are not carried onto themselves by conjugation, and their three fields are exactly the three that are not Galois over Q . *Assumes:* that the two lists are complete, which is the correspondence: six subgroups and no more, hence six intermediate fields and no more, and the second count is never made directly

The failure of $Q(\sqrt[3]{2})$ now has two descriptions. From inside the field, two of the three roots of $x^3 - 2$ are missing. From the group, the subgroup that fixes $Q(\sqrt[3]{2})$ is not carried onto itself by conjugation. They are the same fact, and the second is checked in a table of six rearrangements without computing a single cube root.

Taken on its own, $Q(\sqrt[3]{2})$ over Q also shows why the correspondence needs a Galois extension. Its group has one member and so one subgroup, while there are two fields between Q and $Q(\sqrt[3]{2})$, namely Q and $Q(\sqrt[3]{2})$ itself, and one subgroup cannot be matched with two fields.

* * *

The splitting field of $x^4 - 2$ over Q contains fields that nobody would find by looking at the polynomial. Its roots are $\sqrt[4]{2}$, $i\sqrt[4]{2}$, $-\sqrt[4]{2}$ and $-i\sqrt[4]{2}$, so it is generated by $\sqrt[4]{2}$ and i . The field $Q(\sqrt[4]{2})$ has dimension 4 over Q , because $x^4 - 2$ cannot be factored over the fractions. It has no rational root, so it has no rational factor of degree 1. A rational factor of degree 2 would be $(x - a)(x - b)$ for two of the roots a and b , which is $x^2 - (a + b)x + ab$, so both $a + b$ and ab would be rational; but every pair of the roots has a sum or a product that is not rational. For instance $\sqrt[4]{2}$ and $-\sqrt[4]{2}$ have sum 0 but product $-\sqrt{2}$. Next, $Q(\sqrt[4]{2})$ consists of real numbers and i is not real, so i is not in it; and i is a root of $x^2 + 1$, so adjoining it doubles the dimension to 8. An automorphism is settled by where it sends $\sqrt[4]{2}$, to one of four roots, and where it sends i , to i or to $-i$. That allows at most eight. A splitting field of a polynomial with no repeated root has as many automorphisms as its dimension, which here is 8, so all eight combinations occur. Four roots have twenty-four rearrangements, so only a third of them come from automorphisms.

The eight are exactly the rearrangements that respect negation. An automorphism that sends one root to another must send the negative of the first to the negative of the second. A rearrangement with that property is settled by two choices: where $\sqrt[4]{2}$ goes, to any of the four roots, which also places $-\sqrt[4]{2}$; and where $i\sqrt[4]{2}$ goes, to either of the two roots still free, which also places $-i\sqrt[4]{2}$. That makes eight such rearrangements, and the eight automorphisms give eight different ones, so they give all of them. The other sixteen rearrangements, such as the one exchanging $\sqrt[4]{2}$ with $i\sqrt[4]{2}$

and leaving the other two roots in place, are excluded.

The group can be read off a square. Let r be the automorphism that sends $\sqrt[4]{2}$ to $i\sqrt[4]{2}$ and fixes i , and s the conjugation, which fixes $\sqrt[4]{2}$ and sends i to $-i$. Label the root $i^j\sqrt[4]{2}$ by j , counted modulo 4 because $i^4 = 1$. Take the automorphism that sends $\sqrt[4]{2}$ to $i^k\sqrt[4]{2}$. If it fixes i , it sends $i^j\sqrt[4]{2}$ to $i^j\cdot i^k\sqrt[4]{2}$, so it carries the label j to $j + k$. If it sends i to $-i$, which is i^{-1} , it sends $i^j\sqrt[4]{2}$ to $i^{-j}\cdot i^k\sqrt[4]{2}$, so it carries the label j to $-j + k$.

The four roots are the corners of a square in the plane of pairs. There r is the quarter turn; s is the flip in the real axis, through the corners 0 and 2; r^2s , which carries j to $2 - j$, is the flip through the corners 1 and 3; and rs and r^3s are the flips through the midpoints of opposite sides. So the group is the eight symmetries of a square. That is why it respects negation: the negative of a root is the opposite corner, and a symmetry of a square carries opposite corners to opposite corners.

The turns r and r^3 have order 4, and r^2 and the four flips have order 2, which gives five subgroups of order 2. Take a subgroup of order 4. If it contains r or r^3 , it contains every power of r , and is the four turns. Otherwise its three members other than the identity are chosen from r^2 and the four flips, so at least two of them are flips, say $r^a s$ and $r^b s$. Their product is the turn by $a - b$ quarters. That turn lies in the subgroup, it is not the identity because the two flips are different, and it is not r or r^3 ; so it is r^2 , and $a - b$ is 2. The subgroup is then $1, r^2$, one flip, and the flip two places on, which gives the subgroup of $1, r^2, s$ and r^2s and the subgroup of $1, r^2, rs$ and r^3s . Relabelling the flip $r^k s$ by r gives the flip $r^{k+2}s$, a different flip, which is not in the subgroup of 1 and $r^k s$; so no subgroup generated by a flip is normal. The half turn r^2 commutes with everything, so no relabelling moves it, and the subgroup of 1 and r^2 is normal.

The group of order 8 therefore has ten subgroups: the identity, five of order 2, three of order 4, and the whole. So there are ten fields between Q and the splitting field K . The field of a subgroup of order n has dimension 8 divided by n over Q : the identity gives

K itself, of dimension 8; the five subgroups of order 2 give fields of dimension 4; the three of order 4 give fields of dimension 2; and the whole group gives Q .

The three subgroups of order 4 have two cosets each and so are normal. The four turns fix i , so their field is $Q(i)$. The subgroup made of $1, r^2, s$ and r^2s fixes $\sqrt[4]{2}$, which is the square of $\sqrt[4]{2}$ and survives a change of sign in it, so its field is $Q(\sqrt[4]{2})$. The remaining one fixes $\sqrt{-2}$, the product of i with $\sqrt[4]{2}$, so its field is $Q(\sqrt{-2})$. Each of these three fields has dimension 2 over Q . The fixed field of a subgroup of order 4 also has dimension 2, which is 8 divided by 4, and a field of dimension 2 inside a field of dimension 2 is all of it. That is why finding one element outside Q that the subgroup fixes is enough to name its whole field. Of the five subgroups of order 2 exactly one is normal, the one generated by the automorphism that sends $\sqrt[4]{2}$ to $-\sqrt[4]{2}$ and fixes i ; its field is $Q(\sqrt[4]{2}, i)$, which is Galois over Q . The other four are not normal, and their fields are

$$Q(\sqrt[4]{2}), Q(i\sqrt[4]{2}), Q((1+i)\sqrt[4]{2}), Q((1-i)\sqrt[4]{2})$$

The first two are generated by roots of $x^4 - 2$. The last two are not. Since $(1+i)^2 = 2i$, the fourth power of $1+i$ is -4 , so the fourth power of $(1+i)\sqrt[4]{2}$ is -8 , and $(1+i)\sqrt[4]{2}$ is a root of $x^4 + 8$, a polynomial that appears nowhere in the question; the same holds for $(1-i)\sqrt[4]{2}$. The automorphism sending $\sqrt[4]{2}$ to $i\sqrt[4]{2}$ and i to $-i$, which is rs , fixes $(1+i)\sqrt[4]{2}$: it sends it to $(1-i) \cdot i\sqrt[4]{2}$, and $(1-i) \cdot i = i + 1$.

Such fields are found by adding. Add up the images of one element under a subgroup H . The sum is fixed by H , since a member of H only reorders its terms, so the field the sum generates lies inside the fixed field of H . The fixed field has dimension 8 divided by the order of H . The field generated by the sum has dimension equal to the degree of the minimal polynomial of the sum, and that degree is the number of different images the sum has under the whole group. So whenever the sum has 8 divided by the order of H different images, it generates a field of the full dimension inside the fixed field, which is all of it. For the subgroup of 1 and s ,

the images of $\sqrt[4]{2}$ add up to $2\sqrt[4]{2}$, which has four images, and the field is $Q(\sqrt[4]{2})$. For 1 and rs they add up to $\sqrt[4]{2} + i\sqrt[4]{2}$, which is $(1+i)\sqrt[4]{2}$, again with four images. For 1 and r^2s the images of $\sqrt[4]{2}$ add up to 0, which generates nothing, but the images of $i\sqrt[4]{2}$ add up to $2i\sqrt[4]{2}$, and the field is $Q(i\sqrt[4]{2})$.

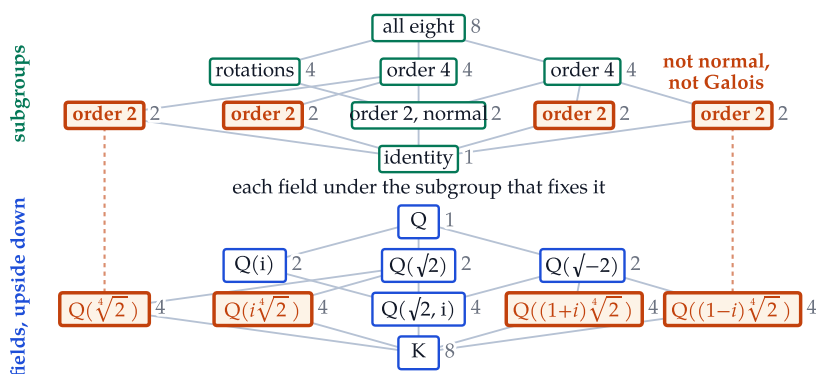


Figure 67.2. Ten subgroups of the group of order 8, above, and the ten fields between Q and the splitting field of $x^4 - 2$ below, drawn on the same geometry with the field side upside down, so that every field sits under the subgroup that fixes it. Rows of one, three, five and one; order times dimension is 8 at every node. The four subgroups in the second colour are not normal and their four fields are not Galois over Q , and two of those fields are generated by no root of $x^4 - 2$. Assumes: that the subgroup list is complete, which is a finite check inside a group of order 8, and that completeness transfers to the field side, which is the theorem and not the drawing

The two fields generated by roots of $x^4 + 8$ are different, although each has dimension 4 over Q . The quotient of $(1-i)\sqrt[4]{2}$ by $(1+i)\sqrt[4]{2}$ is $-i$, so a field containing both would contain i . It would then contain $1+i$, and so $\sqrt[4]{2}$, which is $(1+i)\sqrt[4]{2}$ divided by $1+i$. Containing $\sqrt[4]{2}$ and i , it would be all of K , of dimension 8. Each of the two fields has dimension 4, so neither can contain the other, and they are different fields.

The splitting field of $x^4 + 1$ over Q is generated by any one of its roots. What is its group, and which fields lie between Q and it?

SEVENTEEN POINTS

Seventeen points spaced evenly round a circle can be joined into a regular polygon with a straightedge and a compass. Seven points cannot, and neither can nine.

Put the circle in the plane of pairs, with radius 1 and one corner at 1. The next corner is ζ , the point at angle $2\pi/17$, the corners are the powers of ζ , and $\zeta^{17} = 1$. The coordinates of ζ are the cosine of $2\pi/17$ and the square root of one minus its square, so the polygon can be built exactly when that cosine can. The target will be the double of the cosine, which is $\zeta + \zeta^{-1}$: the point ζ^{-1} is the mirror image of ζ in the horizontal axis, so in the sum the vertical parts cancel and the horizontal parts add. And a number can be built exactly when it lies at the top of a tower of fields, each of them the field before with the square root of one of that earlier field's positive members adjoined.

The route to $\zeta + \zeta^{-1}$ is hidden in the numbers 1 to 16 under multiplication modulo 17. The number 3 generates all of them:

1, 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6

each entry being three times the one before, reduced modulo 17, and all sixteen appearing once. Taking every second entry, then every fourth, then every eighth, gives

$$\{1, 2, 4, 8, 9, 13, 15, 16\}, \{1, 4, 13, 16\}, \{1, 16\}$$

each half the size of the one before, and listed here in increasing order. Each is closed under multiplication modulo 17. The entries in every second place are the even powers of 3, and since $3^{16} \equiv 1$ the list of powers repeats every sixteen places, so a power of 3 lands in an even place exactly when its exponent is even. The product of two even powers of 3 is an even power of 3, so the first set is closed; the same holds for the exponents that are multiples of 4, and for those that are multiples of 8.

Add up the powers of ζ over the first set, and over the eight exponents that remain, which are the first set multiplied by 3:

$$\begin{aligned}\eta_0 &= \zeta + \zeta^2 + \zeta^4 + \zeta^8 + \zeta^9 + \zeta^{13} + \zeta^{15} + \zeta^{16} \\ \eta_1 &= \zeta^3 + \zeta^5 + \zeta^6 + \zeta^7 + \zeta^{10} + \zeta^{11} + \zeta^{12} + \zeta^{14}\end{aligned}$$

Their sum is $\zeta + \zeta^2 + \dots + \zeta^{16}$, the sum of the sixteen roots of $x^{17} = 1$ other than 1. The seventeen roots of $x^{17} - 1$ add to the negative of its coefficient of x^{16} , which is 0. Taking away the root 1 leaves -1 , so $\eta_0 + \eta_1 = -1$. Their product is a sum of sixty-four powers of ζ , one for each choice of an exponent from each set, and among the sixty-four exponent sums, taken modulo 17, each of the numbers 1 to 16 occurs exactly four times and 0 never occurs. To see it, write S for the first set and N for the second, which is 3 times S , and let $c(k)$ be the number of pairs made of an a in S and a b in N with $a + b \equiv k$. First, $c(0) = 0$: if $a + b \equiv 0$ then $b \equiv 16a$, and 16 is in S , which is closed under multiplication, so b would be in S .

Second, take a member h of S and multiply both a and b by it. Since S is closed under multiplication, ha is in S and hb , which is 3 times a member of S , is in N ; and the sum $a + b \equiv k$ becomes $ha + hb \equiv hk$. So every pair counted by $c(k)$ gives a pair counted by $c(hk)$, different pairs give different pairs, and multiplying by the inverse of h , which is also in S , leads back. So $c(hk) = c(k)$. Third, send the pair (a, b) to $(3b, 3a)$. The first entry $3b$ is in $3N$, which is $9S$, and that is S because 9 is in S ; the second entry $3a$ is in $3S$, which is N . So the new pair is again a member of S with a member of N , its sum is $3b + 3a \equiv 3k$, and dividing by 3 leads back; so $c(3k) = c(k)$.

Every number from 1 to 16 is h or $3h$ for some h in S . The second step gives $c(h) = c(1)$, and the third then gives $c(3h) = c(h) = c(1)$. So c takes one value on all sixteen, and sixty-four pairs shared among sixteen numbers give 4 each. The set S has been met before: it is the pattern of acquaintance among seventeen people, which doubling kept and tripling exchanged with the strangers. So the product $\eta_0\eta_1$ contains each of $\zeta, \zeta^2, \dots, \zeta^{16}$ four times, which makes it four times the sum of all sixteen, and that sum was found to be -1 . So $\eta_0\eta_1 = 4 \cdot (-1) = -4$. Two numbers with sum -1 and product -4 are the roots of

$$x^2 + x - 4 = 0$$

namely $(-1 + \sqrt{17})/2$, about 1.5616, and $(-1 - \sqrt{17})/2$, about -2.5616 . One square root has been taken.

Split again. Inside η_0 , the exponents $\{1, 4, 13, 16\}$ and the remaining four, $\{2, 8, 9, 15\}$, give two sums of four powers whose sum is η_0 . Their product is a sum of sixteen powers of ζ , and the sixteen exponent sums $c + d$, with c from the first set and d from the second, are the numbers 1 to 16 modulo 17, each exactly once: modulo 17 the numbers 16 and 13 are -1 and -4 , and 15 and 9 are -2 and -8 , so the first set is ± 1 and ± 4 and the second is ± 2 and ± 8 . The sums are $\pm 1 \pm 2$, $\pm 1 \pm 8$, $\pm 4 \pm 2$ and $\pm 4 \pm 8$, which come to $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 7, \pm 9$ and ± 12 . Modulo 17 the last two are ∓ 8 and ∓ 5 , so the list is $\pm 1, \pm 2$, and so on up to ± 8 : sixteen different numbers modulo 17 and none of them 0. So the product is -1 , and the two sums are the roots of $x^2 - \eta_0x - 1 = 0$, about 2.0495 and -0.4879 . The same split inside η_1 gives the sums over $\{3, 5, 12, 14\}$ and $\{6, 7, 10, 11\}$, about 0.3442 and -2.9057 .

Split a third time. Inside the sum over $\{1, 4, 13, 16\}$, the exponents pair as $\{1, 16\}$ and $\{4, 13\}$. The sums $\zeta + \zeta^{16}$ and $\zeta^4 + \zeta^{13}$ add up to that four-term sum, and their product is $\zeta^5 + \zeta^{14} + \zeta^3 + \zeta^{12}$, which is the sum over $\{3, 5, 12, 14\}$ from the other branch. So $\zeta + \zeta^{16}$ and $\zeta^4 + \zeta^{13}$ are the roots of a quadratic whose coefficient of x is the negative of the four-term sum and whose constant term is the sum over $\{3, 5, 12, 14\}$. The four-term sum came out of the second

square root. The sum over $\{3, 5, 12, 14\}$ came from the split of η_1 , by way of $x^2 - \eta_1 x - 1 = 0$, and that seems to ask for one more square root, that of the discriminant $\eta_1^2 + 4$. It is not a new one. Since $\eta_0 + \eta_1 = -1$ and $\eta_0 \eta_1 = -4$, $\eta_0^2 + \eta_1^2 = (\eta_0 + \eta_1)^2 - 2\eta_0 \eta_1 = 9$ and $\eta_0^2 \eta_1^2 = 16$, so $(\eta_0^2 + 4)(\eta_1^2 + 4) = 16 + 4 \cdot 9 + 16 = 68$. The square root of $\eta_0^2 + 4$ was taken in the second split, and the square root of $\eta_1^2 + 4$ is $\sqrt{68}$, which is $2\sqrt{17}$, divided by it. Both coefficients have been reached with the two square roots already taken. In rounded form the quadratic is

$$x^2 - 2.0495x + 0.3442 = 0$$

with roots about 1.8649 and 0.1845. The larger root is $\zeta + \zeta^{16}$, which is $\zeta + \zeta^{-1}$, and twice the cosine of $2\pi/17$ is 1.86494.

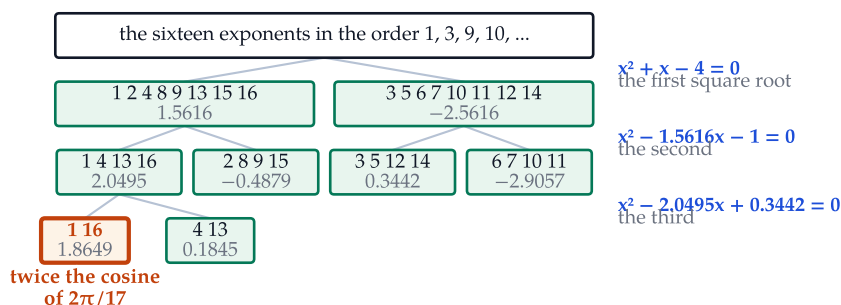


Figure 68.1. The sixteen exponents 1 to 16, arranged in the order given by the powers of 3, split in half three times. At each split the two sums of powers of ζ have a sum and a product lying in the stage before, so each is found by solving one quadratic. Three quadratics, three square roots, and the last number is twice the cosine of $2\pi/17$. *Assumes:* that the sums and products named really do lie in the stage before, which the text computes; the picture records the splitting and the numbers, not the reason

Every one of these quadratics has real coefficients and two real roots, so each step adjoins the square root of a positive number already reached. Three square roots, and the seventeen-sided polygon can be built. Gauss found this in March 1796, a month before his nineteenth birthday.

Five points need a single split. With ζ at angle $2\pi/5$, the number 2 generates 1 to 4 modulo 5 as 1, 2, 4, 3, the sums $\zeta + \zeta^4$ and $\zeta^2 + \zeta^3$ add to -1 , and their product is $\zeta^3 + \zeta^4 + \zeta + \zeta^2$, which is -1 as well. So they are the roots of $x^2 + x - 1 = 0$, and twice the cosine of 72° is the positive root, $(\sqrt{5} - 1)/2$, in agreement with the value found from the angle identities.

Seven points show the halving come to a stop. With ζ at angle $2\pi/7$, the number 3 generates 1 to 6 modulo 7 as 1, 3, 2, 6, 4, 5, and the sets closed under multiplication, apart from $\{1\}$ and all six numbers, are $\{1, 2, 4\}$ and $\{1, 6\}$. The first cuts the six exponents into two blocks of three, itself and $\{3, 5, 6\}$. The second cuts them into three blocks of two, $\{1, 6\}$, $\{2, 5\}$ and $\{3, 4\}$. A chain from all six down to $\{1\}$ either drops there in one step, which makes six blocks, or passes through one of the two. Through $\{1, 2, 4\}$ it halves once and then has to go from three to one; through $\{1, 6\}$ its first step already makes three blocks. So no chain from six down to one can have all its steps of two blocks, and every chain that is not a single jump contains a step of three.

The first split goes as before: $A = \zeta + \zeta^2 + \zeta^4$ and $B = \zeta^3 + \zeta^5 + \zeta^6$ add to -1 , and among their nine exponent sums 0 occurs three times and each of 1 to 6 once, so $AB = 3 - 1 = 2$, and A and B are the roots of $x^2 + x + 2 = 0$, which are $(-1 \pm \sqrt{-7})/2$. They are not real, which is why the argument that builds polygons passes to a field of real numbers before it halves.

The real sums are $y_1 = \zeta + \zeta^6$, $y_2 = \zeta^2 + \zeta^5$ and $y_3 = \zeta^3 + \zeta^4$, twice the cosines of $2\pi/7$, $4\pi/7$ and $6\pi/7$, about 1.2470, -0.4450 and -1.8019 . They add to -1 ; multiplying out gives $y_1y_2 = y_1 + y_3$, $y_2y_3 = y_1 + y_2$ and $y_1y_3 = y_2 + y_3$, which add to -2 ; and $y_3^2 = y_1 + 2$, so $y_1y_2y_3 = (y_1 + y_3)y_3 = y_2 + y_3 + y_1 + 2 = 1$. Multiplied out, $(y - y_1)(y - y_2)(y - y_3)$ has the negative of the sum of the three as its coefficient of y^2 , the sum of their products in pairs as its coefficient of y , and the negative of their product as its constant term. So they are the roots of

$$y^3 + y^2 - 2y - 1 = 0$$

which with $y = 2c$ is the cubic $8c^3 + 4c^2 - 4c - 1$ that refused the seven-sided polygon. The forced step of three is that cubic. All three roots lie in the field of any one of them, since squaring and subtracting 2 sends y_1 to y_2 , y_2 to y_3 and y_3 back to y_1 .

Does 3 generate the numbers 1 to 256 under multiplication modulo 257, and are the exponents in the first of the two sums of 128 powers, in the first split for the regular 257-gon, exactly the squares modulo 257? For a prime p in general, which quadratic do the sum of ζ^k over the nonzero squares k modulo p and the sum over the other nonzero k satisfy, and for which p are the two sums not real?

* * *

Why did the halving work, and for which other n does it work? Each set of exponents was closed under multiplication modulo 17, and each sum of powers was unchanged when every exponent in it was multiplied by a member of its own set. That is a statement about the automorphisms of the field generated by ζ , and it can be made for any n .

Fix n at least 3 and let ζ be the point at angle $2\pi/n$, so that its powers are the n roots of $x^n = 1$. The field $Q(\zeta)$ is the splitting field of $x^n - 1$, which has no repeated root, because its derivative nx^{n-1} vanishes only at 0 and 0 is not a root; so $Q(\zeta)$ is Galois over Q . An automorphism σ sends ζ to a root of $x^n - 1$, say ζ^k , and σ is settled by k , since every element of $Q(\zeta)$ is a polynomial in ζ . The exponent k shares no factor with n . If k and n shared a factor d above 1, then $\sigma(\zeta^{n/d}) = \zeta^{kn/d} = 1$, while $\zeta^{n/d}$ is not 1; but σ sends only 1 to 1. Performing the automorphism with exponent j and then the one with exponent k gives the one with exponent kj . So the group of $Q(\zeta)$ over Q is matched with a subgroup of the numbers modulo n that share no factor with n , under multiplication. Every two of its members commute, a group with that property being called *abelian*, and its order divides the count of those numbers, which is $\phi(n)$, the number of families with partners modulo n .

Conjugation sends ζ to ζ^{-1} , which is ζ^{n-1} , so it carries $Q(\zeta)$ onto

itself and is one of these automorphisms; call it τ . Since ζ is not real, τ is not the identity, and the identity and τ form a subgroup of order 2. Its fixed field K^+ consists of members of $Q(\zeta)$ that conjugation leaves alone, which are real numbers, and it contains $\zeta + \zeta^{-1}$. Every subgroup of an abelian group is normal, so by the theorem on normal subgroups K^+ is Galois over Q , and its group is the collapse of the group of $Q(\zeta)$ by the subgroup of order 2, again abelian, with an order dividing $\varphi(n)$.

Suppose the group of K^+ over Q has a chain of subgroups, each of index 2 in the one before, ending at the identity. The correspondence turns the chain into a tower of fields from Q up to K^+ , each of dimension 2 over the one below. A field of dimension 2 over F is $F(\alpha)$ for any α outside F ; that α satisfies a quadratic over F , and completing the square writes it as $p + q\sqrt{w}$ with p, q and w in F . All these fields consist of real numbers, so w is positive. So $\zeta + \zeta^{-1}$ lies at the top of a tower of square roots of positive numbers, and the n -gon can be built.

Such a chain exists whenever the group has a power of 2 for its order, and the reason is linear algebra over the field with two elements. Let G be abelian with 2^m members, m at least 1, written with its operation as addition, and let $2G$ be the collection of doubles, which is a subgroup because G is abelian. It is not all of G , because doubling sends two different elements to the identity. Take any g other than the identity. By Lagrange's theorem its order is a power of 2, say 2^r with r at least 1. The multiple $2^{r-1}g$ is then not the identity, but its double $2^r g$ is. So that multiple and the identity both double to the identity, and doubling is not one-to-one. A map from a finite set to itself that is not one-to-one cannot be onto, so some member of G is not a double. So the collapse $G/2G$ has more than one member, and each of its members doubles to zero.

An abelian group in which everything doubles to zero is a space over $Z/2$, with the group operation as addition and 0 and 1 as the scalars, so it has a basis, of some length d at least 1. The span of all but one of the basis elements has 2^{d-1} of the 2^d members, so it is a subgroup of index 2 in $G/2G$. The members of G whose image in

$G/2G$ lies in that span form a subgroup of G , and it holds exactly half of G , since it is made of half of the equal-sized classes of the collapse. So G has a subgroup of index 2. That subgroup is again abelian with a power of 2 for its order, and repeating the argument descends to the identity.

So whenever $\varphi(n)$ is a power of 2, the regular n -gon can be built.

* * *

Showing that a polygon cannot be built needs the dimension of $Q(\zeta)$ over Q , and for that the irreducibility of a polynomial. It is enough to treat $n = p^e$, a power of a prime. Put $m = p^{e-1}$, $y = x^m$ and $\Phi = 1 + y + y^2 + \dots + y^{p-1}$. Then $(y - 1)\Phi = y^p - 1 = x^n - 1$, and ζ , the point at angle $2\pi/n$, is a root of Φ , since it is a root of $x^n - 1$ and not of $y - 1$. The degree of Φ is $(p - 1)p^{e-1}$, which is $\varphi(n)$.

Eisenstein's criterion does not apply to Φ , but it applies to $\Phi(x + 1)$. The prime p divides $(p \text{ choose } k)$ for every k from 1 to $p - 1$: that is the fact about the interior of row p of the triangle of choices which made raising to the power p an automorphism. So $(A + B)^p$ and $A^p + B^p$ differ by p times a polynomial with whole-number coefficients, and, applying this again and again, $(x + 1)^{p^j} \equiv x^{p^j} + 1$ modulo p for every j .

Now replace x by $x + 1$ in $(y - 1)\Phi = x^n - 1$ and reduce modulo p . Both m and n are powers of p , so the congruence applies to each. The factor $y - 1$, which is $x^m - 1$, becomes $(x + 1)^m - 1$, and that reduces to $x^m + 1 - 1 = x^m$. The right side becomes $(x + 1)^n - 1$, which reduces to x^n . So x^m times $\Phi(x + 1)$ reduces to x^n . Multiplying by x^m moves every coefficient up m places, so $\Phi(x + 1)$ itself reduces to x^{n-m} , and $n - m = p^e - p^{e-1}$ is the degree $\varphi(n)$ of Φ . So $\Phi(x + 1)$ reduces to $x^{\varphi(n)}$, its leading term alone: every coefficient of $\Phi(x + 1)$ below the leading one is divisible by p . Its constant term is $\Phi(1)$, which is p , and p^2 does not divide it. So $\Phi(x + 1)$ is irreducible over Q by Eisenstein's criterion. Then Φ is irreducible too: replacing x by $x + 1$ throughout a factorisation of Φ would give a factorisation of $\Phi(x + 1)$. For seven points $\Phi(x + 1)$ is $x^6 +$

$7x^5 + 21x^4 + 35x^3 + 35x^2 + 21x + 7$, and for nine, where Φ is $x^6 + x^3 + 1$, it is $x^6 + 6x^5 + 15x^4 + 21x^3 + 18x^2 + 9x + 3$.

So $Q(\zeta)$ has dimension $\varphi(n)$ over Q when n is a power of a prime. Over the field $Q(c)$ of real numbers generated by $c = \zeta + \zeta^{-1}$, the number ζ is a root of $x^2 - cx + 1$ and is not real, so $Q(\zeta)$ has dimension 2 over $Q(c)$, and $Q(\cos 2\pi/n)$ has dimension $\varphi(n)/2$ over Q : 3 for seven points and for nine, 5 for eleven, 6 for thirteen, 10 for twenty-five. A number that can be built lies in a field whose dimension over Q is a power of 2, and by the tower law the dimension of $Q(\cos 2\pi/n)$ divides that. Now suppose the regular n -gon can be built, and let p^e be the exact power of an odd prime p that divides n . Joining every (n/p^e) -th corner builds the regular polygon with p^e sides, so $p^{e-1}(p-1)/2$, the dimension of the field of its cosine, is a power of 2. The odd number p^{e-1} divides a power of 2 only when it is 1, which forces $e = 1$, and then $(p-1)/2$ is a power of 2 and so is $p-1$. With the count $\varphi(n)$ worked out next, that says: the regular n -gon can be built exactly when $\varphi(n)$ is a power of 2.

For seventeen points the dimension is 16, so the group is all sixteen exponents, and $\zeta, \zeta^2, \dots, \zeta^{16}$ are independent over Q , since a vanishing combination of them, divided by ζ , would be one of $1, \zeta, \dots, \zeta^{15}$, which form a basis. The automorphism with exponent j sends the sum of ζ^k over a set H of exponents closed under multiplication to the sum over the set jH , and by independence two such sums are equal only when they run over the same exponents. The sets jH are H and the other blocks into which H cuts the sixteen exponents, and there are 16 divided by the size of H of them. So the sum has exactly that many different images, and it is fixed by the members of H and by no other automorphism. Its images are all roots of its minimal polynomial, so the field it generates has dimension at least 16 divided by the size of H . That field lies inside the fixed field of H , whose dimension is exactly that number, so the sum generates the fixed field of H .

The three sets found by taking every second, fourth and eighth power of 3 are a chain of subgroups; the fields they fix are generated by η_0 , by the sum over $\{1, 4, 13, 16\}$ and by $\zeta + \zeta^{-1}$, with

minimal polynomials $x^2 + x - 4$, $x^4 + x^3 - 6x^2 - x + 1$ and one of degree 8; and the three quadratics solved at the start were the three steps of the correspondence up that chain.

For which n is $\varphi(n)$ a power of 2? Two facts about the count, both found when $\varphi(n)$ was first met, settle it. If m and k share no factor, then $\varphi(mk) = \varphi(m)\varphi(k)$. Send a number modulo mk to the pair of its remainders modulo m and modulo k . This is one-to-one, since a difference divisible by m and by k is divisible by mk . There are mk numbers and mk pairs, so it is also onto. A number shares no factor with mk exactly when it shares none with m and none with k , so the numbers counted by $\varphi(mk)$ are matched with the pairs made of a number counted by $\varphi(m)$ and one counted by $\varphi(k)$, and there are $\varphi(m)\varphi(k)$ such pairs. And $\varphi(p^e) = p^{e-1}(p-1)$ for a prime p , since a number shares a factor with p^e exactly when p divides it, and among the numbers from 1 to p^e those are the p^{e-1} multiples of p , leaving $p^e - p^{e-1} = p^{e-1}(p-1)$.

Splitting n into its prime powers, the two facts make $\varphi(n)$ the product of the numbers $p^{e-1}(p-1)$, one for each prime power p^e in n . The power of 2 in n contributes a power of 2. An odd prime contributes a power of 2 exactly when $e = 1$ and $p-1$ is a power of 2. So $\varphi(n)$ is a power of 2 exactly when every odd prime that divides n divides it only once and is one more than a power of 2. A prime of the form $2^m + 1$ needs m itself to be a power of 2: if m had an odd factor b greater than 1, then $2^{m/b} + 1$, which lies strictly between 1 and $2^m + 1$, would divide $2^m + 1$, because $x + 1$ divides $x^b + 1$ when b is odd.

So the polygons that can be built are exactly those whose number of sides is a power of 2 times a product of distinct primes of the form $2^{2^j} + 1$. Up to 100 they are the polygons with 3, 4, 5, 6, 8, 10, 12, 15, 16, 17, 20, 24, 30, 32, 34, 40, 48, 51, 60, 64, 68, 80, 85 and 96 sides, and no others: the polygons with 18 and with 45 sides are refused because 9 divides both. The first five numbers of that form, 3, 5, 17, 257 and 65537, are prime. The sixth, 4294967297, is 641×6700417 , as Euler found in 1732, and no prime of that form beyond 65537 has ever been found.

OTHER INSTRUMENTS

The buildable polygons have been sorted, and the sorting is not a fact about polygons. It is a fact about a straightedge and a compass. Those two instruments produce lines and circles. Lines and circles meet in points whose coordinates satisfy equations of the first and second degree. A tower of second-degree steps has dimension a power of 2, and that is why $\varphi(n)$ had to be a power of 2 and why 3 could never divide the dimension. The whole obstruction is the number 2. Change the instrument and the obstruction changes with it.

That other instruments reach further was known before it could be proved. Archimedes trisected any angle with two marks on his ruler, by the construction already set against the refusal of the trisection when the tools were first weighed. Here it is again, with A and B exchanged to match the figure. Draw a circle of radius r about the corner O of the angle AOB , with A and B on the circle, and put two marks on the ruler a distance r apart. Lay the ruler through A and slide it until one mark C rests on the circle and the other mark D on the line BO extended beyond O . Then the angle at D is a third of the angle at O . The triangle DCO is isosceles, since DC and CO are both r , so its angles at D and at O are equal; call each of them α . The angle OCA is the outside angle of that triangle at C , so it is $\alpha + \alpha = 2\alpha$. The triangle OCA is isosceles too, since OC

and OA are radii, so the angle OAC is also 2α . Last, the angle AOB is the outside angle at O of the triangle DOA , so it equals the angles at D and A added, $\alpha + 2\alpha = 3\alpha$. A placement of that kind, found by moving something until it fits rather than by crossing two named curves, is called a *neusis*, a verging; Euclid's straightedge has no marks and allows none. Nicomedes, about 200 BC, built a mechanism for the sliding, named the curve it traces the conchoid, and doubled the cube with it as well.

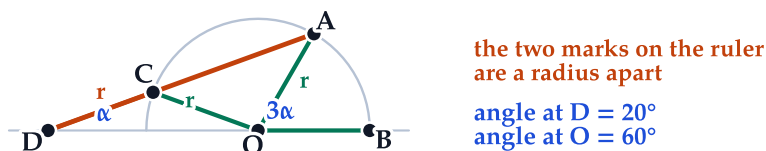


Figure 69.1. Trisection with two marks on the ruler. The circle has radius r and centre O , and the angle to be cut is the angle AOB . The ruler is laid through A and slid until the two marks, a distance r apart, fall at C on the circle and at D on the line BO produced. Then the angle at D is a third of the angle at O , by the two isosceles triangles and the exterior angle of the third. *Assumes:* the angle drawn is 60 degrees, so the angle at D is 20; beyond 135 degrees it is the farther crossing of the ruler with the circle that carries the mark

One step of that proof was passed over, and it repays a look. Keep α for the angle at D , and take the radius as 1. The ruler runs from D through two points of the circle, at distances $4\cos^2\alpha - 1$ and 1 from D ; the first is A and the second is the mark C . The step that read the angle OCA as the exterior angle of the triangle DCO assumed that C lies between D and A . That holds while $4\cos^2\alpha - 1$ exceeds 1. In angles, it holds while α is below 45 degrees, so while the angle at O is below 135. At exactly 135 degrees the two points fall together and the ruler touches the circle. Beyond 135 degrees A lies between D and C . Then the angle OCA is 180 less 2α instead of 2α , and it is the angle OAD that comes to 2α . The angle at O is still the sum $2\alpha + \alpha$. So the conclusion holds for every angle up to two right angles, and only the wording of that one step has to change.

* * *

The second instrument is a pair of conic sections. Menaechmus, about 350 BC, is said to have invented the parabola and the hyperbola for exactly this purpose. Ask for two numbers x and y with

$$x^2 = y \text{ and } xy = 2$$

The first is a parabola, the second a hyperbola. Where they cross, $x^3 = x \cdot x^2 = xy = 2$, so x is the cube root of 2. Two parabolas do it as well: $y = x^2$ and $x = y^2/2$ give $x = x^4/2$, so $x^3 = 2$. What the Greeks could draw, they could solve: the cube was doubled within a century of the problem's being posed, by curves that are not circles, and the objection was only that Euclid's two instruments could not reach the answer.

The same pair of curves reaches every cubic and every quartic. For $x^3 + ax^2 + bx + c = 0$ set $y = x^2$. Then $x^3 = x \cdot x^2 = xy$ and $ax^2 = ay$, so the cubic becomes $xy + ay + bx + c = 0$, and the pair is

$$y = x^2 \text{ and } xy + ay + bx + c = 0$$

a parabola and a second conic, and their real crossings give the real roots, which is all a drawn curve can give; the roots that are not real lie outside the picture, as they did for the tangent method. For a quartic $x^4 + ax^3 + bx^2 + cx + d = 0$ the same substitution, with $x^4 = y^2$, $x^3 = xy$ and $x^2 = y$, gives $y^2 + axy + by + cx + d = 0$, again a conic, so one parabola and one conic settle the quartic too. Descartes made this the organising idea of his *Geometry* in 1637: a problem is of the degree of the curves it needs, and the conics are the instruments of the third and fourth degrees.

* * *

The third instrument is a sheet of paper. A fold makes a crease, and a crease is a line; the paper may be folded so that a marked point lands on a marked line, or so that a marked line lands on

another. The crease of a fold carrying a point P to a point Q is the perpendicular bisector of PQ , so a fold is a reflection, and what the paper can do is decided by which reflections it can make. Most of the folds do no more than a straightedge and a compass would: a crease through two marked points, a crease bisecting an angle, and a crease carrying a point onto a marked line. That last fold can be made in many ways, and its possible creases are the tangents of a certain parabola.

One fold does more. Ask for a single crease that carries a marked point P_1 onto a marked line ℓ_1 and at the same time carries a marked point P_2 onto a marked line ℓ_2 . Beloch found in 1936 what such a fold performs, and the computation is short. Let the crease be the line $y = mx + k$, and write

$$d = (v - mu - k)/(1 + m^2)$$

for a point (u, v) . The reflection of (u, v) in the crease is $(u + 2md, v - 2d)$. To see why, look at the step from the point to its image. It is $2d$ times the vector $(m, -1)$, which has slope $-1/m$ and so crosses the crease at right angles. Halfway along the step is the point $(u + md, v - d)$, and substituting it into $y = mx + k$ shows that it lies on the crease. A step at right angles to the crease with its midpoint on the crease is what a reflection in the crease does. As a check on the length: the vector $(m, -1)$ has length $\sqrt{1 + m^2}$, and d is the distance from the point to the crease divided by the same $\sqrt{1 + m^2}$, so the step covers exactly twice the distance.

Take $P_1 = (a, 1)$ and ℓ_1 the line $y = -1$. The image of P_1 has second coordinate $1 - 2d$, and for that to be -1 the number d must be 1. For this point d is $(1 - ma - k)/(1 + m^2)$, so $1 - ma - k = 1 + m^2$, giving

$$k = -am - m^2$$

Take $P_2 = (c, b)$ and ℓ_2 the line $x = -c$. The image of P_2 has first coordinate $c + 2md'$, with $d' = (b - mc - k)/(1 + m^2)$. For that to be $-c$ we need $2c + 2md' = 0$, that is $c + md' = 0$, and multiplying

by $1 + m^2$ gives $c(1 + m^2) + m(b - mc - k) = 0$. The terms in m^2c cancel, and what is left is $c + mb - mk = 0$, so

$$k = b + c/m$$

Setting the two values of k equal and multiplying by m gives $-am^2 - m^3 = bm + c$, that is

$$m^3 + am^2 + bm + c = 0$$

The slope of the crease is a root of the cubic, and every cubic with real coefficients has a real root, so the fold exists. The paper solves cubics.

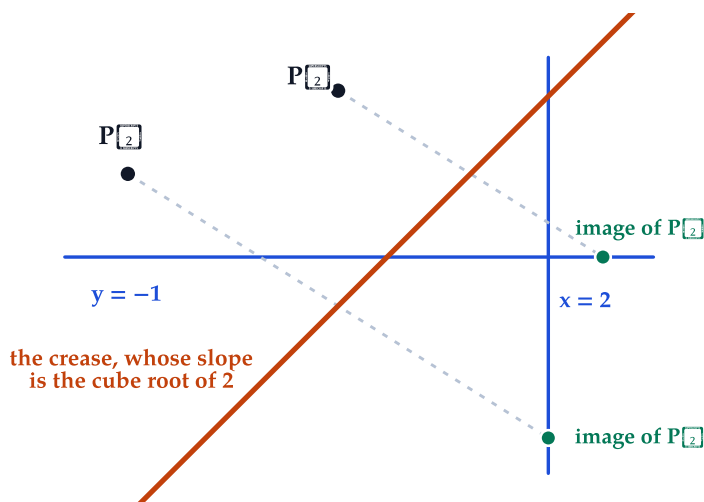


Figure 69.2. One fold of the paper carrying P_1 to the lower line and P_2 to the right-hand line at the same time. The crease is the perpendicular bisector of each point and its landing place, and for $P_1 = (0, 1)$ with the line $y = -1$, and $P_2 = (-2, 0)$ with the line $x = 2$, the slope of the crease comes to the cube root of 2, which no straightedge and compass can build. *Assumes:* the crease drawn at slope 1.2599 and height -1.5874 ; the two dotted segments meet the crease at right angles and are bisected by it

Doubling the cube is the case $a = b = 0$ and $c = -2$: fold $(0, 1)$ onto the line $y = -1$ and $(-2, 0)$ onto the line $x = 2$, and the crease

has slope the cube root of 2, which is 1.2599. Trisecting 60 degrees is the cubic $8x^3 - 6x - 1 = 0$, since the cosine x of 20 degrees satisfies $4x^3 - 3x = 1/2$, the cosine of 60 degrees, and dividing by 8 gives $a = 0$, $b = -3/4$ and $c = -1/8$: fold $(0, 1)$ onto $y = -1$ and $(-1/8, -3/4)$ onto $x = 1/8$, and one of the three creases that do it has slope 0.9397, the cosine of 20 degrees; the other two have for slopes the cosines of 140 and of 260 degrees.

* * *

How far does the paper reach? Every fold except the double fold is settled by conditions of the first or second degree in the coefficients of its crease. For the double fold the elimination above works whatever the two points and the two lines. Each of the two landing conditions is of the second degree in the slope and the intercept of the crease. Removing the intercept between them leaves a cubic in the slope, of which the equation above is one case. So a new point's coordinates lie in a field of dimension 1, 2 or 3 over the field of the numbers already marked. Running the folds in order builds a tower in which every step has dimension 2 or 3. Conversely every such tower can be climbed, a step of dimension 2 by the folds that do the work of a straightedge and compass and a step of dimension 3 by a Beloch fold on the cubic that the step's generator satisfies. So

*a number can be folded exactly when it lies at the top of a tower of fields,
each of dimension 2 or 3 over the one before*

which Beloch's construction and the dimension count together establish, and which Alperin stated in this form in 2000. The dimension of a folded number over the fractions is therefore a product of 2s and 3s, and no other prime can appear.

The consequence for polygons follows the pattern just settled for the compass. The corners of a regular n -gon need the field of $\cos(2\pi/n)$, whose dimension was found to be $\varphi(n)/2$ when n is a power of a prime. Joining every corner at a fixed step reduces any

n to the prime powers that divide it, so the prime powers decide the case. With the straightedge and compass that dimension had to be a power of 2, and the primes dividing n , apart from 2, had to be one more than a power of 2. With paper the dimension may have 3s in it. For a prime p above 3, $\varphi(p)$ is $p - 1$, so $p - 1$ may contain only 2s and 3s; and p may divide n only once, since $\varphi(p^2)$ is $p(p - 1)$ and has p itself as a factor. So the primes allowed are those one more than a product of a power of 2 and a power of 3:

2, 3, 5, 7, 13, 17, 19, 37, 73, 97, 109, 163, 193, 257, 433, 487, 577, ...

called *Pierpont primes*, after the American who listed them in 1895. A regular n -gon can be folded exactly when n is a power of 2 times a power of 3 times distinct Pierpont primes. Below 100 the compass allows the twenty-four polygons already listed, and the paper allows fifty-nine, adding

7, 9, 13, 14, 18, 19, 21, 26, 27, 28, 35, 36, 37, 38, 39, 42, 45, 52, 54, 56,
57, 63, 65, 70, 72, 73, 74, 76, 78, 81, 84, 90, 91, 95, 97

The heptagon, refused by the descent on cubics when the tools were first weighed, is folded in a few creases, and so is the nine-sided polygon refused beside it. The eleven-sided polygon is refused by both, since $\varphi(11)$ is 10 and the 5 in it is neither a 2 nor a 3; the first polygons that no fold reaches are those with 11, 22, 23 and 25 sides.

* * *

So the three famous problems sort themselves by the arithmetic of the instrument. Squaring the circle is refused by every one of these instruments, and by every instrument whose steps are algebraic, since π satisfies no polynomial equation with fraction coefficients. Doubling the cube and trisecting the angle are refused by the straightedge and compass and granted by the marked ruler, the conics and the paper. For the paper the reason is that its steps

may have dimension 3 as well as 2; for the conics, whose crossings give the roots of cubics and quartics, it is that their steps may have dimension 3 or 4 as well as 2.

What neither the conics nor the paper reaches is the fifth degree: a root of a quintic that does not factor has dimension 5 over the fractions. A number reached by a chain of steps of dimension 2, 3 or 4 lies in a field whose dimension is a product of those numbers, which has no prime factor but 2 and 3, and its own dimension must divide that product. Since 5 is neither a 2 nor a 3, no such chain reaches the root, however many steps are allowed. The marked ruler is another matter, since its sliding can lead to equations of higher degree: in 2014 Benjamin and Snyder showed that a marked ruler and a compass build the regular eleven-sided polygon, whose cosine has dimension 5. Whether such an equation can be solved by radicals is a different question, a radical tower allowing steps of every degree, and its obstruction is not the dimension of a field but the shape of a group; the instruments stop being the subject long before the equations do.

Find the pair of conics whose crossings give the roots of $x^3 - 2x - 5 = 0$, and the real root to four places. How many crossings are there, and how many roots?

Which cubic does the fold with $P_1 = (0, 1)$ on $y = -1$ and $P_2 = (-1, -1)$ on $x = 1$ solve, and what is the slope of its crease?

A fold can also take a marked point to a marked line in infinitely many ways. Which curve do the creases of all such folds touch, and why?

For which n below 200 does the regular n -gon need a fold rather than a compass, and which is the first n above 100 that no fold reaches?

The seventeen-sided polygon needs three square roots and no cube root. How many steps of dimension 3, and so how many double folds, do the nine-sided and the thirteen-sided polygons need?

EVERY CYCLOTOMIC POLYNOMIAL

The seventeen points needed the dimension of $\mathbb{Q}(\zeta)$ over $\mathbb{Q}$ for a prime number of points, and the refusal of the nine-sided and the eighteen-sided polygons needed it for a power of a prime. Both came from the irreducibility of one polynomial, proved by Eisenstein's criterion after the substitution of $x + 1$ for x . For a number of points that is not a power of a prime, like 12 or 15, the dimension, and with it the exact group of the field, was never settled; the sorting of the buildable polygons went round it, through the multiplicative count $\varphi(n)$ and the remark that a buildable polygon makes the polygon of each prime-power divisor buildable. The answer is that the group is as large as it could be. An automorphism must send ζ to a point of the same order, and there are only $\varphi(n)$ of those, so the group has at most $\varphi(n)$ members; in fact it has all of them. The group is the whole group of the numbers modulo n that share no factor with n , and its size is $\varphi(n)$. Gauss proved it for primes in 1801, Kronecker for every n in 1854, and the proof that follows is Dedekind's, from 1857.

Let ζ be the point at angle $2\pi/n$. Its n powers are the roots of $x^n - 1$, the n points equally spaced round the circle, and each of them has an *order*, the least positive exponent that brings it to 1. The order

of ζ^k is n divided by the greatest common divisor of k and n , by the argument that counted the families of each order modulo a prime, which used nothing but the rule for powers. So the order of every root divides n . For a divisor d of n , the roots of order d are the ζ^k for which k and n have greatest common divisor n/d . Those k are the numbers $j \cdot n/d$ with j from 1 to d sharing no factor with d , and there are exactly $\phi(d)$ of them, so there are exactly $\phi(d)$ roots of order d . Adding up, n is the sum of $\phi(d)$ over the divisors d of n , which for 12 reads $1 + 1 + 2 + 2 + 2 + 4$.

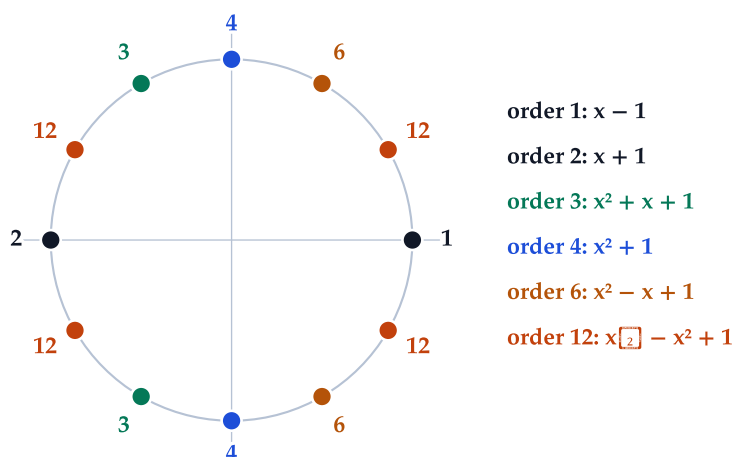


Figure 70.1. The twelve roots of $x^{12} - 1$, each marked with its order, the least power that brings it back to 1. There are $\phi(d)$ of order d for each divisor d of 12: one of order 1 and one of order 2, two each of orders 3, 4 and 6, and four of order 12, the points at angles of 30, 150, 210 and 330 degrees. The points of one order are the roots of one cyclotomic polynomial, and the six polynomials multiply to $x^{12} - 1$. Assumes: the unit circle; the order of the point at angle $2\pi k/12$ is 12 divided by the greatest common divisor of k and 12

For each d , let $\Phi_d(x)$ be the product of the factors $x - \omega$ over the $\phi(d)$ points ω of order d . These are the *cyclotomic polynomials*, from the Greek for cutting the circle. Every root of $x^n - 1$ has one order, so

$x^n - 1 = \text{the product of } \Phi_d(x) \text{ over the divisors } d \text{ of } n$

and the first few are $\Phi_1 = x - 1$, $\Phi_2 = x + 1$, $\Phi_3 = x^2 + x + 1$, $\Phi_4 = x^2 + 1$, $\Phi_5 = x^4 + x^3 + x^2 + x + 1$, $\Phi_6 = x^2 - x + 1$, $\Phi_8 = x^4 + 1$, $\Phi_9 = x^6 + x^3 + 1$, $\Phi_{10} = x^4 - x^3 + x^2 - x + 1$ and $\Phi_{12} = x^4 - x^2 + 1$. For a prime p , Φ_p is $1 + x + \dots + x^{p-1}$, the polynomial of the seventeen points, since every root of $x^p - 1$ other than 1 has order p .

The product formula shows that every Φ_n has whole-number coefficients, working upwards from $\Phi_1 = x - 1$. Suppose all the Φ_d for d below n have them. Then Φ_n is $x^n - 1$ divided by the product of the Φ_d over the proper divisors d of n , and that product is a monic polynomial with whole-number coefficients. Long division by a monic polynomial with whole-number coefficients never divides by anything but its leading coefficient 1, so the quotient has whole-number coefficients too. It also shows how to compute them, by the same divisions: Φ_{15} is $x^8 - x^7 + x^5 - x^4 + x^3 - x + 1$, and Φ_{30} is $x^8 + x^7 - x^5 - x^4 - x^3 + x + 1$. Every coefficient in the list so far is 0, 1 or -1 , and that holds for every n below 105; Φ_{105} , of degree 48, has two coefficients equal to -2 , as Migotti found in 1883, and Schur showed that the coefficients are unbounded as n grows. Nothing later depends on that.

* * *

Is Φ_n irreducible over $\mathbb{Q}$? The seventeen-points argument does not extend: Eisenstein's criterion needs a prime dividing all the lower coefficients, and $\Phi_{15}(x + 1)$ has constant term $\Phi_{15}(1) = 1$, which no prime divides, so no prime will do. The proof that works is by the arithmetic of remainders.

Φ_n is irreducible over $\mathbb{Q}$, for every n .

Let f be the minimal polynomial of ζ over $\mathbb{Q}$. It is monic, it divides Φ_n , since Φ_n vanishes at ζ , and it has whole-number coefficients, since a monic factor of a monic polynomial with whole-number coefficients has them; write $\Phi_n = f \cdot g$ with g monic and

with whole-number coefficients as well. The claim is that every point of order n is a root of f , so that f is all of Φ_n . Every point of order n is a power ζ^k with k sharing no factor with n , and k is a product of primes none of which divides n , so it is enough to show one thing:

if ω is a root of f and p is a prime not dividing n , then ω^p is a root of f .

Repeating this prime by prime carries ζ to every ζ^k : if k is $p_1 p_2 \dots p_r$, then ζ is a root of f , so ζ^{p_1} is, so $(\zeta^{p_1})^{p_2}$ is, and so on until ζ^k . Suppose then that ω is a root of f and ω^p is not. Since ω^p has order n , it is a root of Φ_n , and so of g . That makes ω a root of the polynomial $g(x^p)$. And f , which is irreducible over $\mathbb{Q}$ and has ω as a root, is the minimal polynomial of ω as well, so it divides every polynomial that vanishes at ω : $g(x^p) = f \cdot h$, with h monic and with whole-number coefficients.

Now reduce every coefficient modulo p . Over $\mathbb{Z}/p$, raising to the power p respects sums and products, and fixes every element of $\mathbb{Z}/p$. So if $g(x)$ is $c_0 + c_1 x + \dots + x^m$, the p -th power of its reduction is $c_0^p + c_1^p x^p + \dots + x^{mp}$, which is $c_0 + c_1 x^p + \dots + x^{mp}$ modulo p : the reduction of $g(x^p)$ is the reduction of $g(x)$ raised to the power p .

Let q be any irreducible factor of the reduction of f over $\mathbb{Z}/p$. Then q divides the reduction of $g(x^p)$, since f divides $g(x^p)$, and that is the reduction of g to the power p . An irreducible polynomial over a field that divides a product divides one of the factors, and here every factor is the reduction of g , so q divides the reduction of g . So q^2 divides the reduction of $f \cdot g$, which is the reduction of Φ_n , and therefore divides the reduction of $x^n - 1$.

Write $x^n - 1 = q^2 \cdot r$ over $\mathbb{Z}/p$. By the rule for products its derivative is $2q q' r + q^2 r'$, and q divides both terms: a polynomial divisible by q^2 has derivative divisible by q . The derivative of $x^n - 1$ is $n x^{n-1}$, which over $\mathbb{Z}/p$ is not 0, since p does not divide n ; it is a nonzero number times x^{n-1} , and its only irreducible factor is x . So q would have to be x , up to a constant factor. But x does not divide $x^n -$

1, whose constant term is -1 . That contradiction proves the claim, and with it the theorem.

So $Q(\zeta)$ has dimension $\varphi(n)$ over Q , and the automorphisms $\zeta \mapsto \zeta^k$, one for each k sharing no factor with n , are all of its group: the group of the n -th roots of unity over Q is the group of units modulo n , under multiplication, exactly and for every n . For $n = 15$ it is the eight numbers 1, 2, 4, 7, 8, 11, 13, 14, and the map $\zeta \mapsto \zeta^2$ is an automorphism because 2 shares no factor with 15, which is what the proof needed. The argument works only for primes that do not divide n , and it has to stop there. Take $n = 6$ and the prime 3: modulo 3, the polynomial $x^6 - 1$ is $(x - 1)^3(x + 1)^3$, with the repeated factors the proof had to exclude, and the map $\zeta \mapsto \zeta^3$ sends the point of order 6 to a point of order 2, so it cannot be an automorphism.

* * *

The values of Φ_n at whole numbers have a prime-number secret. The values of $\Phi_8 = x^4 + 1$ at 1, 2, ..., 12 are 2, 17, 82, 257, 626, 1297, 2402, 4097, 6562, 10001, 14642 and 20737, and their prime divisors other than 2 are 17, 41, 257, 313, 1297, 1201, 241, 193, 73, 137, 7321, 89 and 233. Every one of them leaves the remainder 1 on division by 8. For $\Phi_6 = x^2 - x + 1$ the values at 2 to 12 are 3, 7, 13, 21, 31, 43, 57, 73, 91, 111 and 133, with prime divisors 3, 7, 13, 19, 31, 37, 43 and 73; apart from 3 every one leaves the remainder 1 on division by 6.

If a prime p divides $\Phi_n(a)$ for some whole number a and does not divide n , then p leaves the remainder 1 on division by n .

Since Φ_n divides $x^n - 1$, the prime p divides $a^n - 1$, so a is not 0 modulo p , and its order d modulo p divides n . Suppose d were below n . Then a would be a root modulo p of $x^d - 1$, which is the product of the Φ_e over the divisors e of d , so a would be a root modulo p of some Φ_e with e a proper divisor of n . It is also a root of Φ_n modulo p . These are two different factors in the product that makes $x^n - 1$, and each contains $x - a$ over Z/p , so $(x - a)^2$ would

divide $x^n - 1$ over $\mathbb{Z}/p$. Then $x - a$ would divide the derivative nx^{n-1} , that is, na^{n-1} would be 0 modulo p , which it is not, p dividing neither n nor a . So the order of a modulo p is exactly n , and since a^{p-1} is 1 modulo p by Fermat's theorem, n divides $p - 1$.

The converse holds as well. If p leaves the remainder 1 on division by n , then n divides $p - 1$. Take a primitive root g modulo p , which exists; its order is $p - 1$. Then $g^{(p-1)/n}$ has order exactly n : its n -th power is g^{p-1} , which is 1, while for m from 1 up to $n - 1$ its m -th power is g raised to $m(p - 1)/n$, a positive exponent below $p - 1$, and that is not 1. A family of order n is a root of $x^n - 1$ and of no $x^d - 1$ with d below n , so it is a root of no Φ_e with e a proper divisor of n . Since $x^n - 1$ is the product of the Φ_e over the divisors e of n , it must be a root of Φ_n . So, for primes p not dividing n ,

Φ_n has a root modulo p exactly when p leaves the remainder 1 on division by n

and when it has one it has $\varphi(n)$, the powers of that root with exponents sharing no factor with n , so it splits into factors of the first degree. That is a congruence rule of the kind the quadratics obeyed under the law of reciprocity, here for the polynomial whose group is the group of units modulo n . For $x^4 + 1$, the splitting into four factors of the first degree happens, among the odd primes, exactly for 17, 41, 73, 89, ..., those that leave 1 on division by 8. Modulo 2 the polynomial is $(x + 1)^4$; 2 is the one prime the rule sets aside, being the one dividing n .

The theorem produces new primes on demand, in the manner of Euclid. Fix n at least 2, and suppose only finitely many primes left the remainder 1 on division by n , say $p_1, \dots, p_k$. Let a be a multiple of $n \cdot p_1 \cdots p_k$ large enough that $\Phi_n(a)$ is at least 2 in size, which it is once a is 3 or more, each factor $a - \omega$ having size at least $a - 1$. Take a prime p dividing $\Phi_n(a)$; one exists, because $\Phi_n(a)$ is a whole number at least 2 in size.

To see that p is new, look at the constant term $\Phi_n(0)$, which is 1 for every n above 1. Put $x = 0$ in the product formula for $x^n - 1$: the left side is -1 , and the right side is $\Phi_1(0)$, which is -1 , times

the constant terms of the other factors, so those multiply to 1 for every n above 1. Taking $n = 2, 3, 4, \dots$ in turn, the constant terms for the proper divisors above 1 are already known to be 1, so the new factor $\Phi_n(0)$ is 1 as well.

Every term of $\Phi_n(a)$ except the constant term is a multiple of a , so $\Phi_n(a)$ leaves the remainder 1 on division by a , and p does not divide a . Since n and every p_i divide a , p divides neither n nor any p_i . By the theorem, p leaves the remainder 1 on division by n , and it is not among $p_1, \dots, p_k$: it is a new prime.

For every n there are infinitely many primes that leave the remainder 1 on division by n .

Dirichlet proved in 1837 that every progression whose first term and difference share no factor contains infinitely many primes, by the sums over all the whole numbers and the calculus mentioned when the primes in progressions first appeared; for the progression of the numbers 1 more than a multiple of n , no analysis is needed, and the cyclotomic polynomial does the work that Euclid's product of all the primes plus one did for the primes themselves.

* * *

The last use is to build fields with a prescribed group. The group of $Q(\zeta_p)$ for a prime p is the group of units modulo p , and that is the clock with $p - 1$ hours in disguise, a primitive root serving as the hour hand. Every subgroup of a clock with $p - 1$ hours is the multiples of a divisor, and every subgroup of an abelian group is normal. So for every divisor m of $p - 1$ there is a field between Q and $Q(\zeta_p)$, Galois over Q , whose group is the collapse of the clock by the multiples of m . That collapse is the clock with m hours. Given any n , a prime p leaving the remainder 1 on division by n exists. Then n divides $p - 1$, so n can serve as m , and the clock with n hours is the group of some field, and of some polynomial over Q .

For $n = 3$ the prime 7 serves. The subgroup of the six units modulo 7 with two members is the pair 1 and 6, and its fixed field is gen-

erated by $a = \zeta + \zeta^6$, which is $2 \cos(2\pi/7)$, the number whose field was found to have dimension 3 when seven points were refused a ruler and compasses. Its two partners are $b = \zeta^2 + \zeta^5$ and $c = \zeta^3 + \zeta^4$, and the three add up to -1 , the six powers of ζ other than 1 adding to -1 .

Multiplying out and reducing the exponents modulo 7 gives $ab = a + c$, $ac = b + c$ and $bc = a + b$, so the three products add up to twice the sum, which is -2 ; and abc is a times bc , that is $a(a + b) = a^2 + ab$. Here a^2 is $\zeta^2 + 2 + \zeta^{12}$, and ζ^{12} is ζ^5 , so $a^2 = b + 2$, while $ab = a + c$. So abc is $2 + (a + b + c)$, which is 1. The cubic with roots a, b and c is x^3 minus the sum times x^2 , plus the sum of the products in pairs times x , minus the product, so the three satisfy

$$x^3 + x^2 - 2x - 1 = 0$$

and the field of dimension 3 whose group is the clock with 3 hours has been written down.

Two groups make a third: the pairs, one member from each, multiplied place by place, with the pair of identities for identity. A subgroup of each gives a subgroup of the product, and collapsing by it collapses each place on its own. Products of clocks come from products of primes. Given clocks with $n_1, \dots, n_k$ hours, choose different primes $p_1, \dots, p_k$ with p_i leaving the remainder 1 on division by n_i , which the theorem allows, and let N be their product. The group of $Q(\zeta_N)$ is the group of units modulo N . Send each number modulo N to its list of remainders modulo $p_1, \dots, p_k$. This respects multiplication, and it matches the group of units modulo N with the product of the groups of units modulo the p_i , each a clock with $p_i - 1$ hours. Inside each take the multiples of n_i , and let H be the product of these subgroups. It is normal, everything commuting, and the collapse of the product by H is the product of the collapses, the clock with n_1 hours times the clock with n_2 hours and so on. The fixed field of H is Galois over Q with exactly that group.

For two clocks with 2 hours the primes 3 and 5 serve: the group of $Q(\zeta_{15})$ is the product of a clock with 2 hours and a clock with 4,

the subgroup H is the units 1 and 4, and the fixed field, of dimension 4, is $Q(\sqrt{-3}, \sqrt{5})$, generated by $\sqrt{-3} + \sqrt{5}$, whose polynomial is $x^4 - 4x^2 + 64$. Its group is the group of the four sign changes, the same group as that of $Q(\sqrt{2}, \sqrt{3})$, reached inside a field of roots of unity.

So every product of clocks is the group of a polynomial over Q . A finite abelian group is always such a product, which is proved next, so every finite abelian group is the group of a polynomial over Q . The converse, that every polynomial over Q with an abelian group has its roots in some $Q(\zeta_N)$, is the theorem of Kronecker and Weber, and it is relied on where it is quoted and not proved. The group of $Q(\zeta_N)$ is abelian, so every field inside it is Galois over Q , every subgroup being normal, and its group, a collapse of an abelian group, is abelian as well; with the theorem of Kronecker and Weber, the fields of roots of unity, and the fields inside them, are exactly the fields with abelian groups.

Compute Φ_{18} , Φ_{20} and Φ_{21} , and check their degrees against φ .

Show that $\Phi_{2n}(x)$ is $\Phi_n(-x)$ when n is odd and above 1.

Which primes divide some value of $x^2 + x + 1$ at a whole number, and which divide some value of $x^4 - x^2 + 1$?

Which subgroup of the units modulo 105 has a fixed field with group the product of three clocks with 2 hours, and which three square roots generate that field?

Modulo which primes below 100 does $x^4 + 1$ have a root, and modulo which does $x^4 - x^2 + 1$?

TAKING A GROUP APART

Two statements about finite groups are wanted and not yet proved. Every product of clocks was shown to be the group of a polynomial over Q , and the step from there to every finite abelian group leaned on the claim that a finite abelian group is always such a product. Later, the proof that the plane is closed will want a subgroup whose size is the largest power of a prime dividing the size of the group. Neither claim is deep, and both come out of the counting habits already in hand: the size of a subgroup divides the size of the group, the size of an orbit times the size of a stabiliser is the size of the group, a prime dividing the size produces an element of that order, and a normal subgroup can be collapsed.

One more remark about collapsing is needed. Let N be a normal subgroup of G and let S be a subgroup of the collapse. The members of G whose blocks lie in S form a subgroup of G : the identity is there, since its block is the identity block. The block of a product of two of them is the product of their two blocks, which lies in S . The reverse of one has the reverse block, which also lies in S . That subgroup is a union of the blocks in S , each with as many members as N , so its size is the size of S times the size of N . Subgroups of the collapse are subgroups of G seen through the collapsing, with their sizes multiplied by the size of N .

* * *

If p^a is the largest power of a prime p dividing the size of a finite group, the group has a subgroup of exactly p^a members.

Descend on the size n of the group G . When n is 1 there is nothing to prove. Let Z be the collection of elements that commute with everything, a subgroup, and split on whether p divides its size.

Suppose p divides the size of Z . Then Z , being abelian, has an element z of order p , and the p powers of z form a subgroup N . Relabelling leaves N alone, since z commutes with everything, so N is normal and G can be collapsed by it. The collapse has n/p members, and the largest power of p dividing n/p is p^{a-1} . The collapse is smaller than G , so by the descent it has a subgroup of p^{a-1} members. By the remark on collapsing, the members of G whose blocks lie in that subgroup form a subgroup of G with $p^{a-1} \cdot p$ members, which is p^a .

Suppose instead that p does not divide the size of Z . Relabelling splits G into classes, the classes with one member being exactly the elements of Z . Adding up the sizes,

$$n = \text{the size of } Z + \text{the sizes of the classes with more than one member}$$

The left side is divisible by p and the first term on the right is not, so some class with more than one member has a size that p does not divide. The size of that class is the size of G divided by the size of the stabiliser of one of its members, which is the subgroup C of elements commuting with that member. So p does not divide n divided by the size of C . Since n is that quotient times the size of C , and the quotient contributes no factor p , all a factors p in n must sit in the size of C : p^a divides the size of C . No higher power of p can, because the size of C divides n . And C is smaller than G , since the class has more than one member. By the descent C has a subgroup of p^a members, and a subgroup of C is a subgroup of G .

Sylow proved this in 1872, along with two more facts about such subgroups. Call a subgroup of p^a members a *full p -subgroup*.

Any two full p -subgroups are relabellings of each other, and their number leaves the remainder 1 on division by p and divides the part of the size that p does not divide.

Two short remarks carry it. First, if a subgroup P carries a subgroup Q onto itself by relabelling, then the collection PQ of products is a subgroup: it holds the identity. To multiply pq by $p'q'$, slip p' and p' backwards in beside q : $pq \cdot p'q' = pp' \cdot (p' \text{ backwards} \cdot q \cdot p') \cdot q'$. The bracket is a relabelling of q by a member of P , so it lies in Q , and the whole is a member of P times a member of Q , again of that shape. The same trick handles the reverse: the reverse of pq is q backwards $\cdot p$ backwards, which equals p backwards $\cdot (p \cdot q \text{ backwards} \cdot p \text{ backwards})$. The bracket is a relabelling of q backwards by a member of P , so it lies in Q , and the reverse is of that shape again. Second, the size of PQ is the size of P times the size of Q divided by the size of their intersection, because pq equals $p'q'$ exactly when $p' = pd$ and $q' = d \text{ backwards} \cdot q$ for some member d of the intersection. So each product is written in exactly as many ways as the intersection has members, and the number of different products is the size of P times the size of Q divided by that number.

Let P be a full p -subgroup, and let P relabel the collection X of all full p -subgroups. Each orbit has a size dividing the size of P , which is p^a , and every divisor of p^a is a power of p , so each orbit has 1 member or a multiple of p . An orbit of one member is a Q that P carries onto itself. Then PQ is a subgroup, and by the second remark its size is p^a times p^a divided by the size of the intersection, which is a power of p since the intersection is a subgroup of P . No power of p above p^a divides n , so PQ has at most p^a members. It contains P , which already has p^a members, so PQ equals P . Then Q lies inside P , and having the same number of members it is P . So P itself is the only orbit of one member, and the size of X leaves the remainder 1 on division by p .

Now let all of G relabel X ; the aim is to show that every full p -subgroup lies in the orbit of P . Suppose some Q lay outside that orbit, and write Y for the members of X outside it. Every relabelling carries the orbit of P to itself and so carries Y to itself. Let P alone

relabel Y . Its orbits have sizes dividing the size of P , so 1 or a multiple of p , and the only member of X that P leaves alone is P , which is not in Y ; so no orbit within Y has one member and the size of Y is a multiple of p . Now let Q alone relabel Y . By the same count the only member of Y it leaves alone is Q , so the size of Y leaves the remainder 1 on division by p . The two cannot both hold.

So X is a single orbit: every full p -subgroup is a relabelling of P , and so any two are relabellings of each other. The size of an orbit is the size of G divided by the size of a stabiliser, so the size of X divides n . Being 1 more than a multiple of p , it shares no factor with p , so it divides the part of n that p does not divide.

Small groups fall out at once. A group of 15 members has full p -subgroups for 3 and for 5; the number for 3 divides 5 and leaves the remainder 1 on division by 3, so it is 1, and the number for 5 divides 3 and leaves 1 on division by 5, so it is 1. A subgroup that is the only one of its size is carried onto itself by every relabelling, so both are normal; they meet only in the identity, being of different prime sizes, and their 15 products are the whole group.

Members of two normal subgroups meeting only in the identity commute. Take x in one and y in the other, and look at the product $x \cdot y \cdot x$ backwards $\cdot y$ backwards. Grouped as $(x \cdot y \cdot x \text{ backwards}) \cdot y \text{ backwards}$, the bracket is a relabelling of y , so it lies in the second subgroup, and so does the whole. Grouped as $x \cdot (y \cdot x \text{ backwards} \cdot y \text{ backwards})$, the bracket is a relabelling of x backwards, so it lies in the first subgroup, and so does the whole. The product lies in both, so it is the identity, and that says exactly that $x \cdot y = y \cdot x$. In the group of 15 members, then, each member of the subgroup of three commutes with each member of the subgroup of five.

Take a member x of order 3 and a member y of order 5. Since they commute, xy raised to the power k is $x^k y^k$. If that is the identity, then x^k is y to the power $-k$, a member of both subgroups, so it is the identity; then 3 divides k and 5 divides k . So xy has order 15, its fifteen powers fill the group, and every group of 15 members is the clock with 15 hours.

The twelve rotations of the tetrahedron, which were shown to

have no subgroup of six, have four subgroups of three: the number divides 4 and leaves 1 on division by 3, so it is 1 or 4, and it is not 1, since the eight rotations of order 3 could not fit inside one subgroup of three. Among the sixty rotations of the dodecahedron there are 24 elements of order 5, falling into 6 subgroups of five; 20 of order 3, in 10 subgroups of three; and 15 of order 2, lying in 5 subgroups of four. Each count leaves the remainder 1 on division by its prime, as it must.

* * *

Every finite abelian group is a product of clocks whose numbers of hours are powers of primes.

First split by primes. Let A be abelian with n members, and write $n = ab$ with a and b sharing no factor. Let A_a be the members x with x^a the identity, and A_b likewise; both are subgroups. They meet only in the identity, since a member of both has an order dividing a and dividing b . And every x is a product of one from each: there are whole numbers u and v with $ua + vb = 1$, so $x = x^{ua} \cdot x^{vb}$, in which x^{vb} raised to the power a is x to the power va , the identity, so x^{vb} lies in A_a , and likewise x^{ua} lies in A_b . So A is the product of A_a and A_b , and their sizes multiply to n . A prime dividing the size of A_a supplies a member of that order, and every member x of A_a has x^a the identity, so its order divides a ; so only the primes dividing a divide the size of A_a , and likewise for b . The two sizes multiply to ab , the first uses only primes of a and the second only primes of b , and a and b share no prime, so the sizes are a and b themselves. Repeating on the prime powers of n , A is the product of subgroups whose sizes are the prime powers dividing n .

So it is enough to take A abelian with p^k members and split it into clocks. Descend on the size, proving a little more: whenever a has the largest order in A , the group A is the product of the powers of a and some subgroup. That is enough, since the subgroup is again abelian with a power of p for its size, so it splits off a clock in

the same way, and so on until only the identity is left. Let a have the largest order in A , say p^m , and suppose A is not the powers of a alone. Some member lies outside those powers, and raising it to p again and again reaches the identity, which lies inside, so some member b lies outside while b^p lies inside: $b^p = a^t$ for some t . Raising b to the power p^m gives the identity, because the order of b is a power of p no larger than p^m . But b to the power p^m is b^p raised to the power p^{m-1} , which is a to the power $t \cdot p^{m-1}$. So a to the power $t \cdot p^{m-1}$ is the identity, the order p^m of a divides $t \cdot p^{m-1}$, and p divides t .

Write $t = ps$ and put $b' = b \cdot a^{-s}$. Then b' raised to the power p is $a^t \cdot a^{-ps}$, the identity. And b' is still outside the powers of a , since $b = b' \cdot a^s$, and if b' were a power of a then b would be one too. So the p powers of b' form a subgroup N of p members, and it meets the powers of a only in the identity: suppose a member of N other than the identity were a power of a . Since p is prime, that member has order p , so its own powers are all of N , b' among them. A power of a power of a is a power of a , so b' would be a power of a , which it is not.

Collapse A by N . The block of a still has order p^m . A power of a whose block is the identity lies in N , so it lies in the intersection of N with the powers of a , and so it is the identity; hence no power of a below the p^m -th has the identity for its block. And no block has a larger order. Raising the block of x to the order of x gives the identity block, so the order of a block divides the order of its member, which is at most p^m . So the block of a has the largest order in the collapse as well.

By the descent the collapse is the product of the powers of that block and some subgroup; let C be the members of A whose blocks lie in that subgroup, a subgroup of A holding N . Every member x of A is a power of a times a member of C . Indeed, the block of x is some power, say the k -th, of the block of a , times a member of that subgroup; so x times a to the power $-k$ has its block in the subgroup, which puts it in C .

And the only power of a in C is the identity. The block of such

a power is both a power of the block of a and a member of the subgroup, and those two meet only in the identity block; so the power lies in N , and a power of a lying in N is the identity. So A is the product of the powers of a and C . And C is smaller than A : it has the size of the subgroup times p members, while A has the size of the subgroup times p^m times p . The descent finishes it.

How many factors are there? The members x with x^p the identity form a subgroup. In a clock whose number of hours is a power of p , exactly p members have their p -th power the identity. In a product of r such clocks, a member has its p -th power the identity exactly when each of its r parts does, so there are p choices in each factor and p^r members in all. So r is settled by A alone, and no splitting can use a different number of clocks. The sizes are settled too. The p -th powers of the members of A form a subgroup, and it is the product of the same clocks, each with its number of hours divided by p ; a clock of p hours shrinks to the identity alone. That subgroup is smaller than A , so by a descent on the size its clocks are settled by A alone. Multiplying each of their sizes by p recovers the clocks of A with more than p hours, and the number of clocks of p hours is r minus the number of those. For a group of any size split into clocks of prime-power size, the clocks whose numbers of hours are powers of p multiply to the members x with x raised to the power p^e the identity, p^e being the largest power of p dividing the size: every member of such a clock has that property, since p^e is a multiple of its number of hours, and a member of a clock whose number of hours is a power of another prime has it only when it is the identity, its order dividing both. That subgroup is settled by A alone, so the count above settles the clocks one prime at a time.

A finite abelian group is a product of clocks of prime-power size in exactly one way, apart from the order of the factors.

Counting them is now a matter of partitions. For each prime p dividing n with exponent e , the choices are the ways of writing e as a sum of positive whole numbers, with the order of the terms ignored, since the order of the clocks does not matter; and the choices

at different primes are independent. So there are 3 abelian groups of 8 members, the clocks with 8 hours, with 4 and 2 hours, and with 2 and 2 and 2; there are 5 of 16 members; 4 of 36; 4 of 100; and 10 of 720. Take the group of the numbers modulo n that share no factor with n . It decided which regular polygons can be built and which primes give which pattern of factors. It is abelian, so the splitting just proved says exactly which product of clocks it is.

four boxes in every block

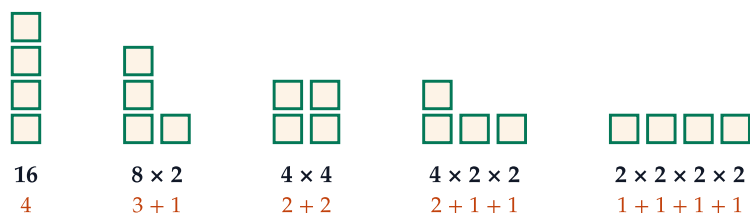


Figure 71.1. The abelian groups of 16 members, one for each way of writing 4 as a sum of positive whole numbers. A column of k boxes stands for a clock with 2^k hours, so the columns of each block multiply to 16. There are five ways and five groups, and no two of the five are the same group. Assumes: the splitting into clocks of prime-power size is unique apart from the order of the factors, which the text proves; for a size with several primes the choices at the different primes are made independently

For which n below 40 is every group of n members abelian? Try 15, 33 and 35 against 8, 12 and 21.

How many abelian groups have 1000 members, and how many have 1024?

The sixty even rearrangements of five things have no normal subgroup other than the identity and the whole group. What does the counting theorem give for the number of subgroups of five members, and why does that not by itself settle the question?

A group whose size is pq , with p below q and p not dividing $q - 1$, has exactly one subgroup of q members and one of p members. Show that it is a clock with pq hours.

WHAT A RADICAL IS

Cardano's formula solves $x^3 = 15x + 4$ as

$$x = \sqrt[3]{2 + \sqrt{-121}} + \sqrt[3]{2 - \sqrt{-121}}$$

and the expression records a sequence of operations. A square root is taken of -121 , a number built from the coefficients. A cube root is taken of $2 + \sqrt{-121}$, a number built from that, and the second cube root is chosen so that the product of the two is 5, a third of the coefficient 15, as the derivation of the formula demands. The cube roots $2 + i$ and $2 - i$ have product 5, and their sum is the root 4.

A formula of that kind builds the roots from the coefficients by the four operations and finitely many root extractions, each applied to something already built. In the language of fields, with every field inside the plane of pairs:

f over Q is solvable by radicals when there is a tower $Q = F_0 \subset F_1 \subset \dots \subset F_k$ with $F_{i+1} = F_i(\alpha_i)$ and $\alpha_i^{n_i}$ in F_i for some positive whole number n_i , and f is a product of factors of degree 1 with coefficients in F_k .

The aim is to turn that definition into a property of the group of f , the group of automorphisms over Q of the field that its roots generate.

The correspondence cannot be applied to such a tower as it stands, because the top of a radical tower need not be Galois over Q : the tower $Q \subset Q(\sqrt[3]{2})$ has a single step, and $Q(\sqrt[3]{2})$ is not Galois over Q , since it contains only one of the three cube roots of 2. Two enlargements repair this. Each produces a tower of radical steps whose top contains the old top, so every root of f that the old tower reached is still reached. The second adjoins roots by the quotient construction, so its fields need not lie inside the plane of pairs. That does no harm. The argument uses only two facts about a tower: each step is a radical step, and each field contains the one before. Both make sense for any fields.

* * *

The first enlargement puts roots of unity at the bottom, and the smallest case shows why. Adjoining a cube root of 2 to Q gives one root of $x^3 - 2$ and not the other two. Adjoining a cube root of 2 to a field that already contains ω , a root of $x^2 + x + 1$, gives all three, since the other two are $\omega\sqrt[3]{2}$ and $\omega^2\sqrt[3]{2}$.

In general, let F be a field that contains a root ζ of $x^m = 1$ whose powers give all m solutions, let a be a nonzero element of F , and let β be a root of $x^m - a$ in some field containing F . The m numbers $\beta, \zeta\beta, \zeta^2\beta, \dots, \zeta^{m-1}\beta$ are roots of $x^m - a$, since $(\zeta^j\beta)^m = (\zeta^m)^j \cdot \beta^m = a$, and they are distinct, since β is not 0, a being not 0, and the powers $1, \zeta, \dots, \zeta^{m-1}$ are distinct. A polynomial of degree m has at most m roots, so these are all of its roots, none of them repeated, and each lies in $F(\beta)$. So $F(\beta)$ is the splitting field over F of a polynomial with no repeated root, and it is Galois over F .

Its group is cyclic: every member is a power of a single member. An automorphism σ of $F(\beta)$ over F sends β to a root of $x^m - a$, hence to $\zeta^j\beta$ for exactly one j from 0 to $m - 1$, and σ is settled by j , since every element of $F(\beta)$ is a polynomial in β . If σ sends β to $\zeta^j\beta$ and τ sends β to $\zeta^l\beta$, then performing τ and then σ sends β to $\sigma(\zeta^l\beta)$, which is $\zeta^l \cdot \sigma(\beta) = \zeta^l \cdot \zeta^j\beta$, because σ leaves ζ alone. So exponents add modulo m , and the group is matched with a sub-

group of the numbers modulo m under addition. Such a subgroup consists of the multiples of its smallest positive member d , by the same division that settled the subgroups of a clock with m hours, so every automorphism is a power of the one that sends β to $\zeta^d \beta$. A radical step taken over a field that contains the right roots of unity is Galois, with a group generated by a single element.

* * *

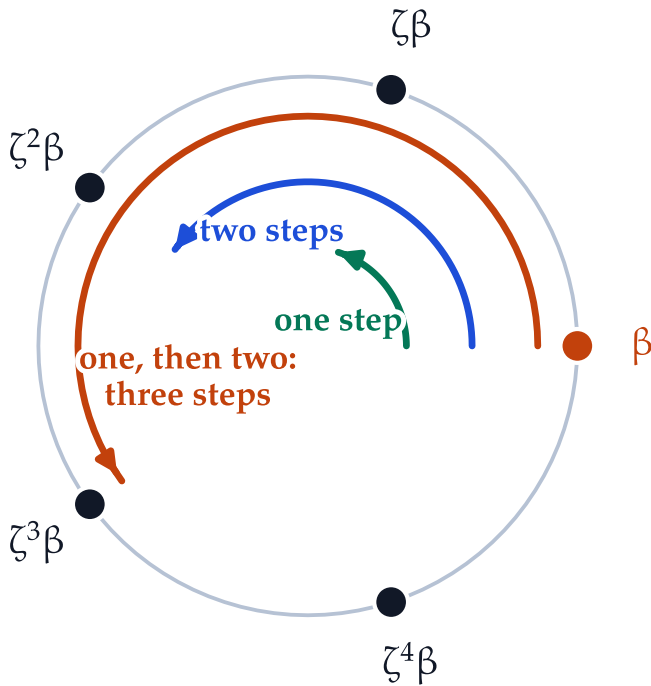
The second enlargement makes the top of the tower Galois over Q , keeping every step a radical step and the roots of unity at the bottom.

Start from a tower as in the definition. Let n be a whole number, at least 3, divisible by every n_i , and let ζ be the point at angle $2\pi/n$, whose powers give all n solutions of $x^n = 1$. For each i let g_i be the minimal polynomial of α_i over Q . Starting from $F_k(\zeta)$, adjoin roots of the g_i by the quotient construction, one at a time, until every g_i splits, and call the resulting field M . It is generated over Q by ζ and the roots of the g_i , since F_k is generated by the α_i and each α_i is a root of g_i . The roots of the minimal polynomial of ζ are powers of ζ , so they lie in M as well. So M is a splitting field over Q of the product of the distinct polynomials among $g_0, \dots, g_{k-1}$ and the minimal polynomial of ζ . That product has no repeated root: each factor is irreducible over Q and so has distinct roots, and two distinct monic irreducible polynomials share no root, since both would be the minimal polynomial of that root. So M is Galois over Q , its group G is finite, and M contains F_k .

Now build a field T inside M by adjoining elements to Q in a fixed order. List the members of G as $\sigma_1, \dots, \sigma_r$. Adjoin ζ first. Then adjoin $\sigma_1(\alpha_0), \sigma_1(\alpha_1), \dots, \sigma_1(\alpha_{k-1})$ in that order; then $\sigma_2(\alpha_0), \dots, \sigma_2(\alpha_{k-1})$; and so on through σ_r . The field reached at the end is T .

T is contained in M , since ζ and every α_i lie in M and every σ in G carries M onto itself.

M is contained in T . M is generated over Q by ζ and the roots



exponents add modulo 5: the group is cyclic

Figure 72.1. Over a field that already contains the fifth roots of unity, the roots of $x^5 - 2$ are one root β times the five powers of ζ , at the corners of a regular pentagon. An automorphism is settled by the corner it sends β to, so it turns the pentagon by some number of steps; turning by one step and then by two is turning by three, and the steps add modulo 5. The group is made of rotations of the pentagon, so it is cyclic; over $\mathbb{Q}(\zeta)$ itself it is all five. *Assumes:* that ζ lies in the base field and so is left alone by every automorphism, which is what makes each one a rotation; over the fractions alone ζ can move too, and the maps that move it are not rotations of this picture

of the g_i , and ζ lies in T . Take a root γ of some g_i . Since M is Galois over Q and α_i lies in M , the minimal polynomial g_i of α_i over Q is the product of the factors $x - \sigma(\alpha_i)$, one for each distinct image $\sigma(\alpha_i)$ with σ in G . So its roots are exactly those images: γ is $\sigma(\alpha_i)$ for some σ in G , and it was adjoined in the block for σ .

So $T = M$, and it remains to check that every adjunction was a radical step. The first was, since $\zeta^n = 1$. Consider the moment at which $\sigma_j(\alpha_i)$ is adjoined, and let W be the field built so far; it contains $\sigma_j(\alpha_0), \dots, \sigma_j(\alpha_{i-1})$, adjoined earlier in the same block. Since σ_j respects multiplication,

$$(\sigma_j(\alpha_i))^{n_i} = \sigma_j(\alpha_i^{n_i})$$

The element $\alpha_i^{n_i}$ lies in F_i , which is $Q(\alpha_0, \dots, \alpha_{i-1})$, so it is obtained from fractions and $\alpha_0, \dots, \alpha_{i-1}$ by the four operations. Apply σ_j to that expression. Since σ_j leaves fractions alone and respects the four operations, the result builds $\sigma_j(\alpha_i^{n_i})$ from fractions and $\sigma_j(\alpha_0), \dots, \sigma_j(\alpha_{i-1})$ by the same operations. All of these lie in W , so $\sigma_j(\alpha_i^{n_i})$ does too. When $i = 0$ the element is a fraction, and σ_j leaves it alone. Each adjunction therefore raises the new element to a power that lands in the field below, which is a radical step.

The argument used one feature of the order of adjunction: for each σ , the elements $\sigma(\alpha_0), \dots, \sigma(\alpha_{i-1})$ come before $\sigma(\alpha_i)$. The order inside the original tower cannot always be changed: α_i may need $\alpha_0, \dots, \alpha_{i-1}$ present before its power lands in the field below. In $Q \subset Q(\sqrt{2}) \subset Q(\sqrt{2}, \sqrt{1 + \sqrt{2}})$, adjoining $\sqrt{1 + \sqrt{2}}$ to Q first is not a radical step, because no positive power of $\sqrt{1 + \sqrt{2}}$ is a fraction. Its even powers are the powers of $1 + \sqrt{2}$, each of the form $a + b\sqrt{2}$ with a and b positive whole numbers, so none of them is a fraction; and if an odd power were a fraction, its square, which is a power of $1 + \sqrt{2}$, would be one too.

Drop every adjunction of an element already present, and write the tower that remains as

$$Q = K_0 \subset K_1 = Q(\zeta) \subset K_2 \subset \dots \subset K_s = M$$

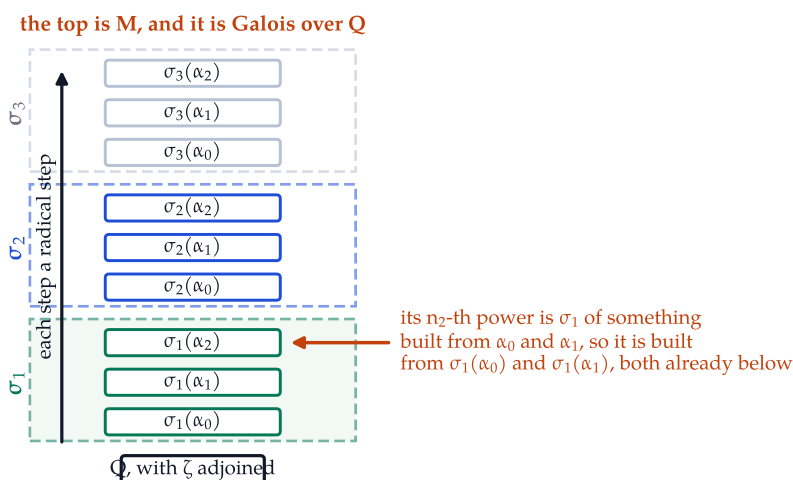


Figure 72.2. The given radical tower, and its images under the automorphisms of the field M built from it, laid end to end in one column. Inside the block for σ , the element $\sigma(\alpha_i)$ is adjoined after $\sigma(\alpha_0)$ through $\sigma(\alpha_{i-1})$; and its n_i -th power is σ of something built from α_0 through α_{i-1} , so it is built in the same way from those earlier images and is already present. Any order in which the images of the earlier generators come first has the same property. *Assumes:* that M is generated by the conjugates of the generators, which is the containment proved in the text; the drawing shows the order of adjunction and nothing about why the two fields coincide

Every K_i with i at least 1 contains ζ . Each later step adjoins an element β with β^{n_i} in the field below, for some n_i dividing n ; β is not 0, since 0 is in every field and adjunctions of elements already present were dropped; and the field below contains ζ^{n/n_i} , whose powers give all n_i solutions of $x^{n_i} = 1$. So every step after the first is a step of the kind the first enlargement describes.

* * *

The chain can now be read off. Let H_i be the group of M over K_i , so that $H_0 = G$ and H_s has one member. M is Galois over every K_i ,

being the splitting field over K_i of the same polynomial, and K_{i+1} is Galois over K_i : for i at least 1 by the first enlargement, and for $i = 0$ because $Q(\zeta)$ is the splitting field of $x^n - 1$. The theorem on normal subgroups, applied to M over K_i , then says that H_{i+1} is normal in H_i and that the collapse of H_i by H_{i+1} is the group of K_{i+1} over K_i . For i at least 1 that group is cyclic. For $i = 0$ it is the group of $Q(\zeta)$ over Q , which is abelian: an automorphism of $Q(\zeta)$ is settled by where it sends ζ , and it sends ζ to some power ζ^a ; performing the one for a and the one for b , in either order, sends ζ to ζ^{ab} . So

$$G = H_0 \supset H_1 \supset \dots \supset H_s = \{\text{identity}\}$$

with each subgroup normal in the one before and every collapse abelian. A group with a chain of that kind is called solvable.

One more step reaches f . The field L generated over Q by the roots of f lies inside F_k , since f is a product of factors of degree 1 with coefficients in F_k , hence inside M , and it is Galois over Q , being the splitting field of the product of the distinct irreducible factors of f . By the theorem on normal subgroups, its group is the collapse of G by the group of M over L . A collapse of a solvable group is solvable. The collapse of G by a normal subgroup is the image of G under the map sending each element to its class, and that map respects the operation; so it is enough to let φ carry G onto a group P , respecting the operation, and show that P is solvable. The images $\varphi(H_i)$ form a chain from P down to the identity. Each is normal in the one before: for h in H_i and x in H_{i+1} , $\varphi(h)\varphi(x)\varphi(h)^{-1} = \varphi(hxh^{-1})$, and hxh^{-1} lies in H_{i+1} . And each collapse is abelian. Two members of $\varphi(H_i)$ are $\varphi(x)$ and $\varphi(y)$ with x and y in H_i ; the element $xyx^{-1}y^{-1}$ lies in H_{i+1} , because the collapse of H_i by H_{i+1} is abelian; so $\varphi(x)\varphi(y)\varphi(x)^{-1}\varphi(y)^{-1}$ lies in $\varphi(H_{i+1})$, which is what it means for the collapse of $\varphi(H_i)$ by $\varphi(H_{i+1})$ to be abelian. So

if f is solvable by radicals, the group of its splitting field over Q is solvable.

The converse holds as well: a polynomial whose group is solvable is solvable by radicals. In its proof, each collapse with a prime

number p of members becomes one radical step, the adjunction of a p -th root, once the p -th roots of unity are present, and those roots of unity must themselves be reached by radicals; both are proved in the chapter that follows. The argument about the quintic uses only the direction just proved.

The splitting field of $x^4 - 2$ is the top of a radical tower. Which chain of subgroups of its group of eight does that tower produce, and are its collapses abelian?

* * *

Cardano's formula fits the theorem once its tower is written out. For $x^3 = px + q$ with p not 0 and $s = q^2/4 - p^3/27$, the derivation of the formula that ran into $\sqrt{-121}$ takes u to be a cube root of $q/2 + \sqrt{s}$ and $v = p/(3u)$, and then $u^3 + v^3 = q$ and $3uv = p$, so $x = u + v$ is a root. With ω the point at a third of a turn, the other two roots are $\omega u + \omega^2 v$ and $\omega^2 u + \omega v$, and the terms of each satisfy the same two relations as u and v , since $\omega^3 = 1$; to write down these roots the formula needs ω as well, and its tower is

$$Q \subset Q(\omega) \subset Q(\omega, \sqrt{s}) \subset Q(\omega, \sqrt{s}, u)$$

Its steps are the roots of unity at the bottom, a square root, and a cube root taken over a field that already contains ω . The formula makes the first enlargement without announcing it.

The cube roots of the formula can be built from the roots of the cubic. With roots r_1, r_2 and r_3 adding to 0, put $u = (r_1 + \omega r_2 + \omega^2 r_3)/3$ and $v = (r_1 + \omega^2 r_2 + \omega r_3)/3$. Since $1 + \omega + \omega^2 = 0$, this gives $u + v = r_1$, $\omega^2 u + \omega v = r_2$ and $\omega u + \omega^2 v = r_3$, and multiplying out gives $3uv = p$ and $u^3 + v^3 = q$. For $x^3 = 15x + 4$, with $r_1 = 4$ and r_2 and r_3 equal to $-2 + \sqrt{3}$ and $-2 - \sqrt{3}$, three times u is $4 + \omega(-2 + \sqrt{3}) + \omega^2(-2 - \sqrt{3}) = 4 + 2 + \sqrt{3}(\omega - \omega^2) = 6 + 3i$, so $u = 2 + i$ and $v = 2 - i$: Bombelli's cube roots, recovered from the roots. Moving the roots round, r_1 to r_2 to r_3 to r_1 , multiplies u by ω^2 and v by ω , and exchanging r_2 with r_3 exchanges u with v . So $u^3 + v^3$ and uv are unchanged by all six rearrangements of the roots,

which is why they are numbers of the ground field, q and $p/3$; u^3 and v^3 themselves are unchanged by the rotations and exchanged by the swaps. Their difference is

$$u^3 - v^3 = (\sqrt{-3/9}) \cdot \delta, \text{ where } \delta = (r_1 - r_2)(r_1 - r_3)(r_2 - r_3)$$

and $\delta^2 = 4p^3 - 27q^2$, so that s , which is $(u^3 - v^3)^2/4$, is $-\delta^2/108$. For $x^3 = 15x + 4$, δ is $66\sqrt{3}$, and $\delta^2 = 13068 = 108 \cdot 121$. The square root in the formula adjoins δ : since $s = -\delta^2/108$, $\sqrt{s}$ is $\delta \cdot \sqrt{-3/18}$, and $\sqrt{-3} = 2\omega + 1$ is already in $Q(\omega)$. When the group of the cubic is all six rearrangements, $Q(\omega, \delta)$ is the fixed field of the three rotations, which fix δ while the swaps change its sign; adjoining u , which the rotations move and whose cube lies in that field, reaches the top. Unless $\sqrt{s}$ already lies in $Q(\omega)$, the square root step matches a collapse with two members; the cube root step matches a collapse with three.

The cubic $x^3 - 3x - 1$, satisfied by twice the cosine of 20° , is the one that rules out the trisection of 60° . Its only candidates for a fraction root, 1 and -1 , give -3 and 1 rather than 0 , so it has none; a cubic that factored over Q would have a factor of degree 1 and hence a fraction root, so it is irreducible over Q , and $Q(r)$ has dimension 3 over Q for any root r . Substituting $2 - r^2$ and reducing with $r^3 = 3r + 1$ gives 0 , so $2 - r^2$ is a root as well, different from r , and the third root is $r^2 - r - 2$, since the three add to 0 . So $Q(r)$ contains all three roots and is Galois over Q , and its group has three members; the automorphism sending r to $2 - r^2$ carries $2\cos 20^\circ$ to $2\cos 140^\circ$ and $2\cos 140^\circ$ to $2\cos 260^\circ$. All three roots are real, and for $x^3 = 3x + 1$ the number s is $1/4 - 1 = -3/4$, so Cardano's formula reaches three real numbers by way of the square root of a negative number, as it did for $x^3 = 15x + 4$.

Neither the group of three members nor the square root of a negative number is an accident. Take a cubic $x^3 - px - q$ with no fraction root. The field of one root has dimension 3 over Q , so 3 divides the order of the group, and the only groups of rearrangements of three roots whose order 3 divides are the three rotations and all six rearrangements. The rotations fix δ and the swaps change its sign,

and δ is not 0, the roots being different; so δ is a fraction exactly when the group is the rotations, which is exactly when the field of one root holds all three. For $x^3 = 3x + 1$, $\delta^2 = 4 \cdot 27 - 27 = 81 = 9^2$; there the square root in the formula, $\sqrt{s} = \sqrt{-3/2}$, adds nothing to $Q(\omega)$, and the cube root is the only step that reaches a root of the cubic, matching the group of three. For $x^3 - 2$, $\delta^2 = -108$, which is not a square, and the group is all six. And when a cubic has three different real roots, δ is a real number other than 0, so $s = -\delta^2/108$ is negative: Cardano's formula passes through the square root of a negative number whenever the three roots are real and different.

Is the field generated by one root of $x^3 - 7x + 7$ its splitting field? And for $x^3 - 4x + 1$?

WHEN THE GROUP ALLOWS IT

A polynomial that can be solved by radicals has a solvable group. The other direction rests on one step, and it can be proved with what is in hand. Let p be a prime, let F contain a root ζ of $x^p = 1$ other than 1, whose powers then give all p solutions, and let K be Galois over F of dimension p . Then $K = F(R)$ for some R with R^p in F , so that the step from F to K is a radical step. The group of K over F has p members. The distinct powers of a member σ other than the identity form a subgroup whose size divides p and is more than 1, so it is p , and the group consists of the powers of σ . For θ in K put

$$R = \theta + \zeta^{-1}\sigma(\theta) + \zeta^{-2}\sigma^2(\theta) + \dots + \zeta^{-(p-1)}\sigma^{p-1}(\theta)$$

Apply σ , which leaves ζ alone. The term $\zeta^{-j}\sigma^j(\theta)$ becomes $\zeta^{-j}\sigma^{j+1}(\theta)$, which is ζ times the next term $\zeta^{-(j+1)}\sigma^{j+1}(\theta)$. The last term becomes $\zeta^{-(p-1)}\sigma^p(\theta) = \zeta^{-(p-1)}\theta$, and since σ^p is the identity and $\zeta^p = 1$ this is ζ times the first term θ . So $\sigma(R) = \zeta R$. Then $\sigma(R^p) = \zeta^p R^p = R^p$, so R^p is fixed by the whole group and lies in F . And R is not 0 for a suitable θ : the automorphisms $1, \sigma, \dots, \sigma^{p-1}$ are distinct, and distinct automorphisms are independent, so their combination with the coefficients $1, \zeta^{-1}, \dots, \zeta^{-(p-1)}$, none of them 0, is not the zero map. Such an R is not in F , since σ moves it, so $F(R)$ is larger than F . Its dimension over F divides p , the dimension of K , and is more than 1, so it is p , and $F(R)$ is all of K . For the cubic x^3

$= 3x + 1$ over $Q(\omega)$, with θ a root r and σ the member of its group sending r to $2 - r^2$, the element R is three times one of Cardano's cube roots.

So over a field that holds the p -th roots of unity, a Galois step of prime dimension p is the adjunction of a p -th root. What remains is to put such steps together, which takes three things: breaking a solvable group into steps of prime size, supplying the roots of unity, and checking that enlarging the bottom field does not spoil the group.

Steps of prime size. Let A be a finite abelian group with more than one member. Some prime p divides its size, so A has an element g of order p . If the powers of g are all of A , then A has exactly p members, so the subgroup with one member has p cosets, a prime number, and there is nothing more to do.

Otherwise collapse A by the powers of g , which is allowed since every subgroup of an abelian group is normal; the collapse is a smaller abelian group. Suppose it has a subgroup with a prime number of cosets. Then so has A : the elements of A whose blocks lie in that subgroup form a subgroup of A , with the same number of cosets. The collapse has more than one member, since the powers of g are not all of A , and it is smaller than A , so the same argument applies to it, and then to its collapse if needed. Sizes cannot shrink for ever, so the process ends in the first case, and the subgroup found there passes back up, one collapse at a time, as just described. So every abelian group with more than one member has a subgroup with a prime number of cosets.

Now take a chain $G = G_0 \supset G_1 \supset \dots$ down to the identity, each subgroup normal in the one before, with abelian collapses. Between G_i and G_{i+1} , the collapse of G_i by G_{i+1} has a subgroup with a prime number of cosets. The elements of G_i whose blocks lie in that subgroup form a subgroup N between G_i and G_{i+1} . It is normal in G_i because every subgroup of the abelian collapse is normal there. Inserting N splits one step into two: the collapse of G_i by N has a prime number of members, and the collapse of N by G_{i+1} is a subgroup of the old abelian collapse, so it is abelian and

smaller. Repeating on the smaller piece, which must stop since the sizes shrink, the chain is refined until every collapse has a prime number of members:

a finite solvable group has a chain, each subgroup normal in the one before, in which every collapse has a prime number of members.

Enlarging the bottom. Let L be Galois over F with group G , and let F' be a larger field, obtained from F by adjoining some numbers, with L' the field obtained from L by adjoining the same numbers. Then L' is Galois over F' , being the splitting field of the same polynomial. An automorphism of L' over F' carries L to itself, since it permutes the roots that generate L over F , and it is determined by what it does to L , since L and F' generate L' . Each such automorphism fixes F , which lies inside F' , so what it does to L is a member of G ; and two different automorphisms of L' over F' do different things to L . So the group of L' over F' can be read as a subgroup H of G . And a subgroup of a solvable group is solvable: intersect each subgroup in the chain of G with H . Each intersection is normal in the one before: if x lies in H and in G_i , and y lies in H and in G_{i+1} , then xyx^{-1} lies in H , and in G_{i+1} because G_{i+1} is normal in G_i . Each block of the new chain lies inside exactly one block of the old chain, and sending it there is one-to-one and respects composition. So each collapse of the new chain is a copy of a subgroup of the corresponding collapse of the old one, which is abelian, and so it is abelian too.

Roots of unity are themselves reached by radicals, and by more than the definition requires. The definition admits the point at angle $2\pi/n$ in one step, since its n -th power is 1. The proof climbs, and reaches that point using only p -th roots for primes p below n . Let n be a whole number, and suppose that for every m below n the point ζ_m at angle $2\pi/m$ lies in a tower of radicals over Q . The group of $Q(\zeta_n)$ over Q is abelian, its members being the maps $\zeta \mapsto \zeta^a$, which commute, and it has at most $n - 1$ members. Refine it to steps of prime size. Every prime p that occurs divides the size of the group, which is at most $n - 1$, so p is below n and ζ_p lies in a

tower of radicals; adjoin all of these first. Over the enlarged bottom the group is a subgroup of the old one, so it is still abelian, and its size divides the old size. Refined to steps of prime size, its primes are therefore among the old ones. Each step now sits over a field that holds the p -th roots of unity for its prime p , so by the step already proved it is the adjunction of a p -th root. So ζ_n lies in a tower of radicals. The climb starts at $n = 1$ and $n = 2$, where ζ is 1 or -1 .

Now the theorem. Let f have a solvable group G over Q , of size n , and let L be its splitting field. Adjoin to Q the numbers of a tower of radicals that reaches ζ_n , which exists as just shown, to get the bottom F' , and let L' be L with the same numbers adjoined. The group of L' over F' is a subgroup of G , so it is solvable, and it has a chain with collapses of prime size, the primes dividing n . By the correspondence the chain is a tower of fields from F' up to L' , each Galois over the one before with a prime number p of automorphisms, and each bottom contains the p -th roots of unity, which are powers of ζ_n since p divides n : ζ_n raised to the power n/p is the point at angle $2\pi/p$. So each step is the adjunction of a p -th root. The roots of f lie in L' , at the top of a tower of radicals.

A polynomial over Q is solvable by radicals exactly when its group is solvable.

That is Galois's theorem. He wrote it down in a memoir of 1831, which the Academy returned as unclear. He summarised it again in a letter on the night before the duel that ended his life in 1832, when he was twenty. Liouville published it in 1846. The proof is a construction: a chain of subgroups is a recipe for a formula.

* * *

For the quartic the recipe can be followed by hand. The group of all 24 rearrangements of four roots has a chain built from subgroups already met: the twelve even ones, then the group V of four, consisting of the identity and the three double swaps, then the identity. Both are normal in the whole group, since relabelling

carries even rearrangements to even ones and double swaps to double swaps. So each is normal in the one before. The three collapses have $24/12 = 2$, $12/4 = 3$ and 4 members. The first task is to find numbers that V leaves alone. For roots r_1, r_2, r_3 and r_4 , the three numbers

$$\theta_1 = r_1 r_2 + r_3 r_4, \theta_2 = r_1 r_3 + r_2 r_4, \theta_3 = r_1 r_4 + r_2 r_3$$

are each unchanged by the members of V , and any rearrangement of the roots rearranges the three among themselves. The size of the orbit of θ_1 says why there are three: the rearrangements that leave θ_1 alone are eight, and since the size of an orbit times the size of the stabiliser is the size of the group, the orbit has $24/8 = 3$ members. Since any rearrangement of the roots only shuffles the three factors of $(y - \theta_1)(y - \theta_2)(y - \theta_3)$, it leaves their product unchanged. So the coefficients of that product are symmetric in the roots, and are therefore polynomials in the coefficients of the quartic. For $x^4 + bx^2 + cx + d$ the cubic comes to

$$y^3 - by^2 - 4dy + (4bd - c^2)$$

A quartic is thereby reduced to a cubic, which Cardano's formula solves, and then to square roots. Ferrari found the reduction in 1540, by completing a square, when he was eighteen and Cardano's pupil, having come to his house four years earlier as a servant, and Lagrange explained it in 1770 by these three numbers.

Take $x^4 + 4x - 1$. The cubic is $y^3 + 4y - 16$, and it has the root 2. So $r_1 r_2 + r_3 r_4 = 2$, while $r_1 r_2 \cdot r_3 r_4$ is the product of all four roots, -1 . The two products $r_1 r_2$ and $r_3 r_4$ are therefore the roots of $z^2 - 2z - 1$, which are $1 + \sqrt{2}$ and $1 - \sqrt{2}$. The sums follow. The quartic has no cube term, so the four roots add to 0; if $r_1 + r_2 = s$, then $r_3 + r_4 = -s$. Take $r_1 r_2 = 1 + \sqrt{2}$ and $r_3 r_4 = 1 - \sqrt{2}$. The coefficient of x in the quartic is $-(r_1 r_2(r_3 + r_4) + r_3 r_4(r_1 + r_2))$, which here is $-(-(1 + \sqrt{2})s + (1 - \sqrt{2})s) = 2\sqrt{2} \cdot s$. It equals 4, so $s = \sqrt{2}$. One pair of roots satisfies $x^2 - \sqrt{2}x + 1 + \sqrt{2} = 0$ and the other $x^2 + \sqrt{2}x + 1 - \sqrt{2} = 0$, and the four roots are

$$(\sqrt{2} \pm \sqrt{(-2 - 4\sqrt{2})})/2 \text{ and } (-\sqrt{2} \pm \sqrt{(4\sqrt{2} - 2)})/2$$

the second pair real, 0.2490 and -1.6633 . Every quartic yields in the same way: one with a cube term loses it when x is replaced by x less a quarter of that coefficient, which shifts all four roots alike and leaves the three numbers to be read off the shifted equation.

Lagrange's analysis of 1770 was the first to see why the known formulas work: in each case some expression in the roots takes fewer values, under all rearrangements, than the degree of the equation, and those values satisfy an equation of lower degree. For the quartic, three values. For the quintic he found that the natural expressions take six values, which is worse than the five roots one started with. He suspected that the search was hopeless. Ruffini argued in 1799 that it was, but left one step unsupported: that the radicals taken along the way lie in the field the roots generate. Abel supplied that step and the proof in 1824. The reason, in the end, is that the sixty even rearrangements of five things cannot be collapsed at all.

* * *

Cardano's formula has an old scandal in it. For a cubic with three different real roots, the formula expresses them through cube roots of numbers that are not real, as with the $\sqrt{-121}$ of Bombelli. For two centuries people tried to rearrange the formula so as to avoid this, and failed. The theory explains the failure.

If a cubic with fraction coefficients has no fraction root and three real roots, then none of its roots lies in any tower of radicals made of real numbers.

Two facts are needed. First, let K be a field of real numbers, p a prime, and a a positive member of K whose real p -th root β is not in K . Then $x^p - a$ is irreducible over K . Its roots in the plane are β times the p -th roots of unity. Suppose it had a factor over K of degree m , with m from 1 to $p - 1$. Up to sign, the constant term of

that factor is a product of m of these roots, which is β^m times a root of unity. That constant term lies in K , so it is real, and a real number of size β^m is β^m or $-\beta^m$. Either way β^m would lie in K . Since m and p have no common divisor, $1 = um + vp$ for whole numbers u and v , and $\beta = (\beta^m)^u \cdot a^v$ would lie in K . Second, the square root δ of the discriminant of the cubic is real, since the three roots are real. Over any field that contains δ and over which the cubic is still irreducible, the group of the cubic has three members: its size is divisible by 3, since one root generates a field of dimension 3, and it contains no swap of two roots, since a swap changes δ to $-\delta$, which differs from δ because the roots are different, while every member of the group leaves δ alone. So the splitting field has dimension 3. One root alone already generates a field of dimension 3, so that field is the whole splitting field, and one root generates all three.

Suppose a real tower existed. Put δ in first, and refine the tower so that each step adjoins, for some prime p , the positive real p -th root of a positive member of the field below. This can be done: an n -th root with $n = pm$ is a p -th root of an m -th root, and a negative real n -th root generates the same field as its negative, which is the positive real n -th root of a positive number: of the same number when n is even, and of its opposite when n is odd.

The cubic is irreducible at the bottom. Its roots have dimension 3 over Q , while the bottom, Q with δ adjoined, has dimension 1 or 2; a root lying in the bottom would make 3 divide 1 or 2. Let K be the last field of the tower over which the cubic is still irreducible, and $K(\beta)$ the next, with β^p in K . Over $K(\beta)$ the cubic is no longer irreducible, and a cubic that factors has a factor of degree 1, so it has a root there. That root has dimension 3 over K and lies in $K(\beta)$, so 3 divides the dimension of $K(\beta)$ over K , which is p by the first fact. So p is 3.

Then $K(\beta)$ has dimension 3, contains a root of the cubic, and so is the splitting field of the cubic over K . A splitting field is Galois, and in a Galois field an irreducible polynomial with one root has all its roots. But $x^3 - a$ is irreducible over K and has the root β in $K(\beta)$, so $K(\beta)$ would contain the other two cube roots of a , which

are β times the cube roots of unity and are not real. A field of real numbers cannot contain them.

The cubic $x^3 - 3x - 1$, whose roots are twice the cosines of 20, 140 and 260 degrees, is of this kind. Its three roots are real, they are expressible by radicals, and every expression for them by radicals passes through numbers that are not real. The numbers of the plane, which were admitted into algebra by Bombelli in 1572 as a trick for getting real answers out of Cardano's formula, were never a trick that better algebra could have avoided.

For $x^4 + x + 1$, what is the cubic with roots θ_1 , θ_2 and θ_3 , and does it have a fraction root? What does that say about the group of the quartic?

Solve $x^4 - 10x^2 + 1 = 0$ by the three numbers θ , and compare with its known roots.

What is the group of $x^4 + 4x - 1$ over Q ? Its cubic has one fraction root.

Which regular polygons have corners that can be written with real radicals alone, and why is the heptagon not one of them?

In the proof that roots of unity are reached by radicals, which primes are needed for $n = 11$, and which for those primes in turn?

THE QUINTIC

The roots of

$$f(x) = x^5 - 6x^3 - 27x - 3$$

cannot be written in terms of its coefficients using the four operations and root extractions. The claim concerns every expression that could ever be written, and the proof examines none of them. It computes the group of f and shows that the group has no chain of the kind a tower of radicals produces.

Two features of f make its group computable: f cannot be factored over $\mathbb{Q}$, and exactly three of its roots are real.

Eisenstein's criterion applies at the prime 3: the coefficients after the leading 1 are 0, -6 , 0, -27 and -3 , all divisible by 3, and 9 does not divide -3 . So f is irreducible over $\mathbb{Q}$.

* * *

The derivative of f is $5x^4 - 18x^2 - 27$. Setting $u = x^2$ gives $5u^2 - 18u - 27 = 0$, so u is $(18 \pm \sqrt{864})/10$, which is 4.7394 or -1.1394 . Only the first is the square of a real number, so the derivative vanishes at exactly two real points, ± 2.1770 . Between two real roots of f the derivative must vanish, by the average-slope theorem. Four real roots would leave three gaps between neighbouring roots, each

needing its own zero of the derivative, and the derivative has only two, so f has at most three real roots. It has at least three: f tends to minus infinity on the left, $f(-2.1770) = 68.79$ is positive, $f(2.1770) = -74.79$ is negative, and f tends to plus infinity on the right, so a curve without breaks crosses zero in each of the three stretches between. The roots are near -2.9859 , -0.1108 and 3.0137 .

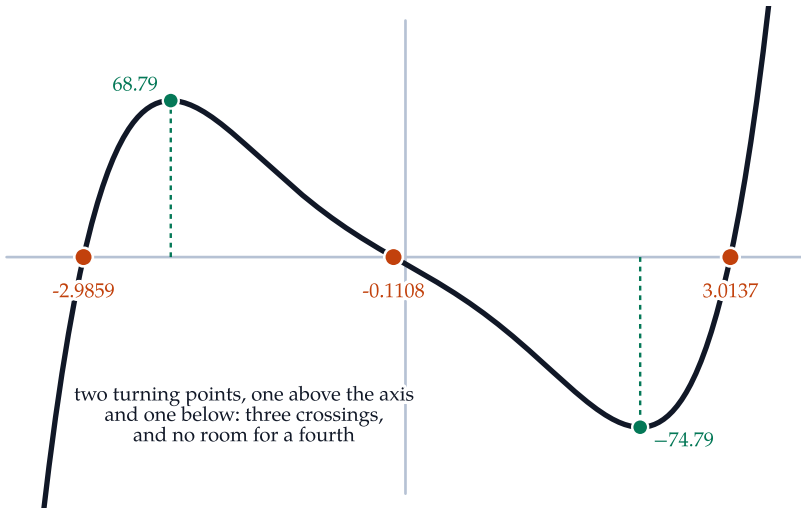


Figure 74.1. The curve of $x^5 - 6x^3 - 27x - 3$. Its derivative vanishes only at ± 2.1770 , so it has exactly two turning points; the left one sits at 68.79, above the axis, and the right one at -74.79 , below it, so the curve crosses three times, near -2.9859 , -0.1108 and 3.0137 . Three real roots and five in all leaves one conjugate pair, and conjugation exchanges that pair and fixes the rest, which is the transposition the argument needs. *Assumes:* that the curve has no further turning points off the drawn range, which the count of the derivative's roots settles, and that a curve crossing the axis has a root there, which is the theorem, proved by halving, that a continuous function takes every value between two of its values

Call the three real roots r_1 , r_2 and r_3 . Dividing f by $(x - r_1)(x - r_2)(x - r_3)$ leaves a quadratic with real coefficients, and that quadratic has no real root: a real root would be a fourth real root of f or a repeated one, and f , being irreducible over $\mathbb{Q}$, has no repeated root. Completing the square writes the roots of the quadratic as

$a + bi$ and $a - bi$, with b not 0. So f has five roots in the plane of pairs, three real and two conjugate to each other.

* * *

Let L be the field generated over Q by the five roots, and G its group. A member of G sends roots of f to roots of f and is settled by what it does to them, since they generate L , so G is a group of rearrangements of the five roots. Write $(a\ b)$ for the rearrangement that exchanges a and b and leaves the other roots alone, a transposition.

G contains a transposition. Conjugation in the plane of pairs, which replaces i by $-i$, sends each root of f to a root of f , so it carries L onto L , and it fixes every rational number, so its restriction to L belongs to G . It fixes the three real roots and exchanges the two others.

The order of G is divisible by 5. For any root α , f is the minimal polynomial of α , since f is monic and irreducible, so $Q(\alpha)$ has dimension 5 over Q . The dimension of L over Q , which is the order of G , is its dimension over $Q(\alpha)$ times 5.

G contains a cycle through all five roots, and a count produces it. Consider the lists $(g_1, g_2, g_3, g_4, g_5)$ of members of G whose product, in that order, is the identity. There are exactly $|G|^4$ of them, because the first four entries may be anything and the fifth is then forced. Moving the first entry to the end produces another such list, since if $g_1 g_2 g_3 g_4 g_5$ is the identity then $g_2 g_3 g_4 g_5 g_1$ is g_1^{-1} times the identity times g_1 . Five such moves return every list to itself. So the number of moves after which a list first comes back divides 5, and since 5 is prime that number is 1 or 5.

Grouping each list with the lists that moving carries it to, the lists fall into families of 5 and families of 1. A family of 1 is a list that moving leaves unchanged, so its five entries are a single g , with g^5 equal to the identity. The total $|G|^4$ is divisible by 5, because $|G|$ is, and so is the number of lists in families of 5, so the number of families of 1 is divisible by 5. One of them is the list of five identities, so there are at least five, and some g other than the identity has g^5

equal to the identity. Such a g is a cycle through all five roots. It moves some root r , and the first return of r among $g(r), g^2(r), \dots$ comes after a number of steps that divides 5, since $g^5(r) = r$; that number is not 1, so it is 5, and $r, g(r), g^2(r), g^3(r)$ and $g^4(r)$ are the five roots.

* * *

How much of G do those two elements force? Let σ be the five-cycle and τ the transposition, exchanging the roots a and b . Conjugating τ by a power of σ gives another transposition:

$$\sigma^k \tau \sigma^{-k} \text{ exchanges } \sigma^k(a) \text{ and } \sigma^k(b)$$

because the compound map performs σ^{-k} , then τ , then σ^k : it sends $\sigma^k(a)$ to a , then to b , then to $\sigma^k(b)$. Any other root is carried by σ^{-k} to a root other than a and b , which τ leaves alone, and σ^k then brings it back, so every other root stays where it was. So G contains the transpositions obtained by carrying the pair $\{a, b\}$ round the cycle.

Draw the five roots as points and join two of them whenever G contains the transposition exchanging them. Label the roots by the numbers modulo 5 so that σ adds 1: give one root the label 0, its image under σ the label 1, the next image 2, and so on round the cycle. The pair $\{a, b\}$ is then a pair of labels differing by some d other than 0, and its images under the powers of σ are the five pairs $\{u, u + d\}$. Starting at any label u and stepping by d visits $u, u + d, u + 2d$, and so on, and each step crosses one of the joined pairs. Five steps add $5d$, which is 0 modulo 5, so the walk first returns to u after a number of steps that divides 5. Because 5 is prime that number is 1 or 5, and it is not 1, since d is not 0. So the walk visits all five labels before returning, every two points are joined by a path, and the picture is connected.

Primality is doing real work, and $x^4 - 2$ shows what happens without it. That polynomial has every other hypothesis: it is irreducible over $\mathbb{Q}$, by Eisenstein's criterion at 2; two of its four roots are real, so conjugation exchanges the other two and is a transposi-

tion; and its group contains a cycle through all four roots, the automorphism sending $\sqrt[4]{2}$ to $i\sqrt[4]{2}$ and fixing i . Label the root $i\sqrt[4]{2}$ by j modulo 4, so that the cycle adds 1. Conjugation fixes 0 and 2 and exchanges 1 and 3, a step of 2, and the walk from 1 by steps of 2 closes at once: the picture falls into the two pieces $\{1, 3\}$ and $\{0, 2\}$. The group does not contain every transposition, and it is not all twenty-four rearrangements; it is the eight that carry each pair of opposite roots, r and $-r$, to such a pair.

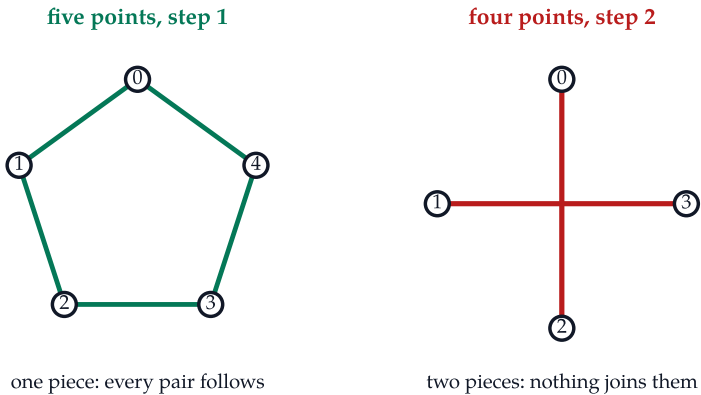


Figure 74.2. Left: five roots at the corners of a pentagon, with the five transpositions got by carrying one pair around the cycle. Every point is reachable from every other along the edges, so the transpositions of all pairs follow and the whole group of rearrangements is forced. Right: the same construction on the four roots of $x^4 - 2$ with a step of two, where the walk closes after two points and the picture falls into two pieces, which is what primality prevents. *Assumes:* that the group contains the five drawn transpositions, which comes from conjugating one of them by the powers of the cycle; the picture records which pairs, not why they are in the group

Connectedness finishes it. If G contains $(u v)$ and $(v w)$, it contains $(u w)$, because performing $(u v)$, then $(v w)$, then $(u v)$ again sends u to w , sends w to u , and returns v to itself. Along a path $u, v, w, \dots$ in the picture, this gives $(u w)$ from $(u v)$ and $(v w)$, then the transposition of u and the next point, and so on, so G contains the transposition of the path's two ends; and in a connected

picture every two points are the ends of a path. So G contains every transposition of two roots. Every rearrangement is a product of transpositions, since one transposition puts the right root in the first place, another the right root in the second, and so on; so G consists of all one hundred and twenty rearrangements of the five roots.

* * *

The group of all rearrangements of five things has no chain with abelian collapses, and the reason is a single subgroup that refuses to shrink.

The even rearrangements were counted when the fifteen puzzle was settled, and four facts from there are used here. Parity is well defined, because a swap changes by an odd number the count of pairs of places whose entries are out of order. Exactly half of the hundred and twenty rearrangements of five things are even. The even ones form a subgroup. And every even rearrangement is a product of cycles of length 3, found by taking its swaps in pairs. Let A be the sixty even rearrangements, and write $(a \ c \ d)$ for the three-cycle that carries a to c , c to d and d to a .

Every three-cycle is a commutator of two members of A . Write $xyx^{-1}y^{-1}$, the commutator of x and y , for the result of performing y backwards, then x backwards, then y , then x . Take $x = (1 \ 2 \ 3)$ and $y = (3 \ 4 \ 5)$. Following 1 through the four stages: y backwards leaves it at 1, x backwards sends it to 3, y sends 3 to 4, and x leaves 4 alone, so 1 goes to 4. Following 4: it goes to 3, then to 2, stays at 2, then goes to 3. Following 3: it goes to 5, stays at 5, goes to 3, and goes to 1. The symbols 2 and 5 return to themselves. So the commutator is $(1 \ 4 \ 3)$. Any three-cycle $(a \ c \ d)$ comes from this one by relabelling: rename 1, 4 and 3 as a , c and d , and 2 and 5 as the two remaining symbols. Relabelling a commutator gives the commutator of the relabelled rearrangements, and it turns the three-cycles x and y into three-cycles, which are even. So $(a \ c \ d)$ is the commutator of two members of A .

So the commutators of members of A generate a subgroup of A that contains every three-cycle, and the three-cycles generate A : A is generated by its own commutators.

Suppose G had a chain

$$G = G_0 \supset G_1 \supset \dots \supset G_m = \{\text{identity}\}$$

with each subgroup normal in the one before and each collapse abelian. G_0 contains A . If G_i contains A , then every commutator of members of G_i lies in G_{i+1} , because the collapse of G_i by G_{i+1} is abelian: the blocks of x and y commute, so the block of $xyx^{-1}y^{-1}$ is the identity block, which is G_{i+1} itself; in particular the commutators of members of A lie in G_{i+1} , and they generate A , so G_{i+1} contains A . Down the chain, every G_i contains A , including G_m , which has one member. A has sixty. There is no such chain.

Four things behave differently. Among the twelve even rearrangements of four things, the identity and the three products of two transpositions sharing nothing form a normal subgroup V , and the collapse of the twelve by V has three members, so it is abelian; hence every commutator of even rearrangements lies in V . In the convention above the commutator of $(1\ 2\ 3)$ and $(1\ 2\ 4)$ is $(1\ 2)(3\ 4)$, and relabelling gives the other two, so the commutators generate V and nothing more. The chain from all twenty-four rearrangements through the twelve even ones and V down to the identity has collapses of two, three and four members, all abelian. With five things there is no such V : if the collapse of A by a normal subgroup is abelian, that subgroup contains every commutator of members of A , and so all of A , and the collapse has a single block.

For $x^4 + x + 1$ with roots r_1, r_2, r_3 and r_4 , the three numbers $(r_1 + r_2)(r_3 + r_4)$, $(r_1 + r_3)(r_2 + r_4)$ and $(r_1 + r_4)(r_2 + r_3)$ are unchanged by the members of V . Which cubic with fraction coefficients has them as roots, and how are the four roots recovered from its roots by square roots?

* * *

So the group of f is not solvable, and a polynomial solvable by radicals has a solvable group. Therefore

the roots of $x^5 - 6x^3 - 27x - 3$ cannot be written in terms of its coefficients using the four operations and root extractions.

Not even one of its roots can be written that way. Suppose one root lay in the top F_k of a radical tower. The field M built from that tower contains F_k , so it would contain that root. Being Galois over Q , M would also contain every root of that root's minimal polynomial, which is f . The roots of f in M generate a splitting field of f . Its group is a collapse of the group of M , and the group of M is solvable, so this collapse is solvable too. Any two splitting fields of a polynomial with no repeated root are matched by an isomorphism that leaves the fractions alone, as the count of extensions of isomorphisms showed, so that group is a copy of the group of all one hundred and twenty rearrangements, which is not solvable. It follows that no formula of that kind solves every polynomial of degree five, since such a formula would solve this one.

Can the roots of $x^5 - 4x + 2$ be written with radicals?

It does not follow that every polynomial of degree five resists. The roots of $x^5 - 2$ are the real fifth root of 2 times the fifth roots of unity. Write ζ for a fifth root of unity other than 1, whose powers give all five; then $Q \subset Q(\zeta) \subset Q(\zeta, \sqrt[5]{2})$ is a tower of radicals. Its top has dimension 20 over Q . The field $Q(\sqrt[5]{2})$ has dimension 5, by Eisenstein's criterion at 2. The field $Q(\zeta)$ has dimension 4, because $x^4 + x^3 + x^2 + x + 1$, whose roots are the fifth roots of unity other than 1, is irreducible. This is the case of the prime 5 in the irreducibility proved for the polygons with a prime power of sides. Both 4 and 5 divide the dimension of the top, and they share no factor, so 20 divides it. Adjoining $\sqrt[5]{2}$ to $Q(\zeta)$ multiplies the dimension 4 by at most 5, so it is at most 20, and it is therefore exactly 20.

Its group can be written down. An automorphism of the top sends ζ to ζ^a , with a from 1 to 4, and $\sqrt[5]{2}$ to $\zeta^b \sqrt[5]{2}$, with b from 0 to 4, and is settled by a and b ; that allows at most twenty, and

the top is a splitting field of dimension 20, so all twenty occur. On the labels k of the roots $\zeta^{k \cdot 5} \sqrt[5]{2}$ the automorphism acts as $k \mapsto ak + b$ modulo 5. A map of that kind other than the identity has $ak + b = k$ only when $(a - 1)k = -b$, which modulo 5 has exactly one solution when a is not 1 and none when a is 1 and b is not 0; so it leaves at most one label in place, while a transposition leaves three, so this group contains no transposition.

Conjugation, which supplied the transposition in the argument above, is here $k \mapsto -k$. It fixes only 0 and exchanges two pairs, because $x^5 - 2$ increases and so has only one real root. The five-cycle $k \mapsto k + 1$ is there, as the count of lists guarantees. The five maps $k \mapsto k + b$ form a normal subgroup, since relabelling $k \mapsto k + c$ by $k \mapsto ak + b$ gives $k \mapsto k + ac$, and the collapse by that subgroup keeps only the multiplier a of each map. Following one map by another multiplies their multipliers modulo 5, and multiplication modulo 5 does not depend on order, so the collapse is abelian. The chain from twenty through five to the identity is the radical tower read off as a chain of groups.

The cubic and the quintic can be set side by side. The group of a cubic has at most six members, the rearrangements of its three roots, and those six descend through the three rotations to the identity, with collapses of two members and of three. The group of this quintic has one hundred and twenty members, and no chain with abelian collapses gets below the sixty even rearrangements, because those sixty are generated by their own commutators. What stops the quintic is not the difficulty of its algebra but a group that cannot be taken apart into abelian pieces.

READING A GROUP OFF REMAINDERS

The group of $x^5 - 6x^3 - 27x - 3$ was found by looking at its graph: three real roots gave an exchange of two roots, and irreducibility gave a cycle of five. That quintic was chosen so that the method would work. Given a polynomial that nobody chose, how is its group found? There is a method that uses no graph and no real numbers, only arithmetic with remainders, and it rests on a theorem of Dedekind's.

Take a polynomial f with whole-number coefficients and leading coefficient 1, and a prime p . Reduce the coefficients modulo p and factor the result over $\mathbb{Z}/p$ into irreducible factors, which can always be done, since over a finite field there are only finitely many candidates for a factor, and a machine does it quickly. Take $x^5 - x - 1$. Modulo 2 it is

$$(x^2 + x + 1)(x^3 + x^2 + 1)$$

Modulo 3 and modulo 5 it is irreducible. Modulo 7 it is again a quadratic times a cubic, modulo 17 a cubic times two factors of the first degree, and modulo 23 a quartic times a factor of the first degree. The degrees of the factors form a pattern, 2 + 3, or 5, or

3 + 1 + 1, or 4 + 1, which changes from prime to prime with no visible rule.

If f has no repeated root modulo p , then the group of f over Q contains a rearrangement of the roots whose cycle lengths are exactly the degrees of the irreducible factors of f modulo p .

Over Z/p the splitting field of the reduced polynomial is a finite field, and its group consists of the powers of the map $a \mapsto a^p$. That map carries each root of an irreducible factor of degree m round a cycle of length m , through the m roots of that factor. So over Z/p the pattern of degrees is the pattern of cycles of one particular rearrangement of the roots. Dedekind showed that the roots of f in its splitting field over Q can be matched one to one with the roots modulo p , and that the matching can be chosen so that this rearrangement, carried back through the matching to the roots over Q , is a member of the group of f over Q . It then has the same cycles as it had modulo p , and those cycles have the lengths of the degrees of the factors. The matching is the hard part. It needs the arithmetic of whole numbers in fields larger than Q , where a prime p can itself break up into several primes. It is relied on here and not proved.

The excluded primes, modulo which f has a repeated root, are the primes that divide the discriminant, since the discriminant of the reduced polynomial is the remainder of the discriminant of f , and it is zero exactly when two roots coincide modulo p . When f has no repeated root its discriminant is a whole number other than zero, so these primes are finitely many: for $x^5 - x - 1$ the discriminant is 2869, which is $19 \cdot 151$.

Modulo 5 the polynomial $x^5 - x - 1$ is irreducible, so its group contains a cycle through all five roots; and a polynomial irreducible modulo a prime is irreducible over Q , since a factorisation over Q into two factors of smaller degree can be taken with both factors monic, and then both have whole-number coefficients, as was shown on the way to Eisenstein's criterion; monic factors keep their degrees modulo the prime, so the reduced polynomial would factor too. Modulo 2 the pattern is $2 + 3$, so the group contains a rear-

rangement that exchanges two roots and cycles the other three. Its cube exchanges the two roots and returns the other three to their places, since three turns of a cycle of three bring each root home, while an exchange done three times is still an exchange. A group of rearrangements of five things that contains a cycle of five and a single exchange is the group of all 120, as the picture of five points showed. So

*the group of $x^5 - x - 1$ is the group of all rearrangements of its roots,
and its roots cannot be written with radicals*

and the whole computation was two factorisations, modulo 2 and modulo 5.

* * *

The patterns explain an oddity met earlier. The polynomial $x^4 - 10x^2 + 1$ is irreducible over $\mathbb{Q}$ and factors modulo every prime. Its group consists of the four sign changes of $\sqrt{2}$ and $\sqrt{3}$, which as rearrangements of the four roots are the identity and three double exchanges. No member of that group moves all four roots in one cycle. By the theorem, then, no prime can leave the polynomial irreducible, for that would put a cycle of four into the group. The factorisations modulo p can only show the patterns $1 + 1 + 1 + 1$ and $2 + 2$, and they do: among the primes from 5 to 3000 the first occurs 101 times and the second 327. In general, a polynomial stays irreducible modulo some prime only if its group contains a cycle through all the roots.

The counts 101 and 327 are close to one in four and three in four, and that is no accident. The group has four members, one with the pattern $1 + 1 + 1 + 1$ and three with the pattern $2 + 2$. Frobenius proved in 1896 that the patterns occur among the primes as often as they occur in the group:

*the proportion of primes that give a certain pattern of degrees approaches
the proportion of members of the group that have that pattern of cycles.*

For $x^5 - x - 1$, with all 120 rearrangements in its group, here are the 4201 primes below 40000 other than 19 and 151.

pattern	primes	share	in the group
5	820	0.195	$24/120 = 0.200$
$4 + 1$	1047	0.249	$30/120 = 0.250$
$3 + 2$	713	0.170	$20/120 = 0.167$
$3 + 1 + 1$	728	0.173	$20/120 = 0.167$
$2 + 2 + 1$	531	0.126	$15/120 = 0.125$
$2 + 1 + 1 + 1$	329	0.078	$10/120 = 0.083$
$1 + 1 + 1 + 1 + 1$	33	0.008	$1/120 = 0.008$

The last row counts the primes modulo which $x^5 - x - 1$ has five different roots. The smallest is 1973. The theorem of Frobenius, which Chebotarev sharpened in 1922, is proved with the same tools as the theorem on primes in progressions, and it is relied on here. It makes the method sharper. The first theorem shows only that each pattern seen among the primes belongs to some member of the group; this one says in what proportion the members of the group have each pattern. So factor a polynomial modulo a few hundred primes and count how often each pattern appears, and the proportions estimate the share of the group with each pattern of cycles, which is often enough to tell which group it is.

* * *

Which primes give which pattern? For some polynomials there is a rule. The quadratic $x^2 - q$ has two roots modulo p when q is a square modulo p and is irreducible otherwise, and by the law of reciprocity that depends only on the remainder of p on division by $4q$. The cubic $x^3 - 3x - 1$ has as its group the three rotations of its roots. A rotation of three things either moves nothing or turns all three round a cycle, so the cubic shows only the patterns $1 + 1 + 1$ and 3. And the primes sort themselves by a congruence: among the primes from 5 to 2000 the cubic has three roots modulo p for exactly the 97 primes that leave 1 or 8 on division by 9, and is irreducible for

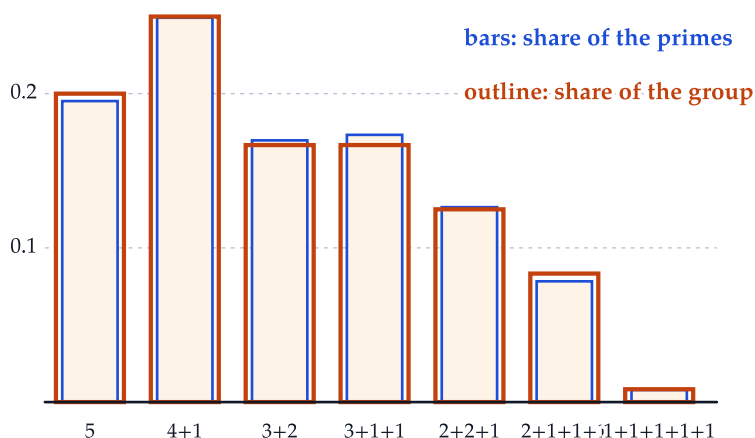


Figure 75.1. For each pattern of degrees, the share of the 4201 primes below 40000 modulo which $x^5 - x - 1$ factors in that pattern, in bars, against the share of the 120 rearrangements of five things whose cycles have that pattern, in outline. The primes 19 and 151, which divide the discriminant, are left out. Assumes: counts made by factoring over $\mathbb{Z}/p$ by machine; the agreement is a theorem only in the limit of all primes, and at 4201 primes a row may be off by a few parts in a hundred

the other 204. That is a reciprocity law for a cubic, and the reason is that its roots, twice the cosines of 20, 140 and 260 degrees, lie in the field of the ninth roots of unity. Every number of that field is a combination of powers of a ninth root of unity ζ with fractions as coefficients, so the map $a \mapsto a^p$ is fixed by what it does to ζ ; it sends ζ to ζ^p , and ζ^p depends only on the remainder of p on division by 9, since $\zeta^9 = 1$. So the whole map, and with it the pattern, can be read off from p modulo 9.

Every polynomial whose group is abelian has such a rule, a congruence on p that decides the pattern. Kronecker stated it in 1853, Weber published a proof in 1886 with a gap in it, and Hilbert closed the gap in 1896: the roots of such a polynomial always lie in a field of n -th roots of unity for some n , and there, as for the cubic, the map $a \mapsto a^p$ depends only on p modulo n . For $x^5 - x - 1$, whose group

is as far from abelian as a group of rearrangements can be, no congruence decides which primes give which pattern. The reason is the converse of the same theorem. If the primes modulo which it has five roots were exactly those in certain progressions modulo n , then its splitting field would sit inside a field of roots of unity, and its group would commute. The group of $x^5 - x - 1$ does not commute, so this cannot happen. The primes 1973, 3769, 5101, 7727, ... modulo which it has five roots obey no rule of that kind. What takes the place of the congruence, for groups that are not abelian, is the subject of a web of conjectures proposed by Langlands in 1967. The proof of Fermat's claim confirmed one instance of them, and most of the web is still open.

A question in the other direction is open too. Every group of all rearrangements of n things is the group of some polynomial over $\mathbb{Q}$, as Hilbert showed in 1892, and every solvable group is too, as Shafarevich proved in 1954; neither proof is given here. Whether every finite group is the group of some polynomial over $\mathbb{Q}$, nobody knows.

Modulo 2 the polynomial $x^5 + x^2 + 1$ is irreducible, modulo 3 it shows the pattern $4 + 1$, and modulo 5 the pattern $3 + 2$. What is its group, which of the three facts could be spared, and can its roots be written with radicals?

Why can $x^4 + 1$ not be irreducible modulo any prime, and which two patterns does it show?

A polynomial of degree 4 shows the patterns 4 , $2 + 2$, $2 + 1 + 1$ and $1 + 1 + 1 + 1$ among the primes, in the proportions 2, 3, 2 and 1 out of 8. How many members has its group, and which group of rearrangements of four things is it?

For which primes does $x^3 - 2$ have three roots, one root, or none, among the primes below 100, and in what proportions should each occur in the long run?

If a polynomial of degree n is irreducible over $\mathbb{Q}$, the average number of its roots modulo p , over all primes p , approaches 1. Why, in terms of the group and the count of orbits?

THE PLANE IS CLOSED

Every polynomial has a root in the plane, and the proof given for it was analysis from end to end: a polynomial grows beyond every bound far from the origin, so the smallest value of its size is taken somewhere on a disc. Proving that a continuous function on a disc that contains its rim takes a smallest value needed the fact that bounded sequences accumulate. And at a point where the value is not 0 a short computation walks downhill, so the smallest value is 0. Nothing in that argument is algebra.

How much analysis does the theorem need? Not none: over the fractions it fails. Adjoin i to the fractions. The field that results meets the line only in the fractions, and the two roots of $x^2 - 2$ lie on the line without being fractions, so $x^2 - 2$ has no root in that field. But the amount can be measured, and it comes to two facts. The first:

a polynomial with real coefficients and odd degree has a real root.

That is the intermediate value theorem applied to its graph. Far from 0 the leading term outweighs all the others, and being a number other than 0 times an odd power, it has opposite signs far to the left and far to the right; so the graph runs between large values of opposite signs and crosses 0 in between. And the second:

every number of the plane has a square root, so every quadratic with coefficients in the plane has its two roots there.

The square root was written down when the plane was built: halve the angle and take the square root of the size. Completing the square does the rest. The halving is algebra, but the square root of a size is the square root of a positive real number, so the second fact leans on the line as well, and the two together are the whole of what the theorem takes from completeness. From those two facts alone the whole theorem follows, by a descent that Laplace found in 1795. Gauss objected to it, because it assumed the roots existed somewhere before finding them; he had himself given in 1799 an argument of a different kind altogether. He then carried the descent through himself in 1816, keeping the auxiliary polynomial the descent builds and working with the symmetric expressions formally, so that no roots had to be assumed. Later the construction of splitting fields supplied the somewhere outright: a field in which the roots are known to exist.

* * *

Let f be a polynomial with real coefficients, of degree n at least 1, which may be taken monic. Write n as a power of 2 times an odd number, $n = 2^v \cdot m$, and argue by descent on v .

When v is 0 the degree is odd and f has a real root, which is a root in the plane. That is the base.

Let v be at least 1, and suppose the theorem holds for every polynomial with real coefficients whose degree has a smaller power of 2 in it. Build a splitting field K of f over the plane, by the construction that adjoins roots one at a time, and let $r_1, \dots, r_n$ be the roots of f in K , listed with repetitions. Now take a real number c , and form the $n(n-1)/2$ elements

$$y_{ij} = r_i + r_j + c \cdot r_i r_j, \text{ one for each pair } i \text{ below } j$$

and the polynomial $g_c(x)$ whose roots they are, the product of the factors $x - y_{ij}$. Its coefficients are, up to sign, the elementary sym-

metric expressions of the y_{ij} . A rearrangement of $r_1, \dots, r_n$ rearranges the pairs, and so rearranges the y_{ij} among themselves. The elementary symmetric expressions of the y_{ij} do not change when the y_{ij} are rearranged, so those coefficients are left alone. So each coefficient of g_c is a symmetric polynomial in $r_1, \dots, r_n$, with coefficients that are whole-number polynomials in c , and by the theorem on symmetric expressions it is a polynomial in the elementary symmetric expressions of the r 's and in c . Those elementary expressions are, up to sign, the coefficients of f : real numbers. So

g_c is a polynomial with real coefficients, of degree $N = n(n-1)/2$

and it can be written down from the coefficients of f without knowing a single root.

Count the powers of 2 in N . Since v is at least 1, n is even, so $n-1$ is odd, and $N = 2^{v-1} \cdot m \cdot (n-1)$, in which $m(n-1)$ is odd. The power of 2 in N is exactly one less than in n . By the supposition of the descent, g_c has a root in the plane. The plane lies inside K , and in K the polynomial g_c is the product of the factors $x - y_{ij}$, so its roots are the y_{ij} and nothing else. So for some pair i below j , the number $r_i + r_j + c \cdot r_i r_j$ lies in the plane.

The pair depends on c , and there are only N pairs. Run c through $N+1$ different real numbers, $0, 1, 2, \dots, N$. Two of them, c and c' , must deliver the same pair i, j . Then both

$$r_i + r_j + c \cdot r_i r_j \text{ and } r_i + r_j + c' \cdot r_i r_j$$

lie in the plane. Subtracting, $(c - c') \cdot r_i r_j$ lies in the plane, and $c - c'$ is a real number other than 0, so the product $r_i r_j$ lies in the plane; and then so does the sum $r_i + r_j$, which is $r_i + r_j + c \cdot r_i r_j$ minus c times that product. But r_i and r_j are the two roots of

$$x^2 - (r_i + r_j)x + r_i r_j$$

a quadratic with coefficients in the plane, and such a quadratic has both its roots in the plane. So r_i lies in the plane, and f has a root there. The descent is complete:

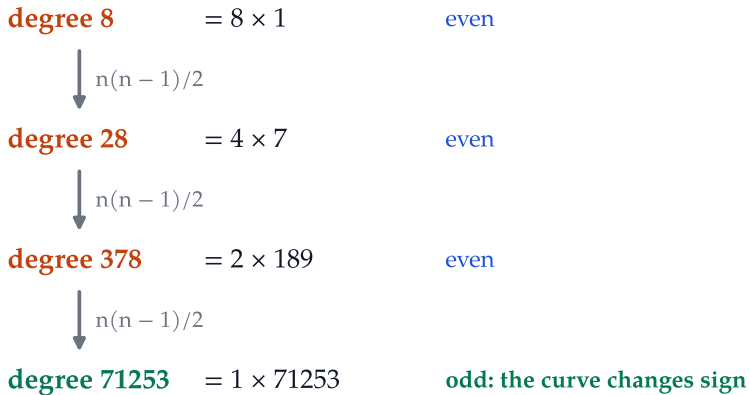


Figure 76.1. The descent that proves the theorem by algebra. A degree n is written as a power of 2 times an odd number, and the auxiliary polynomial has degree $n(n-1)/2$, in which the power of 2 has fallen by one, since $n-1$ is odd. Starting at 8 the degrees are 8, 28, 378 and 71253, with powers of 2 equal to 8, 4, 2 and 1; at the last the degree is odd and a real root exists because the curve changes sign. *Assumes: n even at every step but the last; the auxiliary polynomial's coefficients are symmetric in the roots and so are real*

every polynomial with real coefficients and degree at least 1 has a root in the plane.

Polynomials with coefficients in the plane now follow at no extra cost. Let p be such a polynomial, of degree at least 1, and let p^* be the polynomial whose coefficients are the mirror images of those of p . Mirroring respects sums and products, so mirroring every coefficient of the product $p \cdot p^*$ gives the product of the mirrored factors, which is $p^* \cdot p$, the same product. Its coefficients are therefore their own mirror images, and so are real. It has a root z in the plane. Then $p(z) \cdot p^*(z) = 0$, so one factor vanishes; if $p^*(z) = 0$, then mirroring the whole equation gives $p(z^*) = 0$, and z^* is a root of p .

The plane is closed

Every polynomial of degree at least 1 with coefficients in the plane has a root there, and so, by dividing out one factor at a time, splits into factors of the first degree.

A field with that property is called *algebraically closed*. The plane is one, and it is the smallest one containing the real numbers: any algebraically closed field containing the real numbers contains a root of $x^2 + 1$, and the real numbers with such a root adjoined are already the plane.

* * *

The machinery can be watched on a small case. Take $f = x^4 + 1$, of degree $4 = 2^2 \cdot 1$, so v is 2. The auxiliary polynomial has degree $6 = 2 \cdot 3$, with the power of 2 fallen from 4 to 2, and its own auxiliary would have degree 15, odd. Two steps of the descent, and the third is the intermediate value theorem.

For $f = x^4 + 1$ the elementary symmetric expressions of the roots are 0, 0, 0 and 1. Feeding them into the symmetric-function procedure with $c = 0$ gives

$$g_0(x) = x^6 - 4x^2$$

and with $c = 1$

$$g_1(x) = x^6 - x^4 - 8x^3 - 5x^2 - 8x - 3$$

both with real coefficients, as promised, and both written down from the four coefficients of $x^4 + 1$ alone. The roots of g_0 are 0, 0, $\pm\sqrt{2}$ and $\pm i\sqrt{2}$, and the two zeros say that two pairs of roots of $x^4 + 1$ add up to 0, which is visible after the fact: the four roots are the corners of a square about the origin. The proof does not look at the roots: it runs c through seven values, one more than the six pairs, so some pair is named twice.

* * *

The same two facts can be run through the correspondence instead of through symmetric expressions, and the result is shorter to state and needs more group theory. Given f with real coefficients, multiply it by $x^2 + 1$ and factor the result into monic irreducible factors over the real numbers. Take each different factor once and multiply them together, and let K be a splitting field of that product over the real numbers.

That product has no repeated root: an irreducible polynomial over the real numbers has a derivative that is not the zero polynomial and of lower degree. A root shared by the two would have a minimal polynomial over the real numbers dividing both; dividing the irreducible polynomial, it is that polynomial itself up to a constant factor, and that cannot divide a polynomial other than 0 of lower degree. So the polynomial shares no root with its derivative, and therefore has no repeated root. And two different monic irreducible polynomials share no root at all.

K contains a root of $x^2 + 1$ and so a copy of the plane, it contains every root of f , and it is Galois over the real numbers, being a splitting field of a polynomial with no repeated root. Let G be its group, of size $2^a \cdot m$ with m odd. A group whose size is divisible by p^a , with p^a the largest power of the prime p dividing it, has a subgroup of exactly that size, as was proved when groups were taken apart. Let H be such a subgroup for $p = 2$. By the correspondence, its fixed field has dimension over the real numbers equal to the index of H in G , which is m , an odd number. Every element of that field has a minimal polynomial over the real numbers whose degree divides m and is therefore odd, and an irreducible polynomial of odd degree over the real numbers has degree 1, since an odd degree above 1 would give it a real root and a factor. So the fixed field is the real numbers themselves, m is 1, and G has a power of 2 for its size.

Now look at the group of K over the plane, a subgroup of G , so also of size a power of 2. If it had more than one member it would have a subgroup of index 2, and the field fixed by that subgroup would have dimension 2 over the plane. Any element of that field outside the plane would then have a minimal polynomial over the

plane of degree 2: a quadratic with coefficients in the plane and no root in the plane. That is impossible, since every quadratic over the plane splits. So the group has one member and K is the plane itself, which holds every root of f .

The step that a group whose size is a power of 2, with more than one member, has a subgroup of index 2 can be proved with what the groups have already supplied. Let G have size p^k with k at least 1. Relabelling by each element splits G into classes, each of size dividing p^k ; the elements that form a class by themselves are those that commute with everything, and they make a subgroup Z . Adding up the class sizes gives p^k , and every class not of size 1 has size divisible by p , so the number of classes of size 1 is divisible by p . The identity is one of them, so Z has at least p members. Take an element of order p in Z , which Cauchy's theorem provides; the subgroup it generates lies in Z , so relabelling leaves it alone, and it is normal. Collapse by it; the collapse has size p^{k-1} . If the collapse has one member, then G has p members, and the subgroup holding the identity alone is already of index p . Otherwise, by descent on k , the collapse has a subgroup of index p , and the elements of G that the collapse sends into that subgroup form a subgroup of G , also of index p . Applying the same result to that subgroup, and again to the next, gives the chain down to the identity. So

a group whose size is a power of a prime p , with more than one member, has a subgroup of index p , and a chain down to the identity with every step of index p .

The constructible polygons needed exactly this: a chain of subgroups of index 2. When they were sorted, the chain was proved only for abelian groups, by linear algebra over the field with two elements. It holds for every group whose size is a power of a prime.

* * *

Take any field F , ordered, in which every odd-degree polynomial has a root and every positive element has a square root. Then

$F(i)$, where $i^2 = -1$, is algebraically closed, by the descent above. The descent used nothing else, provided the square root in $F(i)$ can be found without angles, and it can. For $a + bi$ let r be the square root of $a^2 + b^2$, which is at least the size of a . Then $(r + a)/2$ and $(r - a)/2$ are not negative, so they have square roots u and v in F , with u not negative, and choosing the sign of v to match the sign of b makes $(u + vi)^2 = a + bi$. The ordering gave the odd-degree fact its meaning and the square roots gave the quadratics theirs. Artin and Schreier showed in 1927 that the converse holds in a strong form: if a field is not algebraically closed but becomes so on adjoining a root of a single polynomial, then that field can be ordered, it behaves in every algebraic respect like the real numbers, and the closed field is the one reached by adjoining a root of $x^2 + 1$. So a field one step below a closed one is always of this kind. Such fields are called *real closed*, and the algebraic numbers that are real form one, a field whose members can be listed, in which every odd-degree polynomial has a root although it has no completeness at all: huddling sequences in it need not converge.

That last example settles the question the first proof raised: the theorem does not need the continuum, and the reader who follows the descent over the real algebraic numbers finds every step still standing, with the plane replaced by the algebraic numbers of the plane. What the analysis in the first proof supplied is required only by the odd-degree fact and by the square roots of positive numbers, and over the algebraic numbers even those come from counting sign changes in a polynomial, which Sturm made into a procedure in 1829 and which no limit is needed to run.

Every field lies inside an algebraically closed one, as Steinitz proved in 1910 by an argument that adjoins roots without end and needs a principle of choice; the smallest one containing a given field is its *algebraic closure*, and for the real numbers it is the plane. That is the last thing the plane was asked to do, and the reason the search for larger number systems stopped where it did: the equations ran out of complaints.

For a cubic with real coefficients, the descent produces an auxil-

iary polynomial of degree 3. What are its roots in terms of the roots of the cubic, and why is the descent unnecessary in that case?

Write down the auxiliary polynomial of degree 6 with $c = 0$ for $x^4 + px + q$, and check it against the one for $x^4 + 1$.

The proof ran c through the whole numbers $0, 1, 2, \dots, N$. What is the largest number of values of c that could fail to repeat a pair, and what would go wrong over a field in which $N + 1$ different elements cannot be found?

Show that the real algebraic numbers form a field in which every odd-degree polynomial with coefficients in it has a root in it, and find a huddling sequence of real algebraic numbers with no algebraic limit.

Show that both facts fail for the field of fractions and for the field of numbers $a + b\sqrt{2}$ with a and b fractions. For the real numbers that a ruler and compasses reach, exactly one of the two fails; which one, and why does the other survive?

WHERE WAS THE FIELD USED?

The theorem that a finite group G of automorphisms of a field K has exactly as many members as the dimension of K over its fixed field was proved without saying which field K is. Its proof was two counts, one showing the group has no more members than the dimension and one showing it has no fewer. Each count came down to the same move, the first directly and the second through a fact it relied on: suppose something fails, take the shortest failing case, and build from it a shorter one, which is impossible. Both used that a system of linear equations with more unknowns than equations has a solution other than zero, shown by a count of dimensions, and one used that distinct automorphisms are independent, shown by shortening a vanishing combination until it contradicts itself. Neither mentions the fractions, the real numbers, whether some sum $1 + 1 + \dots + 1$ is 0 in the field, or how many elements it has, and neither mentions a polynomial. So the theorem holds for every field whatever.

It can be tried on a field that is not a field of numbers. Let K be $Q(t)$, the ratios of polynomials in a letter t with fraction coefficients, built as the fractions were built from the whole numbers. Replacing t by $1 - t$ in every ratio is an automorphism of K , and so is replacing t by $1/t$; each undoes itself, and they do not commute. Together they generate six substitutions, t , $1 - t$,

$1/t, 1/(1-t), (t-1)/t$ and $t/(t-1)$, closed under composition: a group of six automorphisms of K . The ratio

$$j = (t^2 - t + 1)^3 / (t^2(t-1)^2)$$

is unchanged by both generators, so it lies in the fixed field, and by the theorem K has dimension exactly 6 over that fixed field. Which field is it? The six images of t are roots of the polynomial $(X^2 - X + 1)^3 - j \cdot X^2(X-1)^2$, which is monic of degree 6 with coefficients in $Q(j)$; so K , which is generated over $Q(j)$ by t , has dimension at most 6 over $Q(j)$. The fixed field contains $Q(j)$, so by the tower law the dimension of K over $Q(j)$ is 6 times the dimension of the fixed field over $Q(j)$. That product is at most 6, so the second factor is 1, and the two fields are equal. The fixed field is $Q(j)$. So K , which is $Q(j)(t)$, has dimension 6 over $Q(j)$, the minimal polynomial of t over $Q(j)$ has degree 6, and the monic polynomial of degree 6 above, which has t as a root, must be that minimal polynomial. It is therefore irreducible over $Q(j)$, and nobody had to factor anything to learn it.

Which element generates the fixed field of the three substitutions $t, 1/(1-t)$ and $(t-1)/t$, and what quadratic over $Q(j)$ does it satisfy?

The correspondence built on that theorem is narrower: it concerns splitting fields of polynomials with no repeated root. Where does the narrowing happen?

Separability is the requirement that a polynomial have no repeated root. It was introduced together with a field built to show that it can fail: over the field of ratios of polynomials in t with coefficients in $\mathbb{Z}/2$, the polynomial $x^2 + t$ is irreducible and has a single doubled root. In the proofs that followed, the hypothesis is used in one place: the count of the ways an isomorphism extends upward. The count needs an irreducible factor of degree d to have exactly d distinct roots in the field above, so that a root can be sent to exactly d places. For $x^2 + t$ over the field of ratios there is one place instead of two, and the count fails there, taking with it everything built on the count. Over the fractions, and over any field in which no sum

$1 + 1 + \dots + 1$ is 0, every irreducible polynomial has distinct roots, so the hypothesis holds without being asked for. Working only in those fields, the count never fails, and so no example there shows at which step of the proofs the hypothesis is needed.

* * *

Where were the real numbers used in the proof about the quintic? In placing its five roots in the plane of pairs. The derivative showed that the polynomial has at most three real roots; that rests on the average-slope theorem. The values at the turning points were compared with zero to show it has at least three; that rests on the theorem that a continuous function with values of both signs takes the value zero, proved by repeated halving. The other two roots came from a square root of a positive real number, after the three real roots were divided out and the square completed in the quadratic that remained. With the roots placed there, conjugation, which respects sums and products and fixes the real coefficients, sends each root to a root, so it is an automorphism of the field of the roots. It fixes the three real roots and exchanges the other two, so it is one element of the group, a transposition.

Nothing else in the argument mentions a real number. The five-cycle came from irreducibility, which makes 5 divide the size of the group, and from the count of lists of group elements, which then supplies an element of order 5; among rearrangements of five roots, such an element is a five-cycle. The whole group of rearrangements came from a picture of five points and some edges. The failure of every chain came from three-cycles being commutators. So the statement that a polynomial whose group consists of all rearrangements of five roots is not solvable by radicals contains no analysis at all. The real numbers were an instrument for finding such a polynomial. Other instruments exist. One reduces a polynomial modulo several primes and reads rearrangements off the ways it factors, but it rests on Dedekind's theorem. Take a polynomial with whole-number coefficients and leading coefficient 1, and

a prime modulo which it has no repeated factor. Then the degrees of its irreducible factors are the cycle lengths of some member of its group.

Which steps used the number 5 itself, rather than its being a prime at least 5? For a prime p at least 5, is every polynomial of degree p that is irreducible over Q and has exactly $p - 2$ real roots unsolvable by radicals? Is there an irreducible polynomial of degree 7 over Q with exactly five real roots?

* * *

Was the fundamental theorem of algebra used, the theorem that every polynomial has a root in the plane of pairs? It was proved, by finding where the size of a polynomial is smallest, and none of the arguments about fields needs it. Every root they required was manufactured. Divide the polynomials over a field by an irreducible one: the remainders form a field in which that polynomial has a root, with no analysis and no plane. Splitting fields, including the field that closed up a radical tower, came from repeating that construction. The radical theorem does ask for roots: it was stated with every field inside the plane of pairs, and the group of the quintic was computed from its five roots as numbers of the plane. Those roots were found directly, without the fundamental theorem: halving gave its three real roots, and dividing them out and completing the square gave the other two, which places all five in the plane of pairs.

* * *

Which single step of the whole argument carries the most weight? The candidates are the tower law, which multiplied dimensions and ruled out doubling the cube; the count that forces an equation, which produced minimal polynomials from lists of powers too long to be independent; the fixed-field theorem; and the correspondence. The correspondence rests on the fixed-field

theorem and on the count of extensions, which makes a Galois extension Galois over every field between. The fixed-field theorem is two counts, and one of them rests on the descent that shortens a vanishing combination of automorphisms. Without that descent the fixed-field theorem has no proof, and neither do the correspondence, the theorem on normal subgroups, the polygons built from $\varphi(n)$, the translation of radicals into groups, or the quintic.

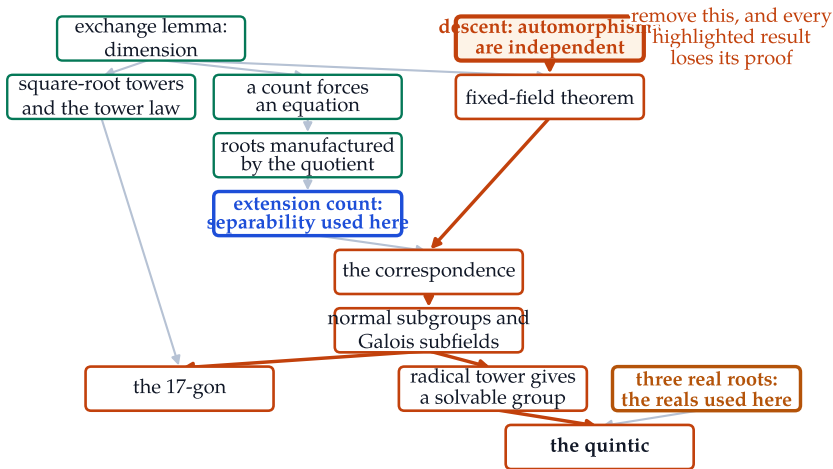


Figure 77.1. The main results from linear maps to the quintic, each drawn below what its proof uses. Everything from the fixed-field theorem down depends on the descent that makes distinct automorphisms independent. Separability is consumed at a single node, the extension count, and the real numbers at a single node, the count of the quintic's real roots, which feeds nothing but the quintic. Assumes: that each arrow records a use actually made in a proof, which is what the text checks by reading the proofs back; arrows for facts used only in examples are left out

For every n , the group of all rearrangements of n things is the group of some polynomial over $\mathbb{Q}$. That is a consequence of Hilbert's irreducibility theorem of 1892. Take n independent letters and the polynomial of degree n whose roots they are, with coefficients, up to sign, the elementary symmetric expressions in the letters. Rearranging the letters gives $n!$ automorphisms of the field they generate, each leaving the coefficients alone, and a

member of the group is settled by what it does to the n roots, so the group of this polynomial over the field its coefficients generate is all $n!$ rearrangements. Hilbert's theorem then lets the coefficients be replaced by suitable whole numbers without shrinking the group. Whether every finite group is the group of some Galois extension of Q is not known.

THE FIFTH SOLID

Near the start of all this a count was made of corners less edges plus faces, and it came out 2 on every solid it was tried on. That one count was enough to list every solid whose faces are alike and whose corners are alike. The list stopped at five, and the fifth on it, with twenty triangular faces, was named as a solid not yet met: the icosahedron. Near the end, an equation of the fifth degree was found to have no solution in radicals. The reason was not a formula but a group: the sixty even rearrangements of five things. No chain of commuting collapses gets below that group. Those two facts stand at opposite ends of one long argument, and they are about one object. The solid the count permitted and the group the quintic ran into are the same thing. That is the claim, and it is proved here.

* * *

The icosahedron has twenty faces and twelve corners; the dodecahedron has twelve faces and twenty corners. The centres of the twenty faces of an icosahedron are the twenty corners of a dodecahedron, and the centres of the twelve faces of a dodecahedron are the twelve corners of an icosahedron, so a rotation carrying either solid onto itself carries the other onto itself as well. The two solids have one group of rotations, and it is easier to see on the

dodecahedron, whose faces are pentagons.

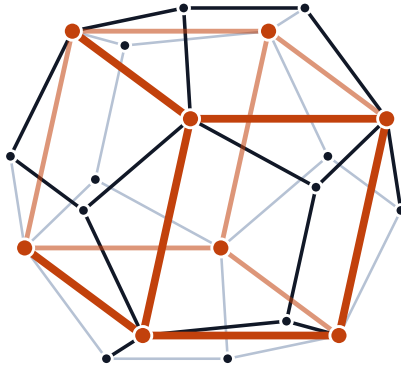
How many rotations? A rotation carries a face to a face, and any of the twelve faces can be the image of a chosen one. The rotations that leave that face where it is are the turns about the axis through its centre, and there are five of them, the pentagon having five positions. So the rotations number $12 \cdot 5$, which is 60, by the count that pairs an orbit with a stabiliser. They can also be listed. Through the centres of opposite faces run 6 axes, each carrying 4 turns other than the identity, of order five: 24. Through opposite corners run 10 axes, each carrying 2 turns, of order three: 20. Through the mid-points of opposite edges run 15 axes, each carrying one half turn: 15. With the identity that is $24 + 20 + 15 + 1$, which is 60. No two of these are the same rotation, since a rotation other than the identity is settled by its axis and its angle, and there are only sixty rotations in all; so the list is complete and every rotation is of one of these four kinds.

Now the solid itself, in coordinates. Let φ be the pentagon's proportion, $(1 + \sqrt{5})/2$, whose reciprocal is $\varphi - 1$. The twenty points

$$(\pm 1, \pm 1, \pm 1), (0, \pm 1/\varphi, \pm \varphi), (\pm 1/\varphi, \pm \varphi, 0), (\pm \varphi, 0, \pm 1/\varphi)$$

are the corners of a regular dodecahedron. Every one of them lies at distance $\sqrt{3}$ from the origin, and the thirty edges are the pairs of them at distance $2/\varphi$: the point $(1, 1, 1)$, for instance, is at that distance from $(0, 1/\varphi, \varphi)$, from $(1/\varphi, \varphi, 0)$ and from $(\varphi, 0, 1/\varphi)$, its three neighbours, and from no other corner. The eight points $(\pm 1, \pm 1, \pm 1)$ are the corners of a cube. Take any of the cube's twelve edges, such as the one from $(1, 1, 1)$ to $(1, 1, -1)$. Its two ends lie on a single face of the dodecahedron, but it is not an edge of that face, so it is a diagonal of the face. The twelve edges of the cube lie one in each of the twelve faces. A cube sits inside the dodecahedron with one diagonal drawn in every pentagon.

Turn the whole solid by a fifth of a turn about the axis through the centre of a face. The cube goes to another cube, inscribed in the same way, with one diagonal in every face. Turning once, twice,



one of the five cubes: one diagonal in each of the twelve faces
 a fifth of a turn about a face axis carries it to the next

Figure 78.1. The dodecahedron with corners $(\pm 1, \pm 1, \pm 1)$, $(0, \pm 1/\varphi, \pm \varphi)$ and the two cyclic rearrangements of the last, and inside it the cube on the eight corners $(\pm 1, \pm 1, \pm 1)$. Each edge of the cube is a diagonal of one pentagon, one diagonal in every face. A fifth of a turn about a face axis carries this cube to the next of the five, and every rotation of the solid rearranges the five cubes; those sixty rearrangements are the sixty even ones. Assumes: a view turned so that no two corners coincide on the page; the drawing is a flat projection, so equal edges do not come out equal in length, and the back edges are drawn lighter to show which is nearer

three times and four times, and not at all, gives five cubes. No two of them are the same. Look at the face the turn is about: that pentagon has five diagonals, and each fifth of a turn carries the diagonal drawn in it to the next one round, so the five cubes draw the five different diagonals of that face.

Two facts about the five cubes come from counting their corners. Five cubes of eight corners make forty memberships among the twenty corners of the dodecahedron, and the memberships fall evenly. A cube with its corners among the twenty and its edges

along diagonals has edges of length 2, so the three neighbours of a corner lie among the corners at distance 2 from it. From $(1, 1, 1)$ those are six: $(1, 1, -1)$, $(1, -1, 1)$, $(-1, 1, 1)$, $(\varphi, 0, -1/\varphi)$, $(0, -1/\varphi, \varphi)$ and $(-1/\varphi, \varphi, 0)$. The three neighbours must also lie at distance $2\sqrt{2}$ from one another, the diagonal of a square of side 2, and the six fall into exactly two threes of that kind, the first three and the last three, with no point in both. The same computation at $(0, 1/\varphi, \varphi)$ gives the same answer, and changing the signs of the coordinates and shifting them round carries those two corners to all twenty, while it keeps the twenty in place and leaves every distance as it was. A cube is settled by one corner and its three neighbours, so at most two such cubes pass through any corner. Forty memberships among twenty corners at no more than two apiece is exactly two apiece: every corner lies in exactly two of the five cubes.

And any two of the five cubes share exactly two corners, a pair of opposite ones. Since each corner lies in exactly two cubes, it belongs to exactly one pair of cubes, the pair that both contain it. Each cube is centred where the solid is centred and holds, with any corner, the corner opposite it, so a corner and its opposite belong to the same pair of cubes. No pair of cubes shares two pairs of opposite corners: of two long diagonals of a cube, an end of one is joined by an edge to an end of the other, and two cubes sharing a corner and one of its neighbours would take the same three of the six at that corner and be one cube. So the ten pairs of opposite corners go to the ten pairs of cubes, one to each.

* * *

Here is the bridge. A rotation of the dodecahedron carries a cube inscribed with one diagonal per face to a cube inscribed the same way. At most two cubes of that kind pass through any corner, and two of the five already pass through each, so every cube inscribed that way is one of the five, and the rotation carries each of the five cubes to one of the five: every rotation rearranges the five cubes. Suppose two rotations rearrange the cubes in the same

way. Do the first and then undo the second: the result is a rotation that leaves every cube where it is. Such a rotation keeps any two cubes where they are, so it carries the two corners they share to themselves, and the line through those two corners to itself. There are ten such lines, one for each pair of cubes, and the rotation carries every one of them to itself. A rotation other than the identity carries few lines through the centre to themselves: the axis it turns about and, when it is a half turn, the lines square to that axis, which all lie in one flat plane. No flat plane through the centre holds more than two of the ten lines. So the rotation is the identity, and different rotations rearrange the cubes differently. Sixty rotations, sixty different rearrangements of five things.

Which sixty? A rearrangement of odd order is even. Suppose one of odd order n were odd. Doing it n times over, an odd number of odd rearrangements in a row, would give an odd rearrangement. But doing it n times gives the identity, which is even. So it cannot be odd. The 24 turns of order five, the 20 of order three and the identity therefore give 45 even rearrangements. The sixty rearrangements the rotations give form a group, and the even ones among them form a subgroup of it, whose size divides sixty and is at least 45. The only divisor of 60 that is 45 or more is 60. So every one of the sixty is even, and since five things have exactly sixty even rearrangements, the rotations of the dodecahedron and the even rearrangements of five things are the same group, one to one.

The rotations of the fifth solid are the sixty even rearrangements.

That the sixty cannot be taken apart was proved when the classes of that group were computed. A normal subgroup is made of whole classes, holds the identity, and has a size dividing 60, and no union of classes holding the identity has a size dividing 60, apart from the identity alone and the whole. Everything the group does, the solid does.

* * *

Corners less edges plus faces is 2. From that, on drawings a child could make, came the list of solids whose faces and corners are all alike, and the list stopped at five. The fifth is the icosahedron.

The fifth solid has sixty rotations, and they are the sixty even rearrangements of five things.

The sixty even rearrangements of five things cannot be taken apart: they have no normal subgroup but the identity alone and the whole.

An equation can be solved by radicals exactly when its group can be taken apart into commuting pieces, and the group of the general equation of the fifth degree contains those sixty, which no chain of commuting collapses gets below.

So there is no formula for the fifth degree. Not because one has not been found. Because the fifth solid exists.

Four steps, and everything that came before lies between the first and the last. For nearly three centuries after the cubic and the quartic fell, the best mathematicians alive searched for the formula of the fifth degree, and each of them was searching for the absence of a solid that was present the whole time, that Euclid had drawn, that a child can hold in one hand and turn sixty ways.

There is a second thing, and it is the reason to have come all this way rather than being told. Told at the start that the solutions of an equation of degree five are governed by the rotations of a twenty-sided solid, a reader could have made nothing of that sentence. Reached, the same sentence has everything that went into reaching it standing behind it and holding it up: the diagonal that refused to be a fraction, the descent that keeps finding a smaller one, the count of dots and edges, the paths reflected across a line, the eight ways of putting down a square, the fields stacked into towers, the classes of a group counted by hand. Each step was small. None was skipped. That is the only way anything in mathematics has ever been understood.

Which leaves the feeling, and it should be named. Not having a result. Relief. The thing that would not yield has a shape, the

shape was in hand all along, and the mind that now sees it is not the mind that opened these pages.

The cube has twenty-four rotations, and they rearrange its four long diagonals. Are all twenty-four rearrangements of four things reached?

The regular tetrahedron has twelve rotations, and they rearrange its four corners. Which twelve of the twenty-four rearrangements are they?

How many rotations of the dodecahedron carry one of the five cubes onto itself, and whose rotations are they?

A half turn about the midpoint of an edge gives an even rearrangement of order two of the five cubes. How many of the cubes does it leave where they are?

AFTERWORD

The mathematics has said its last word.

What remains is the reason for having followed it.

Most of the results in these pages will eventually fade. A formula can be forgotten. A theorem can become hazy. Even a proof can disappear from memory.

What is harder to lose is the way an argument behaves when it stops working.

The diagonal did not produce the fraction that had been expected of it. The count of a solid did not stay at the number that had seemed inevitable. A definition failed to describe the thing it was supposed to describe. A construction reached a boundary. An equation admitted no method of the familiar kind.

Again and again, the useful question was not which procedure came next.

It was what the failure had revealed.

Sometimes the object had been misunderstood. Sometimes the argument had hidden an assumption. Sometimes the definition had been too small for the phenomenon. Sometimes the right response was to change the object being studied, or the language in which it was being described.

That habit is worth more than the results themselves.

So is another one.

A claim should be earned. A proof should say what it depends on. A calculation should be checkable. A counterexample should do more than destroy an assertion: it should explain what the assertion was missing. And when an argument works, the question does not end with the proof. What else does the same idea touch?

That question is how mathematics grows.

The edge of the subject is not the place where the methods stop. It is the place where new methods begin.

Nothing here guarantees that every unanswered question has an answer. The final pages leave problems standing. Mathematics does too. There are questions in these pages whose answers are unknown, and there will be others that have not yet been asked.

That is not a failure of the subject.

It is part of what the subject is.

A procedure can carry a person only as far as the procedure has been carried before. An understanding of how arguments are made gives somewhere to go when the procedure runs out.

That is what these pages were trying to build.

Not a storehouse of mathematics, but a way of meeting it.

WHAT IT COULD HAVE DRAWN ON

The sources listed here are the library this work could have learned from.

They contain proofs, constructions, examples, problems, historical arguments, definitions, competing approaches, and different ways of seeing the same mathematical objects. Some offer a proof, some an example; some expose an assumption; some change the order in which an idea can be understood.

The list is not a catalogue of what was used where. Its purpose is not to tell the reader what to believe, but to show where the mathematics behind these pages can be followed further.

The pages before this one are a single development: a new arrangement, written out of that larger mathematical world into one continuous line.

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